APPLICATIONS OF NEXT GENERATION BIOSURFACTANTS IN THE FOOD SECTOR

Edited by
Inamuddin
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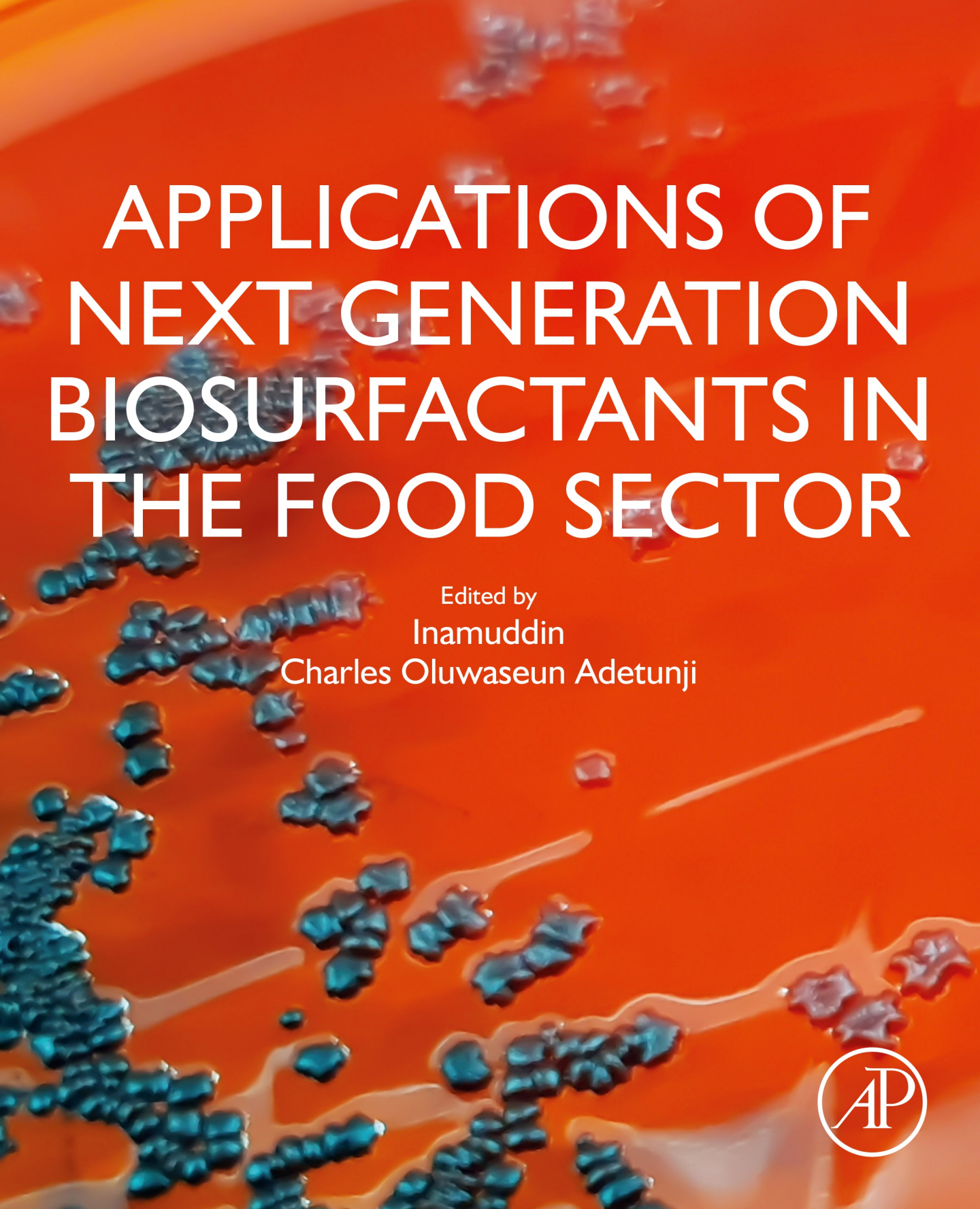

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CHAPTER

1 Characterization and screening of biosurfactant from microorganism with a wide application in food industries

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1 Introduction

Food and food products are very essential energy source for humans to lead a healthy life. Each and every one in the society required food for consumption with good shelf life and high quality parameters. Cappitelli et al. [1] stated that according to World Health Organization, food-borne illness causes are very critical diseases due to its toxicity nature. Many studies revealed that food and water are the major sources for food-borne illness in humans, due to the presence of pathogenic microbes, toxins produced by pathogens, and synthetic chemical in them [2,3]. Nitschke and Silva [4] reported that species of *Escherichia coli, Staphylococcus*, and *Klebsiella* act as primary microbial agents, which cause food-borne illness in humans. Therefore, food processing operation, which ensures the quality and shelf life of food, becomes the most important process in food industrial sector. Most of the food industries started using chemically synthesized food additives, which help to modify the physicochemical and biological properties of food products during manufacturing, processing, packing, and storage to prevent the intoxication of food with pathogens and toxic substances produced by pathogen [5]. Surfactants compounds are surface active compounds, exhibit both hydrophobic (nonpolar) and hydrophilic (polar) nature (Fig. 1) and help to produce oil-in-water or air-in-water-based colloidal emulsion, which helps in decreasing the particle surface and interfacial tensions; this property magnitudes the use of the surfactants in board spectrum of applications [6].

In food-based industries, the surfactants are generally used as emulsifier. Emulsifier is termed as emulgent, which functions as a frame between two immiscible liquids and helps them to be mixed into stable emulsions. Sorbitan esters,

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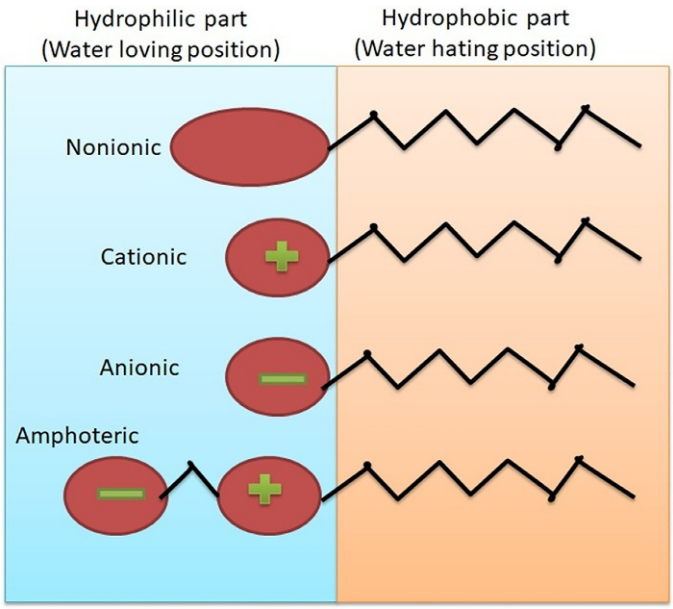

FIG. 1

Surfactant structure and classification.

ethoxylates, glyceryl monostearate, carboxymethyl cellulose, and sucrose esters, etc., (Table 1) are common synthetic emulsifiers used in food application, and this food emulsifier has two parts: one is hydrophilic part (glycerol, sorbitol, sucrose, propylene glycol, or polyglycerol) and the other is lipophilic part (made up of fats and oil) [7]. Food and Drug Administration and other global regulatory authorities implementing more constraint for the use of synthetic emulsifier, and due to environmental and health concern, most of the consumers do not prefer synthetic emulsifier in food processing.

Desai and Banat [8] and Chakrabarti [9] reported that biosurfactant (BS)/emulsifier derived from microbial source is one of the best alternatives to synthetic surfactant used in the food processing industries due to the following reasons:

- **Surface and interface activity**

  The surface activity is a property possessed by certain surfactants or BS substances to influence the surface tension (SFT) of the liquids. Less amount of BSs is enough to influence a surface activity when compared with chemical surfactants because BSs have CMC (critical micelle concentration) 10–40 times lesser than synthetic surfactants [8].

Table 1 Synthetic surfactant commonly used in food industries.

S.No.	Name of synthetic surfactant	*E*-number
1	Lecithins	E 322
2	Polyoxyethylene sorbitan esters(Tweens)	E 432–436
3	Mono- and diglycerides of fatty acids	E 471
4	Acetic acid esters of Monoglycerides	E 472a
5	Lactic acid esters of Monoglycerides	E 472b
6	Citric acid esters of Monoglycerides	E 472c
7	Mono- and diacetyl tartaric acid esters of Monoglycerides	E 472e
8	Sucrose esters of fatty acids	E 473
9	Polyglycerol esters of fatty acids	E 475
10	Polyglycerol polyricinoleate	E 476
12	Propane-1,2-diol esters of fatty acids	E 477
13	Sodium stearoyl-2-lactylate	E 481
14	Calcium stearoyl-2-lactylate	E 482
15	Sorbitan fatty acid esters	E 491–495

- **Tolerance to pH and temperature**

  The surface activities of microbial-derived BSs are more stable and have tolerance toward environmental parameters such as pH and temperature because most of them are isolated from hyperthermophile [10], thermophile [11], and psychrotroph microorganisms [12]. The BSs produced from the microorganisms mostly show stable surface activity toward the pH range of 2–12 according to optimal range of microorganism of interest [11,12]. Cameotra and Makkar [13] explained that current commercial and industrial demand of BSs produced from extremophiles microorganisms is increasing day by day as it can show stable surface activity at extreme environmental (temperature/pH) behaviors, which commonly prevail in industrial applications.
- **Tolerance to ionic strength**

  The ionic strength of a solution represents the measure of the concentration of ions in particular solution. BS produced from *Bacillus licheniformis* JF-2 shows a stable surface activity in a solution having 50 g/L of Nacl and 25 g/L of calcium salt concentration [11]. Similarly, another researcher by Nitschke and Pastore [10] stated that microorganism *B. subtilis* LB5a producing BS shows a stable surface activity in a solution having 20% of NaCl concentration.
- **Biodegradability**

  Gomes and Nitschke [13] stated that biodegradability nature of microbial-derived surfactants or emulsifiers can be used as best replacement for chemical or synthetic surfactants. The change in environment and its related health issue made the customers to prohibit the use of chemical-based products, and their attention changes toward sustainable product in all types of application; this

behavior change among the customers opens a commercial market for the low-cost green additives such as BSs over chemical or synthetic surfactants [14]. Mulligan et al.'s [15] production of BSs using renewable substrates or food processing wastes or agricultural waste made them to fulfill the standard of green chemistry and become one of the important research tools for innovation, ecofriendly, and sustainable application.

- **Low toxicity**

  Toxicity of synthetic surfactants is very high when they reacts with water and also create bioaccumulation in animals and other living organism. The chemical toxicity of surfactants shows LC50 (lethal concentration) to kill *Photobacterium phosphoreum*, and to obtain the same result of killing, 10 times lesser concentration of BSs from microbial source is only required [16]. Flasz et al. [17] demonstrated an experiment to compare the toxicity and mutagenicity profile of BSs and synthetic surfactants showing that BSs resulted with low toxicity and non-mutagenic property when compared with synthetic surfactants. BSs from *Candida bombicola* used in food industrial application show very low toxicity and non-mutagenic property [18].

- **Emulsifying and demulsifying ability**

  Emulsification is a process that results in heterogeneous system called emulsion; for example, one immiscible liquid droplet size of 0.1 mm spread to another liquid will result in oil-in-water or water-in-oil emulsion. The stability of formed emulsions is very low, but which can withstand for years by adding food additives such as BSs [19]. Microorganism *Candida lipolytica* produced a BS (liposan), which can be used for making oil-in-water emulsion and also ensure a stability emulsion for many months or years [20]. In most of the cases, one of food industrial processes produced emulsion. To improve and develop a particular functional property is by another subsequent process called demulsification. Generally, demulsification process commonly used in food processing application is related to oil/fat products and also helps in waste treatment process [21].

- **Antimicrobial activity**

  Antimicrobial activity is nothing but killing microbes (bacteria, fungi, yeast, algae, and viruses) or inhibiting the microbial activity of pathogens. Most of the microbial-derived BSs have this antimicrobial activity (antibacterial, antifungal, algicidal, and antiviral) property, which helps in food processing, transporting, and storage of food materials to a large extent to avoid food spoilage by pathogens (Fig. 2) [22–24].

- **Antiadhesive agents**

  Adhesion is nothing but attachment of microorganism on the surface of food materials. Microbial surfactant majorly helps to prevent adhesion of pathogen on food materials by its antiadhesive property (Fig. 3A and B). The microbial BSs can be used in varying hydrophobicity of the surface, which prevents the adhesion effect of microbes over the surface [25,26]. Hood and Zottola [27] BSs controlling the adherence of microorganisms to food contact surfaces is more essential process in providing safe and high-quality food stuffs to customer.

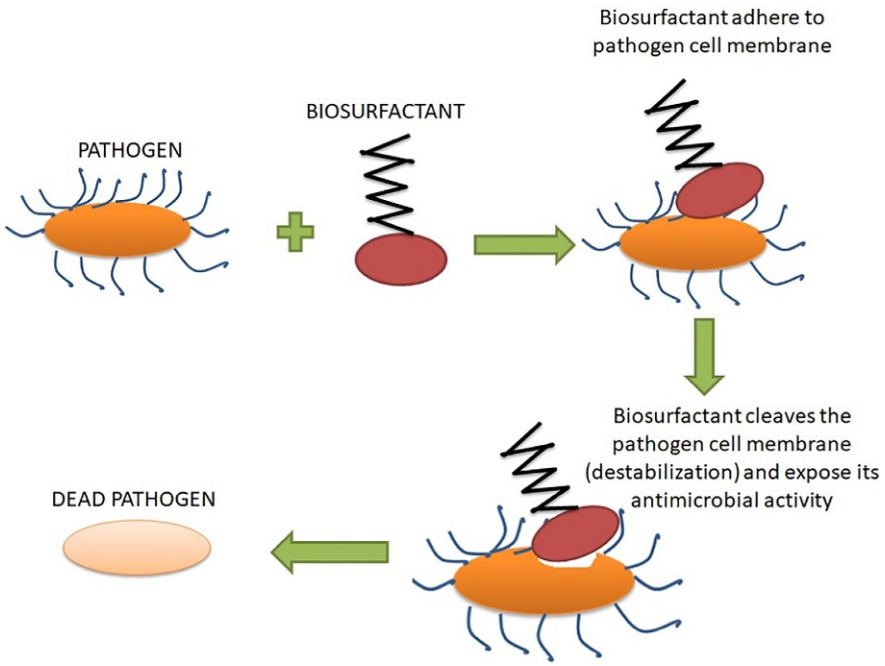

FIG. 2

Antimicrobial activity of biosurfactants.

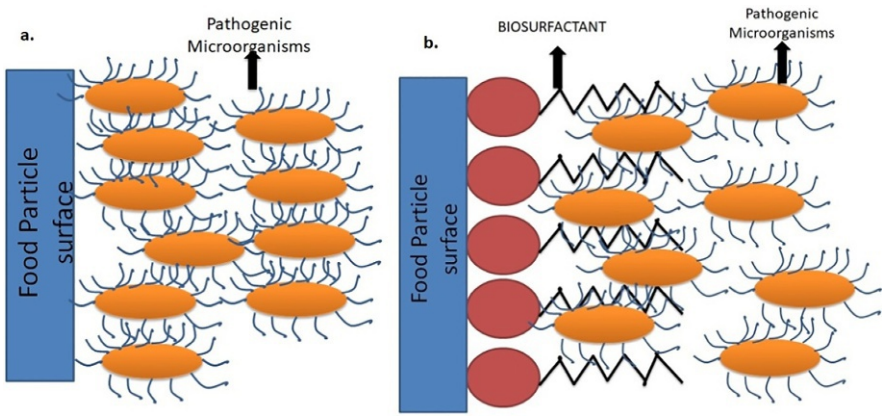

FIG. 3

(A) Adhesion of microbes to the surface in absence of biosurfactants, (B) Anti-adhesion of microbes in the presence of biosurfactants.

All the above-mentioned properties possessed by BSs are derived from microbial source and prove that these can be used in different sectors of food industry. This chapter is mainly formulated to discuss about the methods of characterization and screening of BSs from microbial sources, which are widely used in different sectors of food industry.

2 Classification of biosurfactants

In general, polarity is a major property considered for the classification of synthetic surfactants, whereas the BSs are categorized according to composition of chemical compound and microbial source. BSs are produced from three major classes of microorganism: bacteria, fungi, and yeast [28–31]. Fig. 4 shows the classification of BSs [32] according to the chemical composition.

BSs' production and application obey the concept of green chemistry, which make them to be contemplating under the class of green surfactants or environmentally friendly surfactants [29]. Recently, many researchers working on zero-waste concepts started the production of BSs using preconsumer organic waste as substrate. Preconsumer organic waste is from food processing industries, residential fruit and vegetable peels, commercial organic peels, and agricultural activities, etc., and it's considered as a renewable source or feedstock for the production of various BSs, which have a wide variety of applications related to food industry [9,15,31]. BS is generally used as treatment or cleaning agent of food contact surface, and in some cases, it is utilized as food additive in industrial food processing. Recently, researchers reported the production of BSs from waste material as substrate added an advantage in reduction of waste treatment and management cost [32]. Fig. 5 shows the general flow of BSs' production from different environmental samples.

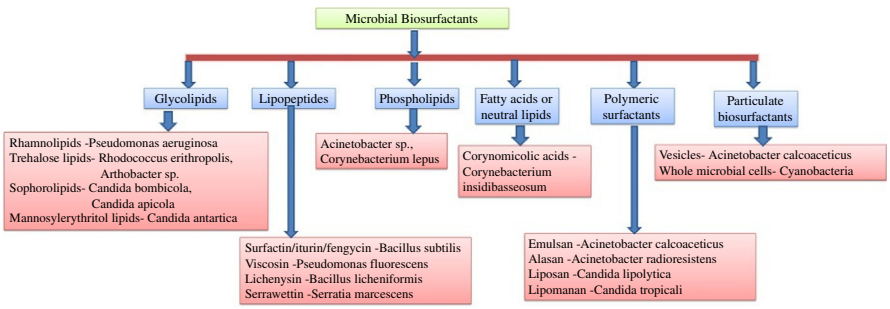

FIG. 4

Classification of biosurfactants and their microbial sources.

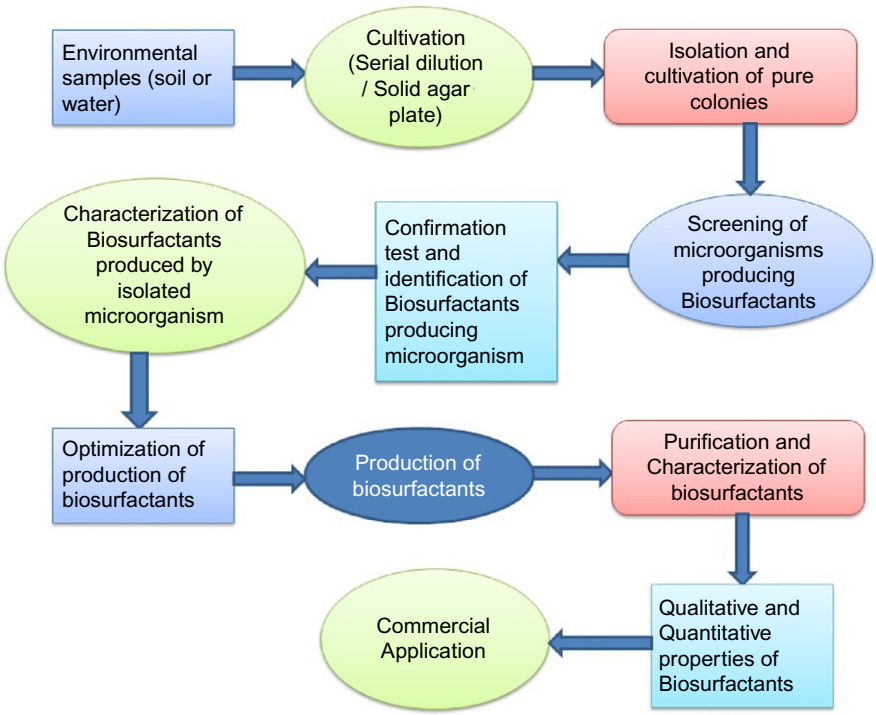

FIG. 5

Process flow diagram of biosurfactants production from different environmental samples.

3 Screening of microorganism producing biosurfactants

The BSs have immense applications in food formulation and food processing. The sampling and isolation of BSs-producing microorganisms are the two important bases for screening and selection of potent microbial strain.

3.1 Sampling and isolation

In general, many researchers collected oil-contaminated soil, water sample from petro-industrial effluent treatment plants, contaminated soil from edible oil mills, activated sludge, costal area and petrol pumps, etc. [33]. The samples from particular location were collected, and the same were preserved in a sterile flask. After reaching the laboratory, 10 g of soil/sludge or 10 mL water/effluent is taken in a flask and diluted with 10 mL of distilled water. Then isolation of pure or single colony of potential BSs-producing microbial flora was isolated using serial dilution method using nutrient agar solid medium incubated at 30°C for 1–5 days [34,35]. The potent microbial strains were isolated from repetitive streaking process using enrichment medium such as M9 minimal salt medium [36] or lipolytic medium [37], and each

distinct morphological strain from group of different population is isolated and stored in nutrient slants at 4°C. Pure culture was isolated, named, and cryopreserved in nutrient medium having 15% glycerol at −80°C [38], which can be used for the screening of potent strains producing BSs.

3.2 Screening and detection of potent strains producing biosurfactants

The screening of potent strains is generally classified into qualitative and quantitative assay methods. Different methods are therefore to identify, screen, and quantify the potential strains producing BSs [39]. Every method used for screening has its own advantages and disadvantages, which are deliberated in the following discussion.

3.2.1 Oil-spreading test

Morikawa et al. [40] state that surface activity of potent organism is found using oil displacement activity. In oil-spreading test method, 20 mL of distilled water was taken into a Petri dish with diameter of 25 cm and then to the water surface 20 μL of crude oil was added. After the abovementioned process, 10 μL of culture broth (cell-free test sample) was then added to the Petri dish. If potent strain of BSs is present in the culture broth, oil displacement activity occurs to form free clearing zone. Diameter of free clearing zone was measured to identify the surfactant activity of give strain. In this process commonly distilled water acted as a negative control and Triton X-100 was used as the positive control in oil-spreading test method [41].

3.2.2 Blood agar method

This qualitative method of screening BSs is also called as hemolytic assay. Carrillo et al. [42] used solid medium (Luria agar with 5% fresh whole blood) to isolate the potent BSs. The streaked plate is incubated at 37°C for 48 h [43].Youssef et al. [43] reported that in hemolytic assay method, the BSs-producing microbial isolate was identified by the formation of clear zone in blood agar plate and the diameter of the clear zone used to confirm the BSs' production [44]. The result was noted as follows (Table 2).

Table 2 Blood agar method.

S. No.	Isolate number[a]	Result(diameter of clear zone in cm)	Remark
1	AR1	Improper zone formation	Incomplete hemolysis
2	AR2	No zone formation	No biosurfactants activity
3	AR3	>1 to 3<	Moderate biosurfactants activity
4	AR4	>3	Shows high biosurfactants activity
5	AR5	<1	very less biosurfactants activity

[a]*AR1–AR5: Isolate from sludge sample.*

Table 3 Drop collapse method.

S. No.	Isolate number[a]	Result	Remark
1	AR1	+	Incomplete collapse spreading—very lesser potent
2	AR2	–	No spreading—absence of BS producing ability
3	AR3	++	Partially collapse spreading
4	AR4	++++	Complete collapse spreading—higher ability to produce BS
5	AR5	++	Partially collapse spreading

[a]*AR1–AR5: Isolate from sludge sample.*

3.2.3 Drop collapse method

In this drop collapse method, 2 mL of mineral oil was taken in micro titer plate with 96 wells. After 1 day of equilibration, 5 mL of culture sample isolated was added on to the surface of mineral oil [45]. Triton X-100 will be commonly used standard solution for comparison. The drop of culture collapse activity and spreading on to the oil surface show that the presence of BSs-producing ability and spreading was noted to determine the potent strain producing BSs (Table 3).

3.2.4 Cetyl trimethyl ammonium bromide CTAB agar method

This method basically helps to screen anionic BSs-producing microbes. In this method, 0.2 g/L of CTAB salt along 0.005 g/L of methylene blue dye was added to agar medium to formulate a new modified mineral agar medium specially used to screen anionic BSs-producing microbes. The 24-h isolated culture broth was added to modify mineral agar medium, and the formation of bluish halo layer around the colonies of microbes shows the presence of BS-producing capability [45].

3.2.5 Penetration assay

In this method, 96-well microplate well was filled with 100 µL of hydrophobic paste (oil + silica gel), over to that 30 µL of oil was added. Then 80 µL of culture with 20 µL of Safranin (1% by volume) was mixed, and 30 µL of above mixture was added over the well filled with hydrophobic paste and kept in an incubator at 300°C. If the culture contains BSs producing microorganisms, then it has the ability to penetrate the oil layer, which changes the color of the solution from red to cloudy white color [45], and the interpretation is shown in Table 4.

3.2.6 Bath test

Rosenberg et al. [46], devised a methodology to determine the hydrophobicity of BS-producing microorganisms by its ability to adhere to the hydrocarbon. The culture broth was centrifuged for 10 min at 10,000 rpm, and the pellet (biomass) was separated. The obtained pellet was washed using sodium or potassium phosphate buffer and then stored in beaker. From the beaker 2 mL of biomass was added to 10 µL of crude oil, and the mixture was vortexed for 10 min, kept for 1 h, after that aqueous

Table 4 Penetration assay.

S. No.	Isolate number[a]	Result	Remark
1	AR1	+	Incomplete penetration—slightly change in color
2	AR2	–	No penetration—absence of BS producing ability
3	AR3	+	Incomplete penetration—slightly change in color
4	AR4	++	Complete penetration and complete change of color to cloudy white—higher ability to produce BS
5	AR5	+	Incomplete penetration—slightly change in color spreading

[a]*AR1–AR5: Isolate from sludge sample.*

solution was separated. The optical density of the abovementioned aqueous solution was determined by using UV–Visible Spectrophotometer. The percentage of biomass adhered to crude oil was measured using the following formula

$$\% \text{ of Biomass adhered to crude oil} = \left\{1 - \left(\frac{OD_c}{OD_i}\right)\right\} * 100$$

OD_C = OD of aqueous solution after crude oil mixing and OD_i = OD of initial biomass in buffer solution.

3.2.7 Surface tension (SFT) measurement/Tensiometeric measurement of SFT

SFT is a common method to screen BSs-producing microorganism. Separated biosurfactant sample from biomass was taken, and the SFT was measured using DuNouy ring-type tensiometer. The measured SFT (dynes/cm) is lesser than that of the distilled water and medium used confirms the presence of biosurfactant strain.

3.2.8 Emulsification assay technique

Separated biosurfactant sample from biomass was taken and to that crude oil, Tris buffer (pH 8) was added in a 50 mL centrifugation tube. Another centrifugation tube with Triton X-100 was taken, and it was used as a positive control. The abovementioned centrifugation tubes were vortexed well for 1 min, and the mixture was made to stand quite for 20 min. Then for the abovementioned solution, OD value was under 610 nm using spectrophotometer. It is easy to screen the BSs-producing biomass by comparing the result with positive control [47].

3.2.9 Emulsification index

This method used to find the stability of BSs produced [48]. In this method, kerosene was added to biosurfactant sample (supernatant) in centrifugation tube. Later the above centrifugation tubes were vortexed well for 1 min and the mixture was made to stand quite for 24 h. The E24 is found using the following formula

$$E24 = \frac{\text{Height of emulsion layer in centrifugation tube}}{\text{Total tube height}} * 100$$

3.2.10 High-throughput biosurfactants-producing microbial screening

In this method, microtiter plated was taken, and each well microbe was grown with glucose as substrate for BSs-producing microorganisms screening. The pure culture producing microorganism was separated from mixed culture by serial dilution (MOST PROBALE NUMBER method-MPN), and simultaneously, the supernatant from the microtiter plated well was assayed for quantification of BSs produced by positive control plates at fast rate with less quantity of sample [47].

4 Identification and characterization of biosurfactants

4.1 Biochemcal characterization methods

Biochemical characterization of BSs-producing microbes helps to identify the classification of microorganism it belongs to (Table 5).

Table 5 Biochemical characterization methods.

S. No.	Method	Inference
1	Catalase—Microorganism able to produce catalase or not tested using chemical hydrogen peroxide as substrate	Positive: Observation of rising bubbles Negative: No Observation of rising bubbles
2	Citrate test—Microbial consume of citrate or not using chemicals Sodium citrate as substrate and indicator—Bromothymol blue	Positive: Change in medium green color solution to blue color Negative: No Change in medium green color solution to blue color
3	Urease test Microorganism able to produce catalase or not tested using chemical urea as substrate	Positive: Change in light yellow color solution to pink color Negative: No Change in yellow color solution to pink color
4	Methyl red test Microorganism able to produce acid or not after 3 days of incubation using glucose phosphate broth as medium	Positive: Change in yellow color solution to red color Negative: No Change in yellow color solution to red color
5	Nitrate test: Microorganism able to produce nitrate reductase or not tested using chemical nitrate as substrate	Positive: Formation of pink color (nitrate reduced) Negative: No formation of pink color (nitrate not reduced)

Continued

Table 5 Biochemical characterization methods. *Continued*

S. No.	Method	Inference
6	Indole test: Microorganism able to breakdown the indole (tryptophan) or not using tryptone broth as medium for growth. Kovac's reagent is used as a indicator solution	Positive: Within Few minutes change of color to cherry red Negative: no change of color to cherry red after giving reaction time 30 min
7	Gelatin test: Microorganism able to proteolytic enzyme or not tested using chemical gelatin as substrate	Positive: Hydrolysis happens—Media is liquefied Negative: No liquefaction occur
8	Starch hydrolysis: Microorganism able to use starch as substrate for carbon source or not	Positive result: Presence of clear halo surrounding colonies Negative result: Absence of clear halo surrounding colonies

4.2 Molecular-level identification of biosurfactants-producing biomass strains

In common, 16s rRNA method was used to identify the BSs-producing biomass strain. First the DNA of the strain producing biosurfactants was separated and amplified using polymerase chain reaction (PCR) methodology. The sequence of DNA was produced using DNA sequencer. Then by BLAST examination of the nucleotide sequence 16S rRNA gene region identified, phylogenetic tree was constructed by comparing the sequence present in NCBI (nucleotide sequence Database) and thus the strain/genus of BSs-producing biomass can be identified [48].

4.3 FTIR and NMR analysis

The functional group and molecular structure of biosurfactants can be characterized by using FTIR and NMR analysis. For example, Sethupathy et al. stated that absorption peaks found in FTIR showed the existence of rhamnolipid functional groups in BSs sample produced by *Pseudomonas pachastrellae* RW43 (screened and isolated strain), and also stated that by analyzing the chemical shift using NMR the molecular structure was determined [49].

5 Conclusion

Nowadays, BSs production from microorganisms is increasing day by day because of its significant application in various fields such as food technology, biomedical, agricultural, and environmental processes in a sustainable way. The review of different biosurfactants analysis, characterization methods show the important points while

selecting the methods for screening and characterization of BSs-producing microorganism, which are as follows: (1) high accurate result, (2) less sample volume requirement, (3) fast and rapid analysis, (4) less expensive, (5) less space for analysis. From the overall earlier reports, the recent development in a microplate and high-throughput screening process satisfies the abovementioned criteria except for the capital cost of the analysis. So, more research work is significantly required according to the current sustainable development principle to overcome the abovementioned disadvantages in the production, characterization, and identification of various BSs.

References

[1] F. Cappitelli, A. Polo, F. Villa, Biofilm formation in food processing environments is still poorly understood and controlled, Food Eng. Rev. 6 (2014) 29–42.

[2] K. Sanchita, P. Pritisnigdha, Production and functional characterization of food compatible biosurfactants, Appl. Food Sci. J. 3 (2019) 1–4.

[3] M.D. Macia, E. Rojo-Molinero, A. Oliver, Antimicrobial susceptibility testing in biofilm growing bacteria, Clin. Microbiol. Infect. 20 (2014) 981–990.

[4] M. Nitschke, S.S. Silva, Recent food applications of microbial surfactants, Crit. Rev. Food Sci. Nutr. 58 (2018) 631–638.

[5] J.M. Campos, T.L.M. Stamford, L.A. Saurbbo, J.M. de Luna, R.D. Rufino, M. Ibrahim, Banat, microbial biosurfactants as additives for food industries, Biotechnol. Prog. 29 (2013) 1097–1107.

[6] N. Ranasalva, R. Sunil, G. Poovarasan, Importance of biosurfactant in food industry, IOSR J Agri. Vet. Sci. 7 (2014) 6–9.

[7] R.K. Sharma, Surfactants: basics and versatility in food industries, Pharma Tutor 2 (2014) 17–29.

[8] J.D. Desai, I.M. Banat, Microbial production of surfactants and their commercial potential, Microbiol. Mol. Biol. Rev. 61 (1997) 47–64.

[9] M. Nitschke, C. SGVAO, Biosurfactants in food industry, Trends Food Sci. Technol. 18 (2007) 252–259.

[10] M. Nitschke, G.M. Pastore, Production and properties of a surfactant obtained from Bacillus subtilis grown on cassava wastewater, Bioresour. Technol. 97 (2006) 336–341.

[11] M.J. Mc Inerney, M. Javaheri, D.P. Nagle, Properties of the biosurfactant produced by Bacillus liqueniformis strain JF-2, J. Ind. Microbiol. Biotechnol. 5 (1990) 95–102.

[12] V. Pruthi, S.S. Cameotra, Production, properties of a biosurfactant synthesized by Arthrobacter protophormiae—an Antartic strain, World J. Microbiol. Biotechnol. 13 (1997) 137–139.

[13] M.Z.V. Gomes, M. Nitschke, Evaluation of rhamnolipid and surfactin to reduce the adhesionand remove biofilms of individual and mixed cultures of food pathogenic bacteria, Food Control 25 (2012) 441–447.

[14] M. Nitsche, S.S. Silva, Recent food applications of microbial surfactants, Crit. Rev. Food Sci. Nutr. (2016), https://doi.org/10.1080/10408398.2016.1208635.

[15] C.N. Mulligan, S.K. Sharma, A. Mudhoo, K. Makhijani, Green chemistry and biosurfactant research, in: Biosurfactants Research Trends and Applications, vol. 2014, CRC Press, Boca Raton, 2014, pp. 1–30.

[16] K. Poremba, W. Gunkel, S. Lang, F. Wagner, Toxicity testing of synthetic and biogenic surfactants on marine microorganisms, Environ. Toxicol. Water Qual. 6 (1991) 157–163.
[17] A. Flasz, C.A. Rocha, B. Mosquera, C. Sajo, A comparative study of the toxicity of a synthetic surfactant and one produced by Pseudomonas aeruginosa ATCC 55925, Med. Sci. Res. 26 (1998) 181–185.
[18] D.A. Cavalero, D.G. Cooper, The effect of medium composition on the structure and physical state of sophorolipids produced by Candida bombicola ATCC 22214, J. Biotechnol. 103 (2003) 31–41.
[19] J. Velikonja, N. Kosaric, Biosurfactants in food applications, in: Biosurfactants:Production: Properties: Applications, CRCPress, New York, 1993, pp. 419–448. Chapter 16, ISBN-13: 9780824788117.
[20] M.C. Cirigliano, G.M. Carman, Purification and characterization of liposan, abioemulsifier from Candida lipolytica, Appl. Environ. Microbiol. 50 (1985) 846–850.
[21] T. Kachholz, M. Schlingmann, Possible food and agricultural applications of microbial surfactants: An assessment, in: N. Kosaric, W.L. Carns, N.C.C. Gray (Eds.), Biosurfactants and Biotechnology, Marcel Dekker, New York, 1987, pp. 183–210.
[22] D. Vollen Broich, M. Ozel, J. Vater, R.M. Kamp, G. Pauli, Mechanism of inactivation of enveloped viruses by the biosurfactant surfactin from Bacillus subtilis, Biologicals 25 (3) (1997) 289–297.
[23] X. Wang, L. Gong, S. Liang, X. Han, C. Zhu, Y. Li, Algicidal activity of rhamnolipid biosurfactants produced by Pseudomonas aeruginosa, Harmful Algae 4 (2005) 433–443.
[24] D.S. Yoo, B.S. Lee, Kim, EK characteristics of microbial biosurfactant as an antifungal agent against plant pathogenic fungus, J. Microbiol. Biotechnol. 15 (6) (2005) 1164–1169.
[25] M.M.C. Velraeds, H.C. Van der Mei, G. Rei, H.J. Busscher, Inhibition of initial adhesion of uropathogenic enterococcus faecalis by biosurfactants from lactobacillus isolates, Appl. Environ. Microbiol. 62 (6) (1996) 1958–1963.
[26] H.J. Busscher, C.G. van Hoogmoed, G.I. Geertsema-Doornbusch, M. van der Kuij-Booij, H.C. van der Mei, Streptococcus thermophilus and its biosurfactants inhibit adhesion by Candida spp. on silicone rubber, Appl. Environ. Microbiol. 63 (1997) 3810–3817.
[27] S.K. Hood, E.A. Zottola, Biofilms in food processing, Food Control 6 (1) (1995) 9–18.
[28] G. Dhanarajan, R. Sen, Amphiphilic molecules of microbial origin: Classification, characteristics, genetic regulations, and pathways for biosynthesis, in: C.N. Mulligan, S.K. Sharma, A. Mudhoo (Eds.), Biosurfactants Research Trends and Applications, CRC Press, 2014, pp. 31–48.
[29] P. Foley, A.K. Pour, E.S. Beach, J.B. Zimmerman, Derivation and synthesis of renewable surfactants, Chem. Soc. Rev. 41 (2012) 1499–1518.
[30] G. Dhanarajan, R. Sen, Cost analysis of biosurfactant production from a scientist's perspective, in: N. Kosaric, F. Vardar-Sukan (Eds.), Biosurfactants Production and Utilization—Processes, Technologies, and Economics, CRC Press, Boca Raton, 2015, pp. 153–161.
[31] R.S. Makkar, S.S. Cameotra, I.M. Banat, Advances in utilization of renewable substrates for biosurfactant production, AMB Express 1 (2011) 5.
[32] M. Nitschke, C. SGVAO, Biosurfactants in the food industry, in: Biosurfactants Research Trends and Applications, Elsevier, 2007, pp. 177–196.
[33] C.N. Mulligan, S.K. Sharma, A. Mudhoo, Biosurfactants, CRC Press, Boca Raton, 2014, pp. 177–196.

[34] A. Pendse, K. Aruna, Use of various screening methods for isolation of potential biosurfactant producing microorganism from oil-contaminated soil samples, J. Pharm. Res. 12 (2018) 599–605.
[35] E.A. Diab, Screening bacterial strains isolated from used motor oil-polluted desert soil for the production of biosurfactants and the possibility of applying the produced biosurfactants for washing and bioremediation of the polluted soil, Int. J. Sci. Res. 4 (2015) 887–895.
[36] N.K. Bordoloi, B.K. Konwar, Microbial surfactant-enhanced mineral oil recovery under laboratory conditions, Colloids Surf. B Biointerfaces 63 (1) (2008) 73–82.
[37] M. Abouseoud, R. Maachi, A. Amrane, S. Boudergua, A. Nabia, Evaluation of different carbon and nitrogen sources in production of biosurfactant by Pseudomonas fluorescens, J. Microbiol. Biotechnol. 223 (2007) 143–151.
[38] S.H. Elwan, E.L.-N. MR, M.S. Ammar, Characteristics of lipase (s) in the growthfiltrate dialysate of Bacillus stearothermophilus cup plate assay, Bull Fac. Sci. 8 (105119) (1977). Riyadh Univ.
[39] A.M. Elazzazy, T.S. Abdelmoneim, O.A. Almaghrabi, Isolation and characterization of biosurfactants production under extreme environmental conditions by alkali-halothermophilic bacteria from Saudi Arabia, Saudi J. Biol. Sci. 22 (2015) 466–475.
[40] S.K. Satpute, A.G. Banpurkar, P.K. Dhakephalkar, I.M. Banat, B.A. Chopade, Methods for investigating biosurfactants and bioemulsifiers: a review, Crit. Rev. Biotechnol. 30 (2) (2010) 127–144.
[41] M. Morikawa, Y. Hirata, T.A. Imanaka, Study on the structure– function relationship of the lipopeptide biosurfactants, Biochim. Biophys. Acta 1488 (2000) 211–218.
[42] P.G. Carrillo, C. Mardaraz, S.I. Pitta-Alvarez, A.M. Giuliett, Isolation and selection of biosurfactants producing bacteria, World J. Microbiol. Biotechnol. 12 (1996) 82–84.
[43] I.M. Banta, The isolation of a thermophilic biosurfactant producing Bacillus sp, Biotechnol. Lett. 15 (6) (1993) 591–594.
[44] N.H. Youssef, K.E. Duncan, D.P. Nagle, K.N. Savage, R.M. Knapp, M.J. McInerney, Comparison of methods to detect biosurfactant production by diverse microorganisms, J. Microbiol. Methods 56 (2004) 339–347.
[45] A. Sethupathy, P. Sivashanmugam, Investigation on ultrasonication mediated biosurfactant disintegration method in sludge flocs for enhancing hydrolytic enzymes activity and polyhydroxyalkanoates, Environ. Technol. (2018), https://doi.org/10.1080/09593330.2018.1481887.
[46] M. Rosenberg, D. Gutnick, E. Rosenberg, Adherence of bacteria to hydrocarbons: a simple method for measuring cell-surface hydrophobicity, FEMS Microbiol. Lett. 9 (2006) 29–33, https://doi.org/10.1111/j.1574-6968.1980.tb05599.x.
[47] C.-Y. Chen, S.C. Baker, C. Richard, Darton the application of a high throughput analysis method for the screening of potential biosurfactants from natural sources, J. Microbiol. Methods 70 (2007) 503–510.
[48] A. Nayarisseri, P. Singh, S.K. Singh, Screening, isolation and characterization of biosurfactant producing Bacillus subtilis strain ANSKLAB03, Bioinformation 14 (6) (2018) 304–314, https://doi.org/10.6026/97320630014304.
[49] N.M. Sidkey, H.F. Mohamed, H.I. Elkhouly, Evaluation of different Screening methods for biosurfactant producers isolated from contaminated Egyptian samples grown on industrial olive oil processing waste, Br. Microbiol. Res. J. 17 (2016) 1–19.

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CHAPTER

2 Optimization and characterization of various biosurfactant and application in food industries

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1 Introduction

Several microorganism products with surface-active characteristics have been found in the past. These products are known as biosurfactants (BSs). They are surface-active molecules with a variety of structural groups, produced by few bacteria (*Actinomycetes*, *Rhodococcus* species, and other microscopic agents, which develop biofilm), various yeasts, and filamentous fungi as well [1,2]. Lipopeptides, glycolipids (low molecular weight), and lipid-containing polymers (high molecular weight) such as lipoproteins (BSs with notable surface-active properties in food processing, as de-emulsifiers, antiviral, antimicrobial, antitumor, and antiadhesive agents), lipopolysaccharide-protein complexes, and polysaccharide-protein-fatty acid complexes are among these [3–5]. BSs have gained a wide interest because of their exceptional properties such as low toxicity, specificity, and comparatively easy to prepare. Exceptional functional properties enabled the use of BSs in several industries such as organic chemicals [6], agrochemicals and fertilizers [7], foods and beverages [8], cosmetics [9], pharmaceuticals [10], medicine [11], and many others. They are utilized as foaming agents, wetting agents, emulsifiers, demulsifiers, spreading agents, functional food components, and detergents, among other things. BSs are utilized in the foodservice industry as emulsion stabilizers, antiadhesives, and antimicrobial/antibiofilm agents. [12]. There is a visible surge in the attention for the applications of microbial emulsifying agents because of their potential claims

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in environmental protection and their distinctive qualities such as high biodegradability, biocompatibility, low toxicity, high foaming capacity, and digestibility [13]. BSs are applied to enhance the surface area of the compounds insoluble in water. This even aids in the bioavailability of hydrophobic substances via a solubilization/desorption mechanism [14]. BSs are with unique property of both hydrophilicity and hydrophobicity. This allows the interface assembly of the BSs between the two immiscible phases. This lowers the surface tension even further, resulting in improved solubility and emulsification of the two immiscible phases. They also operate as a nutrient transport enhancer across biological membranes, influencing a variety of host-microbe interactions [14,15].

The common description of BSs is covered in this chapter, along with its definition. The role of microorganisms in BS generation, their properties, BS types, and BS uses in various segments are all covered in depth.

2 Biological and functional properties of biosurfactant (BS)

The ability of BS to lower surface tension and interfacial tension validates its hydrophilic and hydrophobic properties. Other properties, namely emulsification, cleansing, wetting, phase separation, foaming, aggregation, and surface activity to reduce the oil viscosity enable its exploration in various segments of industries. Microorganisms of various origins (fungi [16,17], bacteria [18], actinomycetes [19])]) supply BS with a wide range of functional and structural features. The key biological and functional features of most BSs for consideration as a surfactant or emulsifier are shown in Fig. 1. It is not stated, however, that all BSs will have the features shown in Fig. 1. The reason for difference is their basic structural organization. The varied properties that enabled BS will certainly find their applications in a wide range and therefore motivate researchers worldwide to find their applications in different unexplored fields.

2.1 Biological properties

2.1.1 Antimicrobial activity

Biosurfactants act as an antimicrobial agent that is efficient against a variety of pathogenic organisms [20]. Glycolipids and lipopeptides are the two types of BS molecules that have been investigated the most. They possess broad-spectrum antimicrobial action and are used as antimicrobials, emulsifiers, and surface-active compounds in a variety of disciplines (cosmetic, food, and pharmaceutical industries) [21].

When compared with traditional antibiotics, BSs' antibacterial properties rely on alternative processes to eliminate target organisms. [22]. They mostly kill bacteria by rupturing the integrity of the plasma membrane or the cell wall [23–25]. Many of the glycolipid and lipopeptide-based BSs were derived from microorganisms found in marine, terrestrial, and hydrocarbon-contaminated environments [26–29]. Surfactin

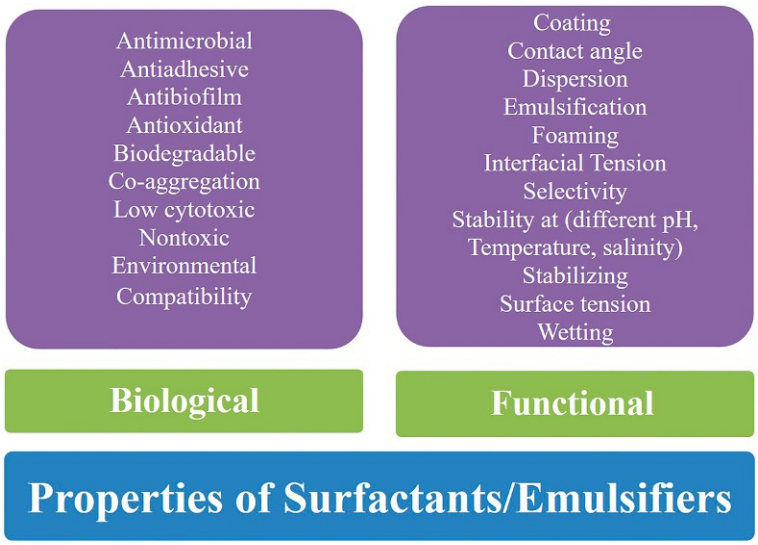

FIG. 1

Various biological and functional properties of biosurfactants.

and iturin (*Bacillus subtilis*) [30], rhamnolipids (*Pseudomonas* species) [26], mannosylerythritol lipids (*C. antarctica*) [31], and fungus prepared BSs [32]. Surfactin (a lipopeptide biosurfactant produced by *Bacillus subtilis* C9) exhibited antibacterial and antiviral activities [33]. Further, C9-surfactin completely wipes out cladoceran grazers (*Daphnia pulex* and *Moina macrocopa*), which are prevalent in the algae cultivation systems [34].

2.1.2 Antiadhesive and antibiofilm activity

BS has been discovered to prevent pathogenic microbes from adhering to infection sites or solid surfaces, as well as preventing biofilm development [35]. The BS adhesive process depends on bacteria adhering to different surfaces being inhibited. Low toxicity and biodegradability enable BS as an excellent substitute for synthetic surface-active agents [35,36]. As a result, prior attachment of BS to solid surfaces may provide a unique and effective approach of preventing harmful microbe colonization and subsequent biofilm development [37,38]. A lipopeptide BS inhibits harmful bacteria (*Pseudomonas aeruginosa* and *Staphylococcus epidermidis*) from forming biofilms (*Bacillus cereus* NK1) [39]. Precoating of the vinyl urethral catheters with surfactin BS solution before media inoculation reduced the amount of biofilm produced by *Proteus mirabilis*, *Salmonella typhimurium*, and *Escherichia coli* [40]. *Lactobacillus cereus* (NK1) has been widely studied as a probiotic for urinary infection prevention [41].

2.1.3 Antioxidant activity

The most efficient strategy to prevent lipid peroxidation is to include antioxidants to your diet. When added to food products, antioxidants can extend shelf life by slowing the process of lipid peroxidation [42,43]. Antioxidants are widely known for their ability to capture the oxidation free radical chain and generate stable free radicals that do not trigger further oxidation. BSs are very famous for their free radical scavenging activities. The BS was found to be a lipopeptide with strong antioxidant action against DPPH [44]. In comparison to the standard ascorbic acid utilized in the study, the antioxidant activity of Bacillus subtilis (MF582633) BS demonstrated less antioxidant capability. Because of its ability to donate hydrogen, increasing the concentration of BS may result in an increased scavenging effect [45]. Sib Sankar Giri et al. [46] explored the potential application of natural BS VSG4 and VS16 isolated from *Bacillus* strains.

2.1.4 Biodegradability

The biodegradability tests (OECD Guidelines for Testing of Chemicals) of sophorolipids BSs (yeast *Candida bombicola*) were performed. From the results, it was noticed that the biodegradation process of BSs was started instantaneously post cultivation. Surfactin and arthrofactin were also examined and have been found to be easily biodegradable as sophorolipids. On the other hand, no sign of biodegradation was observed for synthetic surfactants up to 8 days [47]. Rhamnolipid BS degrades in both aerobic and anaerobic environments. Under the anaerobic conditions, Triton X-100 (synthetic surfactant) is nonbiodegradable; however, it did partially biodegrade under aerobic environment [48]. Mannosylerythritol, a lipid biosurfactant from *Candida Antarctica*, was easily biodegraded in the presence of activated sludge microorganisms. On the other hand, LAS and sodium lauryl sulfate (SDS) showed no sign of biodegradation after 7 days [49]. Bacteria-dependent degradation results were observed for five biological surfactants formed by two *Bacillus* sp., *Arthrobacter oxydans*, *Flavobacterium* sp., and *Dietzia maris* and for SDS [50]. Rhamnolipid BS biodegradation in sandy and loamy soil was comparatively low in the first 2 days of study but was amplified to 92% on 7th day of incubation in both kinds of soil [51]. Another research showed complete degradation after 4 days of cultivation [52].

2.1.5 Toxicity of biosurfactants

Applicational functions are generally supported by toxicity tests for wider research. BSs are commonly believed as low- or nontoxic. Table 1 represents the toxicity data of selected BSs. BSs showed lower hemolytic activity to human erythrocyte in comparison with synthetic cationic surfactants (TTAB, BC, and CTAB) and anionic surfactant SDS. They don't have any harmful effect on the lung, heart, kidney, liver and do not interfere with timing for blood coagulation. Moreover, they showed much lower acute and chronic toxicity as compared with synthetic Triton X-100.

Table 1 Biosurfactants' toxicity and cytotoxicity.

Toxicity test	Biosurfactant (*microbial origin*)	Toxicity, effect	Ref.
Hemolytic activity	Lipopeptide BS (*B. subtilis* ATCC 6633)	Low ability to rupture erythrocytes	[53]
	CTAB (hexadecyltrimethylammonium bromide)	High ability to rupture erythrocytes	
Antimicrobial activity	Biosurfactant M[a] (*P. aeruginosa*)	Microbial growth inhibited	[54]
Hemolytic activity	Biosurfactant NM[b] (*P. aeruginosa*)	More potent for hemolysis	
		Less potent for hemolysis	
In vitro tissue damage		No detrimental effect was observed	
Interference in blood coagulation		Normal clotting time	
Acute toxicity	Glycolipid (*R. ruber* IEGM 231)	No effects on CNS, no weight loss and no change in behavior	[55]
Neutral Red assay	Mannosylerythritol lipid MEL (*Candida* sp. SY16)	Not harmful to human skin and eyes	[49]
Subacute toxicity	Surfactin C (*B. subtilis)*	No deaths observed for rats at any doses	[56]
Histhopathological tests		Weight of liver increased significantly	

[a]*M mucoid strain.*
[b]*NM non-mucoid strain.*

2.1.6 Environmental compatibility

The use of BS is common in environmental protection strategies (oil spills removal, water and soil remediation, etc.) due to its good biodegradability, good physico-chemical properties, and low toxicity. Crude and model oils can be removed from contaminated ground or sand columns using BS in the washing process. Compared with synthetic surfactant saponin [57], synthetic Tween 60 was observed for providing much higher removal [58]. BSs are less harmful, more effective, and biodegradable than chemically synthesized surfactants. As a result, they are less harmful to the environment and may be used instead of synthetic surfactants.

2.2 Functional properties

2.2.1 Coating agents

BSs have been reported to impede the adhesion of infectious agents to infection sites or to solid surfaces. The amount of biofilm formed by *Salmonella enterica,* Proteus mirabilis, Salmonella typhimurium and *E. coli* was reduced when the surfactin solution was passed through vinyl urethral catheters as a precoating agent prior to inoculation with media [40]. Another strategy for using BSs in biomedical applications is to develop appropriate antiadhesion biological coatings for implant materials. The biosurfactant produced by *S. thermophilus* strains desorbs from the glass on its own, leaving a fully nonadhesive layer [59]. The BS surlactin [60] (*Lactobacillus* species) was proposed as an appropriate antiadhesive coating material for catheter. BSs have been potentially used as antiadhesive biological coating material for the medical insertional material. This will aid in the reduction of pathogenic organisms as well as the utilization of artificial chemicals and medications. [11].

2.2.2 Dispersing agents

BS concentration is proportional to the amount of dispersion. The dispersant efficacy is affected by the dispersant/oil ratio. The best dispersion index has arisen with a BS/oil ratio (1:1, *v*/v) with a solution of the BS twice the CMC (50%), while ~25% of the oil was dispersed using crude BS under the similar condition. Increase speed of agitation allowed enhanced interaction between BS and oil. This leads to a higher percentage of dispersion. BS has been potentially applied as a dispersant, but the additives will increase the dispersion efficiency more likely [61].

2.2.3 Emulsification property

Higher surface activity and low toxicity enable the potential and effective applications of BS in the food industry [62]. Customers are increasingly preferring naturally derived components over synthetic ones. The accumulation of bioemulsifiers at the interphase between the two immiscible components lowers the surface tension by further enhancing solubility and emulsification of the non-miscible components. Bioemulsifiers mediate the conversion of insoluble molecules to soluble molecules, which can then be used by microbes for metabolism [63]. BSs with such surface characteristics are excellent choices for boosting oil recovery. Few of the BSs are particularly successful, lowering water surface tension from 72 dynes/cm to 25–30 dynes/cm [64]. Noticeable surface-active properties (food processing, oil recovery, antimicrobial, antitumor, antiviral) are shown by lipopeptide BSs [65]. BSs (made by bacteria such as *Actinomycetes*, *Rhodococcus* species, *Acinetobacter*, and other bio-film forming bacteria) have demonstrated their dispersant properties by efficiently lowering the surface tension of the oil-water interface.

Rhamnolipids exhibited efficient emulsifiers for vegetable oil and water, producing water-in-oil and oil-in-water emulsions [66]. Surfactin (*Bacillus subtilis* LAMI005) showed a superior emulsification index (E_{24} more than 55%) on soybean oil and kerosene [67]. BS molecular characteristics are getting affected by the system

pH and salinity as it controls the BS emulsion behavior. For example, stable emulsions above pH 7.4 were produced using surfactin (*Bacillus subtilis* ATCC 21332). The ionization or protonation of a carboxyl group at the molecular level was attributed to the pH sensitivity of surfactin emulsification [68]. Rhamnolipid (*P. aeruginosa* LBI) was found to be capable of emulsifying a wide range of hydrocarbons above pH 5 [69]. As the pH of a brine kerosene system climbed from 3 to 6, the E24 value for the rhamnolipid grew from 0 (no emulsion) to 70%. [69]. When the Rhamnolipid concentration exceeds salinity, pH range and CMC affect the production of Rhamnolipid structures [70].

2.2.4 Surface and interface activity

An effective BS is the one that is capable of lowering the surface tension of water. Surfactin (*Bacillus subtilis*) and rhamnolipid (*Pseudomonas aeruginosa*) are more effective than other surfactants at lowering water surface tension [71]. BSs are highly functional and very efficient at the CMC of 10–40 times lesser than synthetic surfactants. This is why only a small amount of BS is needed to lower the surface tension. [72].

2.2.5 Temperature, pH, and ionic strength tolerance

Functions and parameters of biosurfactants are generally not getting influenced by change of temperature and pH. Research studies suggested that lichenysin (*B. licheniformis*) was less influenced by pH change (4.5–9.0), temperature change (up to 50°C), and change in NaCl and Ca concentrations. Lipopeptides (*Bacillus subtilis*) were found to be stable when stored for 180 days at 121°C and at −15°C. Activity was also found to be stable at NaCl concentrations (>15% and pH change (4 and 12) [73].

3 Category of biosurfactants

BSs are usually classified based on the composition of chemicals. Chemical composition of BS is directed by the microbial origin and different molecules forming the hydrophobic (saturated or unsaturated fatty acids) and hydrophilic (consists of amino acids, peptides, mono-, di-, or polysaccharides) compounds [74]. According to Rosenberg et al. [75], BSs can be categorized into two types of polymers: i.e., molecules with low molecular mass (effectively decline the surface and interfacial tension) and polymers with high molecular mass (effectively elevate the surface and interfacial tension) (effective as emulsion-stabilizing agents). Glycolipids, lipopeptides, and phospholipids are all examples of low-mass surfactants, whereas polymeric and particulate surfactants are examples of high-mass surfactants. The nutritional environment has been affecting the yield of microbial surfactants for the growing microorganisms. Below are descriptions of the most essential BS groups/classes, as well as some of their classes/subclasses.

3.1 Glycolipids

Among all BSs, glycolipids are known to be used widely. They are the complex of calcium and fatty acids. The bond is mediated by an ether or ester group. Trehalolipids, glycolipids, rhamnolipids, and sophorolipids are the most acknowledged [76].

3.1.1 Rhamnolipids

Rhamnolipids (one or two molecules of rhamnose are connected to one or two molecules of β-hydroxydecanoic acid) are the ideal considered glycolipids. While the –OH group of one acid is engaged in ester formation, the –OH group of the other acids is involved in glycosidic linkage with the rhamnose disaccharide's reducing end [77]. Jarvis and Johnson [78] were the first to describe the production of rhamnose-containing glycolipids in Pseudomonas aeruginosa. As they possess high surface activity, rhamnolipids' physical and chemical characteristics have gained a lot of attention [26,79–83]. The main glycolipids produced by Pseudomonas aeruginosa are L-rhamnosyl-L-rhamnosyl-hydroxydecanoyl-hydroxydecanoate and L-rhamnosyl-hydroxydecanoyl-hydroxydecanoate (di- and mono-rhamnolipids). [84].

3.1.2 Trehalolipids

There have been several structures identified that contain microbiological trehalolipid BS. Trehalose fats are a significant component of rhodococci, and their physicochemical and biological features have piqued interest [85]. Mycolic acids (MAs) are branched hydroxy fatty acids with a lengthy chain. The size, structure, numbers of carbon atoms, and degree of unsaturation of MA are all distinct. Trehalolipids are lipids generated by a variety of bacteria [86]. These chemicals have been proposed for a variety of purposes. Furthermore, lipids containing succinoyl trehalose have been demonstrated in promoting leukemia cell development [87] and inhibiting protein kinase activity [88].

3.1.3 Sophorolipids

These glycolipids are primarily made up of yeast such as *Torulopsis bombicola* [89,90], *Torulopsis apicola*, and *Torulopsis petrophilum*. They comprise a sophorose (dimeric carbohydrate) associated to a long-chain hydroxyl fatty acid through a glycosidic linkage. Sophorolipids are composed of a mixture of six macrolactones and a free acid form. The lactone form of the BS is required or preferred for sophorolipid application [91].

3.2 Lipoproteins and lipopeptides

Microorganisms are producing cyclic lipopetides consisting of decapeptide (gramicidins) and lipopeptide (polymyxins) antibiotics. These products comprise fat connected to a polypeptide chain. Below is the description for two of them.

3.2.1 Surfactin

Surfactin belongs to a group of cyclic lipoheptapeptides and is a significant BS with greater surface [92,93]. The cyclic lipopeptide surfactin, a seven-amino-acid ring structure coupled to a fatty acid chain via lactone linkage, is produced by *Bacillus* species [94].

3.2.2 Lichenysin

BSs produced by *Bacillus licheniformis* acts synergistically and exhibit good stability at different temperature, pH, and salt concentration. Lichenysin is found similar to the surfactin in terms of its structure and physiochemical characteristics [95].

3.3 Phospholipids, neutral lipids, and fatty acids

Numerous yeast and bacteria grow on *n*-alkanes. These microorganisms produce huge quantities of BS containing fatty acids and phospholipid [96]. In *Acinetobacter* sp. Strain produces HO1-N, phosphatidylethanolamine-rich vesicles, which in further forms optically clear isotropic microemulsions of alkanes in water [97]. After growing on n-alkane, *Rhodococcus erythropolis* produces phosphatidylethanolamine, which lowers the interfacial tension between water and hexadecane, which is not lesser than 1mN/m, and a critical micelle concentration of 30mg/L [98].

3.3.1 Corynomycolic acid

Corynomycolic acids are the class of surface-active agents possessing variable number of carbon atoms. Synthesis of biosurfactants with varying chain length is affected by substrate in the growth media. *Corynebacterium lepus* produced a combination of corynomycolic acids with good surfactant capabilities. The application of BS at all pH values between 2 and 10 resulted in significant reductions in surface tension in water and interfacial tension between hexadecane and aqueous phase [99].

3.4 Polymer biosurfactants

High-molecular-weight polymeric BSs have a framework of three or four fatty acid attached repeating sugars [75]. Liposan, emulsan, lipomannan, alasan, and other polysaccharide-protein complexes are the most well-studied polymeric BSs. Liposan is an extracellular water-soluble emulsifier generated by *Candida lipolytica*. It has 83% carbohydrate content and 17% protein content. [96].

3.4.1 Emulsan

Emulsan (a complex acylated polysaccharide, extracellular) is produced by the Gram-negative bacterium *Acinetobacter calcoaceticus* and has a roughly 1000 KD molecular weight [33]. Its industrial applications as an emulsifier have been extensively investigated [100]. This molecule comprises an unbranched polysaccharide backbone with N-acyl and O-acyl-bound fatty acid side chains. [101]. The polysaccharide backbone is made up of three aminosugers in a 1:1:1 ratio:

D-galactosaminouronic acid, D-galactosamine, and a dideoxydiaminohexose [100]. The carbon atoms in side chains of fatty acid range from 10 to 22. It has amino groups that are either acetylated or covalently linked to 3-hydroxybutyric acid by an amide group. It is an efficient emulsifying agent at a concentration as low as 0.001% to 0.01% for hydrocarbons in water [102].

3.5 Particulate biosurfactants

Extracellular vesicles and whole microbial cells are two types of particulate BSs. Microbial cells rely heavily on extracellular membrane vesicles for hydrocarbon absorption. At times, the entire bacterial cell will operate as a surfactant moiety.

3.5.1 Vesicles

When *Acinetobacter* sp. cultured on hexadecane accumulates on extracellular medium, it generates vesicles 20–50 mm in diameter with a buoyant density of 1.158 g/cm^3. These vesicles play a crucial role in *Acinetobacter* sp. alkane's absorption. Protein, phospholipids, and lipopolysaccharide make up these vesicles [97]. *Pseudomonas marginalis* forms vesicles, which acts as surfactants like *Acinetobacter* sp.

3.5.2 Whole microbial cells

Nonhydrocarbon degraders, hydrocarbon-degrading microbes, pathogens, and a few *Cyanobacteria* species all show a strong affinity for water hydrocarbon and water-air interfaces. In these situations, the microbial cells act as surfactants [77].

4 Sources of biosurfactants

4.1 Bacterial biosurfactants

Large number of microbial organisms particularly bacteria are reported to produce BS [5,22]. The chemical nature of BSs is reliant on the BS-fabricating species [121]. Table 2 outlines few bacteria involved in fabrication of BS. In few instances, BSs are produced by microbial cells during their growth on water-insoluble substances. *Rhodoccocus* and *corynebacterium* are the classical representative microorganisms for this instance. Others developed surfactants for water-soluble and oil-soluble substrates. Microorganisms in this group include Pseudomonas aeruginosa and *T. bombicola*. As a result, the primary source of BS is nature and the substrate's constitution. This must be scrutinized to ensure that the right microbial species are chosen [122,123].

Table 2 Biosurfactant types and microbial origin.

Biosurfactant type	Biosurfactant name	Source microorganism	Ref.
Glycolipids	Rhamnolipid	*Pseudomonas* sp.	[82]
	Treahalolipids	*Micrococcus luteus* BN56	[103]
	Sophorolipids	*Torulopsis bombicola*	[89]
	Mannosylerythritollipid	*Candida antartica*	[104]
	Xylolipid	*Lactococcus lactis*	[105]
	Cellobiolipids Flocculosin	*Pseudozyma flocculosa*	[106]
	Glucolipid	*Burkholderia cenocepacia* BSP3	[107]
	Monoacylglycerol	*Candida ishiwadae*	[108]
	Diglycosyl diglycerides	*Lactobacillus fermentii*	[74]
	Polyol lipids	*Rhodotorula glutinous*, *R. graminus*	[74]
lipopeptides and lipoproteins	Surfactin	*B. subtilis*	[94]
	Arthrofactin	*Arthrobacter sp.*	[109]
	Iturin	*B. subtilis*	[110]
	Fengycin	*B. subtilis*	[111]
	Lichenysin	*B. licheniformis*	[112]
	Pumilacidin	*B. pumilus*	[113]
	Peptide lipid	*B. licheniformis*	[95]
	Serrawattin	*Serratia marscescens*	[114]
	Viscosin	*P. fluorescens*	[115]
Fatty acids, neutral lipids and Phospholipids	Fatty acids, Corynomycolic acid	*Corynebacterium leupus*	[116]
	Neutral lipids	*Nocardia erythropolis*	[117]
	Phospholipids	*Thiobacillus thiooxidans*	[118]
Polymeric surfactants	Emulsan	*Acinetobacter calcoaceticus* RAG-	[102]
	Bioemulsan	*Gordinia sp. BS29*	[119]
	Biodispersan	*Acinetobacter calcoaceticus* A2	[120]
	Liposan	*C. lipolytica*	[96]
Particulate biosurfactant	Vesicles and fimbriae Whole cell	*Acinetobacter calcoaceticus*	[97]

4.2 Fungal biosurfactants

On the one hand, the area of BS production by bacteria has been well studied; on the other hand, less fungi are known to produce BS. Among fungi, *Candida batistae*, *Aspergillus ustus*, *Candida bombicola*, *Trichosporon ashi*, *Candida lipolytica*, *Ustilago maydis*, and *Candida ishiwadae* are the noticed fungi for the production of BS (Table 3).

Several of known fungi produce BSs using low-cost raw materials. Sophorolipids (glycolipids) are the major BSs produced by known fungus strains. Many patents have been filled on BSs looking to the wide industrial applications of many BSs (Table 4).

Renewable sources are used by few fungi for the generation of biosurfactants [64]. Use of carbon sources (animal fat, oleic acid, tallow fatty acid residues, and glycerol) in the inoculating media yields 40–120 g/L of fungal BSs [128–130]. Moreover, fungi are known for the generation of higher amount of BSs when compared with bacteria. The higher yield of BS enabled the fungus to be used widely for the industrial applications and also seems to replace surfactant with BS.

Table 3 Biosurfactants and their fungal source.

S.No.	Biosurfactant	Fungi	Ref.
1	Sophorolipid	*Candida bombicola* *Candida batistae* *Trichosporon ashii*	[124,125]
2	Glycolipid	*Ustilago maydis* *Candida ishiwadae*	[108]
3	Glycolipoprotein	*Aspergillus ustus*	[126]
4	Protein-lipid polysaccharide complex	*Candida lipolytica*	[127]
5	Sophorose lipid	*Torulopsis bombicola*	[89]

Table 4 Patents on biosurfactants produced by fungi.

S. No.	Fungi/type of biosurfactant	Title of the patent	Patent No.
1	Sophorolipid producer	Sophorolipids as stimulating agent of dermal fibroblast metabolism	US 6057302 A
2	*C. albicans, C. rugosa, C. tropicalis, C. lipolytica, C. torulopsis*	Microbial biosurfactants as agents for controlling pests	US 20050266036 A1
3	Sophorolipid	Spermicidal and virucidal properties of various forms of sophorolipids	WO 2005089522 A2
4	*Candida bombicola* ATCC 22214	Yeast strains modified in their sophorolipid production and uses thereof	EP 2580321 A1

4.3 Yeast biosurfactants

Nowadays, the number of studied has been increased for BSs produced by yeasts. The studies are mainly focused around the genders *Yarrowia sp*., *Pseudozyma sp*., and *Candida sp*. [17]. GRAS (generally regarded as safe) status for the BSs-producing yeast is greatest advantage for most of the species present, for example, *Kluyveromyces lactis*, *Yarrowia lipolytica*, and *Saccharomyces cerevisiae* [131]. Non-toxicity and absence of pathogenic property are confirmed by the GRAS status of organisms. This status permits their application in the products manufactured by the pharmaceutical and food industries [132]. Widely studied, characterized, and reported yeast BSs in literature have been identified as glycolipids, fatty acids or lipids, protein-carbohydrate complexes, or protein-carbohydrate-lipid. Table 5 presents the BS produced and the corresponding yeast species involved in BS fabrication.

Most interesting glycolipid BSs (long-chain hydroxyaliphatic acids or aliphatic acid containing carbohydrates) are the sophorolipids (long-chain hydroxy fatty acid linked dimeric carbohydrate sophorose). This BS has been discovered in *T. apicola*. [133], *Torulopsis bombicola* [124], and *Trifolium petrophilum* [134]. Sophorolipids

Table 5 Biosurfactants and their fungal source.

S. No.	Biosurfactant	Yeast	Ref.
1	Sophorolipids	*Candida bombicola* *Torulopis petrophilum* *Torulopsis apicola* *Candida bogorienses*	[124,134,140,141]
2	Mannosyl erythritol lipids	*Candida antarctica* *Pseudozyma rugulosa* *Kurtzmanomyces* sp. *I-11*	[136,142,143]
3	Carbohydrate-protein-lipid complex	*Candida lipolytica UCP0988* *Yarrowia lipolytica NCIM 3589* *Debaryomyces polymorphus*	[127,138,144]
4	Carbohydrate-protein complex	*C. lipolytica ATCC 8662* *Yarrowia lipolytica IMUFRJ 50682*	[137,139]
5	Mannano protein	*Saccharomyces cerevisiae* *Kluyveromyces marxianus*	[135,145]
6	Fatty acids	*Candida ingens*	[146]
7	Lipids	*Rhodotorula glutinis*	[147]

are not very effective emulsifiers though they are able to lower surface and interfacial tension [89]. Lactonic and acidic sophorolipids were found to reduce the interfacial tension between n-hexanedecane and water (about 40–5 mN/m) with noticeable stability against variations in pH and temperature. [134]. Exceptional vesicle forming and surface-active properties were reported for mannosylerythritol glycolipid produced by yeasts *Candida antarctica* [135] and *Pseudozyma rugulosa* [136]. Liposan was reported to show its emulsification capability with long-chain hydrocarbons [137]. According to Zinjarde et al. [138], the BS synthesized by *Y. lipolytica* was not capable of expressing its emulsification power on *n*-alkanes. Yansan's excellent emulsifying activity was demonstrated in a variety of aliphatics and aromatic hydrocarbons [139]. High emulsification properties have been reported by many mannoproteins extracted from yeast's wall [135].

5 Applications of biosurfactant

The BS applications have reached more than USD 2 billion [148,149] in 2020. Bioremediation, microbial-aided oil recovery, food, heavy metal contamination removal [148], pharmaceuticals, biomedicine, cosmetics, and nanotechnology are among the potential applications [150].. BSs are a good candidate for commercial applications because of their ability to minimize surface/interfacial tension and low CMC values. Because of its low toxicity and biodegradability, consumers' attitudes regarding the usage of BSs in innovative products may alter as a result of increased environmental consciousness and biotechnological innovation. One of the key applications of BS from several areas is the enhancement for degradation of hydrocarbon in polluted soil and water. Effective use of BSs is also reported for recovery of oil from petroleum sludge [151]. Pharmaceutical sector, from nanomedicine to drug delivery, is showing a significant impact for numerous applications of BSs [152].

5.1 Environmental bioremediation

Beneficial physicochemical properties and low toxicity enabled the BSs as a potential candidate for ecological bioremediation events including waste water treatment [22,74], oil spill cleaning [153], and heavy metal contamination removal [154,155]. Surface hydrophobicity of microorganisms (critical factor for process of biodegradation) is affected by BSs [156]. The hydrocarbon chain of surfactant pollutant (leads to loss of amphiphilicity) is broken down by microbes and uses it as an excellent source of energy and carbon. Pollutants are eventually transformed to carbon dioxide, minerals, and water [50]. Rhamnolipids and other BSs are important in the cleanup of contaminated soil and water. On adding rhamnolipid to deionized water concentrations greater than the CMC point, it boosted petroleum component solubility [157], as well as biomass growth and biodegradation percentages between 1000 and 2500 mg/L and 40%–100%, respectively [158]. Emulsifying contaminants with BSs is one of the ways for biodegradation of hydrocarbons [158]. The BS fabricated

by *Acinetobacter haemolyticus Zn01* showed around 60% index for emulsification. It also has a high potential to make free the polluted water from the diesel oil [159]. Longer stability of emulsion and smaller size of droplet were observed upon emulsifying the oil with rhamnolipid-modified silica nanoparticles in the remediation of marine oil spill. Transformed contact angle and enhanced adsorption of rhamnolipid-modified silica nanoparticles at the oil/sea water interface were also observed [160,161].

Polycyclic aromatic hydrocarbons (PAHs) bioremediation from the soil can be enhanced by the application of BSs. Carcinogenic, teratogenic, and mutagenic properties of PAHs may cause several adverse events [162,163]. Removal efficacy of the heavy metal ions (Fe, Pb, Cd, Cu, Zn) from the contaminated soil upon the use of anionic BSs is affected by particle size, pH of soil, permeability, cation exchange capability, sediment composition, and the existence of different contaminating agents [164–166]. Significant removal efficiency was observed for copper ions from sand surface upon application of the foam-based rhamnolipid solution [167]. Heavy metals were removed effectively during washing [168] upon application of rhamnolipid BS as a washing agent [169].

5.2 Oil recovery

Oil recovery can be improved [170] by injecting a fluid or gas or by using an oil reservoir with surfactants [171]. The mobility of hydrocarbons can be improved by using BSs, which increases their removal from polluted environments and oil extraction during the oil recovery process [172]. As the purity criteria are lower, whole cell broth can also be used for this application. Trehalose lipids derived from the *Nocardia rhodochrous* increased the total recovery of oil from subterranean sandstone by around 30% [74]. The marine bacterium *Aneurinibacillus aneurinilyticus* SBP-11 produced Aneurinifactin BS, which showed oil recovery based on the concentration utilized [172].

5.3 Pharmaceutical applications

A primary focus is on drug delivery methods that can incorporate bigger amounts of medication and excipient, with the possibility of lowering medications volume and adverse effects. [173,174]. Anticancer, antiviral, antimicrobial, hemolytic, antiadhesive, anti-inflammatory, anti-HIV, and immunomodulatory activities are exhibited by BSs [22,175]. Surfactin BS showed more than a few exciting properties including antitumor, antimicrobial and antiviral [267], antiadhesive, apoptotic, hypocholesterolemic, insecticide [175,176]. For instance, surfactin inhibits fibrin coagulation and possesses anticancer properties against Ehrlich's ascites malignant cells [177]. Several additional BSs, including a new lipopeptide isolated from *Paenibacillus ehimensis* strain MA2012, showed antifungal and antibacterial activity. [178], a glycolipid fabricated by oceanic microorganism *Staphylococcus saprophyticus* SBPS [179], rhamnolipid and surfactin take out from *P. aeruginosa* ST5 and *Bacillus*

amyloliquefaciens ST34, respectively [180]. Sophorolipid in combination with amino acids and derivatives of sophorolipid displayed antibacterial (Gram-positive and Gram-negative organisms) properties, anticancer, anti-inflammatory, and immunomodulatory properties [181]. Antimicrobial activity was shown by rhamnolipid against *Mycobacterium tuberculosis* [11]. Additionally, rhamnolipids acquired from the mango kernel oil proved antifungal activity against several pathogens [182]. Antimicrobial activity was shown by BSs extracted from *Lactobacilli* spp. against pathogens present in the urogenital and intestinal tract [183]. The BSs gained from *Lactobacillus paracasei* [188] are used to formulate new green cosmetic products [9,184–186].

5.4 Biosurfactants in food

BSs derived from diverse microorganisms are used as emulsifiers [22], antibacterial agents, and antiadhesive agents in the food sector [187]. Currently, some products are available in international market based on BSs, such as EC-601R manufacture by EcoChem Organics, dispersant of hydrocarbon insoluble in water, aqueous solution of rhamnolipids with unlike purity levels or available in a semisolid dosage form, and the Rhamnolipid Inc. products (www.rhamnolipid.com). Use of rhamnolipids in food products, pharmaceuticals, and cosmetics has been approved by the U. S. Environmental Protection Agency [8]. Emulsifiers are used by food manufacturers to reduce the surface tension between two immiscible phases, allowing for easier mixing [19]. Currently, emulsifiers-based natural or organic ingredients are the prime requirement for the growing functional foods market. There is a tremendous commercial potential for BSs considering their surface-active and emulsifier property. Their use as active BSs in the food industry is based on their surface activity, environmental friendliness [48], low toxicity [62]. Antimicrobial activity has been discovered in a wide range of BSs against all types of microorganisms.

Surfactin produced by *B. subtilis* was the first BS exhibiting antibacterial property [27,175,188]. Iturin, mycosubtilin, fengycin, and bacillomycin are the other lipopeptides produced by *B. subtilis* with the same property [54,165]. BSs act as a multifunctional (emulsifiers, dispersants, wetting agents, and solubilizers) agent in food products (dairy foods, shortenings and margarine, baked goods, salad dressings, etc.) containing higher amount of fats and oils [8,19,148,189].

6 Conclusion

BSs have sparked widespread interest as a result of their distinctiveness qualities, such as low toxicity, selectivity, and convenient preparation, which permit their potential and practical usage in a variety of disciplines, including the food industries, pharmaceutical industries, etc. Emulsifiers-based natural or organic ingredients are the prime requirement for the growing functional foods market. There is a huge potential market for BSs considering their surface-active and emulsifier

property. The antiadhesive activity of BS has a significant impact as a potential method of preventing and disrupting biofilm formation on surfaces that come into direct contact with food. The peculiar combination of many properties such as antimicrobial, emulsifying, and antiadhesive actions offered by BSs recommends their potential applications as a multifunctional additive. BSs have been restricted in food segments/areas due to a lack of information on toxicity and a greater cost of production. The utilization of agro-industrial wastes as a source of media for microorganisms producing BSs can overcome production cost constraints. Additionally, this aids waste treatment and provides a new direction for food and food-related sectors to manufacture BSs using microorganisms. BSs have a status of GRAS additive with potential applications because they are produced by nonpathogenic microorganisms such as lactobacilli and yeasts. Toxicological research on existing and future BSs must be prioritized to promote their wider application in the food business. However, much more study is needed in this field to discover BS-supportive uses in previously untapped fields such as pharmaceuticals, food, and associated industries, etc.

References

[1] A. Peele, V. Ch, K. Vidya Prabhakar, Emulsifying activity of a biosurfactant produced by a marine bacterium, 3 Biotech 6 (2016).
[2] D.E. Otzen, Biosurfactants and surfactants interacting with membranes and proteins: same but different? Biochim. Biophys. Acta Biomembr. 1859 (4) (2017) 639–649.
[3] S. Leonie Asfora, L. de Juliana Moura, C.-T. de Galba Maria, Production and stability studies of the bioemulsifier obtained from a new strain of Candida glabrata UCP 1002, Electron. J. Biotechnol. 9 (4) (2006).
[4] E.Z. Ron, E. Rosenberg, Natural roles of biosurfactants, Environ. Microbiol. 3 (4) (2001) 229–236.
[5] S. Mukherjee, P. Das, R. Sen, Towards commercial production of microbial surfactants, Trends Biotechnol. 24 (11) (2006) 509–515.
[6] S. Jana, S. Mondal, S.S. Kulkarni, Chemical synthesis of biosurfactant succinoyl trehalose lipids, Org. Lett. 19 (7) (2017) 1784–1787.
[7] D.P. Sachdev, S.S. Cameotra, Biosurfactants in agriculture, Appl. Microbiol. Biotechnol. 97 (3) (2013) 1005–1016.
[8] M. Nitschke, S.G.V.A.O. Costa, Biosurfactants in food industry, Trends Food Sci. Technol. 18 (5) (2007) 252–259.
[9] X. Vecino, et al., Biosurfactants in cosmetic formulations: trends and challenges, Crit. Rev. Biotechnol. 37 (7) (2017) 911–923.
[10] A. Varvaresou, K. Iakovou, Biosurfactants in cosmetics and biopharmaceuticals, Lett. Appl. Microbiol. 61 (3) (2015) 214–223.
[11] L. Rodrigues, et al., Biosurfactants: potential applications in medicine, J. Antimicrob. Chemother. 57 (4) (2006) 609–618.
[12] G.S. Kiran, et al., Production of Lipopeptide biosurfactant by a marine Nesterenkonia sp. and its application in food industry, Front. Microbiol. 8 (2017) 1138, https://doi.org/10.3389/fmicb.2017.01138.

[13] N.L. Olivera, et al., Microbial characterization and hydrocarbon biodegradation potential of natural bilge waste microflora, J. Ind. Microbiol. Biotechnol. 30 (9) (2003) 542–548.
[14] S. Vijayakumar, V. Saravanan, Biosurfactants-types, sources and applications, Res. J. Microbiol. 10 (2015) 181–192.
[15] G. Bognolo, Biosurfactants as emulsifying agents for hydrocarbons, Colloids Surf. A Physicochem. Eng. Asp. 152 (1) (1999) 41–52.
[16] R.D. Rufino, et al., Characterization and properties of the biosurfactant produced by Candida lipolytica UCP 0988, Electron. J. Biotechnol. 17 (1) (2014) 34–38.
[17] S.S. Zinjarde, A. Pant, Emulsifier from a tropical marine yeast, Yarrowia lipolytica NCIM 3589, J. Basic Microbiol. 42 (1) (2002) 67–73.
[18] A. Ghasemi, et al., Biosurfactant production by lactic acid bacterium Pediococcus dextrinicus SHU1593 grown on different carbon sources: strain screening followed by product characterization, Sci. Rep. 9 (1) (2019) 5287.
[19] M. Alizadeh-Sani, et al., Bioemulsifiers derived from microorganisms: applications in the drug and food industry, Adv. Pharm. Bull. 8 (2) (2018) 191–199.
[20] K. Harshada, Biosurfactant: a potent antimicrobial agent, J Microbiol Exp 1 (5) (2014) 173–177.
[21] S.M. Mandal, A.E.A.D. Barbosa, O.L. Franco, Lipopeptides in microbial infection control: scope and reality for industry, Biotechnol. Adv. 31 (2) (2013) 338–345.
[22] I.M. Banat, et al., Microbial biosurfactants production, applications and future potential, Appl. Microbiol. Biotechnol. 87 (2) (2010) 427–444.
[23] Y. Sang, F. Blecha, Antimicrobial peptides and bacteriocins: alternatives to traditional antibiotics, Anim. Health Res. Rev. 9 (2) (2008) 227–235.
[24] N.Y. Yount, M.R. Yeaman, Peptide antimicrobials: cell wall as a bacterial target, Ann. N. Y. Acad. Sci. 1277 (2013) 127–138.
[25] M. Agrawal, et al., Recent biomedical applications on stem cell therapy: a brief overview, Curr. Stem Cell Res. Ther. 14 (2) (2019) 127–136.
[26] A. Abalos, et al., Physicochemical and antimicrobial properties of new Rhamnolipids produced by Pseudomonas aeruginosa AT10 from soybean oil refinery wastes, Langmuir 17 (5) (2001) 1367–1371.
[27] P. Das, S. Mukherjee, R. Sen, Antimicrobial potential of a lipopeptide biosurfactant derived from a marine Bacillus circulans, J. Appl. Microbiol. 104 (6) (2008) 1675–1684.
[28] D. Sharma, S.M. Mandal, R.K. Manhas, Purification and characterization of a novel lipopeptide from Streptomyces amritsarensis sp. nov. active against methicillin-resistant Staphylococcus aureus, AMB Express 4 (1) (2014) 50.
[29] D. Sharma, et al., Isolation and functional characterization of novel biosurfactant produced by enterococcus faecium, Springerplus 4 (1) (2015) 4.
[30] F. Ahimou, P. Jacques, M. Deleu, Surfactin and iturin a effects on Bacillus subtilis surface hydrophobicity, Enzyme Microb. Technol. 27 (10) (2000) 749–754.
[31] J.R. Wright, Pulmonary surfactant: a front line of lung host defense, J. Clin. Invest. 111 (10) (2003) 1453–1455.
[32] G. Bai, M.L. Brusseau, R.M. Miller, Influence of a Rhamnolipid biosurfactant on the transport of bacteria through a Sandy soil, Appl. Environ. Microbiol. 63 (5) (1997) 1866.
[33] H.-S. Kim, et al., Production and properties of a lipopeptide biosurfactant from Bacillus subtilis C9, J. Ferment. Bioeng. 84 (1) (1997) 41–46.

[34] J.-H. Yun, et al., Application of biosurfactant from Bacillus subtilis C9 for controlling cladoceran grazers in algal cultivation systems, Sci. Rep. 8 (1) (2018) 5365.
[35] P. Das, S. Mukherjee, R. Sen, Antiadhesive action of a marine microbial surfactant, Colloids Surf. B Biointerfaces 71 (2) (2009) 183–186.
[36] P. Singh, S.S. Cameotra, Potential applications of microbial surfactants in biomedical sciences, Trends Biotechnol. 22 (3) (2004) 142–146.
[37] L. Rodrigues, et al., Kinetic study of fermentative biosurfactant production by Lactobacillus strains, Biochem. Eng. J. 28 (2) (2006) 109–116.
[38] L. Rodrigues, et al., Influence of biosurfactants from probiotic bacteria on formation of biofilms on voice prostheses, Appl. Environ. Microbiol. 70 (7) (2004) 4408–4410.
[39] M.I. Sriram, et al., Biofilm inhibition and antimicrobial action of lipopeptide biosurfactant produced by heavy metal tolerant strain Bacillus cereus NK1, Colloids Surf. B Biointerfaces 85 (2) (2011) 174–181.
[40] J.R. Mireles, A. Toguchi, R.M. Harshey, Salmonella enterica Serovar Typhimurium swarming mutants with altered biofilm-forming abilities: surfactin inhibits biofilm formation, J. Bacteriol. 183 (20) (2001) 5848.
[41] S. Boris, C. Barbés, Role played by lactobacilli in controlling the population of vaginal pathogens, Microbes Infect. 2 (5) (2000) 543–546.
[42] J.F. Mallet, et al., Antioxidant activity of plant leaves in relation to their alpha-tocopherol content, Food Chem. 49 (1) (1994) 61–65.
[43] H. Tsuchihashi, et al., Action of β-carotene as an antioxidant against lipid peroxidation, Arch. Biochem. Biophys. 323 (1) (1995) 137–147.
[44] N. Jemil, et al., Antioxidant properties, antimicrobial and anti-adhesive activities of DCS1 lipopeptides from bacillus methylotrophicus DCS1, BMC Microbiol. 17 (1) (2017) 144.
[45] A.P. Kanakdande, C.N. Khobragade, S. Tabassum, Production of biosurfactant from Bacillus subtilis (MF 582633) and its evaluation for antimicrobial, antioxidant, larvicidal and antitermite activities, Indian J. Chem. Technol. 26 (1) (2019) 76–83.
[46] S.S. Giri, et al., Antioxidant, antibacterial, and anti-adhesive activities of biosurfactants isolated from Bacillus strains, Microb. Pathog. 132 (2019) 66–72.
[47] Y. Hirata, et al., Novel characteristics of sophorolipids, yeast glycolipid biosurfactants, as biodegradable low-foaming surfactants, J. Biosci. Bioeng. 108 (2) (2009) 142–146.
[48] P.K. Mohan, G. Nakhla, E.K. Yanful, Biokinetics of biodegradation of surfactants under aerobic, anoxic and anaerobic conditions, Water Res. 40 (3) (2006) 533–540.
[49] H.-S. Kim, et al., Surface and physico-chemical properties of a glycolipid biosurfactant, mannosylerythritol lipid, from Candidaantarctica, Biotechnol. Lett. 24 (19) (2002) 1637–1641.
[50] T.M.S. Lima, et al., Biodegradability of bacterial surfactants, Biodegradation 22 (3) (2011) 585–592.
[51] X. Pei, X. Zhan, L. Zhou, Effect of biosurfactant on the sorption of phenanthrene onto original and H2O2-treated soils, J. Environ. Sci. 21 (10) (2009) 1378–1385.
[52] R. Fiebig, et al., Biodegradation of polychlorinated biphenyls (PCBs) in the presence of a bioemulsifier produced on sunflower oil, Biodegradation 8 (2) (1997) 67–75.
[53] G. Dehghan-Noude, M. Housaindokht, B.S. Fazly Bazzaz, Isolation, characterization, and investigation of surface and hemolytic activities of a lipopeptide biosurfactant produced by Bacillus subtilis ATCC 6633, J. Microbiol. (Seoul, Korea) 43 (2005) 272–276.
[54] K. Das, A.K. Mukherjee, Characterization of biochemical properties and biological activities of biosurfactants produced by Pseudomonas aeruginosa mucoid and

non-mucoid strains isolated from hydrocarbon-contaminated soil samples, Appl. Microbiol. Biotechnol. 69 (2) (2005) 192–199.
[55] M.S. Kuyukina, et al., In vitro immunomodulating activity of biosurfactant glycolipid complex from Rhodococcus ruber, Bull. Exp. Biol. Med. 144 (3) (2007) 326–330.
[56] Y.-H. Hwang, et al., Subacute (28 day) toxicity of Surfactin C, a Lipopeptide produced by Bacillus subtilis, in rats, J. Health Sci. 55230 (2009) 351–355.
[57] K. Urum, et al., A comparison of the efficiency of different surfactants for removal of crude oil from contaminated soils, Chemosphere 62 (9) (2006) 1403–1410.
[58] M.S. Kuyukina, et al., Effect of biosurfactants on crude oil desorption and mobilization in a soil system, Environ. Int. 31 (2) (2005) 155–161.
[59] H.J. Busscher, T.R. Neu, H.C. van der Mei, Biosurfactant production by thermophilic dairy streptococci, Appl. Microbiol. Biotechnol. 41 (1) (1994) 4–7.
[60] M.M.C. Velraeds, et al., Physicochemical and biochemical characterization of biosurfactants released by Lactobacillus strains, Colloids Surf. B Biointerfaces 8 (1) (1996) 51–61.
[61] D.K.F. Santos, et al., Candida lipolytica UCP0988 biosurfactant: potential as a bioremediation agent and in formulating a commercial related product, Front. Microbiol. 8 (2017) 767.
[62] A. Flasz, et al., A comparative study of the toxicity of a synthetic surfactant and one produced by Pseudomonas aeruginosa ATCC 55925, Med. Sci. Res. 26 (3) (1998) 181–185.
[63] L.R. Rodrigues, et al., Physicochemical and functional characterization of a biosurfactant produced by Lactococcus lactis 53, Colloids Surf. B Biointerfaces 49 (1) (2006) 79–86.
[64] S.K. Satpute, et al., Biosurfactants, bioemulsifiers and exopolysaccharides from marine microorganisms, Biotechnol. Adv. 28 (4) (2010) 436–450.
[65] A.A. Bodour, et al., Structure and characterization of flavolipids, a novel class of biosurfactants produced by Flavobacterium sp. strain MTN11, Appl. Environ. Microbiol. 70 (1) (2004) 114–120.
[66] A. Sarkar, et al., Colloidal aspects of digestion of Pickering emulsions: experiments and theoretical models of lipid digestion kinetics, Adv. Colloid Interface Sci. 263 (2019) 195–211.
[67] D.W. Freitas de Oliveira, et al., Kinetic study of biosurfactant production by Bacillus subtilis LAMI005 grown in clarified cashew apple juice, Colloids Surf. B Biointerfaces 101 (2013) 34–43.
[68] X. Long, et al., Biosurfactant surfactin with pH-regulated emulsification activity for efficient oil separation when used as emulsifier, Bioresour. Technol. 241 (2017) 200–206.
[69] R.B. Lovaglio, et al., Rhamnolipid emulsifying activity and emulsion stability: pH rules, Colloids Surf. B Biointerfaces 85 (2) (2011) 301–305.
[70] M. Abouseoud, et al., Effect of pH and salinity on the emulsifying capacity and naphthalene solubility of a biosurfactant produced by Pseudomonas fluorescens, J. Hazard. Mater. 180 (1) (2010) 131–136.
[71] G.S. Kiran, et al., Rhamnolipid biosurfactants: evolutionary implications, applications and future prospects from untapped marine resource, Crit. Rev. Biotechnol. 36 (3) (2016) 399–415.
[72] A.P. Karlapudi, et al., Role of biosurfactants in bioremediation of oil pollution-a review, Petroleum 4 (3) (2018) 241–249.

[73] L. Anburajan, et al., Heterologous expression, purification, and phylogenetic analysis of oil-degrading biosurfactant biosynthesis genes from the marine sponge-associated bacillus licheniformis NIOT-06, Bioprocess Biosyst. Eng. 38 (6) (2015) 1009–1018.
[74] J.D. Desai, I.M. Banat, Microbial production of surfactants and their commercial potential, Microbiol. Mol. Biol. Rev. 61 (1) (1997) 47.
[75] E. Rosenberg, E.Z. Ron, High- and low-molecular-mass microbial surfactants, Appl. Microbiol. Biotechnol. 52 (2) (1999) 154–162.
[76] K. Muthusamy, et al., Biosurfactants: properties, commercial production and application, Curr. Sci. 94 (6) (2008) 736–747.
[77] N.G.K. Karanth, P.G. Deo, N.K. Veenanadig, Microbial production of biosurfactants and their importance, Curr. Sci. 77 (1) (1999) 116–126.
[78] F.G. Jarvis, M.J. Johnson, A Glyco-lipide produced by pseudomonas aeruginosa, J. Am. Chem. Soc. 71 (12) (1949) 4124–4126.
[79] S.-Y. Chen, et al., Improved production of biosurfactant with newly isolated Pseudomonas aeruginosa S2, Biotechnol. Prog. 23 (3) (2007) 661–666.
[80] R. Cohen, et al., DLVO and non-DLVO forces in thin liquid films from rhamnolipids, J. Adhes. 80 (10 – 11) (2004) 875–894.
[81] R. Cohen, D. Exerowa, Surface forces and properties of foam films from rhamnolipid biosurfactants, Adv. Colloid Interface Sci. 134-135 (2007) 24–34.
[82] A.M. Abdel-Mawgoud, F. Lepine, E. Deziel, Rhamnolipids: diversity of structures, microbial origins and roles, Appl. Microbiol. Biotechnol. 86 (5) (2010) 1323–1336.
[83] O. Pornsunthorntawee, et al., Structural and physicochemical characterization of crude biosurfactant produced by Pseudomonas aeruginosa SP4 isolated from petroleum-contaminated soil, Bioresour. Technol. 99 (6) (2008) 1589–1595.
[84] J.R. Edwards, J.A. Hayashi, Structure of a rhamnolipid from Pseudomonas aeruginosa, Arch. Biochem. Biophys. 111 (2) (1965) 415–421.
[85] F. Lang, et al., Functional significance of cell volume regulatory mechanisms, Physiol. Rev. 78 (1) (1998) 247–306.
[86] C. Asselineau, J. Asselineau, Trehalose-containing glycolipids, Prog. Chem. Fats Other Lipids 16 (1978) 59–99.
[87] K. Sudo, et al., Age-associated characteristics of murine hematopoietic stem cells, J. Exp. Med. 192 (9) (2000) 1273–1280.
[88] H. Isoda, et al., Succinoyl trehalose lipid induced differentiation of human monocytoid leukemic cell line U937 into monocyte-macrophages, Cytotechnology 19 (1) (1995) 79–88.
[89] D.G. Cooper, D.A. Paddock, Production of a biosurfactant from Torulopsis bombicola, Appl. Environ. Microbiol. 47 (1) (1984) 173–176.
[90] R. Hommel, et al., Production of water-soluble surface-active exolipids by Torulopsis apicola, Appl. Microbiol. Biotechnol. 26 (3) (1987) 199–205.
[91] Y. Hu, L.-K. Ju, Purification of lactonic sophorolipids by crystallization, J. Biotechnol. 87 (3) (2001) 263–272.
[92] I.A.H. Namir, et al., Surfactin isoforms from Bacillus subtilis HSO121: separation and characterization, Protein Pept. Lett. 15 (3) (2008) 265–269.
[93] J.-S. Tang, et al., Complete assignments of 1H and 13C NMR spectral data of nine surfactin isomers, Magn. Reson. Chem. 45 (9) (2007) 792–796.
[94] K. Arima, A. Kakinuma, G. Tamura, Surfactin, a crystalline peptidelipid surfactant produced by Bacillussubtilis: isolation, characterization and its inhibition of fibrin clot formation, Biochem. Biophys. Res. Commun. 31 (3) (1968) 488–494.

[95] M.J. McInerney, M. Javaheri, D.P. Nagle, Properties of the biosurfactant produced byBacillus licheniformis strain JF-2, J. Ind. Microbiol. 5 (2) (1990) 95–101.
[96] M.C. Cirigliano, G.M. Carman, Purification and characterization of Liposan, a bioemulsifier from Candida lipolytica, Appl. Environ. Microbiol. 50 (4) (1985) 846–850.
[97] O. Käppeli, W.R. Finnerty, Partition of alkane by an extracellular vesicle derived from hexadecane-grown Acinetobacter, J. Bacteriol. 140 (2) (1979) 707–712.
[98] A. Kretschmer, H. Bock, F. Wagner, Chemical and physical characterization of interfacial-active lipids from Rhodococcus erythropolis grown on n-alkanes, Appl. Environ. Microbiol. 44 (4) (1982) 864–870.
[99] D.G. Cooper, et al., Enhanced production of Surfactin from Bacillus subtilis by continuous product removal and metal cation additions, Appl. Environ. Microbiol. 42 (3) (1981) 408–412.
[100] A. Gorkovenko, et al., Control of unsaturated fatty acid substituents in emulsans, Carbohydr. Polym. 39 (1) (1999) 79–84.
[101] B. Panilaitis, et al., Adjuvant activity of Emulsan, a secreted lipopolysaccharide from Acinetobacter calcoaceticus, Clin. Diagn. Lab. Immunol. 9 (6) (2002) 1240–1247.
[102] Z. Zosim, D. Gutnick, E. Rosenberg, Properties of hydrocarbon-in-water emulsions stabilized by Acinetobacter RAG-1 emulsan, Biotechnol. Bioeng. 24 (2) (1982) 281–292.
[103] B. Tuleva, et al., Isolation and characterization of trehalose tetraester biosurfactants from a soil strain Micrococcus luteus BN56, Process Biochem. 44 (2) (2009) 135–141.
[104] D. Kitamoto, et al., Extracellular accumulation of Mannosylerythritol lipids by a strain of Candida antarctica, Agric. Biol. Chem. 54 (1) (1990) 31–36.
[105] P. Saravanakumari, K. Mani, Structural characterization of a novel xylolipid biosurfactant from Lactococcus lactis and analysis of antibacterial activity against multi-drug resistant pathogens, Bioresour. Technol. 101 (22) (2010) 8851–8854.
[106] B. Mimee, C. Labbé, R. Belanger, Catabolism of flocculosin, an antimicrobial metabolite produced by Pseudozyma flocculosa, Glycobiology 19 (2009) 995–1001.
[107] H.T. Wattanaphon, et al., A biosurfactant from Burkholderia cenocepacia BSP3 and its enhancement of pesticide solubilization, J. Appl. Microbiol. 105 (2) (2008) 416–423.
[108] B. Thanomsub, et al., Monoacylglycerols: glycolipid biosurfactants produced by a thermotolerant yeast, Candida ishiwadae, J. Appl. Microbiol. 96 (3) (2004) 588–592.
[109] M. Morikawa, et al., A new lipopeptide biosurfactant produced by Arthrobacter sp. strain MIS38, J. Bacteriol. 175 (20) (1993) 6459.
[110] L. Thimon, et al., Interactions of bioactive lipopeptides, iturin A and surfactin from Bacillus subtilis, Biotechnol. Appl. Biochem. 16 (2) (1992) 144–151.
[111] T.P. Lin, et al., Functional and transcriptional analyses of a fengycin synthetase gene, fenC, from Bacillus subtilis, J. Bacteriol. 181 (16) (1999) 5060–5067.
[112] I. Grangemard, et al., Lichenysin, Appl. Biochem. Biotechnol. 90 (3) (2001) 199–210.
[113] N. Naruse, et al., Pumilacidin, a complex of new antiviral antibiotics. Production, isolation, chemical properties, structure and biological activity, J. Antibiot. (Tokyo) 43 (3) (1990) 267–280.
[114] T. Matsuyama, et al., A novel extracellular cyclic lipopeptide which promotes flagellum-dependent and -independent spreading growth of Serratia marcescens, J. Bacteriol. 174 (6) (1992) 1769–1776.
[115] M.V. Laycock, et al., Viscosin, a potent peptidolipid biosurfactant and phytopathogenic mediator produced by a pectolytic strain of Pseudomonas fluorescens, J. Agric. Food Chem. 39 (3) (1991) 483–489.

[116] D.G. Cooper, J.E. Zajic, D.E. Gracey, Analysis of corynomycolic acids and other fatty acids produced by Corynebacterium lepus grown on kerosene, J. Bacteriol. 137 (2) (1979) 795.
[117] C.R. Macdonald, D.G. Cooper, J.E. Zajic, Surface-active lipids from Nocardia erythropolis grown on hydrocarbons, Appl. Environ. Microbiol. 41 (1) (1981) 117–123.
[118] J.L. Beebe, W.W. Umbreit, Extracellular lipid of Thiobacillus thiooxidans, J. Bacteriol. 108 (1) (1971) 612–614.
[119] A. Franzetti, et al., Potential applications of surface active compounds by Gordonia sp. strain BS29 in soil remediation technologies, Chemosphere 75 (6) (2009) 801–807.
[120] E. Rosenberg, et al., Production of Biodispersan by Acinetobacter calcoaceticus A2, Appl. Environ. Microbiol. 54 (2) (1988) 317–322.
[121] R.M. Maier, Biosurfactants: evolution and diversity in bacteria, Adv. Appl. Microbiol. 52 (2003) 101–121.
[122] M.E. Mercade, M.A. Manresa, The use of agroindustrial by-products for biosurfactant production, J. Am. Oil Chem. Soc. 71 (1) (1994) 61–64.
[123] K.K. Gautam, V.K. Tyagi, Microbial surfactants: a review, J. Oleo Sci. 55 (4) (2006) 155–166.
[124] J.A. Casas, F. García-Ochoa, Sophorolipid production by Candida bombicola: medium composition and culture methods, J. Biosci. Bioeng. 88 (5) (1999) 488–494.
[125] M. Konishi, et al., Production of new types of sophorolipids by Candida batistae, J. Oleo Sci. 57 (6) (2008) 359–369.
[126] G.S. Kiran, et al., Optimization and production of a biosurfactant from the sponge-associated marine fungus aspergillus ustus MSF3, Colloids Surf. B Biointerfaces 73 (2) (2009) 250–256.
[127] L.A. Sarubbo, C.B. Farias, G.M. Campos-Takaki, Co-utilization of canola oil and glucose on the production of a surfactant by Candida lipolytica, Curr. Microbiol. 54 (1) (2007) 68–73.
[128] M. Deshpande, L. Daniels, Evaluation of sophorolipid biosurfactant production by Candida bombicola using animal fat, Bioresour. Technol. 54 (2) (1995) 143–150.
[129] H.-S. Kim, et al., Extracellular production of a glycolipid biosurfactant, mannosylerythritol lipid, from Candida antarctica, Biotechnol. Lett. 24 (3) (2002) 225–229.
[130] P.A. Felse, et al., Sophorolipid biosynthesis by Candida bombicola from industrial fatty acid residues, Enzyme Microb. Technol. 40 (2) (2007) 316–323.
[131] J.S. Kim, I.K. Lee, B.S. Yun, A novel biosurfactant produced by Aureobasidium pullulans L3-GPY from a tiger lily wild flower, Lilium lancifolium Thunb, PLoS One 10 (2015), https://doi.org/10.1371/journal.pone.0122917, e0122917.
[132] G. Barth, C. Gaillardin, Physiology and genetics of the dimorphic fungus Yarrowia lipolytica, FEMS Microbiol. Rev. 19 (4) (1997) 219–237.
[133] R. Hommel, et al., Production of Sophorose lipid by Candida (torulopsis) apicola grown on glucose, J. Biotechnol. 33 (1994) 147–155.
[134] D.G. Cooper, D.A. Paddock, Torulopsis petrophilum and surface activity, Appl. Environ. Microbiol. 46 (6) (1983) 1426–1429.
[135] D.R. Cameron, D.G. Cooper, R.J. Neufeld, The mannoprotein of Saccharomyces cerevisiae is an effective bioemulsifier, Appl. Environ. Microbiol. 54 (6) (1988) 1420–1425.
[136] T. Morita, et al., Discovery of Pseudozyma rugulosa NBRC 10877 as a novel producer of the glycolipid biosurfactants, mannosylerythritol lipids, based on rDNA sequence, Appl. Microbiol. Biotechnol. 73 (2) (2006) 305–313.

[137] M.C. Cirigliano, G.M. Carman, Isolation of a bioemulsifier from Candida lipolytica, Appl. Environ. Microbiol. 48 (4) (1984) 747–750.
[138] S. Zinjarde, et al., Isolation of an emulsifier from Yarrowia lipolytica NCIM 3589 using a modified mini isoelectric focusing unit, Lett. Appl. Microbiol. 24 (2) (1997) 117–121.
[139] P.F.F. Amaral, et al., Production and characterization of a bioemulsifier from Yarrowia lipolytica, Process Biochem. 41 (8) (2006) 1894–1898.
[140] L. Weber, et al., Oxygenation of hexadecane in the biosynthesis of cyclic glycolipids in Torulopsis Apicola, Biocatal. Biotransformation 5 (2009) 267–272.
[141] A. Tulloch, J. Spencer, M. Deinema, A new hydroxy fatty acid sophoroside from Candida bogoriensis, Can. J. Chem. 46 (2011) 345–348.
[142] D. Kitamoto, et al., Production of mannosylerythritol lipids as biosurfactants by resting cells of Candida Antarctica, Biotechnol. Lett. 14 (4) (1992) 305–310.
[143] K. Kakugawa, et al., Isolation of yeast Kurtzmanomyces sp. I-11, novel producer of mannosylerythritol lipid, Biosci. Biotechnol. Biochem. 66 (1) (2002) 188–191.
[144] M. Singh, J.D. Desai, Hydrocarbon emulsification by Candida tropicalis and Debaryomyces polymorphus, Indian J. Exp. Biol. 27 (3) (1989) 224–226.
[145] T. Lukondeh, N.J. Ashbolt, P.L. Rogers, Evaluation of Kluyveromyces marxianus as a source of yeast autolysates, J. Ind. Microbiol. Biotechnol. 30 (1) (2003) 52–56.
[146] C. Amézcua-Vega, et al., Effect of culture conditions on fatty acids composition of a biosurfactant produced by Candida ingens and changes of surface tension of culture media, Bioresour. Technol. 98 (1) (2007) 237–240.
[147] S.H. Yoon, J.S. Rhee, Lipid from yeast fermentation: effects of cultural conditions on lipid production and its characteristics of *Rhodotorula glutinis*, J. Am. Oil Chem. Soc. 60 (7) (1983) 1281–1286.
[148] S.R. Euston, Molecular simulation of biosurfactants with relevance to food systems, Curr. Opin. Colloid Interface Sci. 28 (2017) 110–119.
[149] R. Jahan, et al., Biosurfactants, natural alternatives to synthetic surfactants: physicochemical properties and applications, Adv. Colloid Interface Sci. 275 (2020), 102061.
[150] L. Sarubbo, et al., Production of a low-cost biosurfactant for application in the remediation of sea water contaminated with petroleum Derivates, Chem. Eng. Trans. 49 (2016) 523–528.
[151] A.A. Befkadu, Q. Chen, Surfactant-enhanced soil washing for removal of petroleum hydrocarbons from contaminated soils: a review, Pedosphere 28 (3) (2018) 383–410.
[152] A.M. Bodratti, P. Alexandridis, Amphiphilic block copolymers in drug delivery: advances in formulation structure and performance, Expert Opin. Drug Deliv. 15 (11) (2018) 1085–1104.
[153] S. Patel, et al., Microbial biosurfactants for oil spill remediation: pitfalls and potentials, Appl. Microbiol. Biotechnol. 103 (1) (2019) 27–37.
[154] A.M. Bailón-Salas, et al., Lead and copper removal from groundwater by spherical agglomeration using a biosurfactant extracted from Yucca decipiens Trel, Chemosphere 207 (2018) 278–284.
[155] K. Guemiza, et al., Treatment technologies used for the removal of As, Cr, Cu, PCP and/or PCDD/F from contaminated soil: a review, J. Hazard. Mater. 333 (2017) 194–214.
[156] E. Kaczorek, A. Olszanowski, Uptake of hydrocarbon by Pseudomonas fluorescens (P1) and pseudomonas putida (K1) strains in the presence of surfactants: a cell surface modification, Water Air Soil Pollut. 214 (1) (2011) 451–459.
[157] C.N. Mulligan, Recent advances in the environmental applications of biosurfactants, Curr. Opin. Colloid Interface Sci. 14 (5) (2009) 372–378.

[158] L.-M. Whang, et al., Application of biosurfactants, rhamnolipid, and surfactin, for enhanced biodegradation of diesel-contaminated water and soil, J. Hazard. Mater. 151 (1) (2008) 155–163.
[159] G. Onur, F. Yilmaz, B. Icgen, Diesel oil degradation potential of a bacterium inhabiting petroleum hydrocarbon contaminated surface waters and characterization of its emulsification ability, J. Surfactant Deterg. 18 (4) (2015) 707–717.
[160] G. Pi, et al., Preparation of oil-in-seawater emulsions based on environmentally benign nanoparticles and biosurfactant for oil spill remediation, ACS Sustain. Chem. Eng. 3 (11) (2015) 2686–2693.
[161] J.B. Prajapati, H. Katariya, R. Patel, Peyer'e patch targeting of Isradipine loaded solid lipid nanoparticles: it's cellular uptake study, J. Drug Delivery Sci. Technol. 43 (2018) 318–326.
[162] S. Lamichhane, K.C. Bal Krishna, R. Sarukkalige, Surfactant-enhanced remediation of polycyclic aromatic hydrocarbons: a review, J. Environ. Manage. 199 (2017) 46–61.
[163] S. Chandra, et al., Application of bioremediation technology in the environment contaminated with petroleum hydrocarbon, Ann. Microbiol. 63 (2) (2013) 417–431.
[164] A. Eivazihollagh, et al., On chelating surfactants: molecular perspectives and application prospects, J. Mol. Liq. 278 (2019) 688–705.
[165] A.J. Das, et al., Bacterial biosurfactants can be an ecofriendly and advanced technology for remediation of heavy metals and co-contaminated soil, Int. J. Environ. Sci. Technol. 14 (6) (2017) 1343–1354.
[166] J.M. Luna, R.D. Rufino, L.A. Sarubbo, Biosurfactant from Candida sphaerica UCP0995 exhibiting heavy metal remediation properties, Process Saf. Environ. Prot. 102 (2016) 558–566.
[167] B. Haryanto, C.-H. Chang, Removing adsorbed heavy metal ions from sand surfaces via applying interfacial properties of Rhamnolipid, J. Oleo Sci. 64 (2) (2015) 161–168.
[168] W. Chen, et al., Heavy metal (Cu, Cd, Pb, Cr) washing from river sediment using biosurfactant rhamnolipid, Environ. Sci. Pollut. Res. 24 (19) (2017) 16344–16350.
[169] G. Dermont, et al., Soil washing for metal removal: a review of physical/chemical technologies and field applications, J. Hazard. Mater. 152 (1) (2008) 1–31.
[170] H. Amani, Study of enhanced oil recovery by rhamnolipids in a homogeneous 2D micromodel, J. Petrol. Sci. Eng. 128 (2015) 212–219.
[171] A. Muggeridge, et al., Recovery rates, enhanced oil recovery and technological limits, Philos. Trans. R. Soc. A Math. Phys. Eng. Sci. 372 (2006) (2014) 20120320.
[172] H. Amani, et al., Comparative study of biosurfactant producing bacteria in MEOR applications, J. Petrol. Sci. Eng. 75 (1) (2010) 209–214.
[173] Y. Zhang, et al., Therapeutic surfactant-stripped frozen micelles, Nat. Commun. 7 (1) (2016) 11649.
[174] M. Torrice, Does nanomedicine have a delivery problem? ACS Cent. Sci. 2 (7) (2016) 434–437.
[175] G. Seydlová, J. Svobodová, Review of surfactin chemical properties and the potential biomedical applications, Open Med. 3 (2) (2008) 123–133.
[176] M. Deleu, et al., Effects of surfactin on membrane models displaying lipid phase separation, Biochim. Biophys. Acta Biomembr. 1828 (2) (2013) 801–815.
[177] E. Ruiz, et al., Antifungal activity and molecular identification of native strains of Bacillus subtilis, Agrociencia 50 (2016) 133–148.
[178] K.W. Naing, et al., Isolation and characterization of an antimicrobial lipopeptide produced by Paenibacillus ehimensis MA2012, J. Basic Microbiol. 55 (7) (2015) 857–868.

[179] P. Mani, et al., Antimicrobial activities of a promising glycolipid biosurfactant from a novel marine staphylococcus saprophyticus SBPS 15, 3 Biotech 6 (2) (2016) 163.
[180] T. Ndlovu, et al., Characterisation and antimicrobial activity of biosurfactant extracts produced by Bacillus amyloliquefaciens and Pseudomonas aeruginosa isolated from a wastewater treatment plant, AMB Express 7 (1) (2017) 108.
[181] F. Haque, et al., Inhibitory effect of sophorolipid on Candida albicans biofilm formation and hyphal growth, Sci. Rep. 6 (1) (2016) 23575.
[182] K. Sathi Reddy, et al., Utilization of mango kernel oil for the rhamnolipid production by Pseudomonas aeruginosa DR1 towards its application as biocontrol agent, Bioresour. Technol. 221 (2016) 291–299.
[183] S.K. Satpute, et al., Biosurfactant/s from lactobacilli species: properties, challenges and potential biomedical applications, J. Basic Microbiol. 56 (11) (2016) 1140–1158.
[184] K.G.O. Bezerra, et al., Saponins and microbial biosurfactants: potential raw materials for the formulation of cosmetics, Biotechnol. Prog. 34 (6) (2018) 1482–1493.
[185] A. Merghni, et al., Antioxidant and antiproliferative potential of biosurfactants isolated from lactobacillus casei and their anti-biofilm effect in oral Staphylococcus aureus strains, Microb. Pathog. 104 (2017) 84–89.
[186] A. Ferreira, et al., Novel cosmetic formulations containing a biosurfactant from lactobacillus paracasei, Colloids Surf. B Biointerfaces 155 (2017) 522–529.
[187] P. Singh, S.S. Cameotra, Enhancement of metal bioremediation by use of microbial surfactants, Biochem. Biophys. Res. Commun. 319 (2) (2004) 291–297.
[188] M. Henkel, et al., Production of microbial biosurfactants: status quo of rhamnolipid and surfactin towards large-scale production, Biotechnol. J. 12 (7) (2017).
[189] B.F. Gibbs, et al., Encapsulation in the food industry: a review, Int. J. Food Sci. Nutr. 50 (3) (1999) 213–224.

CHAPTER

Application of biosurfactant as an emulsifying agent

3

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1 Introduction

Food additives are the chemicals or compounds that are added to food to enhance their organoleptic characteristics and to preserve the food from deterioration by microorganism. Taste, flavor, and nutritional value of food get altered during manufacturing, processing, packaging, storage, transportation, and handling. The majority of the changes in foods are caused by bacteria altering the biochemical properties of the meal. These can be caused by the breakdown of carbohydrates, proteins, lipids, or pectin. Microorganisms digest carbohydrates to create CO_2, H_2, lactate, acetate, butyrate, isovalerate, ethanol, propanol, butanol, diacetyl, acetoin, butanediol, dextran, and other compounds, depending on the carbohydrate and microorganism. Foods that have become slimy and have an unpleasant flavor or odor are referred to as stale foods.

Proteins are degraded by microorganisms into keto acid, organic acid, CO_2, H_2, H_2S, amines, putrescine, mercaptans, cadaverine, etc. To put it another way, proteins putrefy, and foods decay and have a foul odor. Protein breakdown, on the other hand, does not begin until there is a dearth of nitrogen supply. Proteases and peptidases are secreted by bacteria when this occurs, and protein is digested into amino acids. Because of their bitter or sweet flavor, many peptides contribute to deterioration.

Microorganisms breakdown lipids into fatty acids, glycerol, aldehydes, ketones, and other compounds. When food becomes old, it develops a terrible odor and flavor, which is known as rancidity. Oxygen, fat-containing unsaturated fatty acids, light, high temperature, and metal ions are all required for lipid peroxidation. The resulting alcohols, ketones, and aldehydes have a disagreeable odor and taste. β-Oxidation is a prevalent form of fat degradation that creates acetyl coA and fatty acids that are 2C atoms shorter each time they are broken down. Furthermore, some fungi have lipoxydases, which convert unsaturated fatty acids to hydroperoxide, then aldehyde and ketones. Lipid deterioration can be prevented by minimizing O_2 exposure, adding antioxidants, and lowering the temperature.

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Pectins are partly methylated poly-a-(1–4)-D-galacturonic acid polysaccharides. Mechanical stiffness is provided by pectin, which is found in fruits and vegetables. Pectin can be destroyed by pectinases, which are triggered when fruits ripen. It has the potential to soften the fruit. Microbial attack begins after mechanical damage has occurred. Furthermore, food rotting is caused by natural food qualities such as pH, moisture level, and nutritional concentration.

Thus, preservation of food especially packed food is necessary to prevent the food from spoilage. As a result, food preservatives/additives are used to preserve the nutritious value of foods. Food additives, according to the European Food Safety Authority (EFSA), are any compounds that are purposefully added to foods in order to improve its quality. Preservatives, dietary supplements, flavoring agents, coloring and texturing agents, and other additions are among them. Antimicrobials and antioxidants are examples of food preservatives; dietary additives include vitamins, fabrics, amino acids, among others. Textured agents are emulsifiers and stabilizers that are typically used to change the texture or mouthfeel of foods. Flavoring agents are enhancers of sweeteners, spices, and tastes, while textured agents are emulsifiers and stabilizers that are typically used to change the texture or mouthfeel of foods. Chelating agents, lubricants, enzymes, and antifoam agents are just a few of the additives available. The main function of additives is to maintain and improve the nutritional quality of food and the taste and texture of food. The concern of safety of food additive is the fundamental principal behind choosing an additive for foods. Generally, food additives worked by stabilizing the food texture, emulsifying the food, or they can have gelling effect as well. The commonly used food additives are listed in Table 1. As the demand for more natural products increases day by day, the requirement of natural additives in food industry also keeps on increasing.

Surfactants created by microorganisms are environmentally and ecologically safe compounds known as biosurfactants. Biosurfactants are amphiphilic molecules containing both hydrophilic and hydrophobic moieties [1]. The hydrophilic portion can be ionic (cationic or anionic), while the hydrophobic portion is typically a hydrophobic chain (Fig. 1).

Biosurfactants can separate two phases, such as oil and water or air and water. These are also known as bioemulsifiers since they generate emulsions to lessen the interfacial tension between the surfaces. Thus, with the use of surfactants, oil can be solubilized in water. These properties of surfactants make them useful in wide range of industrial applications, namely in detergents, food industry, dairy industry, lubrication, moisture retention, etc. [1–3]. Biosurfactants, because of their antimicrobial, antiadhesive, and emulsification properties, have great potential to be used as food additive.

Surfactants form micelle in solutions, which allow them to partition between the oil and water phase. Depending upon which emulsion (oil in water/water in oil) is to be formed, the hydrophobic and hydrophilic portions arrange themselves inside out. For oil-in-water emulsion, hydrophilic portion arranges itself outward and hydrophilic portion inward and vice versa. The minimum concentration of surfactant needed to form emulsion is called critical micelle concentration (CMC). Addition of surfactant above the CMC has no further effect on emulsion formation. Thus, CMC is the critical factor in determining the rate of formation of emulsion [1].

Table 1 Commonly used food additives.

S. No.	General name	Commercial name	Use	Harmful effect
1	Monosodium glutamate	E621	To enhance flavor of processed foods such as salty foods	Metabolic disorders, weight gain
2	Food colorants	E100-E199	Use to improve the appearance of food products	Allergic reactions, increasing risk of tumors
3	Sodium nitrite	E251	Prevent growth of bacteria, commonly used in processed meats	In the presence of high heat and amino acids, it turns into nitosamine, which can lead to stomach, colorectal, breast and bladder cancer
4	Guar Gum	E412	Used for thickening and binding of foods, e.g., in ice creams, salad dressings, sauces, and soups	High intake can lead to obstruction of esophagus or small intestine
5	High-fructose corn syrup	HFCS42; HFCS 55	Sweetner made from corn; found in soda, candies, breakfast cereals	Weight gain, diabetes and inflammation
6	Artificial sweeteners	Nutrasweet; Equal; SugarTwin	Used in diet foods	Mild side effects like headache
7	Carragenan	E407	Used as thickener, emulsifier and preservative in food	High blood sugar, intestinal ulcers, ulcerative colitis
8	Sodium benzoate	E211	Preservative in food	Increased hypersensitivity
9	Xanthan Gum	Thickening and stabilizing agent	In large amount can cause blotting, gastric	
10	Artificial flavoring agents	E600-E699	Used for giving different flavors to food	Maybe toxic to bone marrow cells

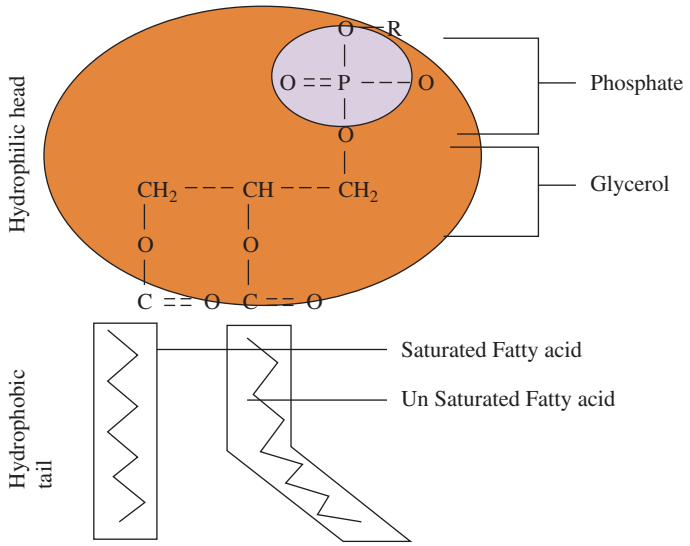

FIG. 1

Structure of biosurfactants showing hydrophobic and hydrophilic moieties.
From Author links open overlay panel, Abraham Peele Karlapudia, TC Venkateswarulua, Jahnavi Tammineedia, Lohit Kanumuria, Bharath Kumar Ravurua, Vijaya ramu Dirisalaa, Vidya Prabhakar Kodali. Role of biosurfactants in bioremediation of oil pollution—a review. Petroleum 4(3) (2018) 241–249. https:/doi.org/10.1016/j.petlm.2018.03.007.

2 Classification of biosurfactants

Biosurfactants can be classified based on their chemical structures and microbial origin. They can be classified into low-mass surfactants such as glycolipids, phospholipids, and lipopeptides or high-mass surfactants such as polymeric and particulate surfactant [4]. Table 2 lists the major classes of biosurfactants with their subclasses and the microorganisms producing them.

3 Glycolipids

Glycolipids are carbohydrates that contain long chains of aliphatic or hydroxyaliphatic acids that are connected together by ester bonds. Most of the biosurfactants are glycolipids by nature. Rhamnolipids, trehalolipids, and sophorolipids are the three primary glycolipids. Rhamnolipids are glycolipids in which rammnose sugar is linked to one or two groups of hydroxydecanoic acid [5] (Fig. 2). They are predominantly produced by *Pseudomonas* sp. including *P. aeruginosa* [6], *Pseudomonas chlororaphis* [7], *Pseudomonas putida* [8], *P. fluorescens* [9] while *Burkholderia*

Table 2 Classification of biosurfactants.

S. No.	Major class	Sub class	Microorganisms
1	Glycolipids	Rhamnolipids	*Pseudomonas aeruginosa, Pseudomonas putida, P. fluroscens, Burkholderia thailandensis*
		Trehalolipids	*Rhodococcus* sp., *Arthobacter* sp.
		Sophorolipids	*Candida bombicola, C. apicola*
		Mannosylerythritol lipids	*C. antartica*
2	Lipopeptides/ lipoproteins	Surfactin/iturin/fengycin	*Bacillus subtilis*
		Lichenysin	*Bacillus licheniformis*
3	Polymeric surfactants	Emulsan	*Acinetobacter calcoaceticus*
		Alasan	*Acinetobacter radioresistens*
		Liposan	*Candida lipolytica*
		Lipomanan	*C. tropicalis*
4	Particulate biosurfactant	Vesicles	*A. calcoaceticus*
		Whole microbial cells	Cyanobacteria
5	Fatty acids, phospholipids and neutral lipids	Phosphatidalethanolamine	Acinetobacter sp.

thailandensis has also been reported to produce rhamnolipids [10]. Predominantly produced by Mycobacterium, Nocardia, and Cornybacterium, trehalolipids are excellent bioemulsifiers [5].

Trehalolipids are used in a variety of fields, including bioremediation, oil recovery, cosmetics, and food. Rhodococcus sp. are also known to produce trehalolipids (Fig. 3).

Sophorolipids are composed of the dimeric carbohydrate sophorose, which is a disaccharide with a -1,2-acetal with or -1 hydroxy acids of chain length C16 or C18, linked via a -glycosidic (ether) connection between the hydroxyl group at the 2-position of sophorose and the OH group of the fatty acid [5] (Fig. 4). They are predominantly produced by nonpathogenic yeasts [11]. *Torulopsis magnolia*, currently known as *Candida apicola*, was the first yeast reported to produce sophorolipids [12]. Several other yeast strains include *C. bogoriensis, C. bombicola* [11,12].

3.1 Lipopeptides/lipoproteins

Lipid molecules are linked to a polypeptide chain in lipopeptide surfactants. Antimicrobial activity is mostly demonstrated by these biosurfactants against bacteria, fungus, algae, and viruses [13]. *Bacillus* sp. produces a wide range of lipopeptide

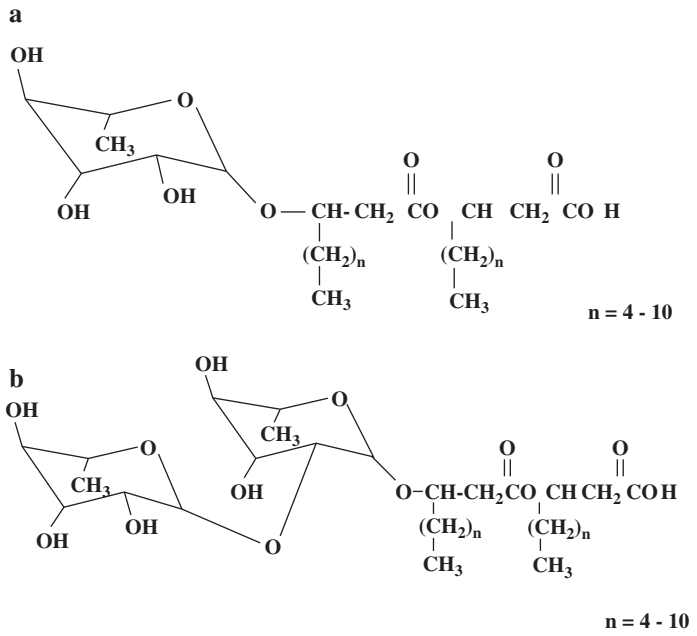

FIG. 2

Chemical structure of rhamnolipid (A) mono-rhamnolipid (B) di-rhamnolipi.
From Nereus W. Gunther IV, Alberto Nuñez, Laurie Fortis, Daniel K.Y. Solaiman. Proteomic based investigation of rhamnolipid production by Pseudomonas chlororaphis *strain NRRL B-30761. J. Ind. Microbiol. Biotechnol. 33 (2006) 914–920. https:/doi.org/doi 10.1007/s10295-006-0169-1.*

biosurfactants, the most well-known of which are the surfactin, iturin, and fengycin families [14]. *B. subtilis, Bacillus licheniformis, B. atropaeus, Bacillus amyloliquefaciens* are some of the species reported to produce lipoproteins [15–17]. Thivaharan and Ramachandra [18] have reported lipoprotein production from *B. siamensis*. Surfactin, which is produced by *B. subtilis*, is the most potent surfactant, lowering surface tension from 72 to 27.9 mN/m [19]. Surfactin is made up of seven amino acid rings connected by lactone connections to a fatty acid chain. Iturin and fengycin have antifungal properties, but surfactin has antibacterial properties [20]. Lichenysin is produced by *B. licheniformis* and is similar in structure to surfactin. Lichenysin is highly stable at high temperature, pH, and salt concentration. The surface tension of water can also be reduced from 72 to 27 mN/m [12].

Fatty acids, phospholipids, and neutral lipids are all types of lipids. During their development on n-alkanes, several bacteria and yeast create these. Phosphatidyl ethanolamine-rich vesicles produced by *Acinetobacter* sp. generate microemulsions of alkanes in water [21]. Some *Aspergillus flavus* [22] and *Thiobacillus thiooxidans* strains have been found to produce phospholipids in large quantities [23].

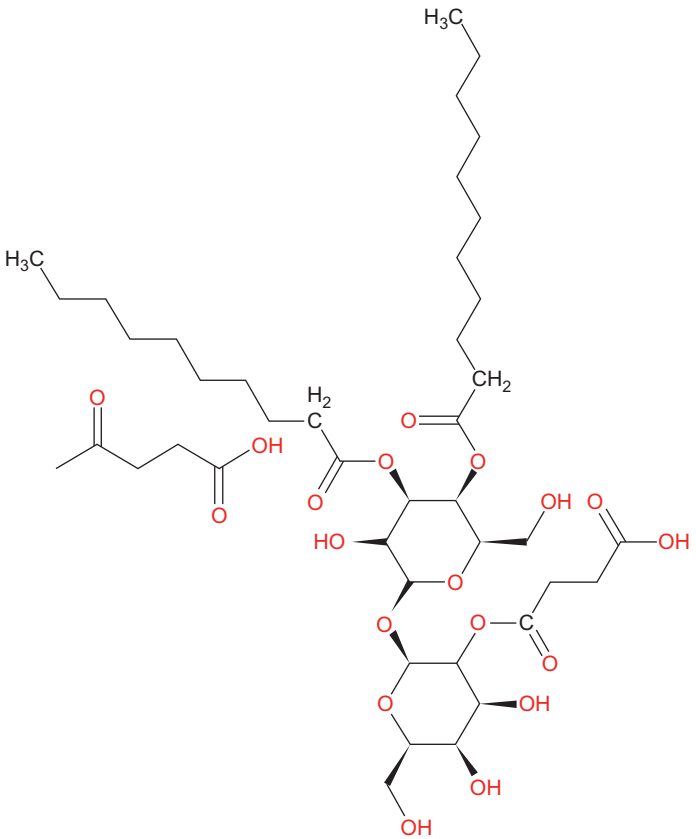

FIG. 3

Chemical structure of trehalolipid.

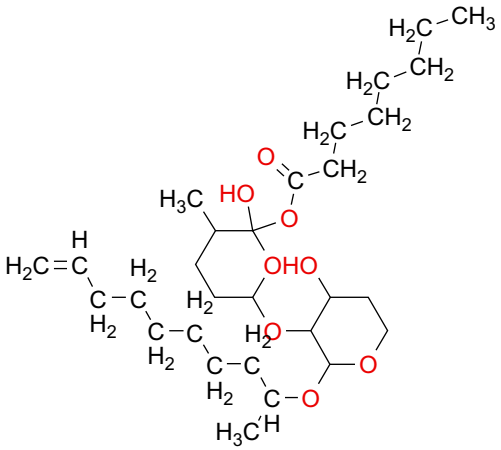

FIG. 4

Chemical structure of sophorolipids

3.1.1 Polymeric biosurfactant

Polymeric biosurfactants are formed of complex ionic heteropolysaccharides, e.g., emulsan, lipomanan, alasan, liposan. Emulsan, produced by *Acinetobacter calcoaceticus*, is a heteropolysaccharide emulsifier consisting of a repeating trisaccharide of *N*-acetyl-D-galactosamine, *N*-acetylgalactosamine uronic acid, and an unidentified *N*-acetyl amino sugar linked with fatty acid molecule by ester linkages [23]. Emulsan is an extremely powerful emulsifier for hydrocarbons in water, even at concentrations as low as 0.001% to 0.1%. Liposan, on the other hand, is an extracellular water-soluble emulsifier made up of 83% carbohydrates and 17% proteins produced by *Candida lipolytica* [24]. Glucose, galactose, galactosamine, and galacturonic acid make up the carbohydrate part of liposan. Biodispersan has been reported to be produced by *A. calcoaceticus* extracellularly. It is an anti-dispersing agent. It is an anionic heteropolysaccharide consisting of four reducing sugars, namely glucosamine, 6-methylaminohexose, galactosamine uronic acid, and an unidentified amino sugar [23].

3.1.2 Particulate biosurfactants

Particulate biosurfactants have a critical function in microbial alkane absorption. These biosurfactants generate a microemulsion by partitioning across extracellular membrane vesicles. *Acinetobacter* spp. has been reported to produce these [5].

4 Biosurfactant in food

Biosurfactants are produced by microorganisms and can be utilized in industry in the same way as conventional surfactants. Surfactants produced by microorganisms are generally regarded as safe (GARS) and therefore can be directly utilized in food industries as preservative. Biosurfactants offer antibacterial and anti-biofilm effects against food pathogens, in addition to their intrinsic surface-active capabilities; thus, biosurfactants can be useful additions or components in food preparation.

4.1 Use of biosurfactants as emulsifier

Emulsifiers and surfactants mixes are not new to the food business and have been routinely utilized in the plan of various food items throughout the hundreds of years. Emulsifiers are primarily used to lower the surface tension between two immiscible phases, allowing the two phases to mix. Emulsions can be oil in water or water in oil. In oil emulsion, small oil droplets form discrete and dispersed phase in water, whereas in water-in-oil phase, small droplets of water distribute in oil [25]. Emulsifier restricts the two phases from separation resulting in the formation of emulsion. Due to this inherent property, biosurfactants have great role in food industry. Although they do not have great surface activity, low-mass surfactants reduce surface tension during emulsification, whereas large-mass surfactants produce and stabilize emulsions [26]. Biosurfactants ensure that nutrients in food are distributed

evenly. Currently, emulsions are used in milk, butter, margarine, mayonnaise, and ice cream. Dairy, fermented products, bread, distilleries routinely utilize manufactured and common emulsifiers and surfactants [25].

In baking industry, use of biosurfactants results in increased specific loaf volume and texture profile of bread. Mnif et al. [27] have studied the effect of *Bacillus subtilis* SPB1 biosurfactant on the quality of bread and compared it with the commercial surfactant, the soya lecithin. They discovered that the loaves were in better shape, had a higher specific volume, and had a lower voided fraction. The bread quality parameters were improved considerably while notable decrease in firmness, chewiness, and adhesiveness was observed. The mechanisms behind the emulsifying effect of biosurfactant in bakery is that the surfactant binds to the gluten protein present in wheat because of its ionic structure and generates a modified elastic structure resulting in increased bread volume [25].

Liposan, a water-soluble emulsifier manufactured by *C. lipolytica*, is commonly used to emulsify edible oils by coating oil droplets and generating stable emulsions. These liposans were widely employed in the cosmetics and food industries to create oil/water emulsions that were stable [28]. In dairy industry, biosurfactants enhance the texture and consistency of foods, e.g., cheese, ice creams [29]. Salad dressings have also been reported to use *Candida utilis* bio-emulsifier. [30]. Thus, property of emulsification makes biosurfactants a good candidate to be used in food industry.

4.2 Use of biosurfactants as antiadhesive agent

Biofilm, a thin layer of microorganisms, adheres to the surface of an organic or inorganic structure with their secreted polymers. Biofilm formation is a basic phenotype of all harmful and useful bacteria. Bacteria form a tight network among themselves, which make them 1000 times more resistant to antibiotics than the free bacteria in circulation [31]. In food industry, biofilm formation is a serious concern as it can lead to contamination resulting in spoilage of food and disease transmission [32]. Food-borne pathogenic bacteria such as *Listeria monocytogenes, Salmonella enteritidis,* and *Staphylococcus aureus* can develop biofilms on a range of surfaces widely used in the food environment, including stainless steel, plastics, rubber, and glass [33,34].

Biosurfactants inhibit the adhesion of microorganisms to surfaces, thus hampering the process of biofilm formation. Adhesion of biosurfactants to surfaces prior to biofilm formation can prevent the colonization of pathogenic microorganisms and hence onset of disease. Zezzi do Valle Gomes and Nitschke [32] evaluated the performance of surfactin from *B. subtilis* and rhamnolipids from *Pseudomonas aeruginosa* against the biofilm formation by food-borne pathogens. Polystyrene was used as the model surface to study individual and mixed cultures of *S. aureus*, *L. monocytogenes*, and *S. enteritidis*. Preconditioning surfaces with 0.25% surfactin reduced *L. monocytogenes* and *S. enteritidis* adhesion to polystyrene by 42.0%, whereas treatment with 1.0% rhamnolipids reduced *L. monocytogenes* adhesion to polystyrene by 57.8% and *S. aureus* adhesion to polystyrene by 67.8%. When compared with individual cultures of bacteria, biosurfactants were less successful at preventing

adherence of mixed cultures. Janek et al. [35] purified trehalose glycolipid from *Rhodococcus fascians* BD8 showing antiadhesive activity of 95% and 70% against *C. albicans* and *Escherichia coli*, respectively, to polystyrene and silicone surfaces. The mechanism behind the antiadhesive property of surfactant is that surfactant modifies the hydrophobicity of solid surfaces, thus affecting the adhesion of bacteria to surface [33,34].

4.3 Use of biosurfactants as antimicrobial agents

Food-borne pathogenic bacteria are always a matter of concern as they cause many diseases to humans. Bacteria such as *Salmonella enterica, E. coli, Staphylococcusaureus* are predominantly responsible for food-borne diseases. Salmonellosis is most commonly associated with the consumption of fresh tomatoes, green leafy vegetables, and rock melon. These directly edible plants get infected during agriculture practices and transfer the pathogen to consumers leading to salmonellosis [1–3]. Despite being a contributor to human disease, pathogens also destroy plants and reduce its yield, e.g., bacterial leaf spot disease of lettuce caused by *Xanthomonas campestris* [3].

Biosurfactants from *Bacillus* such as sufactin, fengicin, iturin, and bacillomicin have been reported to show antimicrobial properties [36]. Lactobacillus and yeast provide the majority of biosurfactants. Biosurfactants from *Lactobacillus jensenii* and *Lactobacillus rhamnosus* were found to have antibacterial activity against *E. coli* and methicillin-resistant bacteria. *S. aureus* is a kind of bacteria. [37]. Yeasts-derived sophorolipid has been reported to act as fruit preservative [28]. Rhamnolipids were found to be effective biosurfactant against *L. monocytogenes* [38]. It is claimed that biosurfactant disturbs interaction between membrane phospholipids as well as permeability of ions channels [39].

4.4 Use of biosurfactant as antioxidant

Antioxidants have a significant function in extending the shelf life of food products by inhibiting the oxidation of lipid content [2]. In food products, lipid oxidation developed rancidity, which leads to change in flavor, aroma and sometime produces toxic by-product. Biosurfactant can also be used as antioxidant to enhance shelf life of food. Antioxidant value of *B. subtilis* RW-I lipopeptide surfactant showed good antioxidant activity and can be an alternative to natural antioxidant [28]. Biosurfactant from *Alcaligenes piechaudii* CC-ESB2 having good antioxidant properties can be used in green and health food products [40].

4.5 Use of biosurfactants for removal of heavy metals

Heavy metals in food are also a matter of serious concern as it poses serious health hazards. The condition of soil in which plant is growing and its surrounding environment are the parameters in determining the uptake of heavy metals. Heavy metal

bioaccumulation in and on food surfaces creates numerous health problems for those who consume contaminated food. Ionic surfactants bind to heavy metals through mechanism of ion exchange and precipitate it resulting in removal of metal in the form of aggregates [41]. Ravindran et al. [41] used 2.0 CMC of the lipopeptide MSI54 generated by *Bacillus* sp. to treat the surface of fresh cabbage, carrot, and lettuce. In cabbage samples treated with the biosurfactant solution, copper contents were reduced by 71.4%, lead by 86.79%, and cadmium by 62.93%. In carrots treated with the biosurfactant solution, copper was reduced by 63.74%, lead by 83.4%, zinc by 62.2%, and cadmium by 69.5%; copper was reduced by 62.4%, lead by 80.77%, cadmium by 66.77%, and nickel by 59% in lettuce. In addition to working as a food additive, biosurfactant could be used as a wash solution to eliminate heavy metal contamination from the surface of vegetables.

5 Conclusion

Biosurfactants are secondary metabolites synthesized by microorganism, which are having great potential as food additive. These surfactants have multifunctional properties making them useful candidate as food ingredient. They can be used as emulsifiers, antiadhesive agents, and antioxidants in foods. Biosurfactant, besides being used as ingredient, can also be used to detoxify food surfaces so that it can be consumed directly. As microbial biosurfactants are safe to use in food, the end use market of these surfactants increases day by day. Further work on cost-effective biosurfactant production at large scale can facilitate more of its use in food industry.

References

[1] J.M. Campos, T.L. Montenegro Stamford, L.A. Sarubbo, J.M. de Luna, R.D. Rufino, I.M. Banat, Microbial biosurfactants as additives for food industries, Biotechnol. Prog. 29 (5) (2013) 1097–1108, https://doi.org/10.1002/btpr.1796.

[2] M. Carocho, I.C.F.R. Ferreira, A review on antioxidants, prooxidants and related controversy: natural and synthetic compounds, screening and analysis methodologies and future perspectives, Food Chem. Toxicol. 51 (1) (2013) 15–25, https://doi.org/10.1016/j.fct.2012.09.021.

[3] M. Rani, J.T. Weadge, S. Jabaji, Isolation and characterization of biosurfactant-producing bacteria from oil well batteries with antimicrobial activities against food-borne and plant pathogens, Front. Microbiol. 11 (2020), https://doi.org/10.3389/fmicb.2020.00064.

[4] N. Ranasalva, R. Sunil, G. Poovarasan, Importance of Biosurfactant in Food Industry, vol. 7, 2014, pp. 6–9.

[5] S. Vijayakumar, V. Saravanan, Biosurfactants-types, sources and applications, Res. J. Microbiol. 10 (5) (2015) 181–192, https://doi.org/10.3923/jm.2015.181.192.

[6] G. Soberón-Chávez, F. Lépine, E. Déziel, Production of rhamnolipids by Pseudomonas aeruginosa, Appl. Microbiol. Biotechnol. 68 (6) (2005) 718–725, https://doi.org/10.1007/s00253-005-0150-3.
[7] N.W. Gunther IV, A. Nuñez, W. Fett, D.K.Y. Solaiman, Production of rhamnolipids by pseudomonas chlororaphis, a nonpathogenic bacterium, Appl. Environ. Microbiol. 71 (5) (2005) 2288–2293, https://doi.org/10.1128/AEM.71.5.2288-2293.2005.
[8] R. Hausmann, C. Syldatk, F. Rosenau, L.M. Blank, J. Hemmerich, C. Müller, R. Wichmann, B. Küpper, M. Zwick, S. Wilhelm, Growth independent rhamnolipid production from glucose using the non-pathogenic pseudomonas putida KT 2440, Microb. Cell Fact. 10 (2011), https://doi.org/10.1186/1475-2859-10-80.
[9] B. Kaskatepe, S. Yildiz, Rhamnolipid biosurfactants produced by pseudomonas species, Braz. Arch. Biol. Technol. 59 (2016) 1–16. http://www.scielo.br/scielo.php?script=sci_issues&pid=1516-8913&lng=en&nrm=iso.
[10] M.A. Díaz De Rienzo, I.D. Kamalanathan, P.J. Martin, Comparative study of the production of rhamnolipid biosurfactants by B. thailandensis E264 and P. aeruginosa ATCC 9027 using foam fractionation, Process Biochem. 51 (7) (2016) 820–827, https://doi.org/10.1016/j.procbio.2016.04.007.
[11] S. Bages-Estopa, D.A. White, J.B. Winterburn, C. Webb, P.J. Martin, Production and separation of a trehalolipid biosurfactant, Biochem. Eng. J. 139 (2018) 85–94, https://doi.org/10.1016/j.bej.2018.07.006.
[12] P.A.J. Gorin, J.F.T. Spencer, A.P. Tulloch, Hydroxy fatty acid glycosides of sophorose from torulopsis magnoliae, Can. J. Chem. (1961) 846–855, https://doi.org/10.1139/v61-104.
[13] M. Healy, C. Devine, Murphy., Microbial production of biosurfactants, Resour. Conserv. Recycl. 18 (1996), https://doi.org/10.1016/S0921-3449(96. pp. 1167–6.
[14] A. Sarwar, G. Brader, E. Corretto, G. Aleti, M. Abaidullah, A. Sessitsch, F.Y. Hafeez, Qualitative analysis of biosurfactants from Bacillus species exhibiting antifungal activity, PLoS One 13 (6) (2018), https://doi.org/10.1371/journal.pone.0198107.
[15] H.S. El-Sheshtawy, I. Aiad, M.E. Osman, A.A. Abo-ELnasr, A.S. Kobisy, Production of biosurfactants by bacillus licheniformis and Candida albicans for application in microbial enhanced oil recovery, Egypt. J. Pet. 25 (3) (2016) 293–298, https://doi.org/10.1016/j.ejpe.2015.07.018.
[16] S.S. Jha, S.J. Joshi, S.J. Geetha, Lipopeptide production by Bacillus subtilis R1 and its possible applications, Braz. J. Microbiol. 47 (4) (2016) 955–964, https://doi.org/10.1016/j.bjm.2016.07.006.
[17] M. Rodríguez, A. Marín, M. Torres, V. Béjar, M. Campos, I. Sampedro, Aphicidal activity of surfactants produced by Bacillus atrophaeus L193, Front. Microbiol. 9 (2018), https://doi.org/10.3389/fmicb.2018.03114.
[18] V. Thivaharan, M.V. Ramachandra, Production of a Lipopeptide biosurfactant by a novel bacillus sp. and its applicability to enhanced oil recovery. *ISRN*, Microbiology (2013) 1–8, https://doi.org/10.1155/2013/621519.
[19] K.K. Gautam, V.K. Tyagi, Microbial surfactants: a review, J. Oleo Sci. 55 (4) (2006) 155–166, https://doi.org/10.5650/jos.55.155.
[20] K.R. Meena, S.S. Kanwar, Lipopeptides as the antifungal and antibacterial agents: applications in food safety and therapeutics, Biomed. Res. Int. 2015 (2015), https://doi.org/10.1155/2015/473050.
[21] D.K.F. Santos, R.D. Rufino, J.M. Luna, V.A. Santos, L.A. Sarubbo, Biosurfactants: multifunctional biomolecules of the 21st century, Int. J. Mol. Sci. 17 (3) (2016), https://doi.org/10.3390/ijms17030401.

[22] U. Ishaq, M.S. Akram, Z. Iqbal, M. Rafiq, A. Akrem, M. Nadeem, F. Shafi, Z. Shafiq, S. Mahmood, M.A. Baig, Production and characterization of novel self-assembling biosurfactants from aspergillus flavus, J. Appl. Microbiol. 119 (4) (2015) 1035–1045,- https://doi.org/10.1111/jam.12929.
[23] J.D. Desai, I.M. Banat, Microbial production of surfactants and their commercial potential, Microbiol. Mol. Biol. Rev. 61 (1) (1997) 47–64.
[24] M.C. Cirigliano, G.M. Carman, Purification and characterization of liposan, a bioemulsifier from Candida lipolytica, Appl. Environ. Microbiol. 50 (4) (1985) 846–850.
[25] M. Alizadeh-Sani, H. Hamishehkar, A. Khezerlou, M. Azizi-Lalabadi, Y. Azadi, E. Nattagh-Eshtivani, M. Fasihi, A. Ghavami, A. Aynehchi, A. Ehsani, Bioemulsifiers derived from microorganisms: applications in the drug and food industry, Adv. Pharm. Bull. 8 (2) (2018) 191–199, https://doi.org/10.15171/apb.2018.023.
[26] I. Kralova, J. Sjöblom, Surfactants used in food industry: a review, J. Dispers. Sci. Technol. 30 (9) (2009) 1363–1383, https://doi.org/10.1080/01932690902735561.
[27] I. Mnif, S. Besbes, R. Ellouze, S. Ellouze-Chaabouni, D. Ghribi, Improvement of bread quality and bread shelf-life by Bacillus subtilis biosurfactant addition, Food Sci. Biotechnol. 21 (4) (2012) 1105–1112, https://doi.org/10.1007/s10068-012-0144-8.
[28] M. Nitschke, S.S.E. Silva, Recent food applications of microbial surfactants, Crit. Rev. Food Sci. Nutr. 58 (4) (2018) 631–638, https://doi.org/10.1080/10408398.2016.1208635.
[29] M.S. Yeh, Y.H. Wei, J.S. Chang, Enhanced production of surfactin from Bacillus subtilis by addition of solid carriers, Biotechnol. Prog. 21 (4) (2005) 1329–1334, https://doi.org/10.1021/bp050040c.
[30] J.M. Campos, T.L.M. Stamford, L.A. Sarubbo, Characterization and application of a biosurfactant isolated from Candida utilis in salad dressings, Biodegradation 30 (4) (2019) 313–324, https://doi.org/10.1007/s10532-019-09877-8.
[31] R.D. Rufino, J.M. Luna, L.A. Sarubbo, L.R.M. Rodrigues, J.A.C. Teixeira, G.M. Campos-Takaki, Antimicrobial and anti-adhesive potential of a biosurfactant Rufisan produced by Candida lipolytica UCP 0988, Colloids Surf. B Biointerfaces 84 (1) (2011) 1–5, https://doi.org/10.1016/j.colsurfb.2010.10.045.
[32] M. Zezzi do Valle Gomes, M. Nitschke, Evaluation of rhamnolipid and surfactin to reduce the adhesion and remove biofilms of individual and mixed cultures of food pathogenic bacteria, Food Control 25 (2) (2012) 441–447, https://doi.org/10.1016/j.foodcont.2011.11.025.
[33] S.C. Marques, J.D.G.O.S. Rezende, L.A.D.F. Alves, B.C. Silva, E. Alves, L.R. De Abreu, R.H. Piccoli, Formation of biofilms by Staphylococcus aureus on stainless steel and glass surfaces and its resistance to some selected chemical sanitizers, Braz. J. Microbiol. 38 (3) (2007) 538–543, https://doi.org/10.1590/S1517-83822007000300029.
[34] T. Meylheuc, C.J. van Oss, M.N. Bellon-Fontaine, Adsorption of biosurfactant on solid surfaces and consequences regarding the bioadhesion of listeria monocytogenes LO28, J. Appl. Microbiol. (2001) 822–832, https://doi.org/10.1046/j.1365-2672.2001.01455.x.
[35] T. Janek, A. Krasowska, Z. Czyznikowska, M. Łukaszewicz, Trehalose lipid biosurfactant reduces adhesion of microbial pathogens to polystyrene and silicone surfaces: an experimental and computational approach, Front. Microbiol. 9 (2018), https://doi.org/10.3389/fmicb.2018.02441.
[36] B. Prasad, A review on applications of BIOSURFACTANTS produced from unconventional inexpensive wastes in food and agriculture industry, World J. Pharm. Res. 4 (2015) 827–842.

[37] K. Sambanthamoorthy, X. Feng, R. Patel, S. Patel, C. Paranavitana, Antimicrobial and antibiofilm potential of biosurfactants isolated from lactobacilli against multi-drug-resistant pathogens, BMC Microbiol. 14 (1) (2014), https://doi.org/10.1186/1471-2180-14-197.

[38] L.V. de Araujo, F. Abreu, U. Lins, L.M. Anna, M.S. Denise, M. Nitschke, D.M.G. Freire, Rhamnolipid and surfactin inhibit listeria monocytogenes adhesion, Food Res. Int. 44 (1) (2011) 481–488, https://doi.org/10.1016/j.foodres.2010.09.002.

[39] A. Zaragoza, F.J. Aranda, M.J. Espuny, J.A. Teruel, A. Marqués, A. Manresa, A. Ortiz, Mechanism of membrane permeabilization by a bacterial trehalose lipid biosurfactant produced by rhodococcus sp, Langmuir 25 (14) (2009) 7892–7898, https://doi.org/10.1021/la900480q.

[40] C. Chen, T. Lin, Y. Shieh, Emulsification and antioxidation of biosurfactant extracts from Chinese medicinal herbs fermentation in vitro, J. Biosci. Bioeng. 120 (4) (2015) 387–395, https://doi.org/10.1016/j.jbiosc.2015.02.010.

[41] A. Ravindran, A. Sajayan, G.B. Priyadharshini, J. Selvin, G.S. Kiran, Revealing the efficacy of thermostable biosurfactant in heavy metal bioremediation and surface treatment in vegetables, Front. Microbiol. 11 (2020), https://doi.org/10.3389/fmicb.2020.00222.

CHAPTER

4 Application of biosurfactant in the production of beverages

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1 Introduction

Beverages (drinks) are liquids purposely formulated for human consumption. The basic functionality of beverages is to make up the thirst and electrolytic balance in the human body. From ancient times, beverages played important roles in different cultures and ethnic groups. The importance of beverages is associated with the dehydration of the human body, which is associated with the craving for fluids called thirst. The hypothalamus of the brain system regulates the thirst in response to the subtle changes in the body's electrolytic levels. The imbalance of electrolytes creates dysfunction of blood circulation in the body system (thus creating oxygen crisis), which may have fatal effects. To maintain a perfect balance of fluid and electrolytes, beverages have profound applications [1,2]. The beverage is an inclusive term for soft drinks, juices, milk derivatives, alcohols, beers, and warm beverages such as coffee, tea, hot chocolate, caffeinated drinks, etc. In modern times, various other types of beverages are also produced for purposes such as immunization, personal care, partying, and social celebrations. Whether formulated for relaxing or nourishing, indulging or energizing, beverages satisfy the desires of the customer with an ever-expanding menu of choices. With the increase in the array of choices for beverages, nuanced new needs are emerging. Consumers around the globe are looking for drinks that are more sophisticated, more natural, and authentic. The definition of new-generation beverage demands more bio-oriented and less synthetic foods for consumption, which includes lowering of addition of sugars, synthetic flavors, and chemicals. The demand for add-on benefits such as promotion of health, fitness,

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immunity, and overall wellness is on top priority. To fulfill all these demands, the formulation of beverages needs a holistic understanding of science and technology of production and also alternative biomolecules that may replace the synthetic chemicals used in beverages. In this context, the most potential candidate modern science has recognized is biosurfactants (BSs). Recently a variety of beverages made with BSs are found locally and globally as well, in different parts of the world [3].

BSs like all other surfactants are interfaced active agents, constituted with a hydrophilic head and a hydrophobic tail and having the supreme power of solubilization in both organic and inorganic solvents [4–6]. Generally, BSs are of two types depending on their origin. One class of BSs are microbial secondary metabolites, and the other class of BSs are plant-based; extracted from different plants [7]. These biologically derived interfaces active amphiphilic agents are either produced extracellularly or as a part of the cell membrane by various filamentous fungi, yeast, and bacteria [8]. Various substrates such as sugar, oil wastes, vegetable oils are potential carbohydrate sources for BSs production [9–12]. BSs are more beneficial for their chemical analogs due to their less harmfulness, much better ecological compatibility, advanced biodegradability, and the best is they are produced from bio-wastes [13]. Because of such properties, these BSs are nowadays popular in bioengineering goods for industrial and medical applications also [14]. BSs are established to be extensively useful edible ingredients in food, especially as wetting and foaming agents, emulsifiers and de-emulsifiers, and as detergents in household goods. BSs are also having versatile applications in petrochemical industries, wastewater treatment, soil treatment, and agro-industries. BSs are termed as "new generation bio-chemical," not only because of their least toxicity, effective environmental compatibility, efficient biodegradability, enhanced foaming property, high selectivity, specific activity at exciting temperature, pH, and salinity but also for their ability to be produced from renewable feedstock [10].

BSs are also amphiphiles comprising a hydrophilic head (containing an alcohol, acid, peptide cations, mono-, di-, or polysaccharides, or anions,) and a hydrophobic body (fabricated with saturated or unsaturated chains of hydrocarbons or fatty acids), the head and the body of BSs are linked by ester linkage (including lactones) or acids/amide chains (single and peptide) or glycosidic chains (sugar-sugar and sugar-hydroxy fats).

The hydrophilic part, i.e., the water-loving part of BSs is accountable for their degree of solubility in water, while the lipophilic part, which likes organic solvents, is accountable for capillary action. The classification by Rosenberg and Ron has divided BSs into two categories based on their molecular mass [15]. High-molecular-weight BSs: effective as emulsion-stabilizing agents. Major classes include polymeric and particulate surfactants; the low-molecular-weight BSs: having a competently lower surface and interfacial tension. Major classes include lipopeptides, glycolipids, and phospholipids [16].

Further in this book chapter, we have narrated and elaborately discussed all the potential roles of different microbial assisted and plant-based BSs on the formulation, production, stability, and other add-on benefits.

2 Applications of biosurfactants in the production of beverages

In the modern era, the change in lifestyle especially in food habits invokes the manipulation of food substances by altering its substances toward more natural products and using some edible additives with several purposes to ensure better health, taste, texture, and safety [17]. In this framework, BSs have proven themselves to be one of the most potential additives.

The contemporary food industry has huge reported applications of BSs. In this section, we are focusing on comprehending the potentiality of BSs in the formulation, production, and stabilization of various types of beverages.

Beverage production includes some technical steps, some are during the production and some after formulation but before to consumption. The step covers: (i) filtration, (ii) pasteurization, (iii) juicing, (iv) infusion, (v) carbonation, (vi) fermentation, (vii) distillation, (viii) mixing and blending, and (ix) packaging.

It is worthy to note that water is the primary most and one of the chief constituents for any beverage. "WHO" has guided for the safety of water and its importance to filtrate to eliminate health hazards [18]. Pasteurization is applied for eliminating microorganisms. Fruit-based beverages need fruit juice as the main ingredient, and thus, juicing step is very important. Infusion is the process by which flavor is extracted from constituents into the beverages.

To produce carbonated soft drinks, dissolution of carbon dioxide is much important; the process is known as carbonation. Fermentation is a metabolic process that converts sugars into alcohols. The fermentation process is most important for the production of beverages such as beers, wines, alcohols, and many more folk traditional fermented beverages [13]. The last steps are mixing and blending to prepare classic beverages followed by their distillation and packaging to make them ready for preservation and consumption [19]. The applications of BSs are mainly found in the pasteurization, infusion, fermentation, and packaging steps.

During study of different kind of literature on BSs, we have come across several potential applications of BSs in the production of beverages following the fore-said steps. For a better understanding of young scholars, we have categorically classified the applications of BSs schemed above (Schemes 1 and 2) [20].

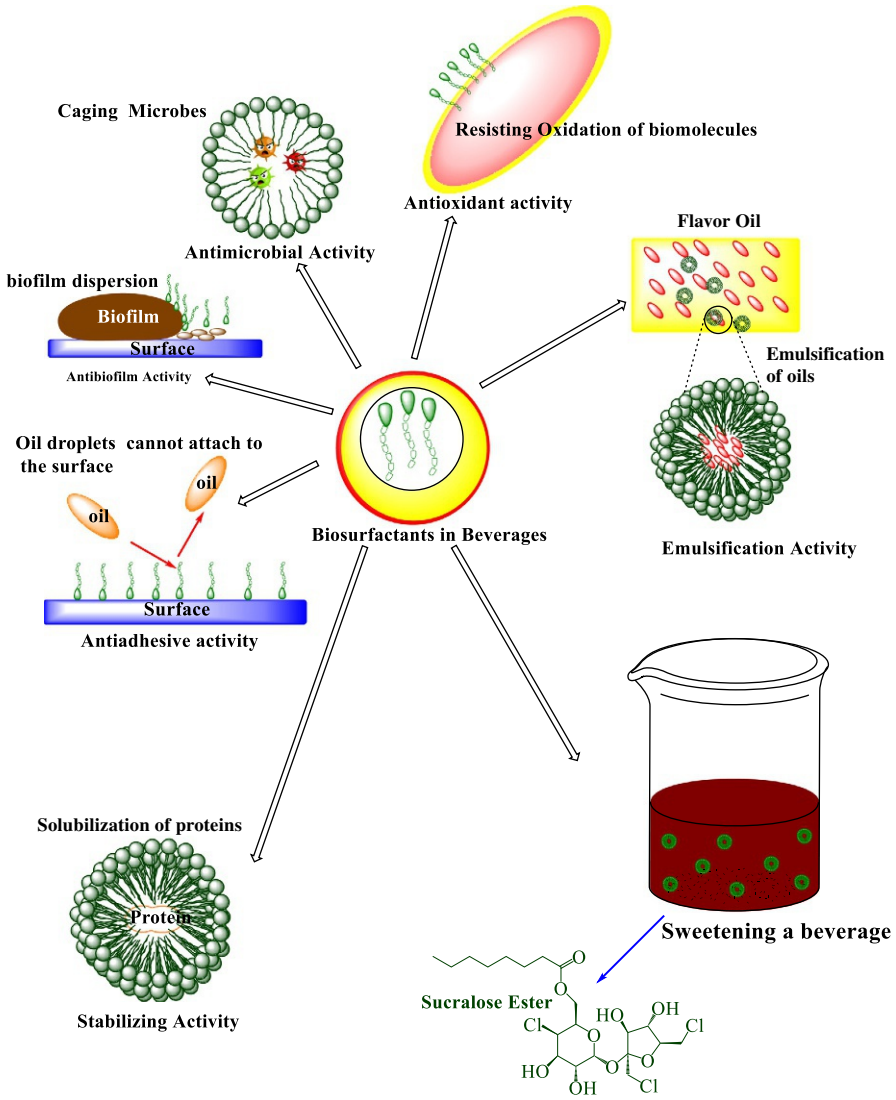
Generally, the main focus of adding edible substances in a food or beverage is to enhance and maintain its quality, appearance, taste, stability, safety, and health benefits. A variety of such additives are incorporated in food substances to retain its soul characteristics associated with the quality and safety of the food for a long time, and these are mixed into food ingredients directly or indirectly during the production, processing, or packaging of food substances [21]. Before incorporation in food products, these additives have to pass through several quality checking procedures, also various rules and regulations to get approved as the food additives by food safety agencies of respective countries. BSs not only play a pivotal technical role in the production, stability, quality control, safety, and enhancement of shelf life of beverages but also add many health and wellness benefits.

SCHEME 1

Applications of BSs in the production of beverages.

2.1 BSs in the emulsification of beverages

Formulation of stable emulsions is one of the most important parts of the production of beverages [19]. The emulsion is a mixture of more than one liquid, which is not miscible. In this case, one type of liquid acts as a dispersion medium through which other liquid is dispersed. Emulsification is the process of emulsion formation, and the



SCHEME 2

Schematic representation of various applications of biosurfactants in beverages.

substance required for this process is known as an emulsifier. In beverage as well as in food industries, the most appreciated property of BSs is their capability to produce stable emulsions. BSs can act as either emulsifying or de-emulsifying agents. BSs help in production, stabilization, and maintaining the properties of emulsions. The use of BSs in food and beverage products can substantially improve their texture,

appearance, creaminess, etc. Emulsifiers help to increase the useful life and stability of the emulsion by minimizing the separation of phases in heterogeneous systems [22].

In general, dairy products (milk-based beverages) consisting of lower stability can be flourished by BSs resulting in the formation of a stable emulsion. Sugar ester BSs obtained through enzymatic synthesis are found to be able to stabilize fresh coconut milk emulsion, which is far better than commercially available coconut milk samples [23].

Microorganisms such as yeasts *Candida valida, Candida utilis, Rhodospiridiumdi obovatum, Hansenula anomala, Rhodotorula graminis*, bacteria *Klebsiella* sp., *Acinetobacter calcoaceticus*, and green algae *Porphiridin cruentum* were identified as major producers of bioemulsifiers with better stabilizing activity than some commercial emulsifiers [24].

In emulsification technology, the most versatile and applied technique is nano-emulsions. Nano-emulsions are emulsions having droplet sizes of 20–500 nm, hence own large surface area [32]. BSs are proven to be eligible candidates for stabilization and formation of nano-emulsion. For instance, Rhamnolipids and Quillaja saponins are reported to show potentiality in this purpose [33].

Nano-emulsions are generally used to encapsulate and deliver any aquaphobic bioactive compounds in beverage products, for instance, fortified milk (vitamin D nano-emulsions added to whole fat milk) [34]. BSs help to solubilize and stabilize various flavor oils and maintain their consistency and appearance [35]. It is worth mentioning that owing to excellent physicochemical features, Quillaja saponins have remarkable application in beverage production, which includes functions as foaming agents, emulsifiers; furthermore, it was reported to reduce cholesterol and also to enhance and maintain beverage flavors [36,37,38].

In a study, Quillaja saponin is reported to be used as a stabilizer of orange oil-flavored nano-emulsion with reduced turbidities, resulting in transparent and clear beverage products [39].

Manoharan et al. 2012 [40] reported that surfactin-sunflower-oil-based nano-emulsion exhibited antibacterial properties against *Salmonella typhi, Listeria monocytogenes, Staphylococcus aureus*, and the fungicide is effective against *Rhizopus nigricans, Penicillium* sp., *Aspergillus niger*, as well as a sporicidal effect against *Bacillus circulans* and *Bacillus cereus*. They studied an in situ evaluation to find out the antimicrobial activity of the surfactin nano-emulsion on mixed vegetables, juice of apple, milk, and raw chicken, which demonstrated a considerable decrease in bacterial and fungal population increase [40].

2.2 BSs as antioxidants

Compounds that prevent the free radical production by inhibiting oxidation reaction are known as antioxidants. Free radicals may damage the cells of any living organisms by chain reaction mechanism; thus to prevent such oxidative deterioration of edible products, antioxidants are very essentials. Many BSs having antioxidant

Table 1 List of biosurfactants with emulsifying and antioxidant properties.

S. No.	Microorganisms	Biosurfactant Type	Function	Ref.
1	*A. piechaudii* CC-ESB2	–	Emulsifier Antioxidant	Chen et al. [25]
2	*Nesterenkonia* sp.	Lipopeptide	Emulsifier Antioxidant	Kiran et al. [26]
3	*Bacillus subtilis* RW-I	Lipopeptide	Antioxidant	Yalçin and Çavuşoğlu [27]
4	*Lactobacillus casei*	–	Antioxidant	Merghni et al. [28]
5	*Lactobacillus pentosus*	Glycolipopeptide	Emulsifier	Vecino et al. [29]
6	*Candida* spp. (*Candida glabrata* CBS138, *Candida albicans* SC5314)	Sophorolipids	Emulsifier	Gaur et al. [30]
7	*Candida utilis*	Carbohydrate-lipid-protein complex	Emulsifier	Campos et al. [31]

property help in maintaining the quality, flavor, color, and aroma and moreover increase the useful life of the product. Foods incorporated with antioxidants are also able to reduce the risk of heart-related and degenerative diseases [9,41].

Some BSs with the antioxidant properties are represented in Table 1.

2.3 BSs as antimicrobial agent

Antimicrobial activity is the most important activity of BSs among their various physicochemical features regarding food safety. A wide range of BSs are found to show activity against various pathogens (such as bacteria, protozoa, fungi, etc.) by inhibiting their growth. The inhibition mechanism includes inactivating the metabolic action, rupturing and changing the permeability of the cell membrane, destabilization of conformation, and other vital actions of microbes. BSs with antimicrobial activity help to protect food from microbial spoilage, thereby increasing the shelf life of food and beverage products [42].

A wide range of lipopeptide BSs are seen to have antimicrobial properties against various pathogens [43a]. Among them, surfactin is the first member established by Arima et al. 1968 [43b] from *Bacillus subtilis*. Iturin, fengycin, mycosubtlins, and bacillomycins are other lipopeptide BSs with antimicrobial activity produced by *B. subtilis* [44]. Other antimicrobial lipopeptide BSs are pumilscidin produced from *Bacillus pumilus* [45] and lichenysin produced from *Bacillus licheniformis* [46]. Bacillus sp. (strains of *B. subtilis* and *Bacillus amyloliquefaciens*) are found to

produce surfactin, which exhibits microbial inhibition against *B. cereus, Streptococcus pneumoniae, Salmonella typhimurium, Staphylococcus aureus, Serratia marcescens, L. monocytogenes, Klebsiella pneumoniae* by Isa et al. [47].

Many rhamnolipid BSs are also found to have antimicrobial actions against various pathogens such as rhamnolipids derived from *Pseudomonas aeruginosa* [48,49]. *Pseudomonas aeruginosa* LBI [50], *Pseudomonas aeruginosa* L2-1 exhibit activity against *Staphylococcus aureus, B. cereus and Micrococcus luteus* [51], *B. pumilus, Penicillium chrysogenum, Micrococcus luteus, Candida albicans,* and *Sarcina lutea* are all susceptible to *Pseudomonas aeruginosa* ATCC-10145 [52]. Sophorolipid BSs derived from *Candida bombicola* ATCC 22214 [53] and produced from yeast strains *Candida glabrata* CBS138 and *C. albicans* SC5314 [30] are found to have antimicrobial activity against different pathogens.

Huang et al. in their two different works have reported the microbial activity of surfactin combined with fengycin and polylysine against *E. coli* [54] and *Salmonella enteritidis* found in milk [55]. Rhamnolipids and nisin combinedly are found to amplify the useful life of soymilk, also inhibit thermophilic spores in ultrahigh-temperature soy milk [56]. We have listed BSs with antimicrobial properties in Table 2.

2.4 BSs as anti-biofilm and antiadhesive agents

In the past years, there was an increasing demand for edible products incorporated or to be covered with substances that resist the biofilm production by certain microorganisms at the surface contaminating the edible substances resulting in health hazards for the consumers [61]. Surfaces associated with edible substances provide the ideal atmosphere for the attachment and growth of such microorganisms resulting in

Table 2 List of biosurfactants with antimicrobial properties.

S. No.	Microorganisms	Biosurfactant type	Target	Ref.
1	*Bacillus subtilis* C19		*Candida albicans*	Yuliani et al. [57]
2	*Bacillus circulans*	Lipopeptide	*Escherichia coli* (NCIM 2931), *Bacillus pumilis* (MTCC 2296), *Micrococcus flavus* (NCIM 2376), *Mycobacterium smegmatis* (NCIM 5138), etc.	Das et al. [43a]
3	*Bacillus* sp. MTCC 5877		*E. coli*	Anjum et al. [58]

Table 2 List of biosurfactants with antimicrobial properties. *Continued*

S. No.	Microorganisms	Biosurfactant type	Target	Ref.
4	*Brevibacteriumcasei*	Glycolipid	*Vibrio alginolyticus, E. coli*	Kiran et al. [59]
5	*Bacillus* spp. (Strains of *B. subtilis* and *Bacillus amyloliquefaciens*)	Surfactin	*Bacillus cereus, Streptococcus pneumoniae, Salmonella typhimurium, Staphylococcus aureus, Serratia marcescens, Listeria monocytogenes, Klebsiella pneumoniae*	Isa et al. [47]
6	*Candida bombicola*ATCC 22,214		*B. subtilis* (KCTC 1028), *Staphylococcus xylosus* (ATCC 35663), *Streptococcus mutans* (ATCC 35668), *Propionibacterium acne* (ATCC 6919)	Kim et al. [53]
7	*Candida* spp. (*Candida glabrata* CBS138 and *C. albicans* SC5314)	Sophorolipids	*B. subtilis* (MTCC 441), *Pseudomonas aeruginosa* (MTCC 424), *E. coli* (MTCC 723), *S. aureus* (MTCC 9886)	Gaur et al. [54]
8	*P. aeruginosa* LBI		*B. cereus, Mucor miehei*	Nitschke et al. [50]
9	*P. aeruginosa* L2-1	Rhamnolipid	*S. aureus, B. cereus* and *Micrococcus luteus*	Costa et al. [51]
10	*P. aeruginosa* PA1		*L. monocytogenes* ATCC 19112 and ATCC 7644	Araújo et al. [60]
11	*P. aeruginosa* ATCC-10145		*Bacillus pumilus, M. luteus, Sarcina lutea, Penicillium chrysogenum, C. albicans*	El-Sheshtawy and Doheim [52]

the formation of biofilm. During manufacturing and processing of food substances, some materials (rubber surfaces, pipes, valves, walls packaging machines, floor, polystyrene and stainless-steel materials, and other parts of processing equipment) become the provider of such suitable environment for pathogens to form biofilm, and to inhibit such biofilm contamination, it must be treated routinely with anti-biofilm agents [62]. Studies for the prevention of such biofilm include several chemical, physical, or biological processes and also improving the packaging materials and technology [63]. Due to having potential surface-active properties, a number of BSs are found to prevent such microbial adherence and thus help to reduce such contamination in food and beverage industries [64].

The antiadhesive property of BS against different microorganisms has made it very useful as either edible additives or as coating agents for surfaces of utensils and packages preventing or decreasing the microbial proliferation in food and beverage industries [65]. Various lactic acid probiotic bacteria can also produce BSs with antiadhesion properties [66]. In this context, an important application of this antiadhesive property of BSs has been suggested is to reduce microbial contamination during dairy processing due to microbial adhesion in heat exchanger [67], and BSs obtained from *Streptococcus thermophilus* have been reported to inhibit such microbial contamination in plates of the heat exchanger within the dairy industry [68].

Pseudomonas aeruginosa PA1 and *B. subtilis* ATCC 21332 show considerable antiadhesive activity against biofilm formation on polystyrene and stainless-steel surfaces by *P. fluorescens* ATCC 13525 (Gram-negative food pathogen) and *L. monocytogenes* ATCC 19112 and ATCC 7644 (Gram-positive food pathogen) [69]. We have listed BSs with anti-biofilm and antiadhesive functions in Table 3.

3 Real-time practical application of BSs and their add-on benefits in beverages

From ancient days, for centuries, people used to believe alcoholic beverages are the blessing of god Dionysus [76]. Modern archeological research says that the history of alcoholic beverage consumption is 8000 years old and has been a large part of socialization for centuries. In ancient times, beverages were used to be plant-based and were made out of plants depending on the availability of plants. The archeological evidence confirms that the earlier productions of wines were in c. 6000 BCE at sites of Georgia [77,78,79] and in c. 5000 BCE in Iran [80]. The domestic-level production of beers is found to be in c. 3000 BCE in Neolithic Europe [81]. The Shang dynasty (1500–1046 BCE) introduced tea as a medicinal drink in the Yunnan province of China [82]. Consumption of alcoholic drinks is considered a part of socializing for centuries in Ancient Greece and Ancient Rome. "Symposium" is a famous social gathering event for wine consumption in Ancient Greece while "Convivium" is the same kind of a social event in Ancient Rome [82].

Table 3 List of biosurfactants with anti-biofilm and antiadhesive functions.

S. No.	Microorganisms	Function	Biosurfactant type	Target	Ref.
1	*Lactobacillus paracasei*	Anti-biofilm Antimicrobial	–	*Candida albicans, Staphylococcus agalactiae, Staphylococcus aureus, Staphylococcus epidermidis*	Gudina et al. [70]
2	*Brevibacteriumcasei*	Anti-biofilm	–	*Pseudomonas* spp., *Vibrio* spp., *E. coli*	Kiran et al. [59]
3	*Candida lipolytica*UCP 0988	Antiadhesive Antimicrobial	Rufisan	*Streptococcus agalactiae, S. mutans, S. mutans* HG, *S. mutans* NS, *S. sanguis* 12, *S. oralis* J22	Rufino et al. [71]
4	*Nocardiopsis* sp.	Anti-biofilm	–	*Vibrio alginolyticus*	Kiran et al. [72]
5	Bacillus sp. MTCC 5877	Antiadhesive Antimicrobial	Glycolipid	*E. coli*	Anjum et al. [58]
6	*Bacillus subtilis* ATCC 21332	Antiadhesive	Surfactin	*Listeria monocytogenes*	Araujo et al. [69]
7	*Pseudomonas aeruginosa* PA1		Rhamnolipids	ATCC 19112 and ATCC 7644 *P. fluorescens* ATCC 13525	
8	*Lactobacillus plantarum*	Antiadhesive	–	*S. aureus, E. coli, Salmonella enterica*	Bakhshi et al. [73]
9	*Bacillus licheniformis* VS-16	Anti-biofilm	Phospho-lipopeptide	*E. coli*	Giri et al. [74]
10	*Pseudomonas* spp.	Anti-biofilm	Glycolipid	*S. aureus*	Silva et al. [63]
11	*Nesterenkonia* sp.	Anti-biofilm Antimicrobial	Lipopeptide	*S. aureus*	Kiran et al. [26]
12	*Lactobacillus casei* ATCC 393	Anti-biofilm Antimicrobial	–	*S. aureus* (ATCC 6538) *S. aureus* (9p and 29P)	Merghni et al. [28]
13	*Acinetobacter indicus* M6	Anti-biofilm Antimicrobial	Glycolipoprotein	*P. aeruginosa* ATCC 9027, *S. aureus* ATCC 6538	Karlapudi et al. [75]

3.1 Biosurfactants in nonalcoholic beverages

Hundreds of nonalcoholic beverages and preparations are known of which the most consumed drinks are soft drinks, mocktails, milk-based beverages, fruit juices, smoothies, etc. The emerging demand for bio-based, organic foods and beverages has elicited the replacement of synthetic color, flavors, emulsifiers, and stabilizers. Consumers, as well as manufacturers, are looking for better alternatives, and thus, the rapid increase in applications of BSs in nonalcoholic beverages such as soft drinks, milk products, and fruit beverages is reported [83,84].

3.1.1 Carbonated soft drinks

To solubilize water-insoluble solids in soft drinks and other nonalcoholic beverages to make it clear and ready to drink, many approaches have been taken. The most notable methods are encapsulation (U.S. patent No: 5871798), microemulsion (U.S. patent No: 4835002 and 6,251,441), emulsion (U.S. patent No: 4946701), etc. Direct application of synthetic surfactant has also been formulated in patents such as (U. S. patent No: 6444253, 6,048,566, 4,136,163, 4,230,688, and 4,296,093), but all these formulations have their problems and drawbacks such as the beverages become optically, visibly unclear, i.e., not like water-clear. Flocculation and precipitation are found after a certain time, which need larger quantities of surfactants; the amalgamation of the huge amount of such synthetic surfactants makes it potentially negative impact on human health. A patent (European Patent No. EP1536701A1), currently used by Pepsico Inc., has reported application of Span (20, 40, 60, and 80) and Tween (20,40,60,65 and 80) as best for beverages to make it clear, stable, and free from crystals, floe, sediment, and any other phase separation phenomena. Both the Span and Tween series of surfactants are synthetic, and excessive consumption of such synthetic chemicals may have a potential negative effects on health especially in children. Recently in a study by Ying yang and coworkers [85], a plant-based natural food-grade saponin called the "Quillaja Saponin" is reported. *Quillaja Saponaria* Molina trees contain it in their bark and have shown many identical physicochemical properties such as Tween 80 synthetic surfactant [85]. The major component of quillaja bark extract is the quillaja saponin consisting of a sugar moiety as the hydrophilic head (xylose, arabinose, rhamnose, galactose, glucuronic acid, and fucose) attached with a steroid aglycone or triterpene (gypsogenic and quillaic acid) as hydrophobic tail [86]. Quillaja saponin was commercialized by Bridgewater, NJ starch company under the trade name Q-Naturale (contains 14% active surfactant ingredient) [85]. The best part of the application of quillaja saponin is that it has been approved for consumption in beverages by many countries such as the United States, EC, and Japan. It has been reported that almost 70% of world consumption of quillaja saponin is a foaming agent in traditional folk alcohols, beverages, modern soft drinks such as root beers at a concentration of 40–100 ppm [87,88]. It is also used in slush-type drinks to improve ice crystal morphology [89]. We have provided the structural presentation of Quillaja saponin extracted from molina tree in Fig. 1.

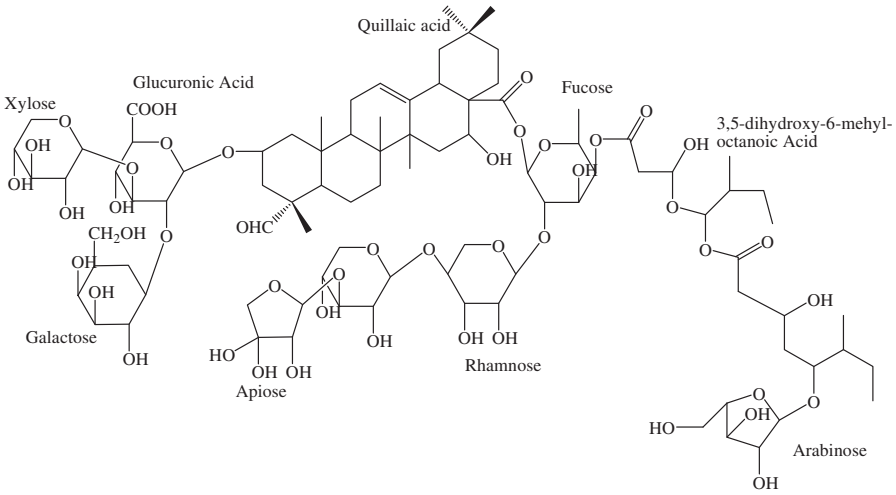


FIG. 1

Structural presentation of quillaja saponin extracted from molina tree [86].

Another such BS obtained from Mohave Yucca (*Yucca schidigera*) extract has been potentially applied in carbonated beverages recently [90]. Yucca saponin is nontoxic, nonmutagenic, neither odor nor taste, and does not affect the color or flavor of the beverages. Under section 21 CFR 172.510, FEMA number 3121, the USFDA has determined that it is safe for human consumption [91].

3.1.2 Milk and milk-derived nonalcoholic beverages

The dairy industry is one of the most advanced food sectors due to its economical contribution to the country's GDP as well as its major role in health and welfare toward people. A great number of milk-based beverages are known all over the world: milkshakes, yogurt, doogh, kefir, ayran, yakult, and many more of these kinds. Milk and dairy beverages are one of the most important and vital sources of animal proteins, vitamins and minerals, and essential fatty acids for women, infants, and young adults [92]. Most milk-based beverages are produced by natural fermentation. Because of the fermentation process and the generation of phenols, flavonoids, bioactive peptides, and BSs, the antioxidant capabilities of those beverages are improved [93,94]. *Bacillus subtilis* natto T-2 and *B. subtilis* natto TK-1, which create cyclic lipopeptides and lipopeptide BSs, are found to be present in practically all fermented milk-based beverages. These BSs prolong the shelf life of beverages while also inducing apoptosis in human leukemia cells and inhibiting the development of human breast cancer cells [95,96]. *B. subtilis* CSY 191 is another microbe that has been shown to produce surfactin BS, which inhibits the growth of human breast cancer (MCF-7) cells [97]. Surfactin has a profound effect on the stability

of beverages also as a nontoxic antibacterial, antiadhesive emulsifier [98]. Yeh et al. (2005) [99] have reported that among the BSs, surfactin has the best emulsion stabilizing effect with a concentration of 0.16 g/L when added to ice cream mix over lecithin (1 g/L). Meena and Kanwar [100] showed the antiadhesive property of surfactin. Surfactin inhibits the biofilm formation in cans and vats of milk-based beverages.

Here in Table 4, a few milk-based fermented beverages are reported, which contain the *B. subtilis* sp. and therefore contain such medicinally important BSs.

3.1.3 Other nonalcoholic beverages

Fruit juices, fruit-milk blend shakes, smoothies, and other nonalcoholic beverages are excellent sources of both microbial and plant-based BSs. Both kinds of BSs regulate the homogenous emulsification of these beverages. The formation of foam is also greatly dependent on such BSs present in the beverages. The antimicrobial property of BSs also protects beverages from yeast or fungal attack and thus enhances the shelf life. Green (detox) juice is a famous health drink among people. A recent study by Fraga et al. (2020) [102] has reported the application of a BS "YlBio" produced by fermentation of cashew apple juice with *Yarrowia lipolytica*, which effectively

Table 4 List of some famous milk-based beverages that contain a large amount of biosurfactants [101].

Product	Common name	Main ingredient	Type of fermentation
Maeshkraj	Kalari/ Kradi	Buffalo milk, cow milk	Following fermentation, acid heat coagulation is used
Phuh	Colostrum milk	Colostrum milk	Acid heat coagulation is employed after fermentation
Zamuthdod	Dahi	Goat milk, buffalo milk, and cow milk	Spontaneous Fermentation
Laes	Lassi	Milk from goats, buffaloes, and cows	Spontaneous Fermentation
Shrikhand	Maska	Milk from buffaloes, cows, and goats	Spontaneous Fermentation
Whey	Whey (Ghol)	Milk from cows, buffaloes, and goats	Spontaneous Fermentation
Kefir	Kefir	Buffalo milk, cow milk, goat milk	Spontaneous Fermentation
Yogurt	Yogurt	Buffalo milk, cow milk, goat milk	Spontaneous Fermentation
Milkshakes	Shakes	Production of milk from goats, buffaloes, and cows	Spontaneous Fermentation

enhanced the stability and consistency of the detox juice without adding any synthetic emulsifier.

3.2 Biosurfactants in alcoholic beverages

The maximum abundance of BSs is found in alcoholic beverages. The fermentation process involved in the production of alcohols and derivative beverages produces varieties of microbial-assisted BSs [40,52–55]. The produced BSs help in stabilization of alcohols, maintain the homogeneous nature of the beverages, act as antimicrobial, antifungal, antioxidant agents, and enhance the shelf life for those beverages [19,24,44,61,63]. Many microbial systems such as bacteria, yeasts are applied during fermentation to produce alcohols, which indeed produce BSs simultaneously. For the production of ethanol and ethanol-containing beverages, the most exploited yeast belongs to the genus *S. cerevisiae* [103]. Thousands of famous, nonfamous, and folk alcoholic beverages can be seen in different parts of the globe. In the section below, we have tried to explain the utility of BSs in the production of different types of alcoholic beverages such as folk beverages, beers, wines, etc.

3.2.1 Traditional -folk fermented beverages

Many regions and cultures across the globe have their traditional fermented beverages. These fermented beverages are proved to have various biomolecules, BSs, and diverse microorganisms to promote BS production during the making of such beverages. The Republic of Congo, South Asian subcontinents especially China, India, and Srilanka, and various parts of Africa are homes to such traditional fermented folk rages. In the republic of Congo, a famous wine called the "plantin wine" or banana wine has been studied to have production of the large amounts of BS during fermentation [104]. It has been reported that 15 bacterial strains were determined, and 100% of those bacteria secreted BSs during the preparation of the wine; 66% BSs were extractible; 33% of the isolates are associated with *lactobacillus plantarum* (Is 2, Is 9, Is 12, and Is 13). These BSs released by *L. plantarum* play a critical role in the preservation of banana wine [104].

In this part of the world, there are also some famous traditional fermented beverages found, which include sugar cane wine (locally called "loungouila"), palm wine ("nsamba"), honey wine ("duma"). Many more such fermented traditional beverages are undocumented and need more scientific studies such as on orange wine, pineapple wine, and grape wine. The plantin wine made of banana is locally called "Mbamvu." In all these fermented traditional beverages, BSs are secreted by various microorganisms. These BSs have their role as antimicrobial, antibiofilm, flavoring, coloring, and stabilization agents. In the Rwanda region, a famous banana wine called "urwagwa" is one of the oldest fermented beverages in Africa [105,106].

In southern Asia, especially in India, there are more than 350 different types of famous, less-familiar, and rare fermented alcoholic beverages that are traditionally prepared. These beverages also contain the high number of microbial BSs and are generally prepared from milk, vegetables, bamboo, meat, fish, and cereal. The

FIG. 2

Image of a very famous Indian folk beverage "Lassi."

representative image of such traditional beverages of Indian folk beverage "Lassi" is in Fig. 2.

Here we present Table 5 containing a large number of identified BSs that are produced during the fermentation of different substrates (vegetables, agro-wastes, and fruits) and their potential application/role in beverages.

Table 5 List of established BSs produced in folk fermented beverages.

Substrate	Microorganism	Biosurfactant type	Cultivation mode	Ref.
Banana	*H. archaeon* AS65	Lipopeptide	Batch shake flask	Chooklin et al. [107]
Orange	*Pseudomonas aeruginosa* MTCC 2297	Rhamnolipids	Batch shake flask	George and Jayachandran [108]
	Bacillus licheniformis KC710973	Lipopeptide	Batch shake flask	Kumar et al. [109]
Carrot	*Bacillus subtilis* l'1a	Iturin	Batch shake flask	Paraszkiewicz et al. [110]
Sugarcane Bagasse	*Corynebacterium aquaticum*	Biosurfactant	Batch shake flask	Martins et al. [111]
Potato Peels and Sugarcane Bagasse	*P. aeruginosa* AJ15	Rhamnolipids	Submerged Fermentation	Das and Kumar [112]

3.2.2 Wines

Wine is classified under the category of alcoholic beverages. Evidence says that it was consumed even thousands of years ago in different cultures across the world. The earliest evidence to date concludes the first production of wine in China (*c*.7000 BCE) [113].

Wine is mainly made from the fermentation of grapes by utilizing yeast. The yeast consumes the sugars of grapes and transforms them into ethanol, CO_2, and heat. Some other wines also rather than the common grape wine are rice wine, fruit wine, palm wine, cherry wine, pomegranate wine, currant wine, and elderberry wine. We've also explained many more traditional folk wines from throughout the world in the folk-fermented beverage section of this book chapter.

Due to the fermentation of fruits and crops using yeast during the production of wines, a large number of different microbial-assisted BSs are also found to be produced. In Table 5**,** we have listed a few reported BSs that are confirmed to be present in wines and other alcoholic fermented beverages. Fig. 3 is the image of three different types of wines used in the market.

3.2.3 Beers

Beers contain with a high amount of bio-originated surface-active agents, i.e., the BSs. The majority of these classes of BSs belong to proteins and peptides liberated during the mashing of constituent grains. The presence of such surface-active compounds is the reason for the formation of foam in champagne and carbonated soda. The stability of foam and shelf life of beers are immensely affected by these peptide-type BSs. The gas in bubbles is hydrophobic. The hydrophobic part of the surfactants interacts with the gas, and the hydrophilic part interacts with the exterior liquid of the bubbles and stabilizes the bubbles for a long time; which indeed enhances the shelf life and quality of beers. Basically, BSs in beer play the role of emulsifier, stabilizer, and foaming agents. Because of its long-lasting foaming property, the BS "mohave

FIG. 3

Image of three different types of wines.

saponin" found in Mohave Yucaa, a native plant of North and Central America, is widely used in carbonated beverages, root beers, standard and low-alcohol beers [114,91].

Quillaja bark and licorice root saponins are commonly used in the manufacturing of beers. Both these surfactants have very efficient emulsifying properties and long-lasting foaming ability [115].

Early studies on beer revealed the presence of a foaming agent Isohumulone. Isohumulone is a surface-active agent and enhances the foam stability of hoped beers [116]. Fig. 4 shows a glass of beer.

3.2.4 Biosurfactants in medicinal beverages

Modern-day medicine is much inclined toward biomolecules rather than synthetic prototypes. The history of human civilization also invokes the application of plants and different foods for the traditional folk treatment of various diseases and also as an energy supplement. Almost in every part of the globe, we can find folk medicines and beverages as supplements and treatment. In a recent scientific article, it has been

FIG. 4

Image of beer.

reported that a saponin BS extracted from Licorice root is extensively used in medicinal beverages [117]. The major saponin glycyrrhizin not only stabilizes the beverage but also acts as an antimicrobial agent. It is also used as a sweetener [117]. The sapindus saponin extracted from *Sapindus mukurossi* has been found to show tremendous anti-yeast activity [5]. The sapindus fruit also called Reetha is extensively used in folk medicines in India, China, and Japan. Saponin is used to treat dermatomycosis and contains anti-tussive, anti-inflammatory, and anthelmintic properties. It is known in Japan as "enmeihi," which means "life-prolonging pericarp," and in China as "wu-huan-zi," which means "non-illness fruit" [117]. These beverages are also used as "energy boosters" in men to increase strength, stamina, and vigor, provide nutrition for overall immunity enhancement. In India, medicinal beverages are consumed in the form of "Sarbat" or "Kadha." Here is Table 6 listed with BS-containing medicinal beverages.

3.3 Biosurfactant in hot beverages (tea/coffee/chocolates)

Beverages that are typically served and consumed hot are called hot beverages. Thousands of regional and cultural hot beverages can be found across the globe. Three most famous hot beverages that are consumed worldwide irrespective of region and culture, namely (i) coffee and caffeine beverages, (ii) tea and derivatives, (iii) hot chocolates.

Coffee is a brewed hot beverage produced from roasted seeds of the coffee shrub. Its first cultivation is reported in Southern Arabia [128].

In the following Table 7, the composition of coffee oil and its high saponification value indicate the presence of BS inconsiderable amount.

This high saponification is associated with the production, stability, and foaming of coffee-based variety of beverages such as Espresso, Americano, Cappuccino, Latte, etc. The coffee oil contains a high amount of surfactant, which due to its surface-active nature, and promotes the formation of emulsions. The antimicrobial activity of coffee-based surfactants also enhances the storage time. For hot chocolate and tea also, their formation, foaming, emulsification, and shelf life are greatly affected by the presence of BSs in the cocoa extract and tea extract, respectively [129,130].

A large number of research studies on tea seeds and leaves established that "tea camellia" is a rich source of saponins and flavonoids, which is effective in anti-inflammatory, anticancer, antiangiogenic, antioxidant, anti-obesity, insecticidal, hepatoprotective, and foam formation [131–144]. A recent study on saponin extracted from green tea seed by Khan et al. [145] reported that the saponin exhibits significant antibacterial activity in both in vitro and in vivo processes. The saponin ineffective in enhancing the activities of lysozyme thus promotes immunity. It is most effective against zoonosis and *salmonella*. Tea seed saponin is thus one of the best candidates for modern-age novel bio-antibiotic formulation.

Table 6 List of medicinal beverages that contain biosurfactants.

Name of biosurfactants		Beverage	Used as	Ref.
A. Microorganism based biosurfactants				
Surfactin		**Milk-based beverages** (*Bacillus subtilis, Staphylococcus aureus*)	Anti-inflammatory agent	[118]
Sophorolipids		**Fermented fruit juices** *C. bombicola, C. bogoriensis*, *C. botistae*, *C. apicola*, etc.	Anti-inflammatory antiviral agent, spermicidal, neuroprotective agent, immunomodulator	[119]
Rhamnolipids		**Wine** *Pseudomonas aeruginosa*	Antimicrobial agent, Bioremediation, biocontrol properties	[120]
Subtilisin		**Milk-based beverages** *B. subtilis*	Antimicrobial agent	[121]
Aminoacids lipid		**Wine and Beer** *Bacillus* sp.	Antimicrobial agent	[122]
Cellobiose lipid		**Folk fermented beverages** *Ustilagomaydis*	Antifungal agent	[123]
Peptidelipid		**Alcoholic beverages** *Bacillus licheniformis*	Antimicrobial agent	[124]
B. Plant-based biosurfactant				
Saponin	QS-21	*Quillaja Saponaria* (soap bark tree) used in **Beers, Soft drinks, Wines**	Adjuvant for vaccine formulation, anti-inflammatory agent, natural shampoo and detergent	[88]
	Saponin DS-1	QS-21 **Medicinal beverages**	Enhancer of nasal and ocular insulin delivery, and mucosal delivery of aminoglycosides	[125]
	Saiko saponin	*Root of Bupleurum marginatum* **Kadha (folk**	Antiviral agent	[126]

Table 6 List of medicinal beverages that contain biosurfactants. *Continued*

Name of biosurfactants		Beverage	Used as	Ref.
		medicinal beverage)		
	Tribulus saponin	*Tribulus terrestris* Gokshura plant **Extensively used in the treatment of sexual disorders in Men**	Immune system booster, antioxidant and anti-inflammatory agent	[127]
	Oleanane saponins	*Calendula officinalis* **Folk boiled beverage**	Anti-spasmodic agent, antiseptic agent, for treating various skin diseases, gastric and menstrual discomfort	[127]
	Saponin glycosides	**Red wine**	heart-protective by lowering LDL cholesterol and increasing HDL cholesterol, prevent RBC(red blood cell) clumping	[127]

4 Market prospect

The ever-increasing demand for these "green" surfactants is driven by rising consumer awareness, resulting in the BS market's expansion. In 2016, the global BS market was valued at USD 3.99 billion [146], USD 4.20 billion in 2017, and within the projected period 2017–2022, the market is estimated to grow at a compound annual growth rate (CAGR) of 5.6%, reaching USD 5.52 billion [146]. However, the global COVID-19 pandemic has had a detrimental influence on the BS market,

Table 7 Composition of coffee oil extract.

Fatty acids	Contents (%)
Palmitic acid (C 16:0)	35.7
Stearic acid (C 18:0)	7.1
Oleic acid (C 18:1)	9.4
Linoleic acid (C 18:2n6c)	43.7
Oil parameters	Value
Saponification value	166.1
Acid value	7.1
Iodine value	70.3
Ester value	158.9

which was valued at USD 2.064 billion in 2020 and is predicted to grow at a CAGR of 5.32% to USD 2.167 billion in 2021 and USD 2.818 billion by 2026 [147].

Many renowned companies such as Ecover, Evonik, BASF Cognis, MG Intobio, Saraya, and many more have ventured into the BS production. Among them, Evonik becomes the first firm in the world that produce BSs on a commercial scale [148]. Some BS production companies and their origin are listed in Table 8.

Table 8 List of commercial houses manufacturing/marketing biosurfactants in different countries [149,150].

Companies	Country of origin	Products
AGAE Technologies LLC	USA	Rhamnolipids
Jeneil Biosurfactant Co. LLC	USA	Rhamnolipids
Paradigm Biomedical Inc.	USA	Rhamnolipids
Rhamnolipids Companies, Inc.	USA	Rhamnolipids
Natsurfact	USA	Rhamnolipids
Tensio Green	USA	Rhamnolipids, Lipopeptides, Trehalolipids
TeeGene Biotech	UK	Rhamnolipids, Lipopeptides
Synthezyme LLC.	USA	Sophorolipids
Ecover	Belgium	Sophorolipids
Groupe Soliance	France	Sophorolipids
MG Intobio Co. Ltd.	South Korea	Sophorolipids
Allied Carbon Solutions (ACS) Ltd.	Japan	Sophorolipids
Kaneka Co.	Japan	Sophorolipids
Saraya Co. Ltd.	Japan	Sophorolipids
Fraunhofer IGB	Germany	Sophorolipids, Cellobiose lipids, Mannosylerythritol lipids
Henkel	Germany	Sophorolipids, Rhamnolipids, Mannosylerythritol lipids.
Evonik	Germany	Sophorolipids, Rhamnolipids.
Glyco Surf	USA	Glycolipids
Kingorigin	China	Glycolipids, phospholipids
Victex	China	Rhamnolipids
BASF	Germany (MNC)	Sophorolipids, glycolipids
BASF-Cognis	Germany	Sophorolipids, glycolipids

5 Future prospects

Versatile and eco-friendly physicochemical properties of BSs have shown to rising interest in the application of BSs in the food and beverage industry, so the quest for discovering such novel eco-surfactant to be used as additives are also increasing day by day replacing the synthetic alternatives. Besides that, using BSs as additives may also enhance the nutraceutical values of the edible products [22]. In a recent market analysis [84], it has been reported that Europe has emerged as the pioneer market for plant-based BSs consumption in the last few years. This has become possible only because of the environmental protection policies and growing health awareness. It has also been expected that continuity in this development would be seen in the coming years. The consumption of BSs is increasing since 2013 in countries such as Sweden, Germany, France, Italy, and the United Kingdom. The Asia Pacific has a very moderate to sluggish growth pace but anticipated a rapid growth by 2025. China and Japan have their dedicated market while India, Thailand, Malaysia, and Indonesia are expected to witness rapid growth by 2025. Central and North America would act as the largest exporter of BSs to North America and thereby aid the global BS industry. Brazil, Argentina, Panama, and Chile are said to dominate their regional markets [84]. The report is giving us an outlook of the demands of BSs in the global market. This modern approach of replacing synthetic food additives with their natural prototypes has also accelerated the pace of research in this field in the last 10 years or so. Hundreds of research reports on microbial assisted production of BSs from different hydrocarbon sources of agro-industrial wastes and their potential application in the food industry are published every year. In spite of having a lot of attractive promises, BSs are still facing many obstacles in the process of their commercialization. The main reasons behind this are the production cost and comparatively lower yield due to inefficient production or bioprocessing methods and use of costlier substrates [151]. Thus, for commercialization of BSs production and its application, it is necessary to discover more efficient processing technologies and strategies to get higher yield and recovery of the obtained product. Moreover, the use of less expensive and renewable substrates such as agro-hazardous wastes such as peels, molasses, bagasse, dairy residues, oil residues, etc., also facilitates such purpose [65].

6 Conclusions

This book chapter allowed showing the effective application of BSs in the production of different classes of beverages. BSs are the new age biomolecules that fulfill all the requirements of modern food science and technology to be considered as edible additives in beverages. BSs are nontoxic, biodegradable, and environmentally friendly, with enhanced foaming properties, high selectivity, and specific action at extreme temperatures, salinity, and pH. However, they can also be extracted from plants or produced from renewable feedstock. A large number of publications, scientific reports have established their effective potentiality as emulsifiers, antioxidants,

and antiadhesives in the production and stability of beverages. From ancient times, ethnic-cultural tribes and groups used BSs in fermented folk beverages. Many BSs such as quillaja Saponin, Mohave Yucca, Sapindus saponin, etc., have been recognized safe by USFDA and FSSAI (India) and have also been commercialized. Europe has emerged as the global leader in the consumption of plant-based BSs. The consumption of BSs is increasing since 2013 in countries such as Sweden, Germany, France, Italy, and the United Kingdom. The Asia Pacific has a very moderate to sluggish growth pace but anticipated a rapid growth by 2025. China and Japan have their dedicated market while India, Thailand, Malaysia, and Indonesia are expected to witness rapid growth by 2025. It is clear from the data of market analysis that BSs are having promising prospects. Although BSs are the best fit for modern science and technology, large-scale manufacturing of BSs is not possible due to expensive recovery and purifying expenses. So, in conclusion, it must be noted that the understanding of food chemistry along with surfactant science and technology will extensively be changed if we emphasize on developing more economy and ecology-friendly methods for BSs productions and their applications in food stuffs. Such exciting development focused on interdisciplinary research in association with technology will facilitate large-scale applications of BSs in the production of beverages to make it a commercial success.

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References

[1] R.H. Cheney, The biology and economics of the beverage industry, Econ. Bot. 1 (3) (1947) 243–275, https://doi.org/10.1007/bf02858570.

[2] J. Burnett, Liquid Pleasures: A Social History of Drinks in Modern Britain, Routledge, 2012, ISBN: 978-1-134-78879-8.

[3] A.K. Das, P.K. Nanda, S. Bandyopadhyay, R. Banerjee, S. Biswas, D.J. McClements, Application of nanoemulsion-based approaches for improving the quality and safety of muscle foods: a comprehensive review, Compr. Rev. Food Sci. Food Saf. 19 (2020) 2677–2700.

[4] M.H. Mondal, S. Malik, A. Roy, R. Saha, B. Saha, Modernization of surfactant chemistry in the age of gemini and bio-surfactants: a review, RSC Adv. 5 (2015) 92707–92718.

[5] M.H. Mondal, S. Malik, A. Garain, S. Mandal, B. Saha, Extraction of natural surfactant saponin from soapnut (*Sapindus mukorossi*) and its utilization in the remediation of hexavalent chromium from contaminated water, Tenside Surfactant Deterg. 54 (2017) 519–529.

[6] M.H. Mondal, M.A. Ali, A. Pal, B. Saha, A review on micellar catalyzed oxidation reactions of organic functional groups in aqueous medium using various transition metals, Tenside Surfactant Deterg. 56 (2019) 516–525.

[7] A. Lozsan, I. Rivas, G. Rodriguez, S. Martinez, M.A. Perez, Determination of surface-active characteristics of a natural surfactant extracted from *Sapindus saponaria*, Tenside Surfactant Deterg. 54 (2017) 109–117.
[8] M. Benincasa, Rhamnolipid produced from agroindustrial wastes enhances hydrocarbon biodegradation in contaminated soil, Curr. Microbiol. 54 (2007) 445–449.
[9] H. Zhang, H. Xiang, G. Zhang, X. Cao, Q. Meng, Enhanced treatment of waste frying oil in an activated sludge system by addition of crude rhamnolipid solution, J. Hazard. Mater. 167 (2009) 217–223.
[10] C.N. Mulligan, S. Wang, Remediation of a heavy metal contaminated soil by a Rhamnolipid foam, Eng. Geol. 85 (2006) 75–81.
[11] M.H. Mondal, S. Malik, S. De, S.S. Bhattacharyya, B. Saha, Employment and resurrection of surfactants in bipyridine promoted oxidation of butanal using bivalent copper at NTP, Res. Chem. Intermed. 43 (2017) 1651–1670.
[12] M.H. Mondal, A. Sarkar, T.K. Maiti, B. Saha, Microbial assisted (pseudomonas sp.) production of novel bio-surfactant rhamnolipids and its characterisation by different spectral studies, J. Mol. Liq. 242 (2017) 873–878.
[13] A. Pal, A. Garain, D. Chowdhury, M.H. Mondal, B. Saha, A comparative spectral study on the interaction of organic dye Congo-red with selective aqueous micellar media of CPC, rhamnolipids and saponin, Tenside Surfactant Deterg. 57 (2020) 401–407.
[14] C.C. Lai, Y.C. Huang, Y.H. Wei, J.S. Chang, Biosurfactant-enhanced removal of total petroleum hydrocarbons from contaminated soil, J. Hazard. Mater. 167 (2009) 609–614.
[15] E. Rosenberg, E.Z. Ron, High- and low-molecular-mass microbial surfactants, Appl. Microbiol. Biotechnol. 52 (1999) 154–162.
[16] J.D. Desai, I.M. Banat, Microbial production of surfactants and their commercial potential, Microbiol. Mol. Biol. Rev. 61 (1997) 47–64.
[17] A.Y. Onaolapo, O.J. Onaolapo, Food additives, food and the concept of 'food addiction': is stimulation of the brain reward circuit by food sufficient to trigger addiction? Pathophysiology 25 (2018) 263–276.
[18] Combating Waterborne Diseases at the Household Level (PDF), World Health Organization, 2007, p. 11, ISBN: 978-92-4-159522-3.
[19] D. DeGroff, The Craft of the Cocktail: Everything You Need to Know to Be a Master Bartender, With 500 Recipes, Clarkson Potter, 2002.
[20] N. Ranasalva, R. Sunil, G. Poovarasan, Importance of biosurfactant in food industry, IOSR J. Agric. Vet. Sci. 7 (2014) 6–9.
[21] G.A. Blekas, Food additives: classification, uses and regulation, in: Encyclopedia of Food and Health, 2016, pp. 731–736.
[22] D.J. McClements, C.E. Gumus, Natural emulsifiers—biosurfactants, phospholipids, biopolymers, and colloidal particles: molecular and physicochemical basis of functional performance, Adv. Colloid Interface Sci. 234 (2016) 3–26.
[23] N.D. Neta, J.C. dos Santos, S. de Oliveira Sancho, S. Rodrigues, L.R. Gonçalves, L.R. Rodrigues, J.A. Teixeira, Enzymatic synthesis of sugar esters and their potential as surface-active stabilizers of coconut milk emulsions, Food Hydrocoll. 27 (2012) 324–331.
[24] F.F.C. Barros, C.P.D. Quadros, M.R. Maróstica Júnior, G.M. Pastore, Surfactina: propriedades químicas, tecnológicas e funcionais para aplicações em alimentos, Quim Nova 30 (2007) 409–414, https://doi.org/10.1590/S0100-40422007000200031.
[25] C. Chen, T. Lin, Y. Shieh, Emulsification and antioxidation of biosurfactant extracts from Chinese medicinal herbs fermentation in vitro, J. Biosci. Bioeng. 120 (2015) 387–395.

[26] G.S. Kiran, S. Priyadharsini, A. Sajayan, G.B. Priyadharsini, N. Poulose, J. Selvin, Production of lipopeptide biosurfactant by a marine Nesterenkonia sp. and its application in food industry, Front. Microbiol. 8 (2017) 1138.
[27] E. Yalçin, K. Çavuşoğlu, Structural analysis and antioxidant activity of a biosurfactant obtained from *Bacillus subtilis* RW-I, Turk J Biochem. 35 (2010) 243–247.
[28] A. Merghni, I. Dallel, E. Noumi, Y. Kadmi, H. Hentati, S. Tobji, A.B. Amor, M. Mastouri, Antioxidant and antiproliferative potential of biosurfactants isolated from *Lactobacillus casei* and their anti-biofilm effect in oral *Staphylococcus aureus* strains, Microb. Pathog. 104 (2017) 84–89.
[29] X. Vecino, L. Barbosa-Pereira, R. Devesa-Rey, J.M. Cruz, A.B. Moldes, Optimization of extraction conditions and fatty acid characterization of *Lactobacillus pentosus* cell-bound biosurfactant/bioemulsifier, J. Sci. Food Agric. 95 (2015) 313–320.
[30] V.K. Gaur, R.K. Regar, N. Dhiman, K. Gautam, J.K. Srivastava, S. Patnaik, M. Kamthan, N. Manickam, Biosynthesis and characterization of sophorolipid biosurfactant by Candida spp.: application as food emulsifier and antibacterial agent, Bioresour. Technol. 285 (2019), 121314.
[31] J.M. Campos, T.L. Stamford, L.A. Sarubbo, Characterization and application of a biosurfactant isolated from Candida utilis in salad dressings, Biodegradation 30 (2019) 313–324.
[32] M.A. Salem, S.M. Ezzat, Nano emulsions in food industry, in: J. Milani (Ed.), Some New Aspects of Colloidal Systems in Foods, Intech Open, 2019, pp. 238–267.
[33] L. Bai, D.J. McClements, Formation and stabilization of nano emulsions using biosurfactants: rhamnolipids, J. Colloid Interface Sci. 479 (2016) 71–79.
[34] I. Golfomitsou, E. Mitsou, A. Xenakis, V. Papadimitriou, Development of food grade O/W nano emulsions as carriers of vitamin D for the fortification of emulsion-based food matrices: a structural and activity study, J. Mol. Liq. 268 (2018) 734–742.
[35] N. Kosaric, Biosurfactants and their application for soil bioremediation, Food Technol Biotechnol. 39 (2001) 295–304.
[36] D. Nayyar, W. Schulok, F. Saleeb, E. Modesta, Dry Mix for Producing a Slush Beverage, US Patent US5853785, 1998.
[37] B. Ozturk, S. Argin, M. Ozilgen, D. McClements, Formation and stabilization of nanoemulsion-based vitamin E delivery system using natural surfactants: quillaja saponin and lecithin, J. Food Eng. 142 (2014) 57–63.
[38] H.P.S. Makkar, K. Becker, Effect of Quillaja Saponins on in vitro rumen fermentation, in: G.R. Waller, K. Yamasaki (Eds.), Saponins Used in Food and Agriculture, Advances in Experimental Medicine and Biology, vol. 405, Springer, Boston, MA, 1996, https://doi.org/10.1007/978-1-4613-0413-5_33.
[39] J. Zhang, L. Bing, G.A. Reineccius, Formation, optical property and stability of orange oil nanoemulsions stabilized by Quallija saponins, LWT-Food Sci. Technol. 64 (2015) 1063–1070.
[40] M.M. Joe, K. Bradeeba, R. Parthasarathi, P.K. Sivakumaar, P.S. Chauhan, S. Tipayno, A. Benson, T. Sa, Development of surfactin based nanoemulsion formulation from selected cooking oils: evaluation for antimicrobial activity against selected food associated microorganisms, J. Taiwan Inst. Chem. Eng. 43 (2012) 172–180.
[41] I.J.B. Durval, I.A. da Silva, L.A. Sarubbo, Application of microbial biosurfactants in the food industry, in: Microbial Biosurfactants, 2021, pp. 1–10.
[42] L. Fracchia, J.J. Banat, M. Cavallo, I.M. Banat, Potential therapeutic applications of microbial surface active compounds, AIMS Bioeng. 2 (2015) 144–162.

[43] **(a)** P. Das, S. Mukherjee, R. Sen, Antimicrobial potential of a lipopeptide biosurfactant derived from a marine *Bacillus circulans*, J. Appl. Microbiol. 104 (2008) 1675–1684. **(b)** K. Arima, A. Kakinuma, G. Tamura, Surfactin, a crystalline peptide lipid surfactant produced by *Bacillus subtilis*: isolation, characterization and its inhibition of fibrin clot formation, Biochem. Biophys. Res. Commun. 31 (1968) 488–494, https://doi.org/10.1016/0006-291x(68)90503-2.

[44] J. Vater, B. Kablitz, C. Wilde, P. Franke, N. Mehta, S.S. Cameotra, Matrix-assisted laser desorption ionization-time of flight mass spectrometry of lipopeptide biosurfactants in whole cells and culture filtrates of *Bacillus subtilis* C-1 isolated from petroleum sludge, Appl. Environ. Microbiol. 68 (2002) 6210–6219.

[45] N. Naruse, O. Tenmyo, S. Kobaru, H. Kamei, T. Miyaki, M. Konishi, T. Oki, Pumilacidin, a complex of new antiviral antibiotics production, isolation, chemical properties, structure and biological activity, J. Antibiot. 43 (1990) 267–280.

[46] M.M. Yakimov, K.N. Timmis, V. Wray, H.L. Fredrickson, Characterization of a new lipopeptide surfactant produced by thermotolerant and halotolerant subsurface *Bacillus licheniformis* BAS50, Appl. Environ. Microbiol. 61 (1995) 1706–1713.

[47] M.H. Mohd Isa, N.H. Shamsudin, N.K. Al-Shorgani, F.A. Alsharjabi, M.S. Kalil, Evaluation of antibacterial potential of biosurfactant produced by surfactin-producing Bacillus isolated from selected Malaysian fermented foods, Food Biotechnol. 34 (2020) 1–24.

[48] A. Abalos, A. Pinazo, M.R. Infante, M. Casals, F. Garcia, A. Manresa, Physicochemical and antimicrobial properties of new rhamnolipids produced by *Pseudomonas aeruginosa* AT10 from soybean oil refinery wastes, Langmuir 17 (2001) 1367–1371.

[49] M. Benincasa, A. Abalos, I. Oliveira, A. Manresa, Chemical structure, surface properties and biological activities of the biosurfactant produced by *Pseudomonas aeruginosa* LBI from soapstock, Antonie Van Leeuwenhoek 85 (2004) 1–8.

[50] M. Nitschke, S.G. Costa, J. Contiero, Structure and applications of a rhamnolipid surfactant produced in soybean oil waste, Appl. Biochem. Biotechnol. 160 (2010) 2066–2074.

[51] S.G. Costa, M. Nitschke, F. Lépine, E. Déziel, J. Contiero, Structure, properties and applications of rhamnolipids produced by *Pseudomonas aeruginosa* L2-1 from cassava wastewater, Process Biochem. 45 (2010) 1511–1516.

[52] H.S. El-Sheshtawy, M.M. Doheim, Selection of *Pseudomonas aeruginosa* for biosurfactant production and studies of its antimicrobial activity, Egypt. J. Pet. 23 (2014) 1–6.

[53] K. Kim, Y. Dalsoo, K. Youngbum, L. Baekseok, S. Doonhoon, K.I.M. Eun-Ki, Characteristics of sophorolipid as an antimicrobial agent, J. Microbiol. Biotechnol. 12 (2002) 235–241.

[54] X. Huang, Z. Wei, G. Zhao, X. Gao, S. Yang, Y. Cui, Optimization of sterilization of *Escherichia coli* in milk by surfactin and fengycin using a response surface method, Curr. Microbiol. 56 (2008) 376–381.

[55] X. Huang, J. Suo, Y. Cui, Optimization of antimicrobial activity of surfactin and polylysine against *Salmonella enteritidis* in milk evaluated by a response surface methodology, Foodborne Pathog. Dis. 8 (2011) 439–443.

[56] N.R. Gandhi, D.V.L.P. Skebba, Rhamnolipid Compositions and Related Methods of Use, W.O. International Application Patent (PCT) 2007/095258 A3, filed Feb. 12, 2007 and issued Aug. 23, 2007, 2007.

[57] H. Yuliani, M.S. Perdani, I. Savitri, M. Manurung, M. Sahlan, A. Wijanarko, H. Hermansyah, Antimicrobial activity of biosurfactant derived from *Bacillus subtilis* C19, Energy Procedia 153 (2018) 274–278.

[58] F. Anjum, G. Gautam, G. Edgard, S. Negi, Biosurfactant production through Bacillus sp. MTCC 5877 and its multifarious applications in food industry, Bioresour. Technol. 213 (2016) 262–269.
[59] G.S. Kiran, B. Sabarathnam, J. Selvin, Biofilm disruption potential of a glycolipid biosurfactant from marine *Brevibacterium casei*, FEMS Microbiol. Immunol. 59 (2010) 432–438.
[60] L.V. De Araujo, F. Abreu, U. Lins, L.M. Santa Anna, M. Nitschke, D.M. Freire, Rhamnolipid and surfactin inhibit *Listeria monocytogenes* adhesion, Food Res. Int. 44 (2011) 481–488.
[61] M.D. da Silva, I.J. Durval, M.E. da Silva, L.A. Sarubbo, Potential applications of anti-adhesive biosurfactants, in: Microbial Biosurfactants, 2021, pp. 213–225.
[62] C. Faille, B. Carpentier, Food contact surfaces, surface soiling and biofilm formation, in: Biofilms in the Food and Beverage Industries, Woodhead Publishing, 2009, pp. 303–330.
[63] S.S.E. Silva, J.W. Carvalho, C.P. Aires, M. Nitschke, Disruption of *Staphylococcus aureus* biofilms using rhamnolipid biosurfactants, J. Dairy Sci. 100 (2017) 7864–7873.
[64] R.M. Donlan, Biofilms: microbial life on surfaces, Emerg. Infect. Dis. 8 (2002) 881.
[65] J.M. Campos, T.L. Montenegro Stamford, L.A. Sarubbo, J.M. De Luna, R.D. Rufino, I. M. Banat, Microbial biosurfactants as additives for food industries, Biotechnol. Prog. 29 (2013) 1097–1108.
[66] X. Yan, S. Gu, X. Cui, Y. Shi, S. Wen, H. Chen, J. Ge, Antimicrobial, anti-adhesive and anti-biofilm potential of biosurfactants isolated from *Pediococcus acidilactici* and *Lactobacillus plantarum* against *Staphylococcus aureus* CMCC26003, Microb. Pathog. 127 (2019) 12–20.
[67] S. Bouman, D.B. Lund, F.M. Driessen, D.G. Schmidt, Growth of thermoresistant streptococci and deposition of milk constituents on plates of heat exchangers during long operating times, J. Food Prot. 45 (1982) 806–812.
[68] H.J. Busscher, M. Vanderkuijlbooij, H.C. Van der Mei, Biosurfactants from thermophilic dairy Streptococci and their potential role in the fouling control of heat exchanger plates, J. Ind. Microbiol. 16 (1996) 15–21.
[69] L.V. de Araujo, C.R. Guimarães, R.L. da Silva Marquita, V.M. Santiago, M.P. de Souza, M. Nitschke, D.M. Freire, Rhamnolipid and surfactin: anti-adhesion/antibiofilm and antimicrobial effects, Food Control 63 (2016) 171–178.
[70] E.J. Gudina, J.A. Teixeira, L.R. Rodrigues, Isolation and functional characterization of a biosurfactant produced by *Lactobacillus paracasei*, Colloids Surf. B. Biointerfaces 76 (2010) 298–304.
[71] R.D. Rufino, J.M. Luna, L.A. Sarubbo, L.R. Rodrigues, J.A. Teixeira, G.M. Campos-Takaki, Antimicrobial and anti-adhesive potential of a biosurfactant Rufisan produced by *Candida lipolytica* UCP 0988, Colloids Surf. B. Biointerfaces 84 (2011) 1–5.
[72] G.S. Kiran, L.A. Nishanth, S. Priyadharshini, K. Anitha, J. Selvin, Effect of Fe nanoparticle on growth and glycolipid biosurfactant production under solid state culture by marine Nocardiopsis sp. MSA13A, BMC Biotechnol. 14 (2014) 1–10.
[73] N. Bakhshi, S. Soleimanian-Zad, M. Sheikh-Zeinoddin, Dynamic surface tension measurement for the screening of biosurfactants produced by *Lactobacillus plantarum* subsp. plantarum PTCC 1896, Enzyme Microb. Technol. 101 (2017) 1–8.
[74] S.S. Giri, S.S. Sen, J.W. Jun, V. Sukumaran, S.C. Park, Role of *Bacillus licheniformis* VS16-derived biosurfactant in mediating immune responses in Carp Rohu and its application to the food industry, Front. Microbiol. 8 (2017) 514.

[75] A.P. Karlapudi, T.C. Venkateswarulu, K. Srirama, R.K. Kota, I. Mikkili, V.P. Kodali, Evaluation of anti-cancer, anti-microbial and anti-biofilm potential of biosurfactant extracted from an Acinetobacter M6 strain, J. King Saud Univ. Sci. 32 (2020) 223–227.
[76] C. Katsigris, C. Thomas, The Bar and Beverage Book, John Wiley and Sons, 2006, ISBN: 978-0-470-07344-5.
[77] D. Keys, Now That's What you Call a Real Vintage: Professor Unearths 8,000-Year-Old Wine, The Independent, 2003.
[78] M. Berkowitz, "World's Earliest Wine". Archaeology, 49, Archaeological Institute of America, 1996.
[79] M. Spilling, W. Wong, Cultures of the World Georgia, Cavendish Square Publishing, first ed., 2008, ISBN: 978-0-7614-3033-9.
[80] A. Ellsworth, "7,000 Year-Old Wine Jar", Penn Museum, Object Number 69–12–15, 2012.
[81] S. Mirsky, Ale's well with the world, Sci. Am. 296 (2007) 102. Bibcode:2007SciAm.296e.102M https://doi.org/10.1038/scientificamerican0507-102.
[82] M.L. Heiss, R.J. Heiss, The Story of Tea: A Cultural History and Drinking Guide, Ten Speed Press, 2007.
[83] A.S. Doost, F. Devlieghere, A. Dirckx, P. Van der Meeren, Fabrication of Origanum compactum essential oil nanoemulsions stabilized using Quillaja Saponin biosurfactant, J. Food Process. Preserv. 2018 (42) (2018) 13668, https://doi.org/10.1111/jfpp.13668.
[84] Plant Based Biosurfactants Market Size, Share & Trends Analysis Report by Application, Regional Outlook, Competitive Strategies, and Segment Forecasts, 2019 To 2025, report ID GVR2077.
[85] Y. Yang, M.E. Leser, A.A. Sher, D.J. McClements, Formation and stability of emulsions using a natural small molecule surfactant: Quillaja saponin (Q-Naturale), Food Hydrocoll. 30 (2013) 589–596.
[86] S. Mitra, S.R. Dungan, Micellar properties of quillaja saponin. 1. Effects of temperature, salt, and pH on solution properties, J. Agric. Food Chem. 45 (1997) 1587–1595.
[87] A. Yoshihiro, Saponin-Containing Effervescent Beverage, 2007. JP2009011200A.
[88] S. Mukai, T. Matsuda, Y. Kanazawa, Manufacture of Foaming Beverages With Saponins, Saccharides and Carbon Dioxide, 1993. JP 05 38,275.
[89] D. Nayyar, W. Schulok, F. Saleeb, E. Modesta, Dry Mix for Producing a Slush Beverage, 1998. US patent No. 5853785.
[90] P.R. Cheeke, Biological effects of feed and forage saponins and their impacts on animal production, in: G.R. Waller, K. Yamasaki (Eds.), Saponins Used in Food and Agriculture, Plenum Press, 1996, pp. 377–385.
[91] Y. Tamura, M. Miyakoshi, M. Yamamoto, Application of saponin-containing plants in foods and cosmetics, in: H. Sakagami (Ed.), Alternative Medicine, IntechOpen, 2012, https://doi.org/10.5772/53333.
[92] P. Walstra, J. Wouters, T. Geurts, Dairy Science and Technology, second ed., CRC Taylor and Francis, 2005.
[93] K.H.S. Farvin, C.P. Baron, N.S. Nielsen, C. Jacobsen, Antioxidant activity of yoghurt peptides: part 1-in vitro assays and evaluation in ω-3 enriched milk, Food Chem. 123 (2010) 1081–1089.
[94] Y. Xiao, L. Wang, X. Rui, W. Li, X. Chen, M. Jiang, M. Dong, Enhancement of the antioxidant capacity of soy whey by fermentation with *Lactobacillus plantarum* B1–6, J. Funct. Foods 12 (2015) 33–44.

[95] X.H. Cao, Z.Y. Liao, C.L. Wang, P. Cai, W.Y. Yang, M.F. Lu, G.W. Huang, Purification and antitumour activity of a lipopeptide biosurfactant produced by Bacillus natto TK-1, Biotechnol. Appl. Biochem. 52 (2009) 97–106.
[96] C.L. Wang, T.B. Ng, F. Yuan, Z.K. Liu, F. Liu, Induction of apoptosis in human leukemia K562 cells by cyclic lipopeptide from *Bacillus subtilis* natto T-2, Peptides 28 (2007) 1344–1350.
[97] J.H. Lee, S.H. Nam, W.T. Seo, H.D. Yun, S.Y. Hong, M.K. Kim, K.M. Cho, The production of surfactin during the fermentation of cheonggukjang by potential probiotic *Bacillus subtilis* CSY191 and the resultant growth suppression of MCF-7 human breast cancer cells, Food Chem. 131 (2012) 1347–1354.
[98] M. Krishnaswamy, G. Subbuchettiar, T.K. Ravi, S. Panchaksharam, Biosurfactants properties, commercial production, and application, Curr. Sci. 94 (2008) 736–747.
[99] M.S. Yeh, Y.H. Wei, J.S. Chang, Enhanced production of surfactin from *Bacillus subtilis* by the addition of solid carriers, Biotechnol. Prog. 2 (2005) 1329–1334.
[100] K.J. Meena, S.J. Kanwar, Lipopeptides as the antifungal and antibacterial agents: applications in food safety and therapeutics, Bio. Med. Res. Int. 1 (2015) 1–9.
[101] R. Akhter, F.A. Masoodi, T.A. Wani, J. Raja, S.A. Rather, Ethnic fermented foods and beverages of Jammu and Kashmir, in: J. Tamang (Ed.), Ethnic Fermented Foods and Beverages of India: Science History and Culture, Springer Nature Singapore Pte Ltd, 2020, https://doi.org/10.1007/978-981-15-1486-9_10.
[102] J.L. Fraga, G.C. Fontes Sant'Ana, K.A. Silva, P.F. Fonseca Amaral, Green (detox) juice physicochemical properties and stabilization effect of naturals emulsifiers, Cienc. Rural 50 (2020) 20190739, https://doi.org/10.1590/0103-8478cr20190739.
[103] B.G. Ribeiro, J.M. Guerra, L.A. Sarubbo, Biosurfactants: production and application prospects in the food industry, Biotechnol. Prog. 36 (2020) 3030.
[104] C.A. Kayath, E. Nguimbi, J. Goma-Tchimbakala, V. Mamonékéné, A.A. Lebonguy, G. Ahombo, Towards the understanding of fermented food biotechnology in Congo Brazzaville advance, J. Food Sci. Technol. 12 (2016) 593–602, https://doi.org/10.19026/ajfst.12.3317.
[105] I. Kanyana, E. Ouma, A., Piet van, quality assessment of banana juice and beer in Rwanda, J. Food Technol. 11 (2013) 38–43.
[106] M. Moukala, C. Kayath, G. Ahombo, N. Dangui, D. Kinavouidi, E. Mouélé, M. Diatewa, Giving more benefits to biosurfactants secreted by lactic acid bacteria isolated from plantain wine by using multiplex PCR identification, Adv. Microbiol. 9 (2019) 917–930, https://doi.org/10.4236/aim.2019.911058.
[107] C.S. Chooklin, S. Maneerat, A. Saimmai, Utilization of Banana Peel as a novel substrate for biosurfactant production by Halobacteriaceae archaeon AS65, Appl. Biochem. Biotechnol. 173 (2014) 624–645.
[108] S. George, K. Jayachandran, Analysis of rhamnolipid biosurfactants produced through submerged fermentation using orange fruit peelings as sole carbon source, Appl. Biochem. Biotechnol. 158 (2009) 694–705, https://doi.org/10.1007/s12010-008-8337-6.
[109] A.P. Kumar, A. Janardhan, B. Viswanath, K. Monika, J.-Y. Jung, G. Narasimha, Evaluation of orange peel for biosurfactant production by *Bacillus licheniformis* and their ability to degrade naphthalene and crude oil, 3 Biotech 6 (2016), https://doi.org/10.1007/s13205-015-0362-x.
[110] K. Paraszkiewicz, P. Bernat, A. Kuśmierska, J. Chojniak, G. Płaza, Structural identification of lipopeptide biosurfactants produced by *Bacillus subtilis* strains grown on the

media obtained from renewable natural resources, J. Environ. Manage. 209 (2018) 65–70, https://doi.org/10.1016/j.jenvman.2017.12.033.
[111] P.C. Martins, V.G. Martin, Biosurfactant production from industrial wastes with potential remove of insoluble paint, Int. Biodeter. Biodegr. 127 (2018) 10–16.
[112] A.J. Das, R. Kumar, Bioslurry phase remediation of petroleum-contaminated soil using potato peels powder through biosurfactant producing *Bacillus licheniformis* J1, Int. J. Environ. Sci. Technol. 15 (2018) 525–532.
[113] H. Li, H. Wang, H. Li, S. Goodman, P. Van Der Lee, Z. Xu, A. Fortunato, P. Yang, The worlds of wine: old, new and ancient, Wine Econ. Policy 7 (2018) 178–182, https://doi.org/10.1016/j.wep.2018.10.002.
[114] P.R. Cheeke, W. Oleszek, Anti-inflammatory and anti-arthritis effects of *Yucca schidigera*: reviw, J. Inflamm. (2006), https://doi.org/10.1186/1476-9255-3-6.
[115] D.C. Setten, G., Werken molecular structures of Saponins from Quillaja saponaria Molina, in: G.R. Waller, K. Yamasaki (Eds.), Saponins Used in Traditional and Modern Medicine, Plenum Press, 1996, pp. 185–193.
[116] C.W. Bamforth, The foaming properties of beer, J. Inst. Brewing 91 (1985) 370–383.
[117] H. Hayashi, H. Sudo, Economic importance of licorice, Plant Biotechnol. 26 (2009) 101–104.
[118] Y. Morita, S. Tadokoro, M. Sasai, D. Kitamoto, N. Hirashima, Biosurfactant mannosylerythritol lipid inhibits secretion of inflammatory mediators from RBL-2H_3 cells, Biochim. Biophys. Acta 2011 (1810) 1302–1308.
[119] P.A. Felse, V. Shah, J. Chan, K.J. Rao, R.A. Gross, Sophorolipid biosynthesis by Candidabombicola from industrial fatty acid residues, Enzyme Microb. Technol. 40 (2007) 316–323.
[120] M. Jadhav, S. Kalme, D. Tamboli, S. Govindwar, Rhamnolipid from Pseudomonas desmolyticum NCIM-2112 and its role in the degradation of Brown 3REL, J. Basic Microbiol. 51 (2011) 1–12.
[121] K.E. Sutyak, R.E. Wirawan, A.A. Aroutcheva, M.L. Chiindas, Isolation of the *Bacillus subtilis* antimicrobial peptide subtilosin from the dairy product derived Bacillus amyliquefaceiens, J. Appl. Microbiol. 104 (2008) 1067–1074.
[122] P.D. Cotter, C. Hill, R.P. Ross, Bacteriocins: developing innate immunity for food, Nat. Rev. Microbiol. 3 (2005) 777–788.
[123] B. Teichmann, U. Linne, S. Hewald, M.A. Marahiel, M. Bölker, A biosynthetic gene cluster for a secreted cellobiose lipid with antifungal activity from *Ustilago maydis*, Mol. Microbiol. 66 (2007) 525–533.
[124] M. Begley, P.D. Cotter, C. Hill, R.P. Ross, Identification of a novel two-peptide lantibiotic, lichenicidin, following rational genome mining for LanM proteins, Appl. Environ. Microbiol. 75 (2009) 5451–5460.
[125] R. San Martin, R. Briones, Industrial uses and sustainable supply of quillaja saponaria (rosaceae) saponins, Econ. Bot. 53 (1999) 302–311.
[126] A. Daverey, K. Dutta, COVID-19: eco-friendly hand hygiene for human and environmental safety, J. Environ. Chem. Eng. 9 (2021), https://doi.org/10.1016/j.jece.2020.104754, 104754.
[127] D. Kregiel, J. Berlowska, I. Witonska, H. Antolak, C. Proestos, M. Babic, L. Babic, B. Zhang, Saponin-based, biological-active surfactants from plants, in: R. Najjar (Ed.), Application and Characterization of Surfactants, IntechOpen, 2017.

[128] B.A. Weinberg, B.K. Bealer, The World of Caffeine: The Science and Culture of the World's Most Popular Drug, Routledge, 2002, ISBN: 978-0-415-92722-2.
[129] R. Ispen, J. Otte, The Relation Between Protein Structure, Interfacial Rheology and Foam Formation for Various Milk Proteins, vol. 21, Nordic Rheology Society, 2004, pp. 143–178.
[130] D. Langevin, Influence of interfacial rheology on foam and emulsion properties, Adv. Colloid Interface Sci. 88 (2000) 209–222.
[131] Z. Xu, X. Li, S. Feng, J. Liu, L. Zhou, M. Yuan, C. Ding, Characteristics and bioactivities of diferent molecular weight polysaccharides from camellia seed cake, Int. J. Biol. Macromol. 91 (2016) 1025–1032.
[132] X. Liu, L. Jia, Y. Gao, B. Li, Y. Tu, Anti-infammatory activity of total flavonoids from seeds of Camellia oleifera Abel, Acta Biochim. Biophys. Sin. 46 (2014) 920–922.
[133] Y.F. Chen, C.H. Yang, M.S. Chang, Y.P. Ciou, Y.C. Huang, Foam properties and detergent abilities of the saponins from Camellia oleifera, Int. J. Mol. Sci. 11 (2010) 4417–4425.
[134] M.N. Uddin, G. Sharma, J.-L. Yang, H.S. Choi, S.-I. Lim, K.W. Kang, W.K. Oh, Oleanane triterpenes as protein tyrosine phosphatase 1B (PTP1B) inhibitors from *Camellia japonica*, Phytochemistry 103 (2014) 99–106.
[135] X. Yang, X. Wang, K. Wang, L. Su, H. Li, R. Li, Q. Shen, The nematicidal effect of camellia seed cake on root-knot nematode Meloidogyne javanica of banana, PLoS One 10 (2015) 0119700.
[136] S. Kijprayoon, V. Tolieng, A. Petsom, C. Chaicharoenpong, Molluscicidal activity of Camellia oleifera seed meal, Science Asia 40 (2014) 393–399.
[137] Y.T. Cheng, C.C. Lu, G.C. Yen, Beneficial effects of Camellia oil (Camellia oleifera Abel.) on hepatoprotective and gastroprotective activities, J. Nutr. Sci. Vitaminol. 61 (2015) S100–S102.
[138] W. Zhao, N. Li, X. Zhang, W. Wang, J. Li, Y. Si, Cancer chemo preventive theasaponin derivatives from the total tea seed saponin of *Camellia sinensis*, J. Funct. Foods 12 (2015) 192–198.
[139] X. Feas, L.M. Estevinho, C. Salinero, P. Vela, M.J. Sainz, M.P.V. Tato, J.A. Seijas, Triacylglyceride, antioxidant and antimicrobial features of virgin camellia oleifera, *C. reticulata* and *C. sasanqua* oils, Molecules 18 (2013) 4573–4587.
[140] J.D. Kim, M.I. Khan, J.H. Shin, H.J. Seo, T.S. Shin, M.Y. Kim, HPLC fractionation and pharmacological assessment of green tea seed saponins for antimicrobial, antiangiogenic and hemolytic activities, Biotechnol. Bioprocess Eng. 20 (2015) 1035–1043.
[141] M. Feng, Z. Zhu, L. Zuo, L. Chen, Q. Yuan, G. Shan, S.-Z. Luo, A strategy for rapid structural characterization of saponins and flavonoids from the testa of Camellia oleifera Abel seeds by ultra-high-pressure liquid chromatography combined with electrospray ionization linear ion trap-orbitrap mass spectrometry, Anal. Methods 7 (2015) 5942–5953.
[142] C.Y. Lin, C.L. Fan, Fuel properties of biodiesel produced from Camellia oleifera Abel oil through supercritical-methanol trans esterifcation, Fuel 90 (2011) 2240–2244.
[143] X. Li, J. Deng, S. Shen, T. Li, M. Yuan, R. Yang, C. Ding, Antioxidant activities and functional properties of enzymatic protein hydrolysates from defatted Camellia oleifera seed cake, J. Food Sci. Technol. 52 (2015) 5681–5690.
[144] H. Chen, C. Yang, T. Ke, Antimicrobial activity of secondary metabolites from Streptomyces sp. K15, an endophyte in Houttuynia cordata Tunb, Nat. Prod. Res. 29 (2015) 2223–2225.

[145] M.I. Khan, A. Ahhmed, J.H. Shin, J. Soo Baek, M. Yong Kim, J.D. Kim, Green tea seed isolated saponins exerts antibacterial effects against various strains of gram positive and gram negative bacteria, a comprehensive study in vitro and in vivo, Hindawi Evid. Based Complement. Alternat. Med. (2018) 3486106, https://doi.org/10.1155/2018/3486106.

[146] Markets and Markets, Biosurfactants Market by Type (Glycolipids (Sophorolipids, Rhamnolipids), Lipopeptides, Phospholipids, Polymeric Biosurfactants), Application (Detergents, Personal Care, Agricultural Chemicals, Food Processing), and Region—Global Forecast to 2022, ID: 4437552, 2017.

[147] 360iResearch, Biosurfactants Market Research Report by Product Type, by Region—Global Forecast to 2026—Cumulative Impact of COVID-19, ID: 4989641, 2021.

[148] Business Standards, 2016. https://www.business-standard.com/content/b2b-chemicals/evonik-becomes-first-firm-to-produce-biosurfactants-on-industrial-scale-116062301145_1.html.

[149] K.K. Sekhon Randhawa, P.K. Rahman, Rhamnolipid biosurfactants—past, present, and future scenario of global market, Front. Microbiol. 5 (2014) 454.

[150] Glam Research, Global Biosurfactant Market By Product Type (Glycolipids, Phospholipid and Fatty Acids) and by End-Users/Application (Environment, Agriculture) Global Market Share, Forecast Data, In-Depth Analysis, and Detailed Overview, and Forecast, 2013–2026, EM-35197925, 2020.

[151] M.H. Mondal, S. Malik, B. Saha, Characterization of pyrene solubilization in selective micellar media of novel bio-degradable natural surfactant saponin (extracted from soap nut) and conventional surfactant SDBS in presence and absence of common salt NaCl, Tenside Surfactant Deterg. 54 (2017) 378–384.

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CHAPTER

Antibiofilm properties of biosurfactants: A tool against the food pathogens

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Abbreviations

BEC	biofilm eradication concentration
BIC	biofilm inhibitory concentration
BS	biosurfactants
CLSM	confocal laser scanning microscopy
CMC	critical micellar concentration
DNA	deoxyribonucleic acid
GCMS	gas chromatography-mass spectrometry
LAB	lactic acid bacteria
LCMS	liquid chromatography-mass spectrometry
MBC	minimum bactericidal concentration
MBD	marine bacterial DNase
MEL	mannosylerythritol lipids
MIC	minimum inhibitory concentration
MRS	den Man, Rogosa, and Sharpe
MRSA	methicillin-resistant *Staphylococcus aureus*
NMR	nuclear magnetic resonance
PPE	phosphatidylethanolamine
SEM	scanning electron microscope
SL	sophorolipid
ST	surface tension
TGA	thermogravimetric analysis
TLC	thin layer chromatography
YPD	yeast extract peptone dextrose

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1 Introduction

Biosurfactants are biological products that have surface-modifying properties such as surfactants. Therefore, these have amphiphilic nature and can lower the surface tension (ST) among the two immiscible surfaces or liquids. The amphiphilic nature is derived from the hydrophilic and hydrophobic parts present there, composed of polar (amino acids, carbohydrates, peptides, etc.) and nonpolar (long carbon chains, fatty acids, etc.) functionalities, respectively.

It's structural assembly has a hydrophilic head (from polar groups), and a hydrophobic tail (from nonpolar groups) which is useful in the emulsification of oil in water (O/W) and water in oil (W/O) via interacting through the similar nature of functional group where oil serves as nonpolar part while water as polar part (Fig. 1) [1–5].

2 Type of biosurfactants

Various types of biosurfactants are known with basic scaffolds such as lipopeptides, phospholipids, glycolipids, neutral lipids, and polymeric compounds. These classes of biosurfactants are produced from various microorganisms and plants, which have wide applicability in food and other industries such as agriculture, cleaning and disinfectant, pharmaceutical, etc. Lipopeptide biosurfactants are composed of lipid and peptide produced by different microorganisms and mainly a cyclic peptide joined with a fatty acid [6]. Mostly obtained from *Bacillus subtilis* and *Serratia marcescens* with good surfactant properties at very low concentrations [6–8], these have different bioactivities, including antibacterial, antiviral, and antimycoplasma. The known lipopeptides are serrawettin, surfactin, subtilisin (a protein-digesting enzyme), polymyxins, viscosin, amphisin, and putisolvin (Fig. 2) [9–17].

Glycolipids biosurfactants are composed of carbohydrates attached to fatty acids. These include Rhamnolipids, Mannosylerythritol lipids, Trehalolipids, and Sophorolipids (Fig. 3), which are obtained from Pseudomonas spp., *Pseudozyma antarctica* [18], *Rhodococcus* sp., *Nocardia* sp., and *Candida* spp., respectively [19–22].

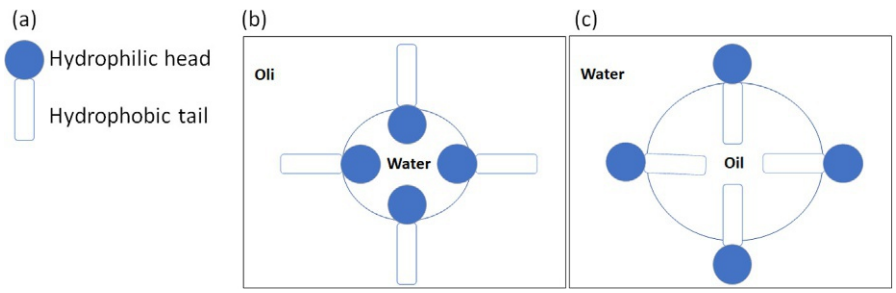

FIG. 1

Biosurfactant, (A) amphiphilic nature, (B) water in oil, and (C) oil in water.

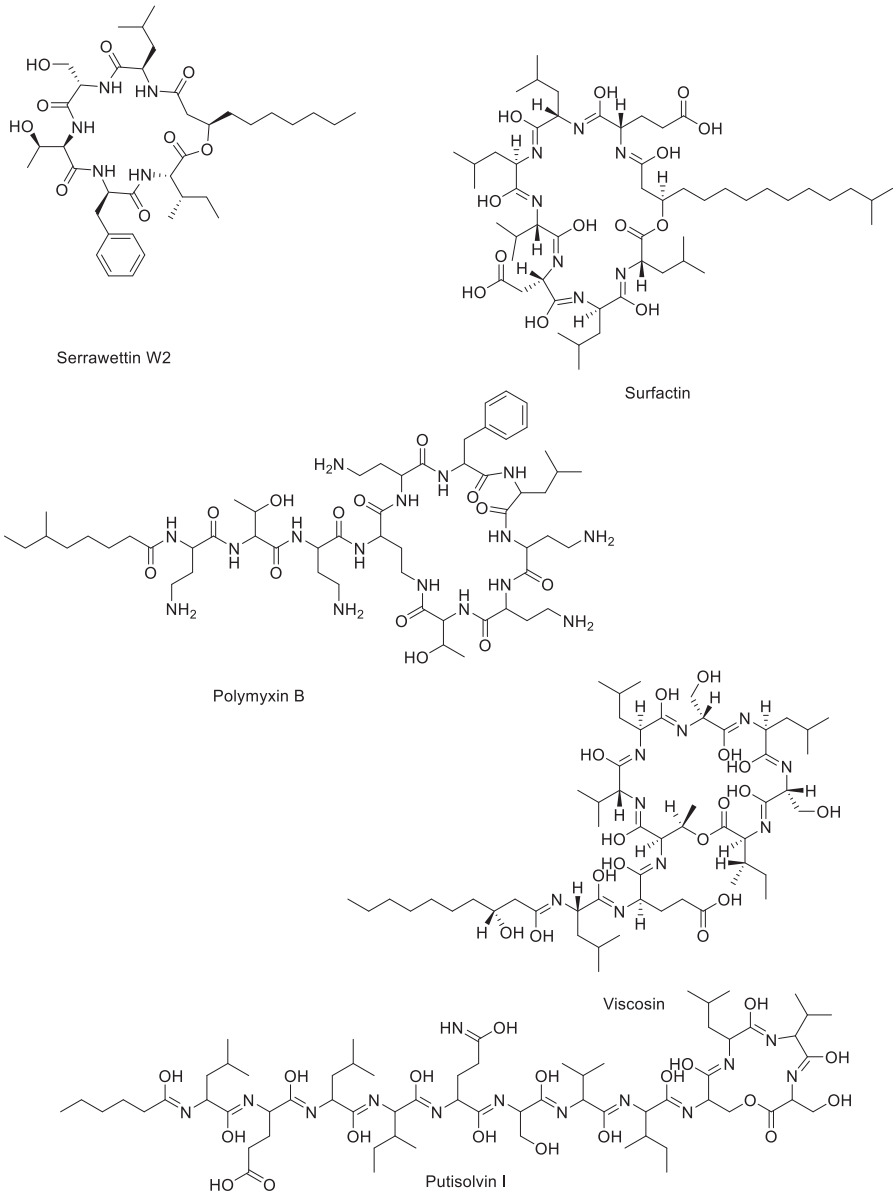

FIG. 2

Lipoprotein biosurfactants.

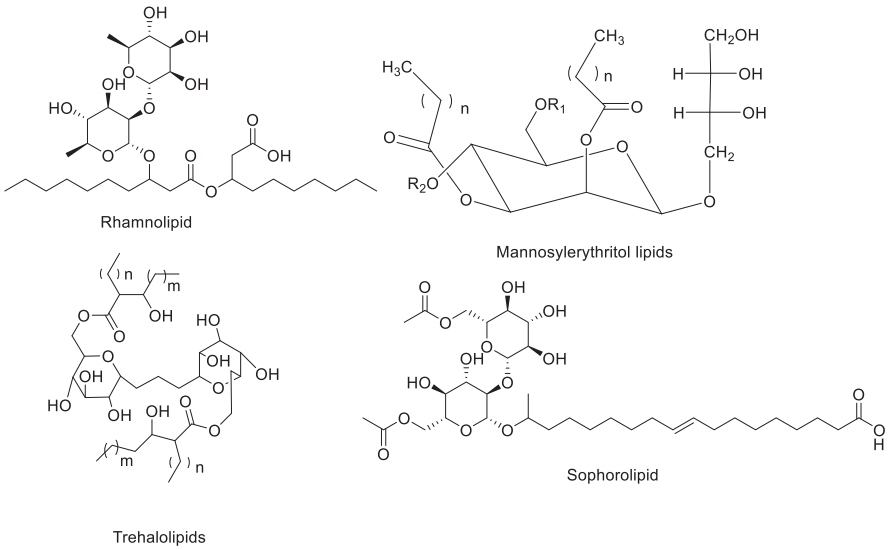

FIG. 3

Glycolipids biosurfactants.

Phospholipid biosurfactants are the lipid-derived phosphate group produced by many microorganisms, including phosphatidylethanolamine (PEP), sphingomyelin, and lecithin (Fig. 4A) [23,24].

Polymeric biosurfactants are composed of different derivatized carbohydrates, lipids, and proteins in the polymeric form, including Emulsan, Liposan, Biodispersan, Mannan-lipid-protein, and Carbohydrate-protein-lipid (Fig. 4B) [25–27].

3 Biofilm and food

Biofilm was described as a unique feature of growth of bacteria, which make the colonization at a higher level of organization to allow them to live and communicate with each other [28]. It is an aggregate of microorganisms forming different and distinct layers or clusters of growing cells and surviving in harsh conditions. The biofilm is surrounded by extracellular polymer substances with high molecular weight and comprised of polysaccharides, lipids, proteins, and extracellular DNA. Biofilms perform a critical part in the pathogenesis of infectious microorganisms [29–31]. It behaves as a barrier and is responsible for tolerance against physicochemical methods and drug resistance or antimicrobial resistance [32,33]. The mechanism lying for the resistance is via poor penetration, efflux pumps, genetic adaptation, and heterogeneity in growth rate inside the biofilms (Fig. 5).

Biofilm-forming microorganisms are generally pathogenic and can grow in food, on the surfaces of apparatus and packaging material used in the food industries [34].

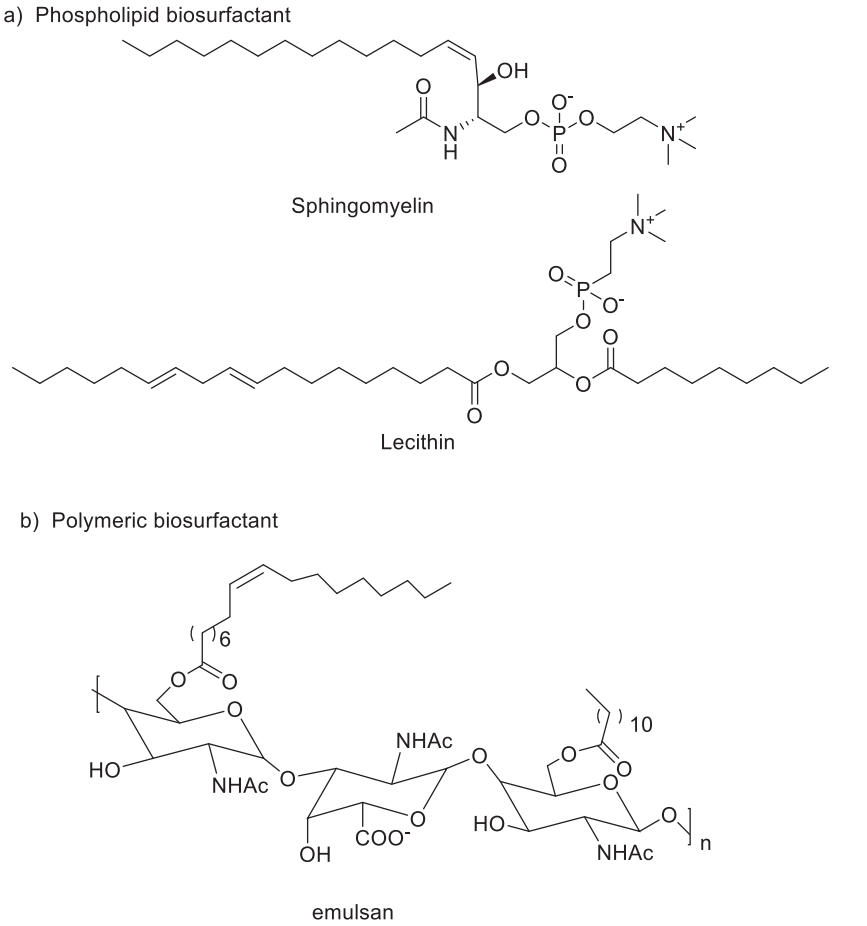

FIG. 4

(A) Phospholipid and (B) polymeric biosurfactants.

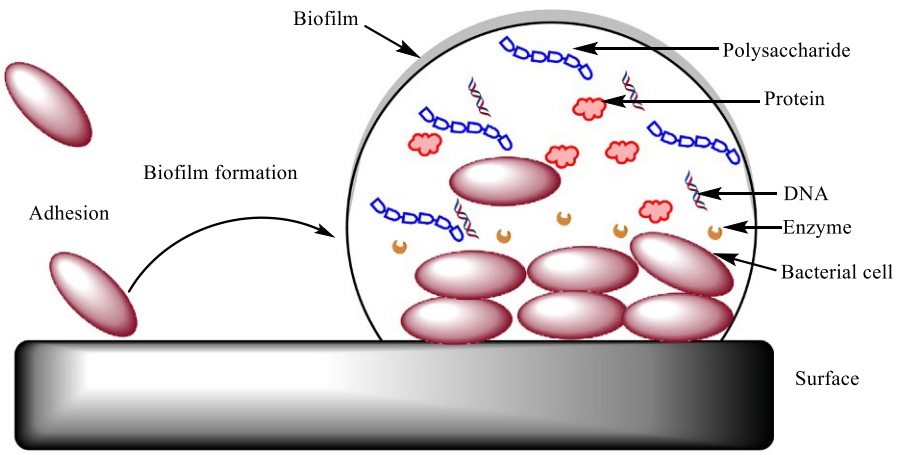

FIG. 5

Biofilm formation and composition.

Biofilms-forming microorganism also causes the alteration of organoleptic properties of food and dairy products, corrosion of metal surfaces leading to several types of health and economic concerns [35]. Thus, inhibiting and destroying the biofilm may be a good strategy to stop or kill the microorganisms [36,37].

4 Biosurfactants against food pathogens

Several biosurfactants have shown potential to control microbial growth and can be used as antibacterial agents and food preservatives. A summary in Table 1 represents the biosurfactants, their sources, and potential applications to control food pathogens.

Table 1 Biosurfactants and their uses.

S. No.	Biosurfactant	Biosurfactant source	Uses (Antibacterial action against food pathogen)	Ref.
1	Mannosylerythritol lipids	*Pseudozyma aphidis* DSM 70725	*Staphylococcus aureus*	Shu et al [38]
2	Sophorolipids	*Candida bombicola*	*Bacillus subtilis, S. epidermis, Pseudomonas acnes*	Dı'az De Rienzo et al [39]
3	Rhamnolipid	*Pseudomonas aeruginosa* PA1	*Listeria monocytogenes, Pseudomonas fluorescens, S. aureus, S. Enteritidis*	Araujo et al [40]
4	Surfactin	*B. subtilis*	*L. monocytogenes, P. fluorescens, S. aureus, Staphylococcus Enteritidis*	Araujo et al [40]
5	Bacterium *Vibrio alginolyticus*	AMS1 (GenBank accession number KX447573	*Candida albicans*	Srikanth et al [41]
6	Lactic acid bacteria (LAB)	*Lactobacillus plantarum* CFR 2194	*Escherichia coli, Serratia marcescens, Bacillus pumilis, Micrococcus luteus*	Madhu et al. [42]

Table 1 Biosurfactants and their uses. *Continued*

S. No.	Biosurfactant	Biosurfactant source	Uses (Antibacterial action against food pathogen)	Ref.
7	Xylolipid	*Enterococcus faecium* MRTL9	*S. aureus, E. coli, C. albicans, P. aeruginosa*	Sharma et al [43]
8	Fengycin	*B. subtilis* strain Bs2504	*Candida divergens*, *Bacillus thermosphacta*	Kourmentza et al [44]
9	Mycosubtilin	*B. subtilis*	*Candida divergens*, *Bacillus thermosphacta*	Kourmentza et al [44]
10	Trehalolipids	*Arthrobacter* spp. and *Rhodococcus erythropolis*	*Enterococcus hirae*, *Proteus vulgaris, E. coli*	
11	Serrawettin W1	*S. marcescens*	*E. coli*	Clements et al [45]
12	Viscosin	*P. fluorescens*	*Mycobacterium species M. luteus*	Niels Geudens et al. [46]
13	Putisolvin	*Pseudomonas putida*	*P. aeruginosa*	Kuiper et al [47]
14	Arthrofactin	*Arthrobacter* sp.	*Pseudomonas species*	Salam et al. [48]
15	Cellobiose lipids	*Ustilago maydis*	*Saccharomyces cerevisiae, Staphylococcus epidermidis*	Rasouli et al. [49]
16	Liamocins	*Aureobasidium pullulans*	*Streptococcus species*, *Streptococcus agalactiae*	Saur et al [50]

Mannosylerythritol lipids (MELs) are known to be glycolipid class of biosurfactants. It shows the wide application in medicine, the cosmetics, food industry, and other fields. The chemical structure of MEL varies by the different acetyl groups and fatty acid chains usually recognized as MEL-A, MEL-B, MEL-C, and MEL-D. These MELs showed antibacterial action against Gram-positive bacteria [51]. It was also revealed these have a strong inhibitory impact on *Bacillus cereus* vegetative cells and spores [52]. Furthermore, MEL-A-modified chitosan nanoparticles showed an antibacterial action against *S. aureus* [53] and *Pseudozyma aphidis*

DSM 70725 was used to manufacture MELs, which were purified using the Fan et al. technique [53]. MELs that are prepared were further characterized by TLC, liquid chromatography-mass spectrometry (LCMS), gas chromatography-mass spectrometry (GCMS), nuclear magnetic resonance (NMR) [54]. The MELs showed minimum inhibitory concentration (MIC) to be 0.625 mg/mL while minimum bactericidal concentration (MBC) to be 2.50 mg/mL, which is calculated by the twofold sequential dilution technique [52]. The MEL's inhibitory effect was evaluated using a bactericidal growth curve, and cell apoptosis activity with double staining apoptosis kit, i.e., Annexin V-FITC/propidium iodide (PI). The crystal violet staining method was used to determine the antibiofilm activity of MELs [55]. MEL-A showed significant antibiofilm activity against *S. aureus* with biofilm eradication concentration (BEC) 32 μg/mL [38]. MELs also showed the antibiofilm activity against a drug-resistant, biofilm-forming bacteria, i.e., Methicillin-resistant *S. aureus* (MRSA), which causes noteworthy morbidity and mortality. It has been commonly found in food products in recent years. The efficacy of MEL-A on biofilm reduction in MRSA was tested using the typical crystal violet staining method [56]. The biofilm elimination effect of MEL-A was examined using SYTO 9 staining and PI doubling labeling by employing Confocal Laser Scanning Microscopy (CLSM). According to author's research, the ideal effective concentration of MEL-A against MRSA was found to be 256 g/mL, when MEL was added to the microbial cells, it showed a decrease in the number of cells. When MEL was used with ultrasound, it showed a synergistic effect in biofilm disruption (Fig. 6) [57].

Sophorolipids (SLs) belong to the glycolipid class of biosurfactants. These are mostly biosynthesized in yeasts such as *Candida bombicola,* which comprises dimeric sugar linked to a hydroxyl fatty acid, a glycosidic bond. SLs can disrupt cell membranes; also, it can inhibit the growth of Gram-positive and Gram-negative bacteria. *Candida bombicola* ATCC 22214 was considered the typical medium for the improvement in SLs production. The SLs have antibacterial action against planktonic cells of demonstrative Gram-negative and Gram-positive bacteria in various carbohydrate-containing media [58]. Two types of SLs were produced S1, crude sophorolipid, and S2 is commercially available Sophorolipids. SLs showed antimicrobial activity against *Bacillus subtilis, Propionibacterium acnes,* and *S. epidermis* at a low MIC concentration of 1.6 mM. Antibiofilm ability of SLs was checked on biofilms of *B. subtilis* BBK006 culture and mixed culture of *S. aureus* ATCC 9144 and *B. subtilis* BBK006. SLs can potentially disrupt the biofilm produced by the above culture at a concentration of over 5% (*v*/v). Regardless of their acid/lactonic composition, SL was good as a bactericidal agent, causing cell death in planktonic cells of Gram-negative and Gram-positive bacteria in a way comparable to predictable antimicrobials with bacteriostatic effects [39].

Biosurfactant rhamnolipid and surfactin possess antibiofilm activity against *Pseudomonas fluorescens* and *Listeria monocytogenes*. *L. monocytogenes* is a Gram-positive microbe linked to diseases caused by food, and *P. fluorescens* [59] (a Gram-negative species typically observed in the water of industrial waste are

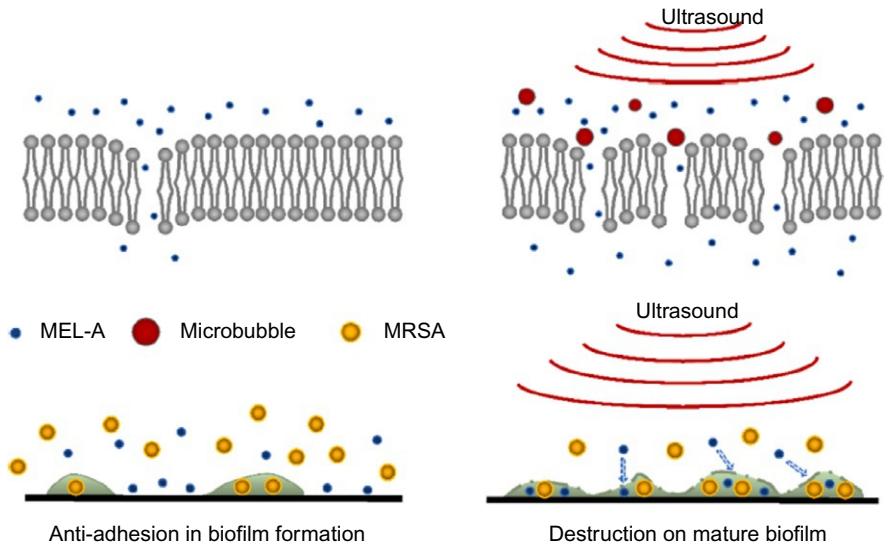

FIG. 6

Proposed synergistic antibacterial and antibiofilm mechanism of ultrasound and MELs against MRSA.

Reproduced from Shu, Q., Lou, H., Wei, T., Zhang, X., Chen, Q. Synergistic antibacterial and antibiofilm effects of ultrasound and MEL-A against methicillin-resistant Staphylococcus aureus. Ultrason. Sonochem. 72 (2021) 105452. https://doi.org/10.1016/j.ultsonch.2020.105452, Copyright Elsevier (2020).

linked to foodstuff deterioration) were chosen for the investigation of antibiofilm activity. As per the method given by Santos et al. [60], *Pseudomonas aeruginosa* PA1 was used to produce crude rhamnolipids [61]. Surfactin was produced from *B. subtilis* ATCC21332 and purified as per described method by Nitschke et al. [62]. By measuring the absorbance at 590 nm, antimicrobial activity was measured [63]. ST and interfacial tension were reduced effectively by biosurfactants, and their critical micellar concentration (CMC) was observed to be lower. The crude rhamnolipids, pure rhamnolipids, and surfactin all had ST values of 28.5 ± 0.3, 25.7 ± 0.3, and 26.8 ± 0.1, respectively. Crude rhamnolipids, surfactin, and purified rhamnolipids showed the CMC of 59.0 ± 2.6, 83.7 ± 0.8, and 110.0 ± 3.0 respectively. Biofilm development on polystyrene surfaces was greatly reduced to 74% and 79% when rhamnolipids (crude and purified) were tested on *L. monocytogenes* ATCC 7644 and *P. fluorescens* ATCC 13525, respectively [40]. Rhamnolipid also showed antimicrobial activity against *B. cereus* with a 19.5 μg/mL MIC and MBC of 39.1 μg/mL [64]. Rhamnolipid and surfactin also showed the antibiofilm activity against *S. aureus* and *S. Enteritidis*, considered food pathogens. Biofilm formed by *S. aureus* was disrupted by rhamnolipid and surfactin 58.5% and 63.7%, respectively, while both biosurfactants disrupt biofilm formed by *S. Enteritidis* to 35.3% and 30.9%, respectively (Fig. 7) [65].

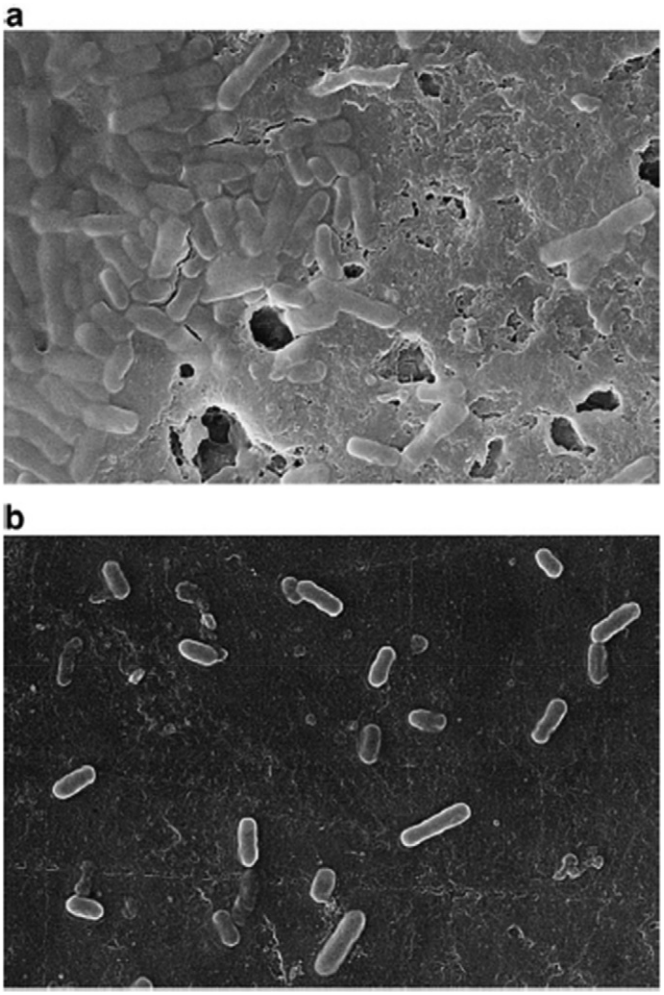

FIG. 7

SEM images (A) without biosurfactant and (B) with surfactin 0.1%; of *Listeria monocytogenes*.

Reproduced from Ref Zezzi do Valle Gomes, M., Nitschke M. Evaluation of rhamnolipid and surfactin to reduce the adhesion and remove biofilms of individual and mixed cultures of food pathogenic bacteria. Food Control 25 (2012) 441–7. https://doi.org/10.1016/j.foodcont.2011.11.025. Copyright Elsevier (2011).

The utmost common pathogen from Candida spp. is *Candida albicans*, which may be causing superficial and invasive candidiasis [66]. Srikanth et al., have tested a biosurfactant's capability from a marine bacterium to disrupt biofilms of *C. albicans*. It has revealed the ability to destroy biofilm and prevent the morphological conversion from yeast to hyphae. Candida sp. biofilms were disrupted by DNase

isolated by an aquatic bacterium *Vibrio alginolyticus* [67]. The researcher explained the synergistic impact of DNase I and marine bacterial DNase (MBD) of *V. alginolyticus* with biosurfactant. The staining technique using crystal violet was utilized to find out the capability of biosurfactant to disrupt biofilm, and absorbance was calculated spectrophotometrically [68]. Biofilm disruption capability of biosurfactant was confirmed by using SEM, and MATH assay to the culture of *C. albicans* to check the hydrophobicity index of the microbial cell. Growth curve of *C. albicans* in yeast extract peptone dextrose (YPD) broth to check the biofilm inhibitory concentration (BIC) of biosurfactants [69]. The MIC of biosurfactant to *Staphylococcus epidermidis, C. albicans,* and other species was 200 g/mL, 160 g/mL, and 600 g/mL, respectively. *C. albicans, S. epidermidis,* and others biofilms showed significant reductions when treated with biosurfactants of about 61%, 72%, and 67%, respectively [41].

Lactic acid bacteria (LAB) are one of the key producers of biosurfactants used in the food industry and medicinal industry widely. LAB was isolated from the cheese samples, and after isolation, it was placed in den Man, Rogosa, and Sharpe agar (MRS) agar. For the production of LAB, an isolated LAB was inoculated in MRS agar for 48 h. Biosurfactant was extracted by centrifugation according to the method described by das et al. [70]. Biosurfactant was tested against Gram-positive bacteria and Gram-negative bacteria. Among Gram-positive bacteria are *Staphylococcus aureus* ATCC 29737, *Micrococcus luteus* ATCC 10240, *Bacillus pumilis* ATCC 14884, while among Gram-negative bacteria, *Pseudomonas aeruginosa* ATCC9027, *Escherichia coli* ATCC10536, *Enterobacter aerogenes* 9805, *Salmonella typhimurium*, and *Serratia marcescens* 98027 were tested by using the good diffusion agar method [71]. The biosurfactants were also tested in vitro anticancer assay defined by Gomha et al. [72], and antiviral assay as described by Vijayan et al. [73]. Glycoprotein produced by LAB showed antimicrobial activity against both types of bacteria. *S. aureus* ATCC 29737 showed the inhibition zone diameter of 20.3 ± 0.25 mm against lab strain *L. plantarum* 60FHE while Gram-negative bacteria *E. coli* ATCC10536 showed an inhibition zone diameter of 38.0 ± 0.10 mm [74]. LAB strain L. *plantarum* 60FHE showed good antibacterial action against the Gram-positive and negative species. *Lactobacillus plantarum* CFR 2194 exhibit some characteristic property such as the ability to produce vitamin B12 [42]. Rice-based ayurvedic fermented food kanjika was used to produce biosurfactants. Further, it was subculture with the *E. coli* MTCC 108, *E. coli* ATCC 31075, *Yersinia enterocolitica* MTCC 859, *Salmonella typhi*, and *S. aureus* F 722 to study the antiadhesive and antimicrobial properties of the biosurfactant. Agar well diffusion method explained by Golek et al. [75] was used to evaluate the antimicrobial property of freeze-dried biosurfactant in solution form with water. *L. plantarum* CFR 2194 showed significant antimicrobial properties against *E. coli* MTCC 108 and *S. aureus* F 722. The antiadhesive property of biosurfactant was assessed according to process describes by Reid et al. [76]. Highest antiadhesive percentage of $64.72 \pm 0.4\%$ was shown by *Y. enterocolitica*, whereas the lowest of $54.43 \pm 0.1\%$ by *E. coli* ATCC. While *E. coli* ETEC and *S. typhi* showed antiadhesive percentages up to $56.78 \pm 0.3\%$ and $61.60 \pm 0.2\%$, respectively. *L. plantarum* CFR 2194 has ability to decrease the ST PBS cell extract,

which indicates the capability to produce the biosurfactant. *L. plantarum* CFR 2194 biosurfactant showed the promising activity against many food pathogens and bacteria [77].

Different bacterial species produce amphiphilic surfactants with high surface activity and emulsifying capabilities. They are frequently used in the recovery of crude oil, soil remediation, and hydrocarbon degradation. Lactobacillus is a species of biosurfactant-producing bacteria that can be found in both human and animal gastrointestinal microflora. Biofilm-forming pathogens colonize hydrophobic silicone rubber surfaces. A biosurfactant applied to a surface alters hydrophobicity and affects microbial adhesion and desorption processes. LAB isolates derived from the buttermilk were used to produce biosurfactants. Biosurfactant (Xylolipid) was derived from *Enterococcus faecium* MRTL9 strain, as shown in Fig. 8. Foodborne pathogenic and food spoilage microorganisms were utilized to assess the antibacterial action of biosurfactants generated from *E. faecium* MRTL9 (a *Bacillus* sp.). Both Gram-positive and Gram-negative bacteria are susceptible to this lipopeptide [78] (Fig. 9). The biosurfactant comprises carbohydrates and lipids, as determined by the FTIR spectrum [79]. The ST of PBS extract was significantly reduced by the biosurfactant produced by strain MRTL9, from 72 to 40.2 $mN\,m^{-1}$ [78].

The hydrophobic part of the biosurfactant chain is a lipid, while the hydrophilic part is mostly sugar. There was a peak with good intensity having a retention rate of 25.29%. The most significant fatty acid synthesized by biosurfactant named C-16 hexadecanoic fatty acid (with 86.6%). TGA analysis was used to examine the thermal degradation of biosurfactants. The activity of partially purified biosurfactant was maintained over a pH of 6–10 having minor changes in values of surface tension, resulting in a weight loss of about 1% [80]. Biofilm formation on silicone tubes was reduced considerably even at a low biosurfactant concentration (25 mg/mL). With a preadsorbed biosurfactant, numerous bacterial cells stuck were reduced to 89% on the silicone rubber [12]. Still, at the maximum concentration of biosurfactant tested, i.e., 25 mg/mL, antibiofilm activity against *C. albicans* was relatively low. At concentrations of 12.5 mg/mL, 25 mg/mL, and 50 mg/mL, the cell viability of mouse fibroblasts was observed [78].

Lipopeptide biosurfactant resembles cyclic structures made up of hydrophilic peptide sequences that are usually 7–10 amino acids long, with a C13–C18 fatty acid chain as the hydrophobic component. They are mainly prepared by pseudomonas and *Bacillus* species [81]. *B. subtilis* strains produced lipopeptide biosurfactants, mainly surfactin, fengycin, and mycosubtilin. Surfactin was produced by the method

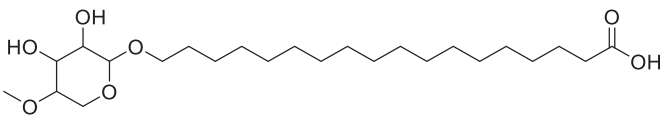

FIG. 8

Structure xylolipid biosurfactant produced by *Enterococcus faecium* MRTL9.

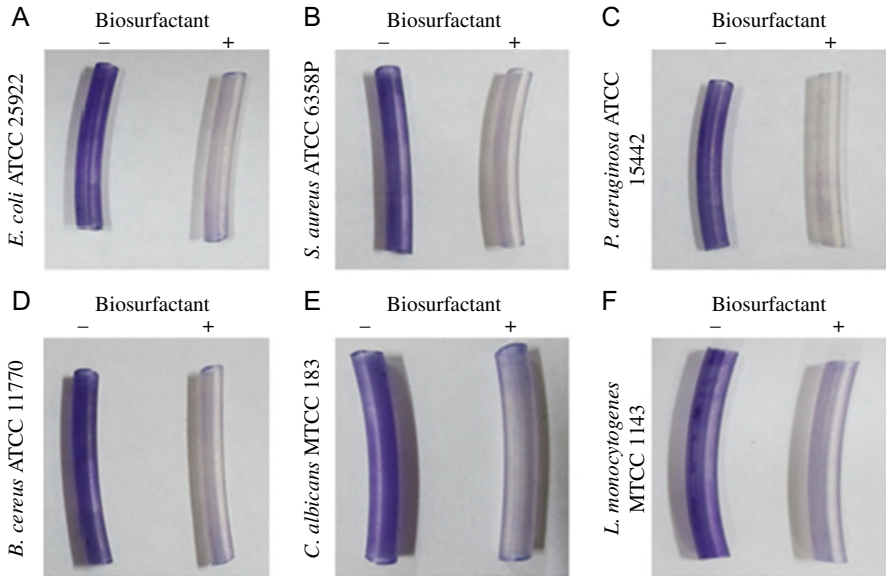

FIG. 9

The antibiofilm capacity of biosurfactant from *Enterococcus faecium* on different bacteria (A–F) (−) without and (+) with biosurfactant at 25 mg/mL concentration. Inhibition was observed by crystal violet staining.

Reproduced from Sharma, D., Saharan, B.S., Chauhan, N., Procha, S., Lal, S. Isolation and functional characterization of novel biosurfactant produced by Enterococcus faecium. Springerplus, 2015:1–14. https://doi.org/10.1186/2193-1801-4-4, Copyright Springer (2015).

described by Coutte et al. [82]. Whereas fengycin was produced from the *B. subtilis* strain Bs2504, with the process defined by Ongena et al. [83]. Most lipopeptide biosurfactants are inactive against bacteria growth but inhibit the growth of Gram-positive bacteria such as *C. divergens* and *Brochothrix thermosphacta*. Surfactin and fengycin were tested against *C. divergens* and *B. thermosphacta*; they showed MIC values of greater than 1000 mg/L while mycosubtilin showed the MIC of 500 and 15.6 mg/L against *C. divergens* and *B. thermosphacta*, respectively. The antimicrobial ability of lipopeptide biosurfactants makes them good biofilm disrupters against food pathogen and Gram-positive bacteria [44].

5 Future directions and conclusion

Biosurfactants are produced from microorganisms and are having good surface-modifying properties similar to surfactants. These have amphiphilic nature and thus can interact with both oil and aqueous surface to reduce the ST. Biosurfactant has been researcher's interest in recent years due to their extensive ability to inhibit

microbes. Production of biosurfactants has reached the optimum stage commercially. Biosurfactants are extensively employed in perfumery, medicine, cosmetics, and other fields owing to their nontoxic nature and biodegradability. These are also commonly utilized in the food industry because of their excellent antibiofilm ability. Different types of biosurfactants include lipopeptides, glycolipids, phospholipids, neutral lipids, and polymeric compounds. Among them, serrawettin, surfactin, subtilisin (a protein-digesting enzyme), polymyxins, viscosin, amphisin, and putisolvin are common antimicrobial biosurfactant used to protect food against food pathogens with good potency without causing significant alternation in palatability. Thus, these open new and alternate ways to combat food pathogens.

Declarations and authors' contributions

G.L.K. and S.C. conceived and designed the concept. A.T.M. carried out all the literature searches. A.T. M., G.L.K., and S.C. carried out the writing of the manuscript. All the authors have read and approved the final version of the manuscript.

Ethics approval

Not applicable.

Competing interests

The authors(s) confirm that this chapter content has no conflicts of interest.

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References

[1] S.B. Oyeleke, M.A. Goro, O. Oyewole, K.S. Okeke, T.T. Ayisha, D. Mohammaed, Potentials of Yeast Isolates for Biosurfactant Production, 2017.

[2] A.P.P. Santos, M.D.S. Silva, E.V.L. Costa, R.D. Rufino, V.A. dos Santos, C.S. Ramos, et al., Production and characterization of a biosurfactant produced by Streptomyces sp. DPUA 1559 isolated from lichens of the Amazon region, Braz. J. Med. Biol. Res. (2017) 51.

[3] A.P. Karlapudi, T.C. Venkateswarulu, J. Tammineedi, L. Kanumuri, B.K. Ravuru, V.R. Dirisala, et al., Role of biosurfactants in bioremediation of oil pollution-a review, Petroleum 4 (2018) 241–249, https://doi.org/10.1016/j.petlm.2018.03.007.

[4] M. Ohadi, H. Forootanfar, G. Dehghannoudeh, T. Eslaminejad, A. Ameri, M. Shakibaie, et al., Antimicrobial, anti-biofilm, and anti-proliferative activities of lipopeptide biosurfactant produced by *Acinetobacter junii* B6, Microb. Pathog. 138 (2020) 103806, https://doi.org/10.1016/j.micpath.2019.103806.

[5] B.G. Ribeiro, J.M.C. Guerra, L.A. Sarubbo, Biosurfactants: production and application prospects in the food industry, Biotechnol. Prog. 36 (2020) e3030, https://doi.org/10.1002/btpr.3030.

[6] K. Arima, A. Kakinuma, G. Tamura, Surfactin, a crystalline peptidelipid surfactant produced by Bacillus subtilis: isolation, characterization and its inhibition of fibrin clot formation, Biochem. Biophys. Res. Commun. 31 (1968) 488–494, https://doi.org/10.1016/0006-291X(68)90503-2.

[7] Y. Chen, S.A. Liu, H. Mou, Y. Ma, M. Li, X. Hu, Characterization of Lipopeptide biosurfactants produced by Bacillus licheniformis MB01 from marine sediments, Front. Microbiol. 8 (2017) 871, https://doi.org/10.3389/fmicb.2017.00871.

[8] T. Matsuyama, T. Murakami, M. Fujita, S. Fujita, I. Yano, Extracellular vesicle formation and biosurfactant production by Serratia marcescens, Microbiology 132 (1986) 865–875.

[9] J. Sung, S. Ahn, N. Kim, S.-K. Jeong, J. Kim, J. Chung, et al., Purification, molecular cloning, and biochemical characterization of Subtilisin JB1 from a newly isolated Bacillus subtilis JB1, Appl. Biochem. Biotechnol. 162 (2009) 900–911, https://doi.org/10.1007/s12010-009-8830-6.

[10] B. Veeraraghavan, J. Seni, P. Baliga, Polymyxins: antimicrobial susceptibility concerns and therapeutic options, Indian J. Med. Microbiol. 29 (2011) 230–242, https://doi.org/10.4103/0255-0857.83905.

[11] P. Das, S. Mukherjee, R. Sen, Genetic regulations of the biosynthesis of microbial surfactants: an overview, Biotechnol. Genet. Eng. Rev. 25 (2008) 165–186, https://doi.org/10.5661/bger-25-165.

[12] J.D. Desai, I.M. Banat, Microbial production of surfactants and their commercial potential, Microbiol. Mol. Biol. Rev. 61 (1997) 47–64, https://doi.org/10.1128/mmbr.61.1.47-64.1997.

[13] M.-S. Yeh, Y.-H. Wei, J.-S. Chang, Enhanced production of surfactin from Bacillussubtilis by addition of solid carriers, Biotechnol. Prog. 21 (2005) 1329–1334, https://doi.org/10.1021/bp050040c.

[14] C.-C. Chang, W.-C. Chen, T.-F. Ho, H.-S. Wu, Y.-H. Wei, Development of natural anti-tumor drugs by microorganisms, J. Biosci. Bioeng. 111 (2011) 501–511, https://doi.org/10.1016/j.jbiosc.2010.12.026.

[15] E.B. Kurbanoglu, M. Ozdal, O.G. Ozdal, O.F. Algur, Enhanced production of prodigiosin by Serratia marcescens MO-1 using ram horn peptone, Braz. J. Microbiol. 46 (2015) 631–637, https://doi.org/10.1590/S1517-838246246220131143.

[16] Y.-H. Wei, L.-F. Wang, J.-S. Changy, S.-S. Kung, Identification of induced acidification in iron-enriched cultures of Bacillus subtilis during biosurfactant fermentation, J. Biosci. Bioeng. 96 (2003) 174–178, https://doi.org/10.1263/jbb.96.174.

[17] Y.-H. Wei, L.-F. Wang, J.-S. Chang, Optimizing iron supplement strategies for enhanced surfactin production with Bacillus subtilis, Biotechnol. Prog. 20 (2004) 979–983, https://doi.org/10.1021/bp030051a.

[18] D. Kitamoto, S. Akiba, C. Hioki, T. Tabuchi, Extracellular accumulation of mannosylerythritol lipids by a strain of Candida antarctica, Agric. Biol. Chem. 54 (1990) 31–36, https://doi.org/10.1080/00021369.1990.10869918.

[19] D.G. Cooper, D.A. Paddock, Production of a biosurfactant from Torulopsis bombicola, Appl. Environ. Microbiol. 47 (1984) 173–176, https://doi.org/10.1128/aem.47.1.173-176.1984.
[20] A. Daverey, K. Pakshirajan, Sophorolipids from Candida bombicola using mixed hydrophilic substrates: production, purification and characterization, Colloids Surf. B Biointerfaces 79 (2010) 246–253, https://doi.org/10.1016/j.colsurfb.2010.04.002.
[21] B. Boothroyd, J.A. Thorn, R.H. Haskins, Biochemistry of the ustilaginales. XII. Characterization of extracellular glycolipids produced by Ustilago sp, Can. J. Biochem. Physiol. 34 (1) (1956) 10–14.
[22] T. Morita, T. Fukuoka, T. Imura, N. Hirose, D. Kitamoto, Isolation and screening of glycolipid biosurfactant producers from sugarcane, Biosci. Biotechnol. Biochem. 76 (2012) 1788–1791, https://doi.org/10.1271/bbb.120251.
[23] A. Kretschmer, H. Bock, F. Wagner, Chemical and physical characterization of interfacial-active lipids from Rhodococcus erythropolis grown on n-alkanes, Appl. Environ. Microbiol. 44 (1982) 864–870, https://doi.org/10.1128/aem.44.4.864-870.1982.
[24] D.G. Cooper, J.E. Zajic, D.F. Gerson, Production of surface-active lipids by Corynebacterium lepus, Appl. Environ. Microbiol. 37 (1979) 4–10, https://doi.org/10.1128/aem.37.1.4-10.1979.
[25] M.C. Cirigliano, G.M. Carman, Purification and characterization of liposan, a bioemulsifier from Candida lipolytica, Appl. Environ. Microbiol. 50 (1985) 846–850, https://doi.org/10.1128/aem.50.4.846-850.1985.
[26] Y. Shabtai, D.L. Gutnick, Exocellular esterase and emulsan release from the cell surface of Acinetobacter calcoaceticus, J. Bacteriol. 161 (1985) 1176–1181, https://doi.org/10.1128/jb.161.3.1176-1181.1985.
[27] O. Käppeli, P. Walther, M. Mueller, A. Fiechter, Structure of the cell surface of the yeast Candida tropicalis and its relation to hydrocarbon transport, Arch. Microbiol. 138 (1984) 279–282, https://doi.org/10.1007/BF00410890.
[28] K.C. Marshall, R. Stout, R. Mitchell, Mechanism of the initial events in the sorption of marine bacteria to surfaces, Microbiology 68 (1970) 337–348.
[29] S.S. Branda, Å. Vik, L. Friedman, R. Kolter, Biofilms: the matrix revisited, Trends Microbiol. 13 (2005) 20–26, https://doi.org/10.1016/j.tim.2004.11.006.
[30] H.-C. Flemming, The perfect slime, Colloids Surf. B Biointerfaces 86 (2011) 251–259, https://doi.org/10.1016/j.colsurfb.2011.04.025.
[31] R. Mishra, A.K. Panda, S. De Mandal, M. Shakeel, S.S. Bisht, J. Khan, Natural antibiofilm agents: strategies to control biofilm-forming pathogens, Front. Microbiol. 11 (2020) 566325, https://doi.org/10.3389/fmicb.2020.566325.
[32] J.W. Costerton, P.S. Stewart, E.P. Greenberg, Bacterial biofilms: a common cause of persistent infections, Science 284 (1999) 1318–1322, https://doi.org/10.1126/science.284.5418.1318.
[33] L. Hall-Stoodley, J.W. Costerton, P. Stoodley, Bacterial biofilms: from the natural environment to infectious diseases, Nat. Rev. Microbiol. 2 (2004) 95–108, https://doi.org/10.1038/nrmicro821.
[34] S. Abdollahi, Z. Tofighi, T. Babaee, M. Shamsi, G. Rahimzadeh, H. Rezvanifar, et al., Evaluation of anti-oxidant and anti-biofilm activities of biogenic surfactants derived from *Bacillus amyloliquefaciens* and *Pseudomonas aeruginosa*, Iran. J. Pharm. Res. 19 (2020) 115–126, https://doi.org/10.22037/IJPR.2020.1101033.
[35] S. Nahar, M.F.R. Mizan, A.J. Ha, S.-D. Ha, Advances and future prospects of enzyme-based biofilm prevention approaches in the food industry, Compr. Rev. Food Sci. Food Saf. 17 (2018) 1484–1502, https://doi.org/10.1111/1541-4337.12382.

[36] M.S. Bader, M. Loeb, A.A. Brooks, An update on the management of urinary tract infections in the era of antimicrobial resistance, Postgrad. Med. 129 (2017) 242–258, https://doi.org/10.1080/00325481.2017.1246055.
[37] L. Dieltjens, K. Appermans, M. Lissens, B. Lories, W. Kim, E.V. Van der Eycken, et al., Inhibiting bacterial cooperation is an evolutionarily robust anti-biofilm strategy, Nat. Commun. 11 (2020) 107, https://doi.org/10.1038/s41467-019-13660-x.
[38] Q. Shu, T. Wei, H. Lu, Y. Niu, Q. Chen, Mannosylerythritol lipids: dual inhibitory modes against Staphylococcus aureus through membrane-mediated apoptosis and biofilm disruption, Appl. Microbiol. Biotechnol. 104 (2020) 5053–5064, https://doi.org/10.1007/s00253-020-10561-8.
[39] M.A. Diaz De Rienzo, I. Banat, B. Dolman, J. Winterburn, P. Martin, Sophorolipid biosurfactants: possible uses as antibacterial and antibiofilm agent, N. Biotechnol. 32 (2015), https://doi.org/10.1016/j.nbt.2015.02.009.
[40] L.V. de Araujo, C.R. Guimarães, R.L. da Silva Marquita, S. VMJ, M.P. de Souza, M. Nitschke, et al., Rhamnolipid and surfactin: anti-adhesion/antibiofilm and antimicrobial effects, Food Control 63 (2016) 171–178, https://doi.org/10.1016/j.foodcont.2015.11.036.
[41] R. Srikanth, S.F. Banu, J. Sowndarya, J.H.S. Parveen, D. Rubini, A. Wilson, et al., Biosurfactant synergized with marine bacterial DNase disrupts polymicrobial biofilms, Folia Microbiol. (Praha) 66 (2021) 831–842, https://doi.org/10.1007/s12223-021-00876-y.
[42] A.N. Madhu, P. Giribhattanavar, M.S. Narayan, S.G. Prapulla, Probiotic lactic acid bacterium from kanjika as a potential source of vitamin B12: evidence from LC-MS, immunological and microbiological techniques, Biotechnol. Lett. 32 (2010) 503–506, https://doi.org/10.1007/s10529-009-0176-1.
[43] V.K. Nuthakki, A. Sharma, A. Kumar, S.B. Bharate, Identification of embelin, a 3-undecyl-1,4-benzoquinone from Embelia ribes as a multitargeted anti-Alzheimer agent, Drug Dev. Res. 80 (2019) 655–665, https://doi.org/10.1002/ddr.21544.
[44] K. Kourmentza, X. Gromada, N. Michael, C. Degraeve, G. Vanier, R. Ravallec, et al., Antimicrobial activity of lipopeptide biosurfactants against foodborne pathogen and food spoilage microorganisms and their cytotoxicity, Front. Microbiol. 11 (2021), https://doi.org/10.3389/fmicb.2020.561060.
[45] T. Clements, T. Ndlovu, W. Khan, Broad-spectrum antimicrobial activity of secondary metabolites produced by Serratia marcescens strains, Microbiol. Res. 229 (2019) 126329, https://doi.org/10.1016/j.micres.2019.126329.
[46] N. Geudens, M.N. Nasir, J. Crowet, J.M. Raaijmakers, K. Fehér, T. Coenye, et al., Membrane interactions of natural cyclic lipodepsipeptides of the viscosin group, BBA-Biomembranes (2016), https://doi.org/10.1016/j.bbamem.2016.12.013.
[47] I. Kuiper, E.L. Lagendijk, R. Pickford, J.P. Derrick, G.E.M. Lamers, J.E. Thomas-Oates, et al., Characterization of two *Pseudomonas putida* lipopeptide biosurfactants, putisolvin I and II, which inhibit biofilm formation and break down existing biofilms, Mol. Microbiol. 51 (2004) 97–113, https://doi.org/10.1046/j.1365-2958.2003.03751.x.
[48] L.B. Salam, Detection of biosynthetic genes of microbially-synthesized secondary metabolites in a contaminated tropical agricultural soil, Biologia (2021), https://doi.org/10.1007/s11756-021-00851-5.
[49] R. Rasouli, M. Navidinia, M.S. Ghahfarokhi, R. Vahabpour, S. Adabian, A.R. Baghestani, Antibiofilm activity of cellobiose dehydrogenase enzyme (CDH) isolated from aspergillus Niger on biofilm of clinical Staphylococcus epidermidis and Pseudomonas

aeruginosa isolates, Arch. Clin. Infect. Dis. 15 (2020) 1–7, https://doi.org/10.5812/archcid.90635.Research.

[50] K.M. Saur, O. Brumhard, K. Scholz, H. Hayen, T. Tiso, A pH shift induces high-titer liamocin production in Aureobasidium pullulans, Appl. Microbiol. Biotechnol. 103 (2019) 4741–4752, https://doi.org/10.1007/s00253-019-09677-3.

[51] J. Nashida, N. Nishi, Y. Takahashi, C. Hayashi, M. Igarashi, D. Takahashi, et al., Systematic and stereoselective total synthesis of mannosylerythritol lipids and evaluation of their antibacterial activity, J. Org. Chem. 83 (2018) 7281–7289, https://doi.org/10.1021/acs.joc.8b00032.

[52] Q. Shu, Y. Niu, W. Zhao, Q. Chen, Antibacterial activity and mannosylerythritol lipids against vegetative cells and spores of Bacillus cereus, Food Control 106 (2019) 106711, https://doi.org/10.1016/j.foodcont.2019.106711.

[53] L. Fan, H. Li, Y. Niu, Q. Chen, Characterization and inducing melanoma cell apoptosis activity of mannosylerythritol lipids-a produced from Pseudozyma aphidis, PLoS One 11 (2016) e0148198, https://doi.org/10.1371/journal.pone.0148198.

[54] M. Onghena, T. Geens, E. Goossens, M. Wijnants, Y. Pico, H. Neels, et al., Analytical characterization of mannosylerythritol lipid biosurfactants produced by biosynthesis based on feedstock sources from the agrofood industry, Anal. Bioanal. Chem. 400 (2011) 1263–1275, https://doi.org/10.1007/s00216-011-4741-9.

[55] J.-R. Bai, K. Zhong, Y.-P. Wu, G. Elena, H. Gao, Antibiofilm activity of shikimic acid against *Staphylococcus aureus*, Food Control 95 (2019) 327–333, https://doi.org/10.1016/j.foodcont.2018.08.020.

[56] Y. Qiu, Y. Wu, B. Lu, G. Zhu, T. Gong, R. Wang, et al., Inhibition of methicillin-resistant Staphylococcus aureus (MRSA) biofilm by cationic poly (D, L-lactide-co-glycolide) nanoparticles, Biofouling 36 (2020) 159–168, https://doi.org/10.1080/08927014.2020.1740687.

[57] Q. Shu, H. Lou, T. Wei, X. Zhang, Q. Chen, Synergistic antibacterial and antibiofilm effects of ultrasound and MEL-A against methicillin-resistant *Staphylococcus aureus*, Ultrason. Sonochem. 72 (2021) 105452, https://doi.org/10.1016/j.ultsonch.2020.105452.

[58] V. Shah, G.F. Doncel, T. Seyoum, K.M. Eaton, I. Zalenskaya, R. Hagver, et al., Sophorolipids, microbial glycolipids with anti-human immunodeficiency virus and sperm-immobilizing activities, Antimicrob. Agents Chemother. 49 (2005) 4093–4100, https://doi.org/10.1128/AAC.49.10.4093-4100.2005.

[59] M. Gershman, D. Kennedy, J. Noble-Wang, C. Kim, J. Gullion, M. Kacica, et al., Multistate outbreak of Pseudomonas fluorescens bloodstream infection after exposure to contaminated heparinized saline flush prepared by a compounding pharmacy, Clin. Infect. Dis. 47 (2008) 1372–1379, https://doi.org/10.1086/592968.

[60] A.S. Santos, A.P.W. Sampaio, G.S. Vasquez, L.M. Santa Anna, N. Pereira, D.M.G. Freire, Evaluation of different carbon and nitrogen sources in production of rhamnolipids by a strain of Pseudomonas aeruginosa, Appl. Biochem. Biotechnol. 98 (2002) 1025–1035, https://doi.org/10.1385/ABAB:98-100:1-9:1025.

[61] L.M. Santa Anna, G.V. Sebastian, E.P. Menezes, T.L.M. Alves, A.S. Santos, N. Pereira, et al., Production of biosurfactants from *Pseudomonas aeruginosa* PA1 isolated in oil environments, Braz. J. Chem. Eng. 19 (2002).

[62] M. Nitschke, G.M. Pastore, Production and properties of a surfactant obtained from Bacillus subtilis grown on cassava wastewater, Bioresour. Technol. 97 (2006) 336–341, https://doi.org/10.1016/j.biortech.2005.02.044.

[63] I. Wiegand, K. Hilpert, R.E.W. Hancock, Agar and broth dilution methods to determine the minimal inhibitory concentration (MIC) of antimicrobial substances, Nat. Protoc. 3 (2008) 163–175, https://doi.org/10.1038/nprot.2007.521.

[64] F.J. de Freitas, E.A. Vieira, M. Nitschke, The antibacterial activity of rhamnolipid biosurfactant is pH dependent, Food Res. Int. 116 (2019) 737–744, https://doi.org/10.1016/j.foodres.2018.09.005.

[65] M. Zezzi do Valle Gomes, M. Nitschke, Evaluation of rhamnolipid and surfactin to reduce the adhesion and remove biofilms of individual and mixed cultures of food pathogenic bacteria, Food Control 25 (2012) 441–447, https://doi.org/10.1016/j.foodcont.2011.11.025.

[66] H. Güdücüoğlu, M. Berktaş, H. Bozkurt, K. Aygül, Y. Bayram, S. Gülmez, et al., Biofilm and fluconazole in Candida strains The biofilm production (SLIME) and fluconazole sensitivity of the Candida strains isolated from the mouth flora of newborns and adults, East. J. Med. 10 (2005) 5–9.

[67] S. Farisa Banu, S. Thamotharan, S. Gowrishankar, S. Karutha Pandian, P. Nithyanand, Marine bacterial DNase curtails virulence and disrupts biofilms of Candida albicans and non-albicans Candida species, Biofouling 35 (2019) 975–985, https://doi.org/10.1080/08927014.2019.1680650.

[68] D. Rubini, S. Farisa Banu, B.N. Veda Hari, D. Ramya Devi, S. Gowrishankar, S. Karutha Pandian, et al., Chitosan extracted from marine biowaste mitigates staphyloxanthin production and biofilms of methicillin- resistant Staphylococcus aureus, Food Chem. Toxicol. 118 (2018) 733–744, https://doi.org/10.1016/j.fct.2018.06.017.

[69] I.A.S.V. Packiavathy, P. Sasikumar, S.K. Pandian, R.A. Veera, Prevention of quorum-sensing-mediated biofilm development and virulence factors production in Vibrio spp. by curcumin, Appl. Microbiol. Biotechnol. 97 (2013) 10177–10187, https://doi.org/10.1007/s00253-013-4704-5.

[70] P. Das, S. Mukherjee, R. Sen, Antimicrobial potential of a lipopeptide biosurfactant derived from a marine Bacillus circulans, J. Appl. Microbiol. 104 (2008) 1675–1684, https://doi.org/10.1111/j.1365-2672.2007.03701.x.

[71] Z.K. Bagewadi, U.M. Muddapur, S.S. Madiwal, S.I. Mulla, A. Khan, Biochemical and enzyme inhibitory attributes of methanolic leaf extract of Datura inoxia Mill, Environ. Sustain. 2 (2019) 75–87, https://doi.org/10.1007/s42398-019-00052-6.

[72] S.M. Gomha, S.M. Riyadh, E.A. Mahmmoud, M.M. Elaasser, Synthesis and anticancer activity of arylazothiazoles and 1,3,4-thiadiazoles using chitosan-grafted-poly(4-vinylpyridine) as a novel copolymer basic catalyst, Chem. Heterocycl. Compd. 51 (2015) 1030–1038, https://doi.org/10.1007/s10593-016-1815-9.

[73] P. Vijayan, A. Godavarthi, P. Vijayan, C. Raghu, G. Ashok, S.A. Dhanaraj, et al., Antiviral activity of medicinal plants of Nilgiris, Indian J. Med. Res. 120 (2004) 24–29.

[74] E.A.E. Sakr, H.A.E. Ahmed, F.A.A. Abo Saif, Characterization of low-cost glycolipoprotein biosurfactant produced by Lactobacillus plantarum 60 FHE isolated from cheese samples using food wastes through response surface methodology and its potential as antimicrobial, antiviral, and anticancer activities, Int. J. Biol. Macromol. 170 (2021) 94–106, https://doi.org/10.1016/j.ijbiomac.2020.12.140.

[75] P. Gołek, W. Bednarski, B. Brzozowski, B. Dziuba, The obtaining and properties of biosurfactants synthesized by bacteria of the genus Lactobacillus, Ann. Microbiol. 59 (2009) 119–126, https://doi.org/10.1007/BF03175608.

[76] G. Reid, C. Heinemann, M. Velraeds, H.C. van der Mei, H.J. Busscher, C. Heinemann, Biosurfactants produced by *Lactobacillus*, Methods Enzymol. 310 (1999) 426–433, https://doi.org/10.1016/S0076-6879(99)10033-8.

[77] A.N. Madhu, S.G. Prapulla, Evaluation and functional characterization of a biosurfactant produced by Lactobacillus plantarum CFR 2194, Appl. Biochem. Biotechnol. 172 (2014) 1777–1789, https://doi.org/10.1007/s12010-013-0649-5.

[78] D. Sharma, B.S. Saharan, N. Chauhan, S. Procha, S. Lal, Isolation and functional characterization of novel biosurfactant produced by *Enterococcus faecium*, Springerplus (2015) 1–14, https://doi.org/10.1186/2193-1801-4-4.

[79] R. Thavasi, S. Jayalakshmi, T. Balasubramanian, I.M. Banat, Production and characterization of a glycolipid biosurfactant from Bacillus megaterium using economically cheaper sources, World J. Microbiol. Biotechnol. 24 (2008) 917–925, https://doi.org/10.1007/s11274-007-9609-y.

[80] R.M. Jain, K. Mody, N. Joshi, A. Mishra, B. Jha, Production and structural characterization of biosurfactant produced by an alkaliphilic bacterium, Klebsiella sp.: evaluation of different carbon sources, Colloids Surf. B Biointerfaces 108 (2013) 199–204, https://doi.org/10.1016/j.colsurfb.2013.03.002.

[81] E. Beltran-Gracia, G. Macedo-Raygoza, J. Villafaña-Rojas, A. Martinez-Rodriguez, Y. Y. Chavez-Castrillon, F.M. Espinosa-Escalante, et al., Production of lipopeptides by fermentation processes: endophytic bacteria, fermentation strategies and easy methods for bacterial selection, in: Fermentation Processes, InTech, 2017, https://doi.org/10.5772/64236.

[82] F. Coutte, V. Leclère, M. Béchet, J.S. Guez, D. Lecouturier, M. Chollet-Imbert, et al., Effect of pps disruption and constitutive expression of srfA on surfactin productivity, spreading and antagonistic properties of Bacillus subtilis 168 derivatives, J. Appl. Microbiol. 109 (2010) 480–491, https://doi.org/10.1111/j.1365-2672.2010.04683.x.

[83] M. Ongena, E. Jourdan, A. Adam, M. Paquot, A. Brans, B. Joris, et al., Surfactin and fengycin lipopeptides of Bacillus subtilis as elicitors of induced systemic resistance in plants, Environ. Microbiol. 9 (2007) 1084–1090, https://doi.org/10.1111/j.1462-2920.2006.01202.x.

CHAPTER

6 Application of biosurfactant as versatile additives or ingredients of food processing

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1 Introduction

1.1 Surfactant

Surfactants are amphiphilic molecules that lower the surface tension at interfaces. Interface can be of solid-liquid, liquid-liquid, or vapor-liquid. The term surfactant refers to as SURFace-ACTive AgeNT. They are the primary components of cleansing agents. They replace bulk molecules at interface, and hence, the free energy of the system decreases. Surfactants are prepared industrially as they have various functions as detergents, emulsifiers, flocculating agent, foaming agents, adhesives, or wetting agents [1]. They can be adsorbed at air-water interface. On addition of surfactant, the surface tension decreases; hence, the spreading and wetting properties increase. Thus, surfactants help in spreading the dye evenly and hence used in textile industry. Surfactants contain both hydrophilic (water loving) and hydrophobic (water hating) portion. They have hydrophobic tail made up of hydrocarbons, which surrounds the dirt or soil by orienting at the surface, and a hydrophilic head surrounding water orienting toward the solution. If surfactant is hydrophilic and at the same time lipophilic (soluble in oils), then it acts as an emulsifying or foaming agent. But if it is less hydrophilic and more lipophilic, then it act as demulsifier or defoaming agent. Surfactants can also be used as food additives, germicides, fungicides, or insecticides. They are used to produce aerosols, corrosion inhibition, and in ore refining. Surfactants are of four types, which are non-ionic, cationic, anionic, or amphoteric based on their composition and polarity [2].

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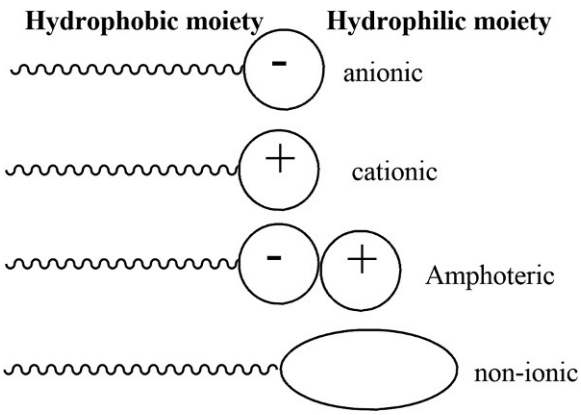


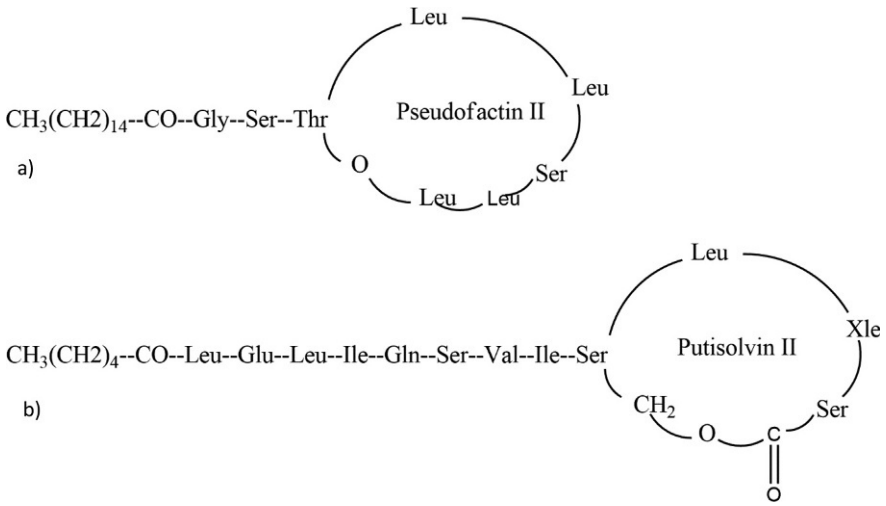
A general structure of a surfactant according to composition of hydrophilic moieties [2].

Surfactants are used widely in oil removal applications. They are also used as antimicrobial agents (generally cationic surfactants serve this purpose). The main function of surfactant is to interact and kill pathogens and also modulate immune response. Thus, the surfactant should enhance solubility, reduce surface tension, and also lower the critical micelle concentration (CMC). Surface tension is the ability to bring a molecule from the solution to the interface and is measured as surface free energy per unit area. Surfactant reduces surface tension, and hence, less work, i.e., energy is required. The lower the CMC, lesser is the surfactant required to reduce the surface tension as a result more efficient surfactant. CMC is the minimum concentration of surfactant in solution above which aggregates of surfactant called as micelles start forming, and it is affected by changes in temperature, ionic strength, and pH. Surfactant is selected on the basis of its charge, physiochemical behavior, adsorption, and solubility. They are used to save energy and to minimize the cost [3]. They have been now used for bioremediation of polluted areas and also for metal removal. Since they can mobilize contaminants, thus they are used for soil scrubbing (for remediation of excavated soil).

1.2 Biosurfactant

Surfactants can be either natural or synthetic. Natural surfactants are also called as biosurfactants and are amphiphilic biological compounds produced by living cells such as a variety of microorganisms such as bacteria, yeast, or fungi. These surfactants are widely under investigation as they fulfill the criteria of green chemistry and are biodegradable, less toxic, renewable, easy large-scale production, function at critical conditions. They are eco-friendly and have low toxicity to any kind of ecosystem. The physical properties of biosurfactants are similar to those of the synthetic surfactants. Biosurfactants, green surfactants, represent a wide variety of

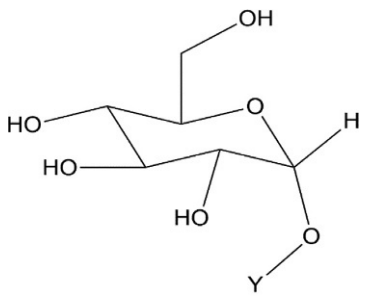
biomolecules such as fatty acids, N-acyl amino acids, mono- and diglycerides, phospholipids, and glycolipids. Biosurfactants being a greener alternative to synthetic surfactant are used in bioremediation to address environmental pollution and can treat pollutants such as industrial waste, polyaromatic hydrocarbons, petroleum products, pesticides, or inorganic heavy metals. Biosurfactants can be produced by microbial oxidation or co-oxidation of sparingly water-soluble carbon substrates. They can be extracellular or associated to the cell wall not excreted in culture broth. They can be produced either by resting or living cells. Various substrates for biosurfactant production are oil waste, dairy whey, starchy waste, molasses, animal fat, soap stock, or by biotransformation (microbial fermentation) [4].



Some examples of biosurfactant (A) Pseudofactin II, (B) Putisolvin II [5].

Biosurfactant production is affected by wide range of factors, which include environmental and nutritional factors. Environmental factors affecting biosurfactants production are pH (pH 8, natural pH of sea water is best suited), temperature (the ambient temperature of 25–30°C is the best) and aeration and agitation (they are best produced by aeration and agitation of oxygen from gas to aqueous phase to maintain 50% saturation of dissolved oxygen (DO) at the flow rate of 1 vvm). Nutritional factors affecting biosurfactants production are the metal ion concentration, incubation time, or C/N (carbon/nitrogen) ratio and C/P (carbon/phosphorous) ratio. It is also dependent on cultivation strategies, which include shake flask, batch, fed-batch, continuous, and integrated microbial/enzymatic processes. Biosurfactants on production were conventionally recovered by solvent extraction, crystallization, precipitation, and centrifugation. But with advancement in technology, modern methods for recovery of biosurfactants include ultrafiltration, adsorption/desorption on polystyrene resins and ion-exchange chromatography [3]. Strain improvement for higher yields is done by cellular and metabolic engineering approaches.

Biosurfactants being cost-effective and environment-friendly are used in various fields such as bioremediation, cosmetics, food, agriculture, oil production, health care, and personal care products [6]. Biosurfactants due to their unique functional properties can be used as for microbial-enhanced oil recovery, in biomedical field (possess antiviral, antifungal, and antibacterial activity), for metal remediation (removal of heavy metals), for soil scrubbing/washing and for bioremediation of crude oil, in agriculture field (as have antimicrobial and insecticidal activity), for laundry industry (with vegetable oil have good emulsion formation ability), in food industry (used as food additives—emulsifiers and are used for fat stabilization in oily food, increasing solubility in instant soups and drinks, for defoaming in production of sugar, for starch complexation in potatoes, and for protective coatings for fruits and vegetables), in cosmetic industry (due to skin-friendly properties), for reduction of CO_2 emissions, and can also be used in ceramics, metal processing, or fixing dyes [2].



Advantages and disadvantages of biosurfactants.

Biosurfactants are better than synthetic surfactants as they are biodegradable, less toxic, biocompatible, lower CMC, raw material is widely available and is cheap, environmental control active at extreme conditions, and are more specific in their action. Some of the disadvantages associated with the biosurfactants are low productivity (for microbes), strong foam formation, expensive large-scale production (in some cases), difficulty in obtaining pure substances.

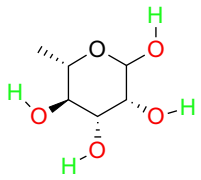
Advantages	Disadvantages
• Low production costs • Biodegradability • In- situ applications • Diversity	• High waste volume • High recovery costs

2 Types of biosurfactants

Major four types of biosurfactants are reported. These are listed below.

2.1 Glycolipids

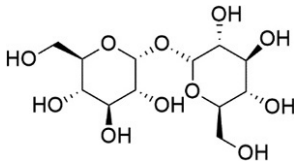
Mainly biosurfactants are glycolipids. These are carbohydrates in which the sugar part is attached with either to a hydroxyaliphatic acid or to a long-chain aliphatic acid. The attachment is through an ester group. Generally mono-, di-, tri-, and tetrasacchrarides such as glucose, mannose, galactose are the constituents of the carbohydrate part of biosurfactants. Trehalolipids, rhamnolipids, and sophorolipids are some of the well-known biosurfactants [7,8].



Structure of glycolipid.

2.1.1 Rhamnolipids

These glycolipids form the widely examined class of biosurfactants. These are characterized by having one or more rhamnose group attached to one or more molecules of decanoic acid. *P. aeruginosa*, which is known to be a Gram-negative, rod-shaped bacteria, is the parent microorganism for this class of biosurfactants [9].

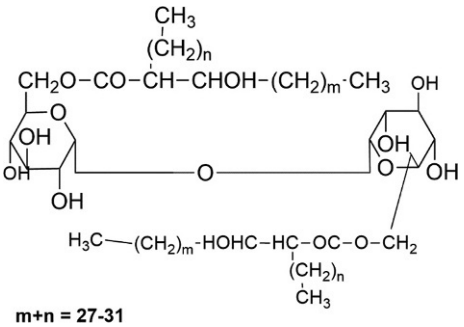


Structure of rhamnose group.

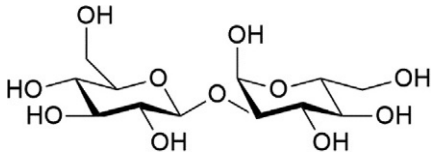
2.1.2 Trehalolipids

These glycolipids are best known to lower down interfacial tension between water and the other interacting surface involved. These are produced by several microorganisms related to mycolates group of bacteria. Some of them are Nocordia, Corynebacterium, Mycobacterium, Arthrobacter, and Gordonia. The structure consists of trehalose hydrophilic group [10]. Here a disaccharide is attached to mycolic acid at C-6 position, and this in turn is linked with various kinds of Mycobacterium. Hence, these are also known as 6,6′-dimycolates of trehalose. Long-chain alpha and beta

hydroxy unsaturated fatty acids with 60–90 carbon atoms are termed as Mycolic acids. Trehalose lipids are able to shift size, number of carbon molecules existing, structure of mycolic acid, and extent of unsaturation tree hollows lipids that are obtained from Arthrobacter and *Rhodococcus erythropolis* did use the interfacial tension in culture stock.



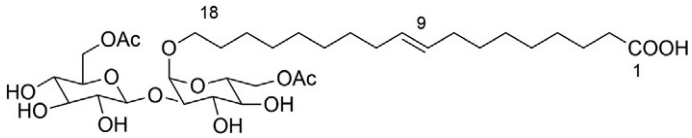
Structure of trehalose group.



Structure of trehalolipids.

2.1.3 Sophorolipids

This class of glycolipids are produced by a nonpathogenic yeast species. In these kinds of glycolipids, a sophorose group, which is a disaccharide having two glucose residues, is attached to a long-chain fatty acid through a glycosidic linkage. These are generally a combination of 6–9 different hydrophobic sophorolipids. Lactone form of these lipids is also used for various applications [11].



Structure of sophorose group.

2.2 Lipoproteins and lipopeptides

These are surface-active agents in which a lipid is attached to a polypeptide chain. These are produced by wide range of microorganisms such as bacteria, fungi, and yeast [8]. Some examples are gramicidin, a decapeptide antibiotic, and polymxin. A lipopeptide

antibiotic shows excellent surface-active properties. These are produced by *Bacillus bravus and Bacillus polymyxa*, respectively [7]. A lipopeptide is almost 35% by weight protein and remaining is acid. *Bacillus subtilis* reduces lipopeptide known as sub tillison, which is an excellent biosurfactant. Its structure consists of a long chain of seven amino acids whose one terminal is linked to a hydroxyl group of beta hydroxyl fatty acid and other terminal to the carbaoxyl functionality of beta hydroxyl fatty acid [12–15]. A hydrocarbon-emulsifying lipopeptide is produced by *Candida petrophilum*. Besides having structural similarities to *B. subtilis*, it contains few more fatty acids. Lipopeptides isolated from *Corynebacterium lepus* can lessen the surface tension of water from 72 to 52 mN/m [16].

2.3 Phospholipids

A group of lipids having two fatty acids, a glycerol unit, and a phosphate group, esterified to an organic molecule is termed as phospholipid. The phosphate group consisting portion is usually polar and hydrophilic. n-Alkanes are used widely as a substrate for growing large varieties of phospholipids and fatty acids through several bacteria and yeast species. Microbial oxidation is the underlying phenomenon that is responsible for the production of fatty-acid-type surfactants. More complex fatty acids, having alkyl side chains and OH groups are also produced along with straight-chain structures. Example includes corynomycolic acid [17]. They have found greater applications in medical field. Phospholipids are produced by almost all microorganisms, but the amount in which they are produced is not appreciably enough and hence has restricted their usage. Scientists have found out methods to induce the extracellular excretion of phospholipids in various microorganisms. For instance, on treatment with penicillin, production of extracellular phospholipids by Corynebacterium alkanolyticum was raised by an order [18,19]. Fatty acids and neutraipids are obtained as extracellular products by various microbial cells. Alcohols, esters, carboxylic acids have shown surface activities to greater extent. Hydrocarbons are often employed for growing microorganisms that produce extracellular fatty acids or neutral lipids. Hence, they become significant for hydrocarbon emulsification. For instance, various hydroxy fatty acids are potential surface-active agents than simple fatty acids (Cooper in press).

2.4 Polymeric surfactants

As the name suggests, these are polymers with surface-active properties. These macromolecules contain both hydrophobic and hydrophilic groups in the main structure. Researchers recently have shown considerable interest in studying the properties of high-molecular-weight surfactants. Emulsan, alasan, and liposan are well-known examples. Emulsan is used majorly for emulsification of hydrocarbons in water even when the concentration is as low as 0.001%–0.01% [20], while liposan synthesized by *Candida lipolytica* is a water-soluble emusifier [21].

3 Properties of biosurfactants

The unique and exceptional characteristics of biosurfactants make them stand out when compared with their synthetic equivalents. Surface activity, ionic strength and temperature, resistance to broad range of pH, low toxicity, antimicrobial activity, biodegradability, and ability to form and break emulsions are some of the distinct features of the surface-active agents responsible for their commercial applications in various disciplines [22].

3.1 Surface and interfacial interaction

Biosurfactants are well known for showcasing their exceptional surface-active properties. Lowering surface tension as well as interfacial tension between different phases (air-liquid, liquid-liquid, solid-liquid) is a remarkable feature of these microbial surfactants. According to Mulligan, an ideal biosurfactant is able to lower down surface tension of water from 72 to 35 mN/m and water/hexadecane interfacial tension from 40 to 1 mN/m. *B. subtilis*, a Gram-positive bacteria, produces a unique surfactant known as surfactin, which can reduce water surface tension from 72 to 25 mN/m and interfacial tension of water/hexadecane to even less than 1 mN/m [14,15]. Biosurfactants are able to achieve these extraordinary results through possessing a very low CMC and form stable micelles when compared with their chemically synthesized counterparts [7].

3.2 Biodegradability

The ability of any material, more specifically organic substance, to get decomposed or broken down into smaller molecules on interactions with biological elements is termed as biodegradability. These biological elements can be microorganisms such as fungi, bacteria. Under the action of these microbes, any material can be disintegrated and gets absorbed into the surrounding environment. Due to increasing environmental concerns, sustainable, eco-friendly products have become the need of the hour. Therefore, synthetic products are being replaced by more efficient and effective bio-originated counterparts [23]. Biosurfactants due to their microbial origin are more biocompatible and consequently show better biodegradability than synthetic equivalents [24]. As a result of their negligible toxicity to the environment, these have found applications such as bioremediation and biosorption [25,26].

3.3 Temperature and pH resistance

Numerous biosurfactants have been produced by various microorganisms. One such is surfactants production from extremophiles. Extremophiles are microorganisms known for their ability to survive under extremely harsh environmental conditions

such as temperature and pH levels. Scientists have incorporated their ability to produce surfactants that would be resistant toward harsh temperature conditions and pH levels. For instance, according to the study of McInerney and others, *Bacillus licheniformis* named bacteria produces a surface-active agent termed as lichenysin. This surface-active agent can survive at high NaCl and Ca concentrations (50 and 25 g/L), broad pH range (4.5–9) and is also insistent to temperatures up to 50°C. Due to their peculiar capability, these have found greater applications in various industrial processes where high temperature and wider pH levels are necessary conditions [23]. Another example showcasing this outlandish feature of biosurfactants produced this way could be of a surfactant generated by *Arthrobacter protophormiae*. The biosurfactant was found to be pH-stable (2 – 12) and has thermal stability (30–100°C) [27].

3.4 Low toxicity

Biosurfactants are usually considered as less or nontoxic surface-active molecules despite of having low data availability related to their toxicity. Hence, they have found applications in various fields such as pharmaceutical industries, food and cosmetic industries. Poremba and team reported that when four synthetic surfactants and two microbial surfactants were compared, it was found that biosurfactants were decomposed at a faster rate. It was also reported that a higher EC50 value that is effective concentration to decrease 50% of population was shown by biosurfactants than synthetic molecules. Marlon A-350, a synthetic surfactant and a biosurfactant derived from *P. aeruginosa*, was analyzed regarding toxicity and mutagenic properties. Flasz and his colleagues found that the chemically synthesized surfactants showed high toxicity and mutagenic properties, whereas biologically derived surfactant came out as nontoxic and nonmutagenic. Acute and chronic toxicity of three microbial surfactants (Emulsan, Rhamnolipids, PES-51) was correlated with three synthetic surfactants (Triton X-100, Corexit-9500, PES-61). The study revealed that synthetic surfactant tritonX-100 showed highest toxicity, whereas PES-51 and Emulsan were least toxic [28].

3.5 Emulsions building and breaking

Emulsions are heterogeneous systems having one or more immiscible liquids. Small droplets of the immiscible liquid are entirely dispersed in another medium. Thus emulsions comprise one dispersed phase and one continuous phase. Due to the nonmutual solubility, emulsions generally form unstable systems. Their stability can be enhanced by adding additives such as surfactants. Synthetic as well as natural surfactants are usually employed for this purpose in various food formulations, cosmetic products. Surfactants of microbial origin are getting more attention due to low toxicity and other benefits. Biosurfactants, besides acting as emulsifiers, could also destabilize the system, hence acting as de-emulsifiers [29]. For instance, *C. lipolytica* named microorganism generates a surface active agent called liposan. Liposan is a

water-soluble emulsifier that has found its utilization in food and cosmetic industries. It covers small droplets of edible oils through a coating formation around them and consequently emulsifies and stabilizes the system [30].

4 Production of biosurfactants

Biosurfactants are natural surfactants, and they are synthesized by microorganisms. During the growth of a variety of microorganisms such as fungi, bacteria, and yeast, biosurfactants can be produced. Because of the presence of rigid cell wall, more amount of biosurfactant is obtained from fungi as compared with bacteria. They can also be produced by the resting cell system. It is always not possible to synthesize them by the main metabolic pathways, as they are considered to be secondary metabolites [31,32]. Their production is favored by the addition of inorganic substrates such as fats, oils, solid or liquid hydrocarbons to water, which helps in sustaining the microorganisms in poor-nutrient or contaminated environments.

During the production of biosurfactants along with the producer strain, culture conditions also play a major role. These parameters are not responsible for the amount of biosurfactant produced but also define the type of biosurfactant produced. The type of microorganism used influences the structure of the produced biosurfactant. From the same carbon source, it is possible to synthesize different type of biosurfactant, which is dependent on the microorganism used. Both the hydrophilic and hydrophobic portions of biosurfactant are produced by the same substrate, but the difference is only in the biosynthetic pathway of substrate metabolism. For example, biosurfactant produced from *Candida guilliermondii* in particular ambient conditions is used as an emulsifying agent in the formation of mayonnaise [33].

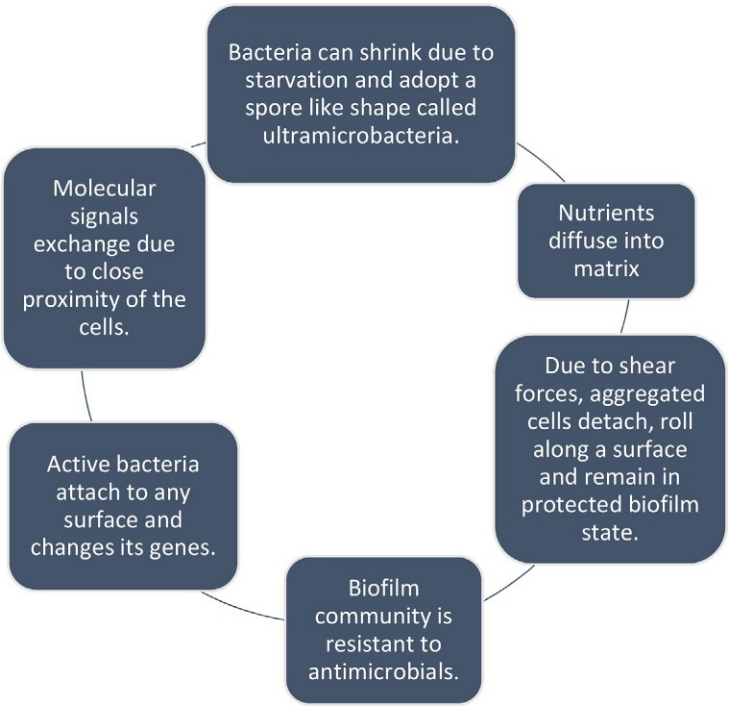
Biosurfactant production is also affected by the operating conditions of the fermentor. Hence, there is a need to control these cultural conditions such as pH, dilution rate, etc. To increase the yield, fermentation conditions should be optimized. Also, recovery of biosurfactant is a crucial part, which depends mainly on its solubility (in water and organic solvents), ionic charge, and also on its location in the cell (extracellular, intracellular, or cell-bound) [3]. The techniques used for the recovery of the biosurfactants are solvent extraction, precipitation, and chromatographic purification. For the isolation of low-molecular-weight biosurfactant, precipitation step is optional, but according to the their hydrophilic-lipophilic balance (HLB) value, different organic solvents are used [34]. The cost of biosurfactant production is very high as compared with their synthetic counterparts because of the downstream processes. Thus, the biosurfactant produced from the inexpensive and waste substrate is highly advantageous due to reduction in the production cost.

5 Application of biosurfactants in food industry

After the development of new food additives, which have thickening and stabilizing abilities such as in gelling emulsifier, the discovery of new microbial polysaccharides and surfactants became popular. To reduce the dependency on the emulsifiers obtained from plants, which were produced by genetically modified crops, and to avail other properties such as antioxidant, antimicrobial, antiadhesive, and as antibiofilm capacities, biosurfactant amphiphilic molecules became common to be used in new and advanced formulations in the food and other industries. Biosurfactant being less toxic can be used in food as it doesn't have any harm on the human body. But then also the toxicity of a particular biosurfactant should be checked before using it.

5.1 Biosurfactants as antibiofilm agents

Biosurfactants are used in the food sanitation and safety of the food items and to make the food hygienic. Sanitization targets the pathogenic microorganisms, which develop biofilms on the food surfaces or on the food. A wide variety of food-borne pathogens such as *Listeria monocytogenes, Salmonella, Escherichia coli* O157:H7 have the ability to attach to food processing surface. Biofilm is a thin layer of microorganisms that adhere to the surface of the food and to each other with the help of extracellularly secreted polymers. They include mixture of microbial extracellular material produced at surface, bacteria or some other material, which is trapped within the resulting matrix. Microorganisms survive due to their ability of biofilm production as it protects them from environmental challenges and also provides nutrients to microorganisms. It protects them from the antibiotic, toxins and prevents them from predation. It helps in upkeep of the extracellular enzyme potential. Antibiofilm agents are used as bacterial biofilms present on food surfaces and are a source of contamination in food industry, and it is associated with spoilage of food and transmission of diseases. Microorganisms are able to sense their own cell density, communicate, and thus, they can behave as a population by cell-to-cell signaling. This phenomenon is known as quorum-sensing. It is responsible for biofilm formation and is dependent on the maturation stage of development, nutrition, and environment of microorganisms. So, for the quality safe products, biofilm production should be minimized by surface sanitation as bacterial biofilms can develop on food processing surfaces, utensils, and tools. Biofilms on various surfaces can lead to the food deterioration, oxidation of metals in storing tanks and food carriage pipelines. It also leads to spread of food-borne pathogens and in decrease in heat transfer to the vessels.



Conceptualization of biofilm development and its dynamic behavior.

Microbial biofilm has a multicellular phenotype with a specific bacterial physiology and is thus different from planktonic bacteria. These are a major cause of incurable and chronic health infections, transmission of communicable diseases, for undesirable marine water and environment. It has also led to the increase in the failure rate of anti-infective therapy. It is found that the biofilm formed of single species is very rare to be found in natural environment, whereas biofilms composed due to microorganisms with complex multispecies are generally found. Thus, biosurfactants that are potentially effective against microbial biofilms, which cause dissemination of diseases, are usually used [5].

Earlier, disinfection, cleaning, and sanitization were done for the removal of biofilm as it prevents the microbial population to aggregate on the surface. Disinfection leads to the increased risk of disinfectant concentration in the production area. Thus, futuristic strategy for the biofilm removal is the surface accustoming by biosurfactants (which were primarily used for the removal of hydrocarbon). It is the safest technique as other physical, chemical, or biological methods are hazardous to food consumers or have certain other drawbacks. Thus, various aspects of biofilm monitoring, control, inhibition, dispersal, and measurement are gaining lot of significance these days. They can disperse microbial biofilms and hence are used in

sanitation as well as clinical sectors. Their dispersal properties compete with the conventional inhibitory agents against yeast and bacterial biofilms. Thus, they are used as microbial dispersal agents in new generations and for removal or suppression of microbial biofilms.

Biofilm formation is very complicated, multi-staged process and is governed by the type and number of microorganisms. The chemical environment is also important for the growth of biofilm. The stages involved in biofilm formation are adhesion, formation of microcolony, and biofilm matriculation. Last step is biofilm dislodging. Biosurfactants can regulate the expression of bacterial genes. For example, biosurfactant obtained from *Lactobacillus reuteri* can regulate the expression of fructosyltransferase and glucosyltransferase genes, which are used for the *Streptococcus mutans* initial adhesion to the surface of tooth. They can create cavities in the center of the biofilm and thus affect the biofilm architecture. Microbial biofilms are affected by following six mechanisms: (a) by loss of intracellular content due to altered cell permeability, (b) membrane degradation causes the disruption of cell membrane structure, (c) biofilm creation decreases due to interference in quorum-sensing, (d) cell surface hydrophobicity decreases and hence cell adhesion changes, (e) energy needs are restricted due to hindrance in electron transport chain, and (f) can also distort the protein structure, which can lead to the disruption of microbial biofilm.

Biosurfactants are used for the pretreatment of food processing surfaces as biofilms developed by pathogens are resistant to traditional antibiotics. Pretreatment of stainless-steel surface is done by biosurfactant produced from *Pseudomonas fluorescens* reduced *L. monocytogenes* population. Also, on preconditioning of stainless-steel surface by *Lactobacillus helveticus* derived surfactant, adhesion of different *Listeria* sp. has been found. Various probiotic bacteria and microbial stains such as rhamnolipids and surfactin have antibiofilm potential against food-associated pathogens of metal surface and polypropylene. They can also delay bacterial adhesion of non-growing and growing cells. Plant biosurfactants and rhamnolipids inhibit complex biofilm production and also enhance the antibiotics microbial inhibitors [35]. Biosurfactant obtained from *Lactobacillus paracasei* exhibits antibiofilm potential against food-borne pathogens such as *S. aureus, Streptococcus agalactiae, Canadian albicans*, and *Staphylococcus epidermidis*.

Biofilm is removed by biosurfactants by tampering biofilm matrix, which in turn decreases the interfacial tension between biofilm and the solid surface. Moreover, the biosurfactants help in the dissolution of exopolysaccharides matrix as they can act as wetting agent. Thus, they can be used for the prevention of biofilm in industrial food processing. Some biosurfactants developed from different microbial stains are used to control biofilm development on medical surfaces and food. They can be used to prevent or delay the microbiological colonization of industrial plant surfaces such as for preparation of foodstuff. They have wide range of applications in sanitation of food surfaces such as polystyrene surfaces, food transporting pipes and lines, stainless steel and heat exchangers. Thus, have application in meat, dairy, and other food processing sectors. Even though these are not hazardous but still their toxicity has to be determined for better and wide range of applications in food processing.

In future, biosurfactants can become an integral part of hygiene products or can be used for food stuff sanitation such as chewing gums. They can be used for reduction and prevention of oral cavities, i.e., the main surface exposed to food in the human body [2].

5.2 Biosurfactants as emulsifiers

Emulsions represent a heterogeneous system of two or more immiscible or non-blendable liquids. Dispersed phase and dispersion medium are the two components of this colloidal system. Dispersed phase is made up of colloids, and it is scattered while dispersion medium represents a system where colloidal particles are dispersed. Therefore the phenomenon of dispersing one liquid into another immiscible liquid is referred as emulsification. Various forces of attraction are involved during this process, namely electrostatic forces of attraction, hydrogen bonding, hydrophilic forces, and many more. Majorly water and oil are the two immiscible liquids forming either oil in water abbreviated as O/W or water in oil abbreviated as W/O emulsions [36,37]. Due to their tendency to rapidly segregate back into their constituent layers, emulsions are thermodynamically unstable [38]. Difference in densities between the aqueous and oil phase makes the contact between the two surfaces unfavorable [39,40].

An emulsifying agent is a surface-active compound incorporated to resolve the issue of immiscibility. An emulsifier or emulsifying agent is usually a substance that allows mixing of two or more non-blendable liquids to form a stable emulsion. Emulsifiers can be classified on the basis of relative spatial distribution of aqueous phase and oils phase or arrangement/nature of emulsifiers. Some examples from daily life include:

Product	Type of emulsion they form
Homogenized milk	O/W
Mayonnaise	O/W
Butter	W/O
Margarine	O/W
Salad dressings	O/W
Ice cream	O/W

The stability of an emulsion is governed by various factors such as particle size, density between the dispersed phase and dispersion medium, particle distribution [41]. One important parameter to identify the kind of emulsion formed by a surfactant is the hydrophilic-lipophilic balance abbreviated as HLB. It is defined as the weight percentage of hydrophilic moities to the weight percentage of lipophilic moities in the given molecule. HLB varies generally from 1 to 20. Surfactants having HLB values between 3 and 9 stabilize the W/O type of emulsion while those having

values greater than 9 promote formation of O/W type emulsions. For example, rhamnolipids are known for their high HLB values and hence stabilize O/W emulsions [36].

Various undesirable phenomena are known to destabilize emulsions. Some of them are flocculation, sedimentation, creaming, coalescence, Ostwald ripening, and phase inversion.

Numerous compounds have been utilized as food emulsifiers to stabilize complex food mixtures. Food emulsion becomes unstable due to various composite ingredients and physical factors such as pH, salt concentration, and/or processing complications. Therefore food emulsifiers have been incorporated in various food formulations since ages. The technique has been proved to be beneficial in many ways. Some of them are listed below.

(1) Food emulsifiers improve consistency, thickness, texture, and appearance of the food formulations.
(2) They promote longer shelf life.
(3) They enhance sensory attributes.

Similarly, while manufacturing cereal-based products, these have shown great benefits such as finer grain size, improved resistance to alteration in flour dough, and greater cell size of the dough.

Synthetic emulsifiers have been exploited for their broad range of applications and unique emulsifying properties for many years. But the demand for sustainable natural products is increasing among the consumers. Therefore there is a need to substitute these synthetic emulsifiers with the natural and environmentally friendly counterparts [2]. Various health authorities and food organizations are consistently emphasizing on regulatory rules to limit the use of synthetically derived products. Numerous natural food emulsifiers are preferentially accepted by various food industries. These not only reduce the pressure on synthetically derived processes and products but also lower down the pollution and risks associated with improper disposal of harsh chemicals and toxic pollutants used or generated during various manufacturing processes.

Natural emulsifiers are either derived from animal cells or taken from plants. Some examples of natural emulsifiers taken from plants are honey, xanthan gum, mustard, agar agar, and so on. Natural emulsifiers derived from animal source are usually seminatural fats. Proteins, phospholipids, polysaccharides, saponins can act as natural emulsifiers.

(1) Lecithin is usually found in egg yolks and soy. It is used for its impressive emulsifying properties in mayonnaise and sauces preparations.
(2) Esters of monoglycerides of fatty acids are made from glycerol, natural fats, and an acid, for instance, tartaric acid, citric acid. The fat part is often extracted from vegetable fats.
(3) Mono- and diglycerides of fatty acids consist of natural fats and glycerol. These can be from animal or plant source [2].

Plant gums are prone to price fluctuations, owing to supply shortages resulting in low crop production and increased energy consumption. Also, some of the emulsifiers are sourced from genetically modified crops (especially soybean) posing fewer challenges for food firms looking to lower down their reliance on food additives obtained from them [2]. These characteristics together with the probability of utilizing emulsifiers that give other beneficial aspects (such as antimicrobial activity) led to a thorough search for better natural emulsifiers with more or less similar or better properties.

Biosurfactants have been studied for their excellent emulsifying activity and have found applications in oil industries and various bioremediation techniques as natural emulsifying agent. *B. subtilis and P. aeruginosa* produce lipopeptide biosurfactants and rhamnolipids, respectively. These natural surface-active agents are potential emulsifiers employed in food industries and are capable of forming stable emulsions with coconut oil, linseed oil, and soybean oil. Glycolipids biosurfactants produced by various yeast and bacteria are stable within broad pH and salinity levels. A glycolipid was produced by *Kluyveromyces marxianus* FRR1586 as a surface-active compound. It was able to form stable oil in water emulsions with coconut oil, soybean oil, and linseed oil at various salinities (2–50 g NaCl/L) and pH levels ($3-11$). Consequently, this surfactant is widely used as a natural emulsifier in mayonnaise, cakes, ice creams, crackers, and sausages. Glycolipid biosurfactants generated by the yeast species *Geotrichum* CLOA 40, *Trichosporon loubieri* CLV20, and *Trichosporon Montevideense* CLOA 70 are potent emulsifiers for vegetable oils. These are stable even at high salt concentrations and high temperatures [42].

Besides acting as excellent emulsifying agents, these surface-active molecules have various other utilizations and applications in the food industry.

5.2.1 Ice cream preparation

For instance, in the preparation of whipped cream and ice creams, biosurfactants are used to control the agglomeration of fat globules, facilitate fat protein association, give smoother texture to the mixture, reduce particle size of ice crystals and air cells, reduce whipping time, enhance fat dispersion, alter meltdown time, incorporate dryness to the final product, and impart resistance to shrinkage. This in turn increases the shelf life of the product and enhances texture. In addition to this, they improve consistency and texture of dairy products.

5.2.2 Bakery products preparation

Biosurfactants play various significant roles in the baking industry as well. They alter the rheological characteristics of the formulas. Some are listed as follows:

(1) Emulsifiers increase the volume of the cake.
(2) They facilitate air incorporation.
(3) They enhance the moisture retention capacity of the batter.
(4) They improve handling properties of the dough.
(5) They promote the homogeneous mixture of ingredients.

A lipopeptide biosurfactant generated by *B. subtilis* was compared with a commercially available synthetic surfactant call soy lecithin. It was concluded that the inclusion of biosurfactant in the dough preparation improved the gas retention capability, the texture of the dough, which resulted in a uniform and cohesive dough formation. Additionally, the quality, consistency, and hence the volume of batter were increased. In order to preserve the batter for later requirements, dough formulations are often preserved by freezing it at lower temperatures. Another method of preservation includes fermentation using varying amounts of yeast. Incorporation of biosurfactants to dough not only reduces the amount of yeast required but also shortens the time needed for fermentation.

Commercially, glycol monostearate is used as an emulsifier in cookie dough for fire softer cookies. Lipopeptide biosurfactants from *B. subtilis* were employed as a replacement to this synthetic emulsifier. Reduction in dough hardness and increased uniformity of the batter were some of the resulting features observed after the biosurfactants addition. Hence, softer cookies were obtained [43].

5.2.3 Muffin and bread preparation

An evaluation done by introducing another biosurfactants obtained from seawater actinobacteria named *Nesterenkonia alba*. This biosurfactant was included in the dough of muffin in order to assess its effect on the refinement of the muffin. Reduced chewiness and gumminess, decreased hardness, improved uniformity of the dough, enhanced cohesiveness, increased softness were some of the advantages observed after this experiment. In a similar way, sophorolipids were used in the dough formulation for bread preparation. It was concluded that the wheat flour dough became more stable. Its texture, consistency, and volume were also improved. During fermentation, sophorolipids when used in bread preparation improved gas retention capacity of the dough, and consequently, the volume of the bread was increased.

5.3 Biosurfactants as antimicrobial agents in food industries

Food safety and hygiene are the two most important parameters to ensure safe consumption of consumers and to protect them from food-borne diseases. According to the World Health Organization, over 200 diseases are caused by pathogenic microorganisms. These disease-causing microorganisms are responsible for 600 million cases of food-borne diseases and 4,20,000 individuals lose their lives each year worldwide. Food-borne disease as defined by WHO is any ailment that is caused by consumption of microbial or chemically contaminated food. *E. coli, L. monocytogenes, Yersinia* spp. are some of the most common disease-causing microbes responsible for various fatal food-borne diseases [44].

Biosurfactants due to their antimicrobial activities are potential surface-active molecules to be used for food safety and hygiene. Numerous biosurfactants are proved to be effective antimicrobial agents against several species of fungi, algae, viruses, and bacteria [45]. Some examples showcasing the unique characteristics against disease-causing pathogens are as follows:

(1) *P. aeruginosa* produces a type of rhamnolipid that displays excellent antimicrobial activity against filamentous fungi and numerous yeast species [46].
(2) Sophorolipids isolation from *S. bombicola* and mannosylerythriol lipids synthesized by *Candida antarctica* act as antimicrobial agents against several pathogenic microbes [47].
(3) *B. polymyxa and Bacillus brevis* produce antimicrobial lipopeptide polymyxins and gramicin S, respectively, which are capable of destroying pathogens and are widely used in food industries.
(4) Other lipopeptide surfactants produced by *B. subtilis*, namely surfactin, bacillomycins, and fengycin exhibit antimicrobial activities [47].

A wide range of microorganisms living in different environments are known to produce numerous biosurfactants. The diversity is so great that innumerable microbes still remain undiscovered and consequently innumerable biosurfactants isolated from them remain unexplored. One such interesting example is shown by a biosurfactant produced by lactic acid bacteria abbreviated as LAB. This bacterium is found in milk and other dairy products. Biosurfactants generated by LAB show structural similarities with glycoproteins, glycolipids, lipopeptides, and lipoproteins. These biosurfactants show excellent antimicrobial activity and are effective against wide range of pathogens including yeasts, bacteria, and fungi. *Streptococcus thermoplulus A, Lactobacillus agilis, Lactobacillus lactis* are some microorganisms to be named that are known for their antimicrobial properties [48]. Lactic acid bacteria are a known probiotic and are present in dairy products. The surfactant molecules produced by the bacteria are considered safe and have been used in food industries for many years. These molecules are potential food decontaminators when directly employed or can be utilized for cleaning and sanitizing surfaces, instruments, and tanks where food ingredients and formulation are stored.

Sophorolipids as explained earlier contain a fatty acid chain (having 16–18 carbon atoms) linked to sophorose group along with varying degrees of unsaturation. These majorly exist in two forms, namely free acid and lactonic form. These two forms show different physicochemical properties when subjected to similar experimental conditions. An experiment was conducted by Zhang and colleagues to study the action of sophorolipids against *E. coli* in the absence and presence of alcohol, preferentially ethanol. A comparison between the free acid form and lactonic form was made. While the free acid form of sophorolipids showed weaker antimicrobial activity against *E. coli*, the results shown by lactonic sophorolipids were impressive. With 20% ethanol, at a concentration of 10 g/L, lactonic form was capable of reducing *E. coli* population to an indistinguishable stage within the time of 2 h. Biosurfactants because of their diverse applications are also employed in various agricultural processes to protect crop plants from pathogenic microbes. For instance, *Nicotiana glutinosa* is a commercially important tobacco plant hybrid. It is easily infected by tobacco mosaic virus and potato virus. Therefore in order to protect the crops and improve quality and yield, plant leaves are treated with emulsions formed by rhamnolipids [7]. Di-rhamnolipids show excellent antifungal activity as demonstrated by

Rodrigues and team. Two fungal contaminants *Aspergillus carbonarins and Aspergillus niger* were found to be responsible for spoilage if crops. The mycotoxins produced by these fungi cause mutagenic, carcinogenic, immunosuppressive deformities even at low concentration intake [46].

5.4 Biosurfactants as antioxidant agent

Antioxidants are the compounds that impede the oxidation reactions (it promotes chain reactions and forms free radicals) of microorganisms cells and which eventually lead to the permanent damage to these cells. These are widely used as food additives to guard the food from deterioration, to prevent the spoilage of food, and also to avoid lipid oxidation, thus increasing the shelf life of the food products. Two main factors responsible for the oxidation of food are exposure to sunlight and air (oxygen). Thus, to prevent oxidation of food, general methods are to store them in the dark or to seal the containers. Some foods are also coated with the layer of wax such as cucumbers [49]. Oxidation can also occur in refrigerated or frozen food [50]. Since oxygen is important for the photosynthesis, thus, storing food in anaerobic conditions makes it rancid (unpleasant flavors) and also the color deterioration takes place. Thus, packaging of food such as fruits and vegetables requires around 8% of oxygen atmosphere. Thus, antioxidants are essential class of food preservatives.

Some of the natural antioxidants are tocopherols (E306), ascorbic acid (AA, E300), or synthetic antioxidants are *tert*. Butylhydroquinone (TBHQ) or propyl gallate (PG, E310) [51]. Due to the toxicity caused by synthetic antioxidants, it became important to search for easily available and beneficial natural antioxidants. Thus, biosurfactants have potential to fulfill this purpose as they have significant antioxidant activity. An example of versatile biosurfactant is mannosylerythritol lipids (MELs) as they have adaptable interfacial and biochemical properties. These have highest antioxidant and protective cell effects and are thus used in antiaging skin care ingredients. Another biosurfactant obtained from *B. subtilis*, RW-I, which is a lipopeptide, is used as an alternative natural antioxidant as it has the good antioxidant capacity to scavenge free radicals. Its antioxidant activity is based on the chelation of iron ions, its redox power and also on the sequestration of DPPH [52] Earlier, it was found that the polysaccharide emulsifier obtained from Klebsiella also has the capacity to inhibit the autooxidation of soybean oil. This polysaccharide was developed in France to be used as a source of rhamnose for the manufacture of a flavor precursor, furaneol. Sequestering of DPPH free radicals is done by the biosurfactants, which are obtained from the strains of *Lactobacillus casei*. Biosurfactants work better if their concentration is increased.

Antioxidant activity of the biosurfactants is considered a very beneficial property and has a wide range of applications in food industry as foods in which antioxidants are present help in combating degenerative diseases, to increase the shelf life of the food as it retards or halts the oxidation reactions taking place. They also reduce the risk of heart disease. Nowadays, most of the people are concerned with eating healthier foods; thus, food formulations with these properties are highly preferred [2].

Seeing the advantage of antioxidant agents in food industry, there are many tests that can be used to determine the antioxidant activity of the biosurfactants. Depending on the method that is employed and its result, a biosurfactant can be more effective at reducing complexes or at sequestering free radicals. The biosurfactant has high antioxidant activity because of the presence of the unsaturated fatty acids. Hence, the chemical structure of biosurfactants is the characteristic property by which they possess these desirable properties.

Antioxidant activity of biosurfactants has a linear relation with the concentration of biosurfactant. Thus, on increasing the concentration, antioxidant activity of biosurfactant increases. On performing three different antioxidant tests of the biosurfactant obtained from *C. utilis*, it was found that it has greater capacity to reduce the complex than its radical sequestration capacity. At a concertation of 5 mg/mL, its values are higher than 70%. On 2,2-diphenyl-1-picrylhydrazyl (DPPH) test, it is found that the biosurfactant obtained from the species of Lactobacillus has 74%–77% radical-sequestering activity at the same concentration of 5 mg/mL [31,32]. Antioxidant properties of the biosurfactants obtained from the Bacillus strains are determined by DPPH and hydroxyl radical scavenging assay or by the Ferric reducing activity [53]. To measure the DPPH-scavenging abilities of the biosurfactant, 1 mL of biosurfactant solution was dissolved in 0.2 mM of freshly prepared DPPH solution (1 mL), and it was left in dark place for around 30 min. The sample that is referred to as blank consists of deionized water. To calculate the percentage of scavenged DPPH radical, value obtained for sample divided by the value for the blank solution is subtracted from 1 and is multiplied by 100. Butylated hydroxytoluene (BHT) is used to compare the reducing power of the biosurfactants. Similarly, to measure the hydroxyl scavenging ability 1 mL of 2.5 mM 1,10-Phenanthroline, 1 mL of 2.5 mM of $FeSO_4$ solution, 1 mL of sample solution, and 1 mL of PBS (pH 7.4) were mixed thoroughly. The reaction was then initiated by the addition of 1 mL of 20 mM H_2O_2 solution and incubating at 37°C for 90 min. The control sample consists of deionized water. To measure the hydroxyl radical activity, value of sample subtracted by blank sample value was divided by the value of control subtracted from blank sample value and multiplied by 100 [54].

The total antioxidant activity (TAC) of the biosurfactants is expressed in relation to the standard ascorbic acid solution with concentration of 1000 μg/mL whose antioxidant activity is considered to be equal to 100%. Thus, the biosurfactant obtained from *Saccharomyces cerevisiae* URM 6670 has potential antioxidant properties at concentrations above 5000 μg/mL and is used as reducing agent in various food formulations such as ascorbic acid [32].

6 Future aspects

Due to the antiadhesive, antibiofilm, and various other properties, biosurfactants are used in many fields of food industry. Also, the combination of properties such as emulsifying agents, antimicrobial and antiadhesive properties has increased the

use of biosurfactants as a multipurpose ingredient or additive in the food. Information regarding their toxicity and high production cost are the major cause for their limited use in food industry. But the problem of high production cost is solved by the use of agroindustrial wastes. They can reduce the waste treatment expense as well as they reduce the production costs. These are the new alternative for food and other related industries as they not only valorize their waste but they are also becoming the producer of biosurfactant. More research is needed toward the ways to calibrate the toxicity of the biosurfactants. Moreover, the biosurfactants that are obtained from the GRAS microorganisms such as lactobacilli and yeasts can have wide range of applications in food and medicine industries. There is a need to study about more and more natural surfactants to detect their structure, properties, and very importantly the toxicological aspects in order to check the safety of these compounds to be utilized in food industry. To develop the sustainable society for the harmony of the environment. Moreover, with the increase in awareness for the clean environment, the introduction of green technology in all fields of industry has become a most important challenge. Due to the technological and social background, utilization of biosurfactant that is eco-friendly and highly functional has become more and more popular.

There is a need to accelerate the knowledge to enhance the applications of biosurfactant in various industries. It is also possible to screen and isolate the biosurfactant-producing microorganisms from hydrocarbon-contaminated soil. About 44 isolates were able to use the waste lubricating oil as substrate but only 10% of them were satisfactorily able to produce the biosurfactants. *Bacillus* sp. strains generally produce lipopeptides, and *Rhodococcus* sp. strains produce trehalose glycolipids. This is the best approach to produce new biosurfactants, which are used for the water and soil bioremediations.

In general, most the discovered biosurfactants include rhamnolipids. Thus, there is a need to investigate other biosurfactants, which may also have these potential applications. Biosurfactants can be used for the in situ remediation and hence the production of the biosurfactants can also be in situ. If this is made possible, then it would be highly advantageous with regard to the technique and cost. Moreover, it could become ecologically acceptable also as this would reduce the labor work and also the transportation required. Since the bioavailability of the contaminant is an important factor for the biodegradation. Thus, the role of biosurfactant production in the subsurface is not yet discovered as it could very effectively enhance the process of natural attenuation. Since, biosurfactant can act as the carrier of the contaminants in the subsurface, thus, there is need to study this area specifically [26]. Only two to three component systems are till now studied to gain basic understanding. Till now, only few studies are done regarding the mixed inorganic and organic contamination. More research is needed for the study of the more complex situations, particularly for the remediation of metal contamination by the biosurfactants. There is need of more information regarding the interaction of biosurfactant and contaminants. The relationship of biosurfactant structure and for the removal of contamination and also for the soil scale-up and reduction of the cost for the ex situ production and also to understand the factors influencing bioremediation of compounds are done

by the increased bioavailability. The economics of the biosurfactant utilization need to be improved. For example, ultrafiltration technique is used for the recovery and subsequent reuse of the biosurfactants. Hence, they decrease the amount of the biosurfactant required. It will very difficult to remove the metals from the biosurfactants, and this is evident from the studies of the effects of soil components such as hydroxides, oxides, and organics, on metal desorption by biosurfactants, which have been initiated in sediments, mining residues and soil. There is also a need to predict and model the efficiency of the enhanced biodegradation, flushing, or washing processes with biosurfactants under the various soil and hydrological conditions [26]. It is also necessary to develop high producing strains and also to check for their wide applications. Moreover, we need to develop the biosurfactants that can use waste substrates and also the knowledge of their genetics and metabolism studies. They can fulfill the need of efficient and low-toxicity products for the future generation. They have potential applications in various industries such as pharmaceutical, food, chemical, cosmetic, and many more.

7 Conclusion

Biosurfactants exhibit extraordinary properties that have attracted scientists' attention toward them. From acting as emulsifiers to showing antimicrobial and antioxidant properties, biosurfactants have become the ultimate alternate to their toxic chemically synthesized synthetic counterparts. Inappropriate data related to their toxicity has limited their use in various food processes. Therefore, toxicological evaluation is necessary in order to ensure food safety. Besides this high costs, low yields and undesired product formation are few reasons that claim more research need to be done in this section. Direct incorporation as additives in food formulations and indirect utilization as detergents to sanitize surfaces that come in contact with the food formulations are major applications of biosurfactants. With adequate knowledge and required information, biosurfactants could be the future of surfactants. However, for its effective use in food industry, factors such as fate and transport of contaminants, in situ production, and reduction of production costs have a scope of research.

References

[1] Encyclopedia Britannica, Surfactant, Definition, Properties, Examples, & Facts, 2021, Available at: https://www.britannica.com/science/surfactant. (Accessed 27 November 2021).

[2] J.M. Campos, T.L. Montenegro Stamford, L.A. Sarubbo, J.M. de Luna, R.D. Rufino, I.M. Banat, Microbial biosurfactants as additives for food industries, Biotechnol. Prog. 29 (5) (2013) 1097–1108.

[3] S. De, S. Malik, A. Ghosh, R. Saha, B. Saha, A review on natural surfactants, RSC Adv. 5 (81) (2015) 65757–65767.

[4] J.-W. Liu, K.-H. Wei, S.-W. Xu, et al., Surfactant-enhanced remediation of oil-contaminated soil and groundwater: a review, Sci. Total Environ. (2020), https://doi.org/10.1016/j.scitotenv.2020.144142. Available at:.
[5] I.M. Banat, M.A.D. De Rienzo, G.A. Quinn, Microbial biofilms: biosurfactants as antibiofilm agents, Appl. Microbiol. Biotechnol. 98 (24) (2014) 9915–9929.
[6] W.R. Finnerty, Biosurfactants in environmental biotechnology, Curr. Opin. Biotechnol. 5 (3) (1994) 291–295.
[7] J. Desai, I. Banat, Microbial production of surfactants and their commercial potential, Microbiol. Mol. Biol. Rev. 61 (1) (1997) 47–64.
[8] K. Muthusamy, S. Gopalakrishnan, T.K. Ravi, Biosurfactants: properties, commercial production and application, Curr. Sci. 94 (2008) 736–774.
[9] J.R. Edwards, J.A. Hayashi, Structure of a rhamnolipid from *Pseudomonas aeruginosa*, Arch. Biochem. Biophys. 111 (1965) 415–421.
[10] C. Asselineau, J. Asselineau, Trehalose-containing glycolipids, Prog. Chem. Fats Other Lipids 16 (1978) 59–99.
[11] K. Gautam, V. Tyagi, Microbial surfactants: a review, J. Oleo Sci. 55 (4) (2006) 155–166.
[12] K. Arima, A. Kakinuma, G. Tamura, Surfactin, a crystalline peptidelipid surfactant produced by *Bacillus subtilis*: isolation, characterization and its inhibition of fibrin clot formation, Biochem. Biophys. Res. Commun. 31 (3) (1968) 488–494.
[13] A. Bernheimer, L. Avigad, Nature and properties of a cytolytic agent produced by *Bacillus subtilis*, J. Gen. Microbiol. 61 (3) (1970) 361–369.
[14] D. Cooper, J. Zajic, C. Denis, Surface active properties of a biosurfactant from Corynebacterium lepus, J. Am. Oil Chem. Soc. 58 (1981) 77–80.
[15] D.G. Cooper, C.R. Macdonald, S.J.B. Duff, N. Kosaric, Enhanced production of surfactin from *Bacillus subtilis* by continuous product removal and metal cation additions, Appl. Environ. Microbiol. 42 (1981) 408–412.
[16] D. Cooper, J. Zajic, D. Gerson, Production of surface-active lipids by Corynebacterium lepus, Appl. Environ. Microbiol. 37 (1) (1979) 4–10.
[17] K. Rahman, E. Gakpe, Production, characterization and application of biosurfactants-review, Biotechnology 7 (2) (2008) 360–370.
[18] M. Kikuchi, T. Kanamaru, Y. Nakao, Relation between the extracellular accumulation of L-glutamic acid and the excretion of phospholipids by penicillin-treated corynebacterium alkanolyticum, Agric. Biol. Chem. 37 (10) (1973) 2405–2408.
[19] Y. Nakao, T. Kanamaru, M. Kikuchi, S. Yamatodani, Extracellular accumulation of phospholipids, UDP-N-acetylhexosamine derivatives and L-glutamic acid by penicillin-treated corynebacterium alkanolyticum, Agric. Biol. Chem. 37 (10) (1973) 2399–2404.
[20] A.A.M. Hatha, G. Edward, K.S.M.P. Rahman, Microbial biosurfactants—review, J. Mar. Atmos. Res. 3 (2007) 1–17.
[21] D.G. Cooper, D.A. Paddock, Production of a biosurfactant from Torulopsis bombicola, Appl. Environ. Microbiol. 47 (1984) 173–176.
[22] S. Chakrabarti, Bacterial Biosurfactant: Characterization, Antimicrobial and Metal Remediation Properties, Ph. D. Thesis, National Institute of Technology, 2012.
[23] S.S. Cameotra, R.S. Makkar, Recent applications of biosurfactants as biological and immunological molecules, Curr. Opin. Microbiol. 7 (2004) 262–266.
[24] P.K. Mohan, G. Nakhla, E.K. Yanful, Biokinetics of biodegradability of surfactants under aerobic, anoxic and anaerobic conditions, Water Res. 40 (2006) 533–540.
[25] M. Deleu, M. Paquot, From renewable vegetables resources to microorganisms: new trends in surfactants, C. R. Chim. 7 (2004) 641–646.

[26] C.N. Mulligan, Environmental applications for biosurfactants, Environ. Pollut. 133 (2005) 183–198.
[27] P. Singh, S.S. Cameotra, Potential applications of microbial surfactants in biomedical sciences, Trends Biotechnol. 22 (2004) 142–146.
[28] K.R. Edwards, J.E. Lepo, M.A. Lewis, Toxicity comparison of biosurfactants and synthetic surfactants used in oil spill remediation to two estuarine species, Mar. Pollut. Bull. 46 (10) (2003) 1309–1316.
[29] J. Velikonja, N. Kosaric, Biosurfactants in food applications, in: N. Kosaric, F.V. Sukan (Eds.), Biosurfactants: Production: Properties: Applications, CRC Press, New York, 1993, pp. 419–448. (Chapter 16), ISBN-13: 9780824788117.
[30] M.C. Cirigliano, G.M. Carman, Purification and characterization of liposan, a bioemulsifier from *Candida lipolytica*, Appl. Environ. Microbiol. 50 (1985) 846–850.
[31] B.G. Ribeiro, J.M. Guerra, L.A. Sarubbo, Biosurfactants: production and application prospects in the food industry, Biotechnol. Prog. 36 (5) (2020), 3030.
[32] B.G. Ribeiro, J.M.C. Guerra, L.A. Sarubbo, Potential food application of a biosurfactant produced by *Saccharomyces cerevisiae* URM 6670, Front. Bioeng. Biotechnol. 8 (2020) 434.
[33] I.R.S. da Lira, E.M.S. da Santos, J.C. dos Santos, R.R. da Silva, Y.A. da Silva, I.J.B. Durval, J.M. Guerra, L.A. Sarubbo, J.M. de Luna, Production of biossurfactant by candida guillhermondii and application in a mayonnaise emulsion, Chem. Eng. Trans. 87 (2021) 259–264.
[34] I.M. Banat, A. Franzetti, I. Gandolfi, G. Bestetti, M.G. Martinotti, L. Fracchia, R. Marchant, Microbial biosurfactants production, applications and future potential, Appl. Microbiol. Biotechnol. 87 (2) (2010) 427–444.
[35] K. Paraszkiewicz, M. Moryl, G. Płaza, D. Bhagat, S.K. Satpute, P. Bernat, Surfactants of microbial origin as antibiofilm agents, Int. J. Environ. Health Res. (2019) 1–20.
[36] I. Kralova, J. Sjöblom, Surfactants used in food industry: a review, J. Dispers. Sci. Technol. 30 (2009) 1363–1383.
[37] D.J. McClements, L. Bai, C. Chung, Recent advances in the utilization of natural emulsifiers to form and stabilize emulsions, Annu. Rev. Food Sci. Technol. 8 (2017) 205–236.
[38] S. Ghosh, D. Rousseau, Fat crystals and water-in-oil emulsion stability, Curr. Opin. Colloid Interface Sci. 16 (5) (2011) 421–431.
[39] S. Kerkhofs, H. Lipkens, F. Velghe, P. Verlooy, J.A. Martens, Mayonnaise production in batch and continuous process exploiting magnetohydrodynamic force, J. Food Eng. 106 (1) (2011) 35–39.
[40] Y. Maphosa, V. Jideani, O. Adeyi, Effect of soluble dietary fibres from Bambara groundnut varieties on the stability of orange oil beverage emulsion, Afr. J. Sci. Technol. Innov. Dev. 9 (1) (2016) 69–76.
[41] P. Given, Encapsulation of flavors in emulsions for beverages, Curr. Opin. Colloid Interface Sci. 14 (1) (2009) 43–47.
[42] A.S. Monteiro, M.R. Bonfim, V.S. Domingues, A. Correa Jr., E.P. Siqueira, C.L. Zani, et al., Identification and characterization of bioemulsifier-producing yeasts isolated from effluents of a diary industry, Bioresour. Technol. 101 (2010) 5186–5193, https://doi.org/10.1016/j.biortech.2010.02.041.
[43] R. Zouari, S. Besbes, S. Ellouze-Chaabouni, D. Ghribi-Aydi, Cookies from composite wheat-sesame peels flours: dough quality and effect of *Bacillus subtilis* SPB1 biosurfactant addition, Food Chem. 194 (2016) 758–769, https://doi.org/10.1016/j.foodchem.2015.08.064.

[44] L. Fonteneau, N.J. Da Silva, L. Fabre, Multinational outbreak of travel-related Salmonella Chester infections in Europe, summers 2014 and 2015, Euro Surveill. 22 (2017) 1–11.
[45] L. Rodrigues, I.M. Banat, J. Teixeira, R. Oliveira, Biosurfactants: potential applications in medicine, J. Antimicrob. Chemother. 57 (2006) 609–618, https://doi.org/10.1093/jac/dkl024.
[46] X. Vecino, J.M. Cruz, A.B. Moldes, L.R. Rodrigues, Biosurfactants in cosmetic formulations: trends and challenges, Crit. Rev. Biotechnol. 37 (2017) 911–923, https://doi.org/10.1080/07388551.2016.1269053.
[47] X. Zhang, R.D. Ashby, D.K.Y. Solaiman, Y. Liu, X. Fan, Antimicrobial activity and inactivation mechanism of lactonic and free acid sophorolipids against *Escherichia coli* O157:H7, Biocatal. Agric. Biotechnol. 11 (2017) 176–182, https://doi.org/10.1016/j.bcab.2017.07.002.
[48] I.M.C. Morais, A.L. Cordeiro, G.S. Teixeira, V.S. Domingues, R.M.D. Nardi, A.S. Monteiro, R.J. Alves, E.P. Siqueira, V.L. Santos, Biological and physicochemical properties of biosurfactants produced by *Lactobacillus jensenii* P6A and *Lactobacillus gasseri* P65, Microb. Cell Fact. 16 (2017) 1–15, https://doi.org/10.1186/s12934-017-0769-7.
[49] A. Kader, D. Zagory, E. Kerbel, C. Wang, Modified atmosphere packaging of fruits and vegetables, Crit. Rev. Food Sci. Nutr. 28 (1) (1989) 1–30.
[50] E. Zallen, M. Hitchcock, G. Goertz, Chilled food systems1: effects of chilled holding on quality of beef loaves, J. Am. Diet. Assoc. 67 (6) (1975) 552–557.
[51] F. Iverson, Phenolic antioxidants: health protection branch studies on butylated hydroxyanisole, Cancer Lett. 93 (1) (1995) 49–54.
[52] Inamuddin, M.I. Ahamed, R. Prasad (Eds.), Microbial biosurfactants, in: Environmental and Microbial Biotechnology, 2021.
[53] M. Basit, M.H. Rasool, S.A.R. Naqvi, M. Waseem, B. Aslam, Biosurfactants production potential of native strains of *Bacillus cereus* and their antimicrobial, cytotoxic and antioxidant activities, Pak. J. Pharm. Sci. 31 (2018).
[54] S. Giri, E. Ryu, V. Sukumaran, S. Park, Antioxidant, antibacterial, and anti-adhesive activities of biosurfactants isolated from Bacillus strains, Microb. Pathog. 132 (2019) 66–72.

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CHAPTER

Application of biosurfactant as a stabilizer in aerated foods

7

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1 Introduction

Biosurfactants are complex molecules produced mainly by microorganisms and are integrated into the composition of the cell membrane, or in adverse conditions are excreted to the external environment. They also are part of a complex chain of molecules having active physiological relevance in microorganisms; biosurfactants are utilized in the defense and maintenance of cell homeostasis in circumstances of high stress. Thus, biosurfactants act together with other molecules, performing functions such as motility, adhesion to substrates, protection of the cell membrane against attacks from toxic elements, acting in the cell-cell interaction, and also acting as an energy storage molecule [1].

Biosurfactants are natural surfactants, that is, they are substances that act by decreasing the surface tension of contact between liquids. They are biomolecules composed of two distinct parts and with opposite chemical properties. One part is hydrophobic, composed of a carbon chain that interacts only with nonpolar molecules, and others are hydrophilic or ionic, which only interacts with polar molecules. The chemical interaction of surfactants with other molecules depends on the size of the molecular chain and the affinity with the medium [2]. We can categorize surfactants as follows.

- **Anionic surfactants**: they are the surfactants that when they dissolve in water, dissociate until the carbon chain is negatively charged. These surfactants include sodium dodecyl sulfate and lauryl, which are commonly found in shampoos.

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- **Cationic surfactant**: They are surfactants that cause the carbon chain to become positively charged when dissociated in water. Compounds such as distearyldimethylammonium chloride, diethyl ester dimethyl ammonium chloride, and glycine betaine are good examples of halogenated compounds that have quaternary ammonium in their chemical structure.
- **Amphoteric surfactant**: They are known as quaternary ammonium surfactants, as they have in their structure a –COOH group and a quaternary ammonium group in the same chain. These surfactants are termed amphoteric as they can behave as both acids and bases. And depending on the pH of the solution, these surfactants can behave in a cationic and anionic way just like betaines, for example.
- **Nonionic surfactants**: They are surfactants that do not ionize when dissolved in water because they do not contain charges that impact the reaction medium. Nonylphenols, amides, and ethoxylated fatty acids are examples.
- **Biotechnology:** Using specific microorganisms, then the biosurfactants are produced using a variety of technical methodologies, including aerated bioreactors and genetic engineering. Submerged, solid, and biphasic fermentation are the most researched production systems, with proven outcomes in the manufacturing industry. In addition to manufacturing methodologies, genetic engineering technologies have resulted in more efficient microbe strains and novel biotechnology routes with commercial potential [3].

Microorganisms' exponential growth cycle speeds up the generation of biosurfactants and other biomolecules in bioreactors, saving time and increasing output. Microorganisms are more efficient, prolific, and adaptable as production conditions than animals and plants when it comes to biosurfactants. There are techniques that make it possible to locate specific strains capable of producing biosurfactants. Microorganisms can decrease a huge quantity of biomass in substances of interest, including the decomposition of food goods [4]. As a result, the purpose of this brief remark is to examine the use of biosurfactants to stabilize aerated food systems. As a result, we will discuss food aeration technologies as well as the fundamentals of biosurfactant food stability.

2 Aerated food product technology

Countless existing products use aeration in food, whereby incorporating air, in the form of bubbles, provides different textures and sensations to the products. Examples of aerated liquid products that can be mentioned are whipped cream, beer, carbonated soft drinks, and sparkling wine, while butter, ice cream, bread, cakes, popcorn, cheese, and carbonated chocolate bars are examples of aerated solid products [5].

The texture of food is highly affected by the incorporation of air into its matrix, firmness, making the product smoother and lighter, greater capacity for absorption of sauces due to increased surface area and decreased flavor intensity, in addition to providing changes in appearance, color, sensation, chewability and provide volume

increase. According to some case studies, aeration in food can improve satiety during hunger, reducing calorie consumption, so controlling how air bubbles and their sizes are distributed in the matrix, in addition to the gas phase's spatial dispersion, can improve product quality during the aeration step [6].

The formation of gas bubbles is a process that begins with the nucleation in a highly viscous liquid; producing a gas through fermentation or chemical change, shaking, or beating the liquid; air processing methods for product development include blowing gas through a narrow nozzle or single orifice; and sparging or blowing gas through a porous plate. There are three categories that underlie the injection of gas into food matrices: the first occurs when the liquid is actively forced around the external gas, the second occurs when gas is formed in situ (e.g., biological, or chemical yeast, formation of steam by applying heat and/or pressure reduction, evaporation of CO_2 on pressure release), and when we have a phenomenon where gas is formed through liquid, it is the third foundation [7–11]. By using one or more methods together, numerous products are produced in the food industry. Table 1 shows detailed several properties of these approaches.

Some characteristics of these processes are highlighted, as they have direct impacts on the final product, as in the case of the gas that will be injected, since aerated products may have their stability affected due to this factor. Foams that use N_2 or air bubbles will provide greater firmness to the product than those that consist of CO_2 bubbles, this fact occurs precisely because of diffusion, which is largely determined by the solubility of the gas. Another way to create microbubbles, which are gaining several studies, is through the phenomenon called acoustic cavitation, which in liquids treated to ultrasound we can have the development, growth, and implosive collapse of small bubbles [12,13].

The application of cavitation generated by ultrasound allows changes in proteins, which can favor the formation of bubbles. The level of changes in structure and function created in proteins may be regulated by altering the equipment's time, assuming that applications in the future, such as lipid alterations at gas interfaces in emulsions in a controlled manner, are feasible. However, the collapse of ultrasound-generated bubbles in a cavitation field stands out, resulting in hot areas that might damage the final structure of product [14].

2.1 Chemical aeration

Food aeration has gotten a lot of attention since it plays such an important role in changing the texture of many goods on the market, such as baked bread. It will have a significant influence during baking and on the texture of the finished product in dishes such as bread. This process starts when air enters the mass system in the form of bubbles during dough mixing. When forming the dough, we must remember that the creation of gas caused by the action of yeasts will result in the diffusion and enlargement of previously integrated bubbles, which is the key to this knowledge [15].

Table 1 Applied methods for incorporating air into food products.

Method	Principle	Examples
Beating or shaking	When applying a mechanical force to medium- and low-viscosity liquids to incorporate air	Egg whites, cake batter, milk shakes
Pressure beating	Apply pressure to dissolve the air in the food	Bakery products (syrup, fat mixture or chocolate)
Mixing of doughs or high viscosity pastes	The entrapment of bubbles occurs as the structure of the mass joins and is trapped, preventing the detachment of the bubbles. Particle size affects aeration	Bread
Entrapment of air	Seizure of air between the layers occurs	Pastries, croissants, between pulled strands and candies
Gas injection	Introduction of nitrogen gas, nitrous oxide, and carbon dioxide	Ice cream, soft drinks and instant whipped cream
Fermentation	Production of carbon dioxide by using yeast	wines, bread, and beer
Steam generation	Application of food cooking in a slow to moderate way, where the generation of steam is often accompanied by the thermal expansion of gases in the bubbles and the evaporation of other dissolved components	Crumpets, pikelets, doughnuts, and most unleavened baked goods
Chemical raising agents	Use of a carbonate or bicarbonate to give the dough its spongy appearance	Cakes
Vacuum expansion	Vacuum expansion followed by rapid cooling to define expanded product	Chocolate bars
Rapid dry heating	Rapid heating of fine or small products to induce blisters or slight puffing	Potato crisps or corn flakes
Frying	Use of an oil at high temperatures so that the internal steam is quickly formed in the food, providing the product to puff	Potato crisps and snacks
Expansion extrusion	Provides the product with a super-heated under pressure that suddenly emerges from an extruder, so that the internal moisture immediately vaporizes into steam bubbles	Crisp snacks, cereals, and sugar confectionery
Puffing	In products that have an overheated moisture content, they are subject to a sudden release of pressure	Cereal and snacks

Cooking is the most significant step in the manufacturing of aerated meals, particularly bread, with the application of heat to the dough playing a key role in changing the sensory qualities of foods as well as the destruction of enzymes and microbes. During cooking, the fiber interacts directly with the structural features of the three-dimensional gluten networks, breaking the starch-gluten matrix and allowing neighboring gaseous cells to communicate [15].

It is critical to recognize that understanding the link between rheology and the aeration process is a necessary step. In addition, a variety of approaches have been utilized to track the fermentation process and investigate dough aeration. Mass density has been frequently utilized as a technique to analyze the aeration that happens during mixing and as a measure of the proportion of gas voids in the mass after mixing [16].

Bakery items are crucial in controlling the density or volume of products to produce products with distinctive gastronomic features, but the information required to achieve this control has, in some ways, been accessible to bakers for generations. It's hard to monitor the progressive development of chemical aeration since the initial application of chemicals in bakeries to release carbon dioxide to enhance bread through aeration is unknown. Managing the relative density or volume of bakery items to produce products with specified culinary attributes is an important role in modern production. Bakers, on the other hand, have had means to accomplish this control for ages [17].

Artificial yeast can be used as a substitute or supplement for yeast in gluten-free bread to improve aeration. The usage of artificial yeast to aerate properly, as with biological yeast, will be dependent on the pace of carbon dioxide generation and the dough's ability to hold gas [7]. Chemical yeast agents are rarely employed in traditional gluten-containing bread, although they are utilized in many gluten-free goods besides yeast. Seven out of 11 gluten-free bread brands in Spain included growth agents [18]. Chemical agents are frequently found in gluten-free and wheat bread recipes [18].

The kind, mix, and concentration of powdered yeast should all be chosen carefully based on the qualities of the goods. According to the same authors, there are hardly no studies evaluating the proper use of chemical yeast agents. In the case of chemical aeration, separate additions of acids with sodium bicarbonate or precisely balanced combinations of these components in a conveyor to produce a composite powdered yeast are becoming more prevalent. Fats can be used to coat the acids, delaying the reaction with the bicarbonate until the cake batter is cooked over the fat's melting point [7].

2.2 Biological aeration

A variety of methodologies have been used to assess the fermentation process and aeration of pasta. The dough density is the most generally used metric for determining the proportion of gas voids in the dough after mixing, as well as an instrument for confirming the aeration achieved during mixing. In total, 89%–95% of the gas in

wheat bread is produced by a combination of fermenting yeast, evaporation, and the expansion of the bubble during baking. During mixing, however, the assimilation of gas into the dough is substantially lower. According to the same authors, mechanical inclusion of gas during the beating of the dough generates 37%–73% of the final gas volume [17].

Aeration by whipping gluten-free pasta is a good chance to ameliorate deficits because gluten-free dough is frequently more liquid and stickier than wheat dough. Furthermore, the quantity, as well as the size dispersion of the first nuclei of air created while the kneading process, is directly connected to the pore structure in breadcrumbs, emphasizing the necessity of the mixing stage for general aeration [17].

The interaction of the substrate with a microbe is critical in the kinetics of gas generation during fermentation in biological aeration. Evaluating the modeling will be dependent on how well the type of gluten-free raw material is understood, as this helps to prevent the formation of these bubbles. One must also understand about yeasts or lactobacilli that are involved in fermentation. According to the same author, the presence of monosaccharides in flours, as well as the activity of enzymes, can have a greater impact on the gas produced by the microorganism. With the use of high gas retention and regulated bacterial synthesis of hydrocolloids, it is feasible to increase performance [7,19,20].

The mixing process has a considerable impact on parameters such as the quantity of gas bubbles, their size distribution, and the temperature of the dough, which are all important for all the elements influencing bubble development and stabilization throughout fermentation and cooking [17]. When the temperature was correlated with the mixture and bread density, it was discovered that higher temperatures in the doughs could be achieved without interrupting the mixers, with a positive linear correlation ($R_2 = 0.8771$, $P = 0.0001$) between the volume of bread and the temperature of the dough. According to the same author, the temperature of the dough had a substantial influence on yeast CO_2 synthesis [17].

Even when the dough densities are similar after mixing, the bread usually has a higher water content and density after baking, which could be due to lower yeast activity, but a bubble stabilization caused by lower viscosity during fermentation and baking has also been observed, this might lead to a decrease in bread volume [17]. Due to the quantity of gas and the amount of bubble size that is dispersed in the matrix, experimental research on wheat dough demonstrated that the energy utilized to carry out the combination can impact the porosity of the end product. To further understand the interaction between mixing settings and bread pore structure, it would be interesting to track bubble sizes during the operation. The amount of gas and the distribution of bubble sizes are affected by the mixing energy, which directly influences the porosity of the final product, making monitoring the bubble size throughout processing very interesting to clarify the interrelationship between mixing parameters and the pore structure of bread [21].

In baking, such as producing Chorleywood (CBP)-type bread, the time of dough mixing is critical. This procedure is broken down into three parts, the first of which involves mixing and homogenizing the substances, which usually goes smoothly.

We have the stimulus for the development of bubbles in the aerated dough stage, which will evolve into the final cells of wheat flour. We will have the optimal rheological properties qualities of this material when mechanical activity happens on top of the dough proteins, resulting in CO_2 retention, gas cell stabilization, and dough aeration [22].

2.3 How gas adds value to food products?

Several parameters can be improved by submitting food matrices to the aerating process, including palatability, the sensory issue, where we have the sensation of softness, lightness, crunchiness, etc., in addition to creating new textures and structures; reducing the intensity of the aftertaste; trapping aroma compounds and delivering them to the retronasal olfactory smell, which can increase flavor perception; increased digestibility, and so on [5].

Carbon dioxide, steam, and air are the three most important gases in food aeration. Nitrogen may also be used in specialized situations, such as pressure beating to generate microaerated chocolate with bubbles that are too small to see or producing a foamy head on beer. Nitrous oxide is the gas that pushes instant whipped cream out a can, chosen for its similar solubility to carbon dioxide but without imparting a sour taste. Most aerated meals, on the other hand, use a combination of carbon dioxide, steam, and air to accomplish aeration [5].

All these foods have a foamy structure created by managing bubbles in the dough when stirring and heating, then turning them into interconnecting gas cells in the powder. Consequently, among the most crucial aspects of the baking process is the creation of foam. Numerous studies have attempted to predict the production of bubbles in dough, but the bulk of them rely on single bubble growth models that ignore the size of the bubble particles inside the dough [23].

When a gas phase is added to a food matrix, it affects not only the texture but also the firmness of the product, bringing out qualities such as lightness, color, and sensory elements. This is since these aerated products lack thermodynamic stability, necessitating the use of surface activators to achieve this stabilization between the air and the liquid, which will preserve the coalescence of the bubbles by boosting the liquid's viscosity in the lamellas between the bubbles. As a result, foods treated with the aeration process can be kept stable if they are physically enclosed in a gel network [6].

The following methods listed here are used to aerate food, according to Campbell [5], and each of them presents its own particularity:

(A) Aeration occurs when yeast produces carbon dioxide during the fermentation process in bread, beer, and wine, for example. Existing bubbles are entrained or attached to lager or wine particles during the blending process, and CO_2 diffuses into them.

(B) Air is trapped in low-medium viscosity liquids, and fat crystals help in the integration of air into cake mixes and ice cream. Bubbles rise and fall out of the

fluid during whipping or shaking, therefore the respective rates of entrainment and distrainment determine the discuss substance and bubble measure conveyance.

(C) When the surfaces are joined, it is assumed that the bubbles are captured from the mixture, with the high consistency of the putty or glue being able to prevent the separation of bubbles caused by the high mechanical action that occurs during the process. As a result, fluids of low to medium consistency, such as cream, thicken during the process due to the fusion of bubbles and the release of protein and fat components, bringing the glue mixture's operation closer at a faster rate.

(D) During cooking, the formation of steam occurs, with the consequent expansion of the gases that are already in that matrix, as well as the dissolution of components that accompany the production of steam, such as CO_2, ethanol, etc.

(E) Cakes and croissants, for example, have the unique property of holding air between their laminated layers.

(F) When products with considered moisture are overheated, high pressure is released, this process is called puffing.

(G) Internal steam rapidly forms when food is fried in extremely hot oil, causing the product to puff.

(H) Other cases

Besides this, some numerical methods can aid in the understanding of these aeration phenomena in food, as in a study where the impact of the change related to various parameters of the output model was investigated, where it was possible to forecast the particle sizes of bubbles in products related to baking and even the increase in oven temperature during the cooking of the products. This mathematical model for bubble growth was discovered to be capable of predicting population balance, reaction kinetics, and calorie and mass transfer equations. Gas is formed in the dough due to the breakdown of sodium bicarbonate, which follows first-order kinetics causing gas formation and expansion, and the same researchers used the confocal laser scanning microscopy technique to observe the experimental distributions in relation to bubble size in breads. The model was created to predict the continuing evolution of foam construction, which is a critical aspect in determining the baked product's quality and is deemed sensitive [23].

These authors suggested that CO_2 and water vapor diffusion were the major factors for bubble formation, and that the dough possessed linear viscoelasticity. They developed a model that predicts three final characteristics in baked goods, such as bread: softness/specific volume, firmness/water content, and acidity/pH. Although the model depicted a single bubble forming, the starting bubble size was chosen by averaging the bubble size dispersion in the dough gained via experimentation [24].

There are many mechanisms that interfere in aerated systems, we can list among them, gravitational drainage, disproportionation, and coalescence. It must be understood that foams are thermodynamically unstable systems that arise as soon as they

are created, which means that the properties of the foam change over time as part of changes in the spatial pattern of gas and liquid in the foam. Gravity and capillary forces cause fluid to flow between the bubbles, causing drainage. The disproportion of bubbles is caused by the differences that occur in the Laplace pressure between the large and small bubbles, the first ones will grow, while the second ones will eventually disappear as a film between the bubbles [25].

3 Principles of food stabilization by biosurfactants

Biosurfactants have several applications in the food area; they can act as emulsifiers helping in the processing of raw materials. The action of emulsifiers is to act in phase dispersion and flavor solubilization, as well as to obtain the correct consistency and texture. The function of an emulsifier is to keep the emulsion stable, controlling the clustering of globules and stabilizing the aerated systems [26].

Air in a food product gives lightness, improved texture, and lower taste intensity, but it also has the potential to shorten the dish's shelf life. Beverages, chocolate and confectionery products, dairy products, baked products, egg products, cereal-based products, in all these applications, maintaining the size distribution of air bubbles and the spatial distribution of the gaseous phase during the product's shelf life, which is measured in months, is critical [27].

Aerated emulsions are three-phase systems that include air in an aqueous phase of serum. Due to the large free energy of the gas-liquid interface, aerated systems are intrinsically thermodynamically unstable. The coupled processes of coalescence and disproportionate bubbles characterize two mechanisms that degrade the aerated product, buoyancy, and Ostwald ripening, as the thermodynamic driving force to decrease the total interfacial area. "Due to buoyancy, bigger air bubbles climb to the top of the container, resulting in a vertical phase separation (cream formation) of the distributed air," they explained. As a result, product quality is inconsistent, which is undesirable. "The polydispersity in the size distribution resulted in a shift in bubble size owing to Ostwald ripening," they said. Under some conditions, the ripening process can help to expedite separation owing to buoyancy since bigger bubbles expand at the cost of smaller ones. "An optimal aerated product, in conclusion, mitigates the impacts of both unstable element" [28].

To improve kinetic stability, substances such emulsifiers, stabilizers, weighing agents, ripening inhibitors, and texture modifiers (thickeners and gelling agents) are added. Proteins (milkshakes, beer foam, wine foam), fat crystals (whipped cream), ice crystals (ice cream, frozen desserts), high viscosity/semisolid (baked goods, batters, bread dough, pastries, cake batters), and solid matrix (baked goods, batters, bread dough, pastries, cake batters) are the primary mechanisms of stabilization of various aerated foods (bread, cakes). Emulsifiers are frequently utilized to improve the stability of other processes [28].

In the confectionery industry, emulsifiers are most commonly used to aerate cake mixes. When emulsifiers are introduced to cake batter, they immediately orient

themselves with their nonpolar portion toward the fat globule interior and their polar portion facing the aqueous phase. This is responsible for lowering the interfacial tension between the fat and the aqueous phase, allowing fat to be distributed more evenly in the bulk as tiny, homogeneous particles. Simultaneously, emulsifiers lower the surface tension between the aqueous phase and the air, allowing for more and quicker air absorption into the bulk. When air is added to cake batter during mixing, the egg protein unfolds, with the lipophilic component going into the gas phase (into the inside of the air bubbles) and the hydrophilic portion staying in the aqueous phase. Together with the emulsifier molecules, this protein coating aids in the creation and stability of the foam. Because emulsifiers inhibit interaction between the fat and the protein, which might destabilize the protein film, their presence at the oil-water interface indirectly aids aeration [29].

One of the parameters that must be considered with certainty is the Gibbs stability, $E > \gamma/2$, which is related to the stability of a bubble when it is subjected to the thickening process, in the equation the term E is the dilatational elasticity of the surface and the surface tension. "Emulsifiers stabilize bubbles through the Gibbs-Marangoni effect," according to the report, "by which, if local thinning occurs, the surfactant quickly diffuses into the depleted area, taking the liquid with it to bind the diluted region." What happens here is possibly the use of short-chain polar-headed fatty acids that are often used as emulsifiers; at a bubble interface, the polar portion that has an affinity for water can assemble in the aqueous liquid while the hydrophobic lipid chain projects into the air [30].

4 Conclusions

Aerated systems are substantially more difficult to stabilize than emulsions. As a result, including biosurfactants as a viable strategy to improve the stability of aerated systems is a significant technical advancement. The physical stabilization mechanism of bubbles by biosurfactants, particularly the combined system of protein and surfactant, should be properly investigated.

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Conflict of interest

All authors agree that there are no conflicts of interest.

References

[1] J. Jeevanandam, C.O. Adetunji, J.D. Selvam, O.A. Anani, A. Inobeme, S. Islam, O.T. Olaniyan, High industrial beneficial microorganisms for effective production of a high quantity of biosurfactant, in: A.M. Inamuddin, S.K. Asiri (Eds.), Green Sustainable Process for Chemical and Environmental Engineering and Science, Elsevier, Amsterdã, 2021, pp. 279–297.

[2] A.R. Markande, D. Patel, S. Varjani, A review on biosurfactants: properties, applications and current developments, Bioresour. Technol. 330 (2021), 124963.

[3] J.R. Barbosa, S.B. da Silva, R.N. de Carvalho Junior, Biosurfactant production by solid-state fermentation, submerged fermentation, and biphasic fermentation, in: A.M. Inamuddin, S.K. Asiri (Eds.), Green Sustainable Process for Chemical and Environmental Engineering and Science, Elsevier, Amsterdã, 2021, pp. 155–171.

[4] J.R. Barbosa, S.B. da Silva, L.H. da Silva Martins, F.W.F. Bezerra, L.C. Freitas, M.C.R. Ferreira, R.N. de Carvalho Junior, Microbial degradation of food products, in: M. Inamuddin, I. Ahamed, R. Prasad (Eds.), Recent Advances in Microbial Degradation, Springer, Singapore, 2021, pp. 155–172.

[5] G.M. Campbell, Aerated foods, in: B. Caballero, P.M. Finglas, F. Toldrá (Eds.), Encyclopedia of Food and Health, Elsevier, 2016, pp. 51–60.

[6] R.N. Zúniga, J.M. Aguilera, Aerated food gels: fabrication and potential applications, Trends Food Sci. Technol. 19 (2008) 176–187.

[7] D. Elgeti, M. Jekle, T. Becker, Strategies for the aeration of gluten-free bread—a review, Trends Food Sci. Technol. 46 (2015) 75–84.

[8] R.H.H. Pinto, J.R. Barbosa, L.C. Freitas, Q.D.S. de Aguiar, F.C.S. Pires, A.P.D.S. Silva, R.N. de Carvalho Junior, Aerogels envisioning future applications, in: Inamuddin, T.A. Rangreez, M.I. Ahamed, R. Boddula (Eds.), Aerogels I Preparation, Properties and Applications, Materiais Research Forum LLC, 2020, pp. 214–229.

[9] J.R. Barbosa, R.H.H. Pinto, L.H. da Silva Martins, R.N. de Carvalho Junior, Bioaerogels: synthesis approaches, biomedical applications and cell uptake, in: Inamuddin, R. Mobin, M.I. Ahamed, T. Altalhi (Eds.), Aerogels II: Preparation, Properties and Applications, Materials Research Forum LLC, 2021, pp. 43–56.

[10] J.R. Barbosa, L.H. da Silva Martins, R.N. de Carvalho Junior, Polymer aerogels: preparation and potential for biomedical application, in: Aerogels II: Preparation, Properties and Applications, Materials Research Forum LLC, 2021, pp. 1–22.

[11] F.W.F. Bezerra, P.D.N. Bezerra, V.M.B. Cunha, M.D.L.A.R. Salazar, J.R. Barbosa, M.P. D. Silva, R.N.D. Carvalho Junior, Supercritical green solvent for Amazonian natural resources, in: Inamuddin, A.M. Asiri (Eds.), Advanced Nanotechnology and Application of Supercritical Fluids, Springer, 2020, pp. 15–31.

[12] B.K. Tiwari, Ultrasound: a clean, green extraction technology, TrAC Trends Anal. Chem. 71 (2015) 100–109.

[13] L. Blanco-Zubiaguirre, N. Arrieta, A. Iturregui, I. Martinez-Arkarazo, M. Olivares, K. Castro, J.M. Madariaga, Focused ultrasound solid-liquid extraction for the determination of organic biomarkers in beachrocks, Ultrason. Sonochem. 27 (2015) 430–439.

[14] S.W. Ohl, E. Klaseboer, B.C. Khoo, Bubbles with shock waves and ultrasound: a review, Interface Focus 5 (2015) 20150019.

[15] A. Ktenioudaki, F. Butler, U. Gonzales-Barron, U. Mc Carthy, E. Gallagher, Monitoring the dynamic density of wheat dough during fermentation, J. Food Eng. 95 (2009) 332–338.

[16] H.M. Elmehdi, J.H. Page, M.G. Scanlon, Monitoring dough fermentation using acoustic waves, Food Bioprod. Process. 81 (2003) 217–223.
[17] D. Elgeti, L. Yu, A. Stüttgen, M. Jekle, T. Becker, Interrelation between mechanical and biological aeration in starch-based gluten-free dough systems, J. Cereal Sci. 76 (2017) 28–34.
[18] M.E. Matos Segura, C.M. Rosell, Chemical composition and starch digestibility of different gluten-free breads, Plant Foods Hum. Nutr. 66 (2011) 224–230.
[19] J.R. Barbosa, R.N. de Carvalho Junior, Occurrence and possible roles of polysaccharides in fungi and their influence on the development of new technologies, Carbohydr. Polym. 246 (2020), 116613.
[20] L.C. de Oliveira, J.R. Barbosa, S.D.C.A. Ribeiro, M.A.M. de Vasconcelos, B.A. de Aguiar, G.V. da Silva Pereira, L.D.F.H. Lourenço, Improvement of the characteristics of fish gelatin-gum arabic through the formation of the polyelectrolyte complex, Carbohydr. Polym. 223 (2019), 115068.
[21] P. Wilde, Foam formation in dough and bread quality, in: S.P. Cauvain (Ed.), Bread Making, Woodhead Publishing, 2012, pp. 370–399.
[22] P.J. Martin, N.L. Chin, G.M. Campbell, C.J. Morrant, Aeration during bread dough mixing, Food Bioprod. Process. 82 (2004) 282–290.
[23] P. Deshlahra, A. Mehra, D. Ghosal, Evolution of bubble size distribution in baked foods, J. Food Eng. 93 (2009) 192–199.
[24] J. Gao, W. Zhou, Oral processing of bread: implications of designing healthier bread products, Trends Food Sci. Technol. 112 (2021) 720–734.
[25] P. Gélinas, Chemical leavening and other baker's yeast substitutes: overview of patents filed between 1833 and 2019, Int. J. Food Sci. Technol. 56 (2021) 1–12.
[26] J.M. Campos, T.L. Montenegro Stamford, L.A. Sarubbo, J.M. de Luna, R.D. Rufino, I.M. Banat, Microbial biosurfactants as additives for food industries, Biotechnol. Prog. 29 (2013) 1097–1108.
[27] A. Dutta, A. Chengara, A.D. Nikolov, D.T. Wasan, K. Chen, B. Campbell, Texture and stability of aerated food emulsions effects of buoyancy and Ostwald ripening, J. Food Eng. 62 (2004) 169–175.
[28] Y. Wang, R.W. Hartel, L. Zhang, The stability of aerated emulsions: effects of emulsifier synergy on partial coalescence and crystallization of milk fat, J. Food Eng. 291 (2021), 110257.
[29] J. Xu, Y. Zhang, W. Wang, Y. Li, Advanced properties of gluten-free cookies, cakes, and crackers: a review, Trends Food Sci. Technol. 103 (2020) 200–213.
[30] E. Dickinson, Food emulsions and foams: stabilization by particles, Curr. Opin. Colloid Interface Sci. 15 (2010) 40–49.

CHAPTER

8

The role of biosurfactants in the improvement of texture and shelf life of starch-containing products

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1 Introduction

The use of biosurfactants as surface-active agents has been well known in many application fields, such as agriculture [1], cosmetic [2], pharmaceutical [3], food and beverages [4], and so forth. Biosurfactants function as additives that aid in reducing the interfacial surface tension between two different phases and prevent microbial attack on the products [5]. In the production of starch-containing products, the addition of biosurfactants can improve the product quality through enhancement of the textural properties and shelf life of the products [6,7]. Furthermore, the ability of biosurfactants to lower the surface and interfacial tension between water molecules and oil/fat globules is known to facilitate the formation and stabilization of emulsions, stable aerated systems, and prevent the agglomeration of fat globules [8,9]. Biosurfactants also have bactericidal activity owing to their ability to penetrate the bacterial cells and induce cell lysis [10]; this property helps to extend shelf life of the products [4,6,11]. Compared with synthetic surfactants, biosurfactants are superior in terms of low toxicity, biodegradability, biocompatibility, and digestibility [5,8,12–14]. Furthermore, due to the increase of consumer awareness toward healthier lifestyles, the use of microbial surfactants (or biosurfactants) is currently

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actively sought to complement and eventually substitute fossil-derived synthetic surfactants [15]. Recent statistics also indicate that the global market for biosurfactants is forecasted to grow at a compound annual growth rate (CAGR) of over 5.5% between 2020 and 2026, reaching a market value of around USD 5.5 billion by the end of 2026 [16].

Biosurfactants are synthesized or secreted by microorganisms (i.e., bacteria, fungi, and yeasts) in the form of low-molecular-weight (LMW) complex molecules, such as glycolipids, phospholipids, fatty acids, lipopeptides, and so forth, or in high-molecular-weight (HMW) biosurfactants, such as polymeric and particulate biosurfactants [17,18]. The latter form of biosurfactant is considered to have better stabilization and emulsification effect than the LMW counterparts [19,20]. The terms of LMW and HWM bacterial molecules have become a debated issue, where LMW molecules are referred to as biosurfactants while HMW ones are referred to as bioemulsifiers. Although both have the ability as an emulsifier, stabilizer, and solubilizer, the differences in their physicochemical properties make the magnitude of their abilities different [11,21,22], see Table 1. Nevertheless, the terms biosurfactant and bioemulsifier are often used interchangeably to describe the surface-active biomolecules from microbial.

Glycolipids, lipopeptides, and lipoproteins are the most common type of biosurfactants that are widely applied in the formulation of some food products due to their economic potentials, i.e., high yield and wide range of applications [7,23,24]. For example, 11.7 g/L of rhamnolipids (a glycolipid biosurfactant) was produced after 144-h fermentation of *Pseudomonas aeruginosa* LBI in soybean oil soapstock as the carbon source [25]. In another study by Das and Kumar, the utilization of bagasse and potato peel waste as mixed carbon sources in submerged fermentation of *Pseudomonas azotoformans* AJ15 produced rhamnolipid with a yield of 1.16 g/L [26].

Table 1 Some of the different properties of biosurfactant and bioemulsifier.

Properties	Biosurfactant	Bioemulsifier
Molecular weight	Low	High
Nature	Amphiphilic	Amphiphilic
Critical micelle concentration	Low	High
Surface activity	Excellent	Low to moderate
Stabilization of emulsion	Low to moderate	Excellent
Mechanism of action	Lowering surface and interfacial tension by adsorption of its molecules into different phases, thus allow the mixing of two different phases	Stabilizing emulsion by increasing the kinetic stability, thus allowing the solubilization of poorly soluble substrates

Rane et al. reported the production of lipopeptide biosurfactant in moderate yields (0.09–0.24 g/L) through fermentation of various sugar-containing agricultural wastes, such as molasses, whey, and fruit peel extracts by *Bacillus subtilis* ANR 88 strain [27]. These studies thus highlight the tremendous opportunities to produce biosurfactants in a cost-effective and sustainable manner, thus offering many advantages over the synthetic ones.

The use of biosurfactants to improve the texture and shelf life of starch-containing food products (such as salad dressings, bread, cakes, and cookies) has been widely implied in many reported studies [5–7,28]. This ability of biosurfactants is strongly related to their properties, i.e., surface and interface activity, chemical and thermal stability, and biological activities. In this chapter, the relation between the properties and the ability of biosurfactants in improving the texture and shelf life of starch-containing products is described. The addition of biosurfactants is known to alter the texture of the food products, including their hardness or firmness, chewiness, cohesiveness, adhesion, and springiness. Furthermore, the antimicrobial activity of biosurfactants plays a major role in preventing microbial contamination on food products, thereby improving their shelf life.

2 Properties of biosurfactants and their role in extending texture profiles and shelf-life of starch-containing products

Biosurfactants possess an emulsifying property, which has been shown to play a critical role in the textural improvement of starch-based products. Emulsification is defined as a dispersion process between two (or more) immiscible liquids, i.e., organic (oil) and aqueous (water) phases, to form a semistable mixture through the addition of surfactant. A schematic illustration of the emulsion formation by biosurfactant is shown in Fig. 1A. Biosurfactant prevents phase separation of food products and forms a long-term or more stable emulsion, which can stand for months or years [29]. In addition to the emulsifying ability, biosurfactants also possess other desirable properties to help in maintaining (and improving) the texture and shelf-life of the food products, such as reducing surface and interfacial tension, high thermal stability, high salt tolerance, and active biological properties. The properties of several biosurfactants are summarized in Table 2.

2.1 Surface and interface activity

In general, surfactant molecules have a variety of polarity degrees and hydrogen bonds, which depend on their molecular structure. Similar to the fossil-derived surfactants, biosurfactants possess an amphiphilic character that arises from the presence of both hydrophilic (polar) head groups (e.g., amino acids, carbohydrates, phosphates, or cyclic peptides) and hydrophobic/lipophilic tail based on long-chain fatty acids or their derivatives. Unlike conventional synthetic surfactants with a well-defined hydrophobic and hydrophilic domain, biosurfactants possess a more

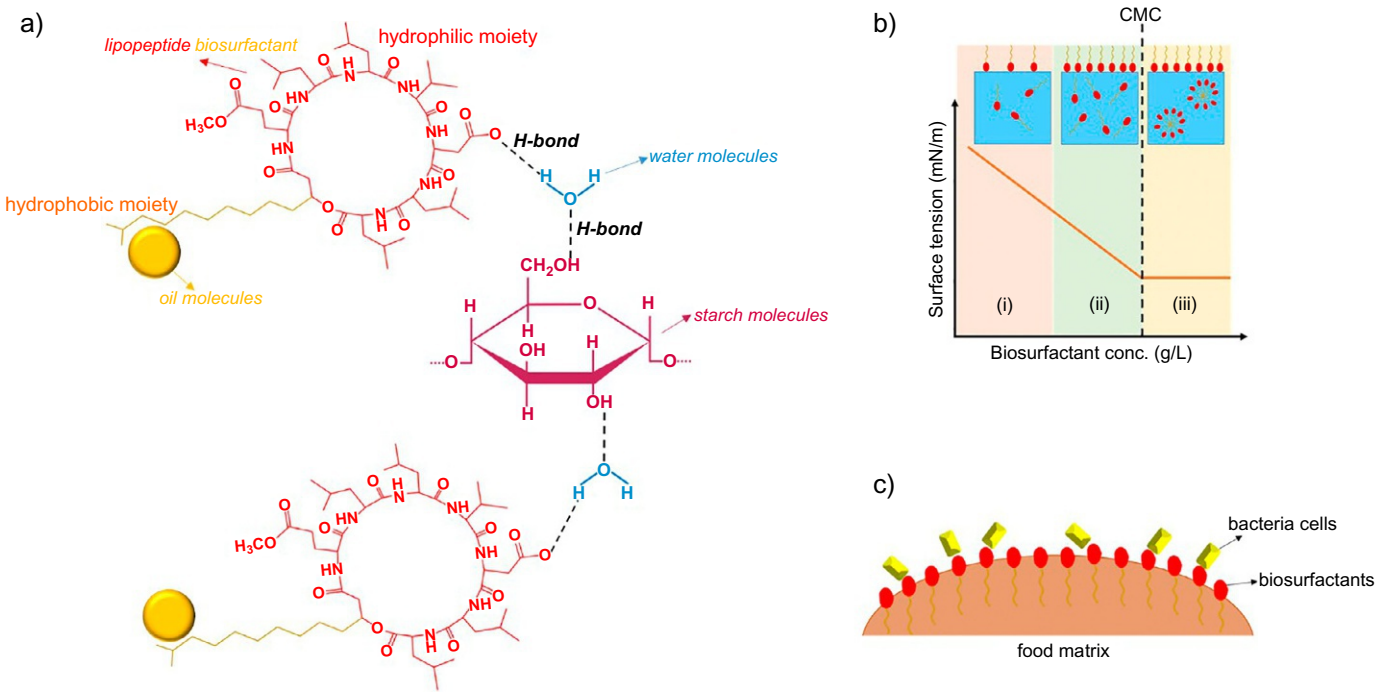


FIG. 1

(A) Interaction between starch–water–biosurfactant–oil, represented by a lipopeptide biosurfactant. (B) Relation of surface tension and biosurfactant concentration: (i) high surface tension at low concentration, (ii) moderate to low surface tension at concentration near critical micelle concentration (CMC), (iii) insignificant change of surface tension at concentration higher than CMC. (C) Illustration of microbial attack prevention by the biosurfactant molecules in the food matrix.

Table 2 Physicochemical properties and stability of several biosurfactants.

Microorganism	Produced biosurfactant and its properties[a]	Stability			Remarks on biological activity[b]	Ref.
		Temp.	pH	Salinity		
Bacillus licheniformis JF-2	Lichenysin biosurfactant ST: 10 mN/m	50°C	4.5–9.0	50 g/L NaCl	No data	[30]
Candida bombicola URM 3718	Contain 70% lipids, 30% carbohydrate ST: 30.79 mN/m IT: 0.73 mN/m CMC: 0.5 g/L EI: 45.81% (@0.5 g/L)	181.76°C	No data	No data	%*I* = 52.42 (1 mg/mL) No potential cytotoxicity against L929 and Vero strains, cell viability >90% at biosurfactant concentration of 50 μg/mL	[31]
Candida lipolytica UCP 0988	Anionic lipopeptide (50% protein, 20% lipids, 8% carbohydrates) ST: 25 mN/m CMC: 0.03% (*w/v*)	No data	No data	No data	Low toxicity against brine shrimp at concentration of 60 mg/L	[32]
Candida utilis UFPEDA1009	Contain 68.63% oleic acid ST: 35.33 mN/m IT: 2.53 mN/m CMC: 0.6 g/L	<200°C	No data	No data	%*I* = 16.52 (20 mg/mL) No potential cytotoxicity against L929 and RAW 264.7 at 0.2 mg/mL, with cell growth inhibition <16%	[33]
Lactobacillus helveticus MRTL91	Glycolipids ST: 39.5 mN/m CMC: 2.5 mg/mL	220°C	No data	No data	No toxicity against *Brassica nigra* and *Triticum aestivum* up to conc, of 5 mg/mL	[34]
Lactobacillus plantarum CFR 2194	Glycoprotein ST: 44.30 mN/m CMC: 6 g/L EI: 1.04 (OD_{540nm})	No data	No data	No data	Effective antimicrobial and antiadhesion activity against *E. coli*, *Yersinia enterocolitica*, and *S. aureus* at 25 mg/mL	[35]
Mucor hiemalis UCP 0039	Glycolipids ST: 32 mN/m EI: 96%	5–100°C	High alkaline (10–12)	17%–25%	No data	[11]

Continued

Table 2 Physicochemical properties and stability of several biosurfactants. *Continued*

Microorganism	Produced biosurfactant and its properties[a]	Stability			Remarks on biological activity[b]	Ref.
		Temp.	pH	Salinity		
Nesterenkonia sp. MSA31	Lipopeptides class ST: 34.6 mN/m CMC: 18.6 μg/mL EI: 85%	240°C	6–10	10%	%*I* = 65% (6 mg/mL) The biosurfactant was effectively inhibits *Staphylococcus aureus* biofilm at concentration of 125 μg/mL	[36]
Pediococcus dextrinicus SHU1593	Lipoprotein class, with ~52% lipids and 47% protein ST: 39.01 mN/m (PBS) EI: 43.04% (hexadecane) CMC: 2.7 mg/mL	121°C	6–10	No data	The biosurfactant (25 mg/mL) show anti-adhesive impact against *B. cereus* (70.50%), *P. aeruginosa* (61.84%), *Salmonella typhimurium* (58.69%), *E. coli* (44.25%), and *Enterobacter aerogenes* (29.87%)	[37]
Pseudomonas otitidis	Glycolipid class ST: 33.4 mN/m CMC: 40 mg/L EI: 63%	280°C	3–11	2%–10%	No data	[38]
Saccharomyces cerevisiae URM 6670	Glycolipids class, with 50.58% linoleic acid ST: 26.64 mN/m IT: 9.12 mN/m	200°C	No data	No data	%*I* = 21.35 (40 mg/mL) No potential cytotoxicity against L929 and RAW 264.7 at 0.2 mg/mL, with cell growth inhibition <15%	[28]
Streptomyces sp. DPUA1559	Glycoproteins (20% proteins, 38% carbohydrates, 12% lipids) ST: 33.76 mN/m EI: 30%–47% CMC: 10 mg/mL	120°C	2–12	0%–12%	No toxicity against *Artemia salina* up to concentration of 150 g/mL	[39]

[a]*Abbreviation:* ST, *surface tension;* IT, *interfacial tension;* CMC, *critical micelle concentration;* EI, *emulsification index.*
[b]%I = *DPPH sequestering activity (%).*

complex chemical structure and a mosaic-like distribution of hydrophobic and hydrophilic sites, making them competitive (or even superior performance) compared to their counterparts in terms of emulsification and surface tension reduction. Such remarkable physical properties of biosurfactants offer a great influence in the modification of textural properties of starch-containing products [4,40].

The ability of biosurfactants to lower interfacial tension of various organic-water systems has been demonstrated in many studies. On average, biosurfactants can decrease the surface tension of water from 72 to about 25 mN/m at 25°C and the interfacial tension of *n*-hexadecane-water interface from 40 to nearly 1 mN/m. For example, rhamnolipid biosurfactant produced by *P. aeruginosa* has also been shown to lower the surface tension to 26 mN/m and the water interfacial tension to <1 mN/m [29]. In another report, surfactin, a biosurfactant obtained from *B. subtilis* grown on cassava, demonstrated the ability to lower water surface tension to 25 mN/m and the hexadecane-water interfacial tension to approximately 1 mN/m [41]. Sophorolipids produced by *Candida bombicola* also reduced the water surface tension to 33 mN/m and the interfacial tension to 5 nM/m [42]. The ability of biosurfactants in lowering the surface tension and interfacial tension is attributed to their ability to bind molecules at different phases, which is affected by their micelle concentrations. Biosurfactants exhibit critical micelle concentrations (CMCs) 10–40 times lower than synthetic surfactants; therefore, less biosurfactants are required to decrease the surface tension compared with synthetic surfactants. The CMC of biosurfactants plays a great role in affecting the surface tension of the medium [5,8]. The relation between CMC and surface tension of the medium is illustrated in Fig. 1B.

2.2 Temperature and pH tolerance

Synthetic surfactants have limited practicability in the food industry, owing to their poor stability against pH, ionic strength, and heat [43]. The moderate to high risk of toxicity, poor degradability and biocompatibility, and poor digestibility also limited the utilization of synthetic surfactants [43–45]. On the other hand, biosurfactants have been reported to display superior tolerance against challenging environmental conditions, such as harsh pH values or high temperatures, without loss of surface activity. The enhanced thermal stability of biosurfactants helps them withstand the processing conditions in the production of starch-containing products, which commonly require a long time and high temperature (e.g., baking). For example, Ribeiro et al. demonstrated that a glycolipid biosurfactant obtained from *Saccharomyces cerevisiae* URM6670 grown in a medium containing 1% waste soybean oil and 1% corn steep liquor is thermally stable up to 200°C [28], thus making it suitable for practical use in the baking industry. In another report from Ribeiro's group, a fatty-acid-type biosurfactant produced through submerged fermentation of *Candida utilis* UFPEDA1009 yeast in a medium supplemented with glucose and waste canola frying oil displays good thermal stability with a decomposition temperature (T_d, corresponding to a 5% weight loss) higher than 320°C [33], which shows great potential

as an animal fat substitute in cookie dough formulations. Notably, thus far, there are only a few examples of biosurfactants that exhibit poor thermal stability. For instance, monorhamnolipid biosurfactant isolated from *Klebsiella* sp. KOD36 strain was found to be thermally stable up to 40°C in terms of the emulsifying activity index (E24%); however, it exhibits relatively good chemical stability in a highly alkaline environment (pH 10) and 10% salinity condition [18]. Moreover, the obtained monorhamnolipid also possesses an emulsification index (E24) above 40%, which is superior to previously reported biosurfactants and conventional synthetic surfactants, such as Triton-X [46]. On the other hand, a rhamnolipid biosurfactant from *P. aeruginosa* shows a better thermal stability over a temperature range of 40–121°C [47]. As described in some of the reported works, synthetic surfactants have failed to improve the quality of starch-based products in terms of texture and shelf life, mainly due to their intolerance to specific pH and ionic strength. Synthetic surfactants are likely to be inactive in the presence of 2% NaCl; meanwhile, biosurfactants have better salt tolerance up to a salt concentration of 10% [8,46]. Furthermore, biosurfactants are also found to be stable over a wide pH range, thus manifesting its potential use for industrial applications.

2.3 Biological activities

The intrinsic antimicrobial activity of biosurfactants plays an essential role in the quality preservation and thus the shelf life of starch-based products during long-term ambient storage or transportation by preventing bacterial adhesion and growth on the food matrices (see Fig. 1C). In addition, biosurfactants are also known to possess antiadhesive and antioxidant properties that help prevent the oxidative deterioration of starch-based products. Numerous studies have reported the growth-inhibiting effects of biosurfactants toward various bacterial strains, and several representative examples from major categories of biosurfactants are presented below:

Glycolipid class. Nitschke et al. reported that the use of rhamnolipids produced by *P. aeruginosa* LBI strain at a concentration of 64 μg/mL showed an effective antimicrobial activity against *Bacillus cereus* and *Mucor miehei*, while some inhibition of other pathogenic organisms, such as *Staphylococcus aureus*, *Neurospora crassa*, and *Micrococcus luteus*, was observed with a much higher minimal inhibitory concentration (MIC) value of 256 μg/mL [25]. Sophorolipid obtained from the yeast *Starmerella bombicola* grown on myristic acid substrate also displayed potent antibacterial activity against Gram-positive *S. aureus* with an MIC of 150 μg/mL. The results showed that the percentage survival of bacterial cells decreased dramatically to ~3% with an exposure time of 30 min and an MIC of 150 μg/mL. At the same time, total growth inhibition was observed with the exposure time of 15 min at concentrations higher than 350 μg/mL [48].

Lipopeptide class. Lipopeptide biosurfactant (i.e., surfactin) produced by *Bacillus* strains has also been reported to possess strong in vitro bactericidal activity against several multidrug-resistant Gram-positive and Gram-negative bacteria, such as *Alcaligenes faecalis*, *P. aeruginosa*, *Escherichia coli*, *Proteus vulgaris*, and

S. aureus with MIC values as low as μg/mL [49]. Furthermore, surfactin is also known to possess other beneficial properties, such as antiadhesive, antitumor, antiviral, and antibiofilm activity, thus demonstrating its versatile potential for use not only in the food industry but also in health, pharmacy, and biotechnology sectors [50]. In another report from Coronel-León et al., lichenysin biosurfactant produced by *Bacillus licheniformis* AL1.1 strain in a molasses medium was found to possess high antiadhesion and activity against food-borne pathogens, such as *Candida albicans* and methicillin-resistant *S. aureus* [51], which may find potential application in the food industry and hospitals.

Others. Biosurfactants BS-B1 and BS-Z9 derived from two *Lactobacillus casei* strains have been shown to exhibit multiproperties, including antioxidant, antiproliferative, and antibiofilm activities [52]. The antioxidant activity of these biosurfactants revealed a radical scavenging activity of 75%–77% and the IC50 values of 2.1–2.2 mg/mL, the test was done against 5 mg/mL of 2,2-diphenyl-1-picrylhydrazyl (DPPH). The radical scavenging activity can prevent oxidative damage to food products. Furthermore, the supplementation of biosurfactants can serve as an effective means for foods that are poor in antioxidants [33]; the excellent antioxidant activity of biosurfactants is attributed to their unsaturated fatty acids content [4]. These *Lactobacillus* strain biosurfactants also show the potential antiproliferative activity of ~67% at an IC50 of 109–129 mg/mL; antibacterial and antiadhesive activity against *S. aureus* at an IC50 of 1.92–2.16 mg/mL and 1.56 mg/mL, respectively. Furthermore, *S. aureus* biofilm formation decreased around 86% after exposure to 25 mg/mL of the biosurfactants from the *Lactobacillus* strain [52].

2.4 Other properties

Low to nontoxic property is one of the advantages of biosurfactants over their synthetic counterparts; this property led them to safe application in food sectors, pharmaceuticals, and cosmetics. Furthermore, their low toxicity allows them to be safely disposed without harmful effects on the environment and the living organisms. The low or nontoxic property of biosurfactants has been reported in many works. For example, the glycolipid biosurfactant from *S. cerevisiae* shows 9.90% and 14.78% inhibition of the noncancerous fibroblast cells (L929) and mice macrophage cells (RAW 264.7), where this value was considered as no cytotoxic potential. Furthermore, these inhibition values were lower than that of phosphate buffer pH 7.4, which showed 12.95% and 20.30% inhibition toward L929 and RAW 264.7, respectively [28]. The *C. utilis* UFPEDAA1009 biosurfactant also displayed low cytotoxicity against mammalian cell lines at a concentration of 200 μg/mL, with 15.83% and 0% inhibition against L929 mouse fibroblasts and RAW 264.7 macrophages, respectively [33]. The biodegradability by water and soil microorganisms adds the ecofriendly properties of biosurfactants over synthetic surfactants [5]. Although these properties may not be directly related to improved food texture or shelf life, the biodegradability properties of biosurfactants can help reduce hazardous waste generation in industrial applications.

3 Effects of biosurfactants in texture and shelf life of some starch-containing products

The ingredients complexity of starch-containing foods is a major obstacle to examine the details of the mechanism of biosurfactants in influencing their textural properties. Nevertheless, alteration in the quality and texture of the final food products after the addition of biosurfactants has been described in several studies. It is also postulated that a lower amount of biosurfactants (compared with synthetic surfactants) is required to give a good emulsification and stabilization effect on the products. The following sections describe the effect of biosurfactants addition in improving of organoleptic properties of breads, muffin, and cookies. The summary of the textural properties of some starch-containing food products is presented in Table 3.

3.1 Salad dressings

Salad dressings represent one of the typical oil-in-water (O/W) emulsions composed of vegetable oil, vinegar, egg yolk, and starch, which can be categorized into three groups based on the oil content: (1) mayonnaise, (2) spoonable, and (3) French-type dressings. Current challenges in this emulsified-semisolid food product are to prevent phase separation of the ingredient between and during the long term, which detrimentally affects the shelf life and rheological properties (i.e., sensory and qualities) of the products. In this regard, the use of biosurfactants as an emulsifier has been reported to improve the consistency and texture of the paste. For example, the addition of 0.7% (*w*/*v*) of HMW biosurfactant derived from *C. utilis*, along with guar gum into the mayonnaise formulation, renders the resultant product good shelf life stability without any observable phase separation after 4 weeks of refrigeration. However, it is worth mentioning that the addition of biosurfactant alone resulted in the formation of segregated phases after 4 weeks, thus suggesting the importance of guar gum as a stabilizing agent. Moreover, the texture of biosurfactant-added salad dressings exhibits better consistency when mixed with spatula. Furthermore, the absence of pathogenic microorganisms after 4 weeks of storage indicates the potential antimicrobial properties of the isolated biosurfactant, whose structure is composed from characterizations. The characterizations of the *C. utilis* derived biosurfactant show the structure of carbohydrate–protein–lipid complex [54].

3.2 Bakery products

3.2.1 Breads

Bread and its variations are among the most widely consumed staple foods around the world and have long been an integral part of the human diet. Breads can be represented as an unstable and elastic solid structure consisting of an elastic network of cross-linked gluten molecules and leached amylose molecules in part and are typically characterized by high starch contents, which vary depending on the origin and

Table 3 Textural properties of some starch-containing products in supplementation of biosurfactants.

Food type	Starch source	Biosurfactant Origin (class)	Texture[a] Control	Texture[a] BS-added	Remarks on shelf-life	Ref.
Bread	Wheat flour	*Bacillus subtilis* SPB1 (lipopeptide)	Bread—final product		Standard plate count test on the bread after 8 days' storage, the number of microbial count (CFU/g): Control • Mold = 7×10^4 • Bacteria = 8×10^4 BS-added • Mold = 2×10^3 • Bacteria = 4×10^2 BS-added bread show lower microbial colony	[6]
			H: 4.445 Ch: 7.196 Co: 0.226 A: 1.006 S: 7.150	H: 1.446 Ch: 1.753 Co: 0.227 A: 0.329 S: 7.324		
Cookies	Wheat-sesame peels flours	*B. subtilis* SPB1 (lipopeptide)	Cookies' dough		No data	[53]
			H: 26.77 Co: 0.486 B: 1.8 A: 1.5 S: 18.627	H: 18.05 Co: 1.205 B: 0.627 A: 2.705 S: 14.225		
Cookies	White flour	*Candida utilis* UFPEDA1009 (contain 68.63% oleic acid)	Cookies' dough		No data	[33]
			H: 63.57 Co: 0.70 A: 1.67 S: 0.77	H: 45.65 Co: 0.46 A: 2.25 S: 0.83		
			Cookies—final product			
			H: 445.59	H: 354.93		

Continued

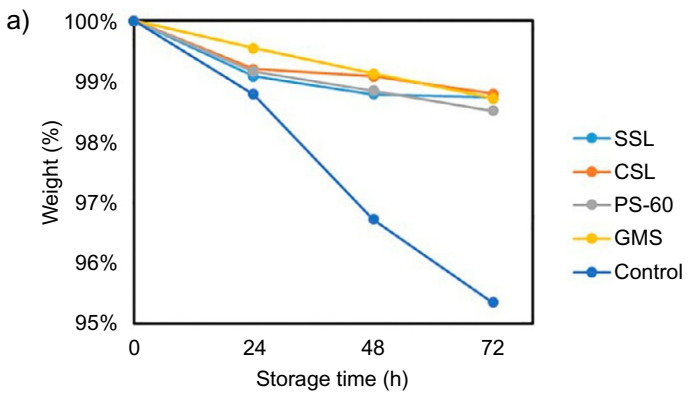
Table 3 Textural properties of some starch-containing products in supplementation of biosurfactants. *Continued*

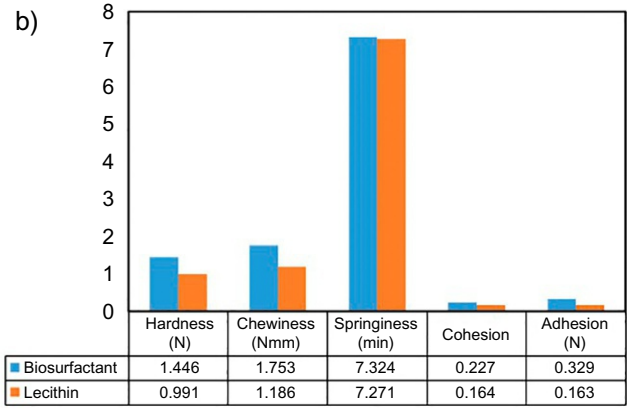
Food type	Starch source	Biosurfactant	Texture[a]		Remarks on shelf-life	Ref.
		Origin (class)	Control	BS-added		
Cookies	White flour	*Saccharomyces cerevisiae* URM 6670 (glycolipids)	Cookies' dough		No data	[28]
			H: 63.57 Co: 0.70 A: 1.67 S: 0.77	H: 44.47 Co: 0.62 A: 1.50 S: 0.83		
			Cookies—final product			
			H: 445.59	H: 426.60		
Muffin	Wheat flour	*Nesterenkonia* sp. MSA31 (lipopeptides)	Muffin—final product		The biosurfactant show antibiofilm activity against *Staphylococcus aureus*	[36]
			H: 19.78 Ch: 11.28 Co: 0.66 S: 0.99	H: 13.45 Ch: 9.00 Co: 0.67 S: 0.98		
Mayonnaise	Instant starch	*C. utilis* UFPEDA1009 (contain 65.88% lipids)	BS-added product exhibits uniform, creamy, and stable consistency		There is no aqueous separation was occurred for product supplemented with biosurfactant + guar gum, after 4 weeks' storage. Meanwhile, aqueous separation was occur for the products supplemented with biosurfactant or guar gum only	[54]

[a]*The texture of food products made with supplementation of biosurfactant (BS added) and without biosurfactant (Control) were compared. The texture properties were including hardness or firmness (H) with unit of N, cohesion (Co), chewiness (Ch) with unit of Nmm, breaking strength (B) with unit of N, adhesion (A), springiness or elasticity (S) with unit of mm.*

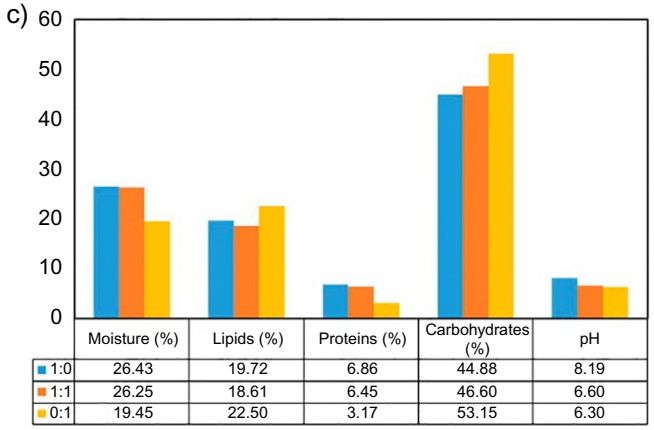
types [55]. Typically, bread (e.g., white bread, bagels, English muffins, etc.) can contain starch up to 50 wt% [56]. Other key ingredients in commercial bread products include sugar, salt, fat, water, and various additives. Among the chemical additives used in bread-making, surfactant plays an important role in producing high-quality breads by influencing dough strength, tolerance, uniform crumble cell size, and gas retention and slicing characteristics [57]. Moreover, the addition of surfactant has been shown to extend the shelf life and retard bread staling (i.e., a common description of changes in the bread crumb structure during storage that leads to a reduced consumer acceptance) by forming a complex network with the periphery of amylose starch granules [58], as illustrated in Fig. 1A. The formation of such surfactant–amylose complexes prevents the over-swelling of starch granules throughout the baking process, thus leading to the lower mobility of water and starch molecules during storage. In this regard, the surfactant molecules are bound to water molecules and subsequently prevent its evaporation, and therefore, the weight loss of the bread during the storage can be slowed down [49], as shown in Fig. 2A. The low water loss of the bread can preserve the texture of the bread and prevent hardening. Similarly, the supplementation of biosurfactant in the process is known to improve the texture of the final bread product without compromising its quality. Biosurfactants are known to have higher thermal stability than synthetic surfactants, which allow them to withstand the high condition during the baking of process without degradation.

Mnif et al. [6] reported that the addition of lipopeptide biosurfactant isolated from *B. subtilis* BS1 strain in bread dough at a concentration of 0.075 wt% yielded bread products with higher-quality attributes (i.e., lower crumb firmness and chewiness and increased cohesiveness) than those produced using commercial soy lecithin, as shown in the Fig. 2B. Moreover, the SPB1 biosurfactant-incorporated breads exhibit a higher specific volume of c. 38.5% and loaf volume (or crumb porosity) compared with that formulated with soy lecithin (24.8%). In this regard, the improved texture profiles of breads supplemented with SPB1 can be attributed to the interaction between biosurfactant and starch molecules, which minimize leaching of amylose chains from gelatinized starch granules during baking. A high degree of starch gelatinization is an unwanted phenomenon during baking since it leads to the deterioration of the bread texture. The use of biosurfactant in bread formulation also leads to a lower staling rate during storage, thereby extending the shelf life of products. In particular, the breads treated with 0.05% SPB1 (a biosurfactant from *B. subtilis*) show a decline in hardening. After 8 days of ambient storage the hardness of the breads increases from approx. 2.72 to 6.85 N; meanwhile, without biosurfactant, the bread had a harder texture in the beginning (approx. 4.31 N) and became harder after 8 days' storage (15.95 N). Furthermore, the reduction of microbial growth is also one indication of longer shelf life, after 8 days' storage, lower mold and bacteria colony were observed on the biosurfactant-added bread compared with the control samples [6].





	Hardness (N)	Chewiness (Nmm)	Springiness (min)	Cohesion	Adhesion (N)
Biosurfactant	1.446	1.753	7.324	0.227	0.329
Lecithin	0.991	1.186	7.271	0.164	0.163



	Moisture (%)	Lipids (%)	Proteins (%)	Carbohydrates (%)	pH
1:0	26.43	19.72	6.86	44.88	8.19
1:1	26.25	18.61	6.45	46.60	6.60
0:1	19.45	22.50	3.17	53.15	6.30

FIG. 2

(A) The effect of surfactant addition to the weight loss of bread during different storage period (*SSL*, steoryl-2-lactylate; *CSL*, calcium steoryl-2-lactylate; *PS-60*, polysorbate-60; *GMS*, glycerol monosorbate). (B) Textural properties of bread made in addition of 0.075% (*w*/w) *Bacillus subtilis* lipopeptide biosurfactant and lecithin. (C) Physicochemical composition of cupcakes prepared using formulation with 100% vegetable fat (1:0), 50% vegetable fat +50% biosurfactant (1:1), and 100% biosurfactant (0:1). The figures were redrawn from the quantitative results of Refs. [6,31,58].

3.2.2 Cookies

Cookies (also referred to as biscuits or crackers) are among the most popular ready-to-eat starch-containing baked goods made of flour, sugar, and fat as the main ingredients. Compared with breads, cookies possess a longer shelf life due to their lower water content, typically below 5 wt%. In addition, some cookies may also contain additional ingredients that can reduce or prevent microbial contamination on the surface, such as ascorbic acid. Several studies have demonstrated that the addition of biosurfactant as an emulsifier can improve the quality, texture, and shelf life of cookies, leading to higher acceptance. For example, Zouari et al. showed that the incorporation of *B. subtilis* SPB1 biosurfactant (0.1% on a flour weight basis) resulted in cookie dough with enhanced texture profiles, such as reduced hardness, springiness, and breaking strength, as well as increased cohesiveness and adhesiveness [53]. This texture improvement can be attributed to the emulsifying activity of biosurfactant, which was found to be more effective than the commercial emulsifier glycerol monostearate.

Ribeiro and coworkers also demonstrated that the *C. utilis* UFPEDA biosurfactant can be utilized as a potential animal fat replacement in the cookie dough formulation without compromising the physicochemical composition (i.e., moisture, lipids, proteins, carbohydrates, and ash content), texture profiles (i.e., adhesiveness, cohesiveness, and hardness), and the calorific value of the end product [33]. Moreover, it was found that complete substitution of pasteurized egg yolk lipid with the biosurfactant gave rise to a significantly lower crumb firmness after baking compared with the control sample, which subsequently leads to the formation of softer and spongier cookies. In addition to decreased crumb firmness, the biosurfactant-treated cookies exhibit a higher spread factor (i.e., an important quality characteristic defined as the ratio of diameter to thickness) due to the high lipid content biosurfactant [33]. Furthermore, the use of biosurfactants rich in omega-9 fatty acids also yields cookies with enhanced health benefits for consumers, particularly for cardiovascular disease prevention [33,60].

3.2.3 Cakes

Cakes are other sweet baked products with starch as one of their main ingredients. Similar to cookies, cakes are also ready-to-eat snacks with a long shelf-life, attributed to their water content. Cakes contain higher water content than cookies, but less than commercial breads. Both cakes and cookies can be made from combination of various compositions and ingredients to provide specific dietary needs and nutritional values and achieve higher consumer acceptance. Followings are some applications of biosurfactants toward improving the overall quality of cakes in terms of shelf life.

Biosurfactant from *Candida bombicola*, which contains 70% lipids and 30% carbohydrates, shows a CMC of 0.5 g/L, the surface tension of 30.79 mN.m, and interfacial tension of 0.73 mN/m. The biosurfactant shows a potential antioxidant activity above 50% at a concentration of 1 mg/mL, and it did not show cytotoxicity potential up to a concentration of 50 μg/mL. The use of this biosurfactant to completely

substitute vegetable fat in the production of cupcakes did not cause considerable change in the physical properties of the final product. The cupcakes, which are prepared by addition of biosurfactant and vegetable fat, also have similar lipids content (Fig. 2C). These results indicate that the use of vegetable fat can be replaced by a biosurfactant. Furthermore, the biosurfactant also shows antioxidant activity, which can help to restrain the oxidative damage of the cupcake and thus help in prolonging the shelf life [31].

3.3 Other starch-based products

A sophorolipid biosurfactant from *Wickerhamomyces anomalus* was able to reduce surface tension from 52.5 to 36.0 mN/m; this biosurfactant is reported to improve the retrodegradation, swelling power, and water-holding capacity of rice flour [61]. The presence of starch in ice cream can help in preventing crystallization, thus improving the texture of the ice cream. Stabilization of the texture and consistency, extension of freshness, and improvement of flavor oil in ice cream can further proceed by supplementation of biosurfactants. Biosurfactants also act as an anti-spattering agents and fat stabilizers during the production process. Unfortunately, no study reports the textural alteration of ice cream induced by biosurfactants. In one study, the use of *P. aeruginosa* Rhamnolipids as one formulation in loaf processing, ice cream, and mayonnaise was reported; the food containing Rhamnolipids shows improvement not only in texture but also in taste, appearance, and viscosity (for mayonnaise). Furthermore, the antibacterial activity of the Rhamnolipids against several tested bacteria (i.e., *B. subtilis, S. aureus*, *Salmonella* sp., *C. albicans*) signifies its potential as food preservative agent [62]. The addition of lactic acid bacteria (LAB), i.e., *Lactobacillus plantarum* to produce fermented foods (including starch-based fermented foods) has also been demonstrated to extend the shelf life of food products. This is due to the bioactive compounds produced by bacteriocin, which can be classified as a peptide biosurfactant [63].

4 Concluding remarks and outlooks

The use of biosurfactants (and/or bioemulsifiers) in food sectors have been a common practice over the centuries. Biosurfactants were not only used in the production of starch-containing food products (i.e., baked goods, creams, salad dressings, cakes and desserts, mayonnaise) but also in other types of food such as cheese, dairy products, etc. [64]. The purpose of biosurfactants supplementation in food products is to improve the flavor, appearance, texture, shelf life, and (additionally) nutritional value. The supplementation of biosurfactants is also known to give stabilization of emulsion and property improvement for bakery and dairy formulations [65].

Biosurfactants produced from microbial strains consist of fatty acids, protein, and carbohydrates with different compositions, which results in a wide variety of biosurfactants with different molecular weights. Biosurfactants with low molecular weight

are more effective in reducing surface and interfacial tension; i.e., glycolipids, fatty acids, lecithins, and monoglycerides. The HMW biosurfactants, which are high in protein and carbohydrate, are more effective to facilitate the formation of stable emulsion [66]. The stability of emulsion is emerging through various types of interactions between the surfactant and starch molecules, such as van der Waals forces, electrostatic forces, hydrogen bonding, and repulsive forces. The emulsification ability of biosurfactant was owing to their amphipathicity, which allows them to form complex with the starch, protein, and fatty fraction of foods [65,67]. The emulsification ability of biosurfactants allows them to alter the textural properties of the food, especially starch-containing foods.

The use of biosurfactant to improve the texture of starch-containing food was patented in 2002 by Haesendonck and Vanzeveren [68]; they claimed that the use of Rhamnolipid can enhance the bakery dough stability, texture, volume, and shape. This improvement is due to the synergistic interaction of the biosurfactants molecules and starch molecules, which produce stable emulsion of the dough. Extension of shelf life is another factor determining the good quality of food products. The addition of biosurfactants can aid in prolonging the shelf life of the food by preventing the adhesion of microbial films. The antiadhesive property of biosurfactants is owing to their hydrophobicity nature, which helps to repel the attachment of microbial (bio) film on food. Besides antiadhesive activity, biosurfactants are also known to have antimicrobial, antioxidant, and antibiofilm properties. Furthermore, the low or no cytotoxicity of biosurfactants allows their safe applications in many sectors, such as food, pharmacy, cosmetics, etc.

Despite the well-known ability of biosurfactants to improve the texture and shelf life of starch-containing products, the actual use of biosurfactants in the industry is still limited. Starch-containing foods have very complex compositions (i.e., combination of carbohydrates, lipids, fibers, proteins, etc.) and various processing conditions (i.e., temperature, pH, and processing steps); thus, it is difficult to fully discover the effectiveness and role of biosurfactants in foods, which may be different for different types of food. Furthermore, the economic aspect still becomes the biggest obstacle for the application of biosurfactants. Although the biosurfactants can be obtained through fermentation of microorganisms in cheap media (such as utilizing the agricultural or industrial wastes), their purification step may require a complex process. Nevertheless, the use of biosurfactants will certainly be popular in the near future, considering the strong demand of consumers for natural or organic food ingredients.

References

[1] D.P. Sachdev, S.S. Cameotra, Biosurfactants in agriculture, Appl. Microbiol. Biotechnol. 97 (3) (2013) 1005–1016.

[2] S.A. Adu, P.J. Naughton, R. Marchant, I.M. Banat, Microbial biosurfactants in cosmetic and personal skincare pharmaceutical formulations, Pharmaceutics 12 (11) (2020).

[3] A. Varvaresou, K. Iakovou, Biosurfactants in cosmetics and biopharmaceuticals, Lett. Appl. Microbiol. 61 (3) (2015) 214–223.
[4] B.G. Ribeiro, J.M.C. Guerra, L.A. Sarubbo, Biosurfactants: production and application prospects in the food industry, Biotechnol. Prog. 36 (5) (2020), e3030.
[5] S. Vijayakumar, V. Saravanan, Biosurfactants-types, sources and applications, Res. J. Microbiol. 10 (5) (2015) 181–192.
[6] I. Mnif, S. Besbes, R. Ellouze, S. Ellouze-Chaabouni, D. Ghribi, Improvement of bread quality and bread shelf-life by *Bacillus subtilis* biosurfactant addition, Food Sci. Biotechnol. 21 (2012) 1105–1112.
[7] I.J.B. Durval, B.G. Ribeiro, J.S. Aguiar, R.D. Rufino, A. Converti, L.A. Sarubbo, Application of a biosurfactant produced by *Bacillus cereus* UCP 1615 from waste frying oil as an emulsifier in a cookie formulation, Fermentation 7 (3) (2021).
[8] D.K.F. Santos, R.D. Rufino, J.M. Luna, V.A. Santos, L.A. Sarubbo, Biosurfactants: multifunctional biomolecules of the 21st century, Int. J. Mol. Sci. 17 (2016) 401.
[9] I.A. Purwasena, D.I. Astuti, M. Syukron, M. Amaniyah, Y. Sugai, Stability test of biosurfactant produced by *Bacillus licheniformis* DS1 using experimental design and its application for MEOR, J. Pet. Sci. Eng. 183 (2019) 106383.
[10] T. Hamouda, J.R. Baker, Antimicrobial mechanism of action of surfactant lipid preparations in enteric gram-negative bacilli, J. Appl. Microbiol. 89 (3) (2001) 397–403.
[11] I.N.S. Ferreira, D.M. Rodríguez, G.M. Campos-Takaki, R.F.d.S. Andrade, Biosurfactant and bioemulsifier as promising molecules produced by Mucor hiemalis isolated from Caatinga soil, Electron. J. Biotechnol. 47 (2020) 51–58.
[12] E.O. Fenibo, G.N. Ijoma, R. Selvarajan, C.B. Chikere, Microbial surfactants: the next generation multifunctional biomolecules for applications in the petroleum industry and its associated environmental remediation, Microorganisms 7 (11) (2019) 581.
[13] I.O. Olasanmi, R.W. Thring, The role of biosurfactants in the continued drive for environmental sustainability, Sustainability 10 (12) (2018).
[14] S. De, S. Malik, A. Ghosh, R. Saha, B. Saha, A review on natural surfactants, RSC Adv. 5 (81) (2015) 65757–65767.
[15] I. Anestopoulos, D.E. Kiousi, A. Klavaris, A. Galanis, K. Salek, S.R. Euston, A. Pappa, M.I. Panayiotidis, Surface active agents and their health-promoting properties: molecules of multifunctional significance, Pharmaceutics 12 (2020) 688.
[16] K. Ahuja, S. Singh, Biosurfactants Market Size By Product (Sophorolipids, Rhamnolipids, Alkyl Polyglucosides [APG], Methyl Ethyl Sulfonates [MES], Sucrose Esters, Sorbitan Esters, Lipopeptides), By Application (Household Detergents, Personal Care, Industrial Cleaners, Food Processing, Oilfield Chemicals, Agricultural Chemicals, Textiles), Industry Analysis Report, Regional Outlook, Application Potential, Price Trend, Competitive Market Share & Forecast, 2020–2026, Delaware, USA, 2020.
[17] E. Rosenberg, E.Z. Ron, Biosurfactants, in: E. Rosenberg, E.F. DeLong, S. Lory, E. Stackebrandt, F. Thompson (Eds.), The Prokaryotes: Applied Bacteriology and Biotechnology, Springer Berlin Heidelberg, Berlin, Heidelberg, 2013, pp. 281–294.
[18] R. Jahan, A.M. Bodratti, M. Tsianou, P. Alexandridis, Biosurfactants, natural alternatives to synthetic surfactants: physicochemical properties and applications, Adv. Colloid Interf. Sci. 275 (2020) 102061.
[19] M. Pacwa-Plociniczak, G.A. Plaza, Z. Piotrowska-Seget, S.S. Cameotra, Environmental applications of biosurfactants: recent advances, Int. J. Mol. Sci. 12 (1) (2011) 633–654.
[20] Q. Liu, J. Niu, Y. Liu, L. Li, J. Lv, Optimization of lipopeptide biosurfactant production by *Bacillus licheniformis* L20 and performance evaluation of biosurfactant mixed system for enhanced oil recovery, J. Pet. Sci. Eng. (2021) 109678.

[21] C. Uzoigwe, J.G. Burgess, C.J. Ennis, P.K.S.M. Rahman, Bioemulsifiers are not biosurfactants and require different screening approaches, Front. Microbiol. 6 (245) (2015).
[22] W. Tao, J. Lin, W. Wang, H. Huang, S. Li, Designer bioemulsifiers based on combinations of different polysaccharides with the novel emulsifying esterase AXE from *Bacillus subtilis* CICC 20034, Microb. Cell Factories 18 (1) (2019) 173.
[23] T. Morita, T. Fukuoka, T. Imura, D. Kitamoto, Glycolipid Biosurfactants, Reference Module in Chemistry, Molecular Sciences and Chemical Engineering, Elsevier, 2016.
[24] B.V. Praveesh, A.R. Soniyamby, C. Mariappan, M. Palniswamy, S. Lalitha, Microbial surfactants: an overview, Int. J. Curr. Res. Rev. 3 (10) (2011) 38–46.
[25] M. Nitschke, S.G.V.A.O. Costa, J. Contiero, Structure and applications of a rhamnolipid surfactant produced in soybean oil waste, Appl. Biochem. Biotechnol. 160 (7) (2010) 2066–2074.
[26] A.J. Das, R. Kumar, Utilization of agro-industrial waste for biosurfactant production under submerged fermentation and its application in oil recovery from sand matrix, Bioresour. Technol. 260 (2018) 233–240.
[27] A.N. Rane, V.V. Baikar, V.R. Kumar, R.L. Deopurkar, Agro-industrial wastes for production of biosurfactant by *Bacillus subtilis* ANR 88 and its application in synthesis of silver and gold nanoparticles, Front. Microbiol. 8 (2017) 492.
[28] B.G. Ribeiro, J.M.C. Guerra, L.A. Sarubbo, Potential food application of a biosurfactant produced by *Saccharomyces cerevisiae* URM 6670, Front. Bioeng. Biotechnol. 8 (2020) 434.
[29] D. Galabova, A. Sotirova, E. Karpenko, O. Karpenko, Role of microbial surface-active compounds in environmental protection, in: M. Fanum (Ed.), The Role of Colloidal Systems in Environmental Protection, Elsevier, 2014, pp. 41–83.
[30] M.J. McInerney, M. Javaheri, D.P. Nagle, Properties of the biosurfactant produced by *Bacillus licheniformis* strain JF-2, J. Ind. Microbiol. 5 (1990) 95–102.
[31] I.A. Silva, B.O. Veras, B.G. Ribeiro, J.S. Aguiar, J.M.C. Guerra, J.M. Luna, L.A. Sarubbo, Production of cupcake-like dessert containing microbial biosurfactant as an emulsifier, Peer J 8 (2020), e9064.
[32] R.D. Rufino, J.M.d. Luna, G.M.d.C. Takaki, L.A. Sarubbo, Characterization and properties of the biosurfactant produced by *Candida lipolytica* UCP 0988, Electron. J. Biotechnol. 17 (1) (2014) 34–38.
[33] B.G. Ribeiro, B.O.d. Veras, J.D.S. Aguiar, J.M.C. Guerra, L.A. Sarubbo, Biosurfactant produced by *Candida utilis* UFPEDA1009 with potential application in cookie formulation, Electron. J. Biotechnol. 46 (2020) 14–21.
[34] D. Sharma, B.S. Saharan, N. Chauhan, A. Bansal, S. Procha, Production and structural characterization of *Lactobacillus helveticus* derived biosurfactant, Sci. World J. 2014 (2014) 493548.
[35] A.N. Madhu, S.G. Prapulla, Evaluation and functional characterization of a biosurfactant produced by *Lactobacillus plantarum* CFR 219, Appl. Biochem. Biotechnol. 172 (4) (2013) 1777–1789.
[36] G.S. Kiran, S. Priyadharsini, A. Sajayan, G.B. Priyadharsini, N. Poulose, J. Selvin, Production of lipopeptide biosurfactant by a marine *nesterenkonia* sp. and its application in food industry, Front. Microbiol. 8 (2017) 1138.
[37] A. Ghasemi, M. Moosavi-Nasab, P. Setoodeh, G. Mesbahi, G. Yousefi, Biosurfactant production by lactic acid bacterium *Pediococcus dextrinicus* SHU1593 grown on different carbon sources: strain screening followed by product characterization, Sci. Rep. 9 (2019) 5287.

[38] P. Singh, B.N. Tiwary, Isolation and characterization of glycolipid biosurfactant produced by a *Pseudomonas otitidis* strain isolated from Chirimiri coal mines, India, Bioresour. Bioprocess. 3 (2016) 42.
[39] A.P.P. Santos, M.D.S. Silva, E.V.L. Costa, R.D. Rufino, V.A. Santos, C.S. Ramos, L.A. Sarubbo, A.L.F. Porto, Production and characterization of a biosurfactant produced by *Streptomyces sp*. DPUA 1559 isolated from lichens of the Amazon region, Braz. J. Med. Biol. Res. 51 (2) (2018).
[40] J.A.V. Costa, H. Treichel, L.O. Santos, V.s.G. Martins, Solid-state fermentation for the production of biosurfactants and their applications, in: A. Pandey, C. Larroche, C.R. Soccol (Eds.), Current Developments in Biotechnology and Bioengineering, Elsevier, 2018, pp. 357–372.
[41] K. Muthusamy, S. Gopalakrishnan, T.K. Ravi, P. Sivachidambaram, Biosurfactants: properties, commercial production and application, Curr. Sci. 94 (6) (2008) 736–747.
[42] M. Nitschke, S.G.V.A.O. Costa, Biosurfactants in food industry, Trends Food Sci. Technol. 18 (2007) 252–259.
[43] C.B.B. Farias, F.C.G. Almeida, I.A. Silva, T.C. Souza, H.M. Meira, R.d.C.F.S. da Silva, J.M. Luna, V.A. Santos, A. Converti, I.M. Banat, L.A. Sarubbo, Production of green surfactants: market prospects, Electron. J. Biotechnol. 51 (2021) 28–39.
[44] G. Pasternak, T.D. Askitosari, M.A. Rosenbaum, Biosurfactants and synthetic surfactants in bioelectrochemical systems: a mini-review, Front. Microbiol. 11 (358) (2020).
[45] G. Płaza, V. Achal, Biosurfactants: eco-friendly and innovative biocides against biocorrosion, Int. J. Mol. Sci. 21 (6) (2020) 2152.
[46] Z. Ahmad, X. Zhang, M. Imran, H. Zhong, S. Andleeb, R. Zulekha, G. Liu, I. Ahmad, F. Coulon, Production, functional stability, and effect of rhamnolipid biosurfactant from *Klebsiella* sp. on phenanthrene degradation in various medium systems, Ecotoxicol. Environ. Saf. 207 (2021) 111514.
[47] R. Khademolhosseini, A. Jafari, S.M. Mousavi, H. Hajfarajollah, K.A. Noghabi, M. Manteghian, Physicochemical characterization and optimization of glycolipid biosurfactant production by a native strain of *Pseudomonas aeruginosa* HAK01 and its performance evaluation for the MEOR process, RSC Adv. 9 (2019) 7932–7947.
[48] I. Abhyankar, G. Sevi, A.A. Prabhune, A. Nisal, S. Bayatigeri, Myristic acid derived sophorolipid: efficient synthesis and enhanced antibacterial activity, ACS Omega 6 (2) (2021) 1273–1279.
[49] P. Das, S. Mukherjee, R. Sen, Antimicrobial potential of a lipopeptide biosurfactant derived from a marine *Bacillus circulans*, J. Appl. Microbiol. 104 (6) (2008) 1675–1684.
[50] K.R. Meena, S.S. Kanwar, Lipopeptides as the antifungal and antibacterial agents: applications in food safety and therapeutics, Biomed. Res. Int. 2015 (2015) 463050.
[51] J.C. León, A. Manresa, A.M. Marques, Lichenysin production and application in the pharmaceutical field, in: D. Muñoz-Torrero, À. Domínguez, À. Manresa (Eds.), Recent Advances in Pharmaceutical Sciences, Research Signpost, Kerala, India, 2016, pp. 147–163.
[52] A. Merghni, I. Dallel, E. Noumi, Y. Kadmi, H. Hentati, S. Tobji, A.B. Amor, M. Mastouri, Antioxidant and antiproliferative potential of biosurfactants isolated from *Lactobacillus casei* and their anti-biofilm effect in oral *Staphylococcus aureus* strains, Microb. Pathog. 104 (2017) 84–89.
[53] R. Zouari, S. Besbes, S. Ellouze-Chaabouni, D. Ghribi-Aydi, Cookies from composite wheat–sesame peels flours: dough quality and effect of *Bacillus subtilis* SPB1 biosurfactant addition, Food Chem. 194 (2016) 758–769.

[54] J.M. Campos, T.L.M. Stamford, R.D. Rufino, J.M. Luna, T.C.M. Stamford, L.A. Sarubbo, Formulation of mayonnaise with the addition of a bioemulsifier isolated from Candida utilis formulation of mayonnaise with the addition of a bioemulsifier isolated from *Candida utilis*, Toxicol. Rep. 2 (2015) 1164–1170.
[55] H.S. Lee, K.H. Kim, S.H. Park, S.W. Hur, J.H. Auh, Amylose-lipid complex as a fat replacement in the preparation of low-fat white pan bread, Foods 9 (2) (2020).
[56] S. Fois, P.P. Piu, M. Sanna, T. Roggio, P. Catzeddu, In vivo and in vitro starch digestibility of fresh pasta produced using semolina-based or wholemeal semolina-based liquid sourdough, Foods 10 (10) (2021).
[57] M.H. Azizi, G.V. Rao, Effect of surfactant gels on dough rheological characteristics and quality of bread, Crit. Rev. Food Sci. Nutr. 44 (7–8) (2005) 545–552.
[58] M. Kurakake, H. Hagiwara, T. Komaki, Effects of various surfactants on rheological properties of maize starch granules, Cereal Chem. 81 (1) (2004) 108–114.
[59] N. Arshad, A. Ahmad, M. Nadeem, R.M. Amir, M.S. Arshad, Reporting the utilization and perspectives of different surface active agents for bread making, Food Sci. Technol. 40 (2020) 312–321.
[60] C.A. Valenzuela, E.J. Baker, E.A. Miles, P.C. Calder, Eighteen-carbon *trans* fatty acids and inflammation in the context of atherosclerosis, Prog. Lipid Res. 76 (2019) 101009.
[61] T. Punrata, J. Thaniyavarn, S.C. Napathorn, J. Anuntagool, S. Thaniyavarn, Production of a sophorolipid biosurfactant by *Wickerhamomyces anomalus* MUE24 and its use for modification of rice flour properties, Sci. Asia 46 (2020) 11–18.
[62] A.K. Al-Asady, W.A. Al-Waely, G.H. Majeed, Study of rhamnolipids cytotoxicity, inhibitory effect on some microorganisms and applying in food products, Int. J. Bio-Technol. Res. 6 (3) (2016) 1–8.
[63] S.S. Behera, R.C. Ray, N. Zdolec, *Lactobacillus plantarum* with functional properties: an approach to increase safety and shelf-life of fermented foods, Biomed. Res. Int. 2018 (2018) 9361614.
[64] N. Kosaric, F.V. Sukan, Biosurfactants: Production and Utilization-Processes, Technologies, and Economics, CRC Press, Boca Raton, 2014.
[65] S.K. Satpute, S.S. Zinjarde, I.M. Banat, Recent Updates on Biosurfactant/s in Food Industry, CRC Press, Boca Raton, 2018.
[66] S.K. Satpute, A.G. Banpurkar, P.K. Dhakephalkar, I.M. Banat, B.A. Chopade, Methods for investigating biosurfactants and bioemulsifiers: a review, Crit. Rev. Biotechnol. 30 (2) (2010) 127–144.
[67] J.M. Campos, T.L.M. Stamford, L.A. Sarubbo, J.M.d. Luna, R.D. Rufino, I.M. Banat, Microbial biosurfactants as additives for food industrie, Biotechnol. Prog. 29 (5) (2013) 1097–1108.
[68] I.V. Haesendonck, E. Vanzeveren, Rhamnolipids in Bakery Products, Puratos NV, United States, 2002.

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CHAPTER

9 The application of biosurfactant for the modification of the rheological properties of wheat dough

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1 Introduction

Surfactants—surface-active agents are the chemical compounds that are usually developed from amphipathic molecules constituting hydrophilic and hydrophobic functional groups that segregate at the juncture between liquid, gas, and solid phases such as oil–water, and air–water interfaces, having varying degrees of polarity and hydrogen bonding. The nonpolar group is designated as a hydrocarbon chain, and the polar group as (a) ionic, that is, cationic or anionic, (b) nonionic, and (c) amphoteric [1]. Surfactants are majorly classified following their capacity for altering the surface properties and interfacial characteristics of a liquid ensuring the formation of micro-emulsions where the oil is solubilized in water and vis-à-vis [2]. Such significant properties facilitate a broad spectrum of applications in distinctive industries, for instance, (a) emulsification, (b) detergent, (c) foaming capacity, (d) lubrication, (e) moisture retention, (f) solubilization, and (g) phase dispersions [3]. Recently, microbial-derived surfactants have grabbed the attention of researchers owing to their ecological adaptability, biodegradability, eco-friendliness, etc. The term biosurfactant is defined as the term significantly obtained from metabolic by-products of yeasts, bacteria, and filamentous fungi, which constitutes emulsification and dispersing agents. Such biosurfactants have significant applications in diverse industries such as petroleum industries, petrochemical industries, food and beverage industries, pharmaceutics [4,5], cosmetics, mining, metallurgical, agrochemical, paper, and pulp industries [6,7].

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Wheat is a substantial cereal crop all over the globe due to the crucial constituent of human nutrition. Durum wheat (*Triticum durum* Desf.) owing to its great quality and economic properties has been extensively used globally. Their grains are larger, heavier, and harder in comparison to the bread wheat. Furthermore, it has high protein content and strength and hence ensures diverse uses [8,9]. Contrarily, the bread wheat (*Triticum aestivum* L.) is used normally for producing different dough-based food items such as bread. Soft wheat flour constitutes nearly 7%–15% protein, 80%–90% gluten, which renders wheat flour the potential to structure a cohesive dough when hydrated and ensures a continuous viscoelastic network during the making of soft dough [10,11]. Apart from analyzing the quality characteristic properties, the rheological properties of dough are significantly emphasized because of their fundamental impact on the (i) technology employed for making dough and (ii) sensory approval of the end product [12]. The adequate management of the process for making dough products is accomplished via frequent measurement of numerous quality-based properties [13]. The measurement of the extensional characteristic properties of flour holds significant importance to evaluate the baking condition of wheat-based flour [14]. There are numerous approaches for scrutinizing the extensional characteristics of dough such as extensograph—mostly exploited and considered as a vital international standard method [15], the Kieffer method—obtained from the basic extensographic procedure, which decreases the sample size and renders good command and hence frequently employed for determining the quality of flour and characteristic properties of dough [16].

For producing baked goods or dough with value additives, it is of sheer significance to choose materials with high content of fiber. The high fiber content has a direct effect on the rheological characteristics of the dough via interaction between fiber components and structural contents of the dough considering both mixing and stretching state, respectively [17]. The structural alterations in the dough due to additions of the fibers have a bad impact on certain attributes of the bread, particularly the (i) volume of the loaf and (ii) structure of the crumbs. Generally, two hypotheses influence the quality of bread and dough, respectively. Firstly, the fibers attach with water and decrease the supply of water accessible to supplementary constituents of the dough. This results in redistributing of the moisture in the dough following the fractional gluten dehydration. Secondly, fibers physically disrupt the network of gluten. Considering reported research data [18], the second hypothesis alters the conformation of the gluten by transforming the β-spiral structure into an intermolecular β-sheet structure. The absence of the elastic β-spiral structure to intermolecular β-sheets in the gluten network decreases the viscoelastic property of the dough [19,20]. This book chapter discusses the impact of diverse biosurfactants on the rheological characteristics of wheat dough. Additionally, this chapter will give a brief insight into the concept of the (1) biosurfactants and their classification, (2) factors affecting the production of biosurfactants, and (3) rheology with an intent to acknowledge the readers with a better understanding of such concepts.

2 Biosurfactants

2.1 Classification

Biosurfactants are majorly classified following the chemical structure of the biosurfactants and the origin of microbes (Fig. 1). Furthermore, they are categorized into molecular weights (500–1500 Da) by Rosenberg and Ron [22], namely (i) high-molecular-weight molecules (MW), which constitute polymeric surfactants and particulate surfactants; and (ii) low MW molecules, which constitute glycolipids, lipopeptides, and phospholipids.

The majority of the biosurfactants are either anionic or neutral; on the other hand, biosurfactants that constitute amine functional groups are designated as cationic [21,23]. Table 1 describes different classifications of biosurfactants and their respective microbial origin.

2.2 Properties

It is of extreme significance to scrutinize the commercial potential of the product by evaluating certain properties of biosurfactants such as surface tension, dispersive properties, toxicity, etc. Table 2 discusses certain properties of biosurfactants that must be considered for commercialization of the biosurfactant in different industries particularly food industries [28].

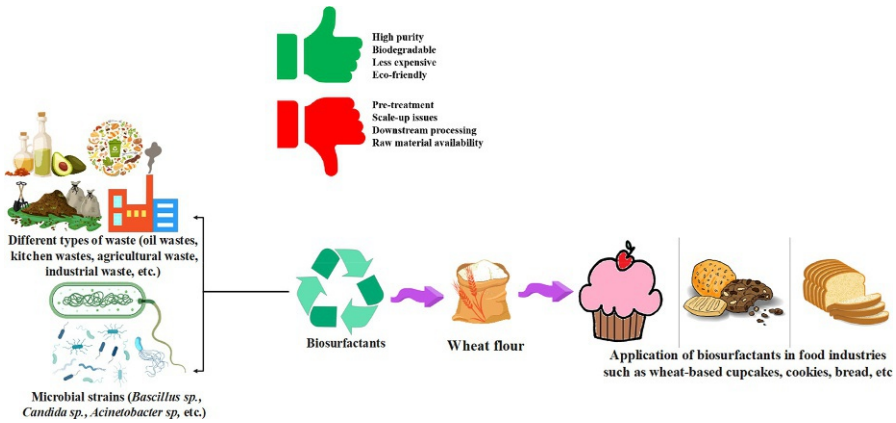

FIG. 1

Diagrammatic representation of the production of biosurfactants from different waste sources in combination with varied microbial sources for their applications in food industries [6,21].

Table 1 Categorization of the biosurfactants and their microbial origin [21].

Classification	Microbial origin
Polymeric surfactants	*Acinetobacter calcoaceticus, Bacillus stearothermophilus, Candida lipolytica, Candida utilis, Halomonas eurihalina, M. thermoautotrophium, Sphingomonas paucimobilis*
Lipopeptides	*Acinetobacter* sp., *Bacillus licheniformis, Bacillus pumilus, Bacillus subtilis, Candida lipolytica, Gluconobacter cerinus, Pseudomonas fluorescens, Serratia marcescens, Streptomyces sioyaensis, Thiobacillus thiooxidans*
Fatty acids	*Arthrobacter paraffineus, Capnocytophaga* sp., *Corynebacterium insidiosum, Corynebacterium lepus, Nocardia erythropolis, Penicillium spiculisporum, Talaramyces trachyspermus*
Particulate surfactants	*Acinetobacter calcoaceticus, Cyanobacteria, Pseudomonas marginalis*
Phospholipids	*Acinetobacter* sp., *Aspergillus, Corynebacterium lepus*
Glycolipids	*Acinetobacter calcoaceticus, Alcanivorax borkumensis, Arthrobacter* sp., *Candida antartica, Candida apicola, Candida batistae, Candida bogoriensis, Candida bombicola, Candida ishiwadae, Candida lipolytica, Lactobascillus fermentum, Nocardia* sp., *Pseudomonas* sp., *Rhodococcus erythropolis, Rhodotorula glutinis, Rhodotorula graminus, Serratia marcescens, Tsukamurella* sp., *Utilago maydis*

Table 2 Important factoring affecting the properties of biosurfactants [21,24–27].

Properties	Description
Surface and interfacial activity	Both efficacy and effectiveness are fundamental properties of an ideal biosurfactant. Compared to chemical surfactants, biosurfactants have 10–40 times fewer CMC values. The efficacy is measured through CMC, on the other hand, competency is affiliated with the surface tensions (ST) and interfacial tensions (IT), respectively. The range of CMC usually occurs between 1 and 2000 mg/L, on the other hand, both the interfacial tension (oil-in-water) and surface tensions ranges nearly 1 mN/m and 30 mN/m. Considering the value, the ideal biosurfactants have the potential to decrease the water ST from 72 to 35 mN/m and IT of water/organic solvent from 40 to 1 mN/m. Biosurfactant from *Bacillus subtilis* could decrease the ST of water to nearly 25 mN/m and IT of water/hexadecane to less than 1 mN/m. The production of rhamnolipids from *Pseudomonas aeruginosa* reduces the ST of water to nearly 26 mN/m and IT to less than 1 mN/m

Table 2 Important factoring affecting the properties of biosurfactants [21,24–27]. *Continued*

Properties	Description
Temperature pH Ionic strength	There are diverse biosurfactants that are exploited at higher (i) temperatures and (ii) pH (ranges between 2 and 12) values. Additionally, they have the potential to tolerate high concentrations of different salts such as 10% or more. Their surface activity is least affected in the presence of extreme environmental conditions. A documented research work signifies that lichenysin from *Bacillus licheniformis* JF-2 has the least effect at higher temperatures (50°C), pH values (4.5–9.0), and concentration of NaCl (50 g/L), respectively. Another research work describes the properties of lipopeptide obtained from *B. subtilis* LB5a. It was steady even at 121°C and after 6 months, the ST was still the same at −18°C, pH 5.0–11.0, and 20% NaCl concentrations, respectively. Diverse industrial processes periodically implicate the exposure of diverse physiological conditions such as temperature, pressure, pH, and ionic strength, resulting in a constant need of isolating novel products obtained from different microbes intending to function under such extreme conditions
Biodegradation	Biosurfactants are rapidly biodegraded by various microbes in water and soil, rendering them capable of numerous environmental applications such as bioremediation, treatment of wastewater, etc.
Toxicity	Less toxicity ensures the exploitation of biosurfactants in the food sector, cosmetics, and pharmaceuticals. Furthermore, biosurfactants can be generated from raw materials that are accessible in bulk or from numerous industrial wastes. A report has suggested that a synthetic anionic surfactant, namely Corexit exhibited 10 times less LC_{50} against *Pseudomonas* sp. as compared to the rhamnolipids which demonstrates the increased toxicity of the synthetic surfactants
Specificity	Biosurfactants are defined as composite molecules constituting specific functional moieties and hence usually have a specific mechanism of action
Biocompatibility and digestibility	Biosurfactants are biocompatible and exhibit good digestibility properties rendering them crucial for their use in food industries
Emulsion forming and breaking	Biosurfactants are considered as (i) emulsifiers and (ii) de-emulsifiers. By definition, emulsion is a composite organization constituting an immiscible liquid that is dissolved in a different liquid in the shape of the droplet. The diameter usually is more than 0.1 mm. Emulsions are least stable but when the biosurfactants are added such emulsions are stable for longer durations like for months or years. For instance, liposan (*Candida lipolytica*) is an emulsifier that is soluble in water and is generally exploited with edible oils forming stable emulsions

2.3 Factors influencing the production process of biosurfactants

The manufacturing process of biosurfactants is either spontaneous/impulsive or induced via lipophilic compounds, varied pH values, temperature, aeration, agitation speed, and growth of cells in stress conditions such as limited nitrogen concentration [29,30]. Numerous physicochemical factors are discussed below [31].

2.3.1 Carbon sources

The carbon source has a fundamental function both in the growth and production of biosurfactants. For example, it has been mentioned that when glucose and vegetable oil are used to produce biosurfactant from the species *Torulopsis bombicola,* very less yield was observed in comparison to the production of the same biosurfactant when both carbon sources were exploited synergistically (70 g/L) [32]. Another research work concluded when both glucose (80 g/L) and soybean oil (40 g/L) were used and resulted in a maximum yield of the sophorose lipids from *T. bombicola* [33]. Even larger yields of sophorolipids (120 g/L) biosurfactants were obtained when both sugar and oil were used altogether [34]. Furthermore, when industrial wastes are exploited for producing biosurfactants from *Candida lipolytica*, it has been observed that the yield of the protein–lipid–carbohydrate complex was 4.5 g/L, with a decrease in the ST of distilled water from 71 to 32 mN/m [35].

2.3.2 Nitrogen sources

The nitrogen source is considered the second most crucial additive for producing biosurfactants via microbes. In various fermentative procedures, the C/N ratio significantly influences the build-up of metabolites. The increased C/N ratio, that is, less level of nitrogen, restricts the growth of bacteria supporting the metabolism of cells toward metabolite production. Contrarily, excess nitrogen results in the synthesis of cell-based material and impedes the building up of products [36]. There are diverse organic and inorganic sources of nitrogen, which are exploited for producing biosurfactants. A group of researchers [37] have interpreted the significance of nitrogen for producing biosurfactants via *Pseudomonas aeruginosa* following the cultivation in a media constituting minerals with glycerol (3%). Since sodium nitrate ($NaNO_3$) is more efficient as compared with ammonium sulfate ($(NH_4)_2SO_4$), but the nutritional restrictions result in the product formation. Another group [38] reported that *P. aeruginosa* employs (i) nitrates, (ii) ammonium, and (iii) amino acids as different sources of nitrogen.

Nitrates (NO_3^-) are initially converted into nitrites (NO_2^-) and then ammonium (NH_4^+). Ammonium is further integrated via glutamate dehydrogenase (GDH) to form glutamate/glutamine synthetase, further resulting in the formation of glutamine. Both lutamine and α-ketoglutarate are then reduced to form glutamine via L-glutamine-2-oxoglutarate aminotransferase. Nonetheless, the formation of lipids instead of sugars is considered as the rate-determining step in the synthesis of rhamnolipids, and the restricted nitrogen source resulted in the agglomeration of formed lipids. Compared with the ammonium, the nitrate assimilation is slower, which

further boosts the restricted nitrogen supply favoring the rhamnolipids production. High production yields of sophorose lipids (obtained from the microbial species *T. bombicola* and *Candida bombicola*) have been accomplished by exploiting yeast extract and urea as the source of nitrogen [39]. Additionally, increased yields of mannosylerythritol lipids (*Candida sp.* SY16, *C. lipolytica, and C. glabrata*) are accomplished by using ammonium nitrate and yeast extract as the source of nitrogen [40].

2.3.3 Growth environment

Diverse growth environment such as temperature, pH, agitation speed, and oxygen (O_2) has a significant impact on the production of biosurfactants[30]. *Candida* sp. produces a maximal yield of biosurfactants in a broad range of pH, for example, pH = 5.7 for *Candida glabrata* UCP 1002, pH = 7.8 for *Candida sp.* SY16, pH = 5.0 for *C. lipolytica,* and pH = 6.0 for *Candida batistae* [41,42]. Additionally, *Pichia anomala* and *Aspergillus ustus* produce maximal yield of biosurfactant at pH 5.5 and 7.0, respectively [41,43]. Even a minute change in the temperature has a significant effect on various microbial production processes. The most desirable temperature for producing biosurfactants by varied species of fungi is 30°C, as observed for diverse *Candida sp.*, namely *Candida sp.* SY16, *C. bombicola*, *C. batistae*, and *T. bombicola* [42]. For *C. lipolytica*, it has been observed that 27°C is the optimal temperature for producing desired biosurfactants. Apart from the aforementioned growth conditions, incubation time also has a crucial influence on the biosurfactant production process. Microbes generally cultivate biosurfactants in different time durations. A documented research report suggests that for producing biosurfactants with maximal yields by fungal species *A. ustus* 5 days of incubation is required on the other hand for *C. bombicola*, the incubation time is 7 days, 8 days, and 11 days, respectively [44]. Additionally, increasing the agitation speed favors the agglomeration of biosurfactants from *P. aeruginosa* UCP 0992 grown in medium glycerol [45]. A group investigated the impact of altering the agitation speed of microbial cultures from 50 RPM to 200 RPM on *Pseudomonas alcaligenes*, which was cultured in palm oil [46]. They have observed that increasing the agitation speed decreases the ST of the cell-free broth to 27.6 mN/m. Contrastingly, another research [47] has observed that agitation speed has an adverse impact considering the decrease of ST by exploiting biosurfactant obtained from *Serratia sp.* SVGG16 was grown in a hydrocarbon-based culture [21].

3 Rheology: A brief description

Rheology—the science describing deformation and flow of matter. It is defined as the study of the response of the materials when stress or strain is applied. Every material exhibits rheological characteristics, which are characterized via rheometers. Numerous rheometers are usually employed for measuring the rheological characteristics of wheat doughs such as penetrometer, consistometer, amylograph, farinograph, mixograph, extensigraph, and alveograph [48].

The applicative domain of rheology has extended into the food sector. There is plenty of research that has been conducted for understanding the rheology of numerous varieties of food such as powder [49], liquids [50], gels [51], emulsions [52], and pastes [53]. Diverse food products exhibit the rheological behavior, which further categorizes them into liquid and solid states meaning varied characteristics in viscous and elastic behaviors both. This particular behavior is described as viscoelasticity induced by the intricate network of long-chain molecules with other molecules [48].

The dough is designated as a viscoelastic and shear-thinning material in combination with the Hookean solid and non-Newtonian viscous liquid. The dough exhibits nonlinear rheological behavior, but extremely small strains display linear property. The small strain is dependent on $G' = \frac{\tau_0 \cos\theta}{\gamma_0}$ $G'' = \frac{\tau_0 \sin\theta}{\gamma_0}$ $tg\ \delta = \frac{G''}{G'}$ various properties such as (1) dough type, (2) mixing, and (3) testing approach. Both storage modulus (G') and loss modulus (G$''$) are two definite terms that are used to describe the rheological properties of the dough as mentioned below:

When the value of strain is $<0.1\%$, storage modulus (G') is greater than the loss modulus (G'') but when it is $>0.1\%$, then the loss modulus (G'') is greater than the storage modulus (G') because of the conversion of viscoelastic solid into elastoviscous liquid [54]. Adding certain ingredients into the dough such as yeast, water, etc., results in the thixotropic behavior of the dough. The inclusion of certain ingredients such as yeast and salt in the mixture of water and flour renders thixotropy. The viscoelastic properties are contributed due to the presence of gluten protein in which the gliadin part symbolizes the viscous property, and glutenin part symbolizes the elastic property owing to the varied molecular sizes of the portions [55]. Increasing the amount of protein in the dough results in larger consistencies. The greater the gluten protein in the dough, the greater the storage modulus than loss modulus [56]. Hydrophobic interactions have a significant role to play in rendering the elastic property of the doughs [57]. An increase in the intermolecular cross-linkage renders increased storage modulus and decreased loss tangent in the wheat dough. The interactions constituting both physical and chemical forces along with the molecules of protein have a crucial function in modulating the rheology of the dough [58].

4 Biosurfactants and rheological properties of wheat-flour-derived doughs

The wheat dough portrays viscoelastic behavior, which means the combination of both viscous fluid and elastic solid, respectively. Wheat flour, water, and yeast are the most crucial ingredients while preparing the dough. Every ingredient has a significant role to play while representing the structural properties and mechanical properties of the dough. The major elements of the flour-like gluten, gas cells, and starch have a fundamental impact on the characteristics of the final product such as texture. The size and number of gas cells vary in consideration with the type of the final product [59]. Another factor, dough aeration impacts the rheological

characteristics via gas bubbles due to mixing, which increases the rate of work input and facilitates the supply of oxygen, and further results in the development of stable morphology [60].

The important mechanical characteristics of dough are very crucial since they alter (a) the handling action of dough amid processing and (b) the interactions encompassed by diverse constituents of wheat dough [61]. Rheology is generally the first set of characterization analysis during the product formulation. The rheological characterization acknowledges the flow characteristics and viscosity profile of the formulated product [62], for instance, if the formulated product requires a shear-thinning behavior, which can be readily studied via the viscosity profile of the product. Nevertheless, not extensive research has been done considering the exploitation of biosurfactants and their impact on the rheological characteristics of the dough.

A research group [63] exploited rhamnolipids-based biosurfactants, which behave like the natural emulsifying agent. The research works aim at developing concentrated emulsions, nearly 50 wt% oil. It has been observed that these biosurfactants were more efficient for producing small-sized droplets. Furthermore, with the increasing concentration of rhamnolipids, there has been a significant reduction in particle size. Such concentrated emulsions exhibit shear-thinning property at a lower shear rate (between 0.1 and $100\,s^{-1}$). This study also studied the stability of these concentrated emulsions, which indicates that biosurfactant behaves like a good emulsifying agent in the system [64].

Another research group exploits *Saccharomyces cerevisiae* URM 6670 biosurfactant intending to improve the rheology of the cookie dough constituting wheat flour and soybean oil, respectively. The biosurfactant from *S. cerevisiae* showed a glycolipid structure having a fatty acid profile, which reveals a high percentage of linoleic acid, 50.58%. The thermal characterization investigation showed better stability at higher temperatures considering industrial applications, with little or no loss of mass at high temperatures ($\geq$200°C). Furthermore, from various cell culture experiments, it has been observed that the biosurfactant was not toxic to the fibroblast cell lines and macrophage cell lines, respectively exhibiting cell inhibition $<$15% [65,66].

Another group [67] developed a cost-effective biosurfactant via *Bacillus cereus* UCP 1615 microbial strain from waste frying soybean oil behaving as the emulsifying agent for forming cookies (Fig. 2). Additionally, the group also investigated the physicochemical properties of the final product at different biosurfactant concentrations such as 0.25%, 0.5%, and 1%, respectively. From the analysis, it has been found out that biosurfactant-based cookies showed the same energy and physical characteristic properties in comparison to the standard protocol for forming cookies but with less moisture content. In addition to this, the presence of biosurfactant resulted in the preservation of the texture of formed cookies even after 45 days of storage. Hence, such results have suggested that biosurfactants are a potential candidate as a green emulsifying agent taking into consideration the demands of the present world of producing biocompatible products with better rheological characteristics and shelf life [65,66].

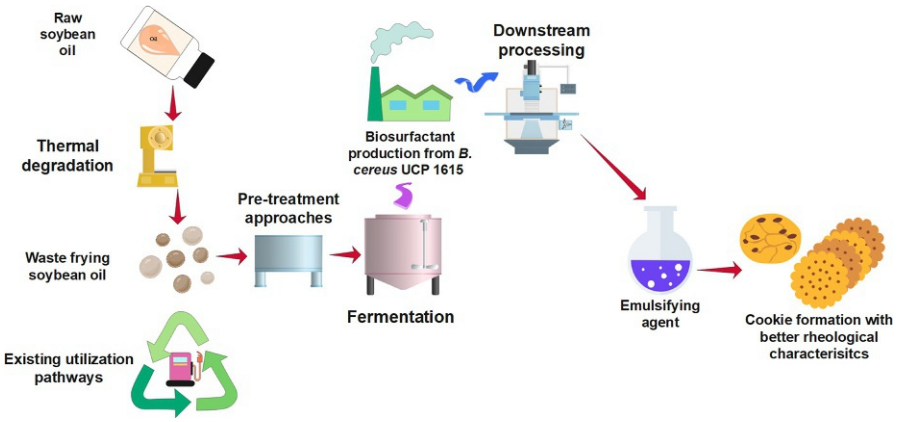

FIG. 2

Diagrammatic description of employment of soybean oil for generating biosurfactant from *Bacillus cereus* UCP 1615 microbial strain [67].

In another work, a team [68] investigated the impact of natural glycolipid biosurfactant, namely mannosythritol lipids (MELs) on the gluten network of frozen dough following the quality and microbial spoilage of the dough. By performing the rheology and other characterization analyses such as differential scanning calorimetry (DSC), it has been observed that MEL had a significant impact on (a) improving the rheological characteristics of frozen dough and (b) decreasing the content of frozen water. In spite of the alterations in the loss and storage modulus values, the value of tan δ is always less than 1 at high frequencies (>20 Hz), meaning the dominant influence of elastic property over the viscous property amidst the complete dynamic scan test. Following the reported data, the decreased viscoelasticity in the frozen dough is because of the mechanical loss of gluten networks during freezing storage conditions, which has a severe impact on forming diverse final products resulting in decreased loaf volume [69]. However, MEL biosurfactants significantly strengthen the gluten networks of frozen dough by conjugating the hydrophobic tails of the emulsifiers to the hydrophobic sites of the gluten protein, which reduces the net charge on the surface of the protein via neutralization of electrostatic repulsive forces resulting in better spongy structure and loaf volume [70,71].

Another research group [72] demonstrates the meal additive (cupcake) utilization of the biosurfactant obtained from microbial strain *C. bombicola* URM 3718 developed in a medium constituting sugarcane molasses (5%), residual soybean oil (5%), and corn steep liquor (3%), respectively. From the analysis, the ST and IT of the biosurfactant were 30.790 ± 0.04 mN/m and 0.730 ± 0.05 mN/m, respectively. The yield obtained was 25 ± 1.02 g/L, and the critical micelle concentration was 0.5 g/L, respectively. The incorporation of the biosurfactant into the cupcake dough replaces 50%, 75%, and 100% of the vegetable fat in the standard cupcake dough (without biosurfactant). The thermal report exhibited stability of the biosurfactant for cooking cupcakes at higher temperatures, that is, 180°C [72].

5 Conclusion and future perspectives

The urgent and dire need for natural foods has forced researchers to explore and identify natural alternatives for their applications in food industries. Biosurfactants, nowadays, are considered as a potential alternative of chemical surfactants because of certain less toxicity, better physicochemical constituting surface tension, interfacial activity, bioavailability, biodegradability, diverse structures, production from a varied renewable substrate, emulsifying activity, antimicrobial properties, and antiadhesive activities. Nevertheless, despite their potential characteristics, their substantial applications in the food industries are a significant obstacle. There is plenty of justifications such as very little information regarding the toxicity, varied composition of the final product, which is affected via nutrients, micronutrients, and environmental factors, respectively. Additionally, in comparison to the chemical surfactants, the high cost of biosurfactant production and issues associated with scale-up were hampered owing to their broad-range applications in different industries. Nevertheless, in the food industries where valuable products are usually generated, but this disadvantage is less consequential in comparison to other nations. There are numerous strategies for decreasing the high cost of production, which constitutes (1) cheap and sustainable resources such as agro-industrial by-products/wastes, (2) optimized production process and downstream processes, and (3) advanced microorganisms. Regardless of such challenges and hindrances, biosurfactants have an unquestionable potential for replacing traditional surfactants and render various perks to human health and the ecosystem. Soon biosurfactants will assuredly become a significant part of the food industry with diverse applications.

References

[1] A. Singh, J.D. Van Hamme, O.P. Ward, Surfactants in microbiology and biotechnology: part 2. Application aspects, Biotechnol. Adv. 25 (1) (2007) 99–121, https://doi.org/10.1016/j.biotechadv.2006.10.004.

[2] T.T.L. Nguyen, A. Edelen, B. Neighbors, D.A. Sabatini, Biocompatible lecithin-based microemulsions with rhamnolipid and sophorolipid biosurfactants: formulation and potential applications, J. Colloid Interface Sci. 348 (2) (2010) 498–504, https://doi.org/10.1016/j.jcis.2010.04.053.

[3] F. Barros, C. de Quadros, G. Pastore, Studies of emulsifying properties and stability of the biosurfactant produced by Bacillus subtilis in cassava wastewater, Food Sci. Technol. 28 (2008) 979–985, https://doi.org/10.1590/S0101-20612008000400034.

[4] L. Fracchia, M. Cavallo, M. Martinotti, I. Banat, Biosurfactants and bioemulsifiers biomedical and related applications—present status and future potentials, in: D. Ghista (Ed.), Biomedical Science, Engineering and Technology, IntechOpen, London, 2012, https://doi.org/10.5772/23821.

[5] P. Saravanakumari, K. Mani, Structural characterization of a novel xylolipid biosurfactant from Lactococcus lactis and analysis of antibacterial activity against multi-drug resistant pathogens, Bioresour. Technol. 101 (22) (2010) 8851–8854, https://doi.org/10.1016/j.biortech.2010.06.104.

[6] J.M. Campos, T.L.M. Stamford, L.A. Sarubbo, J. Moura, R. de Luna, D. Rufino, I.M. Banat, Microbial biosurfactants as additives for food industries, Biotechnol. Prog. 29 (5) (2013) 1097–1108, https://doi.org/10.1002/btpr.1796.
[7] T.J. Smyth, A. Perfumo, R. Marchant, I.M. Banat, M. Chen, R.K. Thomas, J. Penfold, P. S. Stevenson, N.J. Parry, Directed microbial biosynthesis of deuterated biosurfactants and potential future application to other bioactive molecules, Appl. Microbiol. Biotechnol. 87 (4) (2010) 1347–1354, https://doi.org/10.1007/s00253-010-2592-5.
[8] D. Altındal, Effect of seaweed extract (SE) on seed germination characteristics of wheat in salty conditions, Int. J. Agric. For. Life Sci. 3 (2019) 115–120.
[9] N. Zhang, J. Yan, S. Zhang, Rht13 dwarfing gene delays foliar senescence in wheat induced by nitrogen deficiency, Pak. J. Bot. 51 (2019), https://doi.org/10.30848/PJB2019-1(33).
[10] V. Landillon, D. Cassan, M.-H. Morel, B. Cuq, Flowability, cohesive, and granulation properties of wheat powders, J. Food Eng. 86 (2008) 178–193, https://doi.org/10.1016/j.jfoodeng.2007.09.022.
[11] P.R. Shewry, N.G. Halford, P.S. Belton, A.S. Tatham, The structure and properties of gluten: an elastic protein from wheat grain, Philos. Trans. R. Soc. Lond. Ser. B Biol. Sci. 357 (1418) (2002) 133–142, https://doi.org/10.1098/rstb.2001.1024.
[12] B.J. Dobraszczyk, M.P. Morgenstern, Rheology and the breadmaking process, J. Cereal Sci. 38 (3) (2003) 229–245, https://doi.org/10.1016/S0733-5210(03)00059-6.
[13] A. Mondal, A.K. Datta, Bread baking—a review, J. Food Eng. 86 (4) (2008) 465–474, https://doi.org/10.1016/j.jfoodeng.2007.11.014.
[14] A. Miś, D. Dziki, Extensograph curve profile model used for characterising the impact of dietary fibre on wheat dough, J. Cereal Sci. 57 (3) (2013) 471–479, https://doi.org/10.1016/j.jcs.2013.02.004.
[15] D. Indrani, P. Swetha, C. Soumya, J. Rajiv, G. Venkateswara Rao, Effect of multigrains on rheological, microstructural and quality characteristics of north indian parotta—an Indian flat bread, LWT Food Sci. Technol. 44 (3) (2011) 719–724, https://doi.org/10.1016/j.lwt.2010.11.017.
[16] R. Kieffer, M.H. Herbert Wieser, Henderson, and Aris Graveland., Correlations of the breadmaking performance of wheat flour with rheological measurements on a micro-scale, J. Cereal Sci. 27 (1998) 53–60.
[17] A. Miś, Interpretation of mechanical spectra of carob fibre and oat wholemeal-enriched wheat dough using non-linear regression models, J. Food Eng. 102 (4) (2011) 369–379, https://doi.org/10.1016/j.jfoodeng.2010.09.015.
[18] J.E. Bock, S. Damodaran, Bran-induced changes in water structure and gluten conformation in model gluten dough studied by fourier transform infrared spectroscopy, Food Hydrocoll. 31 (2) (2013) 146–155, https://doi.org/10.1016/j.foodhyd.2012.10.014.
[19] D. Šoronja-Simović, J. Zahorec, Z. Šereš, N. Maravić, S.S. Možina, L. Luskar, J. Luković, Challenges in determination of rheological properties of wheat dough supplemented with industrial by-products: carob pod flour and sugar beet fibers, J. Food Meas. Charact. 15 (1) (2021) 914–922, https://doi.org/10.1007/s11694-020-00686-9.
[20] J. Wang, C.M. Rosell, C.B. de Barber, Effect of the addition of different fibres on wheat dough performance and bread quality, Food Chem. 79 (2) (2002) 221–226, https://doi.org/10.1016/S0308-8146(02)00135-8.
[21] D.K.F. Santos, R.D. Rufino, J.M. Luna, V.A. Santos, L.A. Sarubbo, Biosurfactants: multifunctional biomolecules of the 21st century, Int. J. Mol. Sci. 17 (3) (2016) 401, https://doi.org/10.3390/ijms17030401.

[22] E. Rosenberg, E. Ron, High- and low-molecular-mass microbial surfactants, Appl. Microbiol. Biotechnol. 52 (1999) 154–162, https://doi.org/10.1007/s002530051502.
[23] G. Bognolo, Biosurfactants as emulsifying agents for hydrocarbons, Colloids Surf. A Physicochem. Eng. Asp. 152 (1) (1999) 41–52, https://doi.org/10.1016/S0927-7757(98)00684-0.
[24] I.M. Banat, A. Franzetti, I. Gandolfi, G. Bestetti, M.G. Martinotti, L. Fracchia, T.J. Smyth, R. Marchant, Microbial biosurfactants production, applications and future potential, Appl. Microbiol. Biotechnol. 87 (2) (2010) 427–444, https://doi.org/10.1007/s00253-010-2589-0.
[25] F. Barros, C. Quadros, M.M. Jr, G. Pastore, Surfactin: chemical, technological and functional properties for food applications, Quim Nova 30 (2007) 409–414.
[26] M.C. Cirigliano, G.M. Carman, Purification and characterization of liposan, a bioemulsifier from Candida lipolytica, Appl. Environ. Microbiol. 50 (4) (1985) 846–850, https://doi.org/10.1128/aem.50.4.846-850.1985.
[27] M. Nitschke, S.G.V.A.O. Costa, Biosurfactants in food industry, Trends Food Sci. Technol. 18 (5) (2007) 252–259, https://doi.org/10.1016/j.tifs.2007.01.002.
[28] S. Vijayakuma, S. Varatharajan, Biosurfactants-types, sources and applications, Res. J. Microbiol. 10 (2015) 181–192, https://doi.org/10.3923/jm.2015.181.192.
[29] B. Tripathy, Divya, A. Mishra, Sustainable biosurfactants, in: Encyclopedia of Inorganic and Bioinorganic Chemistry, American Cancer Society, 2016, pp. 1–17.
[30] J.D. Desai, I.M. Banat, Microbial production of surfactants and their commercial potential, Microbiol. Mol. Biol. Rev. 61 (1) (1997) 47–64, https://doi.org/10.1128/mmbr.61.1.47-64.1997.
[31] G. Bhardwaj, S. Cameotra, H. Chopra, Biosurfactants from fungi: a review, J. Pet. Environ. Biotechnol. (2013), https://doi.org/10.4172/2157-7463.1000160.
[32] D.G. Cooper, D.A. Paddock, Production of a biosurfactant from Torulopsis bombicola, Appl. Environ. Microbiol. 47 (1) (1984) 173–176, https://doi.org/10.1128/aem.47.1.173-176.1984.
[33] S.Y. Kim, D.K. Oh, K.H. Lee, J.H. Kim, Effect of soybean oil and glucose on sophorose lipid fermentation by Torulopsis bombicola in continuous culture, Appl. Microbiol. Biotechnol. 48 (1) (1997) 23–26, https://doi.org/10.1007/s002530051009.
[34] J.A. Casas, S. García de Lara, F. García-Ochoa, Optimization of a synthetic medium for Candida bombicola growth using factorial design of experiments, Enzym. Microb. Technol. 21 (3) (1997) 221–229, https://doi.org/10.1016/S0141-0229(97)00038-0.
[35] R. Rufino, L. Sarubbo, G.M.J. Campos-Takaki, Enhancement of stability of biosurfactant produced by Candida lipolytica using industrial residue as substrate, World J. Microbiol. Biotechnol. 23 (2007) 729–734, https://doi.org/10.1007/s11274-006-9278-2.
[36] M. Robert, M.E. Mercadé, M.P. Bosch, J.L. Parra, M.J. Espuny, M.A. Manresa, J. Guinea, Effect of the carbon source on biosurfactant production by Pseudomonas Aeruginosa 44T1, Biotechnol. Lett. 11 (12) (1989) 871–874, https://doi.org/10.1007/BF01026843.
[37] L.M. Santa Anna, G.V. Sebastian, N. Jr Pereira, T.L. Alves, E.P. Menezes, D.M. Freire, Production of biosurfactant from a new and promising strain of Pseudomonas aeruginosa PA1, Appl. Biochem. Biotechnol. 91–93 (2001) 459–467, https://doi.org/10.1385/abab:91-93:1-9:459.
[38] C.N. Mulligan, B.F. Gibbs, Correlation of nitrogen metabolism with biosurfactant production by Pseudomonas aeruginosa, Appl. Environ. Microbiol. 55 (11) (1989) 3016–3019.

[39] M. Deshpande, L. Daniels, Evaluation of sophorolipid biosurfactant production by Candida bombicola using animal fat, Bioresour. Technol. 54 (2) (1995) 143–150, https://doi.org/10.1016/0960-8524(95)00116-6.
[40] H.S. Kim, B.D. Yoon, D.H. Choung, H.M. Oh, T. Katsuragi, Y. Tani, Characterization of a biosurfactant, mannosylerythritol lipid produced from Candida Sp. SY16, Appl. Microbiol. Biotechnol. 52 (5) (1999) 713–721, https://doi.org/10.1007/s002530051583.
[41] G. Kiran, T.A. Seghal, R.G. Hema, T. Joseph Selvin, T. Anto Thomas, R. Ravji, K. Natarajaseenivasan, Optimization and production of a biosurfactant from the sponge-associated marine fungus Aspergillus ustus MSF3, Colloids Surf. B: Biointerfaces 73 (2) (2009) 250–256, https://doi.org/10.1016/j.colsurfb.2009.05.025.
[42] M. Konishi, T. Fukuoka, T. Morita, T. Imura, D. Kitamoto, Production of new types of sophorolipids by Candida batistae, J. Oleo Sci. 57 (6) (2008) 359–369, https://doi.org/10.5650/jos.57.359.
[43] J. Thaniyavarn, T. Chianguthai, P. Sangvanich, N. Roongsawang, K. Washio, M. Morikawa, S. Thaniyavarn, Production of sophorolipid biosurfactant by pichia anomala, Biosci. Biotechnol. Biochem. 72 (8) (2008) 2061–2068, https://doi.org/10.1271/bbb.80166.
[44] P.A. Felse, V. Shah, J. Chan, K.J. Rao, R.A. Gross, Sophorolipid biosynthesis by Candida bombicola from industrial fatty acid residues, Enzym. Microb. Technol. 40 (2) (2007) 316–323, https://doi.org/10.1016/j.enzmictec.2006.04.013.
[45] S.N.R.L. Silva, C.B.B. Farias, R.D. Rufino, J.M. Luna, L.A. Sarubbo, Glycerol as substrate for the production of biosurfactant by Pseudomonas aeruginosa UCP0992, Colloids Surf. B: Biointerfaces 79 (1) (2010) 174–183, https://doi.org/10.1016/j.colsurfb.2010.03.050.
[46] F.J.S. Oliveira, L. Vazquez, N.P. de Campos, F.P. de França, Production of rhamnolipids by a Pseudomonas alcaligenes strain, Process Biochem. 44 (4) (2009) 383–389, https://doi.org/10.1016/j.procbio.2008.11.014.
[47] C.D. Cunha, M. do Rosário, A.S. Rosado, S.G.F. Leite, Serratia sp. SVGG16: a promising biosurfactant producer isolated from tropical soil during growth with ethanol-blended gasoline, Process Biochem. 39 (12) (2004) 2277–2282, https://doi.org/10.1016/j.procbio.2003.11.027.
[48] A. Zaidel, D. Norulfairuz, N. Chin, Y.A. Yusof, A review on rheological properties and measurements of dough and gluten, J. Appl. Sci. 10 (2010), https://doi.org/10.3923/jas.2010.2478.2490.
[49] J.A. Grabowski, V.D. Truong, C.R. Daubert, Nutritional and rheological characterization of spray dried sweetpotato powder, LWT Food Sci. Technol. 41 (2) (2008) 206–216, https://doi.org/10.1016/j.lwt.2007.02.019.
[50] Y.W. Park, Rheological characteristics of goat and sheep milk, Small Rumin. Res. 68 (1) (2007) 73–87, https://doi.org/10.1016/j.smallrumres.2006.09.015.
[51] E.A. Foegeding, Rheology and sensory texture of biopolymer gels, Curr. Opin. Colloid Interface Sci. 12 (4) (2007) 242–250, https://doi.org/10.1016/j.cocis.2007.07.001.
[52] M. Corredig, M. Alexander, Food emulsions studied by DWS: recent advances, Trends Food Sci. Technol. 19 (2) (2008) 67–75, https://doi.org/10.1016/j.tifs.2007.07.014.
[53] H.S. Lim, G. Narsimhan, Pasting and rheological behavior of soy protein-based pudding, LWT Food Sci. Technol. 39 (4) (2006) 344–350, https://doi.org/10.1016/j.lwt.2005.02.016.
[54] V. Rasper, M. Pico, R. Fulcher, Alveography in quality assessment of soft white winter wheat cultivars, Cereal Chem. 63 (1986) 395–400.

[55] S.-Y. Shiau, A.-I. Yeh, Effects of alkali and acid on dough rheological properties and characteristics of extruded noodles, J. Cereal Sci. 33 (1) (2001) 27–37, https://doi.org/10.1006/jcrs.2000.0344.
[56] J. Rouillé, G.D. Valle, J. Lefebvre, E. Sliwinski, T. vanVliet, Shear and extensional properties of bread doughs affected by their minor components, J. Cereal Sci. 42 (2005) 45–57, https://doi.org/10.1016/j.jcs.2004.12.008.
[57] J. Lefebvre, An outline of the non-linear viscoelastic behavior of wheat flour dough in shear, Rheol. Acta 45 (2006) 525–538, https://doi.org/10.1007/s00397-006-0093-3.
[58] H. Mirsaeedghazi, Z. Emam-Djomeh, M. Mousavi, Rheometric measurement of dough rheological characteristics and factors affecting it, Int. J. Agric. Biol. 10 (2008).
[59] P. Wilde, Foam formation in dough and bread quality, in: S.P. Cauvain (Ed.), Woodhead Publishing Series in Food Science, Technology and Nutrition, second ed., Woodhead Publishing, 2012, pp. 370–399 (Chapter 15).
[60] N. Chin, L.H. Tan, Y.A. Yusof, R.A. Rahman, Relationship between aeration and rheology of breads, J. Texture Stud. 40 (2009) 727–738, https://doi.org/10.1111/j.1745-4603.2009.00208.x.
[61] R. Upadhyay, D. Ghosal, A. Mehra, Characterization of bread dough: rheological properties and microstructure, J. Food Eng. 109 (1) (2012) 104–113, https://doi.org/10.1016/j.jfoodeng.2011.09.028.
[62] R. Angelico, S. Amin, M. Monduzzi, S. Murgia, U. Olsson, G. Palazzo, Impact of branching on the viscoelasticity of wormlike reverse micelles, Soft Matter 8 (42) (2012) 10941–10949, https://doi.org/10.1039/C2SM26528A.
[63] Z. Li, L. Dai, D. Wang, L. Mao, Y. Gao, Stabilization and rheology of concentrated emulsions using the natural emulsifiers Quillaja saponins and rhamnolipids, J. Agric. Food Chem. 66 (15) (2018) 3922–3929, https://doi.org/10.1021/acs.jafc.7b05291.
[64] C.E. Drakontis, S. Amin, Biosurfactants: formulations, properties, and applications, Curr. Opin. Colloid Interface Sci. 48 (2020) 77–90, https://doi.org/10.1016/j.cocis.2020.03.013.
[65] B.G. Ribeiro, J.M.C. Guerra, L.A. Sarubbo, Biosurfactants: production and application prospects in the food industry, Biotechnol. Prog. 36 (5) (2020), https://doi.org/10.1002/btpr.3030, e3030.
[66] B.G. Ribeiro, J.M. Campos Guerra, L.A. Sarubbo, Potential food application of a biosurfactant produced by Saccharomyces cerevisiae URM 6670, Front. Bioeng. Biotechnol. 8 (2020) 434, https://doi.org/10.3389/fbioe.2020.00434.
[67] I. Durval, B. Ribeiro, J. Aguiar, R. Rufino, A. Converti, L. Sarubbo, Application of a biosurfactant produced by Bacillus cereus UCP 1615 from waste frying oil as an emulsifier in a cookie formulation, Fermentation 7 (2021) 189, https://doi.org/10.3390/fermentation7030189.
[68] Q. Shu, T. Wei, X. Liu, S. Liu, Q. Chen, The dough-strengthening and spore-sterilizing effects of mannosylerythritol lipid-A in frozen dough and its application in bread making, Food Chem. 369 (2022), https://doi.org/10.1016/j.foodchem.2021.131011, 131011.
[69] P. Wang, Z. Jin, X. Xueming, Physicochemical alterations of wheat gluten proteins upon dough formation and frozen storage—a review from gluten, glutenin and gliadin perspectives, Trends Food Sci. Technol. 46 (2, Part A) (2015) 189–198, https://doi.org/10.1016/j.tifs.2015.10.005.
[70] N. Linlaud, E. Ferrer, M.C. Puppo, C. Ferrero, Hydrocolloid interaction with water, protein, and starch in wheat dough, J. Agric. Food Chem. 59 (2) (2011) 713–719, https://doi.org/10.1021/jf1026197.

[71] Z. Sheikholeslami, M. Mahfouzi, M. Karimi, M. Ghiafehdavoodi, Modification of dough characteristics and baking quality based on whole wheat flour by enzymes and emulsifiers supplementation, LWT Food Sci. Technol. 139 (2021), https://doi.org/10.1016/j.lwt.2020.110794, 110794.

[72] I.A. da Silva, B.O. de Veras, B.G. Ribeiro, J.S. Aguiar, J.M. Campos Guerra, J.M. de Luna, L.A. Sarubbo, Production of cupcake-like dessert containing microbial biosurfactant as an emulsifier, PeerJ 8 (2020).

CHAPTER

10 Biofabrication of nanobiosurfactant and their wide application in the food industry

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1 Introduction

Currently, the industrial sector has the most extensive multiplier action in all fields of the economy. Over the last 50 years, the food industry has presented growth quickly due to meet the needs of the citizens and modification lifestyles [1]. In 2020, the global market of foods had a growth rate of about 8.5%, and more than 305 billion dollars are expected [2]. Coulomb [3] has described that the citizens of developed countries who live in cities reached 75%, but it rises to 85% in coming years. So, the industrial food manufacturers increased to satisfy the needs of population and urbanization.

Several studies have confirmed that we are facing a continuous growth in the production of ready-to-eat foods. Bae et al. [4] have shown remarkable growth in sales of ready-to-eat foods. Mercanoglu Taban and Halkman [5] have linked this increasing demand of ready-to-eat into modifying food habits and lifestyles. These ready-to-eat foods contain synthetic surfactants that can produce diverse diseases [6]. On the other hand, this frequent application of synthetic surfactants ends in aquatic ecosystems, leading to significant toxicity of life within this natural environment; this situation has worsened over the past 30 years [7]. So, we assist in increasing demands of foods, principally ready-to-eat foods. However, citizens need food with some characteristics: fresh, lower or without synthetic additive and preservative components, sustainable thumbprint, and no hazards to consuming. These last demands of modern consumers force the food industries to enter a race toward the development of new technologies, which meet the requirements previously. Among the technologies that meet the above requirements are biotechnology-based microbes combined with nanotechnology to produce nanobiosurfactant (NBS) agents.

We talk about food nanobiotechnology; it is an area of emerging interest for the food industry [8]. The NBS is a new product generated from diverse microorganisms

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that are seducing more researchers because it can meet the previous dietary requirements of citizens, and low-cost resources can produce it. This new trend of NBS is not only for its sustainable environmental criteria, which can replace chemically synthesized surfactants. However, it also offers new properties to food either under normal conditions or under extreme conditions in storage and transport into customers. Additionally, Dammak et al. [9] have declared that the health safety and natural approach are two motives of nano-emulsifiers production by food manufacturers. For this, it generated a "label-friendly" marque for replacing the synthetic emulsifiers to encourage consumers to buy them. Santos et al. [10] have called the NBS a multifunctional biomolecule of the new century. It has several applications in the nanobiotechnology of food industries: participating in ready-to-eat foods processing, food packaging, shelf-life extension off, food safety, protection against foodborne pathogens [11–13].

This book chapter summarizes the roles of NBS for diverse uses in the food industry to enhance the functional properties of ready-to-eat foods, provide consumers safe and contamination-free food, and all this with a sustainable thumbprint. We divided this work into three principal axes: Firstly, we presented at the beginning the differences between surfactant, biosurfactant, and NBS, in addition to the characteristic properties of each type. Secondly, we presented the economic situation of the NBS market, its update of the current situation in the whole world, and the raw materials currently used in the production of NBS. After that, we decorated the biofabrication process of NBS. Finally, we discussed diverse applications in food industries; we have highlighted some new applications for improving several proprieties of food such as consistency, texture, the rheology of dough, volume, solubility, emulsion, aroma, foaming of food.

2 Surfactant, biosurfactant, and nanobiosurfactant properties

Surfactant is a surface-active compound, i.e., it is the amphiphilic matter. It has two functional parts: the hydrophobic part is repelled from water, but the other has a strong affinity for water. Diverse microbes can produce a wide range of surfactants called microbial surfactants or biosurfactants. For example, yeasts such as *Starmerella bombicola* and *Pseudozyma antarctica* generate more than 100 g/L of biosurfactants [14]. Bacteria groups such as *Acinetobacter calcoaceticus* produce 1038 unite of biosurfactant per milliliter [15]. The biosurfactant is produced mainly as secondary metabolites in mild production conditions and attached to the microbe's cell surface or liberated outside the microbes [16,17]. Biosurfactant is characterized by more biodegradability and lower toxicity than synthetic ones [7,18]. It is generally used to ensure reduced surface and interfacial tension, critical micelle dilution, and increased solubilize hydrophobic subtracts [19]. So, the application of biosurfactant improves solubility, dispersing, foaming, and critical micelle dilution in water and hydrocarbon solutions.

The criteria chosen made several classifications of biosurfactants. The chemical composition of hydrophobic fraction of biosurfactant leads to four types: glycolipid, fatty acid, lipopeptide, and polymer biosurfactant [14]. Other chemical classification divided them into five groups: glycolipids, lipopeptides, phospholipids, fatty acids, and neutral lipids [15]. For the ionic charge, we can distingue three types: anionic, cationic, or neutral biosurfactants [20]. When we chose the molecular weight as the significant criterion, we can distinguish between two types: high and low-molecular-mass biosurfactants [21]. The size of biosurfactant leads to dividing them into three groups: nanobiosurfactant has size 10–100 nm, microbiosurfactant with the size between 100 and 1000 nm (1 μm), and macrobiosurfactant (0.5–100 μm) [22,23]. So, the categorization of biosurfactants is diversified because it depends on their chemical composition, critical micelle concentration, molecular weight, ionic charge, size molecular, activity, and microbial type.

According to the previous description, we define an NBS as a surface-active compound that does not exceed the size of 100 nm and low molecular mass generated by microorganisms. The NBS can also combine and club biosurfactant-producing microorganisms with nanomaterial such as silver, nano-clay, and nano-zeolite. Several researchers confirmed that the NBS is made principally by glycolipids, lipopeptides, and phospholipids [19,24,25]. We can classify NBS into five principal categories: rhamnolipids, sophorolipids, cellobiose lipids, mannosylerythritol lipids, and lipopeptides (Fig. 1). The comparison of the characteristics of NBS and chemically synthesized surfactant shows that NBS is specific, selective, highly biodegradable, nontoxic, low-cost, and renewable substrates such as organic waste can generate it. Today, the NBS is emerging produce like a sustainable alternative to synthetic surfactant. In food industries, the application of NBS has high potential and opportunities. This natural product has lower critical micellization concentration (CMC) values that improve its efficiency in solubility, dispersion, and foaming in food applications (Fig. 2). When we compared the values of CMC between NBS and synthetic surfactants, we see that these natural alternatives have lower CMC than synthetic surfactants [19]. So, the NBS can give some properties of food as thickening, antibiofilm agents, antioxidant, antibacterial, emulsifier, thermal stabilizing agents, and it can be produced by organic wastes as a way of circular economies.

3 Nanobiosurfactant: Global potentials and biofabrication process

Currently, the biofabrication of NBS shows necessary growing economic attention for these nano-biotechnological products. Geys et al. [24] have declared that the customers of surfactants worldwide have generated 27 billion dollars in 2012. Thus, we are in front of a very profitable market if we propose a label-friendly based NBS for replacing the synthetic surfactant. Arumugam and Furhana Shereen [26] have estimated that biosurfactant portion production will represent 2.69 billion by 2023. The global quantity of biosurfactants is approximately 462,000 tons [27]. The major NBS

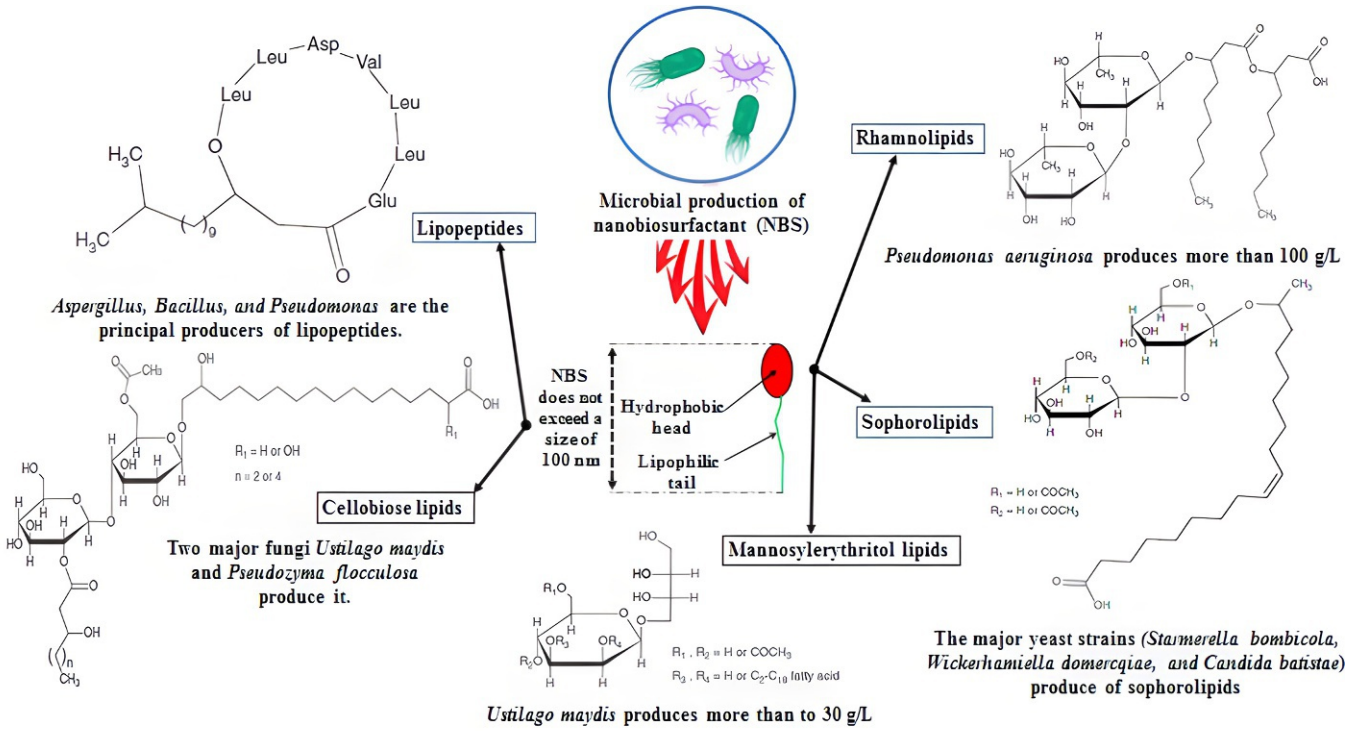

FIG. 1

The principal categories of nanobiosurfactants originated by microbial community.

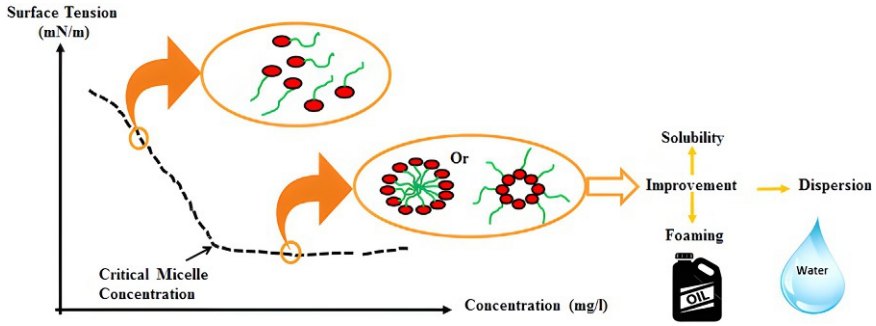

FIG. 2

The main role of NBS in food industries.

is rhamnolipid that mainly built this turnover. Randhawa et al. [28] have identified 17 biosurfactant companies and over 40 patents for rhamnolipids-type NBS. The price of these rhamnolipids is estimated between 0.02 and 0.5 livre sterling per liter of solution [28]. So, the NBS has made a niche for itself in the global market owing to its sustainable alternate, low toxicity, and environment-friendly characteristics.

The processes production of surfactants is excessively high because the raw material, downstream treatment, and purification need high cost and investment. Hence, the need to find less expensive and easily accessible materials to reduce the manufacturing costs of NBS. Some work has successfully produced NBS from abandoned and unused bioresources such as an organic waste Ohno et al. [29] have successfully produced NBS from soybean curd residue by using *Bacillus subtilis*. Radha et al. [17] have used the chicken tallow issued from the slaughterhouse industry as a carbon source of fungal species such as *Yarrowia lipolytica* to produce the NBS. This NBS has a low molecular weight between 75 and 100 kDa and a high potential for industrial applications. Arumugam and Furhana Shereen [26] have used *Calophyllum inophyllum* oilcake issued after the oil extraction process for NBS production by *Enterobacter aerogenes*. Kourmentza et al. [30] have used cooking oil as a low-cost substrate for NBS production by using *Burkholderia thailandensis* bacteria. So, the NBS is produced from a low-cost material, nonedible, and renewable.

We have distinguished two ways for biofabrication of NBS: the first way is natural production, and the second way requires the intervention of an energy tool. The natural process is a spontaneous microbial synthesis that grows on water-immiscible; generally, this synthesis is extracellular. We know that NBS is a secondary metabolite; hence, its production requires installing growth-limiting conditions in different bioreactors (batch, fed batch, continuous, and shake flask) [31]. The stress conditions induced this spontaneous microbial synthesis, i.e., we change the physicochemical parameters of normal incubation. Among these parameters, we have pH, temperature, oxygenation, agitation speed, and source of phosphate or nitrogen [10]. After the production of NBS, purification is necessary. We can distinguish several types

of purification ways such as acid precipitation, evaporation, filtration, and chromatography, crystallization [32–35]. We can use NBS after the purification stage to improve food products. The second method is a method that requires the intervention of energy to decrease the size of NBS [23]. This small particle (<100 nm) can cross the biological membranes, i.e., it increases cell membrane permeability [36]. We distinguish two methods that use energy in industrial food for fabrication of NBS for application in the food industry: (a) The high-energy methods use very high mechanical energy, it is regrouped into three ways (high-shear stirring, high-pressure homogenizers or piston homogenizer, microfluidization, and sonication). (b) The low-energy methods contain several types; we cited spontaneous emulsification, phase inversion composition, phase transition, membrane emulsification, and phase inversion temperature [11,37]. When we have compared these two types, the high-energy method is preferred by the food industry because it generates emulsions with small droplets ($r < 100$ nm) for use in wide food products [38]. We can conclude that NBS biofabrication in food industries requires a good choice of microbe, low-cost biomass for conversion, growth-limiting conditions, and a high-energy tool.

4 Nanobiosurfactant application in the food industry

After the biofabrication of NBS, an application of these sustainable products has become a necessity. Donsì [39] has said that the NBS has several food applications. Numerous ready-to-eat foods are found in it, such as butter, spreads, mayonnaise, salad creams, desserts, and beverages. In the following axes, we will present the new application trends of NBS and the future axes of research that researchers worldwide are excusing.

4.1 Nanobiosurfactant improves the thermal stability of food products

Several foods require cooking to consume; when we use NBS in these food products, they can get denatured and destroyed due to heat. On the other hand, certain foods deteriorate by varying temperatures during packaging, storage, or transport to consumers. Unsuitable temperatures applied to food can generate losses in nutritional, organoleptic qualities, and aesthetic states [40]. In addition, the temperature can affect atom agglomeration and bond formation [41]. These molecular interactions generated by temperature have affected diverse nanotechnologies applications such as the liberation of encapsulated materials, permeability control, and deterioration saving [42]. Furthermore, Souza et al. [43] have confirmed that temperature influences the production of biosurfactants. So, the temperature can destroy both food and NBS.

On the other hand, the thermal stability of biosurfactants is significant propriety for economic demand [44]. In industrial food applications, the choice of successful biosurfactants links to their functionality that should not change with temperature

variations [45]. Besides that NBS are very tiny, they will be very vulnerable to temperature. Reddy et al. [46] have confirmed that temperature affects the morphology and shape of NBS. Thermal degradation is associated with the breakdown of the structure of NBS [44]. So, the good NBS can resist breaking down under heat tension and not present any morphology and shape modification.

We have grouped several works producing NBS to improve foods and recommend agro-industrial application as food additives. Kourmentza et al. [30] have recommended that only accept 5% of surfactants in thermal degradation. Therefore, it is necessary to obtain NBS having a resistance variation of the temperature under this criterion. Arumugam and Furhana Shereen [26] have successfully produced NBS using *E. aerogenes*; this NBS presents the degradation temperature at 280°C until a complete disappearance after 530°C. Ribeiro et al. [47] showed that at 180°C, the NBS generated by *Candida utilis* lost only 0.77% of its mass. This NBS produced only undergoes this type of degradation at the beginning at 320°C. Since cookies are baked at a temperature of 180°C, therefore, NBS will not experience significant mass loss and will remain stable and adequate for the biscuit industry. Ribeiro et al. [47] have recommended using NBS generated by *C. utilis* as a potential ingredient in flour-based sweet food. Other studies by Ribeiro et al. [48] obtained an NBS using *Saccharomyces cerevisiae* URM 6670, which presents a negligible loss of only 0.43% at 200°C; this NBS can replace the egg yolk in muffins and cookies industries. Thus the NBS presented above has shown promising results as an ingredient of a flour-based product that resists temperature rise and allows preserving the properties of the formulated cookies.

4.2 Application of NPS in food packaging

Food packaging is an essential step in the food industry, especially with the new trend of citizens who prefer fresh, hygienic food, which is facilely made and ready to eat without chemical additives. However, during filling, transport, or storage, the food can be contaminated. Hence, there is a need to have very clean packaging without any bacterial contamination and toxins released by different microorganisms. Currently, there is a tendency to use NBS for packaging that meets the above requirements. Therefore, we presented studies that have successfully synthesized NBS with some nanoparticles for use in food packaging.

We know that some NBS generated from microbes can be used for the protection and stabilization of silver nanoparticles. Several studies recommended using silver nanoparticles as antimicrobial compounds in food packages [49,50]. This nanoparticle has bactericide effects against a wide range of highly pathogenic bacteria and fungi such as *Staphylococcus aureus*, *Clostridium*, *Enterococcus*, *Escherichia coli*, *Bacillus cereus*, *Listeria innocua*, *Salmonella choleraesuis* [51]. Silver reacts on bacteria in different ways: At the level of phosphorus in bacterial DNA, membrane proteins causing cell death due to maximum permeability through the membrane, interaction of silvers with divers cells may cause apoptosis, and finally, inhibition of sulfur-containing enzymes in the cytoplasm [52,53]. However, silver

nanoparticles can lose their bactericidal effect and cannot play their protective role of packaging in the presence of sodium chloride salt. This salt reacts with silver nanoparticles to produce AgCl. Saikia et al. [54] have recommended using the combination of rhamnolipid generated by *Pseudomonas aeruginosa* and silver nanoparticles to produce stable NPS against NaCl salt. This NBS has more than 1 month of stability, and it can protect silver nanoparticles more than 30 times.

Other studies have used glycolipids generated by *Brevibacterium casei* for synthase of reverse micelles with silver nanoparticles. This NBS called a green stabilizer of silver nanoparticles for 2 months [55]. Furthermore, Farias et al. [56] have succeeded in synthesizing an NBS with a spherical shape with a uniform distribution of silver nanoparticles; this product has a stability of less than 3 months without adding a passivator. On top of that, Kumar et al. [57] have shown that silver is recommended in the packaging of meat, fish, and fruit juice to inhibit the growth of microbial pathogens. So, using NBS based on silver nanoparticles improves the hygienic quality of packaging to increase the shelf life of food, inhibit bacterial contamination, and prevent or delay the deterioration of food intended for consumers.

4.3 NBS improves emulsion in food industries

The emulsion is the dispersion process of two immiscible liquids (usually oil and water) in suspension by an emulsifier [58]. The emulsifying agent or emulsifier is a stabilizer for emulsion, and it belongs to the chemical class of surfactants [59]. Zhang et al. [60] have confirmed that the gastrointestinal fate of nine food oils droplets is influenced by the emulsions process. Saeedi et al. [23] confirmed that the NBS is also emulsifying agent because it increases the emulsion process. So, this physicochemical character of NBS allows converting insoluble substrate into the soluble matter to lead to several applications; for example, the carriage of nutraceuticals, coloring, and flavoring agents in the food industry (Fig. 2). Aswathanarayan and Vittal [61] have confirmed that nanoemulsions have the capacity to encapsulate active, functional, and flavor compounds, for example, antioxidants, antibrowning agents, flavor and coloring, enzymes, and antimicrobials, in foods. Espitia et al. [62] have described that the nanoemulsion has some characteristics not presented in conventional emulsions; for example, it can enhance kinetic stability, optical transparency, and bioavailability. So, the encapsulation and incorporation of several compounds by NBS can raise its solubility, oxidation, controlled liberation, and absorption via cells, protect food against moisture and gas exchange in ready-to-eat foods.

We have grouped some studies that have succeeded in producing NBS with great potential as an emulsifier in the food industry. Bai and McClements [63] have used microfluidization tools to produce nanoemulsions from natural NBS; this technique has successfully produced an NBS with tiny droplets at low surfactant-to-oil ratios. The rhamnolipids can improve the emulsion of some oil types, medium-chain triglycerides, corn, fish, and lemon oil. So, this NBS can replace commercial emulsion-based products such as quillaja saponins. Thus, to build an NBS that plays the role of an emulsifying agent, we need four components: oleaginous liquid,

aqueous liquid usual water, low-molecular-mass biosurfactants, and energy. Espitia et al. [62] have described several energy tools for nanoemulsion formation. Sałek and Euston [64] have stated that emulsion characteristics of NBS give more stability and bioavailability with optical transparency. These proprieties are suitable for application in some food products that changing color is undesirable, such as juices, water, and meat. Torrego-Solana et al. [65] used genetic engineering to produce a mutant of *P. aeruginosa*; this new bacteria called *P. aeruginosa* 47T2 can increase the NBS yield production by 28%. Applying 20% of this NBS has shown 100% emulsion of olive oil, a significant ingredient in foods preserved by immersion, such as canned fish, meats, and cheese. Kim et al. [66] have successfully produced NBS from *Candida sp.* SY16; this product has a low molecular weight (634 Da) with a high emulsification process of vegetable oil with a crucial hydrophilic–lipophilic balance of 8.8. Kiran et al. [67] have successfully produced an effective emulsifier from the actinomycetes *Nesterenkonia* species that is incorporated into muffins for improved organoleptic attractiveness. Silva et al. [68] have recommended the new NBS produced by *Candida bombicola* URM 3718; this biosurfactant has a significant emulsifying capacity to substitute vegetable fat such as soybean, corn, canola, sunflower, peanut in the formulation of cupcakes. This NBS has demonstrated no significant differences in the physical–chemical composition of cupcakes, which remained indistinguishable. So, the NBS is an emulsifying agent that can assure some roles in food industries: (a) carrying active components, vitamins, natural colorants, antimicrobials, micronutrients, and flavors; (b) maintaining the consistency and texture of food; (c) improving solubilization of fat globules; (d) the rheology of dough, increasing the volume; (e) improving aroma, foaming, and dispersing properties.

4.4 NBS as an antimicrobial agent in ready-to-eat foods

We start this part with a tragedy that shook Canada; in 2008, we woke up to the news of sickened 57 and 24 deaths of Canadian citizens; it is a food-borne listeriosis outbreak by *Listeria monocytogenes* contaminated ready-to-eat foods [69]. This health disaster is estimated to be almost 242 million Canadian dollars [70]. Furthermore, a recent study in the Canadian retail market reports that *L. monocytogenes* is consistently detected in fresh-cut fruits and vegetables. So, the contamination of food by the pathogen microbes is a very hazardous healthy and socioeconomic problem; hence, the need to use natural alternatives with an antimicrobial character such as NBS.

Several studies have shown the effectiveness of specific NBS as antimicrobial agents. de Araujo et al. [71] have successfully produced NBS by *P. aeruginosa* PA1, which can reduce 79% of the biofilm formation of *L. monocytogenes* ATCC 7644. These researchers have recommended using this NBS to decrease adhesion and biofilm formation of a significant agent of listeriosis on food-contact surfaces. Magalhães and Nitschke [72] have combined 156.2 µg/mL of NBS with 320 IU/mL of nisin; this combination can destruct *L. monocytogenes* population entirely after 2 h of application. Haba et al. [32] have produced 11 NBS types from olive and

sunflower waste frying by *P. aeruginosa* 47T2; this product has excellent antibacterial and antifungal activities for several pathogenic bacteria and fungi: *Serratia marcescens*, *E. aerogenes*, *Klebsiella pneumoniae*, *S. aureus*, *Staphylococcus epidermidis*, *B. subtilis, Chaetonium globosum*, *Penicillium funiculosum*, *Gliocadium virens*, and *Fusarium solan*.

The target of NBS is intercalation into the microbial membrane, and it produces destruction of cell permeability. Sotirova et al. [73] have confirmed the loss of permeability of several Gram-positive bacteria such as *E. coli* and *P. aeruginosa* and Gram-negative bacterial strains such as *B. subtilis*. de Freitas Ferreira et al. [74] have also demonstrated that NBS can produce cell lysis by leakage of cellular components. Currently, the NBS used to produce liposomes, which can transport an antibacterial substrate across the bacterial membrane or inside the cytoplasm, where this antibacterial action can be applied. So, the NBS plays nanocarrier roles to enhance the bioactivity of antimicrobial agents. Sanches et al. [75] have loaded effectiveness antibacterial agent called ParELC3 that is a synthetic peptide derived from the ParE toxin against bacterial topoisomerases into the NBS. So, the NBS plays a nanoliposome role in facilitating the crossing of ParELC3 via the bacterial membrane and then reaches the topoisomerases target.

In addition to antimicrobial activities of NBS, Abdel-Mawgoud et al. [76] have described that it can play different roles: antiviral, algicidal, mycoplasmal, and antiamoebic. Currently, with the COVID pandemic, we can use the NBS as an efficient antiviral against the active sites of SARS-CoV-2 [77]. Thus, NBS makes it possible to protect food against several pathogens (virus, bacteria, fungi, algae, mycoplasma, amoeba).

5 Conclusion

The rapidly changing world and the appearance of pandemics necessitate an orientation in our daily consumption. The biofabrication of NBS has become a new trend by food industries. Currently, we are obliged to encourage the application of these natural products in the agro-industrial sector. The NBS has several benefits; it can replace chemically synthesized surfactants, is excellent antibacterial and antifungal in food products, enhances the emulsion of several immiscible liquids, and gives high structural nutritional, textural, and rheological proprieties of ready-to-eat foods.

References

[1] G. Lozano, Y.P. Marleny, J.A. García, S. Gonzalez, C.V.O. Bañuelos, M.P.L. Escareño, N. Balagurusamy (Eds.), Biosensors for food quality and safety monitoring: fundamentals and applications, in: Enzymes in Food Biotechnology, Elsevier, 2019, pp. 691–709, https://doi.org/10.1016/B978-0-12-813280-7.00040-2.

[2] J. Bogue, O. Collins, A.J. Troy, Market analysis and concept development of functional foods, in: Developing New Functional Food and Nutraceutical Products, Elsevier, 2017, pp. 29–45, https://doi.org/10.1016/B978-0-12-802780-6.00002-X.
[3] D. Coulomb, Refrigeration and cold chain serving the global food industry and creating a better future: two key IIR challenges for improved health and environment, Trends Food Sci. Technol. 19 (8) (2008) 413–417, https://doi.org/10.1016/j.tifs.2008.03.006.
[4] H.-J. Bae, M.-J. Chae, K. Ryu, Consumer behaviors towards ready-to-eat foods based on food-related lifestyles in Korea, Nutr. Res. Pract. 4 (4) (2010) 332, https://doi.org/10.4162/nrp.2010.4.4.332.
[5] B. Mercanoglu Taban, A.K. Halkman, Do leafy green vegetables and their ready-to-eat [RTE] salads carry a risk of foodborne pathogens? Anaerobe 17 (6) (2011) 286–287, https://doi.org/10.1016/j.anaerobe.2011.04.004.
[6] K.F. Csáki, Synthetic surfactant food additives can cause intestinal barrier dysfunction, Med. Hypotheses 76 (5) (2011) 676–681, https://doi.org/10.1016/j.mehy.2011.01.030.
[7] T.M.S. Lima, L.C. Procópio, F.D. Brandão, B.A. Leão, M.R. Tótola, A.C. Borges, Evaluation of bacterial surfactant toxicity towards petroleum degrading microorganisms, Bioresour. Technol. 102 (3) (2011) 2957–2964, https://doi.org/10.1016/j.biortech.2010.09.109.
[8] B.S. Sekhon, Food nanotechnology—an overview, Nanotechnol. Sci. Appl. 3 (mai) (2010) 1–15.
[9] I. Dammak, P.J. do Amaral Sobral, A. Aquino, M.A. das Neves, C.A. Conte-Junior, Nanoemulsions: using emulsifiers from natural sources replacing synthetic ones—a review, Compr. Rev. Food Sci. Food Saf. 19 (5) (2020) 2721–2746, https://doi.org/10.1111/1541-4337.12606.
[10] D. Santos, R. Rufino, J. Luna, V. Santos, L. Sarubbo, Biosurfactants: multifunctional biomolecules of the 21st century, Int. J. Mol. Sci. 17 (3) (2016) 401, https://doi.org/10.3390/ijms17030401.
[11] P. Borthakur, P.K. Boruah, B. Sharma, M.R. Das, Nanoemulsion: preparation and its application in food industry, in: Emulsions, Elsevier, 2016, pp. 153–191, https://doi.org/10.1016/B978-0-12-804306-6.00005-2.
[12] M. Nitschke, S.G.V.A.O. Costa, Biosurfactants in food industry, Trends Food Sci. Technol. 18 (5) (2007) 252–259, https://doi.org/10.1016/j.tifs.2007.01.002.
[13] T. Singh, S. Shukla, P. Kumar, V. Wahla, V.K. Bajpai, I.A. Rather, Application of nanotechnology in food science: perception and overview, Front. Microbiol. 8 (août) (2017) 1501, https://doi.org/10.3389/fmicb.2017.01501.
[14] T. Morita, T. Fukuoka, T. Imura, D. Kitamoto, Glycolipid biosurfactants, in: Reference Module in Chemistry, Molecular Sciences and Chemical Engineering, 2016, p. Elsevier, https://doi.org/10.1016/B978-0-12-409547-2.11565-3. B9780124095472116000.
[15] Z. Zhao, J.W.C. Wong, Biosurfactants from Acinetobacter calcoaceticus BU03 enhance the solubility and biodegradation of phenanthrene, Environ. Technol. 30 (3) (2009) 291–299, https://doi.org/10.1080/09593330802630801.
[16] G.G.F. Leite, J.V. Figueirôa, T.C.M. Almeida, J.L. Valões, W.F. Marques, M.D.D.C. Duarte, K. Gorlach-Lira, Production of rhamnolipids and diesel oil degradation by bacteria isolated from soil contaminated by petroleum, Biotechnol. Prog. 32 (2) (2016) 262–270, https://doi.org/10.1002/btpr.2208.
[17] P. Radha, P. Suhazsini, K. Prabhu, A. Jayakumar, R. Kandasamy, Chicken tallow, a renewable source for the production of biosurfactant by Yarrowia lipolytica MTCC9520, and its application in silver nanoparticle synthesis, J. Surfactant Deterg. 23 (1) (2020) 119–135, https://doi.org/10.1002/jsde.12357.

[18] A. Franzetti, I. Gandolfi, C. Raimondi, G. Bestetti, I.M. Banat, T.J. Smyth, M. Papacchini, M. Cavallo, L. Fracchia, Environmental fate, toxicity, characteristics and potential applications of novel bioemulsifiers produced by Variovorax Paradoxus 7bCT5, Bioresour. Technol. 108 (mars) (2012) 245–251, https://doi.org/10.1016/j.biortech.2012.01.005.

[19] R. Jahan, A.M. Bodratti, M. Tsianou, P. Alexandridis, Biosurfactants, natural alternatives to synthetic surfactants: physicochemical properties and applications, Adv. Colloid Interf. Sci. 275 (janvier) (2020) 102061, https://doi.org/10.1016/j.cis.2019.102061.

[20] H.B. Sobrinho, J.M. Luna, R.D. Rufino, A. Porto, L.A. Sarubbo, Biosurfactants: classification, properties and environmental applications, Recent Dev. Biotechnol. 11 (14) (2013) 1–29.

[21] E. Rosenberg, E.Z. Ron, High- and low-molecular-mass microbial surfactants, Appl. Microbiol. Biotechnol. 52 (2) (1999) 154–162, https://doi.org/10.1007/s002530051502.

[22] S.M. Jafari, Y. He, B. Bhandari, Optimization of nano-emulsions production by microfluidization, Eur. Food Res. Technol. 225 (5-6) (2007) 733–741, https://doi.org/10.1007/s00217-006-0476-9.

[23] L.H. Saeedi, M.M. Assadi, S.M. Heydarian, M. Jahangiri, The production and evaluation of a nano-biosurfactant, Pet. Sci. Technol. 32 (2) (2014) 125–132, https://doi.org/10.1080/10916466.2011.574176.

[24] R. Geys, W. Soetaert, I. Van Bogaert, Biotechnological opportunities in biosurfactant production, Curr. Opin. Biotechnol. 30 (décembre) (2014) 66–72, https://doi.org/10.1016/j.copbio.2014.06.002.

[25] M. Pacwa-Płociniczak, G.A. Płaza, Z. Piotrowska-Seget, S.S. Cameotra, Environmental applications of biosurfactants: recent advances, Int. J. Mol. Sci. 12 (1) (2011) 633–654, https://doi.org/10.3390/ijms12010633.

[26] A. Arumugam, M. Furhana Shereen, Bioconversion of Calophyllum inophyllum oilcake for intensification of rhamnolipid and polyhydroxyalkanoates co-production by Enterobacter aerogenes, Bioresour. Technol. 296 (janvier) (2020), https://doi.org/10.1016/j.biortech.2019.122321, 122321.

[27] M.G. Pessôa, K.A.C. Vespermann, B.N. Paulino, M.C.S. Barcelos, G.M. Pastore, G. Molina, Newly isolated microorganisms with potential application in biotechnology, Biotechnol. Adv. 37 (2) (2019) 319–339, https://doi.org/10.1016/j.biotechadv.2019.01.007.

[28] S. Randhawa, K. Kamaljeet, P.K.S.M. Rahman, Rhamnolipid biosurfactantsâ past, present, and future scenario of global market, Front. Microbiol. 5 (septembre) (2014), https://doi.org/10.3389/fmicb.2014.00454.

[29] A. Ohno, T. Ano, M. Shoda, Production of the antifungal peptide antibiotic, iturin by Bacillus subtilis NB22 in solid state fermentation, J. Ferment. Bioeng. 75 (1) (1993) 23–27, https://doi.org/10.1016/0922-338X(93)90172-5.

[30] C. Kourmentza, J. Costa, Z. Azevedo, C. Servin, C. Grandfils, V. De Freitas, M.A.M. Reis, Burkholderia thailandensis as a microbial cell factory for the bioconversion of used cooking oil to polyhydroxyalkanoates and rhamnolipids, Bioresour. Technol. 247 (janvier) (2018) 829–837, https://doi.org/10.1016/j.biortech.2017.09.138.

[31] M.M. Müller, J.H. Kügler, M. Henkel, M. Gerlitzki, B. Hörmann, M. Pöhnlein, C. Syldatk, R. Hausmann, Rhamnolipids—next generation surfactants? J. Biotechnol. 162 (4) (2012) 366–380, https://doi.org/10.1016/j.jbiotec.2012.05.022.

[32] E. Haba, A. Pinazo, O. Jauregui, M.J. Espuny, M.R. Infante, A. Manresa, Physicochemical characterization and antimicrobial properties of rhamnolipids produced by

Pseudomonas aeruginosa 47T2 NCBIM 40044, Biotechnol. Bioeng. 81 (3) (2003) 316–322, https://doi.org/10.1002/bit.10474.

[33] M. Heyd, A. Kohnert, T.-H. Tan, M. Nusser, F. Kirschhöfer, G. Brenner-Weiss, M. Franzreb, S. Berensmeier, Development and trends of biosurfactant analysis and purification using rhamnolipids as an example, Anal. Bioanal. Chem. 391 (5) (2008) 1579–1590, https://doi.org/10.1007/s00216-007-1828-4.

[34] J.C. Mata-Sandoval, J. Karns, A. Torrents, High-performance liquid chromatography method for the characterization of rhamnolipid mixtures produced by Pseudomonas aeruginosa UG2 on corn oil, J. Chromatogr. A 864 (2) (1999) 211–220, https://doi.org/10.1016/S0021-9673(99)00979-6.

[35] J. Mixich, R. Rothert, D. Wullbrandt, Process for the Quantitative Purification of Glycolipids, 1997. US005656747A. Hoechst Aktiengesellschaft. Frankfurt am Main, Germany. US005656747A.

[36] N. Salim, N. Ahmad, S.H. Musa, R. Hashim, T.F. Tadros, M. Basri, Nanoemulsion as a topical delivery system of antipsoriatic drugs, RSC Adv. 6 (8) (2016) 6234–6250, https://doi.org/10.1039/C5RA14946K.

[37] N. Anton, S. Akram, T.F. Vandamme, Transitional nanoemulsification methods, in: Nanoemulsions, Elsevier, 2018, pp. 77–100, https://doi.org/10.1016/B978-0-12-811838-2.00004-7.

[38] Y. Yang, C. Marshall-Breton, M.E. Leser, A.A. Sher, D.J. McClements, Fabrication of ultrafine edible emulsions: comparison of high-energy and low-energy homogenization methods, Food Hydrocoll. 29 (2) (2012) 398–406, https://doi.org/10.1016/j.foodhyd.2012.04.009.

[39] F. Donsì, Applications of nanoemulsions in foods, in: Nanoemulsions, Elsevier, 2018, pp. 349–377, https://doi.org/10.1016/B978-0-12-811838-2.00011-4.

[40] N.N. Potter, J.H. Hotchkiss, Food deterioration and its control, in: N.N. Potter, J.H. Hotchkiss (Eds.), Food Science, Food Science Text Series, Springer US, Boston, MA, 1995, pp. 113–137, https://doi.org/10.1007/978-1-4615-4985-7_7.

[41] Y. Yao, Z. Huang, P. Xie, W. Lianping, L. Ma, T. Li, Z. Pang, et al., High temperature shockwave stabilized single atoms, Nat. Nanotechnol. 14 (9) (2019) 851–857, https://doi.org/10.1038/s41565-019-0518-7.

[42] B.V. Parakhonskiy, D.A. Gorin, H. Bäumler, A.G. Skirtach, Temperature rise around nanoparticles: applications for polyelectrolyte multilayer capsules, J. Therm. Anal. Calorim. 127 (1) (2017) 895–904, https://doi.org/10.1007/s10973-016-5798-7.

[43] E.C. Souza, P.O. Souza de Azevedo, J.M. Domínguez, A. Converti, R.P. de Souza Oliveira, Influence of temperature and PH on the production of biosurfactant, bacteriocin and lactic acid by *Lactococcus Lactis* CECT-4434, CyTA J. Food 15 (4) (2017) 525–530, https://doi.org/10.1080/19476337.2017.1306806.

[44] R. Khademolhosseini, A. Jafari, S.M. Mousavi, H. Hajfarajollah, K.A. Noghabi, M. Manteghian, Physicochemical characterization and optimization of glycolipid biosurfactant production by a native strain of *Pseudomonas Aeruginosa* HAK01 and its performance evaluation for the MEOR process, RSC Adv. 9 (14) (2019) 7932–7947, https://doi.org/10.1039/C8RA10087J.

[45] A. Khopade, R. Biao, X. Liu, K. Mahadik, L. Zhang, C. Kokare, Production and stability studies of the biosurfactant isolated from marine Nocardiopsis Sp. B4, Desalination 285 (janvier) (2012) 198–204, https://doi.org/10.1016/j.desal.2011.10.002.

[46] A.S. Reddy, C.-Y. Chen, S.C. Baker, C.-C. Chen, J.-S. Jean, C.-W. Fan, H.-R. Chen, J.-C. Wang, Synthesis of silver nanoparticles using surfactin: a biosurfactant as stabilizing

agent, Mater. Lett. 63 (15) (2009) 1227–1230, https://doi.org/10.1016/j.matlet.2009.02.028.
[47] B.G. Ribeiro, B.O. de Veras, J. dos Santos Aguiar, J.M.C. Guerra, L.A. Sarubbo, Biosurfactant produced by Candida utilis UFPEDA1009 with potential application in cookie formulation, Electron. J. Biotechnol. 46 (2020) 14–21, https://doi.org/10.1016/j.ejbt.2020.05.001.
[48] B.G. Ribeiro, J.M. Campos Guerra, L.A. Sarubbo, Potential food application of a biosurfactant produced by Saccharomyces cerevisiae URM 6670, Front. Bioeng. Biotechnol. 8 (mai) (2020) 434, https://doi.org/10.3389/fbioe.2020.00434.
[49] M.R. de Moura, L.H.C. Mattoso, V. Zucolotto, Development of cellulose-based bactericidal nanocomposites containing silver nanoparticles and their use as active food packaging, J. Food Eng. 109 (3) (2012) 520–524, https://doi.org/10.1016/j.jfoodeng.2011.10.030.
[50] E.O. Simbine, L. da Cunha, J.L.-G. Rodrigues, E.S. Kamimura, C.H. Corassin, C.A.F. de Oliveira, Application of silver nanoparticles in food packages: a review, Food Sci. Technol. 39 (4) (2019) 793–802, https://doi.org/10.1590/fst.36318.
[51] S.P. Deshmukh, S.M. Patil, S.B. Mullani, S.D. Delekar, Silver nanoparticles as an effective disinfectant: a review, Mater. Sci. Eng. C 97 (avril) (2019) 954–965, https://doi.org/10.1016/j.msec.2018.12.102.
[52] J.R. Morones, J.L. Elechiguerra, A. Camacho, K. Holt, J.B. Kouri, J.T. Ramírez, M.J. Yacaman, The bactericidal effect of silver nanoparticles, Nanotechnology 16 (10) (2005) 2346–2353, https://doi.org/10.1088/0957-4484/16/10/059.
[53] I. Sondi, B. Salopek-Sondi, Silver nanoparticles as antimicrobial agent: a case study on E. coli as a model for gram-negative bacteria, J. Colloid Interface Sci. 275 (1) (2004) 177–182, https://doi.org/10.1016/j.jcis.2004.02.012.
[54] J.P. Saikia, P. Bharali, B.K. Konwar, Possible protection of silver nanoparticles against salt by using rhamnolipid, Colloids Surf. B: Biointerfaces 104 (avril) (2013) 330–332, https://doi.org/10.1016/j.colsurfb.2012.10.069.
[55] G. Kiran, A.S. Seghal, J. Selvin, Synthesis of silver nanoparticles by glycolipid biosurfactant produced from marine Brevibacterium casei MSA19, J. Biotechnol. 148 (4) (2010) 221–225, https://doi.org/10.1016/j.jbiotec.2010.06.012.
[56] C.B.B. Farias, A.F. Silva, R.D. Rufino, J.M. Luna, J.E.G. Souza, L.A. Sarubbo, Synthesis of silver nanoparticles using a biosurfactant produced in low-cost medium as stabilizing agent, Electron. J. Biotechnol. 17 (3) (2014) 122–125, https://doi.org/10.1016/j.ejbt.2014.04.003.
[57] A. Kumar, A. Choudhary, H. Kaur, S. Mehta, A. Husen, Metal-based nanoparticles, sensors, and their multifaceted application in food packaging, J. Nanobiotechnol. 19 (1) (2021) 256, https://doi.org/10.1186/s12951-021-00996-0.
[58] S. Ebnesajjad, Characteristics of adhesive materials, in: S. Ebnesajjad (Ed.), Handbook of Adhesives and Surface Preparation, William Andrew Publishing, Oxford, 2011, pp. 137–183, https://doi.org/10.1016/B978-1-4377-4461-3.10008-2 (Chapter 8).
[59] E. Dickinson, Hydrocolloids and emulsion stability, in: G.O. Phillips, P.A. Williams (Eds.), Handbook of Hydrocolloids, second ed., Woodhead Publishing, 2009, pp. 23–49, https://doi.org/10.1533/9781845695873.23 (Chapter 2).
[60] R. Zhang, Z. Zhang, H. Zhang, E.A. Decker, D.J. McClements, Influence of lipid type on gastrointestinal fate of oil-in-water emulsions: in vitro digestion study, Food Res. Int. 75 (septembre) (2015) 71–78, https://doi.org/10.1016/j.foodres.2015.05.014.

[61] J.B. Aswathanarayan, R.R. Vittal, Nanoemulsions and their potential applications in food industry, Front. Sustain. Food Syst. 3 (novembre) (2019) 95, https://doi.org/10.3389/fsufs.2019.00095.
[62] P.J.P. Espitia, C.A. Fuenmayor, C.G. Otoni, Nanoemulsions: synthesis, characterization, and application in bio-based active food packaging: nanoemulsions in food packaging…, Compr. Rev. Food Sci. Food Saf. 18 (1) (2019) 264–285, https://doi.org/10.1111/1541-4337.12405.
[63] L. Bai, D.J. McClements, Formation and stabilization of nanoemulsions using biosurfactants: rhamnolipids, J. Colloid Interface Sci. 479 (octobre) (2016) 71–79, https://doi.org/10.1016/j.jcis.2016.06.047.
[64] K. Sałek, S.R. Euston, Sustainable microbial biosurfactants and bioemulsifiers for commercial exploitation, Process Biochem. 85 (octobre) (2019) 143–155, https://doi.org/10.1016/j.procbio.2019.06.027.
[65] N. Torrego-Solana, M. José García-Celma, A. Garreta, A.M. Marqués, P. Diaz, A. Manresa, Rhamnolipids obtained from a PHA-negative mutant of pseudomonas aeruginosa 47T2 ΔAD: composition and emulsifying behavior, J. Am. Oil Chem. Soc. 91 (3) (2014) 503–511, https://doi.org/10.1007/s11746-013-2381-2.
[66] H.-S. Kim, J.-W. Jeon, S.-B. Kim, O. Hee-Mock, T.-J. Kwon, B.-D. Yoon, Surface and physico-chemical properties of a glycolipid biosurfactant, mannosylerythritol lipid, from Candidaantarctica, Biotechnol. Lett. 24 (19) (2002) 1637–1641, https://doi.org/10.1023/A:1020309816545.
[67] G.S. Kiran, S. Priyadharsini, A. Sajayan, G.B. Priyadharsini, N. Poulose, J. Selvin, Production of lipopeptide biosurfactant by a marine Nesterenkonia sp. and its application in food industry, Front. Microbiol. 8 (juin) (2017) 1138, https://doi.org/10.3389/fmicb.2017.01138.
[68] I.A. Silva, B.O. Veras, B.G. Ribeiro, J.S. Aguiar, J.M. Campos Guerra, J.M. Luna, L.A. Sarubbo, Production of cupcake-like dessert containing microbial biosurfactant as an emulsifier, PeerJ 8 (avril) (2020), https://doi.org/10.7717/peerj.9064, e9064.
[69] A. Currie, J.M. Farber, C. Nadon, D. Sharma, Y. Whitfield, C. Gaulin, E. Galanis, et al., Multi-province listeriosis outbreak linked to contaminated deli meat consumed primarily in institutional settings, Canada, 2008, Foodborne Pathog. Dis. 12 (8) (2015) 645–652, https://doi.org/10.1089/fpd.2015.1939.
[70] M.K. Thomas, R. Vriezen, J.M. Farber, A. Currie, W. Schlech, A. Fazil, Economic cost of a *Listeria Monocytogenes* outbreak in Canada, 2008, Foodborne Pathog. Dis. 12 (12) (2015) 966–971, https://doi.org/10.1089/fpd.2015.1965.
[71] L.V. de Araujo, C.R. Guimarães, R.L. da Silva Marquita, V.M.J. Santiago, M.P. de Souza, M. Nitschke, D.M.G. Freire, Rhamnolipid and surfactin: anti-adhesion/antibiofilm and antimicrobial effects, Food Control 63 (mai) (2016) 171–178, https://doi.org/10.1016/j.foodcont.2015.11.036.
[72] L. Magalhães, M. Nitschke, Antimicrobial activity of rhamnolipids against Listeria monocytogenes and their synergistic interaction with nisin, Food Control 29 (1) (2013) 138–142, https://doi.org/10.1016/j.foodcont.2012.06.009.
[73] A.V. Sotirova, D.I. Spasova, D.N. Galabova, E. Karpenko, A. Shulga, Rhamnolipid–biosurfactant permeabilizing effects on gram-positive and gram-negative bacterial strains, Curr. Microbiol. 56 (6) (2008) 639–644, https://doi.org/10.1007/s00284-008-9139-3.
[74] J. de Freitas Ferreira, E.A. Vieira, M. Nitschke, The antibacterial activity of rhamnolipid biosurfactant is PH dependent, Food Res. Int. 116 (février) (2019) 737–744, https://doi.org/10.1016/j.foodres.2018.09.005.

[75] B.C.P. Sanches, C.A. Rocha, J.G.M. Bedoya, V.L. da Silva, P.B. da Silva Silva, A.M. Fusco-Almeida, M. Chorilli, J. Contiero, E. Crusca, R. Marchetto, Rhamnolipid-based liposomes as promising nano-carriers for enhancing the antibacterial activity of peptides derived from bacterial toxin-antitoxin systems, Int. J. Nanomedicine 16 (février) (2021) 925–939, https://doi.org/10.2147/IJN.S283400.

[76] A.M. Abdel-Mawgoud, F. Lépine, E. Déziel, Rhamnolipids: diversity of structures, microbial origins and roles, Appl. Microbiol. Biotechnol. 86 (5) (2010) 1323–1336, https://doi.org/10.1007/s00253-010-2498-2.

[77] M.R. Bakkar, A.H.I. Faraag, E.R.S. Soliman, M.S. Fouda, A.M.M. Sarguos, G.R. McLean, A.M.S. Hebishy, G.E. Elkhouly, N.R. Raya, Y. Abo-zeid, Rhamnolipids nano-micelles as a potential hand sanitizer, Antibiotics 10 (7) (2021) 751, https://doi.org/10.3390/antibiotics10070751.

CHAPTER

11

Application of biosurfactants as emulsifiers in the processing of food products with diverse utilization in the baked goods

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1 Introduction

Biosurfactants are amphiphilic molecules produced by living cells that have the same property as synthetic surfactants in that they lower surface and interfacial tension through the same processes. Biosurfactants are synthesized biologically from a variety of substrates including fruit peel wastes, starch-rich wastes, vegetable oil wastes, and other sources, employing microorganisms such as bacteria, fungi, and yeast [1,2]. Biosurfactants have appealing properties such as emulsification, solubilization, dispersion, wetting, foaming, and detergent capacity, and antimicrobial properties. They have a number of advantages over their chemically derived equivalents, including higher biodegradability and biocompatibility, lower toxicity, and raw material availability, all of which combine to make biosurfactants a greener alternative to synthetic surfactants [3]. Eventually, biosurfactants have gained a lot of interest in the last five decades, and multiple studies on biosurfactant generation employing diverse microorganisms have been done. Biosurfactants are now widely used in industries such as oil recovery, food processing, cosmetics, and pharmaceuticals [4]. The market for biosurfactants crossed USD 1.5 billion in 2019 and is

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expected to increase at a rate of over 5.5% during 2020 and 2026. This global analysis emphasizes the growing need for biosurfactant all across the world. Among the so many applications, this book chapter mainly focuses on the use of biosurfactants as emulsifiers in food processing, as well as wide biosurfactant utilization in baked goods production.

Oil-in-water (O/W) emulsions are essential in the production of food products. The addition of lipid droplets to the food processing medium has a significant impact on the physicochemical properties, as well as sensory aspects such as stability, texture, aroma, and taste of the food products [5]. The viscosity and turbidity of the medium are increased when lipid droplets are added. Lipid droplets in emulsion influence the release of nonpolar active substance such as vitamins, nutrients, flavors, antioxidants, and antimicrobials. O/W emulsions are thermodynamically unstable and in the aqueous media appear as tiny lipid droplets. Stabilizers, such as emulsifiers, are now required in the commercial production of food products, with appropriate shelf life and tolerance to environmental changes. Emulsifiers are surface-active compounds that help to form and stabilize the emulsion. By adsorbing at the oil-water interphase [6], several synthetic surfactants, such as spans or tweens, are being utilized in the industry to stabilize oil-water emulsion. But at present consumer demand for commercial items, including food products that are more natural, healthy, environmentally friendly and durable, is increasing [6,7]. As a result, several industries have reformulated their products to exclude synthetic functional ingredients, such as synthetic, which are utilized as emulsifiers in the food industry. In this regard biosurfactants derived from biological sources, generally used as bioemulsifiers, are a great alternative to chemically developed surfactants [8,9].

Now to pick the best natural emulsifier as a substitute of synthetic one, it's important to understand the numerous physicochemical bases for emulsifiers' capacity to generate and stabilize oil-water emulsion. Eventually, a set of examinations may be used to evaluate the relative efficacy of natural and synthetic emulsifiers and determine which ones are the best. The book chapter covered not only all of these previously mentioned aspects, but also the various types of biosurfactants utilized as natural emulsifiers and their biological sources, used in the food industries. Their benefits and limitations are also emphasized.

2 Structural properties of emulsions

Before going over the basics of emulsion, it's vital to go over the many types of systems, such as solution and colloidal solution. A real solution, whether solid, liquid, or gas, is the most basic of all physiochemical states. The size of the particle of real solution is smaller than 0.001 μm. The simplest example of genuine solution is sugar in water [10]. But, in contrast, a colloid is a mixture of liquid, solid, or gaseous compounds in other compounds that may be solid, liquid, or gaseous. In most of the colloids, there are two phases: one that is continuous and the other that is dispersed. The dispersed phase distributes in continuous phase. The particle size of a colloidal

solution ranges from 0.001 to 0.1 μm. Among the various colloidal types, emulsion is created by mixing of two liquids that are immiscible with one another. Therefore, during the production of an emulsion, one liquid (dispersed phase) is allow to disperse in another liquid (continuous phase), resulting in tiny droplets. Now these tiny droplets must have a period of lifetime that might range from a few minutes to several years, depending on intended applications. Therefore, in order to create a stable emulsion of one liquid in the other often, a third component called the emulsifying agent or emulsifier must be introduced. This surface-active third component is commonly known as surfactant, and if it comes from a biological source, it's known as biosurfactant [11].

O/W and water-in-oil (W/O) are the two most common types of emulsions studied so far. Milk is the best example of the former type, and margarine is the latter type. Fig. 1 depicts the basic layout of O/W and W/O emulsions in diagrammatic form. Milk contains about 4% fat, which is distributed in an aqueous phase including protein [10]. Creams are another example of high-fat emulsions with protein in the aqueous medium through to a lesser degree. Emulsions made by blending vegetable oil and water vary from creams or milk. The fat in milk is special in that it is considered within a membrane-enclosed globule [10].

Multiple emulsions, denoted by the symbols water-in-oil-in-water (W/O/W) or oil-in-water-in-oil (O/W/O), can also be created [12]. Because the former offers a broader range of applications, it is examined in greater detail. They are a versatile carrier due to their favorable drug release pathways and/or rate, as well as the in vivo fate of different emulsions. It has a variety of delivery, tailored distribution,

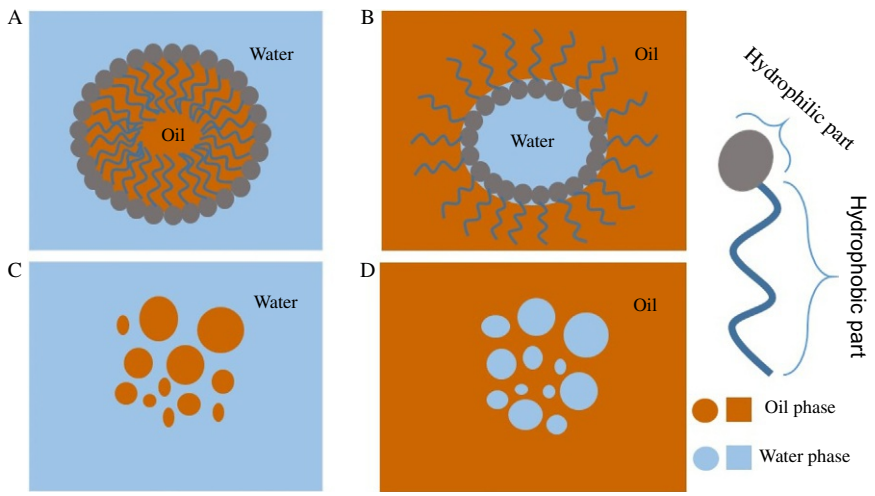

FIG. 1

Schematic representation of oil-in-water (O/W) (A) and water-in-oil (W/O) (B) emulsions, oil droplets in water (C) and water droplets in oil (D).

taste mashing, enzyme immobilization, bioavailability augmentation, and so on. Formation of polymeric gel, and interfacial complex films, introduction of an internal aqueous phase, and surfactant concentration modulation are all factors influencing the stability of multiple emulsions [12]. Electrolytes such as sodium chloride, sodium citrate, ascorbic acid, acetic acid are added to the emulsions to improve their stability.

3 Emulsion formation

As we all know, simply mixing of oil and water does not produce an emulsion. The internal phase droplets are subdivided by mechanical energy until a final average particle diameter of 0.001–0.1 μm is attained by them. Now a liquid drop that has formed a cylindrical shape with a length greater than 1.5 times, when compared its circumference, is unstable and will break down into droplets [13]. Mechanical churning of an oil-water mixture produces liquid droplets, which are subsequently deformed into cylinders and broken down into smaller droplets. The process is continued until the droplets have reached a small, ideal size, which can no longer be deformed or subdivided [14]. Fig. 2 illustrates this occurrence.

4 Role of emulsifiers in emulsion formation

Analyzing the main physicochemical occurrences within a homogenizer can help us to understand the function of the emulsifier in emulsion formation. The homogenizer, a high-pressure valve most typically employed in industry to create tiny lipid droplets, is the source of mechanical energy stated earlier [5]. The emulsifier is normally incorporated in the aqueous phase at first (though that's not always scenario), and afterward the oil and aqueous phases are combined and interleaved using a high-shear mixer, resulting in a coarse emulsion. The coarse emulsion, which has been coated with an emulsion, has rather large droplets with diameters of more than 1 μm. The coarse emulsion is then fed at high pressure via a narrow valve in the homogenizer, causing intense disruptive force such as agitation, cavitation, and shear to split the bigger droplets into small ones [14].

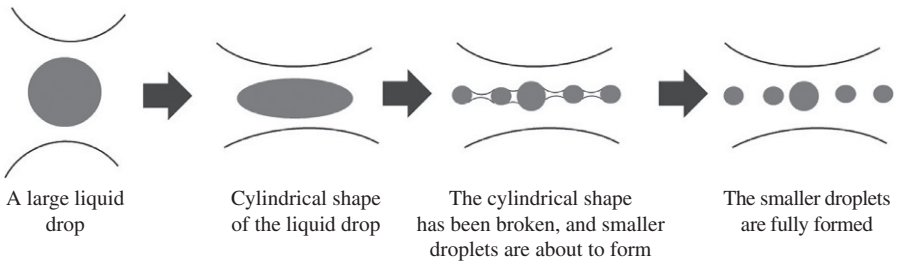

FIG. 2

Schematic representation of subdivision of a liquid drop.

An emulsifier's capacity to lower surface tension is determined by its structure and molecular weight [6]. Furthermore, the amount adsorbed at the contact affects the lowering of interfacial tension, with the equilibrium interfacial tension occurring once the emulsifier concentration exceeds the saturation at the interphase [6]. With increasing an emulsifier's capacity to diminish, γ, the droplets created by employing set halogenation parameters such as number of passes and pressure will be smaller. Because the droplets will not be completely covered with emulsifier before breaking up, the droplet disintegration rate must be slower than the emulsifier adsorption rate [5]. Adding surfactants to an emulsion can result in stable emulsion that lasts months or even a year.

5 Biosurfactant as emulsifier

When biosurfactants are added to aqueous solution, their monomers normally organize spherically into micelle with the hydrophobic part facing the center, forming the nucleus, and the hydrophobic part facing the sphere surface, forming an interface with water. As a result, the biosurfactant lowers the surface tension between water and oil interphase, allowing micelle to form [15]. Therefore, the concentration of biosurfactants at which micelles form is known as a critical micelle concentration (CMC) [1]. Fig. 3 shows the decrease in surface tension when the concentration

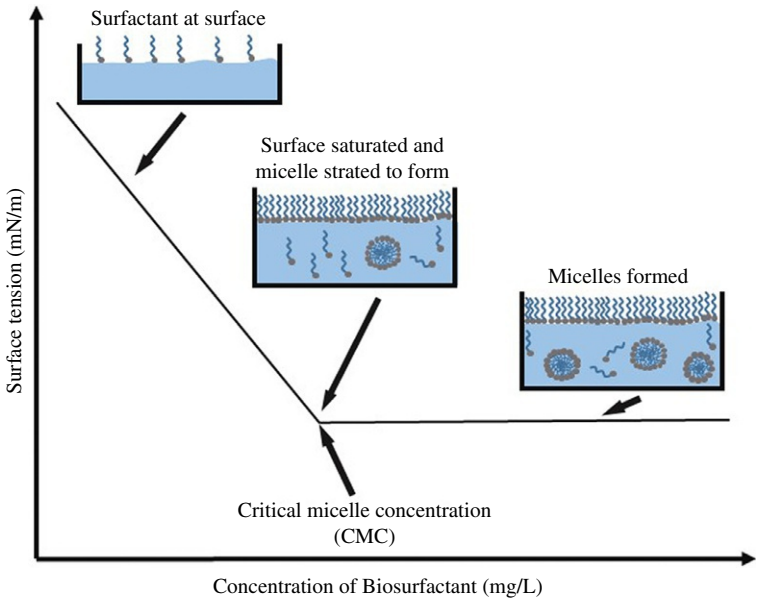

FIG. 3

Schematic representation of change in surface tension with biosurfactant concentration.

of biosurfactants is increased. As an excellent emulsifier, biosurfactant has the potential to bind at the oil-water interface, thereby preventing agglomeration of the fine droplets [16]. Therefore, it functions as a good stabilizer, demonstrating the ability to keep the droplets separated throughout storage.

The addition of biosurfactant to food processing leads to lowering of surface tension at the oil-water interface, which is the pivotal factor that eventually develops droplets in the proper way, stabilizing aerated systems [15]. As a result, these bioemulsifiers improve the reliability and texture of the prepared food while also increasing fragrance solubilization. The use of biosurfactants in food emulsion, including cream, milk, dressings, butter, mayonnaises, and so on, is frequently to the understanding of its creation, characteristics, and stabilization of the structure [17]. It also extends the shelf life of the products.

There are three different forms of emulsions used in food processing industry formed by biosurfactant emulsifier [17]:

(a) Oil-in-water (O/W) emulsion: Oil is the dispersed phase that is spread into the continuous phase of water. It is the cornerstone for water-based food products. The surfactants or other constituents present in the water phase have an impact on their characteristics. They're found in mayonnaise cream, liqueur, ice cream mixtures, milk, and a variety of other products.

(b) Water-in-oil (W/O) emulsion: This is the exact opposite of the previous situation, in which water droplets are suspended in continuous phase of oil. The persistence of these emulsions is determined by the fat characteristics as well as the surfactant utilized. Polyglyceryl oleate, sorbitan, monooleate, lecithin, sorbitan stearate, and lanolin are typical W/O emulsifiers. It's more difficult to create O/W emulsions. W/O emulsions are made by combing oils in O/W emulsions at a high concentration. Butter, cold cream, chord liver oil are examples of this category. Margarine is a true butter alternative with a minimum fat content of 80%, and a maximum water content of 16%; natural emulsifiers such as lecithin or other synthetic emulsifier are used in margarines [13,15,17].

(c) Water-in-oil-in-water (W/O/W) emulsion: Whenever W/O emulsion droplets contain water droplets, this type of emulsions formed [12].

6 Biosurfactant, becoming a significant emulsifier than synthetic surfactants as well as other natural food emulsifiers

Biosurfactants are even more effective than synthetic surfactants at screening the thermodynamically advantageous interaction between the oil and water phases, resulting in lower interfacial tension and tinnier droplets upon homogenization. For example, Zhu et al. compared the efficacy of biosurfactants such as tea saponin and quillaja saponin in decreasing interfacial tension to synthetic surfactants such as Tween 80 [18]. In the absence of surfactant, O/W emulsions comprising 10% medium-chain triglycerides (MCTs) were prepared and the interfacial tension was

measured at 24.42 mN/m. Tea saponin and quillaja saponin when added interfacial tension lowered to 4.8 and 2.8 mN/m, respectively, whereas Tween 80 reduced interfacial tension to 6.4 mN/m. Another research group, Bai and McClement, concluded that increasing a rhamnolipid-type biosurfactant to an MCT O/W emulsion reduced the same to 5 mN/m [19]. All of the findings support that as the biosurfactant concentration increased, more biosurfactant molecules were adsorbed to the oil-water interfaces, lowering interfacial tension and providing stability to the emulsion.

Food emulsions are typically complicated combination of multiple components, making stability problematics. In addition, factors such as pH and salt concentration, as well as manufacturing conditions, have a considerable impact on their stability. They come in a variety of forms such as gels, dispersions, and foams. For millennia, a variety of natural compounds have been utilized as food emulsifiers to aid in the development and stabilization of emulsions. Plant gums (such as arabic, tragacanth, karaya, guar, or locust bean), xanthium gum, lecithin (from soybean or egg yolk), phospholipids, and amphiphilic milk proteins are all instances of naturally produced emulsifiers [20]. Despite their superior emulsifying properties and wide range of applications, these emulsifiers are progressively losing favor among users because of many serious issues. Natural food emulsifiers including lecithin or gum arabic have long been utilized and acknowledged by the food industry. However, they now have considerable functional limits, particularly in methods that involve irradiation or the use of microwaves. Plant gums on the other hand are prone to price changes, owing to supply shortages resulting from low agricultural outputs and raising energy costs and transportation expensive [21]. These characteristics prompted a thorough search for other bioemulsifiers that could perform as well as or better than those already in use. In this scenario, biosurfactant emerged as the preferred option.

It is obvious from the preceding discussion that emulsification plays a critical role in aiding homogenization and enhancing their stability. Because of the potential use in bioremediation and the oil recovery, biosurfactants emulsifying activity has been investigated with various hydrocarbons, although their emulsifying activates with fats or oils used in food industry are less prevalent. Microorganism-produced surfactants successfully participate in this process, assisting in the formation of a stable emulsion. Biosurfactant, an effective emulsifier, is therefore widely used in the food industry.

7 Special attention on nanoemulsion formed by biosurfactants

Depending on the particles size, emulsions are classed as macroemulsion, microemulsion, and nanoemulsion. Nanoemulsions have improved functional qualities over traditional emulsions and have received specific attention for emulsion-based delivery methods in food processing [14,22]. For the encapsulation and optimal distribution of bioactive lipophilic substances, the content and structure of nanoemulsions can be regulated. Nanoemulsions are utilized in the food sector to provide

nutraceuticals, antimicrobials, coloring, and flavoring ingredients [23]. Active ingredient nanoemulsion formulation can be utilized to create biodegradable coatings and packing films that improve functionality, nutritive quality, shelf life, and of course, food quality [20,24].

High-energy and low-energy procedures can be used to make nanoemulsions [14,25]. High-energy technologies use mechanical devices to the oil phase so that it can mix with the water phase and produce smaller oil droplets. To make O/W nanoemulsion, several food sectors employ high-energy technologies. Mechanical devices utilized in high-energy techniques include microfluidizers, ultrasonicators, and high-pressure valve homogenizers [22,25]. Nanoemulsions are made in low-energy techniques by changing the composition or temperature of the oil-water system, with the electricity consumption from the chemical potential of the ingredients [22,26,27].

Nanoemulsions' optical characteristics are critical for its use in the food processing. They are optically clear or mildly murky depending on their droplet size [22]. Their obscurity is measured in terms of turbidity and quantified by transmission measurements. The opacity of nanoemulsions is influenced by average of diameter of particle and restricted particle-size dispersal. As a result, disruptive physical mechanism such as flocculation, coalescence, and gravitational separation, and chemical stability are also influenced by the opacity of nanoemulsions. Because of their opacity, transparent nanoemulsions with microscopic droplets are quickly eroded by UV or visible light.

Nanoemulsions are classified as kinetically stable systems. In the case of nanoemulsions, gravitational separation occurs when the continuous phase has differing relative densities, resulting in creaming or sedimentation. Nanoemulsions, on the other hand, have low droplet agglomeration, including coalescence or flocculation, owing to their small particle size [28].

High volumes of synthetic surfactants are normally intended to generate stable nanoemulsions, and it is well known that such a high surfactant content imposes environmental and economic constraints, limiting the practical applicability of nanoemulsions. Biosurfactants have been offered as replacements to chemical surfactants due to their severe limitation [19,23].

The interfacial characteristics of rhamnolipids at an oil-water interface were studied by Bai and McClements. The effect of rhamnolipid dosage, oily types, and homogenization settings upon microfluidization production of O/W nanoemulsions was investigated [19]. At low surfactant to oil proportions (i.e., 1:10), rhamnolipids were capable of generating O/W nanoemulsions having comparatively tiny droplet diameters. Rhamnolipid-coated droplets have also been found to be stable over a wide temperature range (30°C–90°C), pH conditions (5–9), and salinity (100 mM NaCl).

Rhamnolipid was used by Onaizi to create highly durable crude oil (both heavy and low) emulsions, exhibiting 0% water and oil separation over a month of storage [29]. Despite the stiffness of the crude O/W emulsions generated, they may be readily and inexpensively destabilized by adding an acid such as H_2SO_4.

The addition of a base such as NaOH allows for fast shifting of biosurfactants-stabilized raw O/W emulsions. Although the extent of emulsion switching is influenced by the amount of bioemulsifier used.

8 Hydrophilic-lipophilic balance value for the selection of biosurfactant in food processing

The hydrophilic-lipophilic balance (HLB) within the molecule is one of the parameters that aids in the selection of the appropriate surfactants as emulsifying agents. The proportion of the weight percentage of hydrophilic groups to the weight percentage of lipophilic groups in the surfactant molecule is referred as the HLB value [30]. Griffin proposed the idea of HLB for categorizing surfactants. Stauffer and Ranjith present a method for computing the HLB value. The HLB scale, intended for commercial emulsifying agents, varied from 0 to 20, with the low end indicating an emulsifier that is significantly more soluble in oil than in water and the high end indicating the opposite phenomenon. In other words, a low HLB number implies a strong oil affinity, whereas a high HLB value indicates a high water solubility [10]. Therefore, surfactants having the HLB values between 9 and 12 stabilize O/W emulsion, whereas those HLB values around 3–8 stabilize W/O emulsion [6]. It is now easy to anticipate that when the HLB value rises, the emulsifier becomes more water-soluble and their function shifts from W/O to O/W emulsifiers. There is a relationship between the HLB number and the surfactant's aggregation or phase behavior, and hence, its optical appearance, as shown in Table 1 [31].

The aggregative structure of biosurfactants at different HLB values must be understood in order to better comprehend their role as emulsifiers. Fig. 4 depicts the various types of emulsifier structures at different HLB value ranges.

HLB values are helpful in determining which biosurfactants are the best for usage as emulsifiers in food industry. Once the chemical structure of biosurfactants is determined, HLB analysis can be performed. The HLB value of a trehalose tetraester biosurfactant synthesized by *Rhodococcus* sp. 51T7 was calculated to be 11 by Marques et al., and they also emphasized the outstanding stability of the O/W emulsion whenever this biosurfactant was utilized [6]. Kim et al. found that mannosylerythritol lipid (MEL) generated by *Candida* sp. SY16 has an HLB value of 8.8 and therefore could be employed as a promising O/W emulsifier [32]. The HLB value of a

Table 1 Represents the correlation between the HLB value, optical appearance in solution, and aggregate shape [31].

HLB value	Optical appearance in solution	Shape of aggregate
>13	Clear solution	Micelle
7–10	White dispersion milky appearance	Lamellar phase
<7	Lumps of emulsifier	Reversed micelle

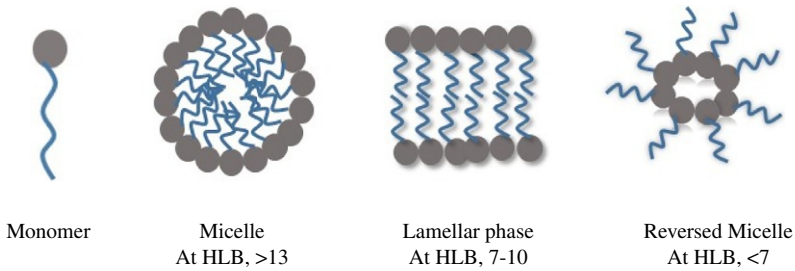

FIG. 4

Models of self-aggregation of biosurfactant molecules (micelle, lamellar phase, and reverse micelle) at different HLB values.

lipopeptide biosurfactant generated by *Bacillus subtilis* ATCC 6633 was reported to be 21.7 by Dehghan-Noudeh. As per Griffin HLB scale, it is quite high and subsequently this molecule is often used as an O/W emulsifier [33]. Ohadi et al. examined and analyzed the HLB value of a lipopeptide biosurfactant produced by *Acinetobacter junii* B6. According to the Griffin HLB scale, biosurfactant has a HLB value of 10, indicating that it is an appropriate O/W emulsifying agent [34].

For the assessment of functional properties, the synergistic effect of several biosurfactants is often investigated. The HLB of a blend of two emulsifiers is therefore determined [35]. Eq. (1) was used to calculate the HLB value of various biosurfactant mixtures,

$$HLB_{AB} = (HLB_A \times A\%) + (HLB_B \times B\%) \quad (1)$$

Here HLB_A and HLB_B are the HLB values of the biosurfactants A and B, respectively. HLB_{AB} represents the HLB value of the biosurfactants mixture. A% and B% are the weight percentage of A and B in the solution, respectively. Boyed et al. created mixes of numerous span (sorbitan fatty acid esters) and Tween (polyoxyethylene 20 sorbitan fatty acid esters) surfactants for covering the HLB ranges of 8.5–16.5 (for O/W emulsion) and 2.0–6.5 (for W/O emulsion) using a sample system consisting of comparable amounts of water and food-grade mineral [13]. Emulsion stability was tested by calculating the rate of oil globule coalescence offering environments, and it was reported that O/W emulsions were most at an HLB of approximately 12 and W/O emulsions becoming most stable at an HLB of about 3.5. To determine the synergistic effect of saponin and rhamnolipid biosurfactants on foam qualities at the air-water interface, Hajimohammadi and Johari-Ahar synthesized 2 wt% rhamnolipid and saponin biosurfactants with various HLB values in 5 mL deionized water [35]. The best HLB value of these biosurfactants was 10.8 for the mixture of 84% rhamnolipid and 16% saponin, according to the findings. However, McClements and Jafari found that utilizing the HLB value to predict efficacy for individual or blended surfactants in emulsion is not always effective [13].

9 Emulsification index (E_{24})

The emulsification index (E_{24}) is a practical and quick way to assess surfactant emulsifying characteristics [36]. Surfactants are known to stabilize emulsion systems including hydrocarbons and/or water-insoluble substances. Although emulsification capability is unrelated to the ability to diminish surface tension, the stability of these emulsions is suggestive of biosurfactant activity. Vegetable oils were chosen for the emulsification testing because of their importance and use in the food sector [36,37]. The higher the biosurfactants emulsification index with vegetable oils, the more likely it is to be used as a bioemulsifier in food processing industry. The biosurfactant's emulsification activity was assessed using the method proposed by Gaur et al. In glass tubes, vegetable oils were mixed in 1:1 ratio (V/V) to an aqueous biosurfactant solution at concentrations of 2.5, 5, and 10 mg/mL and refluxed for 2 min at a frequency of approximately 50 Hz [36]. The emulsification index (E_{24}) was calculated after 24 h using the Eq. (2).

$$E_{24} = h_e/h_t \tag{2}$$

The emulsion layer's height is h_e, while the mixture's total height is h_t.

Soybean oil was used for the most of the studies because of its ease of production and utilization, while other oils such as peanut, corn, canola, and sunflower oil were chosen for their distinct beneficial and functional features for human consumption. Peanut oil, for instance, is high in vitamin E, canola oil is low in saturated fatty acids and includes omega 3, and corn oil is high in necessary acids and deemed to be of high quality [36]. The emulsification activity of the various biosurfactants with vegetable oils is explicitly detailed in the upcoming part, ensuring that all of the oils examined are emulsified satisfactorily.

10 Biosurfactants used as emulsifier in food industry

This section examined that the emulsifying capabilities of biosurfactants with oils or fats used in the food processing have received less attention. It will be determined which biosurfactants are good emulsifiers based on the results of this study. Diverse types of biosurfactants produce different outcomes. Lipopeptides, glycolipids, glycolipopeptides, and glycoproteins are all types of biosurfactants used to investigate emulsifying efficiency. Now it is the time to have a look at it.

10.1 Lipopeptide and lipoprotein-type biosurfactants

Ghojavand et al. used a thermotolerant, halotolerant, and facultative biosurfactants generating bacterial strain *Bacillus mojavensis* to examine the synthesis of a lipopeptide-type biosurfactants [38]. Emulsification studies were carried out to determine the durability of emulsions generated between water and hydrocarbons in the presence of lipopeptide. The lipopeptide emulsification activity was the

strongest at low temperatures without salinity. These results point to it being a good option for emulsification uses in mild environments, and therefore, it is essential to examine the optimized production of this bioemulsifier for its industrial application, especially in food processing.

Kima et al. identified the actinobacterial strain MSA31 from the sea sponge *Fasciospongia cavernosa*, which has strong biosurfactants production activity [39]. The so produced lipopeptide MSA31 was found to be nontoxic in a brine shrimp cytotoxicity experiment. It was also described as a "halo alkali and temperature resistant biosurfactant" that can be employed in the food sector as an emulsifier and emulsion stabilizing agent. Furthermore, when compared with the positive and negative controls, the inclusion of lipopeptide MSA31 in muffins increased organoleptic qualities, and this investigation ultimately gives a useful insight for this lipopeptide to be used in the food sector as an excellent emulsifier [39].

Employing cassava wastewater as substrate, Nitschke and Pastore produced lipopeptide biosurfactants using the *B. subtilis* LB5a strain. Its emulsification activity with the vegetable oil including soybean oil (74.0%) and coconut oil (70.9%) is remarkable [40]. Biosurfactant's high emulsification activity allowed it to be used as an emulsifying agent for oils, resulting in stable emulsions. The ability to produce emulsions with vegetable and fats implies emulsifying agent.

B. pseudomonas BS6 generated and characterized a lipopeptide biosurfactant in fermentation medium, with soybean oil waste as the sole source of carbon, according to Li et al. The surface tension of water could be reduced from 71.6 to 30.2 mN/m with this purified lipopeptide [41]. Its emulsification index with two hydrocarbons and five plant oils ranged from 62.8% to 94.2%, indicating that the biosurfactant can be efficiently used as an emulsifier in food processing.

10.2 Glycolipid-type biosurfactants

Pichia anomala has sparked a lot of attention because it's a good source of vitamins and proteins. It is therefore used in the production of food bioemulsifiers, such as biosurfactant, and has been recognized as a flavoring agent. Utilizing this thermotolerant strain, *Pichia anomala* PY1, obtained from fermented food, Thaniyavarn et al. synthesized sophorolipid-type biosurfactants in a variety of carbon sources (glucose, starch, soybean oil, palm oil, and coconut oil) and nitrogen sources (NH_4NO_3, $(NH_4)_2SO_4$, and $NaNO_3$) rich media. The optimum carbon source was 4% soybean oil, and the ideal nitrogen source was 0.4% $NaNO_3$ for the formation of maximal biosurfactant by *Pichia anomala*. The emulsifying activity of the sophorolipid generated was acceptable [42].

Gaur et al. biosynthesized and characterized a sophorolipid biosurfactant produced by *Candida albicans* SC5314 and *C. glabrata* CBS138 yeast strains. The cell-free broth's surface tension was lowered from 72 to 42 N/m and 55 N/m, respectively. On different vegetable oils, biosurfactant from both strains had variable emulsifying capabilities. *C. albicans* SC5314 had the maximum activity against olive oil (51%) and the lowest activity against soya oil (39%), but *C. glabrata* CBS138 had equivalent emulsifying ability against all the tested oils, namely olive oil (49%),

soybean oil (51%), mustard oil (50%), and almond oil (50%) [43]. The current work implies that yeast-derived biosurfactants could be used in a variety of food formulations that require excellent emulsifying properties.

Radzuan et al. synthesized rhamnolipid from palm oil agricultural refinery using the microorganism *Pseudomonas aeruginosa* PA01. The magnitude of the emulsion index with sunflower oil can be affected by the amount of crude rhamnolipid, and a white emulsion layer has been seen for crude rhamnolipid. When compared with glucose, crude rhamnolipid produced from palm oil waste has a twofold higher emulsification index with sunflower oil [44]. This emulsifying activity may make it easier for rhamnolipid to be used in food processing.

Pseudomonas aeruginosa 47T2 cultivated in submerged culture using waste frying oil as a source of carbon was explored by Haba et al. for the synthesis of rhamnolipid. It has a significant emulsification index with some organic compounds such as linseed oil (90%), crude oil (66.6%), and mineral oil (50%) and therefore offers stability to O/W emulsions [45]. This occurrence suggests that they could be used as emulsifiers in the food processing industry.

Monteiro et al. explored new bioemulsifier-producing yeast from a diary industry's wastewater effluent. *Trichosporon loubieri* CLV 20, *T. montevideense* CLOA 70, and *Geotrichum* sp. CLOA 40 formed the most glycolipid when cultivated in 100 mL liquid mineral medium supplemented with 20 g/L of sunflower oil at 28°C on a reciprocal rotatory shaker for 144 h at 200 rpm. These glycolipid biosurfactants rapidly emulsify vegetable oils under extreme conditions. As a result of these high emulsifying activities, their use in a number of industrial sectors including food has expanded for the production of emulsions [46]. Then they compared the emulsification capability and emulsions stability of two synthetic surfactants (Triton X-100 and Tween 80) with these bioemulsifiers from yeast strains, under investigation using toluene as substrate, finding that the synthetic surfactants had no emulsifying activity in the presence of 30% (W/W) NaCl.

Lactobacillus pentosus was used to produce a glycolipopeptide biosurfactant by Vecino et al. The stability of water and rosemary oil emulsions with a biosurfactant produced by *L. pentosus* was investigated and compared with the stability of emulsion with polysorbate 20. Compared with the polysorbate 20, the biosurfactants from *L. pentosus* produced more stable emulsions with greater emulsion volume [47]. As a result, the biosurfactants studied in this experiment have commercial viability as a biodegradable emulsifier in a variety of applications, primarily in food processing.

Ferreira et al. produced a glycolipopeptide biosurfactant by inoculating *L. paracasei* with 33 g/L glucose, 10 g/L corn step liquor, and 10 g/L yeast extract in a fermentation medium. By mixing essential oils (such as wheat, almond, jojoba, and rosemary) with natural antioxidant extract, the biosurfactant performed a vital function as an emulsifier in O/W emulsion systems [48]. Cell proliferation was above 97% in the presence of biosurfactant or in the addition of O/W emulsions containing the natural antioxidant extract and their biosurfactant, exhibiting better result than SDS. These discoveries bring up new possibilities for biosurfactant application in food processing.

10.3 Glycoprotein biosurfactants

Gutierrez et al. identified and selected *Antarctobacter* sp. TG22, a marine alpha protecobacterium for the synthesis of high-molecular-weight biosurfactants (glycoprotein) with high emulsifying activity. In a low-nutrient seawater media enriched with glucose, this emulsifier was best synthesized. This glycoprotein compared favorably to a gum arabic and xanthan gum as an emulsifying agent for a variety of food oils as according to functional evaluation. Experimental study reported that, with sunflower oil, vegetable oil, and ground nut oil, it produced much stable emulsion. In comparison to gum arabic, xanthan gum, and no emulsifier, the stability of emulsion produced by glycoprotein with powered nut oil was 6.6, 8.8, and 3.1-fold higher, respectively [49].

Lukondeh et al. tested *Kluyveromyces marxianus* FII 510700 cultivated on a lactose-based medium for natural bioemulsifier (glycoprotein) synthesis. With 12 g/L pure bioemulsifier, a maximum of 76% of the corn oil phase could be emulsified [50]. When exposed to wide pH range of 3–11 and NaCl concentration of 2–50 g/L, the glycoprotein formed emulsions were stable for 3 months. This suggests that the bioemulsifier derived from *K. marxianus* keeps O/W emulsions stable under a variety of chemical and physical conditions. It could thus find commercial use in the making of mayonnaise, crackers, cookies, cakes, ice cream and meat products, and other foodstuffs [50].

Biosurfactants of the high-molecular weight glycoprotein category include yasan and liposan. Trindade et al. produced yasan in glucose-based fermentation medium using a strain *Yarrowia lipolytica* IMUFRJ 50682 and employed it as a bioemulsifier for the preparation of O/W emulsions [51]. The emulsion stability was then compared with those made with gum arabic. The evolution of the mean droplet size was examined in O/W emulsions made by mixing of three distinct organic phases, toluene, *n*-hexadecane, perfluoro-*n*-hexane, with two aqueous buffers of varying pH and two bioemulsifiers (yasan and gum arabic). The aging mechanism of all the emulsions examined was found to be independent of the organic phase employed in their formation and to be reliant on the bioemulsifier yasan as well as the medium pH. Gum-arabic-based emulsions often lose stability due to coalescence, whereas yasan-dependent emulsions were stable at low pH, and their stability decreases as pH rises from 3 to 7, indicating that pH is the primary factor affecting the stability of yasan-containing emulsions because of their large surface area, because of their high degree of emulsion forming capability and potential to provide superlative emulsions stability emulsifiers and their use in food processing industry.

10.4 Emulsan

Emulsan is an extracellular heteropolysaccharide composed of two biopolymers: 80% lipopolysaccharide and 20% exopolysaccharide with a large molecular mass [6]. The D-glucose amine developed during the stationary phase is present in the polymeric lipopolysaccharide component of emulsan. In the late 1970s, emulsan

was isolated from hydrocarbon-degrading *Arthrobacter* sp. RAG1 (later named as *Acinetobacter venetianus* RAG-1) for its outstanding emulsion producing and stabilizing characteristics, particularly in regard to hydrocarbon biodegradation [52]. This poly-anionic bioemulsifier established a very thin layer in between water and hydrocarbon droplets, stabilizing the emulsions in the process [53]. Fermentation procedures such as chemo-sat batch, self-cycling fermentation can all be used to produce emulsion.

Acinetobacter calcoaceticus produces emulsan, a bioemulsion that is utilized in the formulation and creation of ice cream and soft cheese and also creams. Apo-alpha-oleo-emulsan, alpha-amyloemulsan, and beta emulsan are three forms of emulsions made by bacteria (*Acinetobacter venetianus* RAG-1, *Streptomyces* NRRLB-15615, NRRLB-15847, or ATCC 31926) that are frequently utilized in the food processing [53].

10.5 Alasan

Alasan is a bioemulsifier made up of proteins and polysaccharides with a large molecular weight (100,000–230,000 Da). *Acinetobacter radioresistant* KA53 bacteria produces this bioemulsifier [53,54]. The bioemulsifier converts into a dense polysaccharide and lacks its emulsifying characteristics if the protein part is destroyed and digested with proteolytic enzymes. Heating polysaccharides with alcohol at 50°C can result in a 2.5-fold rise in polysaccharide concentration, but the protein component and emulsifying capabilities of the molecule stay constant. Heating the mixture to 60°C–90°C decreases viscosity and enhances emulsifying activity by up to 5.8 times its initial value. The emulsifying properties of alasan are also influenced by pH and metal ion concentration [53]. Toren et al. revealed that the protein with a molecular weight of 45,000 Da had the maximum emulsifying ability, and it was even better than the intact alasan. They also found that addition of purified 16,000 and 31,000 Da proteins to 45,000 Da protein increased the emulsifying activity and stability of the O/W emulsion [55,56]. In a fed-batch fermentation process, alasan synthesized from Acinetobacter bacteria yielded 2.2 g of emulsifier per liter of culture fluid. Alasan stabilized a broad range of O/W emulsions, including soybean oil, coconut oil, and raw oils. The emulsions formed in this way are stable over a large range of temperatures (up to 100°C), pH range (3.3–9.2), and magnesium ion concentration range [53]. Because of these outstanding emulsifying properties, this extracellular bioemulsifier is widely employed in the food sector.

Some biosurfactants are poorly characterized, although they have a wide range of applications in the food processing industry. These are now briefly discussed.

Campos et al. identified four yeast species for the synthesis of biosurfactants, which are utilized as bioemulsifiers in the food industry: *C. lipolytica* UCP 0988, *C. tropicalis* UCP 0996, *C. guilliermondii* UCP 0922, and *C. utilis* UFPEDA 1009. When 0.3% (W/V) *C. utilis* was utilized in the cultivation media, 6% (W/V) glucose and 6% (W/V) waste canola frying oil were used as carbon sources, and 0.2% (W/V) ammonium nitrate was employed as nitrogen source, the greatest

biosurfactants were achieved. The biosurfactants obtained by *C. utilis* resulted in the maximum surface tension reduction of 35.53 mN/m among all yeast species. When this biosurfactant was utilized as emulsifier, the highest emulsification index was obtained with various vegetable oils such as soybean oil (43%), sunflower oil (73%), corn oil (73%), rice oil (33%), and motor oil (30%), and therefore, it has the potential to be used as a bioemulsifier in the food sector [57]. Furthermore, employing yeasts has a number of advantages including the "Generally Regarded as Safe" (GARS) designation that most of the species have allowing for the use of by-products from those microorganism in the food industry [58].

In a culture media containing 5% sugar cane molasses, 3% corn steep liquor, and 5% residual soybean oil, Silva et al. developed biosurfactants. The yield was 25 g/L in isolated biosurfactants, with a CMC of 0.5 g/L [37]. The suggested biosurfactant emulsions with the vegetable oils produced good results. The emulsions microphotographs revealed that raising the biosurfactants concentration reduced the oil droplets, boosting the emulsions' stability. The emulsification index of *C. bombicola* URM 3718 biosurfactant with different vegetable oils, such as soybean oil, canola oil, sunflower oil, corn oil, and peanut oil, is 45.81%, 45.44%, 54.60%, 48.28%, and 68.97% [37]. This shows that the bioemulsifier was successful in emulsifying the oils investigated and was later used in the food processing sector.

Durval et al. examined the synthesis of a biosurfactant by *B. cereus* UCP 1615 in a low-cost fermentation medium enriched with waste frying oil. The biosurfactant was able to achieve satisfying emulsification of all the oils (soybean oil, corn oil, peanut oil, and canola oil), according to the emulsification index calculation. According to the findings of the experiment, an increase in biosurfactant concentration resulted in an increase in E_{24}. At biosurfactant concentration of 5.0 and 10.0 mg/mL, maximum E_{24} was reached, with values ranging from 54% to 68% [36]. These studies confirm its use in the food industry as a bioemulsifier.

Few researchers have also looked into the ability of biosurfactants produced by the *Bacillus* genus to emulsify vegetable oils and as a result, the use of these natural substances as bioemulsifiers in food products. For example, in a study utilizing a biosurfactant synthesized by *B. subtilis* ICA56, E_{24} was found to be greater than 50% in soybean oil [59].

Sharma and Saharan isolated a variety of lactic acid bacteria (LAB) and tested them for biosurfactant synthesis and application in biomedical and food processing. The surface tension of a biosurfactant derived from *L. helveticus* MRTL91 was lowered from 72 to 39.5 mN/m, with a CMC of 2.5 mg/mL and an emulsification index of 65% against kerosene oil. After a 15 min heat treatment at 125°C, biosurfactant showed no obvious decrease in surface tension and emulsification efficiency. In the food industries and medical sectors, adsorption of biosurfactants originating from LAB to solid surfaces could develop a suitable technique for reducing microbial adherence and preventing colonization by pathogenic microorganism [60]. It eventually becomes a viable alternative to conventional antibiotics, with the potential to be further exploited in the food industry.

The biosurfactants listed in Table 2 operate as emulsifiers in food processing. Their carbon and nitrogen sources, as well as the microbes that are used to produce

Table 2 Shows a list of biosurfactants with emulsifying properties that are commonly employed in food processing.

Name of the biosurfactant used as bioemulsifier	Microorganism	Substrate used for biosurfactant synthesis	Emulsifying capacity of vegetable oils (E_{24} in percentage)	References
Lipopeptide	*Bacillus subtilis* LB5a	Carbohydrate rich cassava waste water	Soybean oil (74.0%) Coconut oil (70.9%)	[40]
	B. pseudomonas BS6	Soybean oil waste as the sole source of carbon	Soybean oil (>85%) Sesame oil (>85%) Peanut oil (>85%) Olive oil (>65%) Corn oil (>60%)	[41]
	Nesterenkonia sp. MSA31	Olive oil as carbon source, Ammonium nitrate as nitrogen source	N/A	[39]
	B. mojavensis	Carbohydrate rich sucrose as carbon source, ammonium sulphate as nitrogen source	N/A	[38]
Sophorolipid	*Candida albicans* SC5314	Carbohydrate rich glucose as carbon source	Olive oil (51%) Soya oil (39%)	[43]
	C. glabrata CBS138	Carbohydrate rich glucose as carbon source	Olive oil (49%) Soybean oil (51%) Mustard oil (50%) Almond oil (50%)	[43]
	Pichia anomala PY1	4% soybean oil as carbon source and 0.4% $NaNO_3$ as nitrogen source	N/A	[42]
Rhamnolipid	*Pseudomonas aeruginosa* 47T2	Olive and sunflower oil as carbon source	Linseed oil (90%)	[45]
	Pseudomonas aeruginosa PA01	Palm fatty acid distillate	Sunflower oil (30%)	[44]

Continued

Table 2 Shows a list of biosurfactants with emulsifying properties that are commonly employed in food processing. *Continued*

Name of the biosurfactant used as bioemulsifier	Microorganism	Substrate used for biosurfactant synthesis	Emulsifying capacity of vegetable oils (E_{24} in percentage)	References
Glycolipopeptide	*Lactobacillus paracasei*	Glucose and corn steep liquor as sole carbon source	N/A	[48]
Glycoprotein	*Kluyveromyces marxianus* FII 510700	Lactose as carbon source and ammonium sulphate as nitrogen source	Corn oil (76%)	[50]
	Antarctobacter sp. TG22	*n*-Hexadecane as carbon source and ammonium nitrate as nitrogen source	Olive oil (>40%) Sunflower oil (>20%) Ground nut oil (>55%)	[49]
Alasan	*Acinetobacter radiorcsistant*KA53	N/A	Soybean oil Coconut oil	[53]
Not properly characterized	*C. utilis* UFPEDA 1009	Glucose and waste canola frying oil as carbon source, and ammonium nitrate employed as nitrogen source	Soybean oil (43%) Sunflower oil (73%) Corn oil (73%) Rice oil (33%) Motor oil (30%)	[57]
Not properly characterized	*C. bombicola* URM 3718	Sugar cane molasses, corn steep liquor, and residual soybean oil employed as carbon source	Soybean oil (45.81%) Canola oil (45.44%) Sunflower oil (54.60%) Corn oil (48.28%) Peanut oil (68.97%)	[37]

it, are all given in the table. By studying this, future researchers will be able to gain a better understanding of biosurfactants' emulsifying activity.

11 Baked goods

Baked goods are the most regularly foods, which are formed from butter or dough and cooked by baking. Now the question arises as to what is meant by "baking." It's a cooking technique that involves cooking food over a lengthy period of time in dry heat source, such as an ovens, but also hot stones or hot ashes. Bread, cookie, muffin, and cake are the most common baked goods.

The major components responsible for the texture, flavor, and visual appearance of most bakery items are flour, water, and leavening agents. Numerous minor components, such as milk, egg, salt sugar, fats, and oils, can be added to adjust these properties [61]. Leavening can be caused by yeast or bacterial fermentation, chemical interactions, or the dispersal of ambient or injected gasses in the butter. One of the most significant ingredients in baked goods is baking soda. CO_2 from injected sodium bicarbonate, also known as baking soda, leavens cakes, breads, cookies, and other food products. Almost all baking items contain fats and oils, and the most frequent fats used in baked goods are beef fats, lard, and vegetable oils. Soybean oil, corn oil, peanut oil, cottonseed oil, and coconut oil are the most widely used vegetable oils [62]. Yolks, one of the most essential parts of an egg, contain roughly 50% solids, with 60% or more being heavily emulsified fat, and are utilized in bakery dishes for color, texture, and taste [63]. Wheat flour typically contains about 1% sugar, the majority of which are fermentable sugars including glucose, fructose, sucrose, and maltose. Because yeast quickly transforms sucrose to fructose and glucose, invert sugar also is introduced. Because of these important components, bakery products are more delicious and acceptable to consumers, and as a result, demand for bakery products has increased over the world [21].

In the year 2020, the worldwide bakery products industry was worth almost USD 331.37 billion. Between 2021 and 2026, the bakery products sector is predicted to develop at a CGAR of 4.6% reaching a value of around USD 436.91 billion [64]. In the baked goods sector, the United states is a crucial figure, accounting for 20% of worldwide bakery market share. China owns 7% of the worldwide bakery market share, closely behind United States. The Indian baking market is catching up fast. In 2020, it was estimated to be worth roughly USD 7.60 billion. Between 2021 and 2026, the market is expected to increase at an annual rate of 8.5%, reaching a value of USD 12.39 billion [65]. The Indian bakery market is dominated by baked cookies and biscuits. Based on these impressive figures, it's easy to predict that demand for bakery products will continue to rise in the future, benefiting not just the economy but also providing us with the nourishment we require. The global bakery products market in 2021 is visualized in Fig. 5 [64].

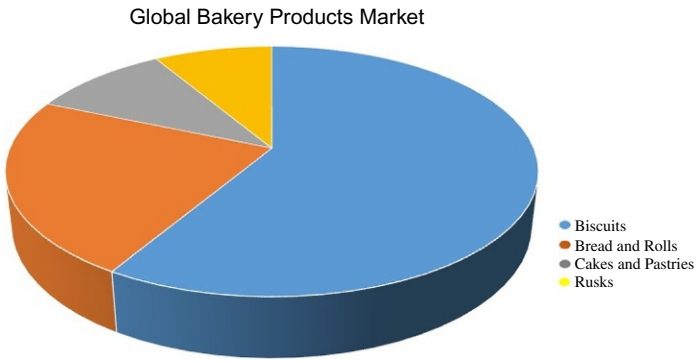

FIG. 5

Global bakery products market [64].

12 Necessity of inclusion of biosurfactants in baked goods

Flour-based goods, such as cakes, cookies, and bread, have risen in prominence as target items for novel healthier formulations in the past few years, satisfying unique dietary needs connected with market preferences [9]. As a result, researchers have looked at the inclusions of various substances that are higher in nutritional value while also being biodegradable, biocompatible, and low in toxicity. Biosurfactants are the most suitable options in this case, and their outstanding application as emulsifiers had already been described. The ability to produce stable emulsions, which can considerably improve the texture, creaminess, and visual appeal of food products, is the most ideal attribute for microbial surfactants in the food processing industry [66]. Biosurfactants can also be employed in baked goods to manage consistency, increase the products usable life, and solubilize aromatic oils during the cooking of oils and fats. In this context, it's worth mentioning that they're a biocompatible solution, which can be applied as antibiofilm, antimicrobial, and antioxidant agents in baked goods formulations [15].

All advancements in texture, dough, volume, and preservation of baked items documented in the last few years are now briefly reviewed. Fig. 6 highlights the applications of biosurfactants in baked goods.

13 Diverse utilization of biosurfactants in baked goods

13.1 Baked goods with a better textural profile

The texture of a food product has a big impact on whether or not people like it. Consumers utilize texture as one of the criteria for judging the quality of food. When we taste the food in our mouth, we use our sense of texture. It can be defined as "solid," "liquid," "hard," "soft," "smooth," "rough," "crispy," "lumpy," "creamy," "crumby," and so on [66,67]. Textural terminology such as surface

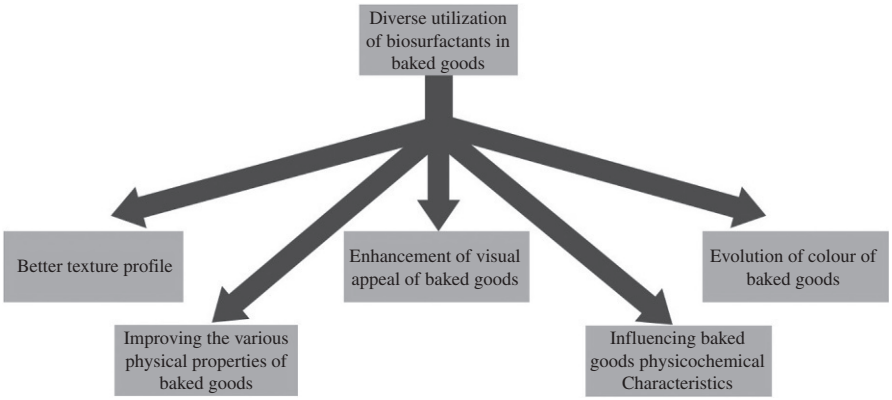

FIG. 6

Representation of various applications of biosurfactant on baked goods.

tension, viscosity, density, and other physical aspects of a food product are intimately related. Because texture is intrinsically tied to food structures at both the micro and macro levels, and is heavily influenced by connections between food biopolymers including polysaccharides, lipids, and proteins, empirical approaches to assess rheological and/or mechanical qualities could be used to define parameters relating to relevant sensory textural properties [67,68].

The biosurfactants can be regarded a possible food additive because it had no noticeable effect on the texture profile of the dough during baking [69]. This test must be repeated with additional formulation to evaluate if the biosurfactant can be included without damaging the food's original texture. Sensory evaluation experiments should also be carried out with the goal of expanding the biosurfactants' usage. Biosurfactants have been shown in the literature to improve the texture profile of a variety of flour-based foods including muffins, cakes, cookies, and crackers as well as breads [70].

According to Pereira et al., fat content in foods such as egg yolk, which is made up of a lipoprotein mixture with a protein content of 16% and 35% lipid content, has a substantial impact on texture. The yolk's lipid content contains 66% triglycerides, 28% phospholipids, 5% cholesterol, and trace amounts of other lipids [71]. Ribeiro et al. looked at how texture affected the inclusion of biosurfactant synthesized by *C. utilis* UFPEDA1009 in cookie formation. They investigated the effects of particularly or completely replacing pasteurized egg yolk with biosurfactant on the physical qualities (firmness, elasticity, and cohesion) of cookies. In comparison to the standard formulation, it was discovered that replacing the yolk with the biosurfactant resulted in minor variations in regard to all variables assessed before cooking. After baking, replacing the yolk with the biosurfactant resulted in a significant drop in firmness, with an even greater reduction (from 445.52 to 354 N) when the substitution was complete [71]. This could be owing to the biosurfactants' high lipid content, which results in a softer, smoother texture product.

Similar results were reported by Zouari et al., who found that adding a bioemulsifier produced by *B. subtilis* SPB1 at concentration over 0.5% resulted in a large decrease in stiffness[72]. The inclusion of a bioemulsifier, on the other hand, resulted in increased cohesion and less elasticity according to the researchers. Supplementary cookie with bioemulsifier produced by *B. subtilis* has been shown to improve the texture profile and reduce microbial proliferation sensitivity. Furthermore, when compared with the commercial emulsifier glycerol monostearate, the introduction of the same bioemulsifier to a wheat flour and sesame seed cookie formulation resulted in a better textural profile with increased smoothness and overall quality [21].

Ribeiro, Guerra, and Sarubbo looked at the potential for a glycolipid biosurfactant produced by *Saccharomyces cerevisiae* URM6670 to be used in baked goods. The partial substitution (2% egg yolk and 2% biosurfactant) and total substitution (4% biosurfactant) of the yolk by the biosurfactant had no effect on the majority of the variables studied in the texture profile analysis. The lone exception was hardness before baking, which dropped from 63.57 to 44.70 N with total substitution. This could be owing to the introduction of biosurfactant and the elimination of egg yolk, which resulted in a higher concentration in the formulation [69]. Overall, the partial and total substitution of egg yolk with the biosurfactant produced by *S. cerevisiae* had no negative impact on the final products' qualities, as the substituted formulations gave similar outcomes to the conventional formulation.

In summary, the biosurfactant displayed functional capabilities similar to egg yolk, indicating that it can be replaced with the biosurfactant examined without impairing final product attributes [71].

Kiran et al. investigated the effect of adding a lipopeptide biosurfactant produced by *Nesterenkonia* sp. MSA31 on muffin texture development. In comparison with positive and negative controls, the lipopeptide-infused muffin exhibits less chewiness, gumminess, and stiffness [39]. Lipopeptide was found to be useful in improving the softness and overall quality of muffins at concentration of 0.75%. When compared with the control, muffins made with 0.75% lipopeptide showed a higher level of springiness. Tess et al. found that a high springiness rating was related with freshness and great muffin quality [73]. As a result, the muffins containing 0.75% lipopeptide were of good quality, with greater springiness value. The compression energy necessary to break down the baked product for swallowing is described as cohesiveness. The muffin with 0.75% lipopeptide had a higher cohesiveness, requiring less compression energy to break it down, indicating the soft character of the muffin. Furthermore, muffins with 0.75% lipopeptide had a very low chewiness index, showing again the soft nature of muffins [39].

Minf et al. investigated the impact of adding a *B. subtilis* SPB1 biosurfactant (lipopeptide) to bread quality. The inclusion of SPB1 bioemulsifier improved the textural profile of bread significantly, especially at a concentration of 0.075% (W/W). In reality, adding emulsifier to bread resulted in a significant decrease in chewiness, firmness, and adhesion indices, as well as a modest rise in cohesion. SPB1 biosurfactant has a considerable antistalling action, according to crumb texture

evolution during storage. These findings showed that the SPB1 biosurfactant could be very useful in the bread-making sector [74].

Sliva et al. investigated the use of *Candida bombicola* URM 3718 biosurfactant as a meal supplement in the form of cupcakes. The biosurfactant was added to the cupcake dessert recipe, substituting 50%, 75%, and 100% of the vegetable fat in the original recipe. The textural profile of the cupcakes did not change significantly when vegetable fat was partially or completely replaced with biosurfactants [37].

Durval et al. used a biosurfactant made from waste frying oil produced by *B. cereus* UCP 1615 as a bioemulsifier in cookie formation [36]. The Turkey test demonstrated that adding the biosurfactant had no effect on the majority of the texture variables studied (firmness, sponginess, and cohesiveness). Furthermore, the consistency of the results for the various formulations is critical for the preservation of cookies' usual qualities; the bioemulsifier demonstrated to be successful in this regard, resulting in an improvement in the cookies' properties when compared with the conventional formulation. Additionally, a substantial difference in hardness was detected for the cookie created with a formulation containing 0.5% biosurfactant, epically when compared with the conventional formulation, implying that utilizing this biosurfactant increased crispness. Chewability should improve as a result of the large increases in cohesion and sponginess. The biosurfactants also ensured that the cookie texture was well presented after 45 days of storage. All of these findings show that the biosurfactants could be used as green emulsifier in the future, given the current markets requirement for biocompatible baked products.

13.2 Improvements to baked goods' different physical properties

Biosurfactants have a significant impact on the increase of making physical qualities such as specific volume, diameter, thickness, weight, and spread factor in baked goods. These improvements have a direct impact on the volume and appearance of baked items.

The Tukey test was used to examine statistical difference in diameter, height, weight, and spread factor between the standard formulation and formulations containing various concentrations of the biosurfactant in the study of *B. cereus* UCP 1615 biosurfactant addition. The spread factor increased linearly when the concentration of the biosurfactant was increased. A formulation comprising 1% biosurfactant yielded the highest spread factor (8.44). The presence of fatty acids in the nonpolar component of the biosurfactant produced by *B. cereus* UCP 1615 supported a rise in the amount of lipids in the cookies, which was immediately represented in the greater spreading rate [36]. While cooking, it is pretty apparent that the spread rate has a direct impact on the height and diameter of cookies. Indeed, there were considerable height disparities among the formulations, with a higher spread factor resulting in a lower height. Following that, when the spread factor was at its maximum at 1% biosurfactant concentration, the height was found to be at a minimum of 5.79 mm.

One of the most essential characteristics in determining the quality of biosurfactant in cookie formulation is the dispersion factor, which is connected to the diameter; furthermore, which can be affected by lipid concentration [71]. As in the previous study, the replacement of egg yolk with biosurfactant produced by *C. utilis* UFPEDA1009 resulted in an increase in lipid concentration, as well as an increase in diameter as a result of the increased mass spreading factor [72]. The diameter and spread factor of cookies were 47.42 mm and 6.20, respectively, at a concentration of 2% biosurfactant. However, when the biosurfactant concentration was increased to 4%, both parameters increased to 48.19 mm and 6.32, respectively, as previously stated [72].

The glycolipid biosurfactant produced by *S. cerevisiae* URM 6670 didn't break the physical properties of the dough after breaking in the cookie recipes. Physical attributes such as weight, diameter, and thickness of cookies after baking for the standard formulation with only 4% egg yolk were 6.92 g, 46.82 mm, and 7.34 mm, respectively. However, substituting the egg yolk with biosurfactant boosted all of the physical properties. The stated physical properties values were 7.08 g, 48.80 mm, and 7.33 mm, respectively, when 2% egg yolk and 2% biosurfactant were present; however, when only 4% biosurfactant was present, the depicted features were 7.13 g, 47.39 mm, and 7.64 mm, respectively [69].

Sliva et al. compared standard formulations with 100% vegetable fats and no biosurfactant to other formulations with varying properties of vegetable fats and biosurfactant, during the baking of cupcake. The results showed that when vegetable oils were replaced with a biosurfactant from *C. bombicola* URM 3718, there was a more dramatic fluctuation in height [37]. Even though there was a noticeable difference in height values, the respective weight measurements did not alter much. This is because reducing the amount of fat used in cupcake preparation reduces the amount of air bubbles integrated into the dough that burst during cooking, leading to an increase in the density of crumbs, and as a result an increase in dough size [75].

Minif et al. investigated the effect of adding a bioemulsifier to a loaf's specific volume and appearance throughout the baking process of the bread. The inclusion of SPB1 biosurfactant resulted in a large rise in specific volume and, as a result, a decrease in density. The efficiency of SPB1 biosurfactant was compared with that of soya lecithin, and it was shown that, with the exception of 0.025%, SPB1 biosurfactant was much more successful in increasing specific volume [74]. Furthermore, adding SPB1 biosurfactant at concentration of 0.05% and 0.075% increased the specific volume of bread by 40.7% and 38.5%, respectively. In comparison with the control test, this is the most significant increase in specific volume. However, with the same soya lecithin concentrations, specific volume improvements are no longer greater than 26%. Now comes the question of why the bread's specific volume was increased [76]. These events could be explained by the fact that emulsifier binds to the hydrophobic surface of gluten proteins in dough, causing them to aggregate. A robust protein network may enhance the volume of bread after cooking by improving the gas storable capacity of dough during fermentation [77]. Moreover, in water, hydrophilic emulsifiers can produce lamellar liquid-crystalline phases that bind with

gliadin. These features may help to increase dough flexibility, permitting gas cells to spread and increase the volume of baked goods.

Similarly, utilizing wheat flour mixed with sophorolipids to make bread enhances gas retention during fermentation and increases loaf volume, giving the bread a better appearance and a longer shelf life.

13.3 Influencing baked goods' physicochemical characteristics

The use of biosurfactants in the production of baked goods has a significant impact on physicochemical properties such as fat, protein, and carbohydrate content, as well as moisture level. The existence of fatty acids in the hydrophobic structure of the biosurfactant causes a large increase lipid content when a lipopeptide or glycolipid-type biosurfactant is used to replace the egg yolk in baked goods. Although egg yolk includes saturated fats, cholesterol, and triglycerides, it also contains potentially hazardous chemicals including trace elements and heavy metals that can induce hypertension, depression, Alzheimer's disease, and gastrointestinal cancer at high quantities [69]. As a result, replacing egg yolk with biosurfactant is quite advantageous. A biosurfactant, on the other hand, can also protect against cardiovascular disease because it is made up of monosaturated fat [78]. Moreover, adding biosurfactants to boost lipid content increases the energy value of food products dramatically [36]. The higher the energy content of baked goods, the more likely it is to be acceptable by the consumer [79]. With this in mind, the bakery industry has recently focused on the large-scale application of biosurfactants in food processing.

Moisture has the potential to shorten the shelf life of baked goods since it is linked to the water activity in the food, which promotes microbial infection. In comparison to the biosurfactant formulation, vegetable and animal fats had a higher moisture content in their compositions [36,37]. Consequently, various studies have been conducted in which biosurfactants have been used to substrate vegetable and animal fats in baked goods to reduce moisture levels. Baking flours are infected with fungi and bacteria, causing baked goods to deteriorate and causing major health problems. Subsequently, the biosurfactant with the ability to lower moisture levels while also providing excellent antimicrobial activity is the most effective [74,80].

The levels of ash, lipids, carbohydrates, and energy content were not changed significantly while compared with the standard formulations in Ribeiro et al.'s evaluation of the potential application of *C. utilis* UFPEDA 1009 biosurfactant in cookie formulation [18]. The moisture content remained low, spanning between 0.02% and 0.07%, with lesser levels when the yolk was removed. Likewise, both alternative formulations had lower protein content than the standard formula, which is likely owing to the unavailability of proteins in the biosurfactant and the existence of proteins in the yolk. Lipid percentage rises from 21.96 in standard formulation to 24.58 and 27.96, respectively, in the alternative formulation, as previously stated. As a result of the higher lipid concentration, the energy value of the alternative formulation comprising partially or totally biosurfactant in place of egg yolk increased. The

energy value enhanced findings suggest that biosurfactants can effectively replace egg yolk in cookie-making process.

The addition of *S. cerevisiae* URM 6670 biosurfactant to the cookie dough had no negative effect on the product's physicochemical attributes after baking. According to experimental outcomes, both partial (2% biosurfactant) and total (4% biosurfactant) substitutions resulted in a considerable shift in moisture [69]. Moreover, as the biosurfactant concentration grew, the lipid content increased dramatically, affecting the energy value of the cookies correspondingly. With no biosurfactant, the energy value was 503.40 cal., which increased to 509.52 cal. in partial substitution and 518.95 cal. in total substitution. This enhancement of energy value and decreasing animal fat content in foods suggest that it is a good egg yolk substitute. But both modifications had a significant negative impact on the protein content, since it decreased in tandem with the reduction in egg yolk concentration [69]. This is due to the biosurfactant's low protein concentration.

The presence of fatty acids in the nonpolar part of the biosurfactant synthesized by *B. cereus* UCP 1615 stimulated an increase in the levels of lipids in the cookies. The biosurfactants composition contained 78% necessary and unsaturated fatty acids, indicating that its inclusion should not impair the cookies' nutritional properties [81]. In comparison to the other formulations with no biosurfactant, 0.5% or 0.25% biosurfactant, the formulation with 1% biosurfactant has a much higher lipid content. The cookies' caloric value increased proportionally as a result of the lipid enhancement. In comparison to the standard formulation, the formulation with 0.5% biosurfactant showed a 24% reduction in moisture [36]. The usage of the biosurfactant was found to be really promising because the reduction in moisture reduces the multiplication of microbes, hence increasing the product's longevity.

Due to the high lipid content of glycolipids, an increase in the percentage of lipid is observed with increase in the proportion of glycolipid biosurfactant produced by *C. bombicola* URM 3718 in cupcakes. It is significant to mention that approximately 75% of the fatty acids in *C. bombicola* glycolipid are unsaturated, while the reaming 25% are saturated. Because this biosurfactant differs from vegetable fat, which may have trans fatty acids that raise cholesterol and cause cardiovascular disease, it is possible to conclude that replacing vegetable fat with biosurfactant improves the quality of the prepared cupcake [78]. When compared with the normal formulation, this increase in lipid content considerably enhanced the calorie value of the cupcake in all formulations incorporating partial or complete replacement with biosurfactant [37]. Furthermore, replacing vegetable fat with biosurfactant resulted in a significant reduction in the moisture value, extending the shelf life of the cupcake.

From an economic and health safety standpoint, the occurrence of mold and fungal deterioration in bread is a major concern. Rope deterioration was caused by bacteria of the species *Bacillus*, while mold spoilage was caused by fungus [82,83]. As a result, the most critical aspect is to protect the bread from these microorganisms. Minif et al. studied the impact of SPB1 biosurfactant addition on molds in order to protect bread from microbial contamination. They found that decreasing the moisture level in bread reduced susceptibility to microbial contamination considerably. Consequently, the SPB1 biosurfactant extends bread's shelf life [74].

13.4 Evolution of color of baked goods

Color is a key aspect in increasing the attention of baked goods, and it is linked to acceptability and taste. In this study, three terms are introduced: L^*, a^*, and b^*, where L^* represents the color range of lightness to darkness as defined by Hunter Lab benchtop color measurement systems' color scale standard, a^* denotes green (−) to red (+), and b^* indicates blue (−) to yellow (+) [39]. The muffin made with lipopeptide from *Nesterenkonia* sp. MSA31 had a small rise in the b^* value, which implies yellow coloration. In comparison with the negative and positive controls, which had b^* values of 32.57 and 34.20, respectively, the lipopeptide muffin had a b^* value of 36.0 [39]. Therefore, the muffins made with lipopeptide were light yellow in color, and this study gave the baked industry novel insight into how adding a bioemulsifier could improve customer acceptability of baked goods.

The replacement of egg yolk with the biosurfactant produced by *S. cerevisiae* improved the textural characteristics of baked goods while also changing their color to a slightly darker side, increasing their visual acceptance [69]. However, more research comprising a comprehensive sensory analysis will be undertaken in the future in order to utilize this discovery in the large industrial sector. When a biosurfactant produced by *C. utilis* UFPEDA 1009 was employed to replace egg yolk during cookie production, similar results were reported [71].

The application of several biosurfactants in baked goods is summarized in Table 3.

14 Challenges in the baked products from a technological and economic standpoint

Despite the benefits that biosurfactants provide to food composition in terms of textural and physical characteristics as well as nutritional value, their usage in the baked industry remains limited. This is owing to the fact that baked goods' compositions are inherently complicated. Baked goods are made up of a variety of constitutes that differ from one another. Different baked foods, for example, include different amounts of fats, proteins, carbohydrates, minerals, vitamins, and fibers [15]. This effect can be caused by a variety of baked processing methods, including temperature variation, pressure changes, pH alteration, viscosity and solubility changes [17]. Thus the interaction between the compounds contained in baked dough and the functional characteristics of products exposed to industrial application remains an impediment that must be overcome for large-scale applications of biosurfactants in baked goods formation [15,21]. Furthermore, the cost of biosurfactant remains expensive, posing a barrier to its widespread industrial use. Following that, the attention must be on the synthesis of these bioemulsifiers from low-cost, environmentally friendly substrates. One of the best examples in this regard is agro-industrial waste [2]. In this regard, ultra-modern technological features must be developed, as well as economic methods, not only to lower the amount of biosurfactant utilized, but also to improve the effectiveness of biosurfactants at low concentrations.

Table 3 Highlights the diverse applications of biosurfactants in baked goods.

Microorganism	Biosurfactant type	Baked goods type	Diverse applications	References
Candida utilis UFPEDA1009	Lipoprotein	Cookie	**(i)** Improve the cookie's textural profile, resulting in a softer, smoother product **(ii)** The inclusion of biosurfactant increased the diameter, which improved the mass spreading factor **(iii)** With the introduction of biosurfactant, lipid concentration rises, increasing the energy value of cookies **(iv)** Because the moisture content of cookies is reduced, their shelf life is extended	[57]
Bacillus subtilis SPB1	Lipopeptide	Cookie	**(i)** Increased smoothness and reduced stiffness in the textural profile **(ii)** Protect bread from microbial contamination	[72]
Saccharomyces cerevisiae URM6670	Glycolipid	Cookie	**(i)** The textural profile of baked goods has improved **(ii)** When biosurfactants are added to cookie dough, physical qualities as weight, diameter, and thickness improve **(iii)** The amount of lipids in the cookies increased considerably, increasing the energy value of the cookies **(iv)** Changing the color of cookie to a slightly darker side and thereby increasing their visual acceptance	[69]
Nesterenkonia sp. MSA31	Lipopeptide	Muffin	**(i)** Less chewiness, gumminess, and stiffness, as well as enhanced softness, have resulted in a dramatic improvement in texture profile **(ii)** The muffins made with lipopeptide were light yellow in color	[39]
B. subtilis SPB1	Lipopeptide	Bread	**(i)** Biosurfactant greatly improved the texture profile of bread **(ii)** The use of biosurfactant greatly increased the bread's specific volume	[74]
C. bombicola URM 3718	Lipopeptide	Cupcakes	**(i)** When vegetable fat was replaced with biosurfactant, the textural profile did not change considerably **(ii)** The inclusion of biosurfactant enhances the density of crumbs, resulting in a larger dough **(iii)** The lipid level of the food was significantly improved by replacing vegetable fat with biosurfactant, which increased the calorie value	[37]
B. cereus UCP 1615	N/A	Cookie	**(i)** The inclusion of biosurfactant helped to boost spread factor, which had a positive impact on cookie height and diameter **(ii)** The cookies' caloric value increased proportionally as a result of the lipid enhancement **(iii)** The moisture level decreases microbial proliferation, extending the cookie's shelf life	[36]

15 Future perspective of biosurfactants in baked industry

Biosurfactants are clearly superior to their chemical equivalents, as evidenced by the preceding explanation. Biosurfactants are used in the baked goods sector to change the textural profile of the products while also improving their physical and physicochemical attributes. Furthermore, it aids in improving the sensory attributes of products, such as color, fragrance, and taste, thereby capturing customers' attention [20]. Biosurfactant's antimicrobial action has a long shelf life and can thus be stored for a long period. The egg yolk, which is a key component of any baked goods and is linked to a variety of health problems, is also replaced by the biosurfactant resulting in healthier products [78]. Because of these healthier modifications, it is now time to apply 21st century research on biosurfactants' potential use on baked goods in the research lab to the large industrial sector. However, as previously indicated, it causes a difficulty with the production of low-cost biosurfactant, which is the principal barrier to its widespread usage. Another severe issue has arisen as a result of the experimental processes for baking biosurfactant-based baked items. There are no optimal approaches for modifying the qualities of a certain type of baked foods with a specific biosurfactant. As a result, efforts must be made to develop reliable, low-cost technologies for obtaining biosurfactants on a wide scale [15,17]. The succeeding biosurfactant will be used in baked products manufacturing by applying a new ideal modified procedure that is still unknown to us, but must be discovered by scientific researchers in the near future. Food biotechnology has already made significant progress. Durval et al., Kiran et al., Minif et al., Ribeiro et al., Sliva et al., and other world-renowned researchers are searching for the best technologies [36,37,39,69,74]. Other key factors that need to be explored are whether all biosurfactants have a regular sensory impact and increase the textural profile of baked goods or not. Furthermore, the interactions of biosurfactants with food components will be examined more closely, as changes in these interactions alter the basic characteristics. If these barriers are addressed, biosurfactants will have the undeniable potential to replace conventional surfactants while also providing health and environmental benefits. Now, as we previously indicated, we must overcome all of the issues that have limited its implementation in a broader range in order to make it the best baked ingredients in the upcoming decades.

16 Conclusion

In 2021, the world's population will be around 7.9 billion people. This enormous number of people requires a minimum amount of food to survive. With the rise in daily population, essential food necessities have just at the technology required for fast food production, but also to keep a close check on the world of research to ensure that these foods are suited for promoting good health. People today rewarded for their health, and they have demonstrated a high demand of "totally natural" foods. Customers' pressure has prompted food industry experts to look for

natural alternatives to the several synthetic additives now utilized in food compositions. Biosurfactants clearly play a key role in this regard, since they are superior to their chemical equivalents in terms of low toxicity, outstanding physicochemical features, biocompatibility, and biosynthesis for renewable resources. In addition to the previously described advantages, biosurfactants have the ability to emulsify, which is discussed in detail throughout this chapter. Because of their excellent emulsifying characteristics, they are being employed as bioemulsifiers in the food. Since the beginning of the 21st century, scientists have been interested in exploring the emulsifying activity of any biosurfactants in any food industry. Biosurfactants such as glycolipid, lipopeptide, lipoprotein, glycolipopeptide are all examples of biosurfactants that have demonstrated outstanding emulsifying action in food processing and are briefly discussed in this chapter. Emulsifying activity is also evidenced by polymeric biosurfactants such as emulsan, alasan, yansan, and glycoprotein. The HLB value of biosurfactants forecasts which biosurfactants are better at emulsifying a specific oil-water interface. The emulsifying index of a biosurfactant toward specific fats or oils indicated how effective it is an emulsifier. The higher the emulsification index, the better it will be at emulsifying a particular oil. The field's research has only recently begun, and it is now being analyzed. More work must now be put into finding stable, low-cost ways for obtaining biosurfactants on a large scale. Not only that, but more efficient techniques are being sought so that biosurfactants can be used as emulsions in a wider range of food processing applications. The sensory impact of adding biosurfactants to baked goods' constituents is also an aspect that needs to be looked into. The efficacy in the processing of baked products, as well as how to improve the corresponding biosurfactant application in baked goods, is briefly discussed above. These biomolecules will undoubtedly be part of the baking processing industry as constituents in formations and surface altering additives in the near future if the focus is given to them.

References

[1] S. De, S. Malik, A. Ghosh, et al., A review on natural surfactants, RSC Adv. 5 (2015) 65757–65767.

[2] M. Fakruddin, Biosurfactant: production and application, J. Pet. Environ. Biotechnol. 3 (2012).

[3] L. Rodríguez-López, M. Rincón-Fontán, X. Vecino, A.B. Moldes, J.M. Cruz, Biodegradability study of the biosurfactant contained in a crude extract from corn steep water, J. Surfactant Deterg. 23 (2019) 79–90.

[4] H. Amani, Evaluation of biosurfactants and surfactants for crude oil contaminated sand washing, Pet. Sci. Technol. 33 (2015) 510–519.

[5] D.J. McClements, C.E. Gumus, Natural emulsifiers—biosurfactants, phospholipids, biopolymers and colloidal particles: molecular and physicochemical basis of functional performance, Adv. Colloid Interf. Sci. 234 (2016) 3–26.

[6] K. Salek, S.R. Euston, Sustainable microbial biosurfactants and bioemulsifiers for commercial exploitation, Process Biochem. 85 (2019) 143–155.

[7] H.P. Kaur, B. Prasad, S. Kaur, A review on applications of biosurfactants produced from unconventional inexpensive wastes in food and agriculture industry, World J. Pharm. Res. 4 (2015) 827–842.
[8] I. Kralova, J. Sjöblom, Surfactant used in food industry: a review, J. Dispers. Sci. Technol. 30 (2009) 1363–1383.
[9] H.K. Sandhu, R. Sehrawat, A. Kumar, P.K. Nema, Overview of food industry and role of innovation in food industry, emerging technologies in food, Science (2020) 3–14.
[10] H.M. Premlal Ranjith, U. Wijewardene, Lipid emulsifiers and surfactants in dairy and bakery products, in: Modifying Lipids in Food, Diotte Consulting and Technology Limited, Cambridge, 2006, pp. 393–428.
[11] M.H. Mondal, A. Sarkar, T.K. Maity, B. Saha, Microbial assisted (*pseudomonas sp*.) production of novel bio-surfactant rhamnolipids and its characterisation by different spectral studies, J. Mol. Liq. 242 (2017) 873–878.
[12] A.Y. Khan, S. Talegaonkar, Z. Iqbal, F.J. Ahmed, R.K. Khar, Multiple emulsions: an overview, Curr. Drug Deliv. 3 (2006) 429–443.
[13] C.E. Stauffer, P. Ambigaipalan, F. Shahidi, Emulsifiers for the food industry, in: F. Shahidi (Ed.), Bailey's Industrial Oil and Fat Products, Wiley, 2020, pp. 1–36.
[14] M.A. Salem, S.M. Ezzat, Nanoemulsions in Food Industry, in: Dispersed Food Systems, IntechOpen, 2018.
[15] I.J.B. Durval, I.A. da Silva, L.A. Sarubbo, Application of microbial biosurfactants in the food industry, in: Inamuddin, et al. (Eds.), Microbial Biosurfactants, Environments and Microbial Biotechnology, Springer, Singapore, 2021.
[16] R.S. Makkar, K.J. Rockne, Comparison of synthetic surfactants and biosurfactants in enhancing biodegradation of polycyclic aromatic hydrocarbons, Environ. Toxicol. Chem. 22 (2003) 2280–2292.
[17] J.M. Campos, I.M. Banat, L.A. Sarubbo, Natural and microbial biosurfactants' use in the food industry, in: R. Thavasi, I. Banat (Eds.), Microbial Biosurfactants and Their Environmental and Industrial Applications, CRC press, 2018, pp. 242–257.
[18] Z. Zhu, Y. Wen, J. Yi, Y. Cao, F. Liu, D.J. McClements, Comparison of natural and synthetic surfactants at forming and stabilizing nanoemulsions: Tea saponin, Quillaja saponin, and Tween 80, J. Colloid Interface Sci. 536 (2019) 80–87.
[19] L. Bai, D.J. McClements, Formation and stabilization of nanoemulsions using biosurfactants: rhamnolipids, J. Colloid Interface Sci. 479 (2016) 71–79.
[20] E.J. Gudiña, L.R. Rodrigues, Research and production of biosurfactants for the food industry, in: G. Molina, V.K. Gupta, B.N. Singh, N. Gathergood (Eds.), Bioprocessing for Biomolecules Productions, Wiley, 2020, pp. 125–144.
[21] B.G. Ribeiro, J.M.C. Guerra, L.A. Sarubbo, Biosurfactants: production and application prospects in the food industry, Biotechnol. Prog. 36 (2020).
[22] J.B. Aswathanarayan, R.R. Vittal, Nanoemulsions and their potential applications in food industry, Front. Sustain. Food Syst. 3 (2019).
[23] J. Jampílek, K. Král'ová, E.V.R. Campos, L.F. Fraceto, Bio-based nanoemulsion formulations applicable in agriculture, medicine, and food industry, in: R. Prasad, V. Kumar, M. Kumar, D. Choudhary (Eds.), Nanobiotechnology in Bioformulations, Nanotechnology in the Life Sciences, Springer, 2019, pp. 33–84.
[24] G. Gundewadi, D.J. Sarkar, S.G. Rudra, D. Singh, Preparation of basil oil nanoemulsion-*Sapindusmukorossi* pericarp extract: Physico-chemical properties and antifungal activity against food spoilage pathogens, Ind. Crop. Prod. 125 (2018) 95–104.

[25] S.M. Jafari, E. Assadpoor, Y. He, B. Bhandari, Re-coalescence of emulsion droplets during high-energy emulsification, Food Hydrocoll. 22 (2008) 1191–1202.
[26] T. Tadros, P. Izquierdo, J. Esquena, C. Solans, Formation and stability of nano-emulsions, Adv. Colloid Interf. Sci. 108 (2004) 303–318.
[27] I. Solè, A. Mastero, C. Pey, C. González, C. Solans, J. Gutiérrez, Nano-emulsion preparation by low energy methods in an non-ionic surfactant system, Colloids Surf. A Physicochem. Eng. Asp. 288 (2006) 138–143.
[28] D.J. McClements, J. Rao, Food-grade nanoemulsions: formulation, fabrication, properties, performance, biological fate, and potential toxicity, Crit. Rev. Food Sci. Nutr. 51 (2011) 285–330.
[29] S.A. Onaizi, M. Alsulaimani, M.K. Al-Sakkaf, S.A. Bahadi, M. Mahmoud, A. Alshami, Crude oil/water nanoemulsions stabilized by biosurfactant: stability and pH-switchability, J. Pet. Sci. Eng. 198 (2021), 108173.
[30] I. Kralova, J. Sjöblom, Surfactants used in food industry: a review, J. Dispers. Sci. Technol. 30 (2009) 1363–1383.
[31] B. Bergenstahl, Physicochemical aspects of an emulsifier functionality, in: G.L. Hasenhuettl, R.W. Hartel (Eds.), Food Emulsifiers and Their Applications, second ed., Springer Science+ Business Media, LLC, 2008, pp. 173–194.
[32] H. Kim, J. Jeon, S. Kim, H. Oh, T. Kwon, B. Yoon, Surface and physico-chemical properties of a glycolipid biosurfactant, mannosylerythritol lipid, from *Candida antarctica*, Biotechnol. Lett. 24 (2002) 1637–1641.
[33] G. Dehghan-Noudeh, M. Housaindokht, B.S.F. Bazzaz, Isolation, characterization, and investigation of surface and haemolytic activities of a lipopeptide biosurfactants produced by *Bacillus subtilis* ATCC 6633, J. Microbiol. 43 (2005) 272–276.
[34] M. Ohadi, G. Dehghannoudeh, H. Forootanfar, M. Shakibaie, M. Rajaee, Investigation of the structural, physicochemical properties, and aggregation behaviour of lipopeptide biosurfactant produced by *Acinetobacter junii* B6, Int. J. Biol. Macromol. 112 (2018) 712–719.
[35] R. Hajimohammadi, S. Johari-Ahar, Synergistic effect of saponin and rhamnolipid biosurfactants system on foam behaviour, Tenside Surfactant Deterg. 55 (2018) 121–126.
[36] I.J.B. Durval, B.G. Ribeiro, J.S. Aguiar, R.D. Rufino, A. Converti, L.A. Sarubbo, Application of a biosurfactant produced by bacillus cereus UCP 1615 from waste frying oil as an emulsifier in cookie formulation, Fermentation 7 (2021).
[37] I.A. Sliva, B.O. Veras, B.G. Ribeiro, J.S. Aguiar, J.A.C. Guerra, J.M. Lunna, L.A. Sarubbo, Production of cupcake-like dessert containing biosurfactant as an emulsifier, PeerJ 8 (2020), e9064.
[38] H. Ghojavand, F. Vahabzadeh, F. Azizmohseni, A. Halotolerant, Thermotolerant, and facultative biosurfactant producer: identification and molecular characterization of a bacterium and evolution of emulsifier stability of a lipopeptide biosurfactant, Biotechnol. Bioprocess Eng. 16 (2011) 72–80.
[39] G.S. Kiran, S. Priyadharsini, A. Sajayan, G.B. Priyadharsini, N. Poulose, J. Selvin, Production of lipopeptide biosurfactant by a marine *Nesterenkonia* sp. and its application in food industry, Front. Microbiol. 8 (2017).
[40] M. Nitschke, G. Pastore, Production and properties of a surfactant obtained from *Bacillus subtilis* grown on cassava wastewater, Bioresour. Technol. 97 (2006) 336–341.
[41] J. Li, M. Deng, Y. Wang, W. Chen, Production and characteristics of biosurfactant produced by *Bacillus pseudomycoides* BS6 utilizing soybean oil waste, Int. Biodeterior. Biodegradation 112 (2016) 72–79.

[42] J. Thaniyavarn, T. Chianguthai, P. Sangvanich, N. Roongsawang, K. Washio, M. Morikawa, S. Thaniyavarn, Production of Sophorolipid Biosurfactant by *Pichia anomala*, Biosci. Biotechnol. Biochem. 72 (2008) 2061–2068.
[43] V.K. Gaur, R.K. Regar, N. Dhiman, K. Gautam, J.K. Srivastava, S. Patnaik, M. Kamthan, N. Manickam, Biosynthesis and characterization of sophorolipid biosurfactant by *Candida* spp.: application as food emulsifier and antibacterial agent, Bioresour. Technol. 285 (2019), 121314.
[44] M.N. Radzuan, I. Banat, J. Winterburn, Production and characterization of rhamnolipid using palm oil agricultural refinery waste, Bioresour. Technol. 225 (2017) 99–105.
[45] E. Haba, A. Pinazo, O. Jauregui, M.J. Espuny, M.R. Infante, A. Manresa, Physicochemical characterization and Antimicrobial properties of rhamnolipids produced by *Pseudomonas aeruginosa* 47T2 NCBIM 40044, Biotechnol. Bioeng. 81 (2002) 316–322.
[46] A.S. Monteiro, M.R.Q. Bonfim, V.S. Domingues, A. Corrêa Jr., E.P. Siqeira, C.L. Zani, V.L. Santos, Identification and characterization of bioemulsifier-producing yeasts isolated form effluents of a dairy industry, Bioresour. Technol. 101 (2010) 5186–5193.
[47] X. Vecino, L. Barbosa-Pereira, R. Devesa-Rey, J.M. Cruz, A.B. Moldes, Optimization of extraction conditions and fatty acid characterization of *L. pentosus* cell bound biosurfactant/bioemulsifier, J. Sci. Food Agric. 95 (2014) 313–320.
[48] A. Ferreira, X. Vecino, D. Ferreira, J.M. Cruz, A.B. Moldes, L.R. Rodrigues, Novel cosmetic formulations containing a biosurfactant from *Lactobacillus paracasei*, Colloids Surf. B 155 (2017) 522–529.
[49] T. Gutiérrez, B. Mulloy, C. Bavington, K. Black, D.H. Green, Partial purification and chemical characterization of a glycoprotein(putative hydrocolloid) emulsifier produced by a marine bacterium *Antarctobacter*, Appl. Microbiol. Biotechnol. 76 (2007) 1017–1026.
[50] T. Lukondeh, N.J. Ashbolt, P.L. Rogers, Evaluation of *Kluyveromycesmarxianus* FII 510700 grown on a lactose-based medium as a source of a natural bioemulsifier, J. Ind. Microbiol. Biotechnol. 30 (2003) 715–720.
[51] J.R. Trindade, M.G. Freire, P.F.F. Amaral, M.A.Z. Coelho, J.A.P. Coutinho, I.M. Marrucho, Aging mechanism of oil-in-water emulsions based on a bioemulsifier produced by *Yarrowialipolytica*, Colloids Surf. A Physicochem. Eng. Asp. 324 (2008) 149–154.
[52] M. Fondi, V. Orlandini, G. Emiliani, M.C. Papaleo, I. Maida, E. Perrin, M. Vaneechoutte, L. Dijkshoorn, R. Fani, Draft genome, sequence of the hydrocarbon-degrading and emulsan-producing strain *Acinetobacter venetianus* RAG-1, J. Bacteriol. 194 (2012) 4771–4772.
[53] M. Alizadeh-Sani, H. Hamishehkar, A. Khezerlou, M. Azizi-Lalabadi, Y. Azadi, E. Nattagh-Eshtivani, M. Fasihi, A. Ghavami, A. Aynehchi, A. Ehsani, Bioemulsifiers derived from microorganisms: applications in the drug and food industry, Adv. Pharm. Bull. 8 (2018) 191–199.
[54] R. Shepherd, J. Rockey, I.W. Sutherland, S. Roller, Novel bioemulsifiers from microorganisms for use in foods, J. Biotechnol. 40 (1995) 207–217.
[55] M. Nitschke, S. Costa, Biosurfactants in food industry, Trends Food Sci. Technol. 18 (2007) 252–259.
[56] A. Toren, S. Navon-Venezia, E.Z. Ron, E. Rosenberg, Emulsifying activates of purified alasan proteins from *Acinetobacter radioresistens* KA53, Appl. Environ. Microbiol. 67 (2001) 1102–1106.
[57] J.M. Campos, T.L.M. Stamford, L.A. Sarubbo, Production of a bioemulsifier with potential applications in the food industry, Appl. Biochem. Biotechnol. 172 (2014) 3234–3252.

[58] P.F.F. Amaral, M.A.Z. Coelho, I.M. Marrucho, J.A.P. Coutinho, Biosurfactants from yeasts: characteristics, production e application, in: R. Sen (Ed.), Biosurfactants, Lands Bioscience, 2008, pp. 428–434.
[59] Ĺ.W.L. França, A.P. Lima, J.A.M. Lemos, C.G.F. Lemos, V.M.M. Melo, H.B. De Sant'ana, L.R.B. Gonçalves, Production of a biosurfactant by *Bacillus subtilis* ICA56 aiming bioremediation of impacted soils, Catal. Today 255 (2015) 10–15.
[60] D. Sharma, B.S. Saharan, Functional characterization of biomedical potential of biosurfactant produced by *Lactobacillus helveticus*, Biotechnol. Rep. 11 (2016) 27–35.
[61] J. Upton, A. Nowak-Wegrzyn, The impact of baked egg and baked milk diets on IgEand non-Ige- mediated allergy, Clin. Rev. Allergy Immunol. (2018).
[62] T. De Pilli, Development of a vegetable oil and egg proteins edible film to replace preservatives and primary packaging of sweet baked goods, Food Control 114 (2020), 107273.
[63] Z. Huang, L. Stipkovits, H. Zheng, L. Serventi, C. Brennan, Bovine milk fats and their replacers in baked goods: a review, Foods 8 (2019) 383.
[64] Bakery Products Market Size, Share, Analysis Report 2021-2026. https://www.expertmarketresearch.com/reports/bakery-products-market.
[65] Indian Bakery Market Size, Share, Report and Forecast 2021-2026. https://www.expertmarketresearch.com/reports/indian-bakery-market.
[66] L. Day, M. Golding, Food structure, rheology, and texture, encyclopedia of, Food Chem. (2016) 125–129.
[67] Y. Pascua, H. Koç, E.A. Foegeding, Food structure: roles of mechanical properties and oral processing in determining sensory texture of soft materials, Curr. Opin. Colloid Interface Sci. 18 (2013) 324–333.
[68] L. Chen, U.L. Opara, texture measurement approaches in fresh and processed food—a review, Food Res. Int. 51 (2013) 823–835.
[69] B.G. Ribeiro, J.M.C. Guerra, L.A. Sarubbo, Potential food application of a biosurfactant produced by Saccharomyces cerevisiae URM 6670, Front. Bioeng. Biotechnol. 8 (2020).
[70] S. Mujumdar, P. Joshi, N. Karve, Production, characterization, and applications of bioemulsifiers (BE) and Biosurfactants (BS) produced by *Acinetobacter* spp.: a review, J. Basic Microbiol. 59 (2019) 277–287.
[71] B.G. Ribeiro, B.O. de Veras, J.D.S. Aguiar, J.M.C. Guerra, L.A. Sarubbo, Biosurfactant produced by Candida utilis UFPEDA1009 with potential application in cookie formulation, Electron. J. Biotechnol. 46 (2020) 14–21.
[72] R. Zouari, S. Besbes, S. Ellouze-Chaabouni, D. Ghribi-Aydi, Cookie from composite wheat-ssame peels flours: dough quality and effect of *Bacillus subtilis* SPB1 biosurfactant addition, Food Chem. 194 (2016) 758–769.
[73] M. Tess, S. Bhaduri, R. Ghatak, K.P. Navder, Physical, textural and sensory characteristics of glutein-free muffins prepared with teff flour (eragrosticstef (zucc) trotter), J. Acad. Nutr. Diet. 9 (2015).
[74] I. Minff, S. Besbes, R. Ellouze, S. Ellouze-Chaabouni, D. Ghribi, Improvement of bread quality and bread shelf-life by *Bacillus subtilis* biosurfactant addition, Food Sci. Biotechnol. 21 (2012) 1105–1112.
[75] S. Zahn, F. Pepke, H. Rohm, Effect of inulin as a fat replacer on texture and sensory properties of muffins, Int. J. Food Sci. Technol. 45 (2010) 2531–2537.
[76] M.H. Azizi, N. Rajabzadeh, E. Riahi, Effect of momo-diglyceride and lecithin on dough rheological characteristics and quality of flat bread, LWT- Food Sci. Technol. 36 (2003) 189–193.

[77] B.S. Kamel, J.G. Ponte, Emulsifiers in baking, in: Advances in Baking Technology, Blackie Academic and Professional, Glasgow, 2004, pp. 179–221.
[78] C.A. Valenzuela, E.J. Baker, E.A. Miles, P.C. Calder, Eighteen-carbon trans fatty acids and inflammation in the context of atherosclerosis, Prog. Lipid Res. 76 (2019).
[79] T.C.S.P. Pires, M.I. Dias, L. Barros, I.C.F.R. Ferreira, Nutritional and chemical characterization of edible petals and corresponding infusions: valorization as new food ingredients, Food Chem. 220 (2017) 337–343.
[80] M. Maleki, R.C. Hoseney, P.J. Mattern, Effect of loaf volume, moisture content, and protein quality on the softness and staling rate of bread, Cereal Chem. (1980) 138–140.
[81] I.J.B. Durval, A.H.R. Mendonça, I.V. Locha, J.M. Luna, R.D. Rufino, A. Converti, L.A. Sarubbo, Production, characterization, evaluation and toxicity assessment of a *Bacillus cereus* UCP 1615 biosurfactant for marine oil spills bioremediation, Mar. Pollut. Bull. 157 (2020), 111357.
[82] J.M. Thompson, C.E.R. Dodd, W.M. Waites, Spoilage of bread by *Bacillus*, Int. Biodeterior. Biodegradation 32 (1993) 55–66.
[83] J.D. Legan, Mould spoilage of bread: the problem and some solutions, Int. Biodeterior. Biodegradation 32 (1993) 33–53.

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CHAPTER

12

Advancements and challenges in production of biosurfactants

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1 Introduction to surfactants

Amphipathic materials are based on the surface-active molecules owing to hydrophilic and hydrophobic groups yearning to absorb at the interfaces of fluids and are termed as surfactants. The hydrophilic and hydrophobic portions allow them to get soluble in polar (water) and nonpolar (oils or fats) solvents. The dual characteristics along with different shapes and sizes of these groups are responsible to perform specific properties and sometimes liable to play a group of properties simultaneously. Like reduction of surface tension, wettability, dispersion/aggregation of solid particles, emulsification/demulsification, foaming/defoaming, and synergistic applications with other surfactants. These properties intrinsically decide their potential applications in different industrial or nonindustrial sectors. Most of daily use commodities (food, beverages, drugs, cosmetics, etc.) either comprise such chemical compounds or involve some proportion of surfactants in their formulations. However, chemical surfactants possess several drawbacks as well, because of potential harmfulness toward environment, ecotoxicity, bioaccumulation, and low biodegradability [1]. For instance, nonylphenol-based industrial surfactants are endocrine disrupters and interfere with the hormonal system of various organisms upon degradation [2]. Research also states that many synthetic surfactants used in drugs or medicines cause skin irritation [3,4]. This context makes them restricted to use and commercialize, withdrawn from markets, and banned in European Union since 2003 [5].

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Therefore, numerous research studies have been stimulating in this regard to preserve nature, shelter human health, and avert environmental degradation via replacing synthetic surfactants with new generational biosurfactants while maintaining their original attributes [6–8].

2 Biosurfactants

Biosurfactants also known as biological or microbial surfactants are the secondary metabolites of either bacteria, yeast, or fungus and are produced at the end phase of their exponential growth. They are the green, natural, and renewable surface-active amphiphilic molecules derived from different microbes, plants, and animals. Biosurfactants consist of two distinct segments with hydrophilic head and hydrophobic tail as shown in Fig. 1A. Head is made up of ester, hydroxyl, phosphate, or carbohydrates with some peptides or proteins while tail consists of fatty alcohols, fatty acids, or hydroxy fatty acids. The biochemical functionalities act as links between both of components such as ethers (C—O—C), amides (N—C═O), and esters (O—C═O) [8]. Micelles and reverse micelles are formed in the presence of polar and nonpolar solvents respectively due to the hydrophilic and hydrophobic nature of head and tail. Likewise, they can aggregate themselves in different structures as shown in Fig. 1B–D.

Biosurfactants are precipitously substituting to conventional surfactants in various industries such as food, petroleum, construction, pharmaceutical, and environmental, because of imparting sufficient biodegradability, less toxicity, bioremediation of pollutants, and offering ecological suitability [9,10]. The production time, growth cycle, and critical micelle concentration (CMC) of biosurfactants are comparatively shorter when synthesized via microorganisms; therefore, commercial appraisal is more profound. For example, *Bacillus subtilis*, *Pseudomonas aeruginosa*, *Acinetobacter calcoaceticus*, and *Candida bombicola* produce surfactin,

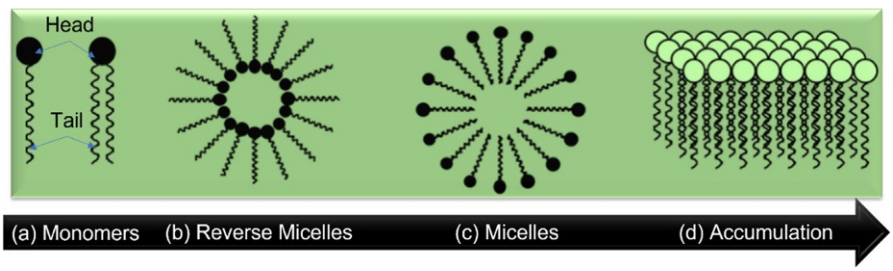

FIG. 1

Transformational structure of biosurfactants with increase in concentration and in different solvents.

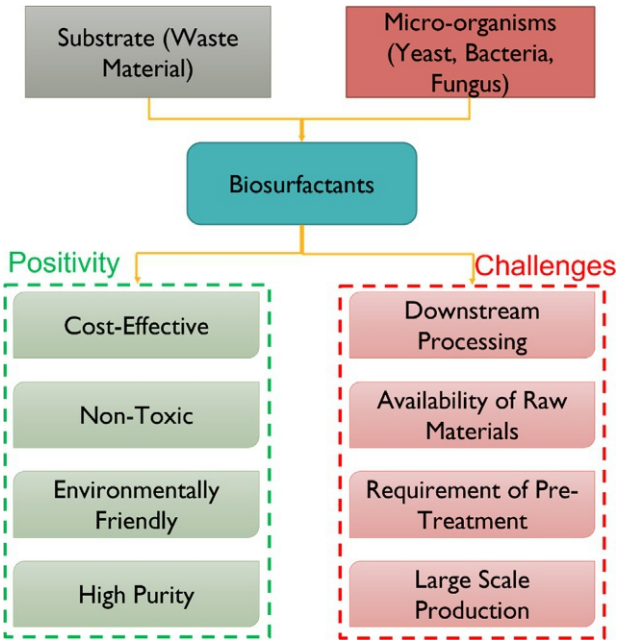

FIG. 2

Positivity and challenges in the production of biosurfactants.

rhamnolipid, emulsan, and sophorolipid surfactants, respectively [11], and have potential to compete at the industrial scale as well.

In food sector, they are demonstrating ample reliability in a variety of ways, e.g., as emulsifiers, foaming, antiadhesive, antimicrobial, wetting agents, and solubilizers as well. However, their production at large scale is uneconomical and hindering successive commercialization because of potential challenges shown in Fig. 2. Researchers are incessantly investigating advanced strategies to reduce initial cost and stimulate large-scale utilization. It has been tried many times to counter such challenges by utilizing low-cost substrates (generated from wastes), microbial strains with higher production rate and adopting simpler extraction processes [12] but still lacking to implement at larger scale. This chapter is exploring the recent advancements, adequateness, and economic viability of new generational biosurfactants and their potential to be employed at industrial level.

3 Classification of biosurfactants

Unlike synthetic surfactants, which are classified according to their polar groups, the biosurfactants are classified based upon their parent microbes, composition, and molecular masses, as represented in Fig. 3 [13]. In the following sections, their different classifications have been discussed in detail.

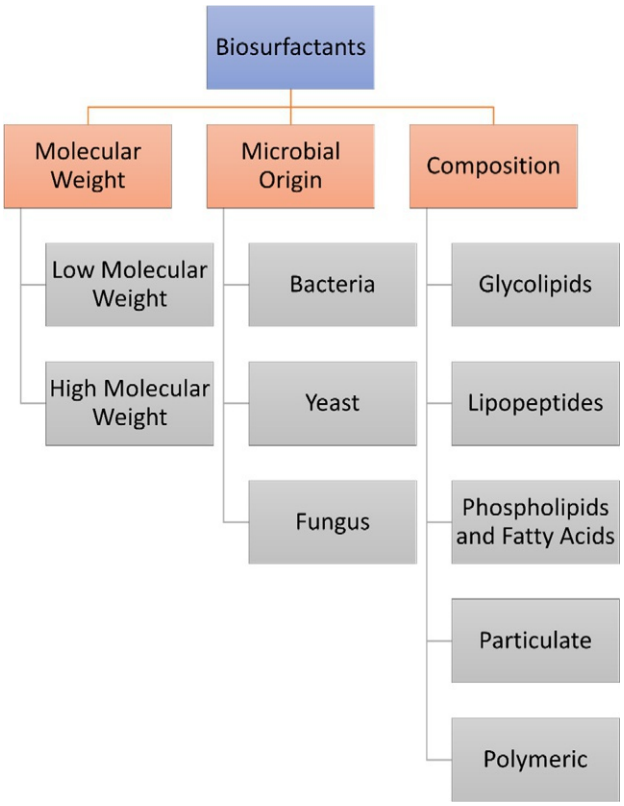

FIG. 3

Classification of biosurfactants according to molecular weight, microbial origin, and composition.

3.1 Classification based on parent microbes

The biosurfactants can be produced from different microbial species such as yeast, fungus, and bacteria; therefore, one of their classifications is based upon the microbial origin. Microbes-based biosurfactants remain assimilated with cell membrane or concealed in culture medium [14,15]. Microbial surfactants are the compounds having strong functional relevance with microorganisms and show quite similar attributes such as motility, cell-to-cell interaction, differentiation, stickiness to substrates, shielding against elemental toxicity, great ionic strength and act as carbon and energy-storing molecules. They present capability to adjust and grow within the boundaries specified for living substances [16]. There is a broad diversity of microbes from distinct genera that have capability to produce biosurfactants with exceptional structural and functional assortments [17]. The screening of microorganisms capable of producing biosurfactants can be done through a variety of methods.

Aximetric drop-shape analysis, colorimetric estimation, rayed drop collapsing test, direct thin layer chromatography (TLC)method, and estimating the emulsification index are the most common adopted strategies for capability analysis of microbial biosurfactants production.

3.2 Classification based on composition

According to compositional or structural classification, the biosurfactants can be categorized as glycolipids, lipopeptides, phospholipids, fatty acids, particulates, and polymeric compounds.

Glycolipids are the very familiar biosurfactants, and they are abundantly available. They consist of mono, di, and polysaccharides. Rhamnolipids, sophorolipids, cellobiolipids, mannosylerythritol, and trehalose lipids are most common types of glycolipids [8,14,18]. These surfactants have higher versatility of commercialization compared with other microbial-based biosurfactants.

Rhamnolipids are a subtype of glycolipids-based biosurfactants. They show excellent physicochemical properties and considered as best bacterial surfactants. Rhamnolipids are mostly produced from *Pseudomonas* and used directly in different industries [19]. The *Pseudomonas aeruginosa* and *Pseudomonas fluorescens* (a methyl pentose monosaccharide strain) are most considered strains for rhamnolipids production. They are produced through microbial fermentation consisting of different rhamnose groups (mono or di) [20]. Disaccharides rhamnolipids are produced through condensation of rhamnose sugar (2 mol) and an acetal group (links the hydrophobic groups) while lipid part comprises ester and carboxyl groups.

According to advanced research, about 60 distinct congeners and homologues of rhamnolipids can be produced depending upon the concentration, type, family, or class of bacteria [21,22]. *Burkholderia* can produce alkyl longer-chain rhamnolipids as contrasted to *Pseudomonas aeruginosa* bacteria [23]. Different types of substrates are being used in such processes such as fructose, glucose, olive oil, alkenes, or citrates, etc., where smaller variations lead to significant changes in physiochemical attributes [24].

Lipopeptides consist of acylated cyclic peptides and fatty acids having variable length and structure. These biosurfactants are produced in the cultural media by numerous bacteria where *Bacillus* and *Pseudomonas* are the most significant species for producing lipopeptides. Besides, various yeast and fungus strains also have the capacity to produce such biosurfactants. Examples of lipopeptides produced by *Bacillus* are surfactin, fengycin, and iturin. Apart from this, polymyxins, wisconsin, and serrawettin can be produced through other microbes [25]. Moreover, it is still persistent to produce lipopeptides from various microbes because of great diversity in this category of biosurfactants. *Brevibacillus*, *Alcaligenes aquatilis*, *Bacillus tequilensis*, and *Bacillus subtilis* are the few examples of microorganisms having potential to produce lipopeptides [26–29].

3.3 Classification based on molecular mass

On average, the molecular mass of biosurfactants is considered between 500 Da and 1000 kDa [30]. Depending upon their masses, they are classified as low- and high-molecular-weight biosurfactants.

Low molecular wight: Generally glycolipids or lipopeptides are considered as the biosurfactants with lower molecular weight. These compounds decrease the interfacial tension at the interfaces of air/water and oil/water. According to composition or structure glycolipids, phospholipids fatty acids, lipopeptides, spiculisporic, and polyketide glycosides are the examples of low-molecular-weight biosurfactants [31].

High molecular wight: The high-molecular-weight biosurfactants are commonly referred as bioemulsifiers; they are most efficiently employed as stabilizers for emulsions of oil in water and bind tightly to surfaces. These emulsifiers are very efficient and work at very low concentrations such as lipopolysaccharide isolated from *Acinetobacter calcoaceticus* in 1997 had about 1000 kDa molecular weight [32]. High-molecular-weight biosurfactants consist of particulates and polymeric compounds [31].

4 Production methods of biosurfactants

As biosurfactants are distinct compounds depending upon microbial origin and composition, therefore it is difficult to outline generic procedure for production. Mostly biosurfactants are incorporated in the culture media during the exponential growth period, and sometimes they are developed through resting cells and immobilizing through biocatalysts. The biosurfactants are produced by using different batch reactors or through incessant fermentations as shown in Fig. 4. For example, the glycolipids are produced from *Torulopsis bombicola* through resting cells involving immobilization through catalysts. Where production can be improved by using adsorption columns or foam fractionation, air-lift fermenter, or aqueous two-phase fermentation.

The biosurfactants are mostly produced through two definite methods, which are solid-state fermentation (SSF) and submerged or liquid-state fermentation (LSF).

4.1 Solid-state fermentation

As indicated by name, SSF takes place on the surfaces of solid-state materials through microbial processes. The solids being employed are capable of absorbing moisture or water (with or without nutrients) in order to sustain growth and metabolism of microorganisms [33,34]. Fungi can grow at lower water concentrations; therefore, it is mostly used in SSF compared with bacteria. In other words, lower water content in SSF medium means that the fermentation will only be carried out by selective microbes such as yeast or fungi, and higher progression of fungi

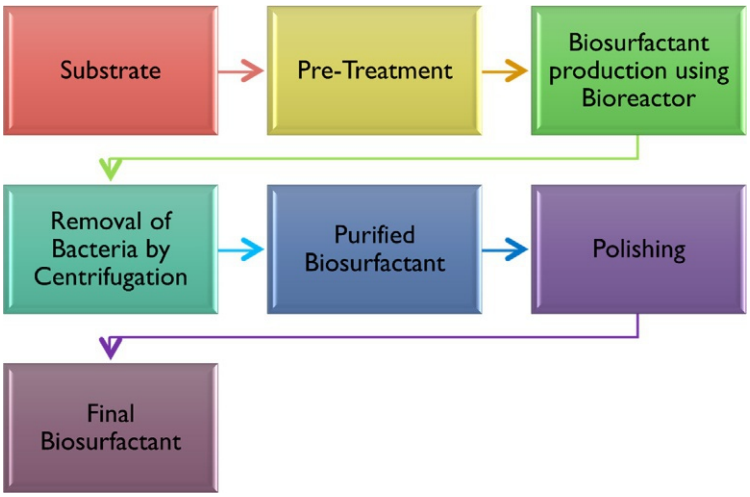

FIG. 4

Synthesis procedure of biosurfactants.

occurs mostly on solid substrates [35]. In SSF technique, mostly unicellular microorganisms are preferred over multicellular because it is optimized for filamentous growth that can percolate through interparticle spaces inside the substrates.

The SSF technique was commenced in the ancient times and has intensively been applied in the agro-industrial applications because of offering low energy consumption, eco-friendly attributes, moreover lesser amount of waste production and elimination of solid wastes. It was neglected during 1940s, later presented enormous outcomes from 1950 to 1970 and onward by supporting steroids transformation through fungal cultures, production of mycotoxin, proteinaceous cattle feed with economical residues and lesser pollutants, bioreactors, modeling and synthesis of microbial products, e.g., food, metabolites, and various bioprocesses [34,36,37].

4.2 Liquid-state fermentation

Submerged or LSF is a biosurfactant cultivation approach that is primarily based upon the culture media comprising high water content by using the bioreactors. The bioreactors used for LSF can be used through two distinct ways, which are intercellular and extracellular routes. Kishore Das et al. conducted a comparative analysis of both LSF and SSF by studying the potential of *Bacillus subtilis* strains to produce lipopeptides biosurfactants through a cheap carbon source [38]. There was about 80 mg/gds and 67 mg/gds yield of crude lipopeptides through LSF and SSF techniques, respectively [38]. Though there was comparatively higher yield through LSF, the qualitative as well as quantitative attributes of biosurfactants produced through both techniques were analogous.

4.3 Comparison of SSF and LSF

If both techniques are compared based on the higher yield of biosurfactants, then SSF is superior even by using same nutrients, their concentration and adopting similar culture conditions, e.g., temperature, shaking speed, etc. According to Kulkarni et al., the product amount of hydrophobin proteins, in mg per gm of biomass, by *Pleurotus ostreatus*, was twofold higher through SSF compared with LSF. These proteins produce during adherence of mycelia to solids because they cause surface hydrophobicity of mycelia when adhering to hydrophobic surface [32]. Similarly, rhamnolipids also produce in higher extent through SSF that is about threefold higher than LSF technique [39]. While iturin A production through SSF was tenfold higher than LSF because iturin A is a secondary metabolite when nutrients availability reduces in the medium [40]. In LSF, the nutrients and O_2 are offered plentifully in the liquid medium; therefore, the bacteria produce biomass instead of iturin A. On the other hand, in SSF, presence of nutritional stress limits the nutrients uptake because of heterogeneities in liquid and stimulates secondary metabolites. Moreover, there is difference between biological structures of both processes. A biofilm forms that is more compatible for iturin A in SSF compared with the planktonic state of liquid culture. But it is not a strict rule that the SSF produces comparatively high amount of biosurfactants. Sitohy et al. stated that LSF shows higher product yield with bacteria and yeast than SSF [41]. This effect depends mostly on the strain and provided culture conditions; thus, a comparison should be drawn between every SSF and LSF processes and through counting every quantitative, qualitative, and economical aspects. Moreover, the studies in favor of SSF were conducted at laboratory scale. Recent technological development is more toward the LSF than the SSF, and also there are some counter cases [41]. LSF is more prevalent for industrial applications, for instance, Novozymes uses LSF for cellulose processing [42]. Whereas SSF is still lacking to be utilized at industrial scale due to the unavailability of mathematical models necessary for bioreactor design and automated controllers feasible for optimizing heterogeneity of bioreactors to the heat and mass transfers [43,44]. At the industrial scale, technical problems occurring during LSF are usually resolvable, such as release of foam at the air-liquid interface. This foam is generated at extreme agitation or aeration because of the presence of some microorganisms supporting dispersion at the interface and elevates probability of contaminations. This foam may reduce the biocatalyst concentration in the bulk liquid and affect the overall performance of bioprocess. Therefore, industrial-scale biosurfactants require antifoaming agents that are uneconomical and increase the overall expenses. On the other hand, the SSF promotes the transfer of oxygen without generating foam and provokes chances of contaminations. Thus, SSF concerning lesser amount of free water can be an alternate method for biosurfactants synthesis [45].

In SSF, the extraction is relatively more problematic than in LSF for downstream process because of complex agro-industrial solid medium that leads toward the residual impurities in the crude extract, where the degree of purity for the product depends upon its intended use [46]. Besides, in downstream processes, the production costs

also limit their applications in certain industries [47,48]. Hence, reduction in total cost is essential to deploy the biosurfactants in every field. In recent investigations, the innovative solutions for integrated fermentation process and extraction have been emerged by introducing new gravity-based separation systems that positively affect to productivity, production cycles, and economics [49]. Even the integration of such separation systems can be foreseen in future for SSF as well especially when using plug-flow bioreactor with fermentation and extraction sections.

5 Factors affecting production of biosurfactants

The biosurfactants production can be influenced by dint of varying culture conditions, such as pH, temperature, dissolved O_2 content, or ionic strength. Moreover nutrients such as carbon, phosphate, nitrogen sources, metallic ions, or other media components may also play a critical role in production and yield of biosurfactants. Some important factors are briefly discussed here in the following sections, also shown in Fig. 5.

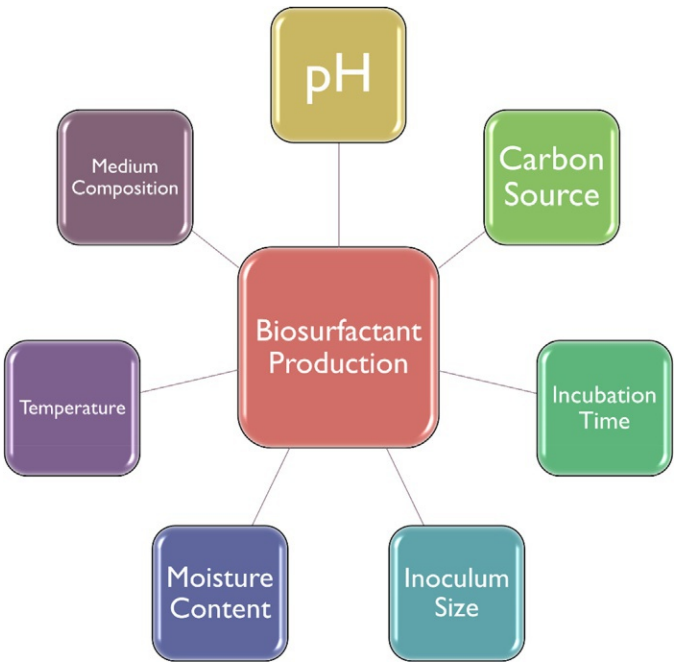

FIG. 5

Factors affecting biosurfactants production.

5.1 Effect of pH

The pH of the medium is very crucial in the production of biosurfactants because it can strongly affect the production of enzymatic secondary metabolites and movement of cell components throughout the cell membrane. Different bacterial species such as *Bacillus subtilis* or *Pseudomonas aeruginosa* are effective around neutral or vaguely basic pH values ranging between 7 and 8 [50]. Fungi can adjust in comparatively lesser pH environments than various bacteria; that is why it is mostly preferred especially in SSF technique. The pH values 4.5 and 5.5 are usually adopted for *Aspergillus fumigatus* and *Pleurotus djamor* fungal strains, respectively [51]. Moreover, a yeast strain, namely *Starmerella bombicola*, can sustain at pH around 6 [52,53].

5.2 Effect of salinity

Presence of salts or salinity of the fermentative medium can also alter the production of biosurfactants. Medium without salts compromises the growth of microbes and produces lesser amount of biosurfactant. Almansoory et al. produced biosurfactants from *Serratia marcescens* with a variety of NaCl concentrations (1%–5%), maximum 1.57 g/L yield was obtained, and there was 28.4 mN/m drop in surface tension when 1% salt concentration was used [54]. Another study showed that without NaCl, the Vibrio specie cannot grow and results in lowest product yield. However, high concentrations reduce the surface tension [55]. Optimum observations were attained at salt concentration of 20 g/L. Similarly, biosurfactants produced by microbial species *Enterobacter cloacae* and *Pseudomonas* at NaCl dosages 0%–15% weight/volume also show that high salinity is a key factor to reduce amount of produced bioproduct, affect biodegradation rate, and disturb superficial activities of biomolecules [56]. Makkar et al. checked the *Bacillus subtilis* ability to grow and produce biosurfactants in various NaCl concentrations (0.01%–4%) [57]. Unlike other studies, they said that even high concentration of NaCl does not alter the quantitative growth parameters and yield of biosurfactants, and there was 0.01% optimal dosage of NaCl. Yet, in other fermentation parameters, there is no developed consensus on the salt concentration and biosurfactants synthesis.

5.3 Effect of agitation and aeration

Aeration and agitation are the essential factors in the synthesis of biosurfactants that facilitate the transition of oxygen from gaseous to aqueous phase [47]. Ghribi et al. performed an experimental study using 2.6 L fermenter to study the effect of aeration on the production of biosurfactants through *Bacillus subtilis* [58]. They concluded that aeration puts a strong impact on production of biosurfactants with varying aeration rates. Oxygen saturation of 30% provides maximum yield of 4.92 g/L of biosurfactants. Silva et al. produced the biosurfactants from *Pseudomonas aeruginosa* and evaluated the effects of agitation and aeration for revolutions 150 and 200 rpm [59].

There was 6.5 g/L amount of bioproduct at agitation of 200 rpm although surface tension remained unaffected even on the 150 rpm while yield reduced to 5 g/L. In same study, aeration effect was also monitored at 60%, 80%, and 90% oxygen saturation and showed no significant change in biosurfactants yield or surface tension. Then *Pseudomonas alcaligenes* was used to check the influence of agitation on biosurfactants production [60]. Agitation speeds were picked between 80 and 220 rpm and demonstrated positive results upon rising speed, whereas 200 rpm was the best suited rpm for optimum yield of biosurfactants.

5.4 Effect of inoculum concentration

Inoculum concentration also significantly affects the production of biosurfactants because bioprocess depends upon cell density and biomass growth [47]. High concentration of microorganism's inoculum increases the growth but upto a specific limit and after that limit, the microbial activity degrades due to interrupted availability of nutrients in the media, while lesser inoculum reduces the concentration of cells and thus reduces the bioproduct's amount as well. Lower cell concentration takes long time to meet the optimal growth. Jimoh et al. performed inoculum of *Paenibacillus* to evaluate the concentration effects on biosurfactants yield [61]. There was g/liter yield of biosurfactants at 1.5% of inoculum concentration. The varying concentration of *Bacillus subtilis* inoculum for biosurfactants production showed a significant dependency upon the initial cell concentration in the media [58]. Another study reported that lessening the microbial growth results in slow but steady production of surfactin biosurfactants through *Bacillus subtilis* by 6% volume/volume concentration of inoculum [62]. Then Nalini et al. also reported that inoculum concentration of Gram-negative bacteria *Serratia rubidaea* substantially affects the amount of bioprocess product [63].

5.5 Effect of temperature

Temperature is also a critical parameter in biosurfactant production technique. As bioprocess proceeds, heat is produced through the metabolic activity of microbes. In SSF, solid media shows lesser conductivity for heat, so heat accumulates in the medium and leads to increase in temperature and thermal gradient as well [34]. Similar to pH, the optimum activity of microbes takes place only at a specific range of temperatures. The optimum biosurfactant production through filamentous fungus and yeast needs temperature range between 25°C and 30°C [52,64]. According to Wu et al. the rhamnolipids are produced between 30°C and 37°C from *Pseudomonas aeruginosa* bacteria [65]. Similarly temperature also affects the production of lipopeptides through *Bacillus cereus* [66].

5.6 Effect of moisture

Just like effect of temperature, the humidity and moisture content of medium also progress as the bioprocess grows and impinges on biosurfactants production. The involved microorganisms utilize the available water content and evaporate through the metabolic activities and end up while establishing a water gradient in the bioreactor. It is a significant factor for the growth of microorganisms and responsible for the products yield; thus, moisture content is critical in SSF. Low humidity leads toward lesser growth and lesser metabolite production. Therefore, appropriate moisture or water content is essential to provide an ultimate microenvironment for endorsement of growth and higher yield of metabolites [67]. If yeast such as *Aspergillus fumigatus* and *Trametes versicolor* are used as biosurfactant producers, then it would have optimal moisture content in the range of 44.7%–50% because this range could prevent the development of prospective impurities during fermentation process [52]. Bacterial fermentation often involves comparatively higher moisture content that can initiate problems related to compaction of solid medium, reduction in gaseous transmissions, and trigger contaminations [68]. Apart from this, in some cases, bacterial SSF supports the biosurfactants production with good results [69].

5.7 Effect of nutrients

Biosurfactants' yield and quality can severely be affected by fermentation medium components or nutrients such as presence, absence, and amount of carbon, nitrogen, metallic ions, and other constituents.

In bioprocesses, the type of carbon source significantly influences the productivity and quality of biosurfactants. For example, sugars as a carbon source extracted from vineyards pruning waste produce glycolipopetide-based biosurfactants while lactose carbon source endorses production of glycoproteins. Moreover, the procedure adopted for extraction of product influences not only its emulsifying capability but also its chemical structure or composition, physical attributes, and quantity of biosurfactants. Composition and amount of carbon source are crucial for biosurfactant congeners and physiochemical attributes. Hydrophilic carbon sources have been proven less promising than the hydrophobic in the context of yield [70]. Because most of the microorganisms used in the production of biosurfactants are isolated from the soils and hydrophobic waste containing waters. Processing of biosurfactants generally needs both hydrophilic and hydrophobic carbon sources in the culture media [71]. Therefore, in the SSF technique, oil-based by-products are used as solid media, e.g., comprising vegetal oils, i.e., crude oils, diesel, or sunflower seed oil [52]. Besides, olive oil mill effluents and peat pressate have also been suggested. Carbon substrate significantly reduces the production cost by using cheap substrate such as rice hull hydrolysate, waste liquor of starch, domestic wastes, potato waste [72,73].

Moreover, consistency, quality, and abundant availability of media also matter. Therefore, supply of solid by-products directly from their originating industries must be ensured for on-time accessibility in bulk quantity and with standard composition.

Nitrogen is also an important nutrient because it plays a key role for desired growth, building of cellular components, and production of biologically active metabolites. Nitrogen can be provided through organic or inorganic sources. There are numerous studies on the selection of appropriate nitrogen source [47]. It can be provided through distinct nitrogen-based compounds, e.g., peptone, yeast extract, NH_4NO_3, KNO_3, and $NaNO_3$ have the potential to produce bacteria-based biosurfactants. Anyhow the optimization of nitrogen concentration is also necessary because overdose of KNO_3 can negatively affect the fermentation process [74]. *Pseudomonas fluorescens* bacteria was comparatively analyzed in fermentation process with three distinct nitrogen sources ($NaNO_3$, NH_4Cl, and NH_4NO_3), and it was said that different sources affect differently the production of biosurfactants, and NH_4NO_3 was superior to other two [75]. But excess of NH_4NO_3 severely affected process yield. Hassan et al. examined the synthesis of biosurfactants through *Pseudomonas aeruginosa* using four nitrogen sources (peptone, NH_4SO_4, $NaNO_3$, and urea) [76]. Optimal findings were with $NaNO_3$ and peptone as nitrogen sources were utilized and yielded similar amount of biosurfactants. $NaNO_3$ provided very significant reduction in surface tension while preparing biosurfactants from *Stenotrophomonas acidaminiphila* [77]. Moreover, $NaNO_3$ was also compared with 12 other nitrogen sources (KNO_3, $(NH_4)_2C_2O_4$, yeast extract, urea, NH_4Cl, NH_4NO_3, peptone, tryptone, corn steep liquor, NH_4Br, $(NH_4)_2CO_3$, and $(NH)_2SO_4$). The results revealed that $NaNO_3$ was the outstanding nitrogen source for surfactin production through *Bacillus subtilis* [62]. Besides, NH_4NO_3 and $(NH_4)_2C_2O_4$ also showed quite similar yield. Similarly, Abushady et al. also checked the performance of *Bacillus subtilis* to produce surfactin in the presence of various organic (urea, yeast extract, casein hydrolysate, and beef extract) and inorganic (NH_4Cl, NH_4NO_3, $(NH_4)_2SO_4$, $(NH_4)_2S_2O_8$, NH_4Mo, $(NH_4)_2HPO_4$, $NH_4H_2PO_4$, NH_4HCO_3, and $NaNO_3$) nitrogen sources [78]. There was comparatively higher surfactin productivity with inorganic sources especially when NH_4NO_3 was used.

Natural nutrient sources owing to complex composition have mostly been studied for microbial growth. Though these sources are typically available in suboptimal concentrations and need to be supplemented with additional nutrients for optimum microbial growth and biosurfactants yield [79]. Copper and iron are the examples of such supplements while iron is considered as an enzyme's activator named as isocitrate lyase enzyme, which contributes to cell growth on substrates that are hydrophobic in nature. For each microbe there are some optimal culture conditions and a certain compound is needed depending upon the desired yielded products. Therefore, experimental studies use various designs to explore the most severely effecting factors and their optimum levels to enhance biosurfactants yield [80].

Moreover, magnesium, potassium, manganese, cobalt, and zinc are also required by the microorganisms in small or large quantities for synthesis of nucleic acid and amino acids or for activation of enzymes and adjusting osmotic pressure [1,62,81]. Thus production of biosurfactants from distinct microbes is also influenced by such nutrients. *Bacillus subtilis* can produce biosurfactants in 3–4 times larger amount in the presence of such metallic salts such as $FeSO_4$, $MnSO_4$, or $MgSO_4$. Metallic ions

individually as well as combinedly behave like cofactors of enzymes and influence the synthesis of surfactins in the fermentation process. Besides, dosage of inorganic ions reduces the productivity of biosurfactants.

As biosurfactants synthesis is generally aerobic and influenced by physical properties of media, thus they should also be compatible for desired microbial growth and yield of metabolites. Hydrophilicity and hydrophobicity of solid particles, mass and energy transfer, porosity of medium, and shape and size of solid particles are interdependent factors and combinedly influence fermentation process. Some studies suggest the incorporation of some endorsing solid mediums as well such as wheat straw and bagasse; however, interesting to note that, these are just the supporting compounds to give an appropriate environment by eliminating porosity and do not act as the microbial nutrients [39,50]. Fungal growth may clog the empty pores of the medium, thus the endorsing material provides less porous and suitable physical environment that is convenient for mass and heat transfers. This environment allows gaseous phase in the interparticle pore spaces and regenerates oxygen that is consumed by the fungus [82].

6 Challenges in production of biosurfactants

There are several investigations for the production of biosurfactants to implement the use of biosurfactants at larger scale because of enormous advantages over chemical surfactants that could allow them to compete at industrial scale and at various environmental applications. Whereas high production cost is the major prohibiting factor [83,84] and apart from this lower yield, inefficient purification and cost of expensive raw materials also the secondary factors. However, it would only be implemented upon decline in the price of raw materials and cost involved in their purification and fermentation methods [85]. Then surfactants produced from microbial fermentation would be commercialized to replace the synthetic ones in the industry. The substrate alone costs about half of the total production cost of biosurfactants [10,86]. According to research, the production cost of biosurfactants is about 10–12 times higher than the production cost of equal amount of synthetic surfactants; that's why they are unable to compete in the current market [87]. Therefore, biosurfactants can be most affected in cosmetics and medicine industries because they can be priced higher than the usual in order to enter in the market. Thus it is still needed to explore more cost-effective production ways by utilizing low-cost raw materials, and their necessity is kept enhancing in research.

Moreover, biosurfactants produced from thermophilic and halophilic bacterial species can show superiority while competing at industrial level because they can withstand extreme environmental condition such as pH, salinity, and temperature. Besides, use of microbial species capable of producing high yield obtained from recombination and mutation methods, use of renewable and cheap substrates, optimization of fermentative processes through statistical models and through the deployment of integrated systems capable of simultaneous production of

biosurfactants along with supplementary products of interest [88,89]. These solutions have the potential to offer less priced, more sustainable, and manageable production approaches of biosurfactants.

7 Conclusion

Green synthesis and tremendous field applications of biosurfactants are contributing to sustainable development and fascinating researchers and industries through innovative solutions to replace chemical surfactants. But utilization of biosurfactants at large scale is still at the research level because of the involvement of large essential production costs. Therefore, the impending replacement of chemical surfactants with innovative new-generational biosurfactants is still a challenge and demands more exploration on reduction of cost through the integration of recyclable waste materials. Anyhow, the insertion of biosurfactants is more likely to happen in smaller proportions through advanced formulations that will contribute to transition of petrochemical industry toward circular economy.

References

[1] I.M.M. Vieira, B.L.P. Santos, D.S. Ruzene, D.P. Silva, An overview of current research and developments in biosurfactants, J. Ind. Eng. Chem. 100 (2021) 1–18, https://doi.org/10.1016/j.jiec.2021.05.017.

[2] A. Soares, B. Guieysse, B. Jefferson, E. Cartmell, J.N. Lester, Nonylphenol in the environment: a critical review on occurrence, fate, toxicity and treatment in wastewaters, Environ. Int. 34 (2008) 1033–1049, https://doi.org/10.1016/j.envint.2008.01.004.

[3] H. Gelderblom, J. Verweij, K. Nooter, A. Sparreboom, Cremophor EL: the drawbacks and advantages of vehicle selection for drug formulation, Eur. J. Cancer 37 (2001) 1590–1598, https://doi.org/10.1016/S0959-8049(01)00171-X.

[4] A. Seweryn, Interactions between surfactants and the skin—theory and practice, Adv. Colloid Interf. Sci. 256 (2018) 242–255, https://doi.org/10.1016/j.cis.2018.04.002.

[5] European Union, Directive 2003/53/EC of the European Parliament and of the Council of 18 June 2003, Off. J. Eur. Union (2003) 24–27. http://faolex.fao.org/docs/pdf/eur39124.pdf.

[6] D.N. Yea, Y.J. Lee, K.H. Park, J.C. Lim, Synthesis of eco-friendly fatty acid based zwitterionic biosurfactants from coconut oil sources and characterization of their interfacial properties, J. Ind. Eng. Chem. 97 (2021) 287–298, https://doi.org/10.1016/j.jiec.2021.02.012.

[7] S. Lang, Biological amphiphiles (microbial biosurfactants), Curr. Opin. Colloid Interface Sci. 7 (2002) 12–20, https://doi.org/10.1016/S1359-0294(02)00007-9.

[8] I.M. Banat, Q. Carboué, G. Saucedo-Castañeda, J. de Jesús Cázares-Marinero, Biosurfactants: the green generation of speciality chemicals and potential production using solid-state fermentation (SSF) technology, Bioresour. Technol. 320 (2021), https://doi.org/10.1016/j.biortech.2020.124222, 124222.

[9] S. Akbari, N.H. Abdurahman, R.M. Yunus, F. Fayaz, O.R. Alara, Biosurfactants—a new frontier for social and environmental safety: a mini review, Biotechnol. Res. Innov. 2 (2018) 81–90, https://doi.org/10.1016/j.biori.2018.09.001.

[10] K. Sałek, S.R. Euston, Sustainable microbial biosurfactants and bioemulsifiers for commercial exploitation, Process Biochem. 85 (2019) 143–155, https://doi.org/10.1016/j.procbio.2019.06.027.

[11] M. Nitschke, S.G.V.A.O. Costa, Biosurfactants in food industry, Trends Food Sci. Technol. 18 (2007) 252–259, https://doi.org/10.1016/j.tifs.2007.01.002.

[12] D.R. Karsa, What are surfactants?: history and applications of surfactants, in: Chemistry and Technology of Surfactants, Blackwell Publishing Ltd, 2007, pp. 1–13, https://doi.org/10.1002/9780470988596.ch1.

[13] M. Henkel, R. Hausmann, Diversity and classification of microbial surfactants, in: Biobased Surfactants, Elsevier, 2019, pp. 41–63, https://doi.org/10.1016/b978-0-12-812705-6.00002-2.

[14] C.N. Mulligan, S.K. Sharma, A. Mudhoo, Biosurfactants research trends and applications, in: Biosurfactants: Research Trends and Applications, 2014, pp. 1–329, https://doi.org/10.1201/B16383.

[15] Biosurfactants: Production and Utilization—Processes, Technologies, and …—Google Books, 2021. https://books.google.com.pk/books?hl=en&lr=&id=PpHaBAAAQBAJ&oi=fnd&pg=PP1&dq=Biosurfactants+Production+and+Utilization++Processes+Technologies+and+Economics&ots=4fF1BPt6-o&sig=EVM45wziLYHWR8kwNcWweKQiSqY&redir_esc=y#v=onepage&q=BiosurfactantsProductionandUtilizationProcesses%2CTechnologies%2CandEconomics&f=false. (Accessed 18 August 2021).

[16] J.D. Van Hamme, A. Singh, O.P. Ward, Physiological aspects: part 1 in a series of papers devoted to surfactants in microbiology and biotechnology, Biotechnol. Adv. 24 (2006) 604–620, https://doi.org/10.1016/J.BIOTECHADV.2006.08.001.

[17] P.L. Gupta, M. Rajput, T. Oza, U. Trivedi, G. Sanghvi, Eminence of microbial products in cosmetic industry, Nat. Prod. Bioprospect. 9 (2019) 267–278, https://doi.org/10.1007/S13659-019-0215-0.

[18] A.M.C.N. Mulligan, S.K. Sharma, Biobased Surfactants Synthesis, Properties, and Applications, 2019. https://scholar.google.com/scholar?q=BiobasedSurfactantsSynthesisPropertiesandApplications. (Accessed 18 August 2021).

[19] H. Chong, Q. Li, Microbial production of rhamnolipids: opportunities, challenges and strategies, Microb. Cell Factories 16 (2017) 1–12, https://doi.org/10.1186/s12934-017-0753-2.

[20] I.M. Banat, A. Franzetti, I. Gandolfi, G. Bestetti, M.G. Martinotti, L. Fracchia, T.J. Smyth, R. Marchant, Microbial biosurfactants production, applications and future potential, Appl. Microbiol. Biotechnol. 87 (2010) 427–444, https://doi.org/10.1007/s00253-010-2589-0.

[21] Ł. Chrzanowski, Ł. Ławniczak, K. Czaczyk, Why do microorganisms produce rhamnolipids? World J. Microbiol. Biotechnol. 28 (2012) 401–419, https://doi.org/10.1007/s11274-011-0854-8.

[22] C.E. Drakontis, S. Amin, Biosurfactants: formulations, properties, and applications, Curr. Opin. Colloid Interface Sci. 48 (2020) 77–90, https://doi.org/10.1016/j.cocis.2020.03.013.

[23] M. Nitschke, S.G.V.A.O. Costa, J. Contiero, Rhamnolipids and PHAs: recent reports on Pseudomonas-derived molecules of increasing industrial interest, Process Biochem. 46 (2011) 621–630, https://doi.org/10.1016/j.procbio.2010.12.012.
[24] Ł. Ławniczak, R. Marecik, Ł. Chrzanowski, Contributions of Biosurfactants to Natural or Induced Bioremediation, 2013, https://doi.org/10.1007/s00253-013-4740-1.
[25] Biobased Surfactants Synthesis, Properties, and … Google Scholar, 2019. https://scholar.google.com/scholar?q=BiobasedSurfactantsSynthesisPropertiesandApplications. (Accessed 30 April 2019).
[26] S. Sharma, L.M. Pandey, Production of biosurfactant by Bacillus subtilis RSL-2 isolated from sludge and biosurfactant mediated degradation of oil, Bioresour. Technol. 307 (2020), https://doi.org/10.1016/J.BIORTECH.2020.123261, 123261.
[27] A.K. Singh, P. Sharma, Disinfectant-like activity of lipopeptide biosurfactant produced by Bacillus tequilensis strain SDS21, Colloids Surf. B: Biointerfaces 185 (2020), https://doi.org/10.1016/J.COLSURFB.2019.110514, 110514.
[28] D. Yalaoui-Guellal, S. Fella-Temzi, S. Djafri-Dib, S.K. Sahu, V.U. Irorere, I.M. Banat, K. Madani, The petroleum-degrading bacteria Alcaligenes aquatilis strain YGD 2906 as a potential source of lipopeptide biosurfactant, Fuel 285 (2021), https://doi.org/10.1016/J.FUEL.2020.119112, 119112.
[29] C. Vigneshwaran, K. Vasantharaj, N. Krishnanand, V. Sivasubramanian, Production optimization, purification and characterization of lipopeptide biosurfactant obtained from Brevibacillus sp. AVN13, J. Environ. Chem. Eng. 9 (2021), https://doi.org/10.1016/J.JECE.2020.104867, 104867.
[30] P. Grazyna, V. Achal, Biosurfactants: eco-friendly and innovative biocides against biocorrosion, sixth ed., Microbial Biosurfactants, Current Research Trends and Applications, vol. 21, International Journal of Molecular Sciences, MDPI, 2020. 2152 pp.
[31] Á. Domínguez Rivera, M.Á. Martínez Urbina, V.E. López y López, Advances on research in the use of agro-industrial waste in biosurfactant production, World J. Microbiol. Biotechnol. 35 (10) (2019) 1–18, https://doi.org/10.1007/S11274-019-2729-3.
[32] S.S. Kulkarni, S.N. Nene, K.S. Joshi, A comparative study of production of hydrophobin like proteins (HYD-LPs) in submerged liquid and solid state fermentation from white rot fungus Pleurotus ostreatus, Biocatal. Agric. Biotechnol. 23 (2020), https://doi.org/10.1016/J.BCAB.2019.101440, 101440.
[33] A. Pandey, C.R. Soccol, D. Mitchell, New developments in solid state fermentation: I-bioprocesses and products, Process Biochem. 35 (2000) 1153–1169, https://doi.org/10.1016/S0032-9592(00)00152-7.
[34] A. Pandey, Solid-state fermentation, Biochem. Eng. J. 13 (2003) 81–84, https://doi.org/10.1016/S1369-703X(02)00121-3.
[35] A.C.S. da Silva, P.N. dos Santos, T.A.L. e Silva, R.F.S. Andrade, G.M. Campos-Takaki, Biosurfactant production by fungi as a sustainable alternative, Arq. Inst. Biol. (Sao Paulo) 85 (2018) 1–12, https://doi.org/10.1590/1808-1657000502017.
[36] A.B.P. Medeiros, A. Pandey, R.J.S. Freitas, P. Christen, C.R. Soccol, Optimization of the production of aroma compounds by Kluyveromyces marxianus in solid-state fermentation using factorial design and response surface methodology, Biochem. Eng. J. 6 (2000) 33–39, https://doi.org/10.1016/S1369-703X(00)00065-6.
[37] B.Z. Han, F.M. Rombouts, M.J.R. Nout, A Chinese fermented soybean food, Int. J. Food Microbiol. 65 (2001) 1–10, https://doi.org/10.1016/S0168-1605(00)00523-7.

[38] K. Das, A.K. Mukherjee, Comparison of lipopeptide biosurfactants production by Bacillus subtilis strains in submerged and solid state fermentation systems using a cheap carbon source: some industrial applications of biosurfactants, Process Biochem. 42 (2007) 1191–1199, https://doi.org/10.1016/J.PROCBIO.2007.05.011.

[39] G.S. El-Housseiny, K.M. Aboshanab, M.M. Aboulwafa, N.A. Hassouna, Rhamnolipid Production by a Gamma Ray-Induced *Pseudomonas aeruginosa* Mutant Under Solid State Fermentation, 2019, https://doi.org/10.1186/s13568-018-0732-y.

[40] S.M. Mizumoto, H.M. Shoda, Biotechnological products and process engineering production of lipopeptide antibiotic iturin a using soybean curd residue cultivated with Bacillus subtilis in solid-state fermentation, Appl. Microbiol. Biotechnol. 72 (2006) 869–875, https://doi.org/10.1007/s00253-006-0389-3.

[41] M.Z. Sitohy, M.M. Rashad, S.F. Sharobeem, A.E. Mahmoud, M.U. Nooman, A.S. Al Kashef, Bioconversion of soy processing waste for production of surfactants, Afr. J. Microbiol. Res. 4 (2010) 2811–2821. http://www.academicjournals.org/ajmr. https://doi.org/10.5897/AJMR.9000128 (Accessed 15 August 2021).

[42] G.H. Hansen, M. Lübeck, J.C. Frisvad, P.S. Lübeck, B. Andersen, Production of cellulolytic enzymes from ascomycetes: comparison of solid state and submerged fermentation, Process Biochem. 50 (2015) 1327–1341, https://doi.org/10.1016/J.PROCBIO.2015.05.017.

[43] B. Pereira, S. Francisco, S. da Silvério, Recent Advances in Sustainable Production and Application of Biosurfactants in Brazil and Latin America, 2016, pp. 31–39, https://doi.org/10.1089/IND.2015.0027. Https://Home.Liebertpub.Com/Ind. 12.

[44] S. Arora, R. Rani, S. Ghosh, Bioreactors in solid state fermentation technology: design, applications and engineering aspects, J. Biotechnol. 269 (2018) 16–34, https://doi.org/10.1016/J.JBIOTEC.2018.01.010.

[45] N. Krieger, D.C. Neto, D.A. Mitchell, Production of microbial biosurfactants by solid-state cultivation, Adv. Exp. Med. Biol. 672 (2010) 203–210, https://doi.org/10.1007/978-1-4419-5979-9_15.

[46] R.R. Singhania, A.K. Patel, C.R. Soccol, A. Pandey, Recent advances in solid-state fermentation, Biochem. Eng. J. 44 (2009) 13–18, https://doi.org/10.1016/J.BEJ.2008.10.019.

[47] A.A. Jimoh, J. Lin, Biosurfactant: a new frontier for greener technology and environmental sustainability, Ecotoxicol. Environ. Saf. 184 (2019), https://doi.org/10.1016/J.ECOENV.2019.109607, 109607.

[48] Z. Najmi, G. Ebrahimipour, A. Franzetti, I.M. Banat, In situ downstream strategies for cost-effective bio/surfactant recovery, Biotechnol. Appl. Biochem. 65 (2018) 523–532, https://doi.org/10.1002/BAB.1641.

[49] B.M. Dolman, F. Wang, J.B. Winterburn, Integrated production and separation of biosurfactants, Process Biochem. 83 (2019) 1–8, https://doi.org/10.1016/J.PROCBIO.2019.05.002.

[50] Z. Zhu, F. Zhang, Z. Wei, W. Ran, Q. Shen, The usage of rice straw as a major substrate for the production of surfactin by Bacillus amyloliquefaciens XZ-173 in solid-state fermentation, J. Environ. Manag. 127 (2013) 96–102, https://doi.org/10.1016/J.JENVMAN.2013.04.017.

[51] Z. Velioglu, R.O. Urek, Physicochemical and structural characterization of biosurfactant produced by Pleurotus djamor in solid-state fermentation, Biotechnol. Bioprocess Eng. 21 (2016) 430–438, https://doi.org/10.1007/s12257-016-0139-z.

[52] P. Jiménez-Peñalver, T. Gea, A. Sánchez, X. Font, Production of sophorolipids from winterization oil cake by solid-state fermentation: optimization, monitoring and effect of mixing, Biochem. Eng. J. 115 (2016) 93–100, https://doi.org/10.1016/J.BEJ.2016.08.006.

[53] A. Cerda, L. Mejias, P. Rodríguez, A. Rodríguez, A. Artola, X. Font, T. Gea, A. Sánchez, Valorisation of digestate from biowaste through solid-state fermentation to obtain value added bioproducts: a first approach, Bioresour. Technol. 271 (2019) 409–416, https://doi.org/10.1016/J.BIORTECH.2018.09.131.

[54] A. Fadhile Almansoory, H. Abu Hasan, M. Idris, S.R. Sheikh Abdullah, N. Anuar, E.M. Musa Tibin, Biosurfactant production by the hydrocarbon-degrading bacteria (HDB) Serratia marcescens: optimization using central composite design (CCD), J. Ind. Eng. Chem. 47 (2017) 272–280, https://doi.org/10.1016/J.JIEC.2016.11.043.

[55] X. Hu, C. Wang, P. Wang, Optimization and characterization of biosurfactant production from marine Vibrio sp. strain 3B-2, Front. Microbiol. 0 (2015) 976, https://doi.org/10.3389/FMICB.2015.00976.

[56] P. Darvishi, S. Ayatollahi, D. Mowla, A. Niazi, Biosurfactant production under extreme environmental conditions by an efficient microbial consortium, ERCPPI-2, Colloids Surf. B: Biointerfaces 84 (2011) 292–300, https://doi.org/10.1016/J.COLSURFB.2011.01.011.

[57] R.S. Makkar, S.S. Cameotra, Effects of various nutritional supplements on biosurfactant production by a strain of Bacillus subtilis at 45°C, J. Surfactant Deterg. 5 (2002) 11–17, https://doi.org/10.1007/S11743-002-0199-8.

[58] D. Ghribi, S. Ellouze-Chaabouni, Enhancement of Bacillus subtilis Lipopeptide biosurfactants production through optimization of medium composition and adequate control of aeration, Biotechnol. Res. Int. 2011 (2011) 1–6, https://doi.org/10.4061/2011/653654.

[59] S.N.R.L. Silva, C.B.B. Farias, R.D. Rufino, J.M. Luna, L.A. Sarubbo, Glycerol as substrate for the production of biosurfactant by Pseudomonas aeruginosa UCP0992, Colloids Surf. B: Biointerfaces 79 (2010) 174–183, https://doi.org/10.1016/J.COLSURFB.2010.03.050.

[60] F.J.S. Oliveira, L. Vazquez, N.P. de Campos, F.P. de França, Production of rhamnolipids by a Pseudomonas alcaligenes strain, Process Biochem. 44 (2009) 383–389, https://doi.org/10.1016/J.PROCBIO.2008.11.014.

[61] A.A. Jimoh, J. Lin, Enhancement of Paenibacillus sp. D9 Lipopeptide biosurfactant production through the optimization of medium composition and its application for biodegradation of hydrophobic pollutants, Appl. Biochem. Biotechnol. 187 (3) (2018) 724–743, https://doi.org/10.1007/S12010-018-2847-7.

[62] A.M. Abdel-Mawgoud, M.M. Aboulwafa, N.A.-H. Hassouna, Optimization of Surfactin production by Bacillus subtilis isolate BS5, Appl. Biochem. Biotechnol. 150 (3) (2008) 305–325, https://doi.org/10.1007/S12010-008-8155-X.

[63] S. Nalini, R. Parthasarathi, Optimization of rhamnolipid biosurfactant production from Serratia rubidaea SNAU02 under solid-state fermentation and its biocontrol efficacy against fusarium wilt of eggplant, Ann. Agrar. Sci. 16 (2018) 108–115, https://doi.org/10.1016/J.AASCI.2017.11.002.

[64] L. Antonio Lourenço, M.D.A. Magina, L.B.B. Tavares, S.M.A.G.U. de Souza, M.D.G. Román, A. Vaz, Environmental technology biosurfactant production by Trametes versicolor grown on two-phase olive mill waste in solid-state fermentation, Environ. Technol. 39 (2018) 3066–3076, https://doi.org/10.1080/09593330.2017.1374471.

[65] J. Wu, J. Zhang, P. Wang, L. Zhu, M. Gao, Z. Zheng, X. Zhan, Production of rhamnolipids by semi-solid-state fermentation with *Pseudomonas aeruginosa* RG18 for heavy

metal desorption, Bioprocess Biosyst. Eng. 40 (2017), https://doi.org/10.1007/s00449-017-1817-8.

[66] S. Nalini, R. Parthasarathi, V. Prabudoss, Production and characterization of lipopeptide from Bacillus cereus SNAU01 under solid state fermentation and its potential application as anti-biofilm agent, Biocatal. Agric. Biotechnol. 5 (2016) 123–132, https://doi.org/10.1016/J.BCAB.2016.01.007.

[67] R. Verhoef, P. de Waard, H.A. Schols, M. Siika-aho, A.G. Voragen, *Methylobacterium* sp. isolated from a Finnish paper machine produces highly pyruvated galactan exopolysaccharide, Carbohydr. Res. 18 (2003) 1851–1859, https://doi.org/10.1016/S0008-6215(03)00261-1.

[68] F.-L. Li, N. Ali, Q. He, H. Peng, M. Sheng, S. Hu, J. Qiu, J. Gu, Humidity control strategies for solid-state fermentation: capillary water supply by water-retention materials and negative-pressure auto-controlled irrigation, Front. Bioeng. Biotechnol. | www.Frontiersin.Org 7 (2019) 263, https://doi.org/10.3389/fbioe.2019.00263.

[69] J.A.V. Costa, H. Treichel, V. Kumar, A. Pandey, Advances in solid-state fermentation, Curr. Dev. Biotechnol. Bioeng. (2018) 1–17, https://doi.org/10.1016/B978-0-444-63990-5.00001-3.

[70] W. Ismail, S. Al Shammary, W.S. El-Sayed, C. Obuekwe, A.M. El Nayal, A.S. Abdul Raheem, A. Al-Humam, Stimulation of rhamnolipid biosurfactants production in Pseudomonas aeruginosa AK6U by organosulfur compounds provided as sulfur sources, Biotechnol. Rep. 7 (2015) 55–63, https://doi.org/10.1016/J.BTRE.2015.03.001.

[71] K.S. Teixeira Souza, E.J. Gudiña, R.F. Schwan, L.R. Rodrigues, D.R. Dias, J.A. Teixeira, Improvement of biosurfactant production by Wickerhamomyces anomalus CCMA 0358 and its potential application in bioremediation, J. Hazard. Mater. 346 (2018) 152–158, https://doi.org/10.1016/J.JHAZMAT.2017.12.021.

[72] S.S. Mohanty, Y. Koul, S. Varjani, A. Pandey, H.H. Ngo, J.-S. Chang, J.W.C. Wong, X.-T. Bui, A Critical Review on Various Feedstocks as Sustainable Substrates for Biosurfactants Production: A Way Towards Cleaner Production, 2021, https://doi.org/10.1186/s12934-021-01613-3.

[73] I.M. Banat, S.K. Satpute, S.S. Cameotra, R. Patil, N.V. Nyayanit, G. Spano, C.L. Hemme, Cost Effective Technologies and Renewable Substrates for Biosurfactants' Production, 2014, https://doi.org/10.3389/fmicb.2014.00697.

[74] C. Vigneshwaran, V. Sivasubramanian, K. Vasantharaj, N. Krishnanand, M. Jerold, Potential of Brevibacillus sp. AVN 13 isolated from crude oil contaminated soil for biosurfactant production and its optimization studies, J. Environ. Chem. Eng. 6 (2018) 4347–4356, https://doi.org/10.1016/J.JECE.2018.06.036.

[75] M. Abouseoud, R. Maachi, A. Amrane, S. Boudergua, A. Nabi, Evaluation of different carbon and nitrogen sources in production of biosurfactant by Pseudomonas fluorescens, Desalination 223 (2008) 143–151, https://doi.org/10.1016/J.DESAL.2007.01.198.

[76] M. Hassan, T. Essam, A.S. Yassin, A. Salama, Optimization of rhamnolipid production by biodegrading bacterial isolates using Plackett–Burman design, Int. J. Biol. Macromol. 82 (2016) 573–579, https://doi.org/10.1016/J.IJBIOMAC.2015.09.057.

[77] T. Onlamool, A. Saimmai, N. Meeboon, S. Maneerat, Enhancement of glycolipid production by Stenotrophomonas acidaminiphila TW3 cultivated in low cost substrate, Biocatal. Agric. Biotechnol. 26 (2020), https://doi.org/10.1016/J.BCAB.2020.101628, 101628.

[78] A.S.B.H.M. Abushady, Molecular characterization of Bacillus subtilis surfactin producing strain and the factors affecting its production, Int. J. Agric. Biol. 7 (2005) 337–344. https://www.researchgate.net/publication/228892817_Molecular_characterization_of_

Bacillus_subtilis_surfactin_producing_strain_and_the_factors_affecting_its_production. (Accessed 18 August 2021).
[79] C.R. Soccol, E.S.F. da Costa, L.A.J. Letti, S.G. Karp, A.L. Woiciechowski, L.P.S. de Vandenberghe, Recent developments and innovations in solid state fermentation, Biotechnol. Res. Innov. 1 (2017) 52–71, https://doi.org/10.1016/J.BIORI.2017.01.002.
[80] P. Jiménez-Peñalver, M. Castillejos, A. Koh, R. Gross, A. Sánchez, X. Font, T. Gea, Production and characterization of sophorolipids from stearic acid by solid-state fermentation, a cleaner alternative to chemical surfactants, J. Clean. Prod. 172 (2018) 2735–2747, https://doi.org/10.1016/J.JCLEPRO.2017.11.138.
[81] Y.H. Wei, I.M. Chu, Enhancement of surfactin production in iron-enriched media by bacillus subtilis ATCC 21332, Enzym. Microb. Technol. 22 (1998) 724–728, https://doi.org/10.1016/S0141-0229(98)00016-7.
[82] Q. Carboué, M. Claeys-Bruno, I. Bombarda, M. Sergent, J. Jolain, S. Roussos, Experimental design and solid state fermentation: a holistic approach to improve cultural medium for the production of fungal secondary metabolites, Chemom. Intell. Lab. Syst. 176 (2018) 101–107, https://doi.org/10.1016/J.CHEMOLAB.2018.03.011.
[83] H. Amani, R. Mohammad, M. Mehrnia, S. Hossein, M. Haghighi, R. Mohammad, Soudi, Scale up and Application of Biosurfactant from *Bacillus subtilis* in Enhanced Oil Recovery, 2010, https://doi.org/10.1007/s12010-009-8889-0.
[84] N.K. Veenanadig, M.K. Gowthaman, N.G.K. Karanth, Scale up studies for the production of biosurfactant in packed column bioreactor, Bioprocess Eng. (2000), https://doi.org/10.1007/S004490050017.
[85] S. Mukherjee, P. Das, R. Sen, Towards commercial production of microbial surfactants, Trends Biotechnol. 24 (2006) 509–515, https://doi.org/10.1016/j.tibtech.2006.09.005.
[86] I.M. Banat, S.K. Satpute, S.S. Cameotra, R. Patil, N.V. Nyayanit, Cost effective technologies and renewable substrates for biosurfactants' production, Front. Microbiol. 5 (2014), https://doi.org/10.3389/FMICB.2014.00697/FULL.
[87] T.B. Lotfabad, N. Ebadipour, R. Roostaazad, Evaluation of a Recycling Bioreactor for Biosurfactant Production by *Pseudomonas aeruginosa* MR01 Using Soybean Oil Waste, 2015, https://doi.org/10.1002/jctb.4733.
[88] P. Singh, Y. Patil, V. Rale, Biosurfactant production: emerging trends and promising strategies, J. Appl. Microbiol. 126 (2019) 2–13, https://doi.org/10.1111/JAM.14057.
[89] A.A.O.S. Prado, B.L.P. Santos, I.M.M. Vieira, L.C. Ramos, R.R. de Souza, D.P. Silva, D. S. Ruzene, Evaluation of a new strategy in the elaboration of culture media to produce surfactin from hemicellulosic corncob liquor, Biotechnol. Rep. 24 (2019), https://doi.org/10.1016/J.BTRE.2019.E00364, e00364.

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CHAPTER

Antibiofilm activities against food pathogens 13

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1 Introduction

Microorganisms' presence in food is natural and is hard to avert its entry, but these microorganisms can be destroyed by cooking or heating to some extent. The uncooked, processed ready-to-eat, and partially cooked food can have pathogenic microbes, which constitute a serious health risk [1]. Recently, there has been growing consumption of fresh products more naturally preserved, minimally processed worldwide because of the concerning of consumers on their health benefits. Meanwhile, food safety problems particularly the pathogenic microorganisms attached to contaminated fresh and processing products have been repeatedly reported [2].

Food-borne pathogens are the main factors behind food-borne diseases and food poisoning and thus pose a great threat to food safety. Many outbreaks have been found to be associated with biofilms [3]. Globally, food-borne pathogens are responsible for 2 billion illnesses and over 1 million deaths annually [4]. Besides the significant incidence of food-borne illness in public health, there are also huge economic costs [5]. Biofilm formation is a complex procedure in which genetic mechanisms and many factors such as the properties of substratum and bacterial cell surfaces are implicated. Biofilm matrix is composed of extracellular polymeric substances (polysaccharides, proteins, extracellular DNA, and lipids). This matrix protects organisms against desiccation, oxidizing or charged biocides, some antibiotics and metallic cations, ultraviolet radiation, many (but not all) protozoan grazers, and host immune defenses [6]. Biofilm cells are more resistant to cleaning and disinfection processes in the food industry compared with the suspension or planktonic cells [7]. Several mechanisms may be involved in disinfectants, sanitizers, and antimicrobial drugs resistance or tolerance of biofilms such as formation of persister cells, limited diffusion of disinfectants and sanitizers through the biofilms, and the expression of certain genes participating in the general stress response during the biofilm mode of growth [8–12].

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For inactivating microorganisms on surface so as to hinder microbial contamination of raw materials and food products, several physical and chemical approaches are frequently used to disinfect and decontaminate. However, their toxicity, inefficiency, and potential for carryover to finished products, and environmentally hostile characteristics, make them inappropriate. Currently, food additives have progressively become one of the most important types of bactericidal agents in the food industry. Excluding for common antiseptics, some food additives, such as colorants, surfactants, and flavorants, possess powerful antibacterial capacity against foodborne pathogens [3]. In addition, some green chemicals (enzymes, phages, and bioregulation) can help to surmount biofilms' drawback in food industry. Nevertheless, there is no one method with absolute efficiency and without side effects in stopping microbial development and biofilm formation [13].

Although biofilm removal is considered important for providing food safety and extending food shelf life, the use of innovative nontoxic antibiofilm agents and antibiofilm surface-coating technologies for food processing environments have been developed recently [14]. Biosurfactants (BSs) are a structurally heterogeneous group of biomolecules that share pronounced surface and emulsifying activities. They can be either located on microbial cell surfaces or released in the extracellular space by different microorganisms [15,16]. Many BSs are known by their antibacterial, antifungal, and antiviral activities, making them good candidates to remove biofilms [17]. As surface-active agents, biosurfactants had strong antiadhesion and antibiofilm properties, preventing bacteria from adhering to surfaces and disrupting biofilm formation by affecting the integrity and viability of bacteria within biofilms.

In this context, the objective of this chapter is to provide an overview on the use of BSs as safe and promising alternative in food industry for controlling and removing biofilms.

2 Biofilms composition

Biofilms are living communities formed of microbial cells in most cases heterogeneous, these communities are anchored on animate and inanimate surfaces, and also between them [18] and protected by an extracellular matrix, this matrix is the key element in establishing and maintaining the structure and properties of the biofilm, it is composed of water, proteins, DNA, and extracellular polymeric substances (EPSs) [19]. The existence of the biofilm is relatively linked to the EPS matrix, this fundamental element for the survival of the micro-community of biofilms makes it possible to maintain the microbial cells in proximity and to ensure prolonged cellular interactions, it also makes it possible to capture nutrients, the components of the biofilms have a certain complexity and dynamics that help protect the microbial micro-community by ensuring mechanical resistance, including increased tolerance to antimicrobials and environmental stresses [20].

The well-defined and structured architecture of the biofilm allows it to provide an adequate environment for the exchange of genetic material between microorganisms;

on the other hand, microorganisms can also communicate with each other via quorum sensing [21]. Because of their role in the persistence of certain infectious diseases and device-related infections, biofilms are of major importance to public health. Their ubiquity in the food industries is hardly overlooked.

3 Stages of formation of microbial biofilms

3.1 Initial contact/attachment to the surface

In complex environments, the ability of microorganisms to adhere and interact with surfaces in order to form a biofilm is essential for their survival. Adhesion to inanimate surfaces is mediated by nonspecific interactions; however, adhesion to animate surfaces involves a specific receptor-ligand interaction [22].

The first step in biofilm formation involves attaching microbial cells to a conditioned surface. Quorum-sensing autoinducers are primarily responsible for biofilm formation; however, a variety of other signals trigger biofilm formation such as antibiotics, pigments, and siderophores [23]. The attachment of cells is provided by the common pili and flagella; moreover, these cellular structures are necessary for the production of mature biofilms. It is noted that in *Erwinia amylovora*, type I fimbriae functioned earlier in attachment, while type IV pilus structures appear to function later in attachment [24]. Attachment may be provided by physical forces such as van der Waal forces, electrostatic interactions, covalent bonds. Strong adhesion is established when bacteria and the surface are able to form covalent bonds. Weaker forces can be as reinforcing as if a high number of contacts are involved (Fig. 1) [22].

3.2 Micro-colony formation

The multiplication of microorganisms is the second step in the formation of biofilms, it occurs directly after the stability of the attachment of cells to surfaces, this step is initiated by specific chemical signals and then to the formation of micro-colonies. Attached bacteria multiply and secrete an extracellular matrix, which leads to the formation of micro-colonies (Fig. 1) [25].

The heterogeneous micro-colonies that make up the biofilm communicate with each other through several processes, the most remarkable of which is quorum sensing, these coordinations between living cells immobile within the biofilm play a key role in the use of nutrients and their digestion and distribution of metabolic products.

3.3 Maturation and architecture

The growth of the biofilm is highly dependent on the transport of nutrients and wastes, the diffusion of which is considered to be the primary transport route [26]. At this stage of formation, the microbial cells reach the recommended levels; they are controlled by quorum sensing. At this stage, several gene products are expressed important for the maturation of the biofilm [27]. Mature biofilm is made

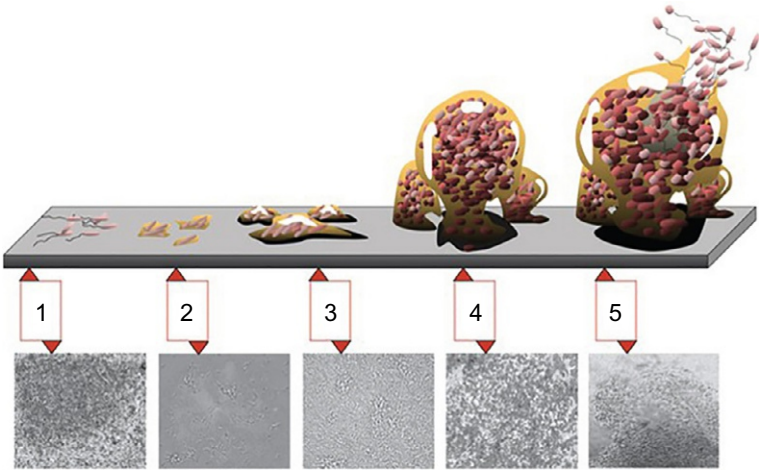

FIG. 1

Stages of biofilm life cycle. 1. Initial attachment; 2. Replication and production of EPS; 3. Formation of micro-colonies; 4. Maturation (development of biofilm architecture); 5. Detachment. Each stage of the life cycle in the diagram is paired with a photomicrograph of a developing *Pseudomonas aeruginosa* biofilm. All photomicrographs are shown in the same scale [31].

up of channels, these channels act as a circulatory system with high permeability to the flow of liquid through the biofilm (Fig. 1) [26].

3.4 Detachment/dispersion of biofilm

The detachment of certain parts of the biofilm is the last stage in the life of the biofilm, this stage occurs following a mechanical or environmental stress, after which, the microorganisms transform from the sessile form to the mobile form [28]. Microbial communities in the biofilm secrete different enzymes that help free parts of the surface for colonization of a new area [29]. In this phase, microbial cells regulate the expression of certain genes linked to the formation of flagella, to allow bacteria to move to new sites [30].

4 Recalcitrance to antibiotics

About 80% of chronic infections and 65% of microbial infections detected in humans are associated with biofilm formation [18], the microbial communities forming these biofilms become recalcitrant to antibiotics, that is to say, up to 1000 times more tolerant to antibiotics than in their planktonic state [32].

The reasons why biofilms are resistant to antimicrobial agents are manifold and yet not understood. This resistance and tolerance to antimicrobial agents use genotypic and phenotypic mechanisms that allow cells to develop or survive. These characteristics of both resistance and tolerance of biofilms vis-à-vis antimicrobial agents are called "recalcitrance" [33]. The genetic mechanisms contributing to recalcitrance can manifest in several ways, for example, by altering the targets of antibiotics on microbial cells and decreasing the affinity of antibiotics [34], the enzymatic inactivation of antimicrobial agents, such as the presence of the plasmid pUM505 gives *Pseudomonas aeruginosa* resistance against ciprofloxacin [35], repellency of antimicrobial agents by active efflux pump [36], and spontaneous mutations in DNA as in the case of the deletion of the "agr" gene in Methicillin-resistant *Staphylococcus aureus* (MRSA) strains contribute to resistance, deletion of the agr gene grants bacteria higher resistance to high concentrations to oxacillin and ampicillin, resistance is strongly due to change in metabolism and overexpression of long-chain hydrophobic fatty acids in the membrane [37].

In contrast, global regulator sarA is a virulence regulator in MRSA, by regulating major virulence factors (e.g., adhesion, toxin production, and biofilm formation) mutations in the sarA gene leads to reductions oxacillin resistance and biofilm formation [38] or exchange of genes between cells in biofilms can confer resistance to antimicrobial agents microbial communities [39]. The contribution of the phenotypic characters of biofilm cells to tolerance to antimicrobial agents is also approved, the limitation of nutrients and the lack of oxygen in the microenvironments of biofilms induce a continuous stress response and reduce the growth rate of microbial cells, the presence of dormant cells such as *Staphylococcus epidermidis* [40], the presence of "persisters cells" known for their ability to escape the effects of antibiotics without undergoing genetic modification [41].

The composition of the biofilm matrices (polysaccharides, nucleic acids, and proteins) [42] also plays a key role in recalcitrance to antimicrobial agents as demonstrated in *Burkholderia pseudomallei* where three genes (bpsI, ppk, and rpoS) were included in the formation of exopolysaccharides [43], and the quorum-sensing signals [44].

5 Microbial food-borne pathogens

Some pathogens are designated as "primary pathogens," which regularly cause disease. Some are classified as "opportunistic pathogens" that infect primarily immune compromised or at-risk populations [45]. Food-borne pathogens could contaminate on fresh produce at many steps along the production from farm to table at point of consumption. The contamination of food-borne pathogen can arise from environmental, animal, or human sources [46].

Many food-borne pathogens such as *Salmonella enterica*, *Listeria monocytogenes* and *Escherichia coli*, *Pseudomonas* spp., *Brochothrix thermosphacta* and *Lactobacillus* spp., viruses (astrovirus, hepatitis A and E viruses, rotavirus, etc.),

and parasites (*Giardia intestinalis*, *Toxoplasma gondii*, and *Trichinella spiralis*) and molds are involved in outbreaks from contaminated food and water. Important number of bacteria are able to develop biofilms in many areas of food processing plants, including floors, walls, and pipes, etc. [47,48,49].

The existence of unwanted biofilms on food-handling contact surfaces may lead to: (1) illnesses transmission; (2) food damage; (3) curtailed time between cleaning events; (4) product pollution by nonstarter bacteria; (5) metal corrosion in pipelines and tanks; (6) decreased heat transfer efficiency or even the heat equipment blockage [50].

6 Biofilm formation in food stuffs

Many food-borne pathogens are ubiquitous in nature and generally found in soil, water, animals, and plants. Pathogens are introduced into a processing plant through the raw materials. Humans and plant equipment may also bring the organisms to plant. Fecal bacteria can contaminate meats. Microbial loads in shrimps, oysters, and clams depend on the quality of the water from which they are harvested. Concerning fruits and vegetables, contamination comes from contaminated irrigation water, animal feces when applied as fertilizers or from grazing domesticated or wild animals in the orchard, untreated manure, contaminated seeds, insect vectors, food handlers, processing environments, and slicing instruments [47].

7 Biofilms removal approaches

7.1 Conventional methods

The most common physical method employed to remove biofilms is manual cleaning of the surface using scrubbers, sonication, and raising temperature of water [10].

The chemical sanitizers used in the food industry are: halogens, peroxygens, acids, and quaternary ammonium compounds [10]. Traditional chemicals used in food preservation include but not limited to sulfites and nitrites. Organic acids such as benzoic, sorbic, acetic, lactic, and propionic acids are commonly used as traditional food preservatives [51]. The existence of soil, water hardness, and temperature of applications restricts the ability of chemical antibiofilms [10].

There is good evidence indicating that biofilm mode of life leads to increase resistance to antimicrobial products. Moreover, the emergence of resistant bacteria to conventional antimicrobials clearly shows the need to new strategies for controlling biofilm in food industry [13]. The effectiveness or efficacy of sanitizers and disinfectants depends on the interval between contamination and sanitizer application, contact time, attachment site, internalization of bacteria, temperature, pH, bacterial load, and age of the produce biofilms [9,12].

7.2 Biosurfactants

The BSs are natural surface-active molecules produced intra or extracellular by microorganisms (bacteria, yeast, and fungi). As these compounds are amphiphilic including hydrophilic and hydrophobic moieties, revealing the capacity to gather between interfaces (air-liquid or liquid-liquid) and decrease the surface and interfacial tension. With chain length between 8 and 18 carbon atoms, the saturated or unsaturated fatty acids or fatty acids derivatives are the main constituent of hydrophobic moiety, while the hydrophilic part is formed by ester, hydroxyl, phosphate or carboxyl groups, or carbohydrates, peptides, or proteins [52,53]. About 2000 different structures of biological origin have been described. These molecules were mostly interesting because of their antibiotics properties [52]. Other biological activities especially the antioxidant capacity of BSs have also been proven [54–56]. The physiological role of BSs is not yet fully understood; however, they play very important roles for microbial producing cells. As surface-active substances, BSs increase the availability of water-insoluble substrates for the growth and maintenance of microorganisms. These BSs are responsible for regulating the properties of the cell surface, promoting adhesion or release of cells from specific sites according to their nutritional needs, or avoiding unfavorable environments. They also facilitate the transport of hydrocarbons at the membrane level, a function attributed to glycolipid-type BSs linked to the cell membrane [53,57].

BSs are classified in accordance with the molecular weight, type of secretion from the cell, the ionic charge of the molecule, and chemical structure. These molecules can be grossly divided into two principal groups: low-molecular-weight components called BSs (lipopeptides, glycolipids, proteins) and high-molecular-weight or bioemulsan (polymers of polysaccharides, lipopolysaccharides, proteins, or lipoproteins) [58,59]. Glycolipids and lipopeptides are the most widely explored groups in the literature due to the high production yields and considerable applications potential in the most diverse fields [59]. Glycolipids consist of sugar and lipids in which carbohydrate moiety is linked to a fatty acid moiety (aliphatic acids or hydroxyl aliphatic acids) [60]. The well-known glycolipids are rhamnolipids, trehalolipids, and sophorolipids [61]. Lipopeptides are composed of amino acid chain (linear or cyclic peptide) linked to fatty acid. Iturin, surfactin, and lichenysin are among the most studied and applied lipopeptides microbial surfactants. Alasan, liposan, lipomanan, emulsan, biodispersan, and some other polysaccharide-protein buildings are the best-contemplated polymeric BSs [60,62]. The particulate BSs are accumulation of extracellular membrane vesicles with diameter of 20–50 nm, buoyant density of 1.158 g/cm^2, and consist of protein, phospholipids, and lipopolysaccharides [63]. The surface tension, the critical micelle concentration (CMC), and the hydrophilic-lipophilic balance (HLB) are the main parameters measuring the surfactants' effectiveness [64]. Various microbial classes have been reported as producers of BSs. The species of genera *Pseudomonas* and *Bacillus* and the species *Candida bombicola*, *Candida lipolytica*, *Yarrowia lipolytica*, *Saccharomyces cerevisiae*, and *Kluyveromyces lactis* are among the most frequently studied species and known as great producers of BSs [65].

Owing to their low toxicity, biodegradability, biodigestibility, biocompatibility, steadiness under a wide range of physicochemical environments, possible reuse through regeneration, and high ability to reduce surface and interfacial surface tension, BSs have aroused considerable interest in recent decades as eco-friendly alternative of nonbiodegradable synthetic surfactants in particular in food industry [59,66–68]. Also, BSs can be applied in food processing to prevent, inhibit, and remove biofilms-associated microorganisms because of their antimicrobial, antiadhesive, and antibiofilms activities [53].

Numerous works showed the microbial surfactants efficiency in the inhibition of pathogens bacteria growth and biofilm formation. The destruction of microbial cells through biosurfactants is done by disrupting the integrity of the plasma membrane or the cell wall. The degree of such cell border damage makes it difficult for any target organism to build biosurfactant resistance [69]. In general, there are two approaches to treat biofilms: (1) biofilm inhibition, which prevents biofilm formation, and (2) biofilm dispersal, which removes biofilms that have already formed [70]. Biosurfactants may act at different levels: showing antibacterial activity by disrupting the integrity and porosity of the membranes or cell walls, altering cell hydrophobicity, regulating or inhibiting quorum sensing signaling, decreasing or inhibiting the production of exopolysaccharides, and adhesion and inhibition of initial adhesion to the surface [71,69,72]. The antiadhesive and antibiofilm abilities of biosurfactants have been shown by several researchers. Rivardo et al. [73] revealed that two biosurfactants produced by *Bacillus* spp. have antiadhesion activity, preventing human bacterial pathogens from forming bacterial biofilms. In the study performed by Sambanthammorthy et al. [17], potential antimicrobial, antiadhesive, and antibiofilm properties of BSs by *Lactobacillus jensenii* and *Lactobacillus rhamnosus* against multidrug-resistant strains of *Acinetobacter baumannii*, *E. coli*, and *S. aureus* were demonstrated. *The produced crude biosurfactants inhibited cell attachment in the presence of host protein, reduced biofilm development, and dispersed preformed biofilm* by damaging the membrane for *A. baumannii* and the cell wall for *S. aureus*.

The biosurfactants of coral-associated bacteria showed antibiofilm activity and stability against biofilm-forming *Pseudomonas aeruginosa* ATCC10145 at various treatment temperatures ranging from 60 to 100 °C [74]. The antibiofilm and dispersal properties of biosurfactants may be due to their antimicrobial and surfactant activities. Diaz De Rienzo et al. [75] demonstrated the antimicrobial activity and biofilm disruption of sophorolipids at different concentrations on *Cupriavidus necator* and *Bacillus subtilis*. A new cyclic lipopeptide coryxin, isolated from a *Corynebacterium xerosis* strain, showed strong antibacterial activity and disruption of preformed biofilm activities of *Staphylococcus aureus*, *Streptococcus mutans*, *Escherichia coli,* and *Pseudomonas aeruginosa* [76]. Kiran et al. [77] revealed that the lipopeptide biosurfactant synthesized by the actinobacterial strain MSA31 isolated from a marine sponge *Fasciospongia cavernosa* was a "halo-alkali and thermal tolerant biosurfactant" that can be employed in the food industry. At low concentration, this lipopeptide proved successful in lowering the biofilm-forming activity of pathogenic multidrug-resistant *Staphylococcus aureus* in a microtiter plate assay [77].

Giordani and coworkers [78] found that the biosurfactant isolated from *Lactobacillus gasseri* BC9 and the loaded liposome exhibited a higher anti *S. aureus* biofilm capacity than free biosurfactant. Two lipopeptides isolated from *Bacillus subtilis* VSG4 and *Bacillus licheniformis* VS16 exhibited notable antibacterial, antiadhesive, and antibiofilm activities against *Staphylococcus aureus*, *Salmonella typhimurium*, and *Bacillus cereus* [79]. The glycolipid biosurfactant fabricated by the endophyte *Burkholderia* sp. WYAT7 isolated from the medicinal plant *Artemisia nilagirica* (Clarke) Pamp. displayed antibacterial activity against bacterial pathogens such as *Pseudomonas aeruginosa* (MTCC 2453), *Escherichia coli* (MTCC 1610), *Salmonella paratyphi*, and *Bacillus subtilis* as well as antibiofilm activity against *Staphylococcus aureus* (MTCC 1430) [80]. Patel et al. [71] have studied the antibiofilm activity of glycolipid BS produced by probiotic lactic acid bacteria (*L. rhamnosus*). The obtained BS exhibited powerful antiadhesion and antibiofilm capacity by inhibiting the bacterial binding to surfaces, leading to the disruption of biofilm emergence by modifying the integrity and viability of bacterial cells within biofilms.

Relative microbial hydrophobicity is thought to be an essential pathogenic feature related to adhesion, which is the initial and crucial stage in biofilm development. Decreasing relative hydrophobicity by altering cell surface properties such as wettability and charge is one of the proposed mechanisms of biofilm formation prevention by biosurfactants [72]. Indeed, the study carried out by Borah et al. [81] investigated the ability of rhamnolipid produced by *Pseudomonas aeruginosa* SS14 utilizing rice-based distillers dried grains with solubles (RL-rDDGS) to remove *Candida tropicalis* biofilm. This study revealed a significant *C. tropicalis* biofilm-disrupting property of the produced RL. In addition, after 24 hours treatment with RL-rDDGS, the cell surface hydrophobicity of *C. tropicalis* decreased significantly and in a concentration-dependent manner (Borah et al., 2019) [88]. It is critical to understand the genetic basis of biofilm development in order to develop efficient biofilm prevention strategies for both disease prevention and engineering applications [82]. Many of the cell-to-cell and cell-to-surface interactions essential for the development, cohesion, and stabilization of bacterial biofilms have been found to be mediated by bacterial extracellular polysaccharides [83]. In fact, various kinds of exopolysaccharides are inherited by organisms which play an important role in biofilm integrity such as adhesion (polysaccharide intercellular adhesion, levan, and vibrio), architectural framework (alginate, colonic acid, and vibrio), and protection (alginate, capsular polysaccharides, α-mannans, and α-glucans) [84]. The regulation of the biosynthetic machinery required for the assembly of these exopolysaccharides is carried out by signaling molecules in many bacteria, with the second messenger cyclic di-guanosine monophosphate (c-di-GMP) playing a particularly important role in exopolysaccharide biosynthesis posttranslational activation [85,86]. Moreover, the expression of some key genes such as glucosyltransferase (*gtfs*) and fructosyltransferase (*ftf*) is required for the sucrose-dependent attachment of microorganism cells to hard surfaces [87]. Biosurfactants are amphipathic molecules with interfacial activity that can change cell-to-surface and cell-to-cell interactions, potentially reducing the bacteria's capacity to cling to surfaces.

Their excessive production, on the other hand, prevents the formation of biofilms [88]. Salehi et al. [89] noticed that biosurfactants were able to inhibit the gene expression of adhesion. In the presence of biosurfactants isolated from *Lactobacillus reuteri*, the expression of glucosyltransferase (*gtfs*) and fructosyltransferase (*ftf*) genes, which play a vital role in the initial attachment of *Streptococcus mutans* to the tooth surface, was suppressed. Tahmourespour et al. [87] found that the biosurfactant produced by *Lactobacillus rhamnosus* reduced significantly the *Streptococcus mutans* biofilm formation and the level of expression of *gtf*B/C and *ftf* genes. Biosurfactants also impact biofilm architecture by producing cavities in the biofilm's center [90]. Single cells can communicate with each other via a quorum sensing (QS) communication process. Quorum sensing is a signaling mechanism that regulates the life activities of bacteria by transmitting signals through the synthesis, release, and acceptance of autoinducers (AIs). In response to variations in population density, quorum sensing (QS) can promote the expression of specific genes and coordinate the behavior of bacteria. Some physiological functions of bacteria that could be regulated by accumulated QS are symbiosis, toxicity, binding, antibiotic production, virulence, sporulation, and biofilm formation [91,92]. Gram-positive bacteria create oligopeptides, while Gram-negative bacteria produce N-acyl homoserine lactones, which are common QS signaling chemicals. The autoinducer-2 (AI-2) is a ubiquitous QS molecule that mediates intra- and interspecies communication. The major QS molecule is synthesized by the spontaneous rearrangement of 4, 5-dihydroxy-2, 3-pentanedione (DPD), which is produced by the enzyme LuxS [93]. The antibacterial, antiadhesive, and antibiofilm properties of BS derived from lactic acid bacteria (LAB), such as *Pediococcus acidilactici* and *Lactobacillus plantarum*, were evaluated *in vitro* against *Staphylococcus aureus* CMCC 26003. In addition to their antibacterial and antiadhesive capacities, the BS fabricated by LAB exhibited a potential inhibitory effect on AI-2 synthesis. In a dose-dependent way, the BS generated by *Pediococcus acidilactici* can drastically inhibit AI-2 synthesis. When the concentration of BS generated by *Lactobacillus plantarum* reached 50 mg/mL, the bioluminescence intensity was substantially lower than that in the case of the positive control [94]. Microbubbles are gas bubbles covered with a shell (surfactant, protein, lipid, or polymer) that ensures a slowing down of disproportionation and preventing coalescence. The MB diameter range is between 10 and 50 mm [68,95–97]. It has been proposed that microbubbles could be used as ingredients in food systems and might be used as fat replacers [97]. Microbubbles technology has found applications in different sectors of industry. It is applied in a washing process, particularly to reduce the microbial contamination on leafy and fresh vegetables [95,98]. It is also applied in the preparation of edible coatings for delivering antimicrobial agents (e.g., essential oils) [99]. The properties of microbubbles depend on their generation method and the surfactant used [68]. The microbubbles can be utilized as carrier system for the delivery of nutrients, antibiotics, ozone, and oxygen. Adsorption of surfactant molecules on gas–water interphase of microbubbles can provide a protective barrier of hydration character, which helps the microbubbles to be more stable [68,100]. For increasing the microbubbles' stability, the BSs are used as substitute

to synthetic surfactants [68]. Currently, the efficiency of biosurfactants was consolidated by the combination of biosurfactants with nanotechnology by using organic or inorganic nanomaterials. The synthesis of rhamnolipid-coated silver (Ag) and iron oxide (Fe_3O_4) nanoparticles demonstrated excellent antibiofilm activity not only during the biofilm formation but also on the preformed biofilms against *P. aeruginosa* and *S. aureus* [101]. Arif et al. [102] studied the use of chitosan-based nanoparticles as carriers of rhamnolipid (RL-CS NPs) in order to evaluate the antimicrobial and antibiofilm activity of this complex against *Helicobacter pylori*. This study revealed the efficiency of RL-CS NPs in inhibiting bacterial growth and in eradicating *Helicobacter pylori* biofilm.

8 Conclusion

As the global need for safe, high-quality, and nutritious food increases, it is necessary to protect these foodstuffs from all types of contamination, in particular by food-borne pathogens capable of forming biofilms on food and food-contact surfaces. Due to its resistance to antimicrobial agents, the formation of biofilm is a major problem for the food industry affecting the shelf life and quality of final products as well as the possibility of causing poisoning and transmitting various kinds of infectious diseases. Biofilm removal is a global challenge that requires developing novel and strong bioactive molecules for controlling, as substitute to chemically synthetized antibiofilms. Additionally, compound selection in the food sector must take into account the temperature at each stage of product processing as well as the pathogens to be controlled. The use of BSs to quarrel food-borne pathogens development and biofilm emergence in food industry is one of the new strategies applied recently.

References

[1] A. Akabar, A.K. Anal, Food safety concerns and food-borne pathogens, *Salmonella*, *Escherichia coli* and *Campylobacter*, FUUAST J. Biol. 1 (1) (2011) 5–17.

[2] W. Mahakarnchanakul, P. Klintham, S. Tongchitpakdee, W. Chinsirikul, Using sanitizer and fine bubble technologies to enhance food safety, in: Conference: FFTC-KU International Workshop on Risk Management on Agrochemicals through Novel Technologies for Food Safety in Asia, 2014. https://www.researchgate.net/publication/308791089.

[3] X. Zhao, F. Zhao, J. Wang, N. Zhong, Biofilm formation and control strategies of food-borne pathogens: food safety perspectives, RSC Adv. 7 (2017) 36670, https://doi.org/10.1039/c7ra02497e.

[4] M.D. Kirk, S.M. Pires, R.E. Black, M. Caipo, J.A. Crump, et al., World Health Organization estimates of the global and regional disease burden of 22 foodborne bacterial, protozoal, and viral diseases, 2010: a data synthesis, PLoS Med. 12 (12) (2015) e1001940, https://doi.org/10.1371/journal.pmed.1001940.

[5] E.E. Giaouris, M.V. Simões, Pathogenic biofilm formation in the food industry and alternative control strategies, in: A.M. Holban, A.M. Grumezescu (Eds.), Food Diseases: Handbook of Food Bioengineering, Academic Press, Elsevier, London, 2018, https://doi.org/10.1016/B978-0-12-811444-5.00011-7.
[6] H. Flemming, J. Wingender, The biofilm matrix, Nat. Rev. Microbiol. 8 (2010) 623–633, https://doi.org/10.1038/nrmicro2415.
[7] X. Shi, X. Zhu, Biofilm formation and food safety in food industries, Trends Food Sci. Technol. 20 (2009) 407–413, https://doi.org/10.1016/j.tifs.2009.01.054.
[8] D. Gutiérrez, P. Ruas-Madiedo, B. Martínez, A. Rodríguez, P. García, Effective removal of staphylococcal biofilms by the endolysin LysH5, PLoS One 9 (9) (2014) e107307, https://doi.org/10.1371/journal.pone.0107307.
[9] I.K. Jahid, S.-D. Ha, A review of microbial biofilms of produce: future challenge to food safety, Food Sci. Biotechnol. 21 (2) (2012) 299–316, https://doi.org/10.1007/s10068-012-0041-1.
[10] K. Myszka, K. Czaczyk, Bacterial biofilms on food contact surfaces—a review, Polish J. Food Nutr. Sci. 61 (3) (2011) 173–180, https://doi.org/10.2478/v10222-01160018-4.
[11] S. Wu, P.-L. Yu, S. Flint, Persister cell formation of *Listeria monocytogenes* in response to natural antimicrobial agent nisin, Food Control 77 (2017) 243–250, https://doi.org/10.1016/j.foodcont.2017.02.011.
[12] L. Yuan, F.A. Sadiq, N. Wang, Z. Yang, G. He, Recent advances in understanding the control of disinfectant-resistant biofilms by hurdle technology in the food industry, Crit. Rev. Food Sci. Nutr. (2020), https://doi.org/10.1080/10408398.2020.1809345.
[13] M. Simões, L.C. Simões, M.J. Vieira, A review of current and emergent biofilm control strategies, LWT Food Sci. Technol. 43 (2010) 573–583, https://doi.org/10.1016/j.lwt.2009.12.008.
[14] X. Bai, L. Xu, A.K. Singh, X. Qiu, M. Liu, A. Abuzeid, T. El-Khateib, A.K. Bhunia, Inactivation of polymicrobial biofilms of foodborne pathogens using epsilon poly-L-lysin conjugated chitosan nanoparticles, Foods 11 (2022) 569, https://doi.org/10.3390/foods11040569.
[15] C. Ceresa, L. Fracchia, E. Fedeli, C. Porta, I.M. Banat, Recent advances in biomedical, therapeutic and pharmaceutical applications of microbial surfactants, Pharmaceutics 13 (2021) 466, https://doi.org/10.3390/pharmaceutics13040466.
[16] V. Walter, C. Syldatk, R. Hausmann, Screening concepts for the isolation of biosurfactant producing microorganisms, in: R. Sen (Ed.), Biosurfactants, Landes Bioscience and Springer Science+Business Media, LLC, New York, 2010, pp. 1–41.
[17] K. Sambanthammorthy, X. Feng, R. Patel, S. Patel, C. Paranavitana, Antimicrobial and antibiofilm potential of biosurfactants isolated from lactobacilli against multi-drug-resistant pathogens, BMC Microbiol. 14 (2014) 197. http://www.biomedcentral.com/1471-2180/14/197.
[18] M. Jamal, W. Ahmad, S. Andleeb, F. Jalil, M. Imran, M.A. Nawaz, T. Hussain, M. Ali, M. Rafiq, M.A. Kamil, Bacterial biofilm and associated infections, J. Chin. Med. Assoc. 81 (1) (2018) 7–11.
[19] P. Di Martino, Extracellular polymeric substances, a key element in understanding biofilm phenotype, AIMS Microbiol. 4 (2) (2018) 274–288.
[20] H.C. Flemming, EPS-then and now, Microorganisms 4 (4) (2016) 41.
[21] R.M. Donlan, Biofilms: microbial life on surfaces, Emerg. Infect. Dis. 8 (9) (2002) 881–890.

[22] C. Berne, A. Ducret, G.G. Hardy, Y.V. Brun, Adhesins involved in attachment to abiotic surfaces by gram-negative bacteria, Microbiol. Spectr. 3 (4) (2015), https://doi.org/10.1128/microbiolspec.MB-0018-2015.
[23] D. López, H. Vlamakis, R. Kolter, Biofilms, Cold Spring Harb. Perspect. Biol. 2 (7) (2010) a000398.
[24] J.M. Koczan, B.R. Lenneman, M.J. McGrath, G.W. Sundin, Cell surface attachment structures contribute to biofilm formation and xylem colonization by *Erwinia amylovora*, Appl. Environ. Microbiol. 77 (19) (2011) 7031–7039.
[25] R. Chandki, P. Banthia, R. Banthia, Biofilms: a microbial home, J. Indian Soc. Periodontol. 15 (2) (2011) 111–114.
[26] J.N. Wilking, V. Zaburdaev, M. De Volder, R. Losick, M.P. Brenner, D.A. Weitz, Liquid transport facilitated by channels in *Bacillus subtilis* biofilms, Proc. Natl. Acad. Sci. U. S. A. 110 (3) (2013) 848–852.
[27] P. Becker, W. Hufnagle, G. Peters, M. Herrmann, Detection of differential gene expression in biofilm-forming versus planktonic populations of *Staphylococcus aureus* using micro-representational-difference analysis, Appl. Environ. Microbiol. 67 (7) (2001) 2958–2965.
[28] P. Stoodley, S. Wilson, L. Hall-Stoodley, J.D. Boyle, H.M. Lappin-Scott, J.W. Costerton, Growth and detachment of cell clusters from mature mixed-species biofilms, Appl. Environ. Microbiol. 67 (12) (2001) 5608–5613.
[29] J.B. Kaplan, C. Ragunath, K. Velliyagounder, D.H. Fine, N. Ramasubbu, Enzymatic detachment of *Staphylococcus epidermidis* biofilms, Antimicrob. Agents Chemother. 48 (7) (2004) 2633–2636.
[30] D.A. Wilkinson, S.J. Chacko, C. Vénien-Bryan, G.H. Wadhams, J.P. Armitage, Regulation of flagellum number by FliA and FlgM and role in biofilm formation by *Rhodobacter sphaeroides*, J. Bacteriol. 193 (15) (2011) 4010–4014.
[31] D. Monroe, Looking for chinks in the armor of bacterial biofilms, PLoS Biol. 5 (11) (2007) e307, https://doi.org/10.1371/journal.pbio.0050307.
[32] D. Lebeaux, J.M. Ghigo, C. Beloin, Biofilm-related infections: bridging the gap between clinical management and fundamental aspects of recalcitrance toward antibiotics, Microbiol. Mol. Biol. Rev. 78 (3) (2014) 510–543.
[33] G. Orazi, G.A. O'Toole, "It takes a village": mechanisms underlying antimicrobial recalcitrance of polymicrobial biofilms, J. Bacteriol. 202 (1) (2019) e00530-19.
[34] T.J. Dougherty, Involvement of a change in penicillin target and peptidoglycan structure in low-level resistance to beta-lactam antibiotics in *Neisseria gonorrhoeae*, Antimicrob. Agents Chemother. 28 (1) (1985) 90–95.
[35] V.M. Chávez-Jacobo, K.C. Hernández-Ramírez, P. Romo-Rodríguez, R.V. Pérez-Gallardo, J. Campos-García, J.F. Gutiérrez-Corona, J.P. García-Merinos, V. Meza-Carmen, J. Silva-Sánchez, M.I. Ramírez-Díaz, CrpP is a novel ciprofloxacin-modifying enzyme encoded by the *Pseudomonas aeruginosa* pUM505 plasmid, Antimicrob. Agents Chemother. 62 (6) (2018) e02629-17.
[36] H.A. Terzi, C. Kulah, I.H. Ciftci, The effects of active efflux pumps on antibiotic resistance in *Pseudomonas aeruginosa*, World J. Microbiol. Biotechnol. 30 (10) (2014) 2681–2687.
[37] H.S. Song, T.R. Choi, Y.H. Han, Y.L. Park, J.Y. Park, S.Y. Yang, S.K. Bhatia, R. Gurav, Y.G. Kim, J.S. Kim, H.S. Joo, Y.H. Yang, Increased resistance of a methicillin-resistant *Staphylococcus aureus* Δagr mutant with modified control in fatty acid metabolism, AMB Express 10 (1) (2020) 64.

[38] L. Li, A. Cheung, A.S. Bayer, L. Chen, W. Abdelhady, B.N. Kreiswirth, M.R. Yeaman, Y.Q. Xiong, The global regulon sarA regulates β-lactam antibiotic resistance in methicillin-resistant *Staphylococcus aureus* in vitro and in endovascular infections, J. Infect. Dis. 214 (9) (2016) 1421–1429.
[39] T. Stalder, E. Top, Plasmid transfer in biofilms: a perspective on limitations and opportunities, NPJ Biofilms Microbiomes 2 (2016) 16022.
[40] V. Carvalhais, A. França, F. Cerca, R. Vitorino, G.B. Pier, M. Vilanova, N. Cerca, Dormancy within *Staphylococcus epidermidis* biofilms: a transcriptomic analysis by RNA-seq, Appl. Microbiol. Biotechnol. 98 (6) (2014) 2585–2596.
[41] T.K. Wood, S.J. Knabel, B.W. Kwan, Bacterial persister cell formation and dormancy, Appl. Environ. Microbiol. 79 (23) (2013) 7116–7121.
[42] E.E. Mann, D.J. Wozniak, Pseudomonas biofilm matrix composition and niche biology, FEMS Microbiol. Rev. 36 (4) (2012) 893–916.
[43] R. Mongkolrob, S. Taweechaisupapong, S. Tungpradabkul, Correlation between biofilm production, antibiotic susceptibility and exopolysaccharide composition in *Burkholderia pseudomallei* bpsI, ppk, and rpoS mutant strains, Microbiol. Immunol. 59 (11) (2015) 653–663.
[44] V. Lazar, A.M. Holban, C. Curutiu, M.C. Chifiriuc, Modulation of quorum sensing and biofilms in less investigated gram-negative ESKAPE pathogens, Front. Microbiol. 12 (2021) 676510.
[45] A.K. Bhunia, Foodborne Microbial Pathogens: Food Science Text Series, Springer Science+Business Media, LLC, part of Springer Nature, New York, USA, 2018, pp. 1–23, https://doi.org/10.1007/978-1-4939-7349-1_1.
[46] M.I. Gil, M.V. Selma, F. López-Gálvez, A. Allende, Fresh-cut product sanitation and wash water disinfection: problems and solutions, Int. J. Food Microbiol. 134 (2009) 37–45, https://doi.org/10.1016/j.ijfoodmicro.2009.05.021.
[47] A. Bhunia, Foodborne Microbial Pathogens: Mechanisms and Pathogenesis, Springer, Berlin, 2009, https://doi.org/10.1007/978-1-4939-7349-1.
[48] E. Giaouris, E. Heir, M. Hébraud, et al., Attachment and biofilm formation by foodborne bacteria in meat processing environments: causes, implications, role of bacterial interactions and control by alternative methods, Meat Sci. (2013), https://doi.org/10.1016/j.meatsci.2013.05.023.
[49] R. Wang, Biofilms and Meat Safety: A Mini-Review, International Association for Food Protection, 2019, pp. 120–127, https://doi.org/10.4315/0362-028X.JFP-18-311.
[50] F. Cappitelli, A. Polo, F. Villa, Biofilm formation in food processing environments is still poorly understood and controlled, Food Eng. Rev. 6 (1–2) (2014) 29–42, https://doi.org/10.1007/s12393-014-9077-8.
[51] A.G. Abdelahamid, N.K. El-Dougdoug, Controlling foodborne pathogens with natural antimicrobials by biological control and antivirulence strategies, Heliyon 6 (2020) e05020, https://doi.org/10.1016/j.heliyon.2020.e05020.
[52] R. Hausmann, C. Syldatk, Type and classification of microbial surfactants, in: N. Kosaric, S.F. Verdar (Eds.), Biosurfactants: Production and Utilization-Processes Technologies and Economics, CRC Press, Taylor and Francis Group, LLC, US, 2015, pp. 3–17.
[53] M. Nitsche, S.S. e Silva, Recent food applications of microbial surfactants, Crit. Rev. Food Sci. Nutr. (2016), https://doi.org/10.1080/10408398.2016.1208635.
[54] A. Kumar, S.K. Singh, C. Kant, H. Verma, D. Kumar, P.P. Singh, A. Modi, S. Droby, M. S. Kesawat, H. Alavilli, et al., Microbial biosurfactant: a new frontier for sustainable

agriculture and pharmaceutical industries, Antioxidants 10 (2021) 1472, https://doi.org/10.3390/antiox10091472.

[55] A. Merghni, I. Dallel, E. Noumi, Y. Kadmi, H. Hentati, S. Tobji, A. Ben Amor, M. Mastouri, Antioxidant and antiproliferative potential of biosurfactants isolated from *Lactobacillus casei* and their anti-biofilm effect in oral *Staphylococcus aureus* strains, Microb. Pathog. (2017), https://doi.org/10.1016/j.micpath.2017.01.017.

[56] S.J. Nadaf, V.M. Kumbar, A.I. Torvi, J.H. Hoskeri, A.K. Shettar, Antioxidant biosurfactants, in: Inamuddin, M.I. Ahamed, R. Prasad (Eds.), Microbial Biosurfactants. Environmental and Microbial Biotechnology, 2021, https://doi.org/10.1007/978-981-15-6607-3.

[57] D. Jiménez Islas, S.A. Medina Moreno, J.N. Gracida Rodríguez, Revisión/review propiedades, aplicaciones y producción de biotensoactivos, Rev. Int. Contam. Ambient. 26 (1) (2010) 65–84. https://www.researchgate.net/publication/285069116.

[58] I.M. Banat, A. Franzetti, I. Gandolfi, G. Bestetti, M.G. Martinotti, L. Fracchia, T.J. Smyth, R. Marchant, Microbial biosurfactants production, applications and future potential, Appl. Microbiol. Biotechnol. 87 (2) (2010) 427–444, https://doi.org/10.1007/s00253-010-2589-0.

[59] B.G. Ribeiro, J.M.C. Guerra, L.A. Sarubbo, Biosurfactants: production and application prospects in the food industry, Biotechnol. Prog. (2020), https://doi.org/10.1002/btpr.3030.

[60] S.J. Varjani, V.N. Upasani, Critical review on biosurfactant analysis, purification and characterization using rhamnolipid as a model biosurfactant, Bioresour. Technol. 232 (2017) 389–397, https://doi.org/10.1016/j.biortech.2017.02.047.

[61] G. Rawat, A. Dhasmana, V. Kumar, Biosurfactants: the next generation biomolecules for diverse applications, Environ. Sustain. (2020), https://doi.org/10.1007/s42398-020-00128-8.

[62] A. Roy, Review on the biosurfactants: properties, types and its applications, J. Fundam. Renew. Energy Appl. 8 (2017) 248, https://doi.org/10.4172/20904541.1000248.

[63] P. Saha, K.V.B. Rao, Biosurfactants—a current perspective on production and applications, Nat. Environ. Pollut. Technol. 16 (1) (2017) 181–188. https://neptjournal.com/upload-images/NL-59-26-(24)B-3387.pdf.

[64] J.M. Campos, T.L. Montenegro Stamford, L.A. Sarubbo, J.M. de Luna, R.D. Rufino, I. M. Banat, Microbial biosurfactants as additives for food industries, Biotechnol. Prog. (2013), https://doi.org/10.1002/btpr.1796.

[65] D.K.F. Santos, R.D. Rufino, J.M. Luna, V.A. Santos, L.A. Sarubbo, Biosurfactants: multifunctional biomolecules of the 21st century, Int. J. Mol. Sci. 17 (3) (2016) 401, https://doi.org/10.3390/ijms17030401.

[66] M. Fakruddin, Biosurfactants: production and applications, J. Pet. Environ. Biotechnol. 3 (2012) 124, https://doi.org/10.4172/2157-7463.1000124.

[67] B.G. Ribeiro, J.M.C. Guerra, L.A. Sarubbo, Potential food application of a biosurfactant produced by *Saccharomyces cerevisiae* URM 6670, Front. Bioeng. Biotechnol. 8 (2020) 434, https://doi.org/10.3389/fbioe.2020.00434.

[68] Q. Xu, M. Nakajima, Z. Liu, T. Shiina, Biosurfactants for microbubble preparation and application, Int. J. Mol. Sci. 12 (2011) 462–475, https://doi.org/10.3390/ijms12010462.

[69] G. Płaza, V. Achel, Biosurfactants: eco-friendly and innovative biocides against biocorrosion, Inter. J. Mol. Sci. 21 (2152) (2020) 2152, https://doi.org/10.3390/ijms21062152.

[70] L. Shrestha, H.-M. Fan, H.-R. Tao, J.-D. Huang, Recent strategies to combat biofilms using antimicrobial agents and therapeutic approaches, Pathogens 11 (2022) 292, https://doi.org/10.3390/pathogens11030292.

[71] M. Patel, A.J. Siddiqui, W.S. Hamadou, M. Surti, A.M. Awadelkareem, S.A. Ashraf, M. Alreshidi, M. Snoussi, S.M.D. Rizvi, F. Bardakci, et al., Inhibition of bacterial adhesion and antibiofilm activities of a glycolipid biosurfactant from *Lactobacillus rhamnosus* with its physicochemical and functional properties, Antibiotics 10 (2021) 1546, https://doi.org/10.3390/antibiotics10121546.

[72] O. Rendueles, J.-M. Ghigo, Multi-species biofilms: how to avoid unfriendly neighbors, FEMS Microbiol. Rev. 36 (2012) 972–989, https://doi.org/10.1111/j.1574-6976.2012.00328.x.

[73] F. Rivardo, R.J. Turner, G. Allegrone, H. Ceri, M.G. Martinutti, Anti-biofilm activity of two biosurfactants produced by *Bacillus* spp. prevents biofilm formation of human bacterial pathogens, Appl. Microbiol. Biotechnol. 83 (2009) 541–553, https://doi.org/10.1007/s00253-009-1987-7.

[74] A.R. Padmavathi, S.K. Pandian, Antibiofilm activity of biosurfactant producing coral associated bacteria isolated from Gulf of Mannar, Indian J. Microbiol. 54 (4) (2014) 376–382, https://doi.org/10.1007/s12088-014-0474-8.

[75] M.A. Diaz De Rienzo, I.M. Banat, B. Dolman, J. Winterburn, P.J. Martin, Sophorolipid biosurfactants: possible uses as antibacterial and antibiofilm agent, New Biotechnol. 32 (6) (2015) 720–726, https://doi.org/10.1016/j.nbt.2015.02.009.

[76] D. Dalili, M. Amini, M.A. Faramarzi, M.R. Fazeli, M.R. Khoshayand, N. Samadi, Isolation and structural characterization of coryxin, a novel cyclic lipopeptide from *Corynebacterium xerosis* NS5 having emulsifying and anti-biofilm activity, Colloids Surf. B Biointerfaces 135 (2015) 425–432, https://doi.org/10.1016/j.colsurfb.2015.07.005.

[77] G.S. Kiran, S. Priyadharsini, A. Sajayan, S. Priyadharsini, N. Poulose, J. Selvin, Production of lipopeptide biosurfactant by a marine *Nesterenkonia* sp. and its application in food industry, Front. Microbiol. 8 (2017) 1138, https://doi.org/10.3389/fmicb.2017.01138.

[78] B. Giordani, P. Emidio Costantini, S. Fedi, M. Cappelletti, A. Abruzzo, C. Parolin, C. Foschi, G. Frisco, N. Calonghi, T. Cerchiara, F. Bigucci, B. Luppi, B. Vitali, Liposomes containing biosurfactants isolated from *Lactobacillus gasseri* exert antibiofilm activity against methicillin resistant *Staphylococcus aureus* strains, Eur. J. Pharm. Biopharm. (2019), https://doi.org/10.1016/j.ejpb.2019.04.011.

[79] S.S. Giri, E. Ryu, V. Sukumaran, S. Chang Park, Antioxidant, antibacterial, and anti-adhesive activities of biosurfactants isolated from *Bacillus* strains, Microb. Pathog. (2019), https://doi.org/10.1016/j.micpath.2019.04.035.

[80] A. Ashitha, E.K. Radhakrishnan, M. Jyothis, Characterization of biosurfactant produced by the endophyte *Burkholderia* sp. WYAT7 and evaluation of its antibacterial and antibiofilm potentials, J. Biotechnol. (2020), https://doi.org/10.1016/j.jbiotec.2020.03.005.

[81] S.N. Borah, S. Sen, L. Goswami, A. Bora, K. Pakshirajan, S. Deka, Rice based distillers dried grains with solubles as a low cost substrate for the production of a novel rhamnolipid biosurfactant having anti-biofilm activity against *Candida tropicalis*, Colloids Surf. B Biointerfaces 182 (2019) 110358, https://doi.org/10.1016/j.colsurfb.2019.110358.

[82] A.F. González Barrios, R. Zuo, Y. Hashimoto, L. Yang, W.E. Bentley, T.K. Wood, Autoinducer 2 controls biofilm formation in *Escherichia coli* through a novel motility

quorum-sensing regulator (MqsR, B3022), J. Bacteriol. 188 (1) (2006) 305–316, https://doi.org/10.1128/JB.188.1.305–316.2006.
[83] O. Rendueles, J.B. Kaplan, J.-M. Ghigo, Antibiofilm polysaccharides, Environ. Microbiol. 15 (2) (2013) 334–346, https://doi.org/10.1111/j.1462-2920.2012.02810.x.
[84] S. Nadar, T. Khan, S.G. Patching, A. Omri, Development of antibiofilm therapeutics strategies to overcome antimicrobial drug resistance, Microorganisms 10 (2022) 303, https://doi.org/10.3390/microorganisms10020303.
[85] E. Maunders, M. Welch, Matrix exopolysaccharides; the sticky side of biofilm formation, FEMS Microbiol. Lett. 364 (2017) fnx120, https://doi.org/10.1093/femsle/fnx120.
[86] M.B. Poulin, L.L. Kuperman, Regulation of biofilm exopolysaccharide production by cyclic di-guanosine monophosphate, Front. Microbiol. 12 (2021) 730980, https://doi.org/10.3389/fmicb.2021.730980.
[87] A. Tahmourespour, R. Kasra-Kermanshahi, R. Salehi, *Lactobacillus rhamnosus* biosurfactant inhibits biofilm formation and gene expression of caries-inducing *Streptococcus mutans*, Dent. Res. J. 16 (2019) 87–94. https://www.drjjournal.net/text.asp?2019/16/2/87/250968.
[88] A. Nickzad, E. Déziel, The involvement of rhamnolipids in microbial cell adhesion and biofilm development—an approach for control? Lett. Appl. Microbiol. 58 (2013) 447–453, https://doi.org/10.1111/lam.12211.
[89] R. Salehi, O. Savabi, M. Kazemi, S. Kamali, A.R. Salehi, G. Eslami, A. Tahmourespour, Effects of *Lactobacillus reuteri*-derived biosurfactant on the gene expression profile of essential adhesion genes gtfB, gtfC and ftf of *Streptococcus mutans*, Adv. Biomed. Res. 3 (2014) 169, https://doi.org/10.4103/2277-9175.139134.
[90] B.R. Boles, M. Thoendel, P.K. Singh, Rhamnolipids mediate detachment of *Pseudomonas aeruginosa* from biofilms, Mol. Microbiol. 57 (5) (2005) 1210–1223, https://doi.org/10.1111/j.1365-2958.2005.04743.x.
[91] Y. Liu, H. Hu, F. Luo, Roles of autoinducer-2 mediated quorum sensing in wastewater treatment, Water Sci. Technol. 84 (4) (2021) 793, https://doi.org/10.2166/wst.2021.278.
[92] K. Jiang, Y. Xu, B. Yuan, Y. Yue, M. Zhao, R. Luo, H. Wu, L. Wang, Y. Zhang, J. Xiao, F. Lin, Effect of autoinducer-2 quorum sensing inhibitor on interspecies quorum sensing, Front. Microbiol. 13 (2022) 791802, https://doi.org/10.3389/fmicb.2022.791802.
[93] H. Li, X. Li, Z. Wang, Y. Fu, Q. Ai, Y. Dong, J. Yu, Autoinducer-2 regulates *Pseudomonas aeroginosa* PAO1 biofilm formation and virulence production in a dose-dependent manner, BMC Microbiol. 15 (2015) 192, https://doi.org/10.1186/s12866-015-0529-y.
[94] X. Yan, S. Gu, X. Cui, Y. Shi, S. Wen, H. Chen, J. Ge, Antimicrobial, anti-adhesive and anti-biofilm potential of biosurfactants isolated from *Pediococcus acidilactici* and *Lactobacillus plantarum* against *Staphylococcus aureus* CMCC26003, Microb. Pathog. 127 (2019) 12–20, https://doi.org/10.1016/j.micpath.2018.11.039.
[95] P. Klintham, S. Tongchitpakdee, W. Chinsirikul, W. Mahakarnchanakul, Combination of microbubbles with oxidizing sanitizers to eliminate *Escherichia coli* and *Salmonella Typhimurium* on Thai leafy vegetables, Food Control 77 (2017) 260–269, https://doi.org/10.1016/j.foodcont.2017.02.030.
[96] T.A.M. Rovers, Microbubble Stability and Applications in Food (Ph.D. thesis), Wageningen University, Wageningen, NL, 2015, p. 138, ISBN: 978-94-6257-475-5.
[97] T.A.M. Rovers, G. Sala, E. van der Linden, M.B.J. Meinders, Potential of microbubbles as fat replacer: effect on rheological, tribological and sensorial properties of model food systems, J. Texture Stud. 47 (2016) 220–230, https://doi.org/10.1111/jtxs.12175.

[98] H. Zhang, R.V. Tikekar, Air microbubble assisted washing of fresh produce: effect on microbial detachment and inactivation, Postharvest Biol. Technol. 183 (2021) 111687, https://doi.org/10.1016/j.postharvbio.2021.111687.

[99] C.G. Otoni, S.F.O. Pontes, E.A.A. Medeiros, N.F.F. de Soares, Edible films from methylcellulose and nanoemulsions of clove bud (*Syzygium aromaticum*) and oregano (*Origanum vulgare*) essential oils as shelf life extenders for sliced bread, J. Agric. Food Chem. 62 (22) (2014) 5214–5219, https://doi.org/10.1021/jf501055f.

[100] K.K. Thi Phan, T. Truong, Y. Wang, B. Bhandari, Nanobubbles: fundamental characteristics and applications in food processing, Trends Food Sci. Technol. (2019), https://doi.org/10.1016/j.tifs.2019.11.019.

[101] H.F. Khalid, B. Tehseen, Y. Sarwar, S.Z. Hussain, W.S. Khan, Z. Ali Raza, S. Zafar Bajwa, A.G. Kanaras, I. Hussain, A. Rehman, Biosurfactant coated silver and iron oxide nanoparticles with enhanced anti-biofilm and anti-adhesive properties, J. Hazard. Mater. 364 (2019) 441–448, https://doi.org/10.1016/j.jhazmat.2018.10.049.

[102] M. Arif, S.M. Samreen, S. Khan, Z. Chi, C.-G. Liu, Chitosan-based nanoparticles as delivery-carrier for promising antimicrobial glycolipid biosurfactant to improve the eradication rate of *Helicobacter pylori* biofilm, J. Biomater. Sci. Polym. Ed. 32 (6) (2021) 813–832, https://doi.org/10.1080/09205063.2020.1870323.

CHAPTER

14 Applications of biosurfactant as solubilizers and wetting agents

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1 Introduction

Biosurfactants or natural surfactants are a group of microbial molecules containing both hydrophobic (nonpolar) and hydrophilic (polar) components. The polar part of biosurfactants may constitute amino acid, carbohydrate, or a phosphate group while the nonpolar part is made by the long-chain saturated or unsaturated fatty acid [1–4]. The presence of two opposite polarities in the same molecule imparts amphiphilic nature to the surfactant molecule (see Fig. 1) [1]. There are two important roles of biosurfactant; firstly, they can lower the interfacial surface tension between oil-aqueous or air-aqueous phases and thus capable of transferring the hydrophobic organic compounds to the mobile phase. Secondly, they can form micelles, which assist in solubilizing hydrophobic organic compounds [5]. Biosurfactants can be produced from a wide range of microorganisms including bacteria, fungi, and yeast. Various microorganisms such as *Bacillus salmalaya*, *Candida lipolytica*, *Pseudomonas aeruginosa*, *Saccharomyces lipolytica*, etc., are used to produce biosurfactant [6–10]. Some examples of biosurfactants and their origin are listed in Table 1 [7].

Biosurfactants have several advantages over chemical surfactants. Various properties such as their eco-friendly nature, effective at low concentrations, low-temperature production, less toxicity, highly selective, tolerance to extreme conditions give them edge for various industrial applications [11]. Although their utility is restricted because of their high production cost, less effective microbial strains, and low yields. The most important property of biosurfactants is to reduce the interfacial tension between two immiscible liquids and formation of micelle, which advocates their use as wetting agents and solubilizers [12]. Many industrial applications need the use of wetting agents and solubilizers. Especially the food industry, pharmaceutical industry, paint industry, and cosmetic industry are highly dependent on

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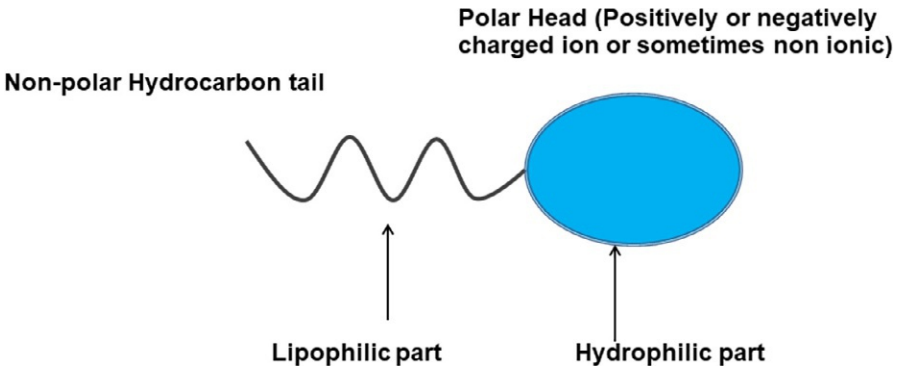

FIG. 1

Structure of a surfactant molecule [1].

Table 1 Examples of biosurfactants and their origin [7].

Types of biosurfactant		Source of microorganism	Type of microorganism
Group	**Class**		
Glycolipids	Rhamnolipids	*Pseudomonas aeruginosa, Burkholderia* sp., *Pseudomonas* sp.	Bacteria
	Sophorolipids	*Torulopsis bombicola, T. apicola*	Yeast
	Trehalolipids	*Rhodococcus erythropolis, Mycobacterium* sp., *Arthrobacter* sp., *Nocardia* sp., *Corynebacterium* sp.	Bacteria
	Mannosyl erythritol lipids (MEL)	*Candida antarctica, Pseudozyma* sp.	Yeast
	Monoacylglycerol	*Candida ishiwadae*	Yeast
	Cellobiolipids	*Candida antartica*	Fungi
	Glucolipid		Bacteria
	Polyol lipids	*Rhodotorulaglutinis, Rhodotorulagraminis*	Yeast
	Flocculosin	*Pseudozymaflocculosa*	Fungi
	Xylolipid	*Lactococcuslactis*	Bacteria
	Lipopolysacharide	*Klebsiellaoxytoca*	
	Glucose, Fructose Sucrose-lipids	*Arthrobacterparaffineus, Corynebacterium, Nocardia* and *Brevibacterium*	Bacteria
	Diglycosyl diglycerides	*Lactobacillus fermenti*	Bacteria

Table 1 Examples of biosurfactants and their origin [7]. *Continued*

Types of biosurfactant		Source of microorganism	Type of microorganism
Group	**Class**		
Lipopeptides and lipoproteins	Surfactin	*Bacillus subtilis*	Bacteria
	Iturin A	*Bacillus subtilis*	Bacteria
	Fengycin	*Bacillus subtilis*	Bacteria
	Lichenysin	*Bacillus licheniformis*	Bacteria
	Viscosin	*Pseudomonas fluorescens*	Bacteria
	Serrawettin	*Serratia marcescens*	Bacteria
	Artrofactin	*Arthrobacter* sp.	Bacteria
	Polymyxin	*Bacillus polymyxa*	Bacteria
	Antibiotic TA	*Myxococcus xanthus*	Bacteria
	Subtilism	*Bacillus subtilis*	Bacteria
	Gramicidin	*Brevibacterium brevis*	Bacteria
	Ornithine	*Pseudomonas* sp.	Bacteria
	Arthrofactin	*Arthrobactor* sp.	Bacteria
	Pumilacidin	*Bacillus pumilus*	Bacteria
	Peptide lipid	*Bacillus licheniformis*	Bacteria
Polymeric surfactants	Emulsan	*Acinetobacter calcoaceticus* RAG-1, *Arthrobacter calcoaceticus*	Bacteria
	Liposan	*Candida lipolytica*	Yeast
	Biodispersan	*Acinetobacter calcoaceticus* A2	Bacteria
	Lipomanan	*Candida tropicalis*	Yeast
	Mannoproteins	*Saccharomyces cerevisiae*	Yeast
	Alasan	*Acinetobacter radioresistens* KA53	Bacteria
	Polysaccharide protein complex	*Acinetobacter calcoaceticus*	Bacteria
	Protein PA	*Pseudomonas aeruginosa*	Bacteria
	Mannan-lipid-protein	*Candida tropicali*	Yeast
	Carbohydrate-protein-lipid	*Pseudomonas fluorescens*	Bacteria
Fatty acids, phospholipids, and neutral lipids	Spiculisporic acid	*Penicillium spiculisporum*	Fungi
	Phosphatidylethanolamine	*Acinetobacter* sp., *Rhodococcus erythropolis*	Bacteria
	Fatty acids	*Corynebacterium lupus*	Bacteria
	Neutral lipids	*Nocardia erythropolis*	Bacteria
	Corynomycolic acid	*Penicillium*	Fungi
	Oleic acid	*Issatchenkiaorientalis*	Yeast
	Phospholipids	*Acidithiobacillusthiooxidans*	Bacteria

Continued

Table 1 Examples of biosurfactants and their origin [7]. *Continued*

Types of biosurfactant		Source of microorganism	Type of microorganism
Group	**Class**		
Particulate biosurfactants	Vesicles	*Acinobactor calcoaceticus, Pseudomonas marginalis*	Bacteria
	Whole microbial cells	*Cyanobacteria*	–
	Bioemulsan	*Gordinia* sp.	Bacteria

wetting agents and solubilizers [13–22]. Their versatile nature and multifunctional applicability make them smart molecules for the future generation (Fig. 2) [11]. In this chapter, the uses of biosurfactants as wetting agents and solubilizers have been explored in the direction of various industrial applications. Their working mechanism and future potential in various sectors are also discussed.

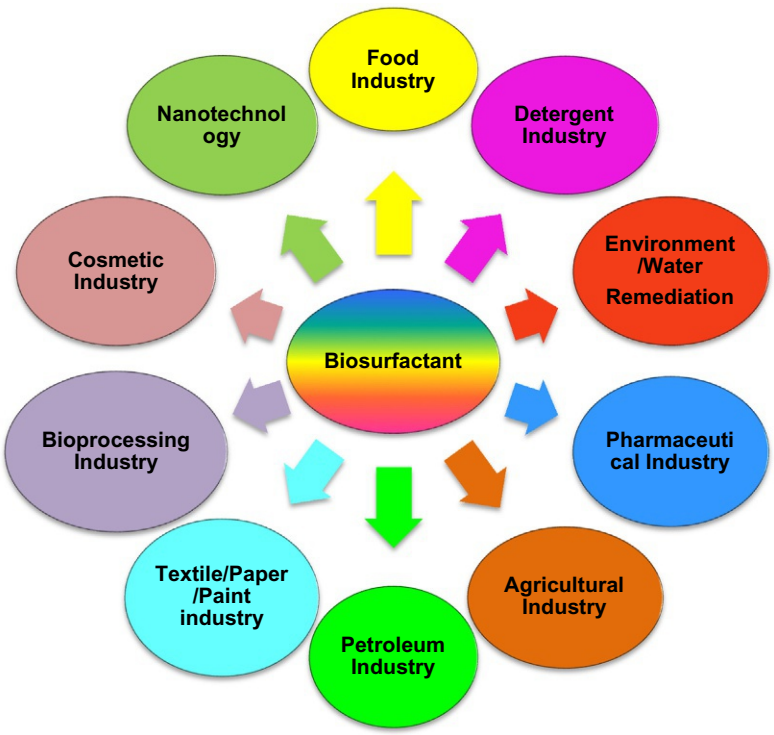

FIG. 2

Versatile application of biosurfactants [11].

2 Types of biosurfactants

The classification of biosurfactants is based on their microbial origin or chemical composition. According to chemical structure, they can be grouped under the category of fatty acids, antibiotics, peptides, glycolipids, phospholipids, lipopeptides, etc. [1,2]. Microbial biosurfactants are classified into two types; where type 1 belongs to the low-molecular-weight biosurfactants, e.g., glycolipids, lipopeptides, phospholipids, etc., which are very effective against reducing surface and interfacial tensions; and type 2 belongs for the high-molecular-weight polymeric and particulate surfactants, which exhibit emulsion-stabilizing properties [3–5].

The amphiphilic biosurfactants contain hydrophilic and hydrophobic moiety in a single molecule. The hydrophobic part is mostly made with long-chain fatty acids and their derivatives, whereas the hydrophilic moiety may include ester, carbohydrates, amino acids, phosphates, peptides, etc. [6]. Based on structure, the biosurfactants are classified into five groups (Fig. 3), which are discussed in the following section [7,8].

2.1 Glycolipids

Glycolipids are the most studied biosurfactants belonging to low-molecular-weight class. They consist of carbohydrates connected with long-chain fatty acids through either ester or ether group [7,8,10]. The best known glycolipids are rhamnolipids, trehalolipids, and sophorolipids. In rhamnolipids, one or two rhamnoses are joined with one or two β-hydroxydecanoic acids. The –OH group of one acid forms glycosidic linkage with the rhamnose disaccharide, the other –OH group on another acid forms ester. The principle source of glycolipid production is *Pseudomonas aeruginosa* [7]. Trehalolipids are formed by linking disaccharide trehalose to mycolic acid. The structure of trehalolipids is decided by the size of mycolic acid, number of carbon atoms, and the degree of unsaturation. The sophorolipids are mainly produced from *Torulopsis* sp. The structure of sophorlipids contains dimeric sophorose attached with long-chain hydroxyl fatty acid by glycosidic linkage [10]. Sophorolipids exist as macrolactones as well as free acid forms although the lactone form is more suitable for various applications. Few well-known glycolipids are shown in Fig. 4.

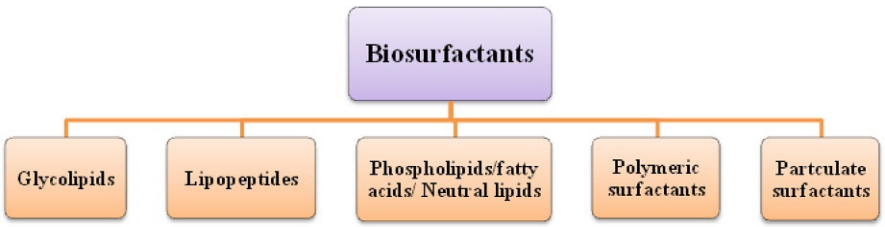

FIG. 3

Classification of biosurfactants.

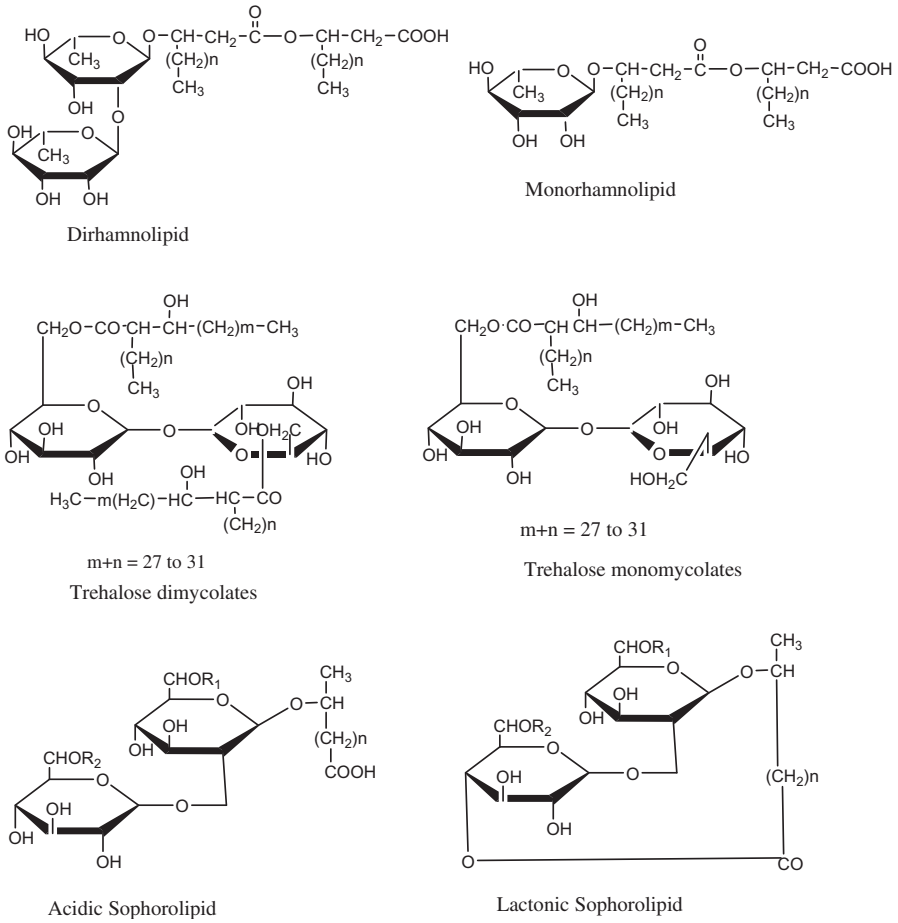

FIG. 4

Few examples of glycolipids.

2.2 Lipopeptides and lipoproteins

These are cyclic compounds containing lipid attached to a polypeptide chain. Surfactin and lichenysin are two important examples in this class. Surfactin contains β-hydroxyl fatty acids and D-/L-amino acid residues and shows superior surface activity [1,2]. Seven amino acids are joined in a ring and coupled with a fatty acid chain by a lactone linkage. The origin of this cyclic lipopeptide is *Bacillus* sp. Lichenysin is structurally similar to surfactin and shows similar physiochemical properties [8]. It is produced from *Bacillus licheniformis* and acts excellently to reduce surface and interfacial tension. The structure of surfactin and lichenysin is depicted in Fig. 5.

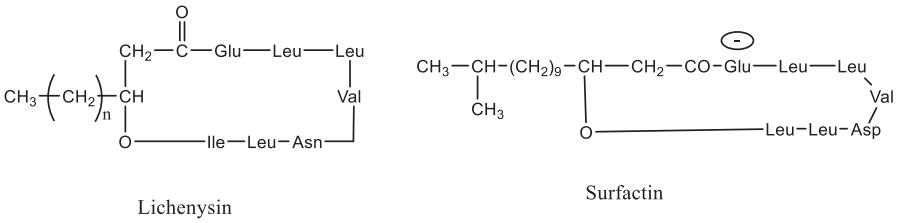

FIG. 5

Structure of lichenysin and surfactin.

2.3 Fatty acids, phospholipids, and neutral lipids

Several microorganisms are known to produce fatty acids and phospholipid surfactants during growth on n-alkanes. Phosphati-dylethanolamine surfactant is produced from *Acinetobacter* sp., and *Rhodococcus erythropolis* has very good surface-active properties [4]. They are excellent in reducing interfacial surface tension and are associated with low critical micelle concentration (CMC) values.

2.4 Polymeric biosurfactants

They belong to high-molecular-weight biosurfactants. Their backbone consists of three or four repeating sugars with attached fatty acids. Some well-known polymeric biosurfactants are emulsan, liposan, alasan, etc. Liposan is synthesized from *Candida lipolytica* consisting of 83% carbohydrate and 17% protein and acts as water-soluble emulsifier [7]. Emulsan has average molecular weight of 1000 KD and is an acylated polysaccharide. It is synthesized using *Acinetobacter calcoaceticus* bacterium. The backbone of emulsan structure consists of unbranched polysaccharide with *O*-acyl and *N*-acyl-linked fatty acid side chains. The 1:1:1 ratio of three aminosugars, e.g., D-galactosaminouronic acid, D-galactosamine, and dideoxydiaminohexose, is the constituent of the polysaccaride backbone. The fatty acid side chains have 10–22 carbon atoms, and the amino groups are either acetylated or covalently linked by an amide group, which is linked to 3-hydroxybutyric acid [8]. Emulsan (Fig. 6) acts as a very effective emulsifier at a concentration as low as 0.01%. Alasan, composed of alanine, polysaccharide, and proteins, is synthesized from *Acinetobacter radioresistens*. It is used for solubilization of aromatic hydrocarbons, crude oil, and paraffins.

2.5 Particulate biosurfactants

Particulate biosurfactants are classified into two categories: (1) extracellular vesicles and (2) microbial cell. Extracellular membrane vesicles are produced from *Acinetobacter* sp. and grown on hexadecane [7]. They consist of proteins, phospholipids, and lipopolysaccharides. The vesicles play important role in hydrocarbon uptake.

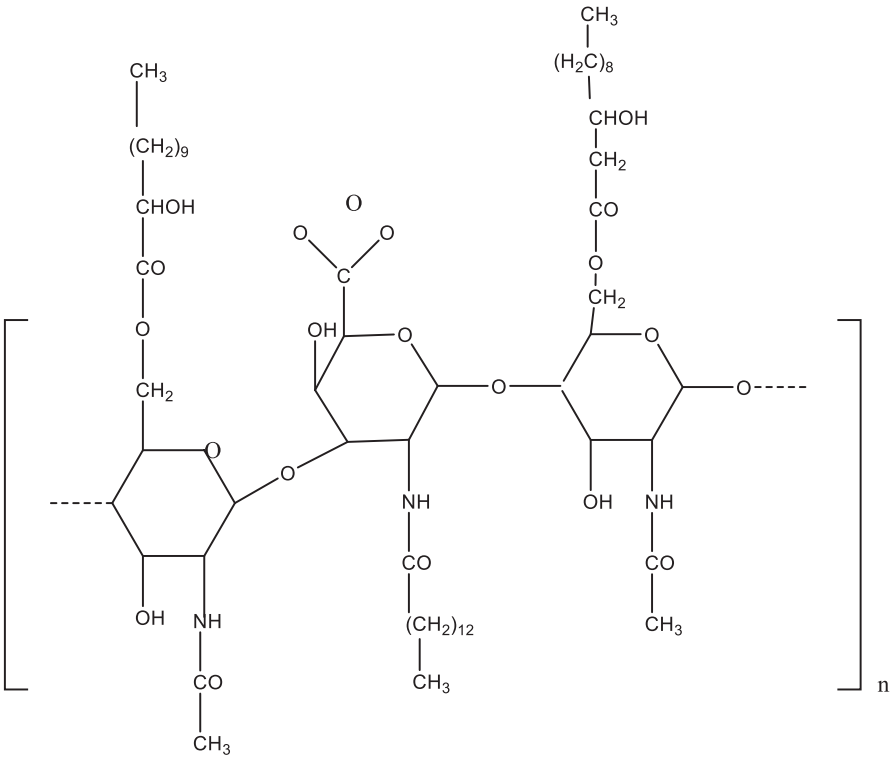

FIG. 6

Structure of emulsan.

Pseudomonas marginalis also form vesicles to work as surfactants. The whole microbial cell acts as a surfactant. These microbial cells have a strong affinity for hydrocarbon-water and air-water interfaces.

3 Properties of biosurfactants

The biosurfactants have some unique and distinct properties compared with their chemically synthesized counterparts. The characteristic features of biosurfactants include their surface activity, biodegradability, emulsifying and demulsifying ability, tolerance to extreme condition of pH, temperature, and ionic strength, low toxicity, etc., made them very effective for commercial application [1,8]. The distinctive features of biosurfactant are briefly discussed below.

3.1 High biodegradibility and low toxicity

Chemically synthesized surfactants are of environmental concern for their nonbiodegradable nature. Alternatively microbial surfactants or biosurfactants are easily degraded by the microorganisms from the ecosystem. The harmful and toxic nature of chemical surfactants limit their use in various applications. On the other hand, biosurfactants are mostly recognized as low or nontoxic substances and are more accepted specially by the food sector, pharmaceutical processing, and cosmetic industries [14,15]. Biosurfactants have shown promising results in removing hydrophobic organic pollutants from aquatic sources.

3.2 Surface tension and interfacial properties

Biosurfactants have the interesting property of lowering both surface and interfacial tension of two heterogeneous phases [16,18]. The standard condition to assume a surfactant to be good when it can reduce the surface tension of water from 72 to 35 mN/m and interfacial tension of water-hexadecane suspension system from 40 to 1 mN/m [23]. Biosurfactants such as surfactins, rhamnolipids, etc., fulfill these criterion and are considered as good biosurfactants. An effective and efficient biosurfactant possesses lower CMC and high surface activity than its chemical counterpart and has the advantage of requirement of less amount. Table 2 contains some selected biosurfactants with their surface tension and CMC values [7,16].

3.3 Environmental stability in extreme conditions

Biosurfactants are extremely stable toward environmental harshness. They remain unaffected by change in temperature, pH, and salinity. Surfactant lichensin produced from *Bacillus licheniformis* is stable in pH range of 4–9, salt concentration of 25–50 g/L, and in a wide range of temperatures [23]. A microbial surfactant derived from *Arthrobactor prophormiae* can withstand temperature from 30°C to 100°C and pH range of 2–12. Many industrial processes require stability of the biosurfactant in a wide temperature, pressure, and pH range where biosurfactants can function effectively and exhibit highly promising compounds.

3.4 Specificity

Biosurfactants are large, complex organic molecules having specific functional groups. Thus they are very specific in their action, which are of great interest in remediation of specific pollutants, de-emulsification, production of specific cosmetic, pharmaceutical, and food industries. They also have superior ability to form molecular assembly and liquid crystal.

Table 2 Surface tension [16] and CMC [7] values of some selected biosurfactants.

Biosurfactant	Source of microorganism	Surface tension (mN/m)	CMC (mg/L)
Rhamnolipids	*Pseudomonas aeruginosa*	29	20
Trehalolipids	*Rhodococcus* sp.	36	–
Sophorolipids	*Torulopsis bombicola*	33	–
Peptide-lipid	*Bacillus licheniformis*	27	–
Serrawettin	*Serratia marcescens*	33	–
Viscosin	*Pseudomonas fluorescens*	26.5	–
Surfactin	*Bacillus subtilis*	27–32	11
Emulsan	*Acinetobacter calcoaceticus*	32	–
Mannan-lipid-protein	*Candida tropicalis*	30	–
Liposan	*Candida lipolytica*	29	–
Carbohydrate-protein-lipid	*Mycobacterium* sp.	27	–
Phosphatic acid		–	70
Phosphatidylethanolamine	*Acinetobacter* sp., *Rhodococcus erythropolis*	–	30
Alkylbenzene sulfonate	–	–	590
Sodium lauryl sulfate	–	–	2000–2900

4 Biosurfactants as wetting agents

Various industries exploit the wetting agents for the developments of materials and products. In textile industry, wetting process is a very important phenomenon as the fibers need a pretreatment in aqueous solution. The fiber needs to get wetted uniformly within a short span of time. Wetting is also required for flow and recovery of petroleum oil, dye and paint industry, detergent industry, pharmaceutical industry, etc. Drugs are needed to be water-soluble for absorption in vivo. Biosurfactants increase the thermodynamic activity in addition to the improved dissolution of the therapeutic molecules.

Decreases of interfacial tension between two immiscible phases allow them to interact more easily. The reduction of interfacial tension by biosurfactants is based on the special property of formation of micelles or aggregations of these molecules. The formation of micelle starts at a concentration known as CMC. At CMC, the surfactants can reduce the interfacial tension very effectively. Once CMC is reached, the solubility and bioavailability of hydrophobic organic compounds are maximized. Lower is the CMC, better is the efficiency of the biosurfactants (Fig. 7) [4]. The solubility of a surfactant in water medium as well as the assessment of surfactant

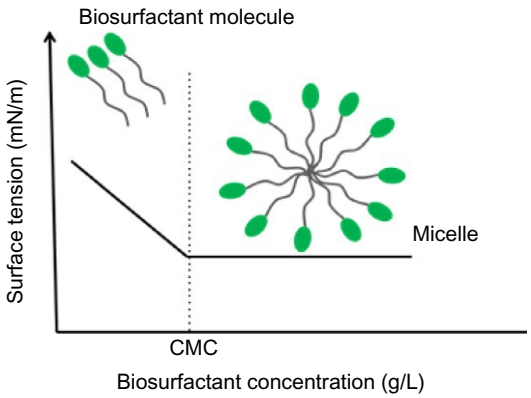

FIG. 7

Micelle formation and CMC of a biosurfactant [4].

Table 3 Application of biosurfactants and their HLB values.

HLB	Applications
1–3	Antifoaming agent
4–6	Water-in-oil emulsifier
7–15	Wetting agent
8–18	Oil-in-water emulsifier
10–15	Detergent
10–18	Solubilizer

activity is measured by CMC. A good surfactant possesses two important qualities: one is efficiency, which is determined by CMC, and the other one is effectiveness, which is related to the surface and interfacial tensions. Another quantity known as hydrophilic-lipophilic balance (HLB) is used to determine a surfactant effectiveness [8]. HLB number ranges from 0 to 20 and decides whether a biosurfactant will form water-in-oil or oil-in-water emulsion. The biosurfactants having low HLB value are lipophilic in nature and prone to form water-in-oil emulsion, whereas high HLB valued biosurfactants have affinity for water (hydrophilic), and they form oil-in-water emulsion [14,19]. Further, ionic surfactants are reported with HLB values of 60 [21]. Thus the HLB numbers are referred for determining the applicability of a biosurfactant in a particular field. Table 3 enlists some application areas of biosurfactants depending on the HLB values.

4.1 Theory of wetting

Wetting phenomenon is generally considered as the ability of a fluid to spread over a surface. The surface may be a solid or a liquid with which the drop forms a heterogeneous phase. More often it is applied for the displacement of air from a liquid

or from a solid surface by water. The degree of wetting is known as wettability, which is dependent on the nature of the drop as well as the surface. The ratio of cohesive force and adhesive force is responsible for the wetting process. The adhesive force between liquid drop and the surface favors wetting, while the cohesive force inside the drop works in opposition to wetting [12,19]. Because of very high surface tension of water (72 mN/m), the spreading of water on covalent solids is not a spontaneous. Wetting takes place when the surface tension of water becomes less than the free energy of the solid surface. Wetting does not take place when the surface tension of the liquid is higher compared with the surface energy of the solid molecules. Wetting agents can increase the spreading ability of a liquid by reducing the surface tension of liquid so that it can spread over the surface. Because of the low surface tension, the liquid drop spreads easily since less energy is required for spreading. In this process, the cohesive force gets weakened while adhesive force gets strengthened. The wetting agents work through the formation of micelles. The hydrophilic part of micelles forms outer layer and surrounds water while the hydrophobic part is attracted toward the oil (Fig. 8) [19]. Thus the process of wetting consists of three consecutive steps of adhesion, spreading, and immersion [19,24].

A liquid drop after falling on a horizontal solid surface reaches equilibrium assuming a shape that minimizes the total free energy of the system. The contact angle and surface tension at this stage are given by Young's equation [12].

$$\gamma_{SV} = \gamma_{SL} + \gamma_{LV} \times \mathrm{Cos}\ \theta_e$$

Where γ_{SV} is the surface tension of solid-vapor phase, γ_{SL} is the interfacial tension of solid-liquid phase, γ_{LV} is the liquid-vapor surface tension, and θ_e denotes the contact angle at the solution interface. The spreading is favored on high-energy surfaces as surface free energy always tries to spread the liquid drop. $\gamma_{LV}\,\mathrm{Cos}\,\theta_e$ is the horizontal component of the surface tension. γ_{LV} acts in opposite direction of the interfacial tension γ_{SL}. At equilibrium the resultant force becomes zero. A liquid drop on a solid surface tries to spread so that the liquid-solid and liquid-vapor interfacial areas increase. At the same time, solid-vapor interfacial area reduces and also the contact angle between the drop and the solid surface decreases. For a low θ value, the water

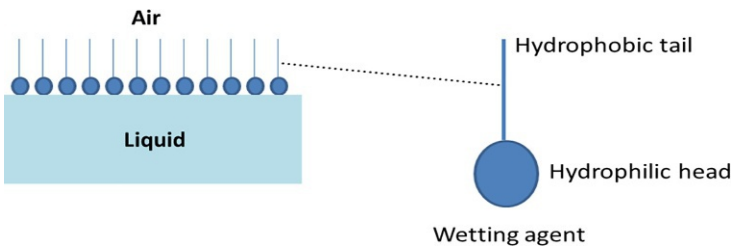

FIG. 8

Biosurfactants at the interface of air and water [19].

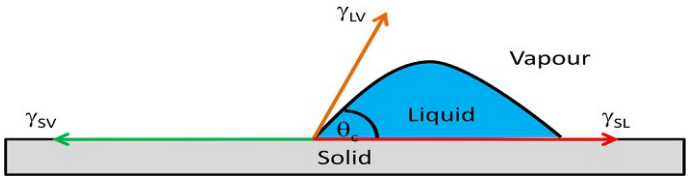

FIG. 9

Relationship between the contact angle and solid-liquid, solid-vapor, and liquid-vapor surface tension [24].

drop spreads readily on the solid surface indicating as a hydrophilic surface (Fig. 9), while for a hydrophobic surface, θ is near to 90 degree or more. For a super hydrophobic surface, the contact angle is highly obtuse [24].

The spreading coefficient S_{SL} is often used to understand the spreading phenomena. It is defined as

$$S_{SL} = \gamma_{SV} - \gamma_{SL} - \gamma_{LV}$$

For a spontaneous spreading to occur, S_{SL} must have a positive value [12]. The addition of surfactant reduces the surface tension of the liquid (γ_{LV}) and the interfacial tension (γ_{SL}) between the solid and the liquid. Thus the spreading coefficient becomes more positive and spreading process becomes feasible.

5 Biosurfactants as solubilizers

Solubilizers are key ingredients in personal care and cosmetic products such as shower gels, fragrances, or tonics [7,8]. They are used to provide a clear and transparent solution by incorporating a lipophilic component in terms of caring oils or fragrances. The ideal formulation of cosmetic applications depends on right amount of solubilizers and the nature of the added lipophilic substances. This process is also important in agriculture, food formulation, and bioremediation process [14].

Solubilization is the process by which a thermodynamically stable isotropic solution of a substance can be prepared by the addition of an amphiphilic compound (surfactant). Without the introduction of an amphilic compound, the substance in normal condition is insoluble or very slightly soluble in the given solvent. The surfactant is introduced at or above the CMC [18].

Solubilization and emulsification, inherently correlated as solubilization, are nothing but formation of a clear emulsion [2]. For cleaning an oil or stabilizing a formulation, an emulsion is necessary. An emulsion can be described as a stable mixture of two or more liquids. The emulsion contains two phases: a continuous phase and a dispersed phase. The oil-in-water system droplets of oils (dispersed phase) are suspended over water, which is the continuous phase [25] while for water-in-oil system, the phases are just reversed. The orientation of the head and tail of the emulsifier changes according to the emulsion type [26](Fig. 10). The oil-in-water emulsions are extensively used in polishers and waxes, fabric softeners, degreasers, whereas the water-

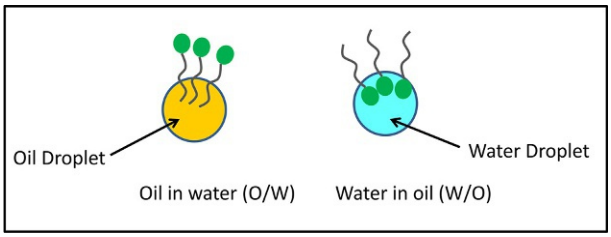

FIG. 10

Types of emulsion: oil-in-water and water-in-oil emulsion [26].

in-oil emulsions are used in aerosol polishes, insecticides, etc. A proper match between HLB values of the surfactant and the target molecule for emulsification is necessary for optimum performance [17,18]. For the formulation of air care products (oil-based perfumes in water medium), furniture care products (surface protection and shining substances, e.g., oils and waxes), etc., are based on HLB values of emulsion technology.

5.1 Bioremediation or environmental application

In environmental application, biosurfactants have shown potential roles in remediation of water, sediments, and soil washing. It is also extensively used in bioremediation of petroleum pollutants. Recovery of oil, reducing oil viscosity for better flow, oil tank cleaning are few properties where biosurfactants are employed in petroleum industry. Biosurfactants obtained from *Bacillus licheniformis* Y-1 are a great emulsifier for crude oil [26]. A rhamnolipid produced from *Pseudomonas aeruginosa* F-2 exhibits good prospect for oil recovery [27].

Polycyclic aromatic hydrocarbons (PAHs) are potential human health hazards. They have ubiquitous existence and are extremely hydrophobic in nature. The poor solubility in aqueous media is the primary reason of their polluting effect of soil and water resources. They have high adsorption ability and low bioavailability. Biosurfactants increase the surface area by reducing the surface tension and interfacial tension of the hydrophobic compounds leading to enhanced bioavailability and mobility. When the concentration of biosurfactant crosses CMC value, the aqueous solubility of PAHs is increased and removed from the source [28]. PAHs also show excellent bioconversion in the presence of biosurfactants because of increased bioavailability and mass transfer rates to cells. Wang et al. have shown that anionic surfactants as well as nonionic surfactants can improve the wetting of coal dust surface significantly [24]. The surfactant-enhanced bioremediation (SER) is becoming a promising method for the removal of hydrophobic contaminants from the polluted groundwater and soil environments. Mao et al. have reported the enhancement of solubility of contaminants using biosurfactants in soil washing technique [29]. The mechanism of interaction of soil particles with biosurfactant during bioremediation process has been proposed [30]. It is reported that mixed surfactants are better for the solubilzation of organic pollutants compared with the single and pure

surfactant. A mixture of cationic-nonionic binary surfactant was found to be more effective for removal of PAHs in comparison with single cationic or nonionic surfactant. A binary mixture of rhamnolipid (RL) and sophorolipid (SL) exhibited greater solubilization efficiency than the individual glycolipids. The RL-SL blends act synergistically to improve the solubility of PAHs. Temperature enhances the solubilization capacity of PAHs while it is diminished with increase of salinity [31]. Tea saponine, a natural biosurfactant, has shown immense potential in removing various hydrophobic organic compounds such as PAH, PBDE, PCB, and heavy metals such as Pb and Ni from electronic waste [32]. Metal remediation process is slightly different from the hydrocarbon remediation process. As the metals are not biodegradable, they are immobilized or removed by transforming into different chemical state by redox reactions or some chemical processes. The biosurfactant aids the desorption of heavy metals from the solid soil surface or aqueous phase by reducing the interfacial tension. Rhamnolipids and surfactin are the most popular biosurfactants used for heavy metal remediation. Some important biosurfactants used for bioremediation process are listed in Table 4 [33,34].

5.2 In food technology

In recent years, biosurfactants are gaining a lot of interest in food technology. Due to nonbiodegradable nature and toxicity associated with the chemically synthesized surfactants, they are avoided for food processing. The distinct structural and functional properties of the microbial surfactants made them ubiquitous in food

Table 4 Application of biosurfactants in bioremediation process [33,34].

Biosurfactant	Origin	Function	Application
Glycolipid	*Nocardiopsis* sp., *Arthrobacter* sp., *Corynebacterium* sp., *R. wratislaviensis* BN38	Emulsifier, solubilizer and dispersant	Biodegradation of hydrocarbons in soil and marine environment
Rhamnolipid	*Pseudomonas aeruginosa*	De-emulsifier	Oil recovery
Rhamnolipid	*Pseudomonas aeruginosa*	Metal sequestering agent	Mining
Sophorolipid	Candida bombicola	Oil mobilization	Enhanced oil recovery
Lipopeptide	*Bacillus licheniformis, Bacillus subtilis, N. alba strain* MSA10	Bioremediation agent	Oil recovery, removal of heavy metals from soil, sediment and water
Polymeric biosurfactants	*Saccharomyces cerevisiae Candida lipolytica*	Emulsifier	Stabilization of hydrocarbon emulsion

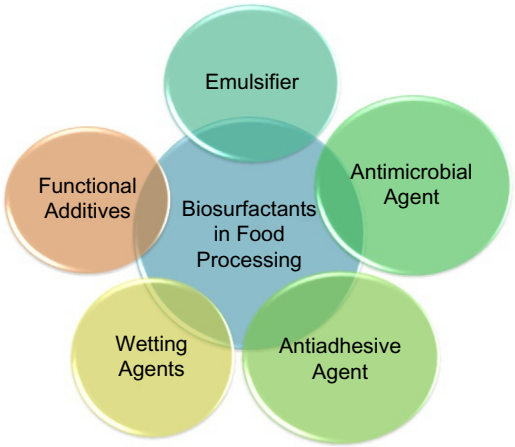

FIG. 11

Role of biosurfactants in food formulation [25].

formulation. They perform various roles as food additives (Fig. 11) [25]. The properties such as foaming agent, emulsifier, solubilizer, dispersants, wetting agent, antimicrobial agent, etc., are exploited in food processing [34,35]. As food additives, they make the food alluring in appearance, taste, and smell as well as maintain the nutritional value and safety of the food. Recent applications of biosurfactants in food processing industries are listed in Table 5.

5.2.1 Role as emulsifier

The biosurfactants are extensively used as emulsifier because of their ability to reduce the surface tension between two immiscible phases and help in mixing. The food texture and consistency are dependent on the quantity and quality of the emulsifier [55,56]. The role of emulsifier in food formulation is to improve the stability by lowering interfacial tension and preventing agglomeration of the heterogeneous phases. Low-molecular-weight surfactants are utilized in the stability of emulsions of beverages, sauces, alcoholic drinks, dressings, etc. For the improvement in the creaminess and cheesiness in salad dressing, oil-in-water type emulsions are used. The important functionalities of a microbial surfactant are to provide stabilization, improvement in the consistency, modification of rheological properties, and enhancement of shelf life. High-molecular-weight surfactant, e.g., liposan, is more effective emulsifier compared with low-molecular-weight sophorolipids. *Bacillus subtilis*, a lipopetide-producing strain, is found to be excellent emulsifying agent for various edible oils [57]. The surfactant produced from *Bacillus subtilis* has shown better prospect as emulsifier compared with the commercially available emulsifier [47]. Mnif et al. have shown that the addition of lipopeptide on bread has improved texture, reduced the rate of microbial proliferation [45]. Bioemulsifier generated from *Candida Utilis* has high potential in food formulation. It has been shown

Table 5 Application of biosurfactants in food processing industries.

Biosurfactants	Target	Application	Ref.
Surfactin/Iturin A	*Bacillus cereus*	Antiadhesive	[36]
Surfactin	*Salmonella enteritidis*	Antimicrobial	[37]
Rhamnolipids and surfactin	*S. aureus*, *Salmonella enteritidis*, and *Listeria monocytogenes*	Antibiofilm	[38]
Surfactin and rhamnolipids	*L. monocytogenes, P. fluorescens*	Antiadhesive	[39]
Surfactin	*Salmonella Typhi, L. monocytogenes, Staphylococcus aureus, B. cereus*	Antimicrobial nanoemulsion	[40]
Rhamnolipids	*L. monocytogenes*	Antimicrobial	[41]
Sophorolipids	Fruits	Preservation	[42]
Rhamnolipids	Fruits and vegetables	Cleaning agent	[43]
B. subtilis MTCC 441	Edible oils	Emulsifier	[44]
Lipopeptide	Bread	Emulsifier	[45]
C. utilis BS	Mayonnaise	Emulsifier	[46]
B. subtilis SPB1	Cookies	Emulsifier	[47]
Xylolipid	*L. monocytogenes, E. coli, B. cereus*	Antiadhesive	[48]
Bacillus subtilis	*L. monocytogenes*	Antibiofilm	[49]
Lactobacillus plantarum CFR 2194	*Escherichia coli*, *Salmonella typhi, Staphylococcus aureus*	Antimicrobial activity	[50]
Sophorolipids	*E. coli*, *Pseudomonas aeruginosa*	Antibiofilm	[51]
Sophorolipids	*E. coli* O157:H7	Antimicrobial activity	[52]
Bacillus licheniformis	Methicillinresistant *Staphylococcus aureus*	Antibiofilm	[53]
Bacillus sp.	*Listeria monocytogenes, Staphylococcus aureus*	Antimicrobial activity	[54]

that incorporation of *Candida Utilis* biosurfactant keeps a mayonnaise product stable for 30 days without any toxic effect [46]. Further the biosurfactants are recently gaining interest for the formulation of micro and nanoemulsion. The antimicrobial properties of nanoemulsions are utilized for food preservation and decontamination of food processing equipment or surface. Surfactin-mixed nanoemulsion of cooking oils, e.g., sunflower, coconut, sesame, castor, groundnut oils, exhibits considerable antimicrobial activities [40]. Very recently, a nano biosurfactant produced from *P. aeruginosa* has shown promising potential in emulsifying property compared with the chemically synthesized surfactant [58]. Thus nanoemulsions can be exercised as safe and cheaper method in food processing.

5.2.2 Role as antimicrobial agent

Many biosurfactants exhibit antimicrobial activity toward yeast, bacteria, algae, virus, and fungi [23]. Lipopeptides from *Bacillus* strain are the most investigated antimicrobial agents. Das et al. have reported surfactin, fengycin, iturin, bacillomycins, and mycosubtilins produced from *B. subtilis* as antimicrobial agents [59]. 1:1 combination of surfactin and polylysin has significantly reduced the population of *salmonella enteritidis* in milk [37], but haemolytic activity of surfactin in red blood cell has restricted their widespread use in food. Rhamnolipids derived from the strain of *Pseudomonas aeruginosa* are also found to act as cleaning agents for fruits and vegetables and exhibit significant antimicrobial activity [41,43]. Biosurfactants from lactobacillus strain are reported to be less toxic and safe for food processing [60]. Sophorolipids originated from *Candida* species also demonstrated antimicrobial activity and are safe to use for sanitization of fruits and vegetables [61].

5.2.3 Role as antioxidant agent

The autoxidation of lipid produces toxic compounds, which degrade the food quality, safety, and flavor. Biosurfactants have the potential to act as antioxidant agents to prevent lipid oxidation to improve shelf life of the food products. Especially, mannosylerythritol lipids (MELs) produced from *Pseudozyma hubeiensis* are reported as versatile antioxidant agent to protect cells. Takahashi et al. have shown from their in vitro studies that MEL-C has good antioxidant property and has potential as antiaging product for skin care treatment [62]. The biosurfactant RW-I obtained from *B. subtilis* strain is known to scavenge free radicals and exhibits significant antioxidant properties [63]. A report by Chen et al. has shown that biosurfactant produced from Chinese medicinal herbs, *Alcaligenes piechaudii* CC-ESB2, is a good emulsifier and antioxidant [64].

5.2.4 Role as antiadhesive agent

Biosurfactants exhibit promising effect as antiadhesive agent to combat foodborne microbes. As the pathogens like to stay in communities, adhesion is the first step during the colonization process. Rossi et al. have studied pathogen-infected lettuce leaves wherein they reported *S. Enteritidis* SE86 to help in adherence and resistance to sanitizers [65]. With the stainless steel and Teflon surfaces when modified with iturin and surfactin, the adhesion of *Bacillus cereus* spores is diminished [36]. Two strains of *Lactobacillus* species are used to produce biosurfactants, which show optimistic antiadhesive properties against *Enterobacteriaceae* [66]. They have promising applications in fermented food sector for prevention of microbes adhesion to intestinal epithelial cells. Biosurfactant derived from *Lactobacillus plantarum* is proved to have both antiadhesive and antimicrobial activity against several food pathogens [50]. A xylose glycolipid originating from *Enterococcus faecium* can stop the adhesion of *E. Coli*, *L. monocytogenes*, and *B. Cereus* very effectively on a polystyrene-conditioned surface [48]. Araujo et al. have explored surfactin and rhamnolipid on steel and

polystyrene surfaces for antiadhesive antibiofilm formation by *L. monocytogenes* and *P. fluorescens* [39]. A report of Abdul et al. has shown that a blend of polyvenyl alcohol and a biosurfactant derived from *Lactobacillus rhamnosus* is effective against biofilm formation and diminished bacterial adhesion [67]. For the disruption of the biofilm produced by *L. monocytogenes,* surfactin was found to be more effective than rhamnolipids [38].

5.2.5 Role as antibiofilm agent

The development of biofilm on food products ruined the quality of the products and caused serious health issues. Thus their presence is a threat to food processing industries. Various microbial bacteria, for example, *Listeria monocytosgenes*, *Escherichia coli*, *Salmonella enteritidis*, etc., are associated with food transmission diseases [68]. Thus controlling biofilm formation is the major concern and challenge for delivering safe and quality food products to consumers. Here biosurfactants are of great help for the prevention and inhibition of biofilm formation [69]. Preconditioning the surface of stainless steel surface with *Pseudomonas fluorescens* surfactant has proved to reduce the adhered *Listeria* cells [70]. Nitschke et al. have studied the effect of rhamnolipids and surfactin on various food microbes for antiadhesive properties on stainless steel, polypropylene, and polystyrene surfaces [71,72]. Inhibition of biofilm formation as well as removal of Cd metal from contaminated vegetables is successfully done by surfactin obtained from *Bacillus* sp. MTCC 5877 and also by *B. licheniformis* VS16 [73,74]. The potential uses of biosurfactants as antibiofilm are exploited in dairy industry for food formulations. Dairy industry faces fouling problems in pipelines, heat exchanger, and ultrafiltration membranes. To reduce the fouling rate of heat exchanger plates, thermophilic biosurfactants from *Streptococcus thermophilus* are observed to act as antibiofilm coating. Biosurfactants derived from *Lactobacillus acidophilus* restrict the adhesion of microbes and stop colonization to act as functional food additive. A recent report by Kiran et al. has shown that a lipopeptide biosurfactant MSA31 obtained from marine sponge *F. cavermosa* is very promising in antibiofilm activity against food pathogen *S. aureus* [75]. It also exhibits good antioxidant activity as well as effective emulsification property, which can be highly beneficial to the food industry.

5.3 In cosmetic formulations

There is increasing demand for the natural, biocompatible, and safe cosmetics for human use. Thus chemical/synthetic surfactants are getting replaced by the biosurfactants in the cosmetic industries. The biousurfactants are associated with several properties, which are crucial for their use in cosmetic formulation. The wetting ability, solubilizing capacity, emulsifying property, dispersing activity, detergency, and foaming ability are the key criteria that are used for preparing various cosmetic products. Some of the properties are as follows:

Wetting: Biosurfactants act as wetting agents to disperse the surface-active molecules on the surface of the skin to remove the oil deposit. The dispersion process becomes easy when the contact angle between the solution and surface decreases with the help of biosurfactants. The wetting ability of the surfactants is utilized to formulate various lotions and cosmetics.

Foaming and detergency: The biosurfactants can form micelles and help in cleaning of oil deposits from the skin and hair. The detergency property is utilized for preparing shampoos, face washes, etc., for cleansing purposes.

Emulsification: Many cosmetic formulations are needed to be stabilized by formation of an emulsion. Face creams and body lotions are oil-based cosmetic formulations, which are beneficial for the skin and hair.

Among the biosurfactants, sophorolipid has multifunctional ability to act as cleanser, antimicrobial, and body odor removal. Thus it is extensively used in skin and hair care cosmetic products [76,77]. Rhamnolipids are also administered in cosmetic formulation of antiaging creams. The antimicrobial properties of rhamnolipids are used for the preparation of toothpaste and deodorants [62]. Various lipopeptide surfactants are used for skin treatments as antiwrinkle cream and skin cleanser lotion [78]. Lipopeptide surfactants are admired in cosmetic industry because of their skin-friendliness and easy washability. Table 6 gives a list of biosurfactants applied in the area of cosmetic formulation.

5.4 In agriculture

Soil quality is an important perspective for crop production in agriculture sector. The biotic and abiotic components of the soil are affected by the presence of various pollutants. Biosurfactants play a major role in improving soil quality. They promote the plant growth by acting as a source of carbon, supplying micronutrients to optimum level, and creating a safe environment from the harmful substances. The antimicrobial properties of biosurfactants are efficient against various plant pathogens to bring sustainable agriculture. A number of strains from *Pseudomonas* strain are reported to be very effective against controlling plant pathogens [79,80]. Lipopeptide is found to act as a biopesticide for *Drosophila melanogaster* fruit fly [81]. Further *Bacillus subtilis* A1 and *Pseudomonas stutzeri* NA3 can inhibit the young *Anopheles stephensi* [82]. Kim et al. have reported the insecticidal activity of di-rhamnolipid against the aphid of peach fruit [83]. The chemical pesticides and herbicides are mainly composed of hydrophobic organic molecules. Thus for agricultural formulations, bioemulsifiers play a vital role in solubilization of such compounds. The dispersions of pesticides and herbicides in the aqueous phase are facilitated by bioemulsifiers.

5.5 In pharmaceutical industry

In medical field, biosurfactants have significant contributions. For improved wetting and dissolution, biosurfactants are used in the formulation of drug. Solubility is the primary condition for a drug to be absorbed in vivo. Further biosurfactants enable the

Table 6 Biosurfactants in cosmetic formulation [25].

Biosurfactants	Application	Cosmetic formulation
Sophorolipid	Detergency	Cleanser
Sophorolipid	Acne treatment, dandruff removal	Cream, lotion
Sophorolipid	Inhibition of elastase and free radicals	Skin care cream
Sophorolipid	Beautification	Lip color, lip cream, eye shadow, compact powder
Sophorolipid	Body odor	Deodorants, perfume and fragrance
Sophorolipid	Anti-inflammatory	Ointment
Rhamnolipids	Beautification, cleaning, body odor	Nail color, toothpaste, deodorants
Rhamnolipids	Antifungal activity	Lotion and cream
Mannosyloerythritol lipids	Antiaging	Face and body cream
Mannosyloerythritol lipids	Hair growth	Hair oil, serum,
Lipopeptides	Foaming	Body wash, face wash
	Antiwrinkle	Cream
	Detergency	Cleanser
	Bleaching and whitening	Skin lightening cream
	Antibacterial and antifungal	Creams
Xylolipids	Oral and dermal application	Toothpaste, face creams
Trehalolipids	Antifungal	Creams and lotions

drugs to penetrate the cell membranes and biological interfaces to work effectively. A lot of investigations are being done on the application of biosurfactants in drug delivery and immunology. Biosurfactants strongly interact with the cellular lipid membrane to alter various biological processes by changing the physicochemical structure of the target cell. A number of biosurfactants are reported to show antiviral, antimicrobial, and anticancer properties [84,85]. The antiproliferative activity and minimal toxicity against nontumor lymphocytes make them interesting in cancer treatment drugs [86]. Biosurfactants act as therapeutic molecules for autoimmune disorder problem by their immunoregulatory power. A recent report has described the potential of various biosurfactants in immune response system [87]. Biosurfactants isolated from *Lactobacillus* strains have demonstrated antibacterial and antimicrobial activity [60,88]. A recent study investigated the interaction of drug and biosurfactant in permeation rate for medical formulation [89].

5.6 Detergent industry

The emulsifying properties of biosurfactants are the key factors in their role in detergency. The principle of detergency is dependent on good wetting property and formation of micelle. The surfactant molecules break the stain particles and remove them from the fabric surface. Oily dirt is emulsified or solubilized by the surfactants and washed away by water. A biosurfactant extracted from strain of *Klebsiella species* exhibited good oil and grease-removing ability [90]. Lipopeptide biosurfactants are very stable and compatible with commercial detergents. They have promising potential as additives in laundry detergent formulation. Various reports highlighted that lipopeptide biosurfactant isolated from *Bascillus subtilis* enhances the cleaning performance of commercial detergents [91,92]. A lipase biosurfactant originated from *Ochrobactrum intermedium* strain displayed excellent stain removal features [93]. Further Sajna et al. reported a glycolipid as detergent additive because of its distinctive hydrophobic character and high temperature stability [94]. Surfactin has exceptional surface-active properties with added advantage of low toxicity and nonirritant nature in detergent formulation and has promising future in detergent industry [95].

6 Challenges and future prospect

Commercialization of biosurfactants is mainly limited by its high production and purification cost. The same amounts of chemical surfactants have the synthetic cost of about 10–12 times lower than the biosurfactants. Thus to bring down the cost is the primary challenge that needs attention. Further research is needed to find out cost-effective method for production of biosurfactants from low-cost raw materials. The yield of biosurfactants varies on varying the microbial strains. Thus selecting the appropriate strain for optimum yield is also important for reducing production cost. Renewable recourses and waste substrate have to be opted to reduce the cost of raw materials. Nonpathogenic strains can be engineered for production of biosurfactant in a much lesser cost [96]. Improved production technology is needed to compete with chemical surfactants in market. Development of biomedical applications of biosurfactants needs more research on structure-activity relationship. A new field of research on emulsification-enhancing proteins (EEPs) is on the rise for the novel application in more affordable way [97]. In spite of their great potential in controlling autoimmune diseases, they are not researched extensively. Immunomodulatory biosurfactants production should reach to the industrial scale in the years to come.

7 Concluding remarks

Biosurfactants are versatile smart molecules having applications in several industrial processes. The demand for environmentally safe, nontoxic substances for human application has increased the market for biosurfactants. In this chapter, types and

properties of biosurfactants have been highlighted. Various applications of biosurfactants are discussed in the light of their wetting and sulubilizing ability. Their unique character of surface activity, emulsifying property, solubilizing and wetting ability has created immense potential in various applications. The antimicrobial and antibiofilm properties of these molecules are of primary importance in food processing industries. In the coming years, it is expected that development of green technology in industrial process will further push the application of biosurfactants to great extent.

References

[1] D.K.F. Santos, R.D. Rufino, J.M. Luna, V.A. Santos, L.A. Sarubbo, Biosurfactants: multifunctional biomolecules of the 21st century, Int. J. Mol. Sci. 17 (2017) 401.

[2] S. Vijaykumar, V.K. Saravanan, Biosurfactants-types, sources and applications, Res. J. Microbiol. 10 (2015) 181–192.

[3] M. Nitsche, S.S. e Silva, Recent food applications of microbial surfactants, Crit. Rev. Food Sci. Nutr. 58 (2016) 631–638.

[4] S. Akbari, N.H. Abdurahman, R.M. Yunus, F. Fayaz, O.R. Alara, Biosurfactants-a new frontier for social and environmental safety: a mini review, Biotechnol. Res. Innov. 2 (2018) 81–90.

[5] B.G. Ribeiro, J.M. Campos, L.A. Sarubbo, Potential food application of a biosurfactant produced by Saccharomyces cerevisiae URM 6670, Front. Bioeng. Biotechnol. 8 (2020), https://doi.org/10.3389/fbioe.2020.00434.

[6] F. Md, Biosurfactant: production and application, J. Pet. Environ. Biotechnol. 3 (2012) 124.

[7] B. Bhattacharya, T.K. Ghosh, N. Das, Application of biosurfactants in cosmetics and pharmaceutical industry, Sch. Acad. J. Pharm. 6 (2017) 320–329.

[8] C.E. Drakontis, S. Amin, Biosurfactants: formulations, properties and applications, Curr. Opin. Colloid Interface Sci. 48 (2020) 77–90.

[9] K. Liu, Y. Sun, M. Cao, J. Wang, J.R. Lu, H. Xu, Rational design, properties and applications of biosurfactants: a short review of recent advances, Curr. Opin. Colloid Interface Sci. 45 (2020) 57–67.

[10] D.G. Hayes, G.A. Smith, Biobased Surfactants: Overview and Industrial State of the Art, 2019, pp. 3–38.

[11] A.A. Jimoh, J. Lin, Biosurfactant: a new frontier for greener technology and environmental sustainability, Ecotoxicol. Environ. Saf. 184 (2019), 109607.

[12] B. Kronberg, K. Holmberg, B. Lindman, Surface Chemistry of Surfactants and Polymers, John Wiley and Sons, 2014.

[13] E.G. Fathabad, Biosurfactants in pharmaceutical industries: (a mini review), Am. J. Drug Discov. Dev. 1 (2011) 58–69.

[14] S.S. Bhadoriya, N. Madoriya, K. Shukla, M.S. Parihar, Biorfactants: a new pharmaceutical additive for solubility enhancement and pharmaceutical development, Biochem. Pharmacol. 2 (2013) 113, https://doi.org/10.4172/2167-0501.1000113.

[15] R. Parthasarathi, K.S. Subha, Prospective biosurfactant applications in food industry-opportunities and challenges, Int. J. Curr. Res. Life Sci. 7 (2018) 1560–1565.

[16] N. Panjiar, S. Ghosh, A. Sachan, Biosurfactants: a multifunctional metabolite, in: Microbial Applications, vol. 2, Springer International Publishing AG, 2017, https://doi.org/10.1007/978-3-319-52669-0_12.

[17] A. Dadrasnia, S. Ismail, Biosurfactant production by Bacillus salmalaya for lubricating oil solubilization and biodegradation, Int. J. Environ. Res. Public Health 12 (2015) 9848–9863.
[18] R. Sharma, H.S. Oberoi, Biosurfactant-aided bioprocessing: industrial applications and environmental impact, in: Recent Advances in Applied Microbiology, Springer Nature Singapore Pte Ltd, 2017, https://doi.org/10.1007/978-981-10-5275-0_3.
[19] M.P. Plociniczak, G.A. Plaza, Z.P. Seget, S.S. Cameotra, Environmental applications of biosurfactants: recent advances, Int. J. Mol. Sci. 12 (2011) 633–654.
[20] A.P. Karlapudi, T.C. Venkateswarulu, J. Tammineedi, L. Kanumuri, B.K. Ravuru, V.R. Dirisala, V.P. Kodali, Role of biosurfactant in bioremediation of oil pollution: a review, Petroleum 4 (2018) 241–249.
[21] E.O. Fenibo, G.N. Ijoma, R. Selvarajan, C.B. Chikere, Microbial surfactants: the next generation mutifunctional biomolecules for applications in the petroleum industry and its associated environmental remediation, Microorganisms 7 (2019) 581.
[22] A.K. Singh, S.S. Cameotra, Efficiency of lipopeptide biosurfactants in removal of petroleum hydrocarbons and heavy metals from contaminated soil, Environ. Sci. Pollut. Res. 20 (2013) 7367–7376.
[23] M. Nitschke, S.G.V.A.O. Costa, Biosurfactants in food industry, Trends Food Sci. Technol. 18 (2007) 252–259.
[24] X. Wang, S. Yuan, B. Jiang, Wetting process and adsorption mechanism of surfactant solutions on coal dust surface, J. Chem. (2019), 9085310. 9 pages https://doi.org/10.1155/2019/9085310.
[25] D. Sharma, Biosurfactants in food, in: Springer Briefs in Food, Health and Nutrition, 2016. http://www.springer.com/series/10203.
[26] B. Liu, J. Liu, M. Ju, X. Li, Q. Yu, Purification and characterization of biosurfactant produced by *Bacillus licheniformis* Y-1 and its application in remediation of petroleum contaminated soil, Mar. Pollut. Bull. 107 (2016) 46–51.
[27] P. Yan, M. Lu, Q. Yang, H.L. Zhang, Z.Z. Zhang, R. Chen, Oil recovery from refinery oily sludge using a rhamnolipid biosurfactant-producing Pseudomonas, Bioresour. Technol. 116 (2012) 24–28.
[28] S.S. Cameotra, R.S. Makkar, Biosurfactant enhanced bioremediation of hydrophobic pollutants, Pure Appl. Chem. 82 (2010) 97–116.
[29] X. Mao, R. Jiang, W. Xiao, J. Yu, Use of surfactants for the remediation of contaminated soils: a review, J. Hazard. Mater. 285 (2015) 419–435.
[30] E. Kaczorek, A. Pacholak, A. Zdarta, W. Smułek, The impact of biosurfactants on microbial cell properties leading to hydrocarbon bioavailability increase, Colloids Interfaces 2 (2018) 35.
[31] D. Song, S. Liang, L. Yan, Y. Shang, X. Wang, Solubilization of polycyclic aromatic hydrocarbons by single and binary mixed rhamnolipid-sophorolipid biosurfactants, J. Environ. Qual. 45 (2016) 1405–1412.
[32] M. Ye, M. Sun, J. Wan, G. Fang, H. Li, F. Hu, X. Xiang, F.O. Kangara, Evaluation of enhanced soil washing process with tea saponin in a peanut oil-water solvent system for the extraction of PBDEs/PCBs/PAHs and heavy metals from an electronic waste site followed by vetiver grass phytoremediation, J. Chem. Technol. Biotechnol. 90 (2015) 2027–2035.
[33] M.M. Usman, A. Dadrasnia, K.T. Lim, A.F. Mahmud, S. Ismail, Application of biosurfactants in environmental biotechnology; remediation of oil and heavy metal, AIMS Bioeng. 3 (2016) 289–304.
[34] J.M. Campos, T.L.M. Stamford, L.A. Sarubbo, J.M. de Luna, R.D. Rufino, I.M. Banat, Microbial biosurfactants as additives for food industries, Biotechnol. Prog. 29 (2013) 1097–1108.

[35] S.K. Satpute, S.S. Zinjarde, I.M. Banat, Recent updates on biosurfactant/s in food industry, in: Chapter in Book, Microbial Cell Factories, Taylor and Francis, 2018, pp. 1–20.
[36] P. Shakerifard, F. Gancel, P. Jacques, C. Faille, Effect of different *Bacillus subtilis* lipopeptides on surface hydrophobicity and adhesion of *Bacillus cereus* 98/4 spores to stainless steel and teflon, Biofouling 25 (2009) 533–554.
[37] X. Huang, J. Suo, Y. Cui, Optimization of antimicrobial activity of surfactin and polylysine against *Salmonella enteritidis* in milk evaluated by a response surface methodology, Foodborne Pathog. Dis. 8 (2011) 439–443.
[38] M.Z.V. Gomes, M. Nitschke, Evaluation of rhamnolipid and surfactin to reduce the adhesion and remove biofilms of individual and mixed cultures of food pathogenic bacteria, Food Control 25 (2012) 441–447.
[39] L.V. Araujo, C.R. Guimarães, R.L.S. da Marquita, V.M.J. Santiago, M.P. Souza, M. Nitschke, D.M.G. Freire, Rhamnolipid and surfactin: anti-adhesion, antimicrobial and anti-corrosion effects, Food Control 63 (2016) 171–178.
[40] M.J. Manoharan, K. Bradeeba, R. Parthasarathi, P.K. Sivakumaar, P.S. Chauhan, S. Tipayno, A. Benson, T. Sa, Development of surfactin based nanoemulsion formulation from selected cooking oils: evaluation for antimicrobial activity against selected food associated microorganisms, J. Taiwan Inst. Chem. Eng. 43 (2012) 172–180.
[41] L. Magalhães, M. Nitschke, Antimicrobial activity of rhamnolipids against *Listeria monocytogenes* and their synergistic interaction with nisin, Food Control 29 (2013) 138–142.
[42] C. Jing, Y. Bingbing, Sophorolipid Fruit Preservative and Use Thereof in Fruit Preservation, Chinese Patent CN 101886047, 2010.
[43] Q. Meng, G. Zhang, Application of Rhamnolipid as Biological Cleaning Agent, Chinese Patent CN 102399644, 2012.
[44] C.R. Suresh Chander, T. Lohitnath, M. Kumar, P.T. Kalaichelvan, Production and characterization of biosurfactant from *Bacillus subtilis* MTCC441 and its evaluation to use as bioemulsifier for food bio-preservative, Adv. Appl. Sci. Res. 3 (2012) 1827–1831.
[45] I. Mnif, S. Besbes, R. Ellouze, S. Ellouze-Chaabouni, D. Ghribi, Improvement of bread quality and bread shelf-life by *Bacillus subtilis* biosurfactant addition, Food Sci. Biotechnol. 21 (2012) 1105–1112.
[46] J.M. Campos, T.L.M. Stamford, R.D. Rufino, J.M. Luna, T.C.M. Stamford, L.A. Sarubbo, Formulation of mayonnaise with the addition of a bioemulsifier isolated from *Candida utilis*, Toxicol. Rep. 2 (2015) 1164–1170.
[47] R. Zouari, S. Besbes, S. Ellouze-Chaabouni, D. Ghribi-Aydi, Cookies from composite wheat-sesame peels flours: dough quality and effect of *Bacillus subtilis* SPB1 biosurfactant addition, Food Chem. 194 (2016) 758–769.
[48] D. Sharma, B.S. Saharan, N. Chauhan, S. Procha, S., Lal isolation and functional characterization of novel biosurfactant produced by *Enterococcus faecium*, Springer Plus 4 (2015) 4.
[49] D.C. Sabaté, M.C. Audisio, Inhibitory activity of surfactin, produced by different *Bacillus subtilis* subsp. *subtilis* strains, against *Listeria monocytogenes* -sensitive and bacteriocin- resistant strains, Microbiol. Res. 168 (2013) 125–129.
[50] A.N. Madhu, S.G. Prapulla, Evaluation and functional characterization of a biosurfactant produced by *Lactobacillus plantarum* CFR 2194, Appl. Biochem. Biotechnol. 172 (2014) 1777–1789.
[51] M.D. De Rienzo, P. Stevenson, R. Marchant, I.M. Banat, Antibacterial properties of biosurfactants against selected gram positive and negative bacteria, FEMS Microbiol. Lett. 224 (2015).

[52] X. Zhang, X. Fan, D.K. Solaiman, R.D. Ashby, Z. Liu, S. Mukhopadhyay, R. Yan, Inactivation of *Escherichia coli* O157: H7 in vitro and on the surface of spinach leaves by biobased antimicrobial surfactants, Food Control 60 (2016) 158–165.
[53] J. Coronel-León, A.M. Marqués, J. Bastida, A. Manresa, Optimizing the production of the biosurfactant lichenysin and its application in biofi lm control, J. Appl. Microbiol. 120 (2016) 99–111.
[54] M.J. Torres, G. Petroselli, M. Daz, R. Erra-Balsells, M.C. Audisio, *Bacillus subtilis* subsp. *subtilis* CBMDC3f with antimicrobial activity against gram-positive foodborne pathogenic bacteria: UV-MALDI-TOF MS analysis of its bioactive compounds, World J. Microbiol. Biotechnol. 31 (2015) 929–940.
[55] D.J. McClements, Food Emulsions: Principles, Practices, and Techniques, CRC Press, Boca Raton, 2016, pp. 1–26.
[56] J.M. Campos, T.L.M. Stamford, L.A. Sarubbo, Production of a bioemulsifier with potential application in the food industry, Appl. Biochem. Biotechnol. 172 (2014) 3234–3252.
[57] M. Nitschke, G.M. Pastore, Production and properties of a surfactant obtained from *Bacillus subtilis* grown on cassava wastewater, Bioresour. Technol. 97 (2006) 336–341.
[58] V. Farheen, S.B. Saha, S. Pyne, B.R. Chowdhury, Production of nanobiosurfactant from *Pseudomonas aeruginosa* and it's application in bakery industry, Int. J. Adv. Res. Biol. Eng. Sci. Technol. 2 (2016). 67 (SI8).
[59] P. Das, S. Mukherjee, R. Sen, Antimicrobial potential of a lipopeptide biosurfactant derived from a marine *Bacillus circulans*, J. Appl. Microbiol. 104 (2008) 1675–1684.
[60] K. Sambanthamoorthy, X. Feng, R. Patel, S. Patel, C. Paranavitana, Antimicrobial and antibiofilm potential of biosurfactants isolated from lactobacilli against multi-drug-resistant pathogens, BMC Microbiol. 14 (2014) 197.
[61] Y. Hirata, M. Ryu, Y. Oda, K. Igarashi, A. Nagatsuka, T. Furuta, M. Sugiura, Novel characteristics of sophorolipids, yeast glycolipid biosurfactants, as biodegradable low foaming surfactants, J. Biosci. Bioeng. 108 (2009) 142–146.
[62] M. Takahashi, T. Morita, T. Fukuoka, I. Tomohiro, D. Kitamoto, Glycolipid biosurfactants, mannosylerythritol lipids, show antioxidant and protective effects against H2O2—induced oxidative stress in cultured human skin fibroblasts, J. Oleo Sci. 61 (2012) 457–464.
[63] E. Yalcin, K. Cavusoglu, Structural analysis and antioxidant activity of a biosurfactant obtained from *Bacillus subtilis* RW-I, Turkish, J. Biochem. 35 (2010) 243–247.
[64] C. Chen, T. Lin, Y. Shieh, Emulsification and antioxidation of biosurfactant extracts from Chinese medicinal herbs fermentation in vitro, J. Biosci. Bioeng. 120 (2015) 387–395.
[65] E.M. Rossi, L. Beilke, M. Kochhann, D.H. Sarzi, E.C. Tondo, Biosurfactant produced by *Salmonella enteritidis* SE86 can increase adherence and resistance to sanitizers on lettuce leaves (*Lactuca sativa* L., *cichoraceae*), Front. Microbiol. 7 (2016) 9.
[66] B. Brzozowski, W. Bednarski, P. Golek, The adhesive capability of two *Lactobacillus* strains and physicochemical properties of their synthesized biosurfactants, Food Technol. Biotechnol. 49 (2011) 177–186.
[67] J. Abdul, S. Salman, M.F.H. Kadhemy, Effect of PVA, PVA/Biosurfactant on some pathogenic bacteria in glass and plastic plates, Int. J. Curr. Microbiol. App. Sci. 3 (2014) 301–309.
[68] A. Bridier, P. Sanchez-Vizuete, M. Guilbaud, J.-C. Piard, M. Naïtali, R. Briandet, Biofilm-associated persistence of food-borne pathogens, Food Microbiol. 45 (2015) 167–178.
[69] M.D. Macià, E. Rojo-Molinero, A. Oliver, Antimicrobial susceptibility testing in biofilm-growing bacteria, Clin. Microbiol. Infect. 20 (2014) 981–990.

[70] T. Meylheuc, M. Renault, M.N. Bellon-Fontaine, Adsorption of a biosurfactant on surfaces to enhance the disinfection of surfaces contaminated with *Listeria monocytogenes*, Int. J. Food Microbiol. 109 (2006) 71–78.
[71] M. Nitschke, L.V. Araujo, S.G. Costa, R.C. Pires, A.E. Zeraik, A.C. Fernandes, D.M. Freire, J. Contiero, Surfactin reduces the adhesion of food-borne pathogenic bacteria to solid surfaces, Lett. Appl. Microbiol. 49 (2009) 241–247.
[72] A.E. Zeraik, M. Nitschke, Biosurfactants as agents to reduce adhesion of pathogenic bacteria to polystyrene surfaces: effect of temperature and hydrophobicity, Curr. Microbiol. 61 (2010) 554–559.
[73] F. Anjum, G. Gautam, G. Edgard, S. Negi, Biosurfactant production through *Bacillus* sp. MTCC 5877 and its multifarious applications in food industry, Bioresour. Technol. 213 (2016) 262.
[74] S.S. Giri, S.S. Sen, J.W. Jun, V. Sukumaran, S.C. Park, Role of *Bacillus licheniformis* VS16-derived biosurfactant in mediating immune responses in carp rohu and its application to the food industry, Front. Microbiol. 8 (2017) 514.
[75] G.S. Kiran, S. Priyadharsini, A. Sajayan, G.B. Priyadharsini, N. Poulose, J. Selvin, Production of lipopeptide biosurfactant by a marine Nesterenkonia sp. and its application in food industry, Front. Microbiol. 8 (2017) 1138.
[76] N. Lourith, M. Kanlayavattanakul, Natural surfactants used in cosmetics: glycolipids, Int. J. Cosmet. Sci. 31 (2009) 255–261.
[77] A. Bhangale, S. Wadekar, S. Kale, D. Bhowmick, A. Pratap, Production of sophorolipids synthesized on castor oil with glucose and glycerol by using *Starmerella bombicola* (ATCC 22214), Eur. J. Lipid Sci. Technol. 116 (2014) 336–343.
[78] K.R. Meena, S.S. Kanwar, Lipopeptides as the antifungal and antibacterial agents: applications in food safety and therapeutics, Biomed. Res. Int. (2015) 1–9.
[79] D.P. Sachdev, S.S. Cameotra, Biosurfactants in agriculture, Appl. Microbiol. Biotechnol. 97 (2013) 1005–1016.
[80] M. Kruijt, H. Tran, J.M. Raaijmakers, Functional, genetic and chemical characterization of biosurfactants produced by plant growth-promoting *Pseudomonas putida* 267, J. Appl. Microbiol. 107 (2009) 546–556.
[81] C.N. Mulligan, Environmental applications for biosurfactants, Environ. Pollut. 133 (2005) 183–198.
[82] P. Parthipan, R.K. Sarankumar, A. Jaganathan, P. Amuthavalli, R. Babujanarthanam, P.K.S.M. Rahman, K. Murugan, A. Higuchi, G. Benelli, A. Rajasekar, Biosurfactants produced by *Bacillus subtilis* A1 and *Pseudomonas stutzeri* NA3 reduce longevity and fecundity of *Anopheles stephensi* and show high toxicity against young instars, Environ. Sci. Pollut. Res. 25 (2018) 10471–10481.
[83] S.K. Kim, Y.C. Kim, S. Lee, J.C. Kim, M.Y. Yun, I.S. Kim, Insecticidal activity of rhamnolipid isolated from Pseudomonas sp. EP-3 against green peach aphid (*Myzus persicae*), J. Agric. Food Chem. 59 (2010) 934–938.
[84] T. Ndlovu, M. Rautenbach, J.A. Vosloo, S. Khan, W. Khan, Characterisation and antimicrobial activity of biosurfactant extracts produced by *Bacillus amyloliquefaciens* and *Pseudomonas aeruginosa* isolated from a wastewater treatment plant, AMB Express 7 (2017) 108, https://doi.org/10.1186/s13568-017-0363-8.
[85] T. Janek, Z. Czyznikowska, M. Lukaszewicz, J. Galezowska, The effect of *Pseudomonas fluorescens* biosurfactant pseudofactin II on the conformational changes of bovine serum albumin: pharmaceutical and biomedical applications, J. Mol. Liq. 288 (2019), https://doi.org/10.1016/j.molliq.2019.111001, 111001.

[86] E.J. Gudina, V. Rangarajan, R. Sen, L.R. Rodrigues, Potential therapeutic applications of biosurfactants, Trends Pharmacol. Sci. 34 (2013) 667–675.
[87] M. Sajid, M.S. Ahmad Khan, S.S. Cameotra, A.S. Al-Thubiani, Biosurfactants: potential applications as immunomodulator drugs, Immunol. Lett. 223 (2020) 71–77.
[88] E.J. Gudiña, E.C. Fernandes, J.A. Teixeira, L.R. Rodrigues, Antimicrobial and anti-adhesive activities of cell-bound biosurfactant from *Lactobacillus agilis* CCUG31450, RSC Adv. 5 (2015) 90960–90968, https://doi.org/10.1039/c5ra11659g.
[89] L. Rodríguez-López, D.S. Shokry, J.M. Cruz, A.B. Moldes, L.J. Waters, The effect of the presence of biosurfactant on the permeation of pharmaceutical compounds through silicone membrane, Colloids Surf. B Biointerfaces 176 (2019) 456–461, https://doi.org/10.1016/j.colsurfb.2018.12.072.
[90] R.M. Jain, K. Mody, A. Mishra, Physicochemical characterization of biosurfactant and its potential to remove oil from soil and cotton cloth, Carbohydr. Polym. 89 (2012) 1110–1116, https://doi.org/10.1016/j.carbpol.2012.03.077.
[91] A.K. Mukherjee, Potential application of cyclic lipopeptide biosurfactants produced by *Bacillus subtilis* strains in laundry detergent formulations, Lett. Appl. Microbiol. 45 (2007) 330–335, https://doi.org/10.1111/j.1472- 765x.2007.02197.x.
[92] M. Bouassida, N. Fourati, I. Ghazala, S. Ellouze-Chaabouni, D. Gharibi, Potential application of *Bacillus subtilis* SPB1 biosurfactants in laundry detergent formulations: compatibility study with detergent ingredients and washing performance, Eng. Life Sci. 18 (2017) 70–77, https://doi.org/10.1002/elsc.201700152.
[93] M. Zarinviarsagh, G. Ebrahimipour, H. Sadeghi, Lipase and biosurfactant from *Ochrobactrum intermedium* strain MZV101 isolated by washing powder for detergent application, Lipids Health Dis. 16 (2017) 177, https://doi.org/10.1186/s12944-017-0565-8.
[94] K.V. Sajna, R.K. Sukumaran, H. Jayamurthy, K.K. Reddy, S. Kanjilal, R.B.N. Prasad, A. Pandey, Studies on biosurfactants from Pseudozyma Sp. NII 08165 and their potential application as laundry detergent additives, Biochem. Eng. J. 78 (2013) 85–92, https://doi.org/10.1016/j.bej.2012.12.014.
[95] D. Fei, G.W. Zhou, Z.Q. Yu, H.Z. Gang, J.F. Liu, S.Z. Yang, R.Q. Ye, B.Z. Mu, Low-toxic and nonirritant biosurfactant Surfactin and its performances in detergent formulations, J. Surfactant Deterg. 23 (2019) 109–118, https://doi.org/10.1002/jsde.12356.
[96] T. Tiso, P. Sabelhaus, B. Behrens, A. Wittgens, F. Rosenau, H. Hayen, L.M. Blank, Creating metabolic demand as an engineering strategy in *Pseudomonas putida*—rhamnolipid synthesis as an example, metabolic engineering, Communications 3 (2016) 234–244, https://doi.org/10.1016/j.meteno.2016.08.002.
[97] D.L. Gutnick, H. Bach, Biosurfactants, in: Reference Module in Life Sciences, 2017, https://doi.org/10.1016/B978-0-12-809633-8.09184-6.

CHAPTER

15

Glycolipid biosurfactants: Biosynthesis and related potential applications in food industry☆

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1 Introduction

Glycolipid biosurfactants are "green" nonionic surfactants derived from renewable agricultural materials and which consist of varying saccharide and nonpolar moieties. The two fractions are usually linked by ester-, ether-, thioester, amine, amide, or lactone bonds [1,2]. The hydrophobic portions are usually fatty acids with or without functional groups [3]. Glycolipids maintain surface-active, odorless, tasteless, and nonirritating qualities; thus, they have high potential for use in food and cosmetic products [4]. Biosurfactants, in general, are grouped with respect to their functionalities into six important categories: (1) cleaning agents, (2) emulsifying agents, (3) wetting agents, (4) solubilizing agents, (5) foam boosting agents, and (6) suspending agents; thus, biosurfactants have enormous potential in many food applications and in agricultural and industrial processes. Throughout the decade from 1990 to 2000, only 207 international patents for biosurfactants were granted by various patent authorities. This trend has changed in the past few years whereby overall patents granted have more than quadrupled to a total of 904 international patents from 2008

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to 2018 [5]. The immense interest in biosurfactants can be attributed to the much-needed awareness for green substitutes for traditional synthetic surfactants. Synthetic surfactants, which are mainly derived from petroleum, have detrimental long-term effects on the global environment and particularly on climate change. Additionally, they are sourced from nonrenewable materials and therefore the production of synthetic surfactants is nonsustainable.

In the recent Conference of the Parties (COP21), several nations agreed to advance a global initiative ensuring that the current mean temperature does not rise more than 2°C above preindustrial levels. Because of their large application potential, the substitution of biosurfactants for petroleum-based surfactants may play a significant role in mitigating carbon dioxide emissions [6] and helping to accomplish this goal. In addition to the numerous favorable performance characteristics of biosurfactants, which will be discussed in this chapter, efforts in mitigation of the above-mentioned global concerns can be realized if synthetic surfactants can be substituted effectively with biosurfactants globally.

However, being "green" and having desirable performance characteristics will not enable large amounts of biosurfactants to be ubiquitously applied in all the relevant industries if they cannot be priced competitively in the market. Lowering costs throughout the biosurfactant manufacturing process must be accomplished to ensure long-term success and acceptance of the bioproduct in the global market. The challenge is in striking a balance between product quality and achieving and maintaining low production costs. Consideration must be given to the entire manufacturing process rather than simply focusing on synthesis costs alone. This allows management of the whole workflow instead of just partial optimization in which improving one aspect of the process can be detrimental to another. The key considerations of entire process optimization are reduced material costs, improved productivity, well managed inventory-carrying costs, and cost-effective downstream processing costs.

This chapter presents a critical review of some of the more well-known glycolipids, (e.g., rhamnolipids (RLs), trehalolipids, sophorolipids (SLs), and mannosylerythritol lipids) and includes references to biosynthesis and their potential applications in the food industry. In addition, a detailed overview of the efficient production of the various glycolipids along with the associated challenges and prospects is presented.

2 Origin and diversity of glycolipids structure

2.1 Rhamnolipids

RLs were first discovered in 1946 as oily glycolipids produced by *Pseudomonas aeruginosa* (formerly known as *Pseudomonas pyocyanea*) from glucose [6]. The oily glycolipid structures were initially characterized as L-rhamnose and β-hydroxydecanoic acid in 1949 [7,8]. In that same year, these same authors discovered the presence of multiple RL congeners including chemical structures representing

two β-hydroxydecanoic acids attached to two rhamnose moieties by a glycosidic bond and between two β-hydroxy fatty acids joined by an ester bond. The linkage between the two rhamnose sugars formed an α-1,2-glycosidic bond [9]. Later, RLs were described as 2-*O*-α-1,2-L-rhamnopyranosyl-α-L-rhamnopyranosyl-β-hydroxydecanoyl-β-hydroxydecanoate and identified as the earliest glycolipid where sugar and hydroxylated fatty acid were chemically linked [10]. Over the years, due to the evolution of more sensitive instrumental techniques, detection of new RL congeners has evolved, which now includes a variety of aliphatic chains, which vary in length from C8 to C12 [11–13].

In another study, *Burkholderia thailandensis* grown on vegetable oil or glycerol yielded mono- and di-RL with C14 fatty acids. Moreover, a small amount of C16 was also detected in the study, showing that *B. thailandensis* could produce a variety of RL congeners with C12–C16 fatty acids [14].

Therefore, rhamnose and lipid moieties that are connected through an *O*-glycosidic bond form RLs. Rhamnose moieties can be present as either mono- (one) or di- (two) units linked by α-1,2-glycosidic bonds [15]. The lipid moieties are linked through the ester bond between β-hydroxy groups and can be saturated, monounsaturated, or polyunsaturated and can have chain lengths between C8 and C16 (Fig. 1). The varieties of lipid moieties contribute the most to the diversity of RL chemical structures and consequently to the wide range of surfactant properties, whereas the number of rhamnose moieties plays a minor part.

2.2 Trehalolipids

Trehalolipids (also known as trehalose lipids) were first discovered in 1933 as alcoholic elements of fat in glycerol produced by *Tubercle bacilli* [16]. Nevertheless, at that time, the authors did not proceed with further investigation into the properties of the new compound. Later in 1956, trehalolipids were successfully purified and their structure characterized as 6,6′-dimycoloyl-α,α′-D-trehalose produced by *Mycobacterium tuberculosis*. Trehalolipids were discovered as an emulsion layer when grown on the hydrocarbon substrate of *Arthrobacter paraffineus* [17]. Subsequently, several species of trehalolipid producers were identified, such as *Mycobacteria, Nocardia*, and *Corynebacteria* [18].

Two glucose molecules linked via an α,α-1,1-glycosidic bond to mycolic acid and a chain of α-branched-β-hydroxy fatty acids compose the general unit of trehalolipids (Fig. 2) [19]. The trehalolipids are typically found as 6, 6′-dimycolate derivatives that consist of an α-branched-chain mycolic acid esterified to glucose at each C6 site. Moreover, the *Rhodococcus* genus produces trehalolipids with a varying chain length of esterified fatty acids ranging between C20 and C90 [20]. The trehalolipids were synthesized as trehalose-6-monocorynomycolates, trehalose-6,6′-diacylates (3-oxo-2-alkyl alkanoic acid), and trehalose-6-acylates (3-oxo-2-alkyl alkanoic acid) in the studies of *Rhodococcus erythropolis* [21,22]. In another study, trehalolipids were presented as acylated compounds containing C35–C40 mycolic acids, C10–C22 saturated and unsaturated fatty acids, hexanedioic and

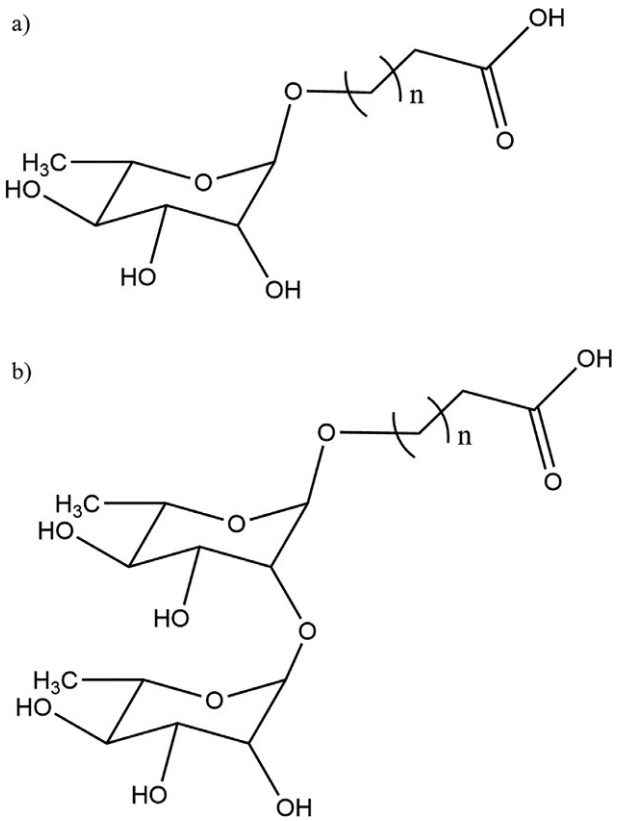

FIG. 1

General structure of (A) mono- and (B) di-rhamnolipids.

dodecanedioic acids [23,24]. Interestingly, nonionic trehalolipids were isolated from *Rhodococcus* [24]. After several years, additional varieties of trehalolipids were discovered including mono-, di-, and tri-corynomycolates [25,26], mono-, di-, tetra-, hexa-, and octa-acylated derivatives of trehalose [19,20], trehalose tetraesters [27,28], and succinoyl trehalose lipids [28].

Mycolic acid moieties primarily determine the structural-functional diversity of trehalolipids, where the number of carbon atoms and degree of unsaturation [29] play significant roles.

2.3 Sophorolipids

SLs were first identified as an extracellular oil (syrup) heavier than water produced by *Torulopsis magnolia* [30]. Several years after its initial discovery, the same authors reported that *T. magnolia* was a species of *T. apicola*. Further investigation

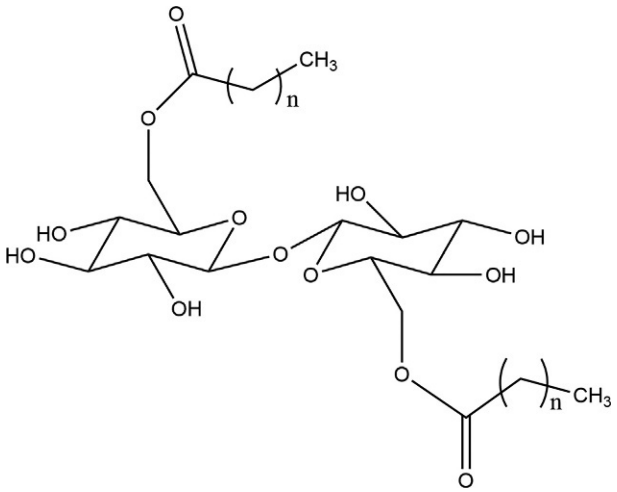

FIG. 2

General structure of trehalolipids.

revealed that the genus *Torulopsis* was synonymous with *Candida*; therefore, *T. apicola* was reclassified as *Candida apicola*. Analysis of the product revealed that the extracellular oil was recognized as a hydroxy fatty acid sophoroside [31]. In 1968, the hydroxy fatty acid sophoroside was explicated as 17-L-hydroxyoctadecanoic acid or 17-L-hydroxy-9-octadecenoic acid linked by a β-glycosidic bond to 2-*O*-β-D-glucopyranosyl-D-glucopyranose [32].

Generally, SLs comprise a complex blend of structural variants. Normally, sophorose moieties (glucose disaccharide) and the hydroxy fatty acids are joined through a β-glycosidic bond at the ω or ω-1 position to form the basic SL structure. These molecules may or may not be acetylated at the 6′ and 6″ positions (Fig. 3) [33], and the hydroxy fatty acids commonly have chain lengths of 16–18 carbon atoms and may be saturated or contain one or more double bonds [34]. In addition, the carboxylic end at the fatty acid side chain can be in the acidic (open) form or in the lactonic (closed) form with esterification typically at the 4″ position but in some rare cases at the 6′- or 6″-position [35].

A wide variation of physicochemical properties within the SL family is due to dissimilarities in structural classes that lead to different target applications. For example, the acidic form is more water-soluble and is a better foam-promoter, whereas the lactonic form displays better surface tension lowering and antimicrobial activity [35]. Furthermore, acetyl groups at the sophorose moiety cater to critical information in various applications [36]. Therefore, all parts of SLs' structures are unique and explicit.

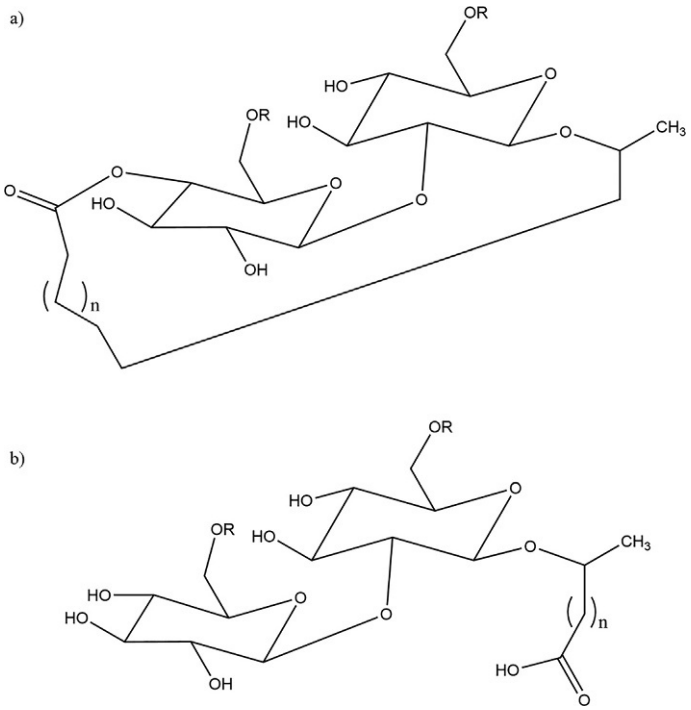

FIG. 3

Common structure of (A) lactonic and (B) acidic sophorolipids. R represents acetyl group or H.

3 Production of microbial biosurfactants: Manufacturing cost optimization

A major consideration for the increased utilization of bio-based glycolipid surfactants is the production economics associated with their synthesis, isolation, and purification. At small scale, production costs can be reduced most effectively by improving the productivity of the synthetic processes but, at larger scales, substrate costs become increasingly significant. Therefore, at industrially relevant capacities, production numbers tend to be reduced on a per weight basis. This was demonstrated in a 2013 paper by Ashby et al. [37]. In that paper, a process economic model was developed for the large-scale (90.7 million kg/year) fermentative synthesis of SL using *Starmerella bombicola* as the producing strain. A fed-batch fermentation strategy was used incorporating both glucose and either high oleic sunflower oil (HOSO) or oleic acid. The results showed remarkably low production costs of US$2.95/kg and US$2.54/kg, respectively. By comparison, the current price of SL is around US$35/kg based on the current production scale economics.

One perceived method of reducing the cost of glycolipid biosynthesis is to use high-volume, low-value coproduct/waste streams (often referred to as second-generation substrates) as fermentation feedstocks. Theoretically, this effort is designed to cut the capital costs of the overall production process, and to date, many studies have been conducted using this rationale. Unfortunately, glycolipid production using such streams is generally recognized as being less efficient when compared with the utilization of first-generation substrates. In addition, the presence of molecular mixtures and contaminants within the coproduct/waste streams can complicate downstream processing. Typically, this problem can be resolved or at least mitigated by applying pretreatment steps to the coproduct/waste streams, but each added processing step increases the overall costs associated with glycolipid production and purification. Alternatively, an advantage of using coproduct/waste streams for glycolipid biosynthesis, aside from substrate cost reduction, is that their use may help to improve the environmental impact of glycolipid production and use as demonstrated in a 2017 paper by Baccile et al for SL [38]. In the next two sections, the discussion will focus on the advantages and disadvantages for using second-generation substrates as feedstocks for glycolipid biosynthesis and the issues involving downstream processing that can affect overall production economics. Specifically, emphasis will be placed on SL, RL, and mannosylerythrytol lipids (MEL), three well-known glycolipid biosurfactants.

3.1 Valorization of agro-industrial wastes and food residues as promising substrates for the economic production of glycolipid biosurfactants

Titer (g/L), yield (g/g), productivity (g/h), and volumetric productivity (g/Lh) are all measures that are vitally important to accurately determine and compare production costs. By maximizing the production numbers, a more efficient process can be obtained, which would effectively reduce the economics of glycolipid biosynthesis. As such, these parameters must be standardized, regardless of feedstock to compare production results successfully and accurately from one study to another. Unfortunately, different laboratories utilize a wide variety of quantification techniques that can result in a broad variation in production numbers. In many instances, crude quantification protocols (e.g., colorimetric assays or gravimetric measurement without knowing the purity of the material) result in overestimated production values, which invalidate production costs. This has been particularly evident in some studies involving SL [39–41] and RL [42] production.

Glycolipids are typically not naturally produced as single congeners but as complex mixtures. Whether the structural variations are the result of different fatty acid length and degree of unsaturation, acetylation pattern or lactonic *vs*. free acid structure as in SL; the presence of mono- *vs*. di-rhamnose or number of and chain length of the β-hydroxy fatty acids in RL; or the degree and location of acetylation or the number, chain length and saturation of the fatty acids associated with MEL, few pure standards are commercially available, which hinders quantitative analysis through

typical chromatographic techniques (e.g., HPLC). Because of this, researchers often attempt to purify glycolipids through selective solvent extraction, precipitation, and evaporation and then determine production numbers gravimetrically. Unfortunately, this method usually results in crude products that contain residual water and/or hydrophobic substrates that may not have been eliminated through the purification process. These contaminants are often not accounted for in the glycolipid product, and thus result in overestimated inaccuracies when measured gravimetrically. To address these discrepancies, Smyth et al [43] published a compilation of standard protocols designed to help accurately identify and measure glycolipid production. In addition, the structural complexity within a single glycolipid family can result in chromatographic retention times that are often closely associated in qualitative analyses. This can generate confusion when attempting to identify specific congeners and application options when the products are supposedly identical. As such, all qualitative data should be accompanied by other means of analysis including (but not limited to) mass spectrometry (MS) and/or nuclear magnetic resonance spectroscopy (NMR) to verify the chemical structures present in the glycolipid mixtures. By doing so, a better comparison can be made from study to study regarding the actual productivity and structural distribution among glycolipid-based fermentations, which will ultimately help standardize the processes and make production economic comparisons more accurate.

3.1.1 Second-generation substrates

Two of the principal considerations for the use of second-generation substrates for glycolipid production involve the perceived environmental benefits associated with utilizing coproduct/waste streams and the idea that cheaper feedstocks will result in lower production costs. This has led to legislation within the European Union emphasizing the utilization of synthetic processes that promote a circular bioeconomy rather than the traditional "cradle to grave" approach [44]. The rationale behind the use of cheaper feedstocks is based on the understanding that raw material costs can account for as much as 50% of the entire manufacturing cost [45–47]. These second-generation substrates are indeed available in higher quantities [48,49] and at a lower price, but the hidden expenses (e.g., purification costs) and lower conversion efficiencies generally mute the expected benefits of using these feedstocks. Therefore, while utilization of these feedstocks may provide inexpensive alternatives to first-generation feedstocks, for noticeable benefits, productivity needs to be at least maintained when compared with first-generation feedstocks, which may require further advances in process engineering or genetic manipulation approaches and even better, a combination of both.

3.1.2 Sophorolipids (SL)

Starmerella bombicola is the most well known of all SL-producing strains. As such, this yeast strain has been tested repeatedly for its ability to utilize second-generation substrates for SL biosynthesis under both submerged and solid-state fermentation conditions. Table 1 shows many of the coproduct/waste streams used for SL biosynthesis. Under submerged fermentation conditions using either batch or fed-batch

Table 1 Review of the submerged and solid-state fermentation studies for SL biosynthesis from second-generation feedstocks.

Entry	Hydrophobic substrate	Hydrophilic substrate	Type of fermentation	Ref.
Submerged fermentation:				
1	Oleic acid	Soy molasses	Fed-batch	[50]
2	Tallow or coconut fatty acid	Glucose	Fed-batch	[51]
3	Oleic acid	Deproteinized whey+glucose	Batch	[52]
4	Soybean oil	Sugarcane molasses	Batch	[53]
5	N/A	Sugarcane molasses	Batch	[54]
6	Jatropha, Karanja or Neem oil	Glycerol	Batch	[55]
7	Waste cooking oil	Glucose	Fed-batch	[56]
8	Castor oil	Glycerol	Fed-batch	[57]
9	Olive oil	Pretreated corn cob hydrolysate	Batch	[58]
10	Oleic acid	Pretreated rice straw	Batch	[59]
11	Waste cooking oil	Pretreated corn stover	Fed-batch	[60]
12	Cottonseed oil	Rice bran and/or Glucose	Batch	[61]
13	Tapis, Melita, or Ratawi oil	Glucose	Batch	[62]
14	Chicken fat	Glucose	Batch	[63]
Solid-state fermentation:				
15	Pretreated mango kernel fat	Glucose (wheat bran powder as support)	Batch	[64]
16	Soybean oil	Pretreated sunflower oil cake	Batch	[65]
17	Motor oil	Pretreated sunflower oil cake	Batch	[66]
18	N/A	Pretreated winterization oil cake + sugar beet molasses as hydrophilic substrate	Batch	[67]
19	Soybean oil	Pretreated safflower oil cake	Batch	[68]

processes, such low-value substrates as waste cooking oil, animal fat, lignocellulosic biomass, molasses, and exotic oils (e.g., karanja, neem, melita, ratawi, tapis, castor, or jatropha oil) have been used with incubation times ranging from 4 to 10 days. In contrast, solid-state fermentation processes have been used to evaluate such feedstocks as mango kernel oil with wheat bran powder as a support material [64],

sunflower and safflower oil cakes in combination with soybean oil or motor oil [66,68], and winterization oil cake together with sugar beet molasses [67]. The results of these studies showed that to date, it is not feasible to achieve analogous productivities by using second-generation feedstocks as primary carbon sources when compared with using the most used glucose and oleic acid. This may be due to the existence of inhibitory components within the coproduct/waste streams [51,58,69] or that the substrate is not ideal for the CYP52M1 monooxygenase [70], both of which would limit SL production. Even efforts designed to increase cell permeability toward these substrates, specifically by using ultrasound pulsation during exponential growth phase, resulted in very little improvement in the end titers even after 10 days of incubation [56].

Another intriguing side-stream feedstock that has been evaluated for SL biosynthesis is crude glycerol. Crude glycerol is the by-product of the chemical transesterification reaction in the biodiesel production process. With the ongoing emphasis for bio-based fuels, biodiesel has quickly become a focal point for biofuels, and its increased production has resulted in an increase in crude glycerol. To assess the feasibility of utilizing crude glycerol as a feedstock for SL production, it was fed to *S. bombicola* and the productivity determined after a total of 7 days [71]. In that study, the crude glycerol was composed of 40% glycerol, 34% hexane-solubles (made up of 92% fatty acid soaps/fatty acid methyl esters and 6% mono- and di-acylglycerols) and 26% water. It was found that with pure (99%) glycerol, SL titers of only 9 g/L were realized. With crude glycerol, the titer improved to 60 g/L, which indicated that in this situation, the presence of fatty acids or fatty acid derivatives in the crude glycerol preparation helped in the production of SL since the fatty acid side chains did not have to be synthesized de novo from pure glycerol. Another effect of using crude glycerol as opposed to pure glycerol was in the lactone to free acid ratio among the SL congeners. When synthesized from pure glycerol, approximately 99% of the SL was in the lactonic conformation, whereas, from crude glycerol, only 25% of the SL was in the lactone form and 75% was in the open chain form (32% free fatty acid form and 43% fatty acid methyl ester form). This result further indicated that the composition of the crude glycerol has a noticeable effect on the synthesis and structural content of the resulting SL. In another study, the actual alkyl esters derived from the methyl, ethyl, and propyl transesterification of soybean oil were used as cofeeds in combination with pure glycerol to assess their effect on production and chemical structure [72]. In that study, SL titers were achieved up to 46 g/L, but the most interesting result was that the open-chain conformation was enhanced with a total open-chain content of up to 80% for the SL synthesized in the presence of propyl-soyate. Therefore, it is evident that the composition of the crude glycerol feedstock does influence the synthetic process and composition of the resulting SL.

3.1.3 Rhamnolipids (RL)

Another class of important glycolipid biosurfactants is RL. RL is known to be synthesized by several different members of the *Pseudomonas* genus, particularly strains of *P. aeruginosa*. Like SL, efforts to improve the economics of RL production have

Table 2 Second-generation feedstocks for RL production using various strains of *Pseudomonas aeruginosa*.

P. aeruginosa strain	Feedstock	Ref.
PAO1	Olive mill waste	[73]
MTCC 2297	Waste frying coconut oil	[74]
	Orange peel	[75]
LBI	Used soybean oil	[76]
	Soybean oil soapstock	[73,76,77]
	Soapstock	[77]
AT10	Waste fatty acid	[78]

proceeded along a couple of lines, one of which is the use of inexpensive industrial coproducts and waste streams as feedstocks (Table 2) [48]. Among these streams, the coproducts from oil industries have been demonstrated to be the most effective second-generation feedstocks used for RL production. The presence of free fatty acids in these feedstocks provides the precursors necessary for their efficient conversion to the component β-hydroxy fatty acids through the β-oxidation process [79].

RL production is generally dependent on the specific bacterial strain and process conditions used in the synthetic process. Yields as high as 60% have been obtained from *P. aeruginosa* LBI when soapstock was used as the substrate [76]. In that study, RL concentrations were determined by measuring the rhamnose concentration within the production media and multiplying that value by an experimentally determined conversion factor. While utilizing specific conversion factors for determining overall RL production may cause inaccuracies in overall production values, these factors do help in determining the estimated yields under any growth conditions. Unfortunately, as mentioned in the previous section, several protocols are used to measure RL production, which undoubtedly contributes to a disparity in documented RL yields. However, regardless of the quantitative method used, it can be unequivocally determined that *P. aeruginosa* LBI clearly produced larger RL yields from soapstock, than from vegetable oils [77].

3.1.4 Mannosylerythritol lipids

MELs are another class of glycolipids that have garnered increased attention for their surfactant properties. As is the case with SL and RL, efforts are underway to assess the possibility of utilizing agro-industrial coproduct/waste streams as feedstocks to improve the production economics of MEL. Specifically, various studies have focused on the use of milk by-products, soapstock, crude glycerol, waste frying oil, refining waste, and lignocellulosic sugars as feedstocks for MEL production. Initial work in this area was conducted by Bednarski et al [80], who utilized soapstock and postrefinery fatty acid waste to improve MEL biosynthesis using *Moesziomyces (M) antarcticus* (formerly *Pseudozyma antarctica*) as the producing strain. In both instances, the addition of the coproduct/waste feedstock improved the MEL yield

when compared with fermentations that used glucose alone as the carbon source. Subsequently, the same research group tested a deproteinized and hydrolyzed lactose solution as a substitute for glucose and used postrefinery fatty acids, animal fat, or fish oil as hydrophobic carbon sources [81]. Combinations of the lactose and the hydrophilic waste streams resulted in MEL titers of between 12 and 27 g/L.

Later, another study utilized a series of waste products including crude glycerol, waste frying oil, postrefinery waste, and soapstock for MEL biosynthesis using *M. antarcticus* and *M. aphidis* (formerly *Pseudozyma aphidis*) as the producing strains [82]. In that study, soapstock, postrefining waste, and waste frying oil proved to be efficient substrates for MEL production. At about the same time, Faria et al tested MEL production from hydrophilic substrates using *M. antarcticus*, *M. aphidis*, and *M. rugulosus* (formerly *Pseudozyma rugulosa*) [83–85]. Comparable MEL production was detected from glucose, xylose, and/or a 1:1 mixture of glucose and xylose. The best producing strain was *M. antarcticus*, which produced approximately 5 g/L MEL from each of the sugars tested.

Utilization of cellulose as the substrate for MEL biosynthesis has been described using two different processes to liberate the individual sugars from the cellulosic and hemicellulosic material. The first protocol involved the pretreatment of the cellulosic material using cellulolytic enzymes prior to use as feedstock while in the second protocol, the hydrolytic enzymes were incorporated directly into the fermentation media along with the yeast culture. This strategy was adopted to help protect against product inhibition of the enzymes. A comparison of the production efficiency between *M. antarcticus* and *M. aphidis* showed that *M. antarcticus* synthesized larger MEL titers than *M. aphidis* and that including the cellulolytic enzymes directly into the media improved productivity when compared with the pretreatment protocol. The direct utilization of cellulose or hemicellulose (xylan), without prior hydrolysis, has also been investigated [85]. Interestingly, *M. antarcticus* grew on xylan, but not on cellulose, signifying xylanase activity within this strain. This result implied that at least some MEL-producing yeasts can degrade lignocellulosic biomass and convert the resulting sugars into MEL. A more recent research effort involved the use of a two-stage approach where lignocellulosic sugars were employed along with plant oils or other hydrophobic waste streams to improve MEL biosynthesis. In that study, the sugars were used to increase cellular biomass followed by a continuous or repeated feeding strategy of hydrophobic carbon sources, which resulted in improved conversion proficiencies.

In conclusion, use of second-generation feedstocks is a worthwhile possibility to reduce processing costs and redesign glycolipid biosynthesis to produce a more sustainable and circular bioeconomy. Unfortunately, there are limitations that must be considered to maximize the benefits of using these substrates. The first consideration is that their usage cannot generally compete with the higher productivities obtained with traditional first-generation substrates. In addition, pretreatments may be necessary to render the substrate useable, added isolation and purification processes may be needed, and the compositional inconsistency among different agro-industrial coproducts can cause variations in production efficiency and differences in structural

content, which will complicate repeatability. However, since substrate cost is one of the main drivers for development of an economic production process, it is important to undertake additional research to realize a cheaper production process that can be combined into a biorefinery platform in a cost-effective manner.

3.2 Effect of downstream processing optimization on cost reduction

Downstream processing is another major consideration for optimizing the economical production of glycolipids as these processes (including isolation and purification expenses among others) can be costly and often make up as much as 60%–80% of the overall manufacturing costs [86]. Purification is the single most important downstream activity relating to glycolipid biosynthesis and application. Understandably, the required purity for any specific application is dependent upon the desired properties needed for the anticipated use. However, the physical characteristics of the materials and evaluation of their potential uses should be performed using the purest material possible as even minor contaminants can dramatically alter the performance of the glycolipid and result in misleading information pertaining to possible applications. This is especially challenging for molecules such as glycolipids that are normally produced as structural mixtures. For SL that might constitute a method to separate the acidic congeners from the lactonic congeners, which display completely different behavior [87] or for RL to separate mono-RL from di-RL [88].

For most glycolipids, the major challenges for product purification revolve around the elimination or removal of residual media components (primarily unutilized feedstocks), the low concentrations of the target compounds, the separation of different structural variants, and developing an infrastructure that can handle large volumes of fermentation broth. To accomplish these goals, many different operations have been utilized including filtration, precipitation, solvent extraction, and adsorption or other forms of column chromatography [89].

Precipitation is typically the first and easiest method of isolating glycolipids from fermentation broth. For SL and MEL, this may be through the process of selective solvent extraction (typically in ethyl acetate or in some cases methyl-tert-butyl ether; MTBE for MEL) [90], water removal using anhydrous sodium sulfate, filtration, concentration via solvent evaporation, precipitation into n-hexane [82,91] followed by a second filtration or further extraction and drying. The problem with this method is that large volumes of waste organic solvent are generated, which increases overall costs because of the necessity for waste solvent removal. For RL, precipitation may begin by acidifying the fermentation broth to pH 2–3, which reduces product solubility in aqueous solutions and allows the RL to fall out of solution [92]. The efficiency of this process can be enhanced when the RL is initially extracted in ethyl acetate, which can give purities as high as 90% [93].

Additional downstream processing protocols designed to purify glycolipids have involved various chromatographic methods. For example, RL has been purified using adsorption chromatography [94] in which the column was packed with a polystyrene resin, and methanol was used as the mobile phase to elute the adsorbed RL.

Other techniques involved column chromatography using activated silica gel [95] to separate the RL into mono- and di-RL congeners. In this process, chloroform was used initially to remove neutral lipids followed by a sequence of chloroform:methanol mobile phases in varying ratios and volumes to elute the mono-RL followed by di-RL at purities ranging from 95% to 100%. In an extension of this work, Miao et al [88] used a silica gel column with a mobile phase consisting of chloroform/methanol/water for purification of a crude RL extract. Upon elution, two separate peaks appeared: the first representing mono-RL and the second di-RL.

For MEL purification, flash column chromatography has been proven effective [91]. In this procedure, the crude MEL-containing extract is solubilized in chloroform and loaded onto a silica-packed column. MEL was eluted using a gradient of acetone and chloroform, beginning with higher ratios of chloroform to elute fatty acids and acylglycerols, followed by increasing acetone ratios to acquire clean MEL. In addition, this method allows the elution and separation of the specific structural variants present in the crude mixture [96].

Overall, the downstream processing procedures needed for isolating and purifying glycolipids are presently predicated upon solvent extraction and column chromatography and while each purification need is subject to the composition of the desired product for intended application, these additional processes do contribute additional cost to the overall expenditures of the procedure. At industrial scale, the use of solvents will result in large amounts of waste, which is costly to dispose of, and chromatographic methods tend to be expensive and only viable at laboratory scale. Therefore, practical application of second-generation feedstocks for glycolipid biosynthesis still requires advancement to make the processes economically viable. Future advancements in process engineering for both the synthesis and purification efforts as well as genetic engineering to either increase productivity or enhance specific congener synthesis will be necessary to maximize the economic benefits to their use.

4 Potential application of biosurfactants in food industry

4.1 Glycolipids as disrupting agents and antiadhesives: Control of biofilm

Biofilms are microbial aggregates that can be attached to a surface or may form layers without adhering to a surface. The biofilm layers can be monolayer or multilayer depending on how the bacteria interact with each other and the colonized surface. Microbial biofilms can withstand extremely hostile conditions such as starvation and desiccation and can impede physical and chemical agents from contacting the inner microbial layers. This prevention leads to a healthier microbial consortium and, depending on the types of bacteria present, may cause a rise in the incidence of food-borne illness outbreaks initiated by pathogenic microorganisms such as *E. coli*, *S. aureus*, and *Salmonella* sp. [97–100]. The development of biofilms

has been a genuine concern for food manufacturers and consumers because the biofilm layers can also cause product deterioration and pose a risk to public health. Therefore, scientists have been working on finding antibiofilm agents to disrupt biofilm aggregation.

Antibiofilm materials have an active mechanism to detach biofilm microorganisms, thus relegating them to a planktonic (free-living) state, whereas antiadhesive materials prevent the adherence of microorganisms from the initial sessile state (Fig. 4) [101].

RLs have antiadhesive and disrupting properties against the formation of biofilm-producing pathogenic microorganisms. The preformed biofilms of *E. coli* I5, *P. aeruginosa* E26, and *Enterococcus faecalis* I27 on a borosilicate surface were effectively inhibited in the initial adhesion by RLs [102]. This indicated that RLs can successfully restrict the adhesion of microorganisms other than the genus *Pseudomonas* to surfaces. Moreover, RLs were tested as antibiofilm agents against several food-borne pathogens in another study. The antibiofilm potential of the RLs has exhibited promising results [103]. For example, da Silva, I.F., Neves, N.A., e Silveira, B.M.P., Vespermann, K.A.C., Valente, M.E.R. [104] stated that RLs have excellent antibiofilm and antiadhesive properties in distinct stages of food processing, including storage, packaging, and transport.

De Rienzo, M.A.D., Stevenson, P., Marchant, R., Banat, I.M. [105] reported that SLs interrupted the growth of *P. aeruginosa* PAO1 and *E. coli* NCTC 10418 on

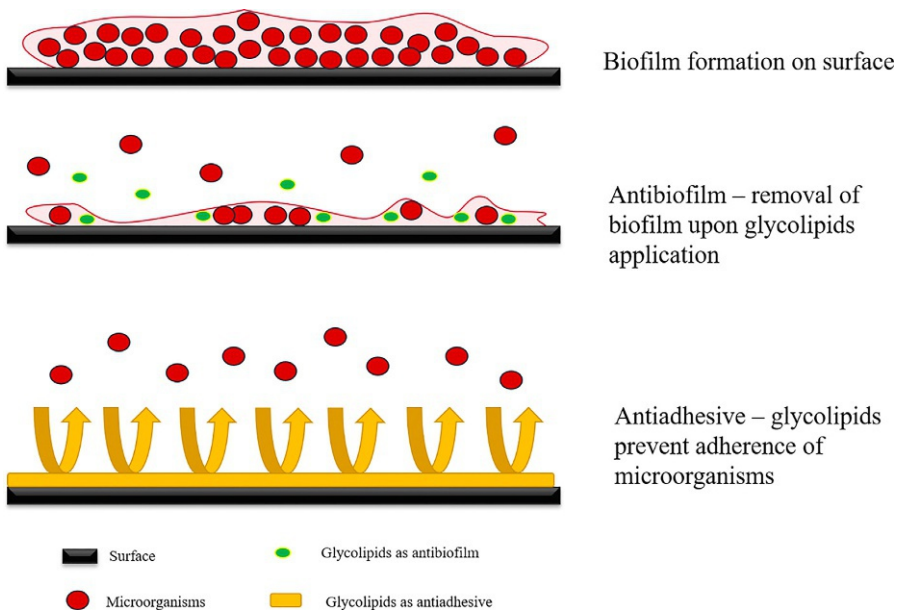

FIG. 4

Biosurfactants as antibiofilm and antiadhesive on surfaces.

preformed biofilms at 5% minimal inhibitory concentration (MIC) values. In addition, the disruption of biofilms of *B. subtilis* BBK006 and *S. aureus* ATCC 9144 with SLs was also accomplished at a 5% MIC [106]. Conversely, Zhang, X., Fan, X., Solaiman, D.K., Ashby, R.D., Liu, Z. [99] evaluated the antiadhesive activity of SLs against *E. coli* O157:H7 in concentration and time-dependent approaches. The authors revealed that the cytoplasmic membrane of *E. coli* was physically disrupted in the presence of SLs. This phenomenon was observed under transmission electron microscopy (TEM). Subsequently, similar authors revealed that SLs inhibited the growth of food-borne pathogens such as *B. subtilis*, *Cupriavidus necator*, and *S. aureus* [99,107].

Recently, trehalolipids produced by *Rhodococcus fascians* showed significant antiadhesive properties against *C. albicans* and *E. coli* at 95% and 70%, respectively [108]. The investigation was completed on polystyrene and silicone surfaces. In addition, the inhibition of biofilms was also observed on Gram-positive and Gram-negative bacteria at approximately 11%–34% inhibition.

Interestingly, the emergence of antibiofilm and antiadhesive compounds has exquisite potential in dealing with biofilm formation of both fungi and bacteria, especially in the food industry. Antibiofilm material possesses added chemical and structural diversity compared to synthetic analogues. Thus, it is safer for humans and stringent in this industry.

4.2 Glycolipids as food preservatives

4.2.1 Antimicrobial activity against food pathogens

Both enzymatically synthesized and microbially derived biosurfactants have been demonstrated to possess antimicrobial qualities. The sugar-fatty esters and methyl sugar derivatives have been shown to have antimicrobial activity toward some pathogenic and spoilage bacteria, such as *Listeria monocytogenes* [109], *Bacillus cereus* [110], *Staphylococcus aureus* [99], spores of *Clostridium perfringenes* [111], *Escherichia coli*, etc. [99]. The MIC can be used to compare the antimicrobial effect. However, caution must be taken when using MIC as an indicator, as MIC determination methods can vary between laboratories. The structures of sugar moieties and fatty acid moieties and the levels of esterification reportedly affect the inhibition effect. For example, Habulin et al. [112] reported that sucrose esters possessed greater antimicrobial activity when compared with fructose fatty acid esters due to higher solubility. The medium and short-chain fatty acid moieties were more effective against bacteria [113] while diesters were less effective as they have limited solubility [110]. The food compositions also serve as factors to determine the antimicrobial efficacy [109].

The mechanisms of biosurfactants' antimicrobial properties appear to be related to their surface activity, rather than to a specific pathway or target. Some biosurfactants are able to insert themselves into the lipid bilayer of the cytoplasmic membrane [114] and alter the rigidity and fluidity of the cell [115]. As the biosurfactants' concentration increases, they begin to form mixed micelles with the membrane

phospholipid, which causes pores to develop within the cell envelope, causing leakage of the cellular content [116]. In general, biosurfactants have been demonstrated more effective against Gram-positive bacteria than Gram-negative bacteria [117]. However, the reasons have not been definitively elucidated.

4.2.2 Antioxidant activity

The antioxidant activity of biosurfactants is a relatively new research area compared with other properties. Lipopeptides and glycolipids produced from various microorganisms have been studied with respect to their antioxidant activity. The antioxidant activity for biosurfactants relies on certain specific structural characteristics including the presence of tyrosine and proline within some peptides, unsaturated fatty acids that can act as electron donors to free radicals [118], and the presence of methionine in the peptides that can impart reducing power [119]. Because of these characteristics, glycolipids, in theory, should have lower antioxidant activity. Therefore, studies were conducted that focused more on lipopeptides than glycolipids. In one case, Basit et al. [120] reported that the lipopeptides from *B. cereus* had very good ferric-reducing activity and DPPH scavenging activity at 2 mg/mL (similar to BHT). Zouari et al. [121] found that lipopeptides from *Bacillus subtilis* showed radical-scavenging activity and facilitated wound healing on female rats. The antioxidant activity could be the presence of specific residues in the peptide ring, which reacted with the DPPH free radicals. Abdollahi et al. [122] compared surfactin (lipopeptide) and RL activity, the reducing power for surfactin was almost twice that of RL. Both biosurfactants' antioxidant activities were much weaker than Vitamin C. The surfactin's scavenging activity was similar to that of BHA and greater than RL. Compared with Vitamin E, both biosurfactants were much weaker. The glycolipid mannosylerythritol-C that was produced by *Pseudozyma hubeiensis* was shown to have high DPPH radical- and superoxide anion-scavenging activity in cultured human skin fibroblasts, due to the existence of unsaturated fatty acids, the number and location of the acetyl group, and the chirality of the erythritol moiety [123]. Because of these properties, both lipopeptides and glycolipids are highly preferred in dermatological products.

4.3 Glycolipids as food emulsifiers

Glycolipids can serve as emulsifiers by accumulating at the oil-water interface, reorienting themselves, and forming viscoelastic films to stabilize the new surface [124]. The emulsification properties in food have been primarily directed toward sugar-fatty acid esters, instead of microbially derived molecules. The earliest family of sugar-fatty acid esters that were approved by FDA for use in the food industry is sucrose esters [125]. The emulsification properties of these molecules are closely related to the HLB values. By changing the degree of esterification, the fatty acid chain length, and the ratio of sucrose esters with different HLB values, the biosurfactants (although chemically synthesized) had a broad range of applications. For example, they can be utilized in oil-in-water emulsions, the fatty acid chain length

is vital for stabilizing the dispersed phase: longer chain lengths stabilize the oil droplets for a longer period due to stronger hydrophobic interactions [126–128]. The solubility of the molecules also determines the stabilizing function. It has been shown that sucrose esters can retain the emulsion droplet size for a longer time than glucose esters, due to the better solubility in aqueous phase [127] that can be better absorbed on the interface.

Sugar-fatty acids have been used to create and stabilize foam by preventing liquid drainage, disproportionation, collapse by lamellar cleavage and Ostwald ripening [129,130]. The foamability is an important function in food such as ice cream, marshmallow, and bakery products. Liang et al. [131] reported that the glucose esters' concentration has positive relation with the foamability, as higher concentrations can cover more of the interface. Unlike stabilizing oil-in-water emulsions, the foamability had a weaker correlation with the HLB values. The medium-chain length fatty acid esters were shown to have the best foamability [128,132]. As for the capacity to stabilize the foam, the glucose esters were much weaker than sucrose esters, probably due to their smaller size and their inability to form viscoelastic films at the interface.

5 The efficiency of glycolipids in comparison with chemical surfactants

Chemical surfactants are derived from petroleum-based products that are composed of 10 or more alkyl groups. In contrast, glycolipid biosurfactants are produced from renewable sources via fermentation, which is more environmentally friendly and safe. Glycolipids possess shorter alkyl groups than chemical surfactants. Despite a structural similarity to chemical surfactants, the short alkyl groups of glycolipid biosurfactants promote aqueous solubility that directly lowers the critical micelle concentration (CMC) values [133]. Low CMC values indicate that a small amount of surfactant is essential to minimize the surface tension.

Polysorbate, sodium dodecyl sulfate (SDS), and polyoxyethylene sorbitan esters of monoglycerides (Tweens) are chemical surfactants commonly used in the food industry [134]. However, glycolipids supplemented with soybean oil have exhibited enhanced performance as food emulsifiers than chemical surfactants, namely Tween 80, polyoxyethylene, SDS, and lauryl benzene sulfonate (LAS) [135]. A review by Sharma, D. [136] highlighted that glycolipids can be utilized in bakery products with outstanding improvement in terms of quality and properties of foods. The author also mentioned that glycolipids are an excellent substitute for the currently used chemical surfactants. Another study also supported this statement by mentioning that because of their physicochemical and biological properties, glycolipids are promising alternatives to chemical surfactants [137].

Glycolipids exhibit outstanding properties compared with chemical surfactants that can be useful in different food industries, from processing to packaging. Furthermore, the combination of glycolipids enhances their characteristics and performance

as multipurpose ingredients or applications. Therefore, substituting glycolipid surfactants over chemical surfactants can reduce toxicity, increase biodegradability, and encourage trust from customers. Overall, glycolipids are considered a safer choice to be utilized, especially in food industries. In addition, biosurfactant use is aligned with the increasing demand for natural ingredients.

6 Challenges and future prospective

As increased efforts emerge to combat greenhouse gas emissions, climate change, and improve environmental quality, biorenewable feedstocks/precursors are being continually sought that might drive research toward a more circular bioeconomy. This has been no more evident than in bioprocesses where microbes are used to produce value-added products. Glycolipids are biobased amphiphilic molecules that have received a lot of attention because of their range of potential applications and their propensity to be synthesized in large amounts (particularly SL). Many of the known glycolipids show surfactant properties that mimic chemical-based surfactants, which makes them particularly intriguing as potential alternatives to petroleum-based analogues. In addition, glycolipids have been demonstrated to have good antimicrobial qualities (particularly against Gram + bacteria) and in the case of at least one glycolipid class, can modulate the mammalian taste response [138,139]. These, as well as other glycolipid-specific applications have increased the research and application interest in all classes of glycolipids. In fact, SL has already been adopted by Ecover as a formulary in surface cleaning solutions [132], by Saraya as an additive (SOFORO) to their Happy Elephant line of dishwashing detergents [133], and by Givaudan (Sopholiance S) as an ingredient in their cosmetic lines [134]. Additionally, with the ever-increasing problem of antibiotic resistance, glycolipids may find an additional application base as a substitute for some of the more commonly used antimicrobial compounds. With all these possibilities, it is becoming increasingly beneficial to find the least expensive routes to synthesize these valuable materials. Efforts to address this issue have been focused on three areas including improving production yields, utilizing high-volume, low-value renewable second-generation substrates, and improving downstream processing. Unfortunately, research has shown that the use of coproduct/waste feedstocks typically results in depressed product yields and introduces additional complications into the production process, the effect of which is to increase overall costs or at least limit the benefits of using inexpensive starting materials. These complications can include compositional nonuniformity among the substrates, which can hinder reproducibility from one feedstock to another. Typically, coproduct/waste streams are not identical depending on the specific process used to generate the coproduct and the efficiency of that process. Since glycolipids are typically synthesized as structural mixtures, any variation in the content of the substrate can result in differences in the congener ratios, which may hinder application potential. This is problematic from the standpoint of cost control.

Another issue in using coproduct/waste streams as feedstocks is the probability for contaminants, which complicates production and downstream processing. For example, crude glycerol typically contains many different components depending on the efficiency of the transesterification process. These components can be alkyl esters, free fatty acids, acylglycerols, ash, water, salts etc. While some of these materials can be utilized by the producing strain to synthesize glycolipids, many cannot and will remain after the production process completes. This results in the need to effectively remove any residual contaminants in a postproduction purification process, which can be laborious and costly.

Another issue pertaining to cost control of the glycolipid production process involves the cost of downstream processing, particularly as it relates to isolation and purification of the glycolipids. Each step in the purification process will increase the overall cost to produce a pure product. Presently, most purification is centered on solvent extraction and precipitation and column chromatography. The problems associated with these actions include the generation of large amounts of waste solvent, which can be costly to dispose of and chromatography processes whose costs rise as larger scales are required. The use of coproduct/waste substrates is a worthwhile effort but currently, there are too many hidden costs that hinder their efficient utilization for cost reduction in glycolipid biosynthesis. In fact, to effectively maximize the process for optimum production at minimal cost, improvements in the production systems either through genetic engineering of the producing strains or process engineering of the fermentation method will be vital. In addition, beneficial developments must also be realized in the purification associated with downstream processes whether that includes methods to simply remove unwanted contaminants or separate different congeners from the glycolipid mixture for specific applications. At any rate, additional work is vital to generate an economical process for glycolipid biosynthesis.

References

[1] C. Stubenrauch, Sugar surfactants—aggregation, interfacial, and adsorption phenomena, Curr. Opin. Colloid Interface Sci. 6 (2001) 160.

[2] G.O. Reznik, et al., Use of sustainable chemistry to produce an acyl amino acid surfactant, Appl. Microbiol. Biotechnol. 86 (2010) 1387.

[3] J. Grüninger, A. Delavault, K. Ochsenreither, Enzymatic glycolipid surfactant synthesis from renewables, Process Biochem. 87 (2019) 45.

[4] V.K. Recke, et al., Lipase-catalyzed acylation of microbial mannosylerythritol lipids (biosurfactants) and their characterization, Carbohydr. Res. 373 (2013) 82.

[5] H. Hosney, E.G. Al-Sakkari, A. Mustafa, I. Ashour, I. Mustafa, A. El-Shibiny, A cleaner enzymatic approach for producing non-phthalate plasticiser to replace toxic-based phthalates, Clean Technol. Environ. Policy 22 (2020) 73–89, https://doi.org/10.1007/s10098-019-01770-5.

[6] S. Bergstrom, H. Theorell, H. Davide, Pyolipic acid, a metabolic product of pseudomonas-pyocyanea, active against mycobacterium-tuberculosis, Arch. Biochem. 10 (1946) 165.

[7] G. Hauser, M.L. Karnovsky, Studies on the production of glycolipide by Pseudomonas aeruginosa, J. Bacteriol. 68 (1954) 645.
[8] F. Jarvis, M. Johnson, A glyco-lipide produced by Pseudomonas aeruginosa, J. Am. Chem. Soc. 71 (1949) 4124.
[9] J.R. Edwards, J.A. Hayashi, Structure of a rhamnolipid from Pseudomonas aeruginosa, Arch. Biochem. Biophys. 111 (1965) 415.
[10] N. Shaw, Bacterial glycolipids, Bacteriol. Rev. 34 (1970) 365.
[11] N.B. Rendell, G.W. Taylor, M. Somerville, H. Todd, R. Wilson, P.J. Cole, Characterisation of Pseudomonas rhamnolipids, Biochim. Biophys. Acta, Lipids Lipid Metab. 1045 (1990) 189.
[12] E. Déziel, F. Lépine, D. Dennie, D. Boismenu, O.A. Mamer, R. Villemur, Liquid chromatography/mass spectrometry analysis of mixtures of rhamnolipids produced by Pseudomonas aeruginosa strain 57RP grown on mannitol or naphthalene, Biochim. Biophys. Acta Mol. Cell Biol. Lipids 1440 (1999) 244.
[13] E. Déziel, F. Lépine, S. Milot, R. Villemur, Mass spectrometry monitoring of rhamnolipids from a growing culture of Pseudomonas aeruginosa strain 57RP, Biochim. Biophys. Acta Mol. Cell Biol. Lipids 1485 (2000) 145.
[14] D. Dubeau, E. Déziel, D.E. Woods, F. Lépine, Burkholderia thailandensis harbors two identical rhl gene clusters responsible for the biosynthesis of rhamnolipids, BMC Microbiol. 9 (2009) 1.
[15] A.M. Abdel-Mawgoud, F. Lépine, E. Déziel, Rhamnolipids: diversity of structures, microbial origins and roles, Appl. Microbiol. Biotechnol. 86 (2010) 1323.
[16] R. Anderson, M. Newman, The chemistry of the lipids of tubercle bacilli: XXXIII. Isolation of trehalose from the acetone-soluble fat of the human tubercle bacillus, J. Biol. Chem. 101 (1933) 499.
[17] T. Suzuki, K. Tanaka, I. Matsubara, S. Kinoshita, Trehalose lipid and α-branched-β-hydroxy fatty acid formed by bacteria grown on n-alkanes, Agric. Biol. Chem. 33 (1969) 1619.
[18] C. Asselineau, J. Asselineau, Trehalose-containing glycolipids, Prog. Chem. Fats Other Lipids 16 (1978) 59.
[19] L. Shao, X. Song, X. Ma, H. Li, Y. Qu, Bioactivities of sophorolipid with different structures against human esophageal cancer cells, J. Surg. Res. 173 (2012) 286.
[20] A. Franzetti, I. Gandolfi, G. Bestetti, T.J. Smyth, I.M. Banat, Production and applications of trehalose lipid biosurfactants, Eur. J. Lipid Sci. Technol. 112 (2010) 617.
[21] A. Kretschmer, F. Wagner, Characterization of biosynthetic intermediates of trehalose dicorynomycolates from Rhodococcus erythropolis grown on n-alkanes, Biochim. Biophys. Acta, Lipids Lipid Metab. 753 (1983) 306.
[22] E. Ristau, F. Wagner, Formation of novel anionic trehalosetetraesters from Rhodococcus erythropolis under growth limiting conditions, Biotechnol. Lett. 5 (1983) 95.
[23] R. Kurane, et al., Chemical structure of lipid bioflocculant produced by Rhodococcus erythropolis, Biosci. Biotechnol. Biochem. 59 (1995) 1652.
[24] M.V. Singer, W. Finnerty, Physiology of biosurfactant synthesis by Rhodococcus species H13-a, Can. J. Microbiol. 36 (1990) 741.
[25] J.C. Philp, M.S. Kuyukina, I.B. Ivshina, S.A. Dunbar, N. Christofi, S. Lang, V. Wray, Alkanotrphic *Rhodococcus ruber* as a biosurfactant producer, Appl. Microbiol. Biotechnol. 59 (2002) 318.
[26] T. Pirog, T. Shevchuk, I. Voloshina, E. Karpenko, Production of surfactants by *Rhodococcus erythropolis* strain EK-1, grown on hydrophilic and hydrophobic substrates, Appl. Biochem. Microbiol. 40 (2004) 470.

[27] J.-S. Kim, M. Powalla, S. Lang, F. Wagner, H. Lünsdorf, V. Wray, Microbial glycolipid production under nitrogen limitation and resting cell conditions, J. Biotechnol. 13 (1990) 257.
[28] Y. Uchida, R. Tsuchiya, M. Chino, J. Hirano, T. Tabuchi, Extracellular accumulation of mono-and di-succinoyl trehalose lipids by a strain of *Rhodococcus erythropolis* grown on n-alkanes, Agric. Biol. Chem. 53 (1989) 757.
[29] K. Gautam, V. Tyagi, Microbial surfactants: a review, J. Oleo Sci. 55 (2006) 155.
[30] P. Gorin, J. Spencer, A. Tulloch, Hydroxy fatty acid glycosides of sophorose from Torulopsis magnoliae, Can. J. Chem. 39 (1961) 846.
[31] A. Tulloch, J. Spencer, P. Gorin, The fermentation of long-chain compounds by torulopsis magnoliae: I. structures of the hydroxy fatty acids obtained by the fermentation of fatty acids and hydrocarbons, Can. J. Chem. 40 (1962) 1326.
[32] A. Tulloch, J. Spencer, Fermentation of long-chain compounds by Torulopsis apicola. IV. Products from esters and hydrocarbons with 14 and 15 carbon atoms and from methyl palmitoleate, Can. J. Chem. 46 (1968) 1523.
[33] D.W. Develter, S.J. Fleurackers, Sophorolipids and rhamnolipids, in: Surfactants From Renewable Resources, vol. 213, 2010.
[34] A. Davila, R. Marchal, N. Monin, J. Vandecasteele, Identification and determination of individual sophorolipids in fermentation products by gradient elution high-performance liquid chromatography with evaporative light-scattering detection, J. Chromatogr. A 648 (1993) 139.
[35] I.N. Van Bogaert, K. Saerens, C. De Muynck, D. Develter, W. Soetaert, E.J. Vandamme, Microbial production and application of sophorolipids, Appl. Microbiol. Biotechnol. 76 (2007) 23.
[36] V. Shah, et al., Sophorolipids, microbial glycolipids with anti-human immunodeficiency virus and sperm-immobilizing activities, Antimicrob. Agents Chemother. 49 (2005) 4093.
[37] R.D. Ashby, A.J. McAloon, D.K. Solaiman, W.C. Yee, M. Reed, A process model for approximating the production costs of the fermentative synthesis of sophorolipids, J. Surfactant Deterg. 16 (2013) 683.
[38] N. Baccile, et al., Development of a cradle-to-grave approach for acetylated acidic sophorolipid biosurfactants, ACS Sustain. Chem. Eng. 5 (2017) 1186.
[39] H.-J. Daniel, R.T. Otto, M. Reuss, C. Syldatk, Sophorolipid production with high yields on whey concentrate and rapeseed oil without consumption of lactose, Biotechnol. Lett. 20 (1998) 805.
[40] H.-J. Daniel, M. Reuss, C. Syldatk, Production of sophorolipids in high concentration from deproteinized whey and rapeseed oil in a two stage fed batch process using Candida bombicola ATCC 22214 and Cryptococcus curvatus ATCC 20509, Biotechnol. Lett. 20 (1998) 1153.
[41] G. Pekin, F. Vardar-Sukan, N. Kosaric, Production of sophorolipids from Candida bombicola ATCC 22214 using Turkish corn oil and honey, Eng. Life Sci. 5 (2005) 357.
[42] A.M. Abdel-Mawgoud, G. Stephanopoulos, Simple glycolipids of microbes: chemistry, biological activity and metabolic engineering, Synth. Syst. Biol. 3 (2018) 3.
[43] T.J. Smyth, M. Rudden, K. Tsaousi, R. Marchant, I.M. Banat, Protocols for the detection and chemical characterisation of microbial glycolipids, in: Hydrocarbon and Lipid Microbiology Protocols, Vol. 29, Springer, 2014.

[44] S.K. Satpute, G.A. Płaza, A. Banpurka, Biosurfactants' production from renewable natural resources: example of innovativeand smart technology in circular bioeconomy, in: Management Systems in Production Engineering, 2017.
[45] I. Mnif, D. Ghribi, Glycolipid biosurfactants: main properties and potential applications in agriculture and food industry, J. Sci. Food Agric. 96 (2016) 4310.
[46] C. Mulligan, Factors influencing the economics of biosurfactants, Biosurfactants 339 (1993).
[47] A. Mustafa, F. Niikura, C. Pastore, H.A. Allam, O.B. Hassan, M. Mustafa, A. Inayat, S. A. Salah, A. Abdel Salam, R. Mohsen, Selective synthesis of alpha monoglycerides by a clean method: techno-economic and environmental assessment, Sustain. Chem. Pharm. 27 (2022) 100690, https://doi.org/10.1016/j.scp.2022.100690.
[48] I.M. Banat, S.K. Satpute, S.S. Cameotra, R. Patil, N.V. Nyayanit, Cost effective technologies and renewable substrates for biosurfactants' production, Front. Microbiol. 5 (2014) 697.
[49] S.M. Ibrahim, A. Mustafa, Synthesis and characterization of new bifunctional SnZrSi oxide catalysts for biodiesel production, J. Mol. Liq. 354 (2022) 118811, https://doi.org/10.1016/j.molliq.2022.118811.
[50] D.K. Solaiman, R.D. Ashby, J.A. Zerkowski, T.A. Foglia, Simplified soy molasses-based medium for reduced-cost production of sophorolipids by Candida bombicola, Biotechnol. Lett. 29 (2007) 1341.
[51] P.A. Felse, V. Shah, J. Chan, K.J. Rao, R.A. Gross, Sophorolipid biosynthesis by Candida bombicola from industrial fatty acid residues, Enzym. Microb. Technol. 40 (2007) 316.
[52] A. Daverey, K. Pakshirajan, Sophorolipids from Candida bombicola using mixed hydrophilic substrates: production, purification and characterization, Colloids Surf. B: Biointerfaces 79 (2010) 246.
[53] A. Daverey, K. Pakshirajan, Kinetics of growth and enhanced sophorolipids production by Candida bombicola using a low-cost fermentative medium, Appl. Biochem. Biotechnol. 160 (2010) 2090.
[54] M. Takahashi, et al., Production of sophorolipid glycolipid biosurfactants from sugarcane molasses using Starmerella bombicola NBRC 10243, J. Oleo Sci. 60 (2011) 267.
[55] A.P. Bhangale, S.D. Wadekar, S.B. Kale, A.P. Pratap, Sophorolipids synthesized using non-traditional oils with glycerol and studies on their surfactant properties with synthetic surfactant, Tenside Surfactant Deterg. 51 (2014) 387.
[56] G.L. Maddikeri, P.R. Gogate, A.B. Pandit, Improved synthesis of sophorolipids from waste cooking oil using fed batch approach in the presence of ultrasound, Chem. Eng. J. 263 (2015) 479.
[57] V.K. Bajaj, U.S. Annapure, Castor oil as secondary carbon source for production of sophorolipids using Starmerella bombicola NRRL Y-17069, J. Oleo Sci. 64 (2015) 315.
[58] M. Konishi, Y. Yoshida, J.-I. Horiuchi, Efficient production of sophorolipids by Starmerella bombicola using a corncob hydrolysate medium, J. Biosci. Bioeng. 119 (2015) 317.
[59] X.-G. Liu, X.-J. Ma, R.-S. Yao, C.-Y. Pan, H.-B. He, Sophorolipids production from rice straw via SO 3 micro-thermal explosion by Wickerhamiella domercqiae var. sophorolipid CGMCC 1576, AMB Express 6 (2016) 1.

[60] A. Samad, J. Zhang, D. Chen, X. Chen, M. Tucker, Y. Liang, Sweet sorghum bagasse and corn Stover serving as substrates for producing sophorolipids, J. Ind. Microbiol. Biotechnol. 44 (2017) 353.
[61] F. Haque, M. Sajid, S.S. Cameotra, M.S. Battacharyya, Anti-biofilm activity of a sophorolipid-amphotericin B niosomal formulation against Candida albicans, Biofouling 33 (2017) 768.
[62] M.U.H. Shah, M. Sivapragasam, M. Moniruzzaman, M.M.R. Talukder, S.B. Yusup, M. Goto, Production of sophorolipids by Starmerella bombicola yeast using new hydrophobic substrates, Biochem. Eng. J. 127 (2017) 60.
[63] P. Jiménez Peñalver, Sophorolipids Production by Solid-State Fermentation: From Lab-Scale to Pilot Plant, Universitat Autònoma de Barcelona, 2017.
[64] V.J. Parekh, V.B. Patravale, A.B. Pandit, Mango kernel fat: a novel lipid source for the fermentative production of sophorolipid biosurfactant using Starmerella Bombicola NRRL-Y 17069, Ann. Biol. Res. 3 (2012) 1798.
[65] M.M. Rashad, M. Nooman, M. Ali, A. Al-Kashef, A. Mahmoud, Production, characterization and anticancer activity of Candida bombicola sophorolipids by means of solid state fermentation of sunflower oil cake and soybean oil, Grasas Aceites 65 (2014), e017.
[66] M.M. Rashad, A. Al-Kashef, M. Nooman, A. El-din-Mahmoud, Co-utilization of motor oil waste and sunflower oil cake on the production of new sophorolipids by Candida bombicola NRRL Y-17069, Res. J. Pharm., Biol. Chem. Sci. 5 (2014) 1515.
[67] P. Jiménez-Peñalver, T. Gea, A. Sánchez, X. Font, Production of sophorolipids from winterization oil cake by solid-state fermentation: optimization, monitoring and effect of mixing, Biochem. Eng. J. 115 (2016) 93.
[68] M. Nooman, M. Mahmoud, A. Al-Kashef, M. Rashad, Hypocholesterolemic impact of newly isolated sophorolipids produced by microbial conversion of safflower oil cake in rats fed high-fat and cholesterol diet, Grasas Aceites 68 (2017), e212.
[69] A. Mustafa, Lipase catalyzed reactions: a promising approach for clean synthesis of oleochemicals, in: Sustainable Solutions for Environmental Pollution: Waste Management and Value-Added Products, vol. 417, 2021.
[70] V. Morya, J.-H. Park, T.J. Kim, S. Jeon, E.-K. Kim, Production and characterization of low molecular weight sophorolipid under fed-batch culture, Bioresour. Technol. 143 (2013) 282.
[71] R.D. Ashby, A. Nuñez, D.K. Solaiman, T.A. Foglia, Sophorolipid biosynthesis from a biodiesel co-product stream, J. Am. Oil Chem. Soc. 82 (2005) 625.
[72] R.D. Ashby, D.K. Solaiman, T.A. Foglia, The use of fatty acid esters to enhance free acid sophorolipid synthesis, Biotechnol. Lett. 28 (2006) 253.
[73] I.M. Ramírez, et al., Rhamnolipid and surfactin production from olive oil mill waste as sole carbon source, Bioresour. Technol. 198 (2015) 231.
[74] S. George, K. Jayachandran, Production and characterization of rhamnolipid biosurfactant from waste frying coconut oil using a novel P seudomonas aeruginosa D, J. Appl. Microbiol. 114 (2013) 373.
[75] S. George, K. Jayachandran, Analysis of rhamnolipid biosurfactants produced through submerged fermentation using orange fruit peelings as sole carbon source, Appl. Biochem. Biotechnol. 158 (2009) 694.
[76] M. Nitschke, S.G. Costa, R. Haddad, L.A.G. Gonçalves, M.N. Eberlin, J. Contiero, Oil wastes as unconventional substrates for rhamnolipid biosurfactant production by Pseudomonas aeruginosa LBI, Biotechnol. Prog. 21 (2005) 1562.

[77] M. Benincasa, J. Contiero, M. Manresa, I. Moraes, Rhamnolipid production by Pseudomonas aeruginosa LBI growing on soapstock as the sole carbon source, J. Food Eng. 54 (2002) 283.
[78] A. Abalos, A. Pinazo, M. Infante, M. Casals, F. Garcia, A. Manresa, Physicochemical and antimicrobial properties of new rhamnolipids produced by Pseudomonas a eruginosa AT10 from soybean oil refinery wastes, Langmuir 17 (2001) 1367.
[79] A.M. Abdel-Mawgoud, F. Lépine, E. Déziel, A stereospecific pathway diverts β-oxidation intermediates to the biosynthesis of rhamnolipid biosurfactants, Chem. Biol. 21 (2014) 156.
[80] W. Bednarski, M. Adamczak, J. Tomasik, M. Płaszczyk, Application of oil refinery waste in the biosynthesis of glycolipids by yeast, Bioresour. Technol. 95 (2004) 15.
[81] W. Bednarski, M. Narwojsz, M. Adamczak, R. Nawotka, Carbon-source-dependent synthesis and composition of biosurfactant synthesized by Pseudozyma Antarctica, Environ. Biotechnol. 2 (2006) 31.
[82] E. Dzięgielewska, M. Adamczak, Evaluation of waste products in the synthesis of surfactants by yeasts, Chem. Pap. 67 (2013) 1113.
[83] N.T. Faria, M. Santos, C. Ferreira, S. Marques, F.C. Ferreira, C. Fonseca, Conversion of cellulosic materials into glycolipid biosurfactants, mannosylerythritol lipids, by Pseudozyma spp. under SHF and SSF processes, Microb. Cell Factories 13 (2014) 1.
[84] N.T. Faria, M.V. Santos, P. Fernandes, L.L. Fonseca, C. Fonseca, F.C. Ferreira, Production of glycolipid biosurfactants, mannosylerythritol lipids, from pentoses and D-glucose/D-xylose mixtures by Pseudozyma yeast strains, Process Biochem. 49 (2014) 1790.
[85] N.T. Faria, S. Marques, C. Fonseca, F.C. Ferreira, Direct xylan conversion into glycolipid biosurfactants, mannosylerythritol lipids, by Pseudozyma Antarctica PYCC 5048T, Enzym. Microb. Technol. 71 (2015) 58.
[86] S. Mukherjee, P. Das, R. Sen, Towards commercial production of microbial surfactants, Trends Biotechnol. 24 (2006) 509.
[87] S.L. Roelants, et al., Towards the industrialization of new biosurfactants: biotechnological opportunities for the lactone esterase gene from Starmerella bombicola, Biotechnol. Bioeng. 113 (2016) 550.
[88] S. Miao, N. Callow, S.S. Dashtbozorg, J.-L. Salager, L.-K. Ju, Ethylation of Di-rhamnolipids: a green route to produce novel sugar fatty acid nonionic surfactants, J. Surfactant Deterg. 17 (2014) 1069.
[89] J. Hubert, et al., New perspectives for microbial glycolipid fractionation and purification processes, C. R. Chim. 15 (2012) 18.
[90] U. Rau, L.A. Nguyen, H. Roeper, H. Koch, S. Lang, Downstream processing of mannosylerythritol lipids produced by Pseudozyma aphidis, Eur. J. Lipid Sci. Technol. 107 (2005) 373.
[91] D. Kitamoto, S. Akiba, C. Hioki, T. Tabuchi, Extracellular accumulation of mannosylerythritol lipids by a strain of Candida antarctica, Agric. Biol. Chem. 54 (1990) 31.
[92] A. Weber, T. Zeiner, Purification of biosurfactants, in: Biosurfactants: Production and Utilization—Processes, Technologies, and Economics, vol. 159, CRC Press, 2014, p. 129.
[93] Y.-H. Wei, C.-L. Chou, J.-S. Chang, Rhamnolipid production by indigenous Pseudomonas aeruginosa J4 originating from petrochemical wastewater, Biochem. Eng. J. 27 (2005) 146.

[94] E. Haba, A. Pinazo, O. Jauregui, M. Espuny, M.R. Infante, A. Manresa, Physicochemical characterization and antimicrobial properties of rhamnolipids produced by Pseudomonas aeruginosa 47T2 NCBIM 40044, Biotechnol. Bioeng. 81 (2003) 316.
[95] L. Sim, O. Ward, Z. Li, Production and characterisation of a biosurfactant isolated from Pseudomonas aeruginosa UW-1, J. Ind. Microbiol. Biotechnol. 19 (1997) 232.
[96] T. Fukuoka, T. Morita, M. Konishi, T. Imura, D. Kitamoto, Characterization of new types of mannosylerythritol lipids as biosurfactants produced from soybean oil by a basidiomycetous yeast, Pseudozyma shanxiensis, J. Oleo Sci. 56 (2007) 435.
[97] P. Fernandes, J. São José, E. Zerdas, N. Andrade, C. Fernandes, L. Silva, Influence of the hydrophobicity and surface roughness of mangoes and tomatoes on the adhesion of Salmonella enterica serovar typhimurium and evaluation of cleaning procedures using surfactin, Food Control 41 (2014) 21.
[98] K.E. Heiman, R.K. Mody, S.D. Johnson, P.M. Griffin, L.H. Gould, Escherichia coli O157 outbreaks in the United States, 2003–2012, Emerg. Infect. Dis. 21 (2015) 1293.
[99] X. Zhang, et al., Inactivation of Escherichia coli O157: H7 in vitro and on the surface of spinach leaves by biobased antimicrobial surfactants, Food Control 60 (2016) 158.
[100] E.J. Gudiña, L.R. Rodrigues, Research and production of biosurfactants for the food industry, Bioprocess. Biomol. Prod. 125 (2019).
[101] K. Paraszkiewicz, M. Moryl, G. Płaza, D. Bhagat, S.K. Satpute, P. Bernat, Surfactants of microbial origin as antibiofilm agents, Int. J. Environ. Health Res. 31 (2021) 401.
[102] J. Carrazco-Palafox, B.E. Rivera-Chavira, J.R. Adame-Gallegos, L.M. Rodríguez-Valdez, E. Orrantia-Borunda, G.V. Nevárez-Moorillón, Rhamnolipids from Pseudomonas aeruginosa Rn19a modifies the biofilm formation over a borosilicate surface by clinical isolates, Coatings 11 (2021) 136.
[103] D. Sharma, E. Gupta, J. Singh, P. Vyas, D.S. Dhanjal, Microbial biosurfactants in food sanitation, in: Sustainable Food Systems From Agriculture to Industry, vol. 341, Elsevier, 2018.
[104] I.F. da Silva, N.A. Neves, B.M.P. e Silveira, K.A.C. Vespermann, M.E.R. Valente, Microbial biosurfactants for contamination of food processing, in: Microbial Biosurfactants, 11, Springer, 2021.
[105] M.A.D. De Rienzo, P. Stevenson, R. Marchant, I.M. Banat, Antibacterial properties of biosurfactants against selected gram-positive and-negative bacteria, FEMS Microbiol. Lett. 363 (2016).
[106] M.A.D. De Rienzo, I.M. Banat, B. Dolman, J. Winterburn, P.J. Martin, Sophorolipid biosurfactants: possible uses as antibacterial and antibiofilm agent, New Biotechnol. 32 (2015) 720.
[107] X. Zhang, R.D. Ashby, D.K. Solaiman, Y. Liu, X. Fan, Antimicrobial activity and inactivation mechanism of lactonic and free acid sophorolipids against Escherichia coli O157: H7, Biocatal. Agric. Biotechnol. 11 (2017) 176.
[108] T. Janek, A. Krasowska, Ż. Czyżnikowska, M. Łukaszewicz, Trehalose lipid biosurfactant reduces adhesion of microbial pathogens to polystyrene and silicone surfaces: an experimental and computational approach, Front. Microbiol. 9 (2018) 2441.
[109] Y. Chen, B. Nummer, M. Walsh, Antilisterial activity of lactose monolaurate in milk, drinkable yogurt and cottage cheese, Lett. Appl. Microbiol. 58 (2014) 156.
[110] M. Ferrer, et al., Synthesis of sugar esters in solvent mixtures by lipases from Thermomyces lanuginosus and Candida antarctica B, and their antimicrobial properties, Enzym. Microb. Technol. 36 (2005) 391.

[111] L. Thomas, E. Davies, J. Delves-Broughton, J. Wimpenny, Synergist effect of sucrose fatty acid esters on nisin inhibition of Gram-positive bacteria, J. Appl. Microbiol. 85 (1998) 1013.
[112] M. Habulin, S. Šabeder, Ž. Knez, Enzymatic synthesis of sugar fatty acid esters in organic solvent and in supercritical carbon dioxide and their antimicrobial activity, J. Supercrit. Fluids 45 (2008) 338.
[113] J.J. Kabara, D.M. Swieczkowski, A.J. Conley, J.P. Truant, Fatty acids and derivatives as antimicrobial agents, Antimicrob. Agents Chemother. 2 (1972) 23.
[114] S. Thennarasu, et al., Membrane permeabilization, orientation, and antimicrobial mechanism of subtilosin A, Chem. Phys. Lipids 137 (2005) 38.
[115] K. Ren, P. Patra, B.P. Lamsal, A. Mendonca, Antimicrobial mechanism of "fatty acyl glutamic acid" biosurfactant against *Escherichia coli* O157: H7 and *Listeria monocytogenes* involves disruption of the cytoplasmic membrane, ACS Food Sci. Technol. (2021).
[116] A. Helenius, K. Simons, Solubilization of membranes by detergents, Biochim. Biophys. Acta 415 (1975) 29.
[117] S.Y. Shao, et al., Lipase-catalyzed synthesis of sucrose Monolaurate and its antibacterial property and mode of action against four pathogenic Bacteria, Molecules 23 (2018) 1118.
[118] M. Valko, et al., Free radicals and antioxidants in normal physiological functions and human disease, Int. J. Biochem. Cell Biol. 39 (2007) 44.
[119] O. Tabbene, et al., Antioxidative and DNA protective effects of bacillomycin D-like lipopeptides produced by B38 strain, Appl. Biochem. Biotechnol. 168 (2012) 2245.
[120] M. Basit, M.H. Rasool, S.A.R. Naqvi, M. Waseem, B. Aslam, Biosurfactants production potential of native strains of *Bacillus cereus* and their antimicrobial, cytotoxic and antioxidant activities, Pak. J. Pharm. Sci. 31 (2018).
[121] R. Zouari, et al., Evaluation of dermal wound healing and in vitro antioxidant efficiency of *Bacillus subtilis* SPB1 biosurfactant, Biomed. Pharmacother. 84 (2016) 878.
[122] S. Abdollahi, et al., Evaluation of anti-oxidant and anti-biofilm activities of biogenic surfactants derived from *Bacillus amyloliquefaciens* and *Pseudomonas aeruginosa*, Iran. J. Pharm. Res. 19 (2020) 115.
[123] M. Takahashi, T. Morita, T. Fukuoka, T. Imura, D. Kitamoto, Glycolipid biosurfactants, mannosylerythritol lipids, show antioxidant and protective effects against H2O2-induced oxidative stress in cultured human skin fibroblasts, J. Oleo Sci. 61 (2012) 457.
[124] D. McClements, Emulsion stability, in: D. McClements (Ed.), Food Emulsions Principles, Practices and Techniques, CRC Press, Boca Raton, 2004.
[125] FDA, The Generally Recognized as Safe (GRAS) Status of Sucrose Monoesters of Lauric Acid, Palmitic Acid and Stearic Acid as Emulsifying Agents for Flavors Used in Fruit Flavored Beverages, 2008.
[126] M. Ferrer, et al., Comparative surface activities of di-and trisaccharide fatty acid esters, Langmuir 18 (2002) 667.
[127] K. Ren, B.P. Lamsal, Synthesis of some glucose-fatty acid esters by lipase from Candida antarctica and their emulsion functions, Food Chem. 214 (2017) 556.
[128] D. An, X. Zhang, F. Liang, M. Xian, D. Feng, Z. Ye, Synthesis, surface properties of glucosyl esters from renewable materials for use as biosurfactants, Colloids Surf. A Physicochem. Eng. Asp. 577 (2019) 257.

[129] D. Carp, G. Bartholomai, P. Relkin, A. Pilosof, Effects of denaturation on soy protein-xanthan interactions: comparison of a whipping-rheological and a bubbling method, Colloids Surf. B Biointerfaces 21 (2001) 163.
[130] L. Panizzolo, L. Mussio, M.C. Añon, A kinetic description for the destabilization process of protein foams, Int. J. Food Prop. 15 (2012) 60.
[131] M.-Y. Liang, Y. Chen, M.G. Banwell, Y. Wang, P. Lan, Enzymatic preparation of a homologous series of long-chain 6- O-acylglucose esters and their evaluation as emulsifiers, J. Agric. Food Chem. 66 (2018) 3949.
[132] T.M. Mata, F. Pinto, N. Caetano, A.A. Martins, Economic and environmental analysis of animal fats acidity reduction by enzymatic esterification, J. Clean. Prod. 184 (2018) 481.
[133] S. De, S. Malik, A. Ghosh, R. Saha, B. Saha, A review on natural surfactants, RSC Adv. 5 (2015) 65757.
[134] A. Molet-Rodríguez, L. Salvia-Trujillo, O. Martín-Belloso, Beverage emulsions: key aspects of their formulation and physicochemical stability, Beverages 4 (2018) 70.
[135] A.L.S. Coelho, P.E. Feuser, B.A.M. Carciofi, C.J. de Andrade, D. de Oliveira, Mannosylerythritol lipids: antimicrobial and biomedical properties, Appl. Microbiol. Biotechnol. 104 (2020) 2297.
[136] D. Sharma, Industrial applications of biosurfactants, in: Biosurfactants: Greener Surface Active Agents for Sustainable Future, vol. 115, Springer, 2021.
[137] A.D.J.S. Cortã, Surfactants of microbial origin and its application in foods, Sci. Res. Essays 15 (2020) 11.
[138] M.H. Ozdener, et al., Sophorolipid biosurfactants activate taste receptor type 1 member 3-mediated taste responses and block responses to bitter taste in vitro and in vivo, J. Surfactant Deterg. 22 (2019) 441.
[139] D. Solaiman, R.D. Ashby, M.H. Ozdener, A. Bachmanov, Compositions Containing a Bitter Tastant and at Least One Sophorolipid, and Methods of Reducing Bitter Taste Attributed to a Bitter Tastant in an Edible Composition, Google Patents, 2018.

CHAPTER

Future applications of biosurfactant in food industry

16

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1 Introduction

Biosurfactant (BST) is an important microbial compound of plant and animal origins, which has several health benefits including antioxidant, antimicrobial, and antiadhesive properties [1]. It is biodegradable and has great resistance to various degree of temperatures, hence can effectively be applied in the food industry [2–5]. In addition, it has high surface tension and forming ability, which make them excellent materials for skin care formulations [6–8]. This uniqueness makes it promising material for many food processing operations in the industry [9]. Currently, some food products based on BST blend of rhamnolipid and enzymes in aqueous solutions with different purity levels or in a semisolid form can be found in the international market [4,10,11]. However, based on the U.S. Environmental Protection Agency approval, the rhamnolipids from *P. aeruginosa* are the most promising class of BSTs used in food products and cosmetics [10,12].

Structurally, according to chemical contents, BST is grouped into two, namely the particulate and the polymeric forms. The first classification includes the phospholipids, glycolipids, and lipoproteins, while the second has the structure of a typical fatty acid [13–16]. In terms of application, the particulate type has great antioxidant potentials and is regularly exposed to food surfaces, while the polymeric type has great antimicrobial properties, thus avoiding food spoilage or deterioration [11]. The commercial application many of the BSTs is well known in the food processing, especially in the formulation of new products with certain desirable sensory features, textures, and other quality characteristics. For example, the BST obtained from *Bacillus subtilis* has proven to be effective in enhancing the quality of dough for cookies production [17]. In fact, the inclusion of surfactants made from carrot [18] into the dough will even improve the flavor and pasting properties of the food product. The material has found other similar applications in cake, salad dressing, ice-cream, bread, and other baked foods with an emphasis on quality improvement

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as well as shelf life extension. Thus, the commercial application of the BST in the food industry cannot be overemphasized.

The current chapter presents a novel approach of applications of the BST in the food industry. It also presents an overview of the application of the material in the processing and formulation of food product, diary food, and packaging. The prospect of the materials for other industrial applications, such as water purification, tubber and grains storage, biofilm synthesis for water packaging, was suggested. The potential application of the BST in the design of biodegradable packaging materials for packaging liquid foods is suggested here as a way-forward attempt to replace non-degradable polyethylene in similar applications. Also, the application of some microorganisms, such as r*hodococcus*, *pseudomonas*, and *burkholderia*, which can prevail in cold condition to produce novel BST for biotechnological modification of food products during freezing is presented in this chapter.

2 Production of biosurfactant

The BST can be produced through biological and biochemical approaches. The biological method involves the use of bacteria to act on an organic product to generate the material. They are commonly produced by hydrocarbon-degrading bacteria acting on the solution of sucrose, glucose, ethanol, or glycerol. The production of the BST is either associated with growth limitation or resting condition or immobilized cells or is produced with the addition of precursors. Generally, they are produced by the influence of carbon, nitrogen, phosphate sources and other environmental factors. Different types of microorganisms have been used in the synthesis of biological surfaces for various applications in the food industry [19–24]. A typical biological method for producing plant-derived BST used in the formulation of cosmetic products involves extraction of *Malpighia ermaginata*, glycine max, and *Chenopodium quinoa* BSTs from the seeds of quinoa, soybeans, and acerola using hydroalcoholic methodology with 50% alcohol and assisted by ultrasound [8]. The biochemical method of synthesis may involve the application of candida antarctica lipase b enzymes to produce sorbitol-based BSTs through fermentation [25]. This involves enzymic reaction between the sorbitol and fatty acids at temperature of 50°C in an orbital shaker incubator operating at 200 rpm for 48 h [24]. Other biochemical approaches may involve batch or continuous fermentation processes [25]. This can be achieved by fractionating the adsorption column or foam, applying an air-lift fermenter, and using two-phase aqueous fermentation, as shown in Fig. 1 [27].

3 Applications of biosurfactant in food industry

3.1 Application in food products processing and formulation

The BST has been applied successfully in the processing and formulation of food, including cookies, salad dressing, cakes, ice-cream, and bread. The material

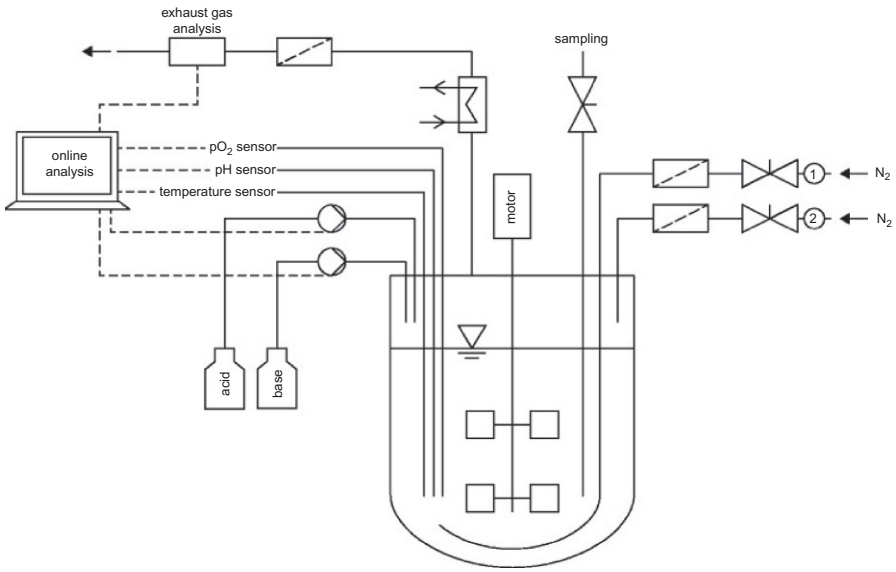

FIG. 1

Fermentation procedure used for production of biosurfactant [26].

is mainly applied in food for surface sterilization to prevent the multiplication or curtail the growth of spoilage microorganisms [28]. The material assists in heaping fat molecules together, improving texture and rheological behavior of oil/fat-based foods. The BSTs such as rhamnoipids and sophrolipods have been demonstrated to improve the properties of confectionaries and salad [29]. In meat-based foods, the surfactant aids in emulsification of the incomplete processed fat tissue. However, production of the material involves microbial colonization from different biomaterials such as polysaccharides [30]. Although biomaterials comprise pathogenic and nonpathogenic microorganisms, the use of nonpathogenic microorganisms is practically acceptable, while prevention of pathogenic microorganisms to the nourishment surfaces of food items has been a major concern [30]. Thus, the application of those synthesized by nonpathogenic organisms has attracted much attention, particularly those applicable to the food industries [31,32]. *Lactobacillus*, in particular, is considered generally as safe microorganism for food applications [33]. The BST produced by *Bacillus cereus* from waste frying oil is also a nonpathogenic material used as an emulsifier in cookie preparations [3,34]. The addition of this BST will normally promote an increase in the quantity of lipids in the cookies due to the presence of fatty acids in its nonpolar portion, which is a biochemical characteristic of the biomolecule [35,36], as shown in Fig. 2.

The BSTs applied to foods have helped preserve the product or extend its shelf life [4,11,37–39]. Typically, Zouari et al. [17] reported the application of the BST into dough to improve its texture and consistency. Similarly, the chemical, sensory,

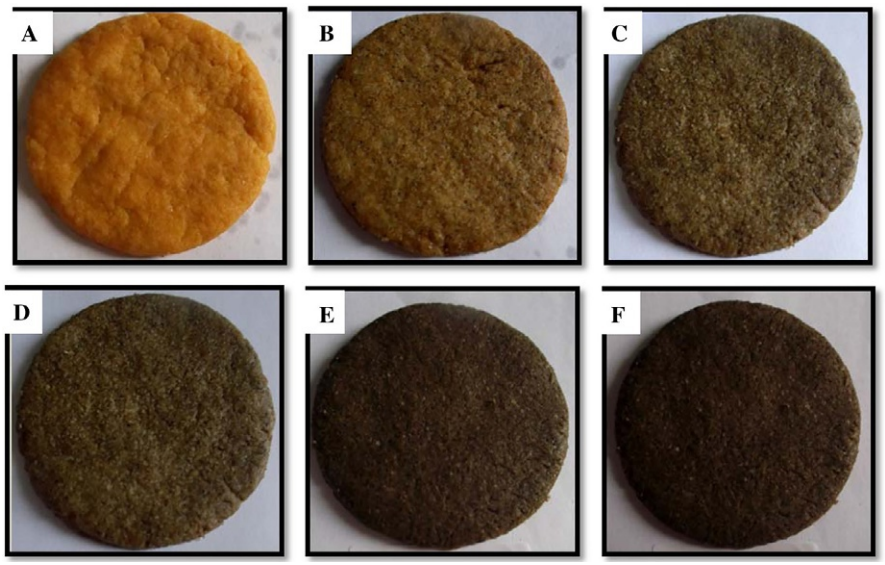

FIG. 2

Effect of biosurfactant on cookies texture [17].

and physical characteristics of biscuits can be improved by the addition of the BST made from *Zizyphus lotus* L. [40]. Also, Ahmed et al. [18] incorporated BST made from carrot pomace into wheat flour to improve the quality of cookie products produced from it. The BST produced by *Candida utilis* was reported by Ribeiro et al. [41] to significantly improve the quality of cookies. Also, Sharama and Sharma [42] reviewed the application of microbial surfactant as an active food ingredient. The authors concluded that the material could help enhance quality and safety when added to food. The role of the BST derived from *Bacillus licheniformis* in preventing the progress of heavy metals in leafy vegetable in agricultural products processing was studied by Giri et al. [43]. The authors reported that the material can effectively remove heavy metals in vegetables such as carrot, radish, ginger, and potato. The quality of fruit juices has also been improved by the addition of a glycolipid-based BST obtained from the strain of *Wickerhamomyces anomalus* yeast, which effectively inactivates spoilage microorganisms in the food [44].

Molecules that are used to prevent other biomolecules from oxidizing or changing from one state to another are called antioxidants [9,45–47]. When biomolecules are oxidized, structural or cellular damage may occur due to the formation of free radicals in the process [47–50]. The detrimental effect is so enormous that ordinary cellular antioxidants may not be sufficient to mitigate this process due to oxidative stress maybe caused by environmental factors such as the UV light [5,49,50]. For this reason, the use of the BSTs, which have improved physical, chemical, and structural characteristics compared with the ordinary antioxidants, has been recommended.

The BSTs isolated from *Bacillus* strains have been reported to exhibit strong antioxidation ability on the cell structures of food products, thereby effectively preserving the nutrients that maybe lost due to oxidation [48]. Takahashi et al. [6] demonstrated the ability of gycolipid BST to show active antioxidizing behavior against fat in scare products by preventing oxidation of the product. They seem to act like an oxygen scavenger to prevent oxidation, thus implying that it could be used as a substitute for natural antioxidants.

3.2 Application in dairy food process equipment maintenance

The formulation of dairy products such milk, meat, and cheese is done with the objective of meeting the consumer demand while ensuring that nutrients [51] and other ingredients remain preserved in the process [52]. The potentials of the BST in diary food formulation are enhanced by its high surface-to-volume ratio, which will reduce the surface tension of the product. In the dairy industry, the BST is used to address the incidence of pipeline fouling, which may occur due to the accumulation of microorganisms from equipment overuse [53]. The dairy equipment such as the heat exchangers have often experienced defect from fouling, and this has often affected the processing capacity of the equipment for long-time usage. Thus, an effective fouling management or maintenance is critical for continuous operation of the equipment [2]. The BSTs produced from hemophilic dairy *Streptococci* were examined by Tang et al. [54] to see if they may help control fouling of heat-exchanger plates, as shown in Fig. 3. Extracellular BSTs produced by *Streptococcus thermophilus* cause adherent cells to detach and act as an anti-BST layer on a

FIG. 3

Typical case of fouling in the diary food industry [54].

substrate [55]. There are, however, no enough studies to showcase the effect of the BST on the microorganisms causing fouling and basic equipment breakdown in the dairy food industry. Consequently, the commercial application of the biological surfactants in this technology remains for the future.

3.3 Application in food storage facility maintenance

The BSTs have found application in food storage facility design [51] and maintenance. It is used as a cleaning agent for heavy oils that can interfere with the normal stability of the stored products [56]. The storage systems can be designed using biological surfactants, which could improve on their ability to control the multiplication of microorganisms in the food [10,57–61]. The material has an inherent ability to reduce the viscosity of the oils within the facilities, thereby effectively aiding the cleaning of the system. This ability was demonstrated by Wei et al. [62] and Banat et al. [63], who reported that the BST will normally reduce the viscosity of the oil to cleanup a sorbent storage tank. Similarly, the viscosity reduction ability of the material was reported by Makkar and Cameotra [64], in their work on the application of unconventional substrates production of the material. Knowledge on the commercial application of this technology is greatly limited, as only few reports are available in literature. Further research is needed especially in the design of storage system for studying the interrelationship between the micro- and macro-storage environments as they may affect food storage stability and the influence of the BSTs.

3.4 Application in food packaging

The BST has also found application in the design of food packaging materials because of its capacity to improve the shelf life of the food products [65,66]. The material exhibits similar behavior to the nanoparticles commonly used to improve the physical characteristics of the biodegradable packaging material [24]. The BST can be added to thermoplastic during synthesis to cause a uniform dispersion of the nanoparticles in the film matrix to create a surface tension capable enhancing the hydrophilic nature of the packaging material. The addition of the material into active cassava starch biodegradable has been reported to enhance the keeping ability of the material [67]. The effect of plasticizer and surfactants on the stability of polyvinyl alcohol/chitosan and corn starch films was studied by Stachowiak et al. [68] and Ortega-Toro et al. [69], respectively. Also, the role of BSTs in the structural development of biofilms using *Pseudomonas aeruginosa* was reported by Pamp et al. [12]. The authors noted that the presence of the BST in the film enhances the surface hydrophilicity and roughness of the resulting thermoplastic, as shown in Fig. 4. Although further test may be needed to improve the film strength and appearance, it however has high potential for application in liquid food packaging [70–72]. Also, study is needed to design a BST-based biodegradable system for packaging liquid foods, which can effectively contain the food product while

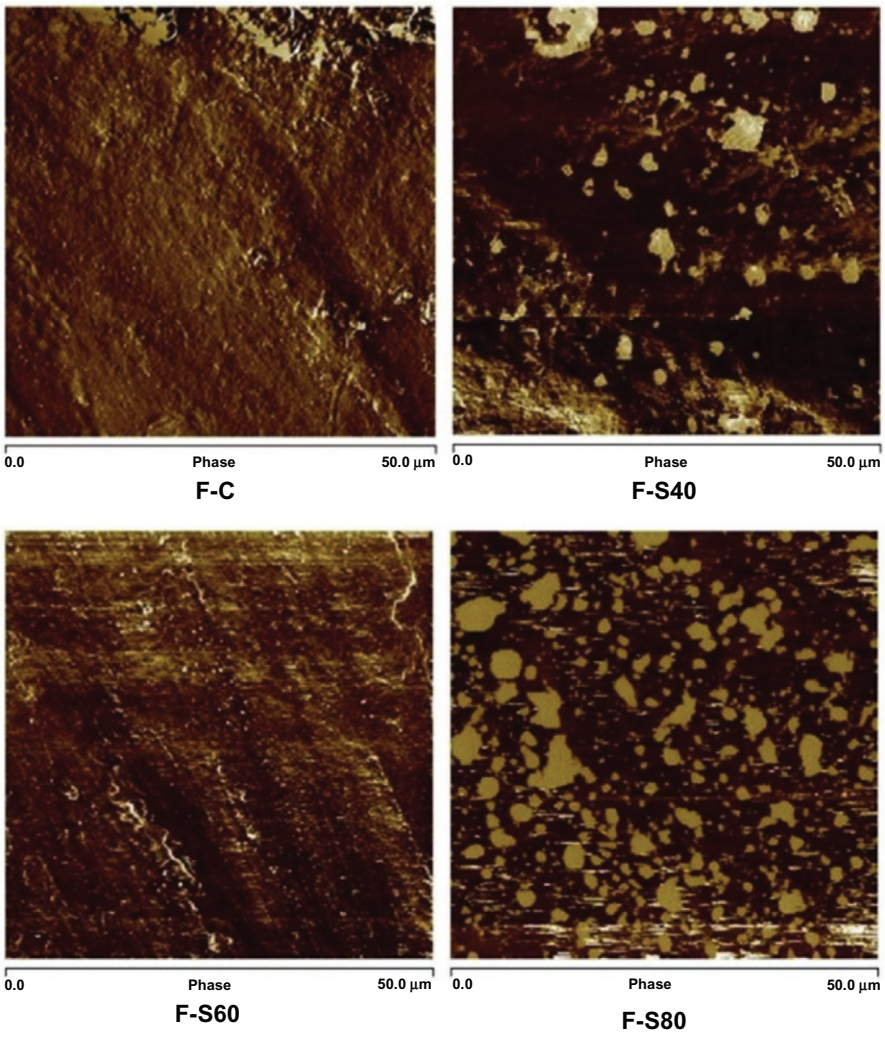

FIG. 4

Morphology of starch with and without biosurfactant at different span levels [69].

addressing the environmental challenges posed by the nondegradable materials such as nylon, low- and high-density polyethylene. It is essential to study the effect of the BST inclusion on the hydrophilicity, surface roughness, toughness, and strength of the thermoplastic material. To the best of the authors' knowledge, there is no available research on BST-based thermoplastic film for commercial application in the packaging of water and liquid foods.

3.5 Application in food freezing

Freezing of food is an essential aspect of food processing and preservation. This is usually done by lowering the temperature of the system to approximately <5°C with a view to inhibiting the growth of the microorganisms and promoting the shelf life of a food product. The commercial application of this technology is widely known especially for the temperate food products. However, it is not suitable for tropical foods because of the long-term effect it can have on the product. For this reason, certain degree of pretreatment is usually needed to protect the food and enhance the efficiency of the system. For instance, fruits such as mango and banana can be stored conveniently in frozen environment for a long period when pretreated with glucose solution. The fruits are frozen in syrup and in dry sugar pack to exclude air and to prevent desiccation and oxidation. The glucose material will essentially spread the cell of the product and decrease its freezing point to accommodate the temperature of the system. This will help avoid the occurrence of chilling injury and maintain the quality after storage. The application of the BST for this application is not well known, but the material has the potential to act on the product cell membrane and even improve its storability because of its emulsifying and antioxidation abilities. These properties will in fact aid the BST to prevent the oxidation of the food product under freezing condition. Certain microorganisms that can prevail in the cold environment, such as *Rhodococcus* [73], *Pseudomonas* [74], *Bacillus* [73], *Sphingomonas*, and *Burkholderia* [75], can be used to produce the BSTs that effectively adapt for food freezing applications. Accordingly, Perfumo et al. [76] and Gesheva et al. [77] reported that the BSTs from cold adapted microorganisms can interact with water, ice, hydrophobic compounds, and gases at freezing temperatures and be used to prolong the shelf life of food product. To the best of the authors' knowledge, the commercial application of the BST freezing technology for food processing and formulation is not yet exploited.

4 Challenges and prospects

After extensive literature search, it is obvious that the application of the BSTs in the food industry is mostly as antioxidants, antiadhesives, and antimicrobial agents in the food and dairy processing and formulation. However, the material has potential for application in water purification, tubber and grains storage, and biofilm synthesis for liquid food packaging. The potential application of the material in the design of biodegradable systems for packaging liquid foods, which can effectively contain the product while addressing the environmental challenges posed by the nondegradable materials such as nylon, low- and high-density polyethylene, has not yet been reported. Further research is needed especially in the design of storage system for studying the interrelationship between the micro- and macro-storage environments as they may affect food storage stability and the influence of the biosurfactants. Other future area of applications of the BST is in cold and freeze storage of food products because of its high emulsifying and antioxidant ability, which can help prevent the oxidation of the food products and extend their shelf lives at low temperatures.

5 Conclusions

The promising future applications of the BSTs in food processing and formulation cannot be overemphasized. There are many commercial applications of the material yet untapped, and only a few cases were reported in the literature. This paper presented some potential applications of the BST including the design of biodegradable film liquid food packaging, storage and processing of food products, dairy products equipment maintenance, and food freezing technology. This is possible because of the ability of the material to resist high temperature, pH, together with their emulsification, antioxidation, and antimicrobial abilities. To the best of the authors' knowledge, the commercial application of the BST in the abovementioned future needs has not yet been fully exploited in food processing and formulation. The current chapter presents the application of BST in the food industry.

References

[1] M. Basit, M. Rasool, S.A.R. Naqvi, M. Waseem, B. Aslam, Biosurfactants production potential of native strains of *Bacillus cereus* and their antimicrobial, cytotoxic and antioxidant activities, Pak. J. Pharm. Sci. 31 (2018) 251–256.

[2] V.C. Akubude, S. Sule, D.C. Chinweuba, V.C. Okafor, Application of biosurfactant in food industry, in: Green Sustainable Process for Chemical and Environmental Engineering and Science, 2021, pp. 109–125, https://doi.org/10.1016/B978-0-12-823380-1.00004-6.

[3] I.J.B. Durval, I.A. da Silva, L.A. Sarubbo, Application of microbial biosurfactants in the food industry, in: Microbial Biosurfactants, Springer, 2021, pp. 1–10, https://doi.org/10.1007/978-981-15-6607-3_1.

[4] I. Mnif, D. Ghribi, Glycolipid biosurfactants: main properties and potential applications in agriculture and food industry, J. Sci. Food Agric. 96 (2016) 4310–4320, https://doi.org/10.1002/JSFA.7759.

[5] G.S. Kiran, S. Priyadharsini, A. Sajayan, G.B. Priyadharsini, N. Poulose, J. Selvin, Production of lipopeptide biosurfactant by a marine Nesterenkonia sp. and its application in food industry, Front. Microbiol. (2017) 8, https://doi.org/10.3389/FMICB.2017.01138/FULL.

[6] T. Morita, T. Fukuoka, T. Imura, D. Kitamoto, Production of mannosylerythritol lipids and their application in cosmetics, Appl. Microbiol. Biotechnol. 97 (2013) 4691–4700,- https://doi.org/10.1007/S00253-013-4858-1.

[7] A. Moldes, X. Vecino, L. Rodríguez-López, M. Rincón-Fontán, J.M. Cruz, Biosurfactants: the use of biomolecules in cosmetics and detergents, in: New and Future Developments in Microbial Biotechnology and Bioengineering, 2020, pp. 163–185, https://doi.org/10.1016/B978-0-444-64301-8.00008-1.

[8] K.G.O. Bezerra, I.G.S. Silva, F.C.G. Almeida, R.D. Rufino, L.A. Sarubbo, Plant-derived biosurfactants: extraction, characteristics and properties for application in cosmetics, Biocatal. Agric. Biotechnol. 34 (2021), https://doi.org/10.1016/J.BCAB.2021.102036, 102036.

[9] I. da Silva, K. Bezerrac, K. Bezerra, I.J. Batista Durval, C. Bronzo Barbosa Farias, C.J. Galdino Junior, et al., Evaluation of the emulsifying and antioxidant capacity of the

biosurfactant produced by Candida bombicola URM 3718, Chem. Eng. Trans. 79 (2020) 67–72, https://doi.org/10.3303/CET2079012.

[10] U. Sood, D. Singh, P. Hira, J. Lee, V. Kalia, L.-J. Rup, et al., Rapid and solitary production of mono-rhamnolipid biosurfactant and biofilm inhibiting pyocyanin by a taxonomic outlier *Pseudomonas aeruginosa* strain CR1, J. Biotechnol. 307 (2020) 98–106. Elsevier.

[11] S. Nalini, P. Rengasamy, I. Dhinakarasamy, Biosurfactant in food and agricultural application, Environ. Biotechnol. 2 (2020) 75–94.

[12] S. Pamp, T. Tolker-Nielsen, Multiple roles of biosurfactants in structural biofilm development by *Pseudomonas aeruginosa*, J. Bacteriol. 189 (2007) 2531–2539.

[13] T. Clements, T. Ndlovu, S. Khan, W. Khan, Biosurfactants produced by Serratia species: classification, biosynthesis, production and application, Appl. Microbiol. Biotechnol. 103 (2019) 589–602.

[14] E. Shoeb, F. Akhlaq, U. Badar, A. Jameela, I. Samina, Classification and industrial applications of biosurfactants, Acad. Res. Int. 11 (2013) 243–251.

[15] S. Shekhar, A. Sundaramanickam, T. Balasubramanian, Biosurfactant producing microbes and their potential applications: a review, Crit. Rev. Environ. Sci. Technol. 45 (2015) 1522–1554, https://doi.org/10.1080/10643389.2014.955631.

[16] L. Sung-Chyr, Biosurfactants: recent advances, J. Chem. Technol. Biotechnol. 66 (1996) 109–120.

[17] R. Zouari, S. Besbes, S. Ellouze-Chaabouni, D. Ghribi-Aydi, Cookies from composite wheat-sesame peels flours: dough quality and effect of *Bacillus subtilis* SPB1 biosurfactant addition, Food Chem. 194 (2016) 758–769, https://doi.org/10.1016/J.FOODCHEM.2015.08.064.

[18] M. Ahmad, T.A. Wani, S.M. Wani, F.A. Masoodi, A. Gani, Incorporation of carrot pomace powder in wheat flour: effect on flour, dough and cookie characteristics, J. Food Sci. Technol. 53 (2016) 3715–3724, https://doi.org/10.1007/S13197-016-2345-2.

[19] R. Makkar, S. Cameotra, Production of biosurfactant at mesophilic and thermophilic conditions by a strain of *Bacillus subtilis*, J. Ind. Microbiol. Biotechnol. 20 (1998) 48–52. Springer.

[20] S. Mukherjee, P. Das, C.S.R. Sen, Enhanced production of biosurfactant by a marine bacterium on statistical screening of nutritional parameters, Biochem. Eng. J. 42 (2008) 245–260. Elsevier 2018.

[21] C. Chen, S. Baker, R.C. Darton, Batch production of biosurfactant with foam fractionation, J. Chem. Technol. Biotechnol. 81 (2006) 1923–1931. Wiley online Library.

[22] S. Silva, C. Farias, R. Rufino, J.M. Luna, L.A. Sarubbo, Glycerol as substrate for the production of biosurfactant by *Pseudomonas aeruginosa* UCP 0992, Colloids Surf. B Biointerfaces 79 (2010) 174–183, https://doi.org/10.1016/j.colsurfb.2010.03.050. Elsevier 2010.

[23] F. Bicca, L. Fleck, M.A.-R. de Microbiologia, Production of biosurfactant by hydrocarbon degrading *Rhodococcus ruber* and *Rhodococcus erythropolis*, Rev. Microbiol. 30 (1999). SciELO Brasil 1999.

[24] A. Fadeyibi, Z.D. Osunde, Effect of thickness and matrix variability on properties of a starch-based nanocomposite supple film, Food Res. 5 (2021) 416–422, https://doi.org/10.26656/FR.2017.5(4).281.

[25] A. Patil, G. Usmani, P.D. Meshram, Synthesis and characterization of sorbitol based biosurfactants from renewable sources by using Candida antarctica lipase-B enzyme, Int. J. Adv. Eng. Res. Technol. 2 (2014).

[26] J. Willenbacher, J.-T. Rau, J. Rogalla, C. Syldatk, R. Hausmann, Foam-free production of surfactin via anaerobic fermentation of *Bacillus subtilis* DSM 10T, AMB Express 5 (2015) 1, https://doi.org/10.1186/S13568-015-0107-6.

[27] J. Willenbacher, C. Syldatk, R. Hausmann, Production of surfactin with *Bacillus subtilis* in a fermentation process with integrated foam fractionation, Chem. Ing. Tech. 86 (2014) 1596, https://doi.org/10.1002/CITE.201450032.

[28] M.S. Kuyukina, I.B. Ivshina, T.A. Baeva, O.A. Kochina, S.V. Gein, V.A. Chereshnev, Trehalolipid biosurfactants from nonpathogenic Rhodococcus actinobacteria with diverse immunomodulatory activities, N. Biotechnol. 32 (2015) 559–568, https://doi.org/10.1016/J.NBT.2015.03.006.

[29] S.J. Geetha, I.M. Banat, S.J. Joshi, Biosurfactants: production and potential applications in microbial enhanced oil recovery (MEOR), Biocatal. Agric. Biotechnol. 14 (2018) 23–32, https://doi.org/10.1016/J.BCAB.2018.01.010.

[30] A. Panchabhai, Application of biosurfactant for effective production of biocides from sulfate-reducing bacteria, in: Green Sustainable Process for Chemical and Environmental Engineering and Science, 2021, pp. 367–377, https://doi.org/10.1016/B978-0-12-823380-1.00010-1.

[31] T. Mouafo, A. Mbawala, N. Robert, Effect of different carbon sources on biosurfactants' production by three strains of lactobacillus spp, Biomed. Res. Int. (2018), 5034783.

[32] A. Ghasemi, M. Moosavi-Nasab, P. Setoodeh, G. Mesbahi, G. Yousefi, Biosurfactant production by lactic acid bacterium *Pediococcus dextrinicus* SHU1593 grown on different carbon sources: strain screening followed by product, Sci. Rep. 9 (2019).

[33] E.J. Gudiña, J.A. Teixeira, L.R. Rodrigues, Isolation and functional characterization of a biosurfactant produced by *Lactobacillus paracasei*, Colloids Surf. B Biointerfaces 76 (2010) 298–304, https://doi.org/10.1016/j.colsurfb.2009.11.008.

[34] I.J.B. Durval, B.G. Ribeiro, J.S. Aguiar, R.D. Rufino, A. Converti, L.A. Sarubbo, Application of a biosurfactant produced by *Bacillus cereus* UCP 1615 from waste frying oil as an emulsifier in a cookie formulation, Fermentation 7 (2021) 189, https://doi.org/10.3390/FERMENTATION7030189.

[35] A. Fadeyibi, Z.D. Osunde, E.C. Egwim, P.A. Idah, Performance evaluation of cassava starch-zinc nanocomposite film for tomatoes packaging, J. Agric. Eng. 48 (2017) 137–146, https://doi.org/10.4081/jae.2017.565.

[36] A. Fadeyibi, Z.D. Osunde, M.G. Yisa, Prediction of some physical attributes of cassava starch-zinc nanocomposite film for food-packaging applications, J. Packag. Technol. Res. 3 (2019) 35–41, https://doi.org/10.1007/S41783-018-0046-1.

[37] J.M. Campos, T.L. Montenegro Stamford, L.A. Sarubbo, J.M. de Luna, R.D. Rufino, I.M. Banat, Microbial biosurfactants as additives for food industries, Biotechnol. Prog. 29 (2013) 1097–1108, https://doi.org/10.1002/BTPR.1796.

[38] M. Nitschke, S.S. Silva, e., Recent food applications of microbial surfactants, Crit. Rev. Food Sci. Nutr. 58 (2018) 631–638, https://doi.org/10.1080/10408398.2016.1208635.

[39] A.B. Moldes, A.M. Torrado, M.T. Barral, J.M. Domínguez, Evaluation of biosurfactant production from various agricultural residues by *Lactobacillus pentosus*, J. Agric. Food Chem. 55 (2007) 4481–4486, https://doi.org/10.1021/JF063075G.

[40] M. Masmoudi, H. Yaich, M. Borchani, R. Mbarki, H. Attia, Chemical, physical and sensory characteristics of biscuits enriched with jujube (Zizyphus lotus L.) flour and fiber concentrate, J. Food Sci. Technol. 58 (2021) 1411–1419, https://doi.org/10.1007/S13197-020-04652-7.

[41] B.G. Ribeiro, B.O. de Veras, A.J. dos Santos, J. Medeiros Campos Guerra, L.A. Sarubbo, Biosurfactant produced by *Candida utilis* UFPEDA1009 with potential application in cookie formulation, Electron. J. Biotechnol. 46 (2020) 14–21, https://doi.org/10.1016/J.EJBT.2020.05.001.

[42] V. Sharma, D. Sharma, Microbial biosurfactants: future active food ingredients, in: Microbial Bioprospecting for Sustainable Development, 2018, pp. 265–276, https://doi.org/10.1007/978-981-13-0053-0_14.

[43] S.S. Giri, S.S. Sen, J.W. Jun, V. Sukumaran, S.C. Park, Role of *Bacillus licheniformis* VS16-derived biosurfactant in mediating immune responses in carp rohu and its application to the food industry, Front. Microbiol. (2017) 8, https://doi.org/10.3389/FMICB.2017.00514.

[44] K.S.T. Souza, E.J. Gudiña, Z. Azevedo, V. de Freitas, R.F. Schwan, L.R. Rodrigues, et al., New glycolipid biosurfactants produced by the yeast strain Wickerhamomyces anomalus CCMA 0358, Colloids Surf. B Biointerfaces 154 (2017) 373–382, https://doi.org/10.1016/J.COLSURFB.2017.03.041.

[45] M. Ohadi, H. Forootanfar, H.R. Rahimi, E. Jafari, M. Shakibaie, T.T. Eslaminejad, et al., Antioxidant potential and wound healing activity of biosurfactant produced by Acinetobacter junii B6, Curr. Pharm. Biotechnol. 18 (2017) 900–908.

[46] B. Ribeiro, J.M. Campos Guerra, Leonie Asfora Sarubbo. Potential food application of a biosurfactant produced by *Saccharomyces cerevisiae* URM 6670, Front. Bioeng. Biotechnol. 1 (2020).

[47] A. Doost, J. van Camp, K. Dewettinck, P. van der Meeren, Production of thymol nanoemulsions stabilized using Quillaja Saponin as a biosurfactant: antioxidant activity enhancement, Food Chem. 293 (2019) 134–143.

[48] S. Giri, E. Ryu, V. Sukumaran, S. Park, Antioxidant, antibacterial, and anti-adhesive activities of biosurfactants isolated from bacillus strains, Microb. Pathog. 132 (2019) 66–72.

[49] A. Merghni, I. Dallel, E. Noumi, Y. Kadmi, H. Hentati, S. Tobji, A. Ben Amor, M. Mastouri, Antioxidant and antiproliferative potential of biosurfactants isolated from *Lactobacillus casei* and their anti-biofilm effect in oral *Staphylococcus aureus* strains, Microb. Pathog. 104 (2017) 84–89.

[50] B.G. Ribeiro, J.M.C. Guerra, L.A. Sarubbo, Potential food application of a biosurfactant produced by *Saccharomyces cerevisiae* URM 6670, Front. Bioeng. Biotechnol. (2020) 8, https://doi.org/10.3389/FBIOE.2020.00434/FULL.

[51] M. Sunmonu, A. Fadeyibi, O. Olabanjo, Moringa ve farklı depolama malzemeleri kullanarak toz Irvingia gabonensis'in kalitesi ve mikrobiyal inaktivasyonu, Harran Tarım ve Gıda Bilimleri Dergisi (2020) 391–400. https://doi.org/10.29050/HARRANZIRAAT.733598.

[52] Y.T.H. Dang, S. Gangadoo, P. Rajapaksha, V.K. Truong, D. Cozzolino, J. Chapman, Biosensors in food traceability and quality, in: Comprehensive Foodomics, Elsevier, 2021, pp. 308–321, https://doi.org/10.1016/B978-0-08-100596-5.22853-9.

[53] A. Fadeyibi, M.G. Yisa, F.A. Adeniji, K.K. Katibi, K.P. Alabi, K.R. Adebayo, Potentials of zinc and magnetite nanoparticles for contaminated water treatment, Agric. Rev. (2018), https://doi.org/10.18805/ag.r-113.

[54] S.Z. Tang, Y.L. He, F.L. Wang, Q.X. Zhao, Y. Yu, On-site experimental study on fouling and heat transfer characteristics of flue gas heat exchanger for waste heat recovery, Fuel 296 (2021), https://doi.org/10.1016/J.FUEL.2021.120532, 120532.

[55] S. da MGC, M. de AO, D.G. Almeida, S. da MEP, H.M. Meira, L.A. Sarubbo, Application of green surfactants to control barnacle recruitment, Reg. Stud. Mar. Sci. 45 (2021) 101854, https://doi.org/10.1016/J.RSMA.2021.101854.
[56] I.M. Banat, R.S. Makkar, S.S. Cameotra, Potential commercial application of biosurfactants, Appl. Microbiol. Biotechnol. 53 (2000) 495–508.
[57] A.K. Pradhan, N. Pradhan, G. Mall, H.T. Panda, L.B. Sukla, P.K. Panda, et al., Application of lipopeptide biosurfactant isolated from a halophile: *Bacillus tequilensis* ch for inhibition of biofilm, Appl. Biochem. Biotechnol. 171 (2013) 1362–1375, https://doi.org/10.1007/S12010-013-0428-3.
[58] S. Satpute, N. Mone, P. Das, A.G. Banpurkar, I.M. Banat, *Lactobacillus acidophilus* derived biosurfactant as a biofilm inhibitor: a promising investigation using microfluidic approach, Appl. Sci. 8 (2018), https://doi.org/10.3390/app8091555. MdpiCom.
[59] F. Rivardo, R. Turner, G. Allegrone, H. Ceri, M.G. Martinotti, Anti-adhesion activity of two biosurfactants produced by bacillus spp. prevents biofilm formation of human bacterial pathogens, Appl. Microbiol. Biotechnol. 83 (2009) 541–553, https://doi.org/10.1007/s00253-009-1987-7. Springer.
[60] A. Pradhan, N. Pradhan, G. Mall, H.T. Panda, L.B. Sukla, P.K. Panda, B.K. Mishra, Application of lipopeptide biosurfactant isolated from a halophile: *Bacillus tequilensis* CH for inhibition of biofilm, Appl. Biochem. Biotechnol. 17 (2013) 1362–1375. Springer.
[61] L. Fracchia, M. Cavallo, G. Allegrone, M. Martinotti, A Lactobacillus-derived biosurfactant inhibits biofilm formation of human pathogenic *Candida albicans* biofilm producers, Appl. Microbiol. Biotechnol. 2 (2010). ResearchgateNet.
[62] Q. Wei, R. Mather, A.F. Fotheringham, Oil removal from used sorbents using a biosurfactant, Bioresour. Technol. 96 (2005) 331–334.
[63] I. Banat, N. Samarah, M. Murad, Biosurfactant production and use in oil tank clean-up, World J. Microbiol. Biotechnol. 7 (1991) 80–88, https://doi.org/10.1007/BF02310921.
[64] R. Makkar, Cameotra., An update on the use of unconventional substrates for biosurfactant production and their new applications, Appl. Microbiol. Biotechnol. 58 (2002) 428–434, https://doi.org/10.1007/s00253-001-0924-1.
[65] A. Fadeyibi, Z. Osunde, M. Yisa, Optimization of processing parameters of nanocomposite film for fresh sliced okra packaging, J. Appl. Packag. Res. 11 (2019).
[66] A. Fadeyibi, Z. Osunde, M. Yisa, Effects of period and temperature on quality and shelf-life of cucumber and garden-eggs packaged using cassava starch-zinc nanocomposite film, J. Appl. Packag. Res. 12 (2020).
[67] B.C. Maniglia, D.A. Laroque, L.M. de Andrade, B.A.M. Carciofi, J.A.S. Tenório, C.J. de Andrade, Production of active cassava starch films; effect of adding a biosurfactant or synthetic surfactant, React. Funct. Polym. 144 (2019), https://doi.org/10.1016/J.REACTFUNCTPOLYM.2019.104368, 104368.
[68] N. Stachowiak, J. Kowalonek, J. Kozlowska, Effect of plasticizer and surfactant on the properties of poly(vinyl alcohol)/chitosan films, Int. J. Biol. Macromol. 164 (2020) 2100–2107, https://doi.org/10.1016/J.IJBIOMAC.2020.08.001.
[69] R. Ortega-Toro, A. Jiménez, P. Talens, A. Chiralt, Effect of the incorporation of surfactants on the physical properties of corn starch films, Food Hydrocoll. 38 (2014) 66–75, https://doi.org/10.1016/J.FOODHYD.2013.11.011.
[70] M. Shokouhfard, R.K. Kermanshahi, R.V. Shahandashti, M.M. Feizabadi, The inhibitory effect of a *Lactobacillus acidophilus* derived biosurfactant on biofilm producer *Serratia marcescens*, Iran. J. Basic Med. Sci. 18 (2015) 1001–1007. NcbiNlmNihGov.

[71] J. Coronel-León, A.M. Marqués, J. Bastida, A. Manresa, Optimizing the production of the biosurfactant lichenysin and its application in biofilm control, J. Appl. Microbiol. 120 (2016) 99–111, https://doi.org/10.1111/jam.12992. Wiley online Library.

[72] G. Kiran, B. Sabarathnam, J. Selvin, Biofilm disruption potential of a glycolipid biosurfactant from marine *Brevibacterium casei*, FEMS Immunol. Med. Microbiol. 59 (2010) 432–438. AcademicOupCom.

[73] J. Coronel-León, G. de Grau, A. Grau-Campistany, M. Farfan, F. Rabanal, A. Manresa, et al., Biosurfactant production by AL 1.1, a *Bacillus licheniformis* strain isolated from Antarctica: production, chemical characterization and properties, Ann. Microbiol. 65 (4) (2015) 2065–2078, https://doi.org/10.1007/S13213-015-1045-X.

[74] T. Janek, M. Łukaszewicz, A. Krasowska, Identification and characterization of biosurfactants produced by the Arctic bacterium *Pseudomonas putida* BD2, Colloids Surf. B Biointerfaces 110 (2013) 379–386, https://doi.org/10.1016/J.COLSURFB.2013.05.008.

[75] R.E. Vollú, D. Jurelevicius, L.R. Ramos, R.S. Peixoto, A.S. Rosado, L. Seldin, Aerobic endospore-forming bacteria isolated from Antarctic soils as producers of bioactive compounds of industrial interest, Polar Biol. 37 (2014) 1121–1131, https://doi.org/10.1007/S00300-014-1505-Y.

[76] A. Perfumo, I.M. Banat, R. Marchant, Going green and cold: biosurfactants from low-temperature environments to biotechnology applications, Trends Biotechnol. 36 (2018) 277–289, https://doi.org/10.1016/J.TIBTECH.2017.10.016.

[77] V. Gesheva, E. Stackebrandt, E. Vasileva-Tonkova, Biosurfactant production by halotolerant *Rhodococcus fascians* from Casey Station, Wilkes Land, Antarctica, Curr. Microbiol. 61 (2) (2010) 112–117, https://doi.org/10.1007/S00284-010-9584-7.

CHAPTER

17 Next generation biosurfactants and their practical application in the food sector

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1 Introduction

Unorganized and unplanned exponential growth in industrialization and the rapid increase in the global population are responsible for air, water, and soil pollution. Rapid growth in population and swift changes in their lifestyles increase the demand for various novel products and force entrepreneurs to produce new products. The production of synthetic chemicals was increased to meet this incredible demand, as these are economically cheap, efficient, and attractive. However, the use of synthetic chemicals produces many health hazards. Moreover, they are not environmentally friendly. In recent years, awareness of people about natural products and stringent environmental policies of government agencies has created a demand for natural products. In this current scenario, there is an imperative demand to investigate innovative, effective, and inexpensive natural foods for the help of humanity [1–3].

This chapter addresses a brief description of biosurfactants followed by reasons that made them potential alternatives for replacing synthetic surfactants in the food industry, which are given in the first part of this chapter. Description about biosurfactants' antioxidant properties, current research studies on their antioxidant abilities are also discussed in the second part of this chapter. Finally, conclusive remarks about the application of antioxidants in the food industry are given.

2 Biosurfactants

The majority of surfactants are produced from the synthetic origin; it is reported that around 70%–75% surfactants are produced from petrochemicals alone. Synthetic surfactants are toxic, highly unstable, not biodegradable, pose many harmful effects on all life forms, and contribute toward environmental pollution. Moreover, these petrochemicals are nonrenewable and are quickly exhausting. Thus, there is a great

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challenge in developing new technologies and strategies to replace unrenewable supplies such as coal, petrochemicals, and other fossil fuels produced with naturally available renewable resources [4,5].

Sustainable production and the use of biosurfactants are required in this current scenario to minimize environmental pollution and its harmful impacts. Biosurfactants are natural products produced from sustainable sources such as inexpensive carbon sources and industrial wastes. They are evolving as suitable substitutes for nonsustainable artificial surface-active agents such as petroleum-based products. Biosurfactants are nontoxic, biodegradable, eco-friendly, stable in various temperatures, pH, salinity, etc. They have high selectivity and can be produced abundantly; thus, their usage has increased in the food, cosmetics, pharmaceutical, and petroleum industries. They are widely used to recover aquaphobic composites and emulsifying agents in the food and cosmetic industries [6].

Biosurfactants reduce the interfacial surface tension, thus widely used as domestic and industrial cleaning agents such as detergents, foaming agents, etc. Due to their excellent cleaning abilities, they are widely used in textile industries. They are also used as emulsification agents, plasticizers, solubilizing agents, etc. In contemporary times, the requirement for biosurfactants has risen considerably in the agriculture, mining, paper industries [7]. Fig. 1 shows some of the essential uses of biosurfactants. Biosurfactants have good surface activity characteristics and lower the surface tension of water [7–9].

Biosurfactants are produced from wide-ranging microbial sources such as bacteria, fungus, and yeast. Some of the categories of biosurfactants include

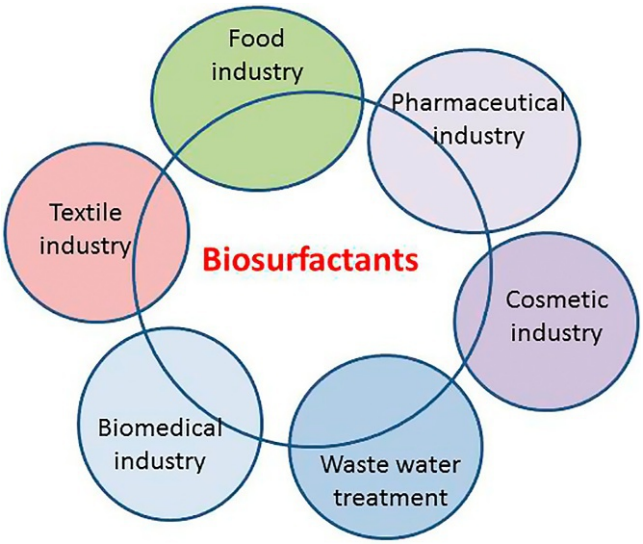

FIG. 1

Important uses of biosurfactants.

glycolipid-type, lipopeptide-type, fatty-acid-type, polymer-type. They may have hydrophobic, hydrophilic, or amphiphilic characteristics [10]. Biosurfactants are emerging as suitable alternatives to synthetic surfactants that can be used to remediation of water, soil, and overall environmental health [11].

2.1 Biosurfactant in food applications

Biosurfactants are one of the unique products used in the food processing, medical, and therapeutic applications. They are mainly applying as either food additives or cleansing agents in the food industry. The primary concern in the food processing industry is to enhance taste causing characteristics and safety to the food and consumer. These surfactant additives in the food processing applications incorporate or enhance the product's nutritional value, texture, appearance, taste, and safety. In addition to this, biosurfactants are used as emulsifying agents, thickeners, and humectants [12].

Biosurfactants are gaining significant interest as they are biodegradable, nontoxic, and have abundant availability, diversity in structures, produced from a variety of substrates, and, most importantly, the ability to compete with synthetic surfactants in minimizing surface tension and interfacial tension and in emulsion stabilization [13]. In the food industry, biosurfactants are used as antioxidants, antimicrobial, antibiofilm, cleaning agents. They are also used as food additives in the preparation of various food formulations [12].

3 Antioxidants in the food industry

One of the most important aims of any technology used in the food industry is safety and enhancing taste, color, texture, appearance, nutritional value, and shelf life. Food additives are generally added to enhance the abovementioned characteristics of food. Biosurfactants are considered versatile food additives as they possess emulsification, adhesive, antioxidant, and antimicrobial characteristics. These food additives are classified as food preservatives such as antimicrobial and antioxidants, flavor enhancers such as sweeteners, color, and texture enhancers such as emulsifiers and stabilizers, nutritional value enhancers, etc. [14]

Antioxidants perform an essential role in the food processing industry in maintaining quality, taste, aroma, texture, shelf life, nutritional value and providing safety to the consumer. In most food formulations, fats are added for extra taste. During the storage, fats, oils, and lipids containing foods' qualities spoil due to several degradation reactions. Oxidation reactions are one of the prevalent reactions responsible for food quality spoilage. Fats present in food products are one of the potential sources of oxidation reactions of food products. Lipids present in the fats are considered responsible for most oxidation reactions; these reactions are responsible for the rancidification, unpleasant flavors, and reduced shelf life and nutritional quality of foods.

During the food products storage and processing, oxidation of lipids in the food produces toxic compounds, unpleasant taste and odors, and reduces the nutritional value and quality of the food products. These toxic compounds may create health problems for the consumer. Thus, controlling the oxidation process is a must in the food industry. Oxidation of food products can be controlled in many ways, i.e., inactivation of enzymes responsible for oxidation reactions, controlling temperature, pressure, and other physical conditions, using good packaging, using additives such as antioxidants, etc. Out of this, adding antioxidants is efficient in overcoming or minimizing these harmful effects that arise from the oxidation process.

Antioxidants are commonly called oxidation inhibitors. Antioxidants react with free radicals produced in the food products, convert them into inactive products, and inhibit oxidation reactions. Their antioxidant activity is affected by structural features, chemical configuration, physical environments used such as temperature, pressure, pH, etc., the concentration of antioxidants, presence of other chemicals in the food, etc.

Antioxidants provide hydrogen groups to free radicals and restrict in vivo and in vitro oxidative chain reactions. They are used as free radical scavengers [15]. Antioxidants are employed in the food processing, cosmetic, and drug industries for the preservation of numerous products. These additives hinder the oxidation reactions of lipids and reduce the spoilage of products.

Various unstable oxidizable species such as hydroxyl anions and superoxide anions are constantly generated in tissues and are removed regularly. If they are not removed regularly, they may damage the cell tissue and cause various skin problems. Antioxidants perform a crucial part in controlling damage triggered by oxidative stress and protect humans against harmful effects. In recent years, oxidative stress caused by ROS has been increasing due to changes in lifestyles, excessive smoking, ultraviolet radiation, environmental pollution, etc. Human inherent antioxidants may not be sufficient to eliminate the adverse consequences of ROS. Thus, external antioxidants such as vitamins, polyphenols, carotenoids reduce skin damage due to excessive oxidative stress. Thus, the application of antioxidants is gaining importance in the cosmetic industry [16–18].

3.1 Synthetic antioxidants and natural antioxidants

Antioxidants are mainly classified into two types: synthetic antioxidants and natural antioxidants. The most common synthetic antioxidants are BHA, BHT, TBHQ, ascorbic acid, and gallic acid esters. Synthetic antioxidants are cheap, stable, and effective and have undergone several toxicological tests. However, synthetic antioxidants are toxic, carcinogenic, nonbiodegradable and pose environmental problems. In addition, new regulations on applying synthetic antioxidants restrict their usage. Natural antioxidants are considered safer and healthier compared with their synthetic origin. A large number of natural antioxidants, more than 980, have been identified as an alternative to their synthetic counterparts two decades ago [19]. Most of the naturally available ones are phenolic compounds. Antioxidants having phenolic

groups are the most widely used ones. Natural antioxidants can be produced from vegetables, fruits, spices, herbs, nuts, oilseeds, cereals, legumes, animal and microbial sources.

3.2 Biosurfactants as antioxidants

In aerobic systems, molecules with unpaired electrons such as superoxide anion and hydroxyl radicals' formation during respiration are inevitable. These free radicals are highly reactive with other compounds in the body and may damage cells or tissues [20,21]. The human inherent defense system may not be efficient to eliminate these oxidation reactions. Hence, foods containing antioxidants may help to reduce oxidation stress.

Microorganisms are one of the most abundantly available species on earth. The exploitation of microorganisms in the preparation of various food products has been in practice since ancient times. However, industrial production of antioxidants derived from microbial sources started in the early 1980s. Biosurfactants are natural antioxidants; they are made of both polar and nonpolar structures. The presence of both water-attraction and water-repellent functional groups in the molecular structure of the biosurfactant is responsible for their surface activity. Biosurfactants consist of positive, negative, and amphoteric charged ions and long fatty acid chains [22].

Biosurfactants alter the surface characteristics and minimize the bio-adhesion properties of microorganisms. Biosurfactants restrict oxidative chain reactions and show good antioxidant properties. They have effective vegetable oils solubilizing ability and fats stabilizing ability during cooking. Their organoleptic properties make their use in bread and various food formulations such as ice creams, cakes, cookies, salad dressings, etc. They are also used to increase the rheological properties of food materials. Biosurfactants are considered green surfactants as their synthesis meets many of green chemistry principles. Producing biosurfactants from renewable sources and wastage materials such as agricultural and industrial wastage minimizes environmental pollution, which helps in energy and environmental sustainability [23,24].

4 Current research on biosurfactants as antioxidants

The global demand for food production has been increasing exponentially due to the increase in population. Still, it is reported that most of the countries are not meeting their food demands, and a large number of deaths are occurring due to food scarcity. The yield of food production crops decreases with time due to the accumulation of heavy metals, pesticides, hydrocarbons, and other pollutants in the agricultural soil. Thus biosurfactants are used to eliminate these contaminants from the soil [25]. Some reports say that the glycolipid biosurfactants have the ability in regulating the harmful effects of phytopathogenic fungi. They were also found to have

stimulation ability in plant signaling and immunity [26]. They are used as additives in various food formulations to improve the shelf life, texture, consistency, stability, aggregation of fat globules, rheological properties, etc. They are added to increase the flavor, texture, oil solubilizing purposes and reduce the aging of food products in bakehouse and ice cream products [27].

Biosurfactants are secondary metabolites; they are generated from various microbial variants. The biosurfactant characteristics and yield depend on various parameters such as nature of carbon source, nitrogen source, incubation time, incubation temperature, pH, aeration agitation, type of process, strain, etc. [28]. Traditionally biosurfactants are produced from microbial strains isolated from the contaminated site, fermentation process, etc. But due to the advancement of technology, biosurfactants are synthesized from microorganisms isolated from numerous microbial sources. Recent studies on the antioxidant ability of biosurfactants suggest alternative or even the complete replacement of synthetic antioxidants with biosurfactants. Table 1 shows some of the necessary research works on the ability of biosurfactants as antioxidants.

Abdollahi et al. derived glycolipid biosurfactant (rhamnolipids) and lipopeptide biosurfactant (surfactin) and evaluated antiadhesive, antioxidant, and biofilm formation properties. Antioxidative properties were evaluated by DPPH, FRAP, and FTC methods [29]. Results demonstrated that unsaturated fatty acids present in lipids could able to remove free radicals and restrict the peroxidative reactions of lipids. Both rhamnolipid and surfactin biosurfactants showed good antioxidant properties. Due to the intrinsic surface activity, *Bacillus amyloliquefaciens* NS6 demonstrated very good antioxidative characteristics.

Table 1 Recent literature on biosurfactants as antioxidants.

S. No.	Biosurfactant microbial source	Ref.
1	*Bacillus amyloliquefaciens* NS6 *Pseudomonas aeruginosa* MN1	[29]
2	*Saccharomyces cerevisiae* URM 6670	[13]
3	*Lactobacillus casei* MRTL3	[30]
4	*Marinobacter litoralis* MB15	[31]
5	*Mannosylerythritol* lipids (MEL)	[18]
6	*Basillus subtilis* (MF 582633)	[32]
7	*Bacillus subtilis* RW-I	[33]
8	*Candida bombicola* URM 3718	[34]
9	*B. subtilis* VSG4 and *Bacillus licheniformis* VS16	[35]
10	*Leuconostoc mesenteroides* ssp. *cremoris* strains, *Lactobacillus jensenii* (ATCC 25258), and *Lactobacillus acidophilus* (ATCC 4356)	[36]
11	Lactic acid bacteria	[37]
12	*lactobacillus casei*	[38]

Biosurfactants derived from yeasts are nontoxic, have high thermal resistance and antioxidant properties, thus finding numerous applications in food industries. Biosurfactant synthesized from *Saccharomyces cerevisiae* URM 6670 was used to study the feasibility of replacing egg yolk with surfactant in cookie formulation. The studies showed no significant difference in physical, chemical, and textural properties [13]. Lactic acid bacteria have great preservation applications in food processing; it is very effective against food spoilage and pathogenic bacteria. Sharma et al. derived biosurfactants from the *Lactobacillus casei* MRTL3 strain and studied their antioxidant properties [30]. The experimental results report the ability of *L. casei* biosurfactant as an antioxidant in the food processing industries.

In the food industry, surfactants having good antioxidant properties are very important as good antioxidants that fight against degenerative diseases and coronary heart diseases. Biosurfactant, rhamnolipid homologs, produced from isolating nonpathogenic *Marinobacter litoralis* MB15 bacteria from seawater, showed remarkable antimicrobial and antioxidant activities [31]. This research finding revealed that these glycolipid biosurfactants could be used as an alternative to synthetic preservatives in various formulations. Their studies suggest rhamnolipid biosurfactant can be used to reduce bio-adhesion and film development; thus find a promising future in food and biomedical fields as an antioxidant.

Biosurfactants are produced from various microbial strains such as bacteria, yeast, and fungi. Their yield and efficiency are affected by many parameters such as microbial strain, type of nutrients used in the growth of microbial strains, various physicochemical conditions present during synthesis [39,40]. The microbial strain source also significantly influences the yield and quality of biosurfactants. Reports identified that source from which microbial strain is isolated is generally suitable for the growth of that microorganism and has significantly improved that stain's performance.

Antioxidant activity of mannosylerythritol lipids (MELs) produced from the *Pseudozyma* yeast strain was tested by Makoto Takahashi et al. [18]. Antioxidant activity of three types of MEL derivatives MEL-A,-B,-C was tested, and all three derivatives showed antioxidant properties. MEL biosurfactants have excellent antitumor and cell differentiation activities against leukemia. Their skin and hair moisturizing ability finds applications in the cosmetics and pharmaceutical industries. Human skin fibroblast cells were examined for the antioxidant property of MEL-C under H_2O_2 oxidative stress, and results showed significant protective activity against oxidative stress. Amruta and Chandrahasya have produced biosurfactants from *Bacillus subtilis* (MF 582633) from contaminated soil and evaluated its antimicrobial, antioxidant, larvicidal, and antitermite activities [32]. This biosurfactant, surfactant, showed good antioxidant activity in DPPH scavenging in antioxidant activity studies.

Some biosurfactants are considered potential antioxidants. Antioxidant activity of *B. subtilis* RW-I biosurfactant derived from the petroleum industry effluent was evaluated by Yalçın and Çavuşoğlu using three methods [33]. Functional groups

characterization of the biosurfactant was done using FTIR studies. The studies reported that *B. subtilis* RW-I biosurfactant has antimicrobial properties and could be an alternative to synthetic oxidant. Emulsifying and antioxidative ability of *Candida bombicola* URM 3718 was produced from low-cost agro-industrial residue substrates, which was studied by da Silva et al. [34]. The antioxidant activity tested by the phosphomolybdenum method reported that *Candida bombicola* URM 3718 biosurfactant has significant antioxidant properties and has potential emulsifying and microbial antagonist industrial applications.

B. subtilis VSG4 and *Bacillus licheniformis* VS16 biosurfactants were evaluated for their adverse properties against oxidants, microbial growth, and microbial adhesion [35]. DPPH and hydroxyl free ions methods were applied to study the antagonist properties against the oxidative activity of both biosorbents. The results reported that biosurfactants' free radical scavenging ability ranges in 69.1%–73.5% and 63.3%–69.8% for *B. subtilis* VSG4 and *B. licheniformis* VS16 biosorbents, respectively. These microbial surfactants also displayed significant antibacterial and antibiofilm formation characteristics.

The oxidation process is very critical for the supply of energy for many biological activities in living things. However, oxidative stress can damage tissues and lead to many diseases, including cancer, gastrointestinal tract problems, atherosclerosis, arthritis, cirrhosis, etc. The human inherent defense system may repair the damaged systems to some extent. But for efficient control, external antioxidant supplements are needed to inhibit oxidative stress [20,41,42]. Virtanen et al. reported that lactic acid bacteria could scavenge superoxide anion, hydroxyl radicals, and hydrogen peroxide free radicals [36]. In total, 25 lactic-acid-producing bacteria were tested, and four strains exhibited high scavenging abilities in two methods.

The effect of the antioxidative activity using lactic acid bacteria was evaluated by Kaizu et al. in vitamin-*E*-deficient rats [37]. They have examined 570 strains, and 19 strains showed significant antioxidant activities. *L. casei* biosurfactant was selected by Abderrahmen Merghni et al. for assessing their adverse characteristics against oxidative and proliferative properties [38]. In vitro scavenging ability on the DPPH radical method was used to assess antimicrobial activity. The experimental results showed the antimicrobial and the above two properties confirmed the potential application against oral diseases.

Boutheina Gargouri et al. have reported that biosurfactant produced from B-2 strain has potential antioxidant properties and could be used for various industrial purposes [43]. Stenotrophomonas strain B-2 separated from contaminated wastewater collected from Tunisian oil fields was characterized for their functional groups. FTIR analysis reveals the presence of lipids and peptides. DPPH method was used for determining antioxidant activity. Biosurfactant produced from crude oil was evaluated for emulsification, hydrophobicity, antimicrobial, and antioxidant abilities. They reported that this biosurfactant could be used in food, cosmetics, and detergents industries.

5 Conclusions

The food industry is experiencing huge demand to produce antioxidants synthesized from biosurfactants. Even though biosurfactants have wide-ranging applications, their use is still not up to the mark due to the high cost of production. The fermentation media's cost is crucial in biosurfactant production and is around 50% of total manufacturing expenditure. Hence, to make this process economically feasible, research should be focused on alternative substrates such as agricultural waste, industrial waste, etc. New methods and processing techniques should be developed to make these natural products more competitive in future. Using biosurfactants as a total replacement to synthetic surfactants in the food industry requires addressing cost, availability, and efficiency challenges. The viability of any process depends on the abundant and economic availability of raw materials. Hence, efforts should be made to identify industrial and agricultural waste materials as raw materials, which would offset the waste disposal cost by producing valuable biosurfactants. Technological developments in the synthesis of biomolecules can help progress the commercial use of biosurfactants. Studies show that biosurfactants have great potential in food industry applications as antioxidants. Efforts to understand the free radical reactions are needed to understand various oxidative reactions. Currently, an enormous effort is being kept on biosurfactant applications as antioxidants in the food industry. More investigations are necessary to support the usage of biosurfactants in the food processing industries for better commercial feasibility and a lesser amount of garbage generation.

References

[1] K.V. Narayana Saibaba, R.V. Kandisa, Adsorption isotherm studies on methylene blue dye removal using naturally available biosorbent, Rasayan J. Chem. 12 (4) (2019) 2176–2182.

[2] K.V. Narayana Saibaba, Removal of dyes from industrial effluents using bioremediation technique, in: J. Aravind, M. Kamaraj, M. Prashanthi, S. Rajakumar (Eds.), Strategies and Tools for Pollutant Mitigation- Avenues to a Cleaner Environment, vol. 9, Springer Nature, 2021, pp. 173–194.

[3] A. Kumar, S.K. Singh, C. Kant, H. Verma, D. Kumar, P.P. Singh, A. Modi, S. Droby, M. S. Kesawat, H. Alavilli, S.K. Bhatia, G.D. Saratale, R.G. Saratale, S.-M. Chung, M. Kumar, Microbial biosurfactant: a new frontier for sustainable agriculture and pharmaceutical industries, Antioxidants 10 (9) (2021) 1472, https://doi.org/10.3390/antiox10091472.

[4] S. Akbari, N.H. Abdurahman, R.M. Yunus, F. Fayaz, O.R. Alara, Biosurfactants—a new frontier for social and environmental safety: a mini review, Biotechnol. Res. Innov. 2 (1) (2018) 81–90.

[5] J.M. Campos, T.L. Montenegro Stamford, L.A. Sarubbo, J.M. De Luna, R.D. Rufino, I. M. Banat, Microbial biosurfactants as additives for food industries, Biotechnol. Prog. 29 (5) (2013) 1097–1108.

[6] M. Garg, C.M. Priyanka, Isolation, characterization and antibacterial effect of biosurfactant from Candida parapsilosis, Biotechnol. Rep. 18 (2018), e00251, https://doi.org/10.1016/j.btre.2018.e00251.
[7] A.D.J. Cortes-Sanchez, Surfactants of microbial origin and its application in foods, Sci. Res. Essays 15 (1) (2020) 11–17.
[8] D. Sourav, S. Malik, A. Ghosh, R. Saha, B. Saha, A review on natural surfactants, RSC Adv. 5 (81) (2015) 65757–65767.
[9] J.D. Chávez González, M.Á. Rodríguez Barrera, Y. Romero Ramírez, A. Ayala Sánchez, J.C. Ruvalcaba Ledezma, J. Toribio Jiménez, Pantoea agglomerans isolated from producer surfactant pasture rhizosphere Tanzania and Llanero, Rev. Mex. Cienc. Agrícol. 7 (4) (2016) 961–968.
[10] A.A. Aslam, M. Ishtaiq, R. Badar, M.S. Nazir, Z. Tahir, M.A. Abdullah, Chapter 9—applications of biosurfactants in the production of industrially relevant bioproducts, in: Inamuddin, C.O. Adetunji, A.M. Asiri (Eds.), Green Sustainable Process for Chemical and Environmental Engineering and Science, Elsevier, 2021, pp. 173–201, https://doi.org/10.1016/B978-0-12-823380-1.00012-5.
[11] E.B. Manga, P.A. Celik, A. Cabuk, I.M. Banat, Biosurfactants: opportunities for the development of a sustainable future, Curr. Opin. Colloid Interface Sci. 56 (2021), 101514, https://doi.org/10.1016/j.cocis.2021.101514.
[12] M. Nitschke, S.S. Silva, E, recent food applications of microbial surfactants, Food Sci. Nutr. 58 (2018) 631–638, https://doi.org/10.1080/10408398.2016.1208635.
[13] R.B. Galdino, G.J.M. Campos, S.L. Asfora, Potential food application of a biosurfactant produced by Saccharomyces cerevisiae URM 6670, Front. Bioeng. Biotechnol. 8 (2020) 434, https://doi.org/10.3389/fbioe.2020.00434.
[14] A.L. Branen, R.J. Haggerty, Introduction to food additives, in: A.L. Branen, P.M. Davidson, S. Salminen, J.H. Thorngate (Eds.), Food Additives, Marcel Dekker, New York, NY, 2002, pp. 1–9.
[15] M. Valko, D. Leibfritz, J. Moncol, M.T. Cronin, M. Mazur, J. Telser, Free radicals and antioxidants in normal physiological functions and human disease, Int. J. Biochem. Cell Biol. 39 (1) (2007) 44–84.
[16] T. Morita, M. Kitagawa, S. Yamamoto, A. Sogabe, T. Imura, T. Fukuoka, D. Kitamoto, Glycolipd biosurfactants, Mannosylerythritol lipids, repair the damaged hair, J. Oleo Sci. 559 (2010) 267–272.
[17] S.K. Heo, H.S. Yi, H.J. Yun, C.H. Ko, J.W. Choi, S.D. Park, Ethylacetate extract from Draconis Resina inhibits LPS-induced inflammatory responses in vascular smooth muscle cells and macrophases via suppression of ROS production, Food Chem. Toxicol. 48 (2010) 1129–1136.
[18] M. Takahashi, T. Morita, T. Fukuoka, I. Tomohiro, D. Kitamoto, Glycolipid biosurfactants, Mannosylerythritol lipids, show antioxidant and protective effects against H2O2-induced oxidative stress in cultured human skin fibroblasts, J. Oleo Sci. 61 (8) (2012) 457–464.
[19] J. Pokorny, N. Yanishlieva, M. Gordon, Antioxidants in Food Practical Applications, Woodhead Publishing Limited, Abington Hall, Abington Cambridge, England, 2001.
[20] M. Zommara, N. Tachibana, M. Sakomo, Y. Suzuki, T. Oda, H. Hashida, K. Imaizumi, Whey from cultured skim milk decreases serum cholesterol and increases antioxidant enzymes in liver and red blood cells in rats, Nutr. Res. 16 (1996) 293–302.

[21] T. Oxman, M. Shapira, A. Diver, R. Klein, N. Avazov, B. Rabinowitz, A new method of long-term preventive cardioprotection using Lactobacillus, Am. J. Physiol. Heart Circ. Physiol. 278 (2000) H1717–H1724.
[22] P. Bhatt, A. Verma, S. Gangola, G. Bhandari, S. Chen, Microbial glycoconjugates in organic pollutant bioremediation: recent advances and applications, Microb. Cell Factories 20 (2021) 1–8.
[23] P. Foley, A.K. Pour, E.S. Beach, J.B. Zimmerman, Derivation and synthesis of renewable surfactants, Chem. Soc. Rev. 41 (2012) 1499–1518.
[24] C.N. Mulligan, S.K. Sharma, A. Mudhoo, K. Makhijani, Green chemistry and biosurfactant research, in: C.N. Mulligan, S.K. Sharma, A. Mudhoo (Eds.), Biosurfactants Research Trends and Applications, CRC Press, Boca Raton, FL, 2014, pp. 1–30.
[25] D.P. Sachdev, S.S. Cameotra, Biosurfactants in agriculture, Appl. Microbiol. Biotechnol. 97 (3) (2013) 1005–1016.
[26] I. Mnif, D. Ghribi, Glycolipid biosurfactants: main properties and potential applications in agriculture and food industry, J. Sci. Food Agric. 96 (13) (2016) 4310–4320.
[27] M. Nitschke, C. SGVAO, Biosurfactants in food industry, Trends Food Sci. Technol. 18 (2007) 252–259.
[28] B.S. Saharan, R.K. Sahu, D. Sharma, A review on biosurfactants:fermentation, current developments and perspectives, Genet. Eng. Biotechnol. J. 1 (2011) 1–14.
[29] S. Abdollahi, Z. Tofighi, T. Babaee, et al., Evaluation of antioxidant and anti-biofilm activities of biogenic surfactants derived from Bacillus amyloliquefaciens and Pseudomonas aeruginosa, Iran. J. Pharm. Res. 19 (2) (2020) 115–126, https://doi.org/10.22037/IJPR.2020.1101033.
[30] D. Sharma, S.B. Singh, Simultaneous production of biosurfactants and Bacteriocins by probiotic Lactobacillus casei MRTL3, Int. J. Microbiol. (2014) 698713.
[31] E. Haque, K. Kayalvizhi, S. Hassan, Biocompatibility, antioxidant and anti-infective effect of biosurfactant produced by Marinobacter litoralis MB15, Int. J. Pharm. Investig. 10 (2020) 173–178, https://doi.org/10.5530/ijpi.2020.2.32.
[32] P. Amruta Kanakdande, N. Chandrahasya Khobragade, Production of biosurfactant from Basillus subtilis (MF 582633) and its evaluation for antimicrobial, antioxidant, larvicidal and anti-termite activities, Ind. J. Chem. Technol. 26 (2019) 76–83.
[33] E. Yalçın, K. Çavuşoğlu, Structural analysis and antioxidant activity of a biosurfactant obtained from Bacillus subtilis RW-I, Turkish J. Biochem. 35 (2010) 243–247.
[34] S. da Ivison Amaro, K.G.O. Bezerrac, I.J. Batista, Evaluation of the emulsifying and antioxidant capacity of the biosurfactant produced by Candida bombicola URM 3718, Chem. Eng. 79 (2020).
[35] S.S. Giri, E.C. Ryu, V. Sukumaran, S.C. Park, Antioxidant, antibacterial, and anti-adhesive activities of biosurfactants isolated from Bacillus strains, Microb. Pathog. 132 (2019) 66–72.
[36] T. Virtanen, A. Pihlanto, S. Akkanen, H. Korhonen, Development of antioxidant activity in milk whey during fermentation with lactic acid bacteria, J. Appl. Microbiol. 102 (2007) 106–115.
[37] H. Kaizu, M. Sasaki, H. Nakajima, Y. Suzuki, Effect of antioxidative lactic acid bacteria on rats fed a diet deficient in vitamin E, J. Dairy Sci. 76 (9) (1993) 2493–2499.
[38] A. Merghni, I. Dallel, E. Noumi, Y. Kadmi, H. Hentati, S. Tobji, A.B. Amor, M. Mastouri, Antioxidant and antiproliferative potential of biosurfactants isolated from

Lactobacillus casei and their anti-biofilm effect in oral Staphylococcus aureus strains, Microb. Pathog. 104 (2017) 84–89, https://doi.org/10.1016/j.micpath.2017.01.017.

[39] Z. Tahir, M.S. Nazir, A.A. Aslam, S. Bano, Z. Ali, M.N. Akhtar, K. Azam, M.A. Abdullah, Active metabolites and biosurfactants for utilization in environmental remediation and eco-restoration of polluted soils, in: Green Sustainable Process for Chemical and Environmental Engineering and Science, Elsevier, Amsterdam, The Netherlands, 2021, pp. 31–51.

[40] M.Y. Lin, F.J. Chang, Antioxidative effect of intestinal bacteria Bifidobacterium longum ATCC 15708 and Lactobacillus acidophilus ATCC 4356, Dig. Dis. Sci. 45 (2000) 1617–1622.

[41] N.K. Prasad, R. Jayakumar, P. Wagdevi, S. Ramesh, Application of biosurfactants as bioabsorption agents of heavily contaminated soil and water, in: Green Sustainable Process for Chemical and Environmental Engineering and Science, Elsevier, Alpharetta, GA, USA, 2021, pp. 21–30.

[42] K.B. Beckman, B.N. Ames, The free radical theory of aging matures, Physiol. Rev. 78 (1998) 547–581.

[43] B. Gargouri, M. del Mar Contreras, S. Ammar, A. Segura-Carretero, M. Bouaziz, Biosurfactant production by the crude oil degrading Stenotrophomonas sp. B-2: chemical characterization, biological activities and environmental applications, Environ. Sci. Pollut. Res. 24 (2017) 3769–3779, https://doi.org/10.1007/s11356-016-8064-4.

CHAPTER

18 Next-generational biosurfactant and their practical application in the food industry

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1 Introduction

Biosurfactants are amphiphilic chemicals produced in surfaces of mostly of microbial cells or have extracellular hydrophobic and hydrophilic sides that bestow the ability to accumulate between fluid phases, which reduces surface and interfacial tension at the surface and interface level [1]. In a broad category, surfactants are tensio-active biomolecules with multifunctional properties used in various sectors and processes of industry. Surfactants are produced in both ways synthetically and biologically [2]. The amphiphilic substance that can produce micelles can be actively used for surface chemical actions, which are designated as SURFace ACTive AgeNTS or SURFACTANTS [3].

Based on the chemical structure and their microbial origin, biosurfactants are mainly classified as phospholipids, glycolipids, polymeric biosurfactants, and lipopeptides (surfactin). The widely identified glycolipids include rhamnolipids, sophorolipids, and trehalolipids in which most of the research has completed [4,5]. Among the domain of bacteria, *Pseudomonas, Bacillus*, and *Acinetobacter* genera are excellent producers of biosurfactants, as literature of researcher informed [6].

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Biosurfactant can contend with synthetic type of surfactants in reference to the category of biodegradability, low toxicity, performance level, environment-friendliness, and stability over different environmental factors [2]. [7,8] respectively researched on a fungal species *Candida bombicola* and a yeast species *Pseudozyma rugulosa*. Biosurfactants are categorized into two divisions such as low-molecular-weight (LMW) and high-molecular-weight (HMW) biosurfactants in terms of their biochemical properties. Biosurfactants were applied to increase the surface area of hydrophobic substrates as well as the bioavailability of hydrophobic substances through solubilization/desorption. They also control the attachment and deattachment of microorganisms from the surfaces [9].

Depending on the polarity of the head group, they are divided into anionic, cationic, nonionic, and zwitterionic [10].

(1) The anionic surfactants bear a negative charge, which are the commonly occurring surfactants in chemically and naturally also [11,12]. These negatively charged surfactants have important use in personal care products and soaps due to the effectiveness as cleansing agents. Furthermore, biosurfactants are also applied in the health, cosmetics, remediation, oil industry, and agriculture because they possess wide range of values of hydrophilic-hydrophobic balance (HLB), emulsification property, and surface tension reducer [13].

(2) The positively charged surfactants (cationic) are sound appropriate for cationic charge, that is why they are applied as anticorrosion/antistatic agents, hair conditioners, fabric softeners, flotation separators, and bactericides [14].

(3) The nonionics are surfactants comprising uncharged hydrophilic head groups. Because of their low irritating effects, these are excellent in low-temperature detergents and emulsifiers [15].

(4) The zwitterionics are amphoteric surfactants, possess excellent dermatological properties, and are compatible for the skin. These surfactants are used in the preparation of shampoos and cosmetics [11,16].

The first commercial biosurfactants on the market were sophorolipids. As per the information of currently known biosurfactants, rhamnolipids possess the highest potential for becoming the next generation of biosurfactants and can be introduced on the market. Biosurfactants research is still in the infant condition of initial commercialization. However, there are still considerable obstacles in the process development of the production of the next-generation rhamnolipids [17]. Rhamnolipids are primarily made of one or two units of rhamnose linked in a 1,2-glycosidic bond linkage to 2 β-hydroxy fatty acids ranging between 8 and 18 carbons in lengthwise.

The widely studied rhamnolipids are mostly produced by the Gram-negative bacteria, such as *Pseudomonas aeruginosa* [18]. Rhamnolipids extracted from a species of *Marinobacter* have industrial prospects due to cheap, renewable feed stocks, reported by [19].

Microbial-derived surfactants are gaining their momentum in the market as an alternative to their chemical counterparts for the application in industries. Although

it has created a worldwide market for these compounds, the actual production process has many limitations such as factors, the high cost of production [20].

2 Concept of next-generation biosurfactant

Synthetic surfactant has earned good popularity in the modernized society due to its diverse range of applications such as in agro-chemical industry, pharmaceutical industry, food industry, etc. But, referring to the fact, in spite of having good marketability position in the ameliorated society, surfactants developed synthetically turned out to be an issue of criticism for a group of environmentalist, the primary reason for such disapproval might be due to their nonbiodegradability and highly toxic nature. On top of that, they are synthesized from petrochemicals and its derivatives, which ensured the inclusion of extended hydrocarbon chains that may have the potency to restrict biodegradation process, thereby causing environmental setbacks, for instance, eutrophication of rivers, increment of phosphate concentration, etc. [21,22]. Therefore, it is of utmost requirement in reference to the present context to replace the petroleum, coal, and natural gas-based products with renewable, biodegradable, and sustainable products, and developing such strategies is to be novel challenge for human and environmental protection agencies [23–25]. In consideration of tackling environmental problems caused by synthetic surfactants, production of biosurfactant has gained its popularity among the researchers and industrialists all over the world. It is dominating the market of surfactant, as many manufacturers are going for eco-friendly production, which is possibly done from natural and sustainable resources [21]. Biosurfactant is able to substitute its chemical counterpart because of its various advantageous properties; they have comparably simpler structure [26], salinity tolerance capacity is much higher than their chemical counterparts [27]. Also, they have high biodegradability, low toxicity, and environmental compatibility, practicably synthesizable from agro-industrial wastes and have the capacity to maintain their properties even at the extreme conditions such as temperature, pH, etc., which the chemical equivalents lack of [28,29].

They are primarily synthesized by microorganisms, plants, and higher organisms such as vertebrates as well [30]. Multitude number of people has attempted to produce biosurfactant in an innovational way using agro-based raw materials, which is however has not made it to the commercial level at present. In order to reduce production cost, attempts have been made to produce these substances using low-cost, easily available raw materials such as sugarcane molasses, corn steep liquor, agro-industrial waste, etc. [31–34], and their ventures have been found successful. The mode of action of the biosurfactant is equivalent to chemical surfactant, which includes minimizing the surface tension of the medium by means of forming micelles, thereby increasing solubility and favoring the occurrence of microemulsions [35]. Surface tension is indirectly proportional to the concentration of biosurfactant, for instance, more the number of biosurfactants, less will be the surface tension [36]. Another factor that is worth to mention is that biosurfactants with lower

molecular weight are found to be more efficient in reducing surface tension, but, at the same time, those compounds with higher molecular weight have considerable advantages as they show more efficiency and efficacy in micelles formation [37].

Considering its profitable and beneficial properties, majority of the world population is embracing, and the graph of studies being done on biosurfactant can be seen growing exponentially in the last two decades [38,39], and the fact is strengthened with displaying rapid growth of the numbers of patents on biosurfactant from around 250 in 2006 to more than 850 in 2019 [40]. Given that, the market value of biosurfactant is expected to cross 2.4 billion US dollars by 2025 [41].

At present, they are being employed in cosmetic and beauty products [42], pharmaceutical products (antimicrobial, antibiofilm) [43,44], petroleum industry for enhanced oil recovery (EOR) [26], and food industry in particular [45].

3 Rhamnolipids: The next-generation biosurfactants for the food industry

Rhamnolipids, which are glycolipid biosurfactants, are derived primarily from several strains of the bacterium *P. aeruginosa* [46]. Additionally, *Burkholderia* sp. produces them [47]. The production of rhamnolipids by bacteria is summarized in Table 1. They occur naturally as dirhamnolipid (DRL, rhamnosyl-rhamnosyl-β-hydroxydecanoyl-β-hydroxydecanoate) and monorhamnolipids (MRL, rhamnosyl-β-hydroxydecanoyl-β-hydroxydecanoate) congeners [46].

Table 1 Some rhamnolipids-producing strains and rhamnolipids as antimicrobial agents.

Source of rhamnolipids	As antifungal agent	As antibacterial agent
Pseudomonas aeruginosa	*Alternaria alternata*	*Enterobacter aerogens*
Pseudomonas chlororaphis	*Verticillium dahlia*	*Klebsiella pneumoniae*
Pseudomonas fluorescens	*Mucor circinelloides*	*Serratia marcescens*
Pseudomonas indica	*Mucor meihei*	*Bacillus* sp.
Pseudomonas nitroreducens	*Aspergillus niger*	*Micrococcus* sp.
Pseudomonas putida	*Aspergillus carbonarius*	*Staphylococcus* sp.
Pseudomonas luteola	*Phytophthora* sp.	*Streptococcus* sp.
Burkholderia thailandensis	*Campylobacter* sp.	*Erwinia carotovora*
Burkholderia plantarii	*Fusarium* sp.	*Proteus mirabilis*
Burkholderia glumae	*Botrytis* sp.	*Salmonella* sp.
Burkholderia kururiensis	*Pythium* sp.	*Clostridium* sp.
Acinetobacter calcoaceticus	*Plasmopara* sp.	*Listeria* sp.
Enterobacter sp.	*Rhizoctonia* sp.	–

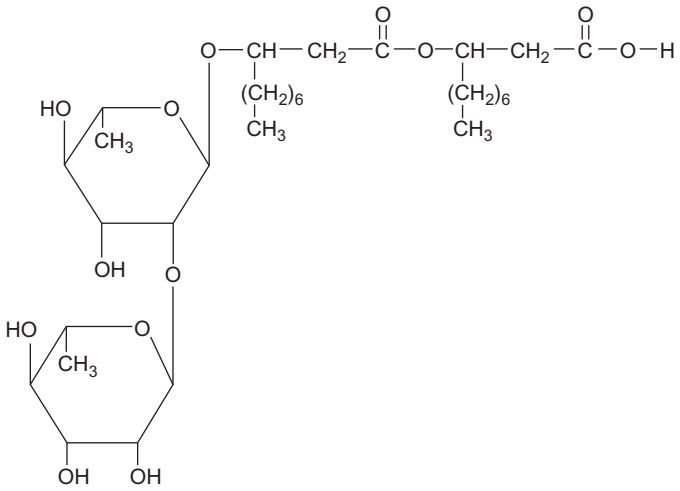

FIG. 1

The major rhamnolipid form (Rha2-C10-C10) synthesized by *Pseudomonas aeruginosa*.
Source: P. Vatsa, L. Sanchez, C. Clement, F. Baillieul, S. Dorey, Rhamnolipid biosurfactants as new players in animal and plant defense against microbes, Int. J. Mol. Sci. 11 (2010) 5095–5108.

Rhamnolipids are composed of two moieties: rhamnose (glycon/hydrophilic/lyophilic component) and lipid (aglycon/hydrophobic/lyophilic component). The rhamnose moiety is composed of mono- or di (L)-rhamnose sugars linked together via α-1, 2-glycosidic bond. The lipid component is composed of one or more β-hydroxy saturated/unsaturated fatty acid chains (primarily β-hydroxydecanoate) containing C8-C24 carbon atoms bonded together by an ester linkage. The two moieties are connected through a glycosidic bond. *P. aeruginosa* is the source of four commonly occurring rhamnolipids: 3-[3-(α-L-Rhamnopyranosyloxy) decanoyloxy] decanoic acid (Rha-C10-C10); 3-[(6-Deoxy-α-L-mannopyranosyl) oxy]decanoic acid (Rha-C10); 3-[2-*O*-α-L-Rhamnopyranosyl-α-L-rhamnopyranosyl)oxy)] decanoic acid (Rha2-C10); and 3-[3-(2-*O*-α-L-Rhamnopyranosyl-α-L-rhamnopyranosyloxy) decanoyloxy] decanoic acid (Rha2-C10-C10) [48]. The major rhamnolipid form (Rha2-C10-C10) synthesized by *P. aeruginosa* [47] is shown in Fig. 1.

3.1 Potential applications of rhamnolipids

Rhamnolipids are important industrial bioactive agents due to their high emulsification potential, antimicrobial action, and antibiofilm formation properties. They emulsify oil and significantly decrease the surface tension of water from the magnitude of 72 mN/m to a value around 25–30 mN/m [46]. Fig. 2 illustrates potential applications of rhamnolipids.

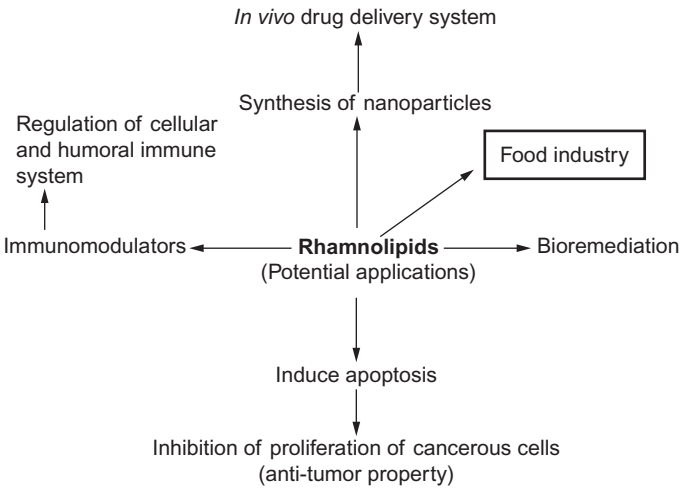

FIG. 2

Some potential applications of rhamnolipids

Source: P. Thakur, N.K. Saini, V.K. Thakur, V.K. Gupta, R.V. Saini, A.K. Saini, Rhamnolipid the glycolipid biosurfactant: emerging trends and promising strategies in the field of biotechnology and biomedicine, Microb. Cell Fact. 20 (2021) 1–15.

3.2 Rhamnolipids in food processing industry

Rhamnolipids played a major role as emulsifier and stabilizer in food industry (for example, processing sector), and it is due to factors [49] mentioned in Fig. 3. These second-generation biosurfactants are green alternatives, having ecofriendly biological properties, based mainly on sustainable production from microorganisms as renewable resources [48,50]. For example, rhamnolipids are obtained from *P. aeruginosa* strains using low-cost agro-based substrates and waste products [49]. Rhamnolipids can be used to check and control contamination of food due to their wide-spectrum antibacterial and antifungal activities (see Table 1 and Fig. 4). For example, along with nisin, they can control growth of *Listeria monocytogens* in foods without affecting their quality. Their antifungal activities get enhanced in combination laurel oil or NaCl, which is a type of synergism [48]. Rhamnolipids show direct biocide action against pathogenic bacteria and fungi. Bacterial cell surface structure and shape are altered, which includes the reduction of lipopolysaccharide concentration and alteration of outer membrane protein in the pathogen. They cause direct lysis of zoospores by intercalating within their cell membranes and reduce mycelia growth [47]. Some common applications of rhamnolipids in food industry are depicted in Fig. 5.

Biofilms are an issue of great concern in food industry, since they can not only contaminate, spoil, and decay food but also cause transmission of diseases impacting health of consumers as well as economy. Biofilm formation can be altered by changes in the surface's hydrophobicity, and these changes occur due to the presence

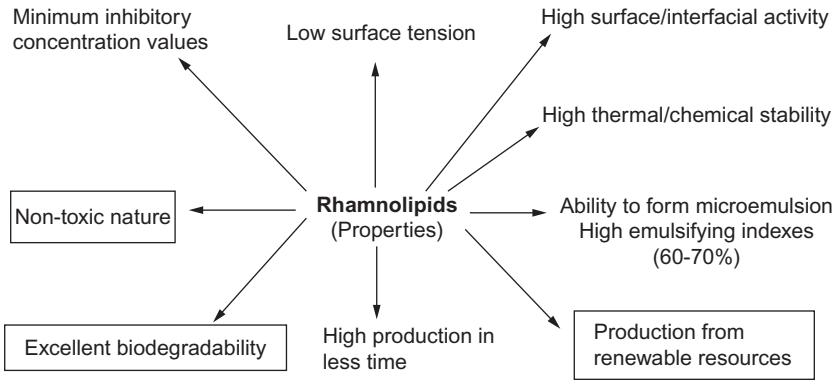

FIG. 3

Some important properties of rhamnolipids for food industry.

Source: J.P. Sinumvayo, N. Ishimwe, Agriculture and food applications of rhamnolipids and its production by Pseudomonas aeruginosa, J. Chem. Eng. Process Technol. 6 (2015) 2–9; P. Thakur, N.K. Saini, V.K. Thakur, V.K. Gupta, R.V. Saini, A.K. Saini, Rhamnolipid the glycolipid biosurfactant: emerging trends and promising strategies in the field of biotechnology and biomedicine, Microb. Cell Fact. 20 (2021) 1–15.

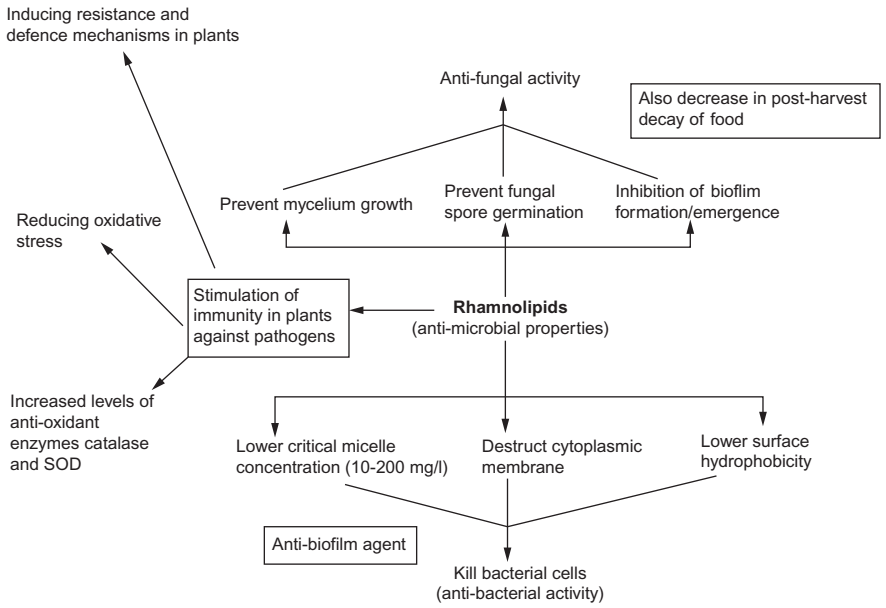

FIG. 4

Antimicrobial properties of rhamnolipids essential for food industry.

Source: P. Thakur, N.K. Saini, V.K. Thakur, V.K. Gupta, R.V. Saini, A.K. Saini, Rhamnolipid the glycolipid biosurfactant: emerging trends and promising strategies in the field of biotechnology and biomedicine, Microb. Cell Fact. 20 (2021) 1–15.

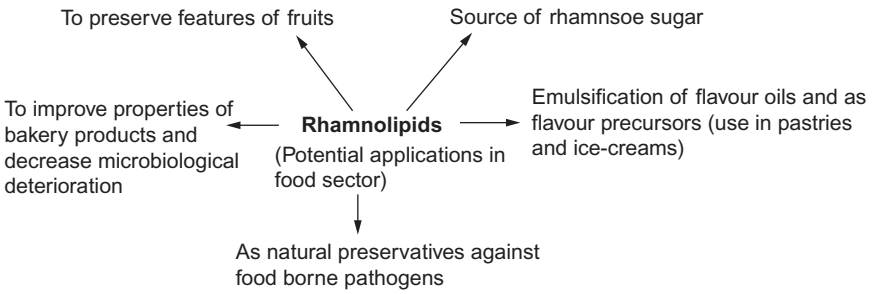

FIG. 5

Some common applications of rhamnolipids in food industry.
Source: https:/patents.google.com/patent/US20060233935A1/en.

of surface-bound biosurfactants. The addition of rhamnolipids to the surface as a preconditioner significantly reduces the likelihood of food-borne pathogenic bacteria building biofilms by eliminating adhesion [51]. Rhamnolipids alter lipid bilayer of cell membrane of pathogens and destroy formation of biofilms—thus acting as antiadhesive and antibiofilm agents in food industry [48].

Due to their antibacterial properties and other physicochemical properties, rhamnolipids can extend the shelf life of food goods without affecting consumer health, avoiding the need for synthetic chemicals, which are typically harmful to human and environmental health. Rhamnolipids with different combination of molecules such as niacin, natamicin, and nisin can syngerstically extend shelf life of food products by preventing spore germination or by inhibiting growth of bacteria and molds [49]. As natural food preservatives or food additives against food-borne pathogens causing food-borne diseases, rhamnolipids also prevent pathogens from developing resistance against chemical preservatives [48,49].

Another potential application of rhamnolipids in food industry is the decrease in postharvest decay of food, especially due to their antifungal activity, either alone or synergistically with other substances. In this way, they also induce inhibition of resistance against chemical fungicides [48]. Applications of synthetic fungicides have many eco-toxicological risks such as contamination of food products with residues and its harmful impacts on human health, resistance development against chemical fungicides, and environmental pollution. Natural antimicrobial substances can be one of the most promising strategies to control phytopathogens [52].

4 Application of next-generation biosurfactant with special reference to food processing

Food processing is the process where the fresh foods are turned into consumable food products [53]. The main reason behind this is to ensure safety, quality, and availability of perishable foods. This may involve one or many different processes including

washing, chopping, pasteurizing, freezing, fermenting, packaging, and cooking [54]. Also, it includes alteration or adding of ingredients to food products to polish the essence and the shelf life [55,56].

Emulsifiers such as lecithin or gum Arabic, which are derived from nature, have been widely used in food processing industries as an ingredient to food products. However, in modern food processing technology, which involved irradiation or microwaves, they showed some functional limitations. Also, plant gums, due to shortage of supply, which may happen because of low crop yields and transportation costs, are often subjected to price fluctuation. On top of that, in order to enhance the productivity, genetically modified crops are used, which due to biological and biomedical concerns give rise to some limitations to be used in food products [57]. Biosurfactants, due to their fascinating properties, gain a diverse application in the food industry. They are being used as emulsifying agents, wetting agents and to improve the mixing of food ingredients, as natural preservatives in various food formulations since they have antimicrobial and antibiofilm properties, as cleaning agent to remove unwanted dirt from industrial equipment as they have antiadhesive properties too [45,57].

In food processing industries, they can be utilized in two distinct ways: a) indirect utilization to clean/treat food contact surfaces; b) direct utilization as food additives/ingredients. In other way, food industries can utilize the wastes as substrate for biosurfactant production contributing in reduction of waste treatment cost [45]. As demonstrated by Dubey and Juwarkar, biosurfactants are produced from the wastes of distillery and curd processing and from the waste effluents of cassava and potato processing [58,59]. Numbers of biosurfactants derived from microorganisms have been reported to be used in emulsifying diverse range of oils, bread dough, ice-cream, mayonnaise, bakery products, and many more. Besides having the notable role as emulsifier, biosurfactants also have other functions such as stabilizing agent in food products, thereby promoting qualities in terms of texture, rheological properties, and shelf life of starch-containing products and fat-based products [45]. Some of the applications of biosurfactant in various types of food processing industries have been mentioned in Table 2.

A great number of food items are being deteriorated by natural events and are in need of protection from spoilage. Microbes played a major role in the spoilage of food products and contaminate many ready-to-eat fresh products, which on consuming can cause several food-borne diseases, thereby causing great economic losses to the food industries [80,81]. Various preservation techniques, such as heat treatment, acid treatment, salting, drying, and chemical preservatives have been extensively using to prevent the spoilage, but their safety and impact on human health are arguable. Several studies are being conducted to find good alternatives to the presently used preservative methods focusing on natural antimicrobials in particular. Biosurfactants can also be used as natural preservatives since good number of scientific analyses have verified them as antimicrobial agent [45,82,83]. Following their antimicrobial properties, they can be viewed as powerful antimicrobial agents alias natural preservatives for food industry, which will increase the food shelf life without

Table 2 Applications of biosurfactant in different food processing industries.

S. No.	Types of food industry	Role of biosurfactant	Ref.
1	Bakery	When added in dough formulation, improved its texture properties, make it more adherent dough, increase the bread volume	[60,61]
		When added in muffin preparation, displayed decreased hardness, chewiness, and gumminess, along with enhancing softness, springiness and cohesiveness and promote longer shelf life	[62,63]
2	Oil processing	To stabilize emulsions with soybean oil, coconut fats, and linseed oil	[64,65]
		To emulsify soybean oil, olive oil, sunflower oil, mustard oil	[66,67]
		Used to develop emulsion-based fish oil delivery systems to incorporate into foods	[68]
3	Salad dressing	To prevent crystal formation of salad oils	[69,70]
		As antimicrobial agent to inhibit mold growth during salad dressing	[71]
4	Margarine and ice cream	As anti-spattering agent, to stabilize the freeze-thaw process in whipped toppings and coffee whiteners	[45]
		When added to ice-cream mix, it increases quality, improves overrun and organoleptic attributes	[72]
5	Chocolate and sweets	As anti-sticking agents in candies, to modify viscosity of chocolate, to enhance glossiness in confectionary coatings	[45,69]
6	Dairy	As natural preservatives to prevent contamination of food products by food-borne pathogens	[73–75]
		To prevent/avoid fouling condition of heat exchanger plates of pasteurizer	[76,77]
		When added to milk, it reduced the count of *Salmonella enteritidis* (food-borne pathogen found commonly in dairy products)	[78]
		When used in cleaning equipment, it reduced the biofilm formation of *S. aureus* and *E.coli* by 97% in dairy cans and vats	[79]
7	Cheese production	When formulations containing rhamnolipids and natamycin are used in cottage cheese preparation, it inhibits the growth of thermophilic spores	[71]

concern to consumer health, excluding the need of using synthetic preservatives. Another important matter to worry in food processing industry is biofilm, as they can be a source of persistent contamination causing food spoilage and the transmission of diseases. Limiting the microbial adherence to surfaces in contact with food may be the crucial way to provide quality safe products to consumers. Pretreatment

of the surfaces that involve in food manufacturing processes with biosurfactant or other microbial compounds, which are able to modify physicochemical properties of surfaces, is requisite approach to make the final food products contamination-free.

5 Biosurfactant as an emulsifying agent, antimicrobial agent, Solubilizers, and wetting agents

Biosurfactants are considered to be amphipathic compounds produced on the living surfaces, mostly on the microbial cell surface, or are excreted by them that gives them the ability to accumulate between the fluid phases. This allows them to reduce the surface and interfacial tension, respectively [1]. These substances have both hydrophilic and hydrophobic moieties partitioned at the interphase within solid, liquid, and gas, thus forming the phases such as oil-water, air-water interphase. The hydrophobic moieties are nonpolar, whereas hydrophilic moieties are polar [57].

Classification of biosurfactants: Most of the biosurfactants are either anionic or neutral, and those that are cationic contain amine group and hydrophilic moiety and carbohydrate, cyclic compounds, aminoacids, phosphate carboxylic acids, or alcohol, their molecular mass ranges between 500 and 1500 Da [84]. Biosurfactants are generally categorized as follows (Table 3).

Table 3 Classification of biosurfactants [85].

Group	Subgroup	Class	Microorganisms
(I) Low-molecular-weight surface-active agents	Glycolipids	Rhamnolipids Sophorolipids Trehalolipids Cellobiolipids	*P. aeruginosa*, *Pseudomonas* sp., *T. bombicola*, *T. apicola*, *U. zeae*, *Ustilago maydis*
	Lipopeptides and lipoproteins	Surfactin Viscosin Peptide-lipid	*B. subtilis, P. fluorescens, Bacillus licheniformis*
	Fatty acids, neutral lipids, and phospholipids	Fatty acids Neutral lipids Phospholipids	*Chiromantis lepus, N. erythropolis, Thiobacillus thiooxidans, Acinetobacter calcoaceticus, Acinetobacter radioresistens*
	Polymeric surfactants	Emulsan Alasan Liposan	*Candida lipolytica, A. calcoaceticus*
(II) High-molecular-weight surface-active agents	Particulate surfactants	Vesicles Fimbriae Whole cells	Variety of bacteria

5.1 Uses of biosurfactants in food industry

The fundamental reason behind food processing is not only about enhancing shelf life of food product, but also to improve its quality in terms of look, taste, and smell as desirous as to freshly prepared as possible. This approach (food processing) uses various ingredients to maintain or improve the final quality of products, thereby bringing diversity in food items, which the people can even access very easily and enjoy. Food additives become necessary ingredients in the food supply, as people are attracted to food that doesn't taste, look, or smell good. Biosurfactant has diverse beneficial characteristics such as emulsifying, antiadhesive, and antimicrobial, which makes them multipurpose ingredient or additive. These biomolecules if we developed in proper way would represent irreplaceable food additives in food industries in the near future [86].

In general, they are used in food industries as food additives (emulsifiers). For instance, fatty acid esters containing glycerol, lectin and its derivatives, sorbitan, or ethylene glycol are being used in large scale as they enhance the organoleptic properties of food with minimal health hazards [87]. These biomolecules have prospective application in food industry, for instance, fat stabilization in food oils, reducing foam formation in sugar production, increasing the solubility of the ingredients in drink and soups, protective coatings for vegetables, fruits, etc. The food industries are emphasizing on strict microbial control measures for production purposes, in order to ensure the safety of consumers. Also, it is of utmost need to consider the end products to have a good quality and acceptance by the consumers [82,88].

Another very important issue the food industry has to deal with is biofilms, which have the potential to spoil and contaminate the food products, resulting in serious food safety issues and crumbling the economic status of food industries. Establishment of biofilms can bring damages to industrial equipment, e.g., blockage of pipelines, corrosion of equipment, weakening temperature transfer systems [89]. Disinfection procedures are not followed in most of the food industries, which is why residues are adhered to different surfaces, thereby providing a place for microbial growth. Microorganisms attached to surfaces can remain and survive for prolonged periods and under certain conditions, they formed biofilms, which are able to aggregate nutrients, residues, and other microorganisms [90].

5.2 Antimicrobial agent

Biosurfactants also can be used as natural preservatives as massive number of scientific studies have proven to act as antimicrobial agents against bacteria, yeast, fungi, algae, and viruses [45]. Among biosurfactants, lipopeptides have been widely reported to have antimicrobial action. It is found that genus *Bacillus* is responsible for producing lipopeptides. Biosurfactant called surfactin, which is the first biosurfactant found to have antimicrobial property, is reported to be produced by *B.subtilis*. Fengycin, iturin, bacillomycins, and mycosubtilins are also lipopeptide compounds with same property that are found to be produced by the same *Bacillus* [91]. Gharaei-Fathabad [92]

reported that biosurfactants can be use in treating number of diseases as well as therapeutic and probiotic agents as they found them to play antiadhesive agents to pathogens; in addition to that, they also have found them to have antibacterial, antifungal, and antiviral activity. For instance, *B. circulans* produce a biosurfactant, which showed strong antimicrobial activity against Gram-positive and Gram-negative pathogens, and semipathogenic microbial strains including MDR strain.

5.3 Solubilizers

In the cosmetic industry, biosurfactants are able to replace their chemical counterpart due to their advantageous properties, such as emulsification, foaming, water-binding capacity, spreading, and wetting properties. They are vastly used in insect-repellent products, antacids, bath products, acne pads, contact lens solutions, lipsticks, toothpaste, and many more products [87].

6 Biochemical properties of various next-generation biosurfactant for food industries

Biosurfactants have diverse applications in many industries, and out of these, their application in food industry has been intensely expanding in recent years [93,94]. They have some interesting biochemical properties [95,96] and therefore, can be explored for potential use in the food industries. Considering their reports in the food sector, the different properties of biosurfactants are briefly outlined below (Fig. 6).

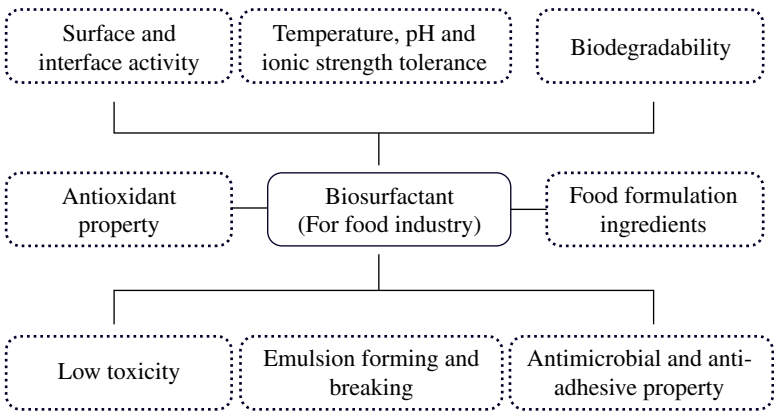

FIG. 6

Various biochemical properties of biosurfactants with reference to food industry.

6.1 Surface and Interface activity

Like surfactin [97] and rhamnolipid [98], a good-quality surfactant can lower surface tension (ST) of water from 72 to 35 mN/m and the interfacial tension (IT) of water/hexadecane from 40 to 1 mN/m [99]. The biosurfactant (Glycolipid), produced *by Saccharomyces cerevisiae* in a low-cost medium (waste soybean oil and corn steep liquor), was able to reduce the ST of culture medium from 58 to ~27 mN/m, with an IT against n-hexadecane of ~9 mN/m [100]. Largely, biosurfactants are better in efficiency and effectiveness, and their CMC (critical micelle concentration) is ~10–40 times lower than the chemical surfactants, which means maximum decrease of ST can be done by minimum concentration of biosurfactant [101].

6.2 Temperature, pH, and ionic strength tolerance

Several biosurfactants (e.g., lichenysin, lipopeptide, etc.) and their surface activity remained unchanged by dynamic environmental condition, such as temperature, ionic strength, and pH [64,102]. *Arthrobacter protophormiae*, an Antarctic psychrophilic strain, produced a thermostable biosurfactant that can withstand 30–100°C and pH of 2–12 [103]. Recently, a biosurfactant, isolated from *Candida bombicola,* demonstrated stability in a range of temperatures (4–120°C), pH values (2–12), and tolerance to high concentrations of NaCl (2%–10%) [104]. Industrial processes often expose to extreme environments, hence there is an indispensable need to isolate new extremophiles-derived products so as to function uninterruptedly under these conditions of commercial interest [105].

6.3 Biodegradability

Compounds such as biosurfactants are easily degradable in contrast to the synthetic surfactants [106], and these are particularly applicable for environmental bioremediation process [99,107]. A recent study of biosurfactant (Bio-mRL) has found the inherent biodegradable nature of the molecule ranging from 34% to 92% mineralization [108]. The rising environmental concern, among consumers and the regulatory authorities, has forced the industries to search for substitute eco-friendly bio-based products [105]. In a comparative study among six biosurfactants, four synthetic surfactants, and two commercial dispersants, it was found that most biosurfactants were quickly degraded, except for a synthetic (sucrose-stearate) that degraded more rapidly than the biogenic glycolipids (rhamnolipids, trehalose lipids, sophorose lipids) [109].

6.4 Low toxicity

Few literatures are available on the toxicity of biosurfactants; however, they are usually measured low or nontoxic products and therefore, appropriate for food industry. Synthetic anionic surfactant (Corexit) displayed higher toxicity than rhamnolipids [110]. In a comparative study among six biosurfactants, four synthetic surfactants,

and two commercial dispersants, biosurfactants showed higher EC50 (effective concentration to decrease 50% of test population) values than the synthetic dispersants [109]. In a comparative study, more toxicity and mutagenic effect were observed from the chemical-derived surfactant (Marlon A-350), whereas the biosurfactant (from *P. aeruginosa*) was slightly nontoxic and nonmutagenic [111]. Recently, rhamnolipids (produced by *Marinobacter* MCTG107b and *Pseudomonas* MCTG214 (3b1)) showed a lack of cytotoxicity and the absence of any significant mutagenic/antimutagenic potential [112].

6.5 Emulsion formation and breaking

Biosurfactants are able to stabilize (emulsifiers) or destabilize (de-emulsifiers) the emulsion [45]. Coating the oil's droplets, polymeric surfactants create very stable emulsions that never coalesce [45]. This property is particularly useful for producing oil/water emulsions for cosmetics/food. In dairy products (i.e., soft cheese and ice creams), emulsifier's addition improves the texture and creaminess that impart a special value [113]. There were few attempts in food industry to evaluate the emulsion formation by biosurfactants [103,114,115]. In food processing, de-emulsification can be appealing related to the fat and oil products and waste treatment [116]. They (biosurfactants) display a multiple of properties that certainly furnish the final quality of foods such as mayonnaise and salad dressings [70], baked goods [117], ice cream [118,119], and fats and oils [120,121] and allow greater tolerance to the adverse conditions routinely found in food processing activities [122,123].

6.6 Antimicrobial and antiadhesive property

Apart from having emulsifying and stabilizing properties, biosurfactants have antimicrobial and antiadhesive properties [57], which can be put to use to extend the useful life of products. Significant contribution of these biogenic compounds is the development of strategies for prevention of microbial proliferation by controlling the coherence of microorganisms to the exposed surfaces of foods and equipment that ensures the safety and quality of products [70]. For example, potential use of biosurfactants, as antiadhesive agent, has been suggested in the processing of dairy products, during which the encrustations' formation is usual as the microorganisms adhere to the heat exchanger [39,57]. Prominent antimicrobial and antiadhesive properties can also been seen in some recent studies on *Wickerhamomyces anomalus* CCMA 0358 [124] and *Cyberlindnera saturnus* SBPN-27 [125].

6.7 Antioxidant property

Antioxidant activity is of great importance in the packaged food industry as it decreases the threat of heart disease, resists degenerative diseases, and increases the products' life by impeding or retarding oxidation reactions [45]. Food formulations, with antioxidant property, favor sales because of their increasing concern for

safe and healthier foods [126,127]. Based on the method, a biosurfactant can sequester free radicals or deplete the complexes, and its application directly depends on the experimental results [128]. High antioxidant activity can also be attributed to the unsaturated fatty acids, and therefore, the chemical structure of biosurfactant contributes strongly to the desired properties [62]. Higher level of antioxidant activity of biosurfactants can be observed in some recent reports such as from *C. utilis* [128] and *Lactobacillus* sp., [129].

7 Scale-up of biosurfactants production: Important parameters toward commercialization

The overall process economics of biosurfactant production and the product quality are greatly influenced by the parameters such as the media components (i.e., carbon, nitrogen source), physicochemical characteristics of the microbial community, growth conditions in the bioreactor, product isolation stages, etc. [130]. Involvement of these parameters makes the biosurfactant production process a very sensitive and tedious job. For example, it requires maintaining stringent environmental conditions for culture of microorganism. In addition to that, there is a high chance of contamination to a certain level during the isolation of biosurfactants from the medium. And such contamination may affect the performance of biosurfactants after all the critical micelle concentration (CMC) is a function of purity of the biosurfactants [131]. Therefore, more intensive and in-depth investigations are very imperative toward commercialization of the biosurfactant production processes.

Strenuous efforts have been made in the recent past for large-scale commercial production of biosurfactants. In this regard, two (2) strategies have been explored for sustainable production of biosurfactant, namely (a) reducing the overall cost of production and (b) increasing the yield of product (biosurfactant) [132].

In the current market scenario, a biosurfactant cannot compete with its synthetic counterpart, as the cost of production of the former is 10–12 times higher than that of the equal amount of the latter [133,134]. On the flipside, introduction of external supplements such as nutrients to amplify the production (yield) would also result in higher cost of biosurfactant production. However, biosurfactant can replace the synthetic surfactants in the large-scale industrial applications if the cost of raw materials, fermentation processes, and purification steps decreases significantly. Alternative strategy such as carbon reincorporation can be a commercially viable option in this regard to reduce the cost as well as increasing the biosurfactant yield [131,132]. Almost half of the total production cost for rhamnolipid production through microbial fermentation is required to be spent on raw materials. This production process uses hydrophobic substrates for achieving higher product yield, but this category of substrates are excessively higher in cost than sugar-based ones. Apart from this, fermentation conditions and purification become expensive steps especially when large-scale rhamnolipid yield is focused. Moreover, the production yield depends on the type and composition of the strain species. Therefore, the selection of the most

relevant strain species so as to optimize or increase the rhamnolipid production yield is very important as well [132]. The wastes and renewable resource-based substrates can undoubtedly be used as a replacement for expensive traditional substrates, which also would optimize the fermentation cost of rhamnolipid production in a very effective way [135]. Agricultural, food, dairy industry, and municipal solid waste could be utilized effectively as a substrate in this regard [136,137].

Metabolic engineering of nonpathogenic strains is considered as one of the most productive strategies toward biosurfactant production and increasing the biosurfactant yield. Metabolic engineering on certain bacterial species could convert them into rhamnolipid-producing strain, which otherwise could not have produced rhamnolipid. There is possibility to convert monorhamnolipid-producing strain into di-rhamnolipid producing one by changing the content of the strain or metabolic engineering of the strain [132]. The marine environment harbors a wide and largely untapped microbial biodiversity. It is the established treasure house of many powerful molecules that are being used for food, cosmetics, and pharmacological applications at present. Therefore, it can be a rewarding source for discovering novel biosurfactant-producing microorganisms. Exploration of newer biosurfactant-producing microbial species particularly bacteria, yeast (and other fungi) would increase the variability of these biomolecules to be easily accessible for large-scale commercial production and concurrently would decrease the dependence on some of the microbial species of known human pathogens such as *Pseudomonas aeruginosa*, *Candida*, and *Bacillus* [138].

The recent years have witnessed with increasing concern on technoeconomic feasibility analysis for scale-up of biosurfactant production. The technoeconomic analysis on the industrial-scale production of rhamnolipid using glycerol as carbon source suggested that the variable cost shares 83.5% of the total production cost. The commercial surfactants are obtainable at USD 2 per kg, whereas the minimum selling price (MSP) for the proposed scenario is $ 59.6 per kg for rhamnolipid [34,135]. Further, the electricity consumption constitutes the major share of the variable cost. The scaling-up of production necessities the use of different power consuming devices and equipment such as air compressors and stirrers (large scale), which are essential for aeration and centrifugation, respectively. Moreover, this study highlighted the significance of raw materials along with other utilities [135]. Another study highlighted the convenient and possible way of replacing costly carbon sources with low-cost materials in the production of sophorolipids by making use of glucose and oleic sunflower oil/oleic acid. The production cost is estimated as USD 2.54 per kg. The cost of raw materials constitutes 87–89% of the total production cost [139]. Ekpenyong et al. [140] presented a kinetic modeling and quasi-economic analysis for production of a glycolipopeptide biosurfactant in a medium co-optimized by response surface and neural network approaches. A selected logistic model showed the likeliness of having maximum biosurfactant concentration of 132.52 g/L, maximum specific growth rate of 0.292 h^{-1}, and specific biosurfactant yield of 4.413 g/g. Also, the quasi-economic evaluation reported the unit cost of € 0.57/g glycolipopeptide [140]. Elias et al. [141] carried out a technoeconomic environmental analysis on sophorolipid biosurfactant production from sugarcane bagasse. They estimated the probability of having a minimum biosurfactant selling

price (MBSSP) below $ 20 per kg for two different methods of production. The reaction time in the bioreactor, maximum unitary volume of the bioreactor, biosurfactant concentration at the end of the cultivation, reaction time and maximum unitary volume of the pretreatment reactor, ultrafiltration permeate flux, and the membrane operation time are the most important process variables for improvising the performances of the reactors and scale-up of the technology [141]. Another study was carried out recently, where the commercial production of a biosurfactant from *Pseudomonas cepacia* was investigated in a semi-industrial 50 L capacity bioreactor to separate hydrocarbons from oily effluents. A biosurfactant concentration of 40.5 g/L was achieved, and the surface tension was found to be reduced to 29 m N/m. The cost of the formulated and isolated biosurfactants was estimated as US$ 0.14–0.15/L and US$ 0.02/g, respectively [142]. Technical developments in reactor design and process control are also essential toward achieving sustainable industrial-level process economics. Biosurfactant life cycle analysis, considering all the phases and aspects, starting from production, consumption, applications, and to toxicity-related issues, carbon footprint generation, and depletion of resources is also the need of the hour [136].

8 Demographic impact of next-generation biosurfactants with special reference to food

Surfactants are a varied group of synthetic and natural compounds that have a common tension active property and are used in everyday life [13]. Surfactants can be formed synthetically and biologically. They have a multipurpose functionality that has application in various industrial processes. Synthetic surfactants efficiently lower the surface and interfacial tensions, whereas biosurfactants are in terms of emulsion-stabilizing agent [2]. Biosurfactants are considered to be the potential substitutes for various synthetic surfactants in processes such as lubrication and fixing dyes, various pharmaceutical industries, and also in organic and nonorganic processes of food [143]. Various studies have also found that the bio-based surfactants could be applied for a better treatment of contaminated soil and water and suitable for treatment of a range of diseases [13].

Considering its functionality, biosurfactants can compete better in terms of performance, biodegradability, and stability in wide range of environmental factors. Unlike biosurfactants, synthetic surfactants are more demanding in terms of cost-effectiveness and their availability in the market but have led to several undesired results, thus affecting the environment. The current society is facing several challenges of implementation, enforcing environmental protection and climate change for the next generations. Thus, studies on human and environmental safety to develop the efficiency of sustainable technologies on environmental remediation are being investigated [13]. Tremendous emphasis been given to various environmental impacts that have caused due to several toxic substances leading to difficulty in degradation. The increase in environmental issues and the advancement of technology are pushing the researchers toward progressive solutions for nonbiodegradable substances, thus aiding toward industrial growth and *environmental sustainability*.

Biosurfactants have multipurpose properties with vast applications in the cleanup of *environmental contaminants* through *bioremediation*. In particular, there is a growing discussion around environmental sustainability, and the significant role that biosurfactants will progressively play in future in terms of waste reduction and prospective recycle of the treated waste. Various studies have found that the use of industrial waste and by-products can foster environmental preservation, which in turn will boost the *environment and economy* when value is added to the waste streams [144]. According to the United Nations Environment Programme, an approximate of 93.1 crore tons of food waste were formed in 2019, 61% from residences, 26% from food delivery service, 13% from retail, meaning 17% of total world food production could be wasted. Therefore, food waste is critical for those engaged in production, distribution, and marketing in order to safeguard our environment [137]. In terms of global biosurfactants market, it is expected to reach 476,512.2 tons by 2025, due to increasing demand from various countries in the world, which will account to 21% of net income [143]. Over the years, the usages of biosurfactants have proved to be eco-friendly, biocompatible, and low toxic against various changing environmental factors. The usefulness and identified benefits have provided a platform and promising future in various agricultural and industrial applications leading to more favorable competitiveness (socially and economically) and greener environment.

The developments of alternatives for replacing toxic substances with suitable green energies are the prime challenges for environment. In relation to this, the concern toward human health is growing at a rapid rate associated with toxic substances released from chemical surfactants that affect human body. Unlike synthetic surfactants, biosurfactants are potentially rich and environment friendly, which makes desirable for sustainable and better health conditions [145]. The applications of biosurfactants are essential in the field of medicine and cosmetics, thus impacting human health. Protecting the ecosystem and the environment is crucial and is of interest in the formation of surfactants as an alternative to chemical substances [146]. Therefore, there is a need to give utmost attention and importance of biosurfactant-producing mechanisms for high yielding and market availability for greater sustainability and environment [12].

Innovation and research ideas are essential to substitute the chemically produced surfactants by green resolutions that are eco-friendly and helpful for the sake of environment and human beings. Researching toward better alternative and nature-friendly has become a bigger challenge considering the cost and yield factors, thus ensuring viability in the market.

9 Conclusion and future projection

Researchers in food industries are giving attention toward natural alternatives for synthetic ingredients that are used currently in food formulation. The primary cause of such approaches is to satisfy the consumers (most of the world's population) who are demanding "completely natural" food products. As of now, biosurfactant stands

in topmost position as an alternative to chemical counterparts due to its numerous favorable and beneficial properties. However, the production of biosurfactant at large scale for commercialization is unfeasible at this point of time due to production costs, reduced yields, and divergence in the characteristics of the product. Because, the profitable production of biosurfactant is ascertained by some factors, which include cost of raw materials, economic and viable production procedure, and the product yield of microorganisms. One can work on using cheaper and waste substances to produce the biosurfactant, if it is to bring to the level of commercialization. Apart from production, other factors are to be concern before using it in food products, specifically safety, since some biosurfactants are produced with making use of opportunistic bacteria. Therefore, there is a need of more research studies to identify nonpathogenic biosurfactant-producing strains. Another important aspect that needs the attention of future researchers is the assessment of the sustainability and effectiveness of biosurfactants life cycle, which should spotlight all the phases, starting from production to till application. Also, it should consider toxicity-related issues, generation of carbon footprint, and lessening of resources. Despite having all the shortcomings and frailty, biosurfactants have the paramount ability to take over synthetic counterpart and deliver benefits to human and the environment. Soon or later, these bio-based surfactants will be part and parcel of every food processing industries as an ingredient for food formulations and surface-modifying agents.

References

[1] C.D. Cunha, M. Do Rosario, A.S. Rosado, S.G.F. Leite, *Serratia* sp SVGG 16: a promising bio-surfactant producer isolated from tropical soil during growth with ethanol-blended gasoline, Process Biochem. 39 (2004) 2277–2282.

[2] E.O. Fenibo, G.N. Ijoma, R. Selvarajan, C.B. Chikere, Microbial surfactants: the next generation multifunctional biomolecules for applications in the petroleum industry and its associated environmental remediation, Microorganisms 7 (2019) 581.

[3] E. Gakpe, P.K.S.M. Rahman, A.A. Mohamed Hatha, Microbial biosurfactants—review, J. Mar. Atmos. Res. 3 (2007) 1–17.

[4] H. Yin, Y. Qiang, Y. Jia, J. Ye, H. Peng, et al., Characteristics of biosurfactant produced by *Pseudomonas aeruginosa* S6 isolated from oil containing wastewater, Process Biochem. 44 (2009) 302–308.

[5] R.S. Makkar, K.J. Rockne, Comparison of synthetic surfactants and biosurfactants in enhancing biodegradation of polycyclic aromatic hydrocarbon, Environ. Toxicol. Chem. 22 (2003) 2280–2292.

[6] H.B. Sobrinho, J.M. Luna, R.D. Rufino, A.L.F. Porto, L.A. Sarubbo, Biosurfactants: classification, properties and environmental applications, Rec. Dev. Biotechnol. 11 (2013) 1–29.

[7] G. Bhardwaj, S.S. Cameotra, H.K. Chopra, Biosurfactants from fungi: a review, J. Pet. Environ. Biotechnol. 4 (2013) 1–6.

[8] T. Morita, M. Konishi, T. Fukuoka, T. Imura, D. Kitamoto, Physiological differences in the formation of the glycolipid biosurfactants, mannosylerythritol lipids, between *Pseudozyma antarctica* and *Pseudozyma aphidis*, Appl. Microbiol. Biotechnol. 74 (2007) 307–315.

[9] S. Vijayakumar, V. Saravanan, Biosurfactants-types, sources and applications, Res. J. Microbiol. 10 (2015) 181–192.
[10] N. Dave, T.A. Joshi, Concise review on surfactants and its significance, Int. J. Appl. Chem. 13 (2017) 663–672.
[11] A. Bratovcic, S. Nazdrajic, A. Odobasic, I. Sestan, The influence of type of surfactant on physicochemical properties of liquid soap, Int. J. Mater. Chem. 8 (2018) 31–37.
[12] J. Sil, P. Dandapat, S. Das, Health care applications of different biosurfactants: review, Int. J. Sci. Res. 6 (2017) 41–50.
[13] S. Akbari, N.H. Abdurahman, R.M. Yunus, F. Fayaz, O.R. Alara, Biosurfactants—a new frontier for social and environmental safety: a mini review, Biotechnol. Res. Innov. 2 (2018) 81–90.
[14] L. Rhein, Surfactant action on skin and hair: Cleansing and skin reactivity mechanisms, in: I. Johansson, P. Somasundaran (Eds.), Handbook for Cleaning/Decontamination of Surfaces, Elsevier, Amsterdam The Netherlands; Boston, MA, USA, 2002, pp. 305–369.
[15] N. Alwadani, P. Fatehi, Synthetic and lignin based surfactants: challenges and opportunities, Carbon Resour. Convers. 1 (2018) 126–138.
[16] M. Lukic, I. Pantelic, S. Savic, An overview of novel surfactants for formulation of cosmetics with certain emphasis on acidic active substances, Tenside Surfactant Deterg. 53 (2016) 7–19.
[17] M.M. Markus, J.H. Kügler, M. Henkel, M. Gerlitzki, B. Hörmann, M. Pohnlein, C. Syldatk, R. Hausmann, Rhamnolipids—next generation surfactants? J. Biotechnol. 162 (2012) 366–380.
[18] A.M. Abdel-Mawgoud, F. Lepine, E. Deziel, Rhamnolipids: diversity of structures, microbial origins and roles, Appl. Microbiol. Biotechnol. 86 (2010) 1323–1336.
[19] L. Tripathi, M.S. Twigg, A. Zompra, K. Salek, V.U. Irorere, T. Gutirrez, G.A. Spyroulias, R. Marchant, I.M. Banat, Biosynthesis of rhamnolipid by a Marinobacterspecies expands the paradigm of biosurfactant synthesis to a new genus of the marine microfora, Microb. Cell Factories 18 (2019) 164.
[20] P. Singh, Y. Patil, V. Rale, Biosurfactant production: emerging trends and promising strategies, J. Appl. Microbiol. 126 (2018) 2–13.
[21] R. Marchant, I.M. Banat, Microbial biosurfactants: challenges and opportunities for future exploitation, Trends Biotechnol. 30 (2012) 558–565.
[22] N. Kosaric, F. Vardar-Sukan, Biosurfactants: Production and Utilization—Processes, Technologies, and Economics, CRC Press, London, Boca Raton, FL, 2015.
[23] O.E. Ajala, F. Aberuagba, T.E. Odetoye, A.M. Ajala, Biodiesel: sustainable energy replacement to petroleum-based diesel fuel: a review, Chem. Bio Eng. Rev. 2 (2015) 145–156.
[24] S. Fantechi, Nanotechnology for sustainable economy Euro-pean and international forum on nanotechnology, Eur. Res. Area 338 (2009).
[25] S.M. Shaban, A.A. Abd-Elaal, Studying the silver nanoparticles influence on thermodynamic behavior and antimicrobial activities of novel amide Gemini cationic surfactants, Mater. Sci. Eng. C 76 (2017) 871–885.
[26] S. Shekhar, A. Sundaramanickam, T. Balasubramanian, Biosurfactant producing microbes and their potential applications: a review, J. Environ. Sci. Technol. 45 (2015) 1522–1554.
[27] D.K.F. Santos, R.D. Rufino, J.M. Luna, V.A. Santos, L.A. Sarubbo, Biosurfactants: multifunctional biomolecules of the 21st century, Int. J. Mol. Sci. 17 (2016) 401.

[28] S. Sudhanshu, S. Arumugam, B. Tangavel, Biosurfactant producing microbes and their potential applications: a review, Crit. Rev. Environ. Sci. Technol. 45 (2015) 1522–1554.
[29] S.S. Cameotra, P. Singh, Potential applications of microbial surfactants in biomedical sciences, J. Trends Biotechnol. 22 (2004) 142–146.
[30] M. Nitschke, G.M. Pastore, Biossurfactantes: propriedades e aplicacoes, Quim Nova 25 (2002) 772–776.
[31] M.T. Hippolyte, M. Augustin, T.M. Hervé, N. Robert, S. Devappa, Application of response surface methodology to improve the production of antimicrobial biosurfactants by *Lactobacillus paracasei* subsp. tolerans N2 using sugar cane molasses as substrate, Bioresour. Bioprocess. 5 (2018) 48.
[32] A. López-Prieto, H. Martínez-Padrón, L. Rodríguez-López, A.B. Moldes, J.M. Cruz, Isolation and characterization of a microorganism that produces biosurfactants in corn steep water, CyTA J. Food 17 (2019) 509–516.
[33] R. Kumar, A.J. Das, Utilization of agro-industrial waste for biosurfactant production under submerged fermentation and its application in oil recovery from sand matrix, Bioresour. Technol. 260 (2018) 233–240.
[34] D.A. Rivera, M. Urbina, M.A. López, V.E. López, Advances on research in the use of agro-industrial waste in biosurfactant production, World J. Microbiol. Biotechnol. 35 (2019) 155.
[35] F.C.P. Rocha e Silva, N.M.P.R.e. Silva, J.M. Luna, R.D. Rufino, V.A. Santos, L.A. Sarubo, Dissolved air flotation combined to biosurfactants: a clean and efficient alternative to treat industrial oily water, Rev. Environ. Sci. Biotechnol. 17 (2018) 591–602.
[36] S.C.S. da Araújo, R.C.B. Silva-Portela, D.C. de Lima, M.M.B. Fonsêca, W.J. Araújo, U.B. da Silva, A.P. Napp, E. Pereira, M.H. Vainstein, L.F. Agnez-Lima, MBSP1: a biosurfactant protein derived from a metagenomic library with activity in oil degradation, Sci. Rep. 10 (2020) 1340.
[37] A. Fariq, A. Saeed, Production and biomedical applications of probiotic biosurfactants, Curr. Microbiol. 72 (2016) 489–495.
[38] S.J. Varjani, V.N. Upasani, Critical review on biosurfactant analysis, purification and characterization using rhamnolipid as a model biosurfactant, Bioresour. Technol. 232 (2017) 389–397.
[39] P.R.F. Marcelino, F. Gonçalves, I.M. Jimenez, B.C. Carneiro, B.B. Santos, S.S. da Silva, Sustainable production of biosurfactants and their applications, in: A.P. Ingle, A.K. Chandel, S.S. da Silva (Eds.), Lignocellulosic Biorefining Technologies, John Wiley & Sons Ltd, Pondicherry, India, 2020, pp. 159–184.
[40] EPO, European Patent Office: Espacenet Patent Search Platform, 2019, Munich, The Hague, Berlin, Vienna, Brussels http://worldwide.espacenet.com/?locale=enEP.
[41] Global Market Insights, Inc, 2019. https://www.prnewswire.com/news-releases/biosurfactants-market-value-worth-2-4-billion-by-2025-global-market-insights-inc-300968770.html.
[42] A. Ferreira, X. Vecino, D. Ferreira, et al., Novel cosmetic formulations containing a biosurfactant from *Lactobacillus paracasei*, Colloids Surf. B: Biointerfaces 155 (2017) 522–529.
[43] P.S. Stewart, Mechanisms of antibiotic resistance in bacterial biofilms, Int. J. Med. Microbiol. 292 (2002) 107–113.
[44] D. Sharma, B.S. Saharan, Functional characterization of biomedical potential of biosurfactant produced by *Lactobacillus helveticus*, Biotechnol. Rep. 11 (2016) 27–35.

[45] M. Nitschke, S. Costa, Biosurfactants in food industry, Trends Food Sci. Technol. 18 (2007) 252–259.
[46] P. Das, X.-P. Yang, L.Z. Ma, Analysis of biosurfactants fromin dustrially viable Pseudomonas strain isolated from crude oil suggests how rhamnolipids congeners affect emulsification property and antimicrobial activity, Front. Microbiol. 5 (2014) 696.
[47] P. Vatsa, L. Sanchez, C. Clement, F. Baillieul, S. Dorey, Rhamnolipid biosurfactants as new players in animal and plant defense against microbes, Int. J. Mol. Sci. 11 (2010) 5095–5108.
[48] P. Thakur, N.K. Saini, V.K. Thakur, V.K. Gupta, R.V. Saini, A.K. Saini, Rhamnolipid the glycolipid biosurfactant: emerging trends and promising strategies in the field of biotechnology and biomedicine, Microb. Cell Factories 20 (2021) 1–15.
[49] J.P. Sinumvayo, N. Ishimwe, Agriculture and food applications of rhamnolipids and its production by *Pseudomonas aeruginosa*, J. Chem. Eng. Process. Technol. 6 (2015) 2–9.
[50] J.K. Madsen, R. Pihl, A.H. Møller, A.T. Madsen, D.E. Otzen, K.K. Andersen, The anionic biosurfactant rhamnolipid does not denature industrial enzymes, Front. Microbiol. 6 (2015) 292.
[51] M.Z.V. Gomes, M. Nitschke, Evaluation of rhamnolipid and surfactin to reduce the adhesion and remove biofilms of individual and mixed cultures of food pathogenic bacteria, Food Control 25 (2012) 441–447.
[52] K. Manoj, An Outlook on the Bio-Fungicides, Malini (a Multidisciplinary Peer Reviewed Research & Creative Annual Journal Published by Marwari Mahavidyalaya, Bhagalpur), 2020, pp. 33–42.
[53] C. Monteiro, R. Levy, R. Claro, et al., A new classification of foods based on the extent and purpose of their processing, Cad. Saude Publica 26 (2010) 2039–2049.
[54] J. Floros, R. Newsome, W. Fisher, et al., Feeding the world today and tomorrow: the importance of food science and technology, Compr. Rev. Food Sci. Food Saf. 9 (2010) 572–599.
[55] J. Dwyer, V. Fulgoni, R. Clemens, et al., Is 'processed' a four-letter word? The role of processed foods in achieving dietary guidelines and nutrients recommendations, Adv. Nutr. 3 (2012) 536–548. American Society for Nutrition.
[56] C. Weaver, J. Dwyer, V. Fulgoni, et al., Processed food: contributions to nutrition, Am. J. Clin. Nutr. 99 (2014) 1525–1542.
[57] J.M. Campos, T.L.M. Stamford, L.A. Sarubbo, et al., Microbial biosurfactants as additives for food industries, Biotechnol. Prog. 29 (2013) 1097–1108.
[58] K. Dubey, A. Juwarkar, Determination of genetic basis for biosurfactant production in distillery and curd whey wastes utilizing *Pseudomonas aeruginosa* strain BS2, Indian J. Biotechnol. 3 (2004) 74–81.
[59] K.S. Noah, D.F. Bruhn, G.A. Bala, Surfactin production from potato process effluent by *Bacillus subtilis* in a Chemostat, in: B.H. Davison, B.R. Evans, M. Finkelstein, J.D. McMillan (Eds.), Twenty-Sixth Symposium on Biotechnology for Fuels and Chemicals. ABAB Symposium, Humana Press, USA, 2005, pp. 465–474.
[60] I. Mnif, S. Besbes, R. Ellouze-Ghorbel, et al., Improvement of bread dough quality by *Bacillus subtilis* SPB1 biosurfactant addition: optimized extraction using response surface methodology, J. Sci. Food Agric. 93 (2013) 3055–3064.
[61] V. Haesendonck, I.P. Hilda, E.C. Vanzeveren Albert, Rhamnolipids in Bakery Products, European Patent: 1415 538 A1, 2006.
[62] G.S. Kiran, S. Priyadharsini, A. Sajayan, et al., Production of lipopeptide biosurfactant by a marine *Nesterenkonia* sp. and its application in food industry, Front. Microbiol. 8 (2017) 1138.

[63] S. Akira, Y. Akira, Method of Modifying Quality of Wheat Flour Product, Patent No JP 61205449, 1986.
[64] M. Nitschke, G.M. Pastore, Production and properties of a surfactant obtained from *Bacillus subtilis* grown on cassava wastewater, Bioresour. Technol. 97 (2006) 336–341.
[65] E. Haba, A. Pinazo, O. Jauregui, et al., Physicochemical characterization and antimicrobial properties of rhamnolipids produced by *Pseudomonas aeruginosa* 47T2 NCBIM 40044, Biotechnol. Bioeng. 81 (2002) 316–322.
[66] D.A. White, L.C. Hird, S.T. Ali, Production and characterization of a trehalolipid biosurfactant produced by the novel marine bacterium *Rhodococcus* sp., strain PML026, J. Appl. Microbiol. 115 (2013) 744–755.
[67] A. Daverey, K. Pakshirajan, Production of sophorolipids by the yeast *Candida bombicola* using simple and low cost fermentative media, Food Res. Int. 42 (2009) 499–504.
[68] F. Liu, Z. Zhu, C. Ma, et al., Fabrication of concentrated fish oil emulsions using dual-channel microfluidization: impact of droplet concentration on physical properties and lipid oxidation, J. Agric. Food Chem. 64 (2016) 9532–9541.
[69] G.L. Hasenhuettl, R.W. Hartel, Food Emulsifiers and Their Applications, second ed., Springer, New York, 2008.
[70] J.M. Campos, T.L.M. Stamford, L.A. Sarubbo, Characterization and application of a biosurfactant isolated from *Candida utilis* in salad dressings, Biodegradation 30 (2019) 313–324.
[71] N.R. Gandhi, V.L.P. Saikia, Rhamnolipid Compositions and Related Methods of Use, W. O. International Application Patent (PCT) 2007/095258 A3, 2007.
[72] M.S. Yeh, Y.H. Wei, J.S. Chang, Enhanced production of surfactin from *Bacillus subtilis* by the addition of solid carriers, Biotechnol. Prog. 2 (2005) 1329–1334.
[73] P. Golek, W. Bednarski, B. Brzozowski, B. Dziuba, The obtaining and properties of biosurfactants synthesized by bacteria of the genus *Lactobacillus*, Ann. Microbiol. 59 (2009) 119–126.
[74] E.J. Gudina, V. Rocha, J.A. Teixeira, L.R. Rodrigues, Antimicrobial and antiadhesive properties of a biosurfactant isolated from *Lactobacillus paracasei* ssp *paracasei* A20, Lett. Appl. Microbiol. 50 (2010) 419–424.
[75] E.J. Gudina, E.C. Fernandes, J.A. Teixeira, L.R. Rodrigues, Antimicrobial and anti-adhesive activities of cell-bound biosurfactant from *Lactobacillus agilis* CCUG31450, RSC Adv. 5 (2015) 90960–90968.
[76] H.J. Busscher, M. van der Kuij-Booij, H.C. van der Mei, Biosurfactants from thermophilic dairy streptococci and their potential role in the fouling control of heat exchanger plates, J. Ind. Microbiol. Biotechnol. 16 (1996) 15–21.
[77] H.J. Busscher, C.G. van Hoogmoed, G.I. Geertsema-Doornbusch, M. van der Kuij-Booij, H.C. van der Mei, *Streptococcus thermophilus* and its biosurfactants inhibit adhesion by Candida spp. on silicone rubber, Appl. Environ. Microbiol. 63 (1997) 3810–3817.
[78] X.F. Huang, D.H. Yang, Q. Zhou, Evaluation of screening methods for demulsifying bacteria and characterization of lipopeptide bio-demulsifier produced by Alcaligenes sp, Bioresour. Technol. 100 (2009) 1358–1365.
[79] K.J. Meena, S.J. Kanwar, Lipopeptides as the antifungal and antibacterial agents: applications in food safety and therapeutics, BioMed. Res. Int. 1 (2015) 1–9.
[80] A. Lopez-Malo, S. Maris Alzamora, E. Palou, Aspergillus flavus growth in the presence of chemical preservatives and naturally occurring antimicrobial compounds, Int. J. Food Microbiol. 99 (2005) 119–128.

[81] T. Rydlo, J. Miltz, A. Mor, Eukaryotic antimicrobial peptides: promises and premises in food safety, J. Food Sci. 71 (2006) 125–135.
[82] L.R. Rodrigues, I.M. Banat, J.A. Teixeira, R. Oliveira, Biosurfactants: potential applications in medicine, J. Antimicrob. Chemother. 57 (2006) 609–618.
[83] L.R. Rodrigues, Inhibition of bacterial adhesion on medical devices, in: D. Linke, A. Goldman (Eds.), Bacterial Adhesion. Advances in Experimental Medicine and Biology, Springer, Berlin, 2011, pp. 351–367.
[84] G. Bognolo, Biosurfactants as emulsifying agents for hydrocarbons, Colloids Surf. A Physicochem. Eng. Asp. 152 (1999) 41–52.
[85] S. Dey, S. Malik, A. Ghosh, R. Saha, B. Saha, A review on natural surfactants, RSC Adv. 5 (2015) 65757–65767.
[86] N. Ranasalva, R. Sunil, G. Poovarasan, Importance of biosurfactant in food industry, IOSR J. Agric. Vet. Sci. 7 (2014) 6–9.
[87] G. Bloomberg, Designing proteins as emulsifiers, Lebensmittel. Technol. 24 (1991) 130–131.
[88] S.N. Garcia, B.I. Osburn, M.T. Jay-Russell, One health for food safety, food security, and sustainable food production, Front. Sust. Food Syst. 4 (2020) 1.
[89] A.S. Lynch, G.T. Robertson, Bacterial and fungal biofilm infections, Annu. Rev. Med. 59 (2008) 415–428.
[90] D. Bagge-Ravn, N.G.Y. Hjelm, M. Christiansen, J.N.C. Johansen, L. Gram, The microbial ecology of processing equipment in different fish industries—analysis of the microflora during processing and following cleaning and disinfection, Int. J. Food Microbiol. 87 (2003) 239–250.
[91] P. Das, S. Mukherjee, R. Sen, Antimicrobial potential of a lipopeptide biosurfactant derived from a marine *Bacillus circulans*, J. Appl. Microbiol. 104 (2008) 1675–1684.
[92] E. Gharaei-Fathabad, Biosurfactants in pharmaceutical industry: a mini-review, Am. J. Drug Discov. Dev. 1 (2011) 58–69.
[93] P. Dikit, S. Maneerat, A. Saimmai, The effective emulsifying property of biosurfactant-producing *Marinobacter hydrocarbonoclasticus* ST1 obtained from palm oil contaminated sites, Appl. Biochem. Microbiol. 55 (2019) 615–625.
[94] E.J. Gudina, L.R. Rodrigues, Research and production of biosurfactants for the food industry, in: Bioprocessing for Biomolecules Production, 2019, pp. 125–143.
[95] M. Garg, M. Chatterjee, Isolation, characterization and antibacterial effect of biosurfactant from *Candida parapsilosis*, Biotechnol. Rep. 18 (2018), e00251.
[96] A.K.N. Felix, J.J. Martins, J.G.L. Almeida, M.E.A. Giro, K.F. Cavalcante, V.M.M. Melo, O.D.L. Pessoa, M.V.P. Rocha, L.R.B. Gonçalves, R.S. de Santiago Aguiar, Purification and characterization of a biosurfactant produced by *Bacillus subtilis* in cashew apple juice and its application in the remediation of oil-contaminated soil, Colloids Surf. B: Biointerfaces 175 (2019) 256–263.
[97] D. Cooper, C. Macdonald, S. Duff, N. Kosaric, Enhanced production of surfactin from *Bacillus subtilis* by continuous product removal and metal cation additions, Appl. Environ. Microbiol. 42 (1981) 408–412.
[98] C. Syldatk, S. Lang, F. Wagner, V. Wray, L. Witte, Chemical and physical characterization of four interfacial-active rhamnolipids from Pseudomonas spec. DSM 2874 grown on n-alkanes, Z. Naturforsch. C 40 (1985) 51–60.
[99] C.N. Mulligan, Environmental applications for biosurfactants, Environ. Pollut. 133 (2005) 183–198.

[100] B.G. Ribeiro, J.M.C. Guerra, L.A. Sarubbo, Potential food application of a biosurfactant produced by *Saccharomyces cerevisiae* URM 6670, Front. Bioeng. Biotechnol. 8 (2020) 434.
[101] J.D. Desai, I.M. Banat, Microbial production of surfactants and their commercial potential, Microbiol. Mol. Biol. Rev. 61 (1997) 47–64.
[102] M.J. McInerney, M. Javaheri, D.P. Nagle Jr., Properties of the biosurfactant produced by *Bacillus licheniformis* strain JF-2, J. Ind. Microbiol. Biotechnol. 5 (1990) 95–101.
[103] V. Pruthi, S.S. Cameotra, Production and properties of a biosurfactant synthesized by *Arthrobacter protophormiae*—an Antarctic strain, World J. Microbiol. Biotechnol. 13 (1997) 137–139.
[104] M. Pinto, B. Ribeiro, J. Guerra, R. Rufino, L. Sarubbo, J. Luna, Production in bioreactor, toxicity and stability of a low-cost biosurfactant, Chem. Eng. Trans. 64 (2018) 595–600.
[105] S.S. Cameotra, R. Makkar, Synthesis of biosurfactants in extreme conditions, Appl. Microbiol. Biotechnol. 50 (1998) 520–529.
[106] P.K. Mohan, G. Nakhla, E.K. Yanful, Biokinetics of biodegradation of surfactants under aerobic, anoxic and anaerobic conditions, Water Res. 40 (2006) 533–540.
[107] M. Deleu, M. Paquot, From renewable vegetables resources to microorganisms: new trends in surfactants, C. R. Chim. 7 (2004) 641–646.
[108] D.E. Hogan, F. Tian, S.W. Malm, C. Olivares, R.P. Pacheco, M.T. Simonich, A.S. Hunjan, R.L. Tanguay, W.T. Klimecki, R. Polt, Biodegradability and toxicity of monorhamnolipid biosurfactant diastereomers, J. Hazard. Mater. 364 (2019) 600–607.
[109] K. Poremba, W. Gunkel, S. Lang, F. Wagner, Toxicity testing of synthetic and biogenic surfactants on marine microorganisms, Environ. Toxicol. Water Qual. 6 (1991) 157–163.
[110] K. Poremba, W. Gunkel, S. Lang, F. Wagner, Marine biosurfactants, III. Toxicity testing with marine microorganisms and comparison with synthetic surfactants, Z. Naturforsch. C 46 (1991) 210–216.
[111] A. Flasz, C. Rocha, B. Mosquera, C. Sajo, A comparative study of the toxicity of a synthetic surfactant and one produced by *Pseudomonas aeruginosa* ATCC 55925, Med. Sci. Res. 26 (1998) 181–185.
[112] G.-P. Voulgaridou, T. Mantso, I. Anestopoulos, A. Klavaris, C. Katzastra, D.-E. Kiousi, M. Mantela, A. Galanis, K. Gardikis, I.M. Banat, Toxicity profiling of biosurfactants produced by novel marine bacterial strains, Int. J. Mol. Sci. 22 (2021) 2383.
[113] E. Rosenberg, E.Z. Ron, High-and low-molecular-mass microbial surfactants, Appl. Microbiol. Biotechnol. 52 (1999) 154–162.
[114] T. Lukondeh, N.J. Ashbolt, P.L. Rogers, Evaluation of Kluyveromyces marxianus FII 510700 grown on a lactose-based medium as a source of a natural bioemulsifier, J. Ind. Microbiol. Biotechnol. 30 (2003) 715–720.
[115] R. Shepherd, J. Rockey, I.W. Sutherland, S. Roller, Novel bioemulsifiers from microorganisms for use in foods, J. Biotechnol. 40 (1995) 207–217.
[116] K. Traudel, S. Merten, Possible food and agricultural application of microbial surfactants: an assessment, in: Biosurfactants and Biotechnology, Routledge, 2017, pp. 183–210.
[117] K. Bandyopadhyay, C. Chakraborty, S. Bhattacharyya, Fortification of mango peel and kernel powder in cookies formulation, J. Acad. Ind. Res. 2 (2014) 661–664.
[118] F. Anjum, G. Gautam, G. Edgard, S. Negi, Biosurfactant production through Bacillus sp. MTCC 5877 and its multifarious applications in food industry, Bioresour. Technol. 213 (2016) 262–269.

[119] M. Kieliszek, A.M. Kot, A. Bzducha-Wróbel, S. BŁażejak, I. Gientka, A. Kurcz, Biotechnological use of Candida yeasts in the food industry: a review, Fungal Biol. Rev. 31 (2017) 185–198.
[120] L. Bai, D.J. McClements, Formation and stabilization of nanoemulsions using biosurfactants: Rhamnolipids, J. Colloid Interface Sci. 479 (2016) 71–79.
[121] J. Zhang, L. Bing, G.A. Reineccius, Comparison of modified starch and Quillaja saponins in the formation and stabilization of flavor nanoemulsions, Food Chem. 192 (2016) 53–59.
[122] S. Gallo, T. Brochado, L. Brochine, D. Passareli, S. Costa, I.S. Bueno, J.C. de Balieiro, R.F. Neto, L. Tedeschi, Effect of biosurfactant added in two different oil source diets on lamb performance and ruminal and blood parameters, Livest. Sci. 226 (2019) 66–72.
[123] J.M. Campos, T.L. Stamford, R.D. Rufino, J.M. Luna, T.C.M. Stamford, L.A. Sarubbo, Formulation of mayonnaise with the addition of a bioemulsifier isolated from *Candida utilis*, Toxicol. Rep. 2 (2015) 1164–1170.
[124] K.S.T. Souza, E.J. Gudiña, Z. Azevedo, V. de Freitas, R.F. Schwan, L.R. Rodrigues, D.R. Dias, J.A. Teixeira, New glycolipid biosurfactants produced by the yeast strain Wickerhamomyces anomalus CCMA 0358, Colloids Surf. B: Biointerfaces 154 (2017) 373–382.
[125] S.S. Balan, C.G. Kumar, S. Jayalakshmi, Physicochemical, structural and biological evaluation of Cybersan (trigalactomargarate), a new glycolipid biosurfactant produced by a marine yeast, Cyberlindnera saturnus strain SBPN-27, Process Biochem. 80 (2019) 171–180.
[126] A. Amiri, A. Mousakhani-Ganjeh, Z. Amiri, Y.G. Guo, A.P. Singh, R.E. Kenari, Fabrication of cumin loaded-chitosan particles: characterized by molecular, morphological, thermal, antioxidant and anticancer properties as well as its utilization in food system, Food Chem. 310 (2020) 125821.
[127] A. Hameed, S.A. Hussain, M.U. Ijaz, S. Ullah, Z. Muhammad, H.A.R. Suleria, Y. Song, Antioxidant activity of polyphenolic extracts of filamentous fungus Mucor circinelloides (WJ11): extraction, characterization and storage stability of food emulsions, Food Biosci. 34 (2020) 100525.
[128] B.G. Ribeiro, B.O. de Veras, J. dos Santos Aguiar, J.M.C. Guerra, L.A. Sarubbo, Biosurfactant produced by *Candida utilis* UFPEDA1009 with potential application in cookie formulation, Electron. J. Biotechnol. 46 (2020) 14–21.
[129] A. Merghni, I. Dallel, E. Noumi, Y. Kadmi, H. Hentati, S. Tobji, A.B. Amor, M. Mastouri, Antioxidant and antiproliferative potential of biosurfactants isolated from *Lactobacillus casei* and their anti-biofilm effect in oral *Staphylococcus aureus* strains, Microb. Pathog. 104 (2017) 84–89.
[130] A. Karnwal, Biosurfactant production using bioreactors from industrial byproducts, in: H. Sarma, M.N.V. Prasad (Eds.), Biosurfactants for a Sustainable Future: Production and Applications in the Environment and Biomedicine, John Wiley & Sons Ltd., 2021, pp. 59–78.
[131] S.T. Malkapuram, V. Sharma, S.P. Gumfekar, S. Sonawane, G. Boczkaj, M.M. Seepana, A review on recent advances in the application of biosurfactants in wastewater treatment, Sustain. Energy Technol. Assess. 48 (2021) 101576.
[132] C.E. Drakontis, S. Amin, Biosurfactants: formulations, properties, and applications, Curr. Opin. Colloid Interface Sci. 48 (2020) 77–90.
[133] B.G. Ribeiro, J.M.C. Guerra, L.A. Sarubbo, Biosurfactants: production and application prospects in the food industry, Biotechnol. Prog. 36 (2020), e3030.

[134] L.T. Bagheri, N. Ebadipour, R.R. Azad, Evaluation of a recycling bioreactor for biosurfactant production by *Pseudomonas aeruginosa* MR01 using soybean oil waste, J. Chem. Technol. Biotechnol. 91 (2015) 1368–1377.
[135] L.F. Moutinho, F.R. Moura, R.C. Silvestre, A.S. Romao-Dumaresq, Microbial biosurfactants: a broad analysis of properties, applications, biosynthesis, and techno-economical assessment of rhamnolipid production, Biotechnol. Prog. 37 (2021), e3093.
[136] G. Rawat, A. Dhasmana, V. Kumar, Biosurfactants: the next generation biomolecules for diverse applications, Environ. Sustain. 3 (2020) 353–369.
[137] S.S. Mohanty, Y. Koul, S. Varjani, A. Pandey, H.H. Ngo, J.-S. Chang, J.W.C. Wong, X.-T. Bui, A critical review on various feedstocks as sustainable substrates for biosurfactants production: a way towards cleaner production, Microb. Cell Factories 20 (2021) 120.
[138] C. Nikolova, T. Gutierrez, Biosurfactants and their applications in the oil and gas industry: current state of knowledge and future perspectives, Front. Bioeng. Biotechnol. 9 (2021) 626639.
[139] R.D. Ashby, A.J. Mcaloon, A process model for approximating the production costs of the fermentative synthesis of sophorolipids, J. Surfactant Deterg. 16 (2013) 683–691.
[140] M. Ekpenyong, A. Asitok, S. Antai, B. Ekpo, R. Antigha, N. Ogarekpe, A. Antai, U. Ogbuagu, N. Ayara, Kinetic modeling and quasi-economic analysis of fermentative glycolipopeptide biosurfactant production in a medium co-optimized by statistical and neural network approaches, Prep. Biochem. Biotechnol. 51 (2021) 450–466.
[141] A.M. Elias, A.A. Longati, H.R. Ellamla, F.F. Furlan, M.P.A. Ribeiro, P.R.F. Marcelino, J.C. dos Santos, S.S. da Silva, R.C. Giordano, Techno-economic-environmental analysis of Sophorolipid biosurfactant production from sugarcane bagasse, Ind. Eng. Chem. Res. 60 (2021) 9833–9850.
[142] R.C.F.S. da Silva, D.G. de Almeida, P.P.F. Brasileiro, R.D. Rufino, J.M. de Luna, L.A. Sarubbo, Production, formulation and cost estimation of a commercial biosurfactant, Biodegradation 30 (2019) 191–201.
[143] R.S. Reis, G.J. Pacheco, A.G. Pereira, D.M.G. Freire, Biosurfactants: production and applications, in: Biodegradation-Life of Science, vol. 2, 2013, pp. 31–61.
[144] I.O. Olasanmi, R.W. Thring, The role of biosurfactants in the continued drive for environmental sustainability, Sustainability 10 (12) (2018) 4817.
[145] M. Elshikh, R. Marchant, I. Banat, Biosurfactants: promising bioactive molecules for oral-related health applications, FEMS Microbiol. Lett. 363 (2016) 213.
[146] M. Benincasa, Rhamnolipid produced from agro industrial wastes enhances hydrocarbon biodegradation in contaminated soil, Curr. Microbiol. 54 (2007) 445–449.

Further reading

[147] Rhamnolipids in Bakery Products, https://patents.google.com/patent/US20060233935A1/en, Accessed 13 Sept 2021.
[148] H. Chong, Q. Li, Microbial production of rhamnolipids: opportunities, challenges and strategies, Microb. Cell Factories 16 (2017) 1–12.
[149] S. De Malik, S. Ghosh, A.R. Saha, B. Saha, A review on natural surfactants, RSC Adv. 5 (2015) 65757–65767.

[150] E.J. Gudiña, E.C. Fernandes, A.I. Rodrigues, J.A. Teixeira, L.R. Rodrigues, Biosurfactant production by *Bacillus subtilis* using corn steep liquor as culture medium, Front. Microbiol. 6 (2015) 59.
[151] S.A. Morris, R.T. Thompson, R.W. Glenn, K.P. Ananthapadmanabhan, G.B. Kasting, Mechanisms of anionic surfactant penetration into human skin: investigating monomer, micelle and submicellar aggregate penetration theories, Int. J. Cosmet. Sci. 41 (2019) 55–66.
[152] A. Iyer, K. Mody, B. Jha, Emulsifying properties of a marine bacterial exopolysaccharide, Enzym. Microb. Technol. 38 (2006) 220–222.
[153] N. Kosaric, Biosurfactants and their application for soil bioremediation, Food Technol. Biotechnol. 39 (2001) 295–304.
[154] I. Van Haesendonck, E.C.A. Vanzeveren, Rhamnolipids in Bakery Products, International Application Patent (PCT) WO, 2004, p. 040984.
[155] R.J. Linhardt, R. Bakhit, L. Daniels, F. Mayerl, W. Pickenhagen, Microbially produced rhamnolipid as a source of rhamnose, Biotechnol. Bioeng. 33 (1989) 365–368.
[156] R. Jahan, A.M. Bodratti, M. Tsianou, P. Alexandridis, Biosurfactants, natural alternatives to synthetic surfactants: physicochemical properties and applications, Adv. Colloid Interf. Sci. 275 (2020) 102061.

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CHAPTER

19 Applications of biosurfactant as an antioxidant in foods

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1 Introduction

Food additives, namely surfactants, are nonnutritive ingredients included to food to change physicochemical, sensory, or biological properties through treatment, manufacture, processing, storage, packing, handling, and transportation. Because one of the most important principles of additive usage is safety, an additive must satisfy an acceptable toxicological review before being approved for use, taking into consideration any combined, synergic, or protective effects that may arise from its utilization. Surfactants are very profitable amphiphilic chemicals that form micelles to lower interfacial strain between various phases, such as gases and liquids or solids and liquids, by congregating only at interface [1]. The risks of using chemical-based surfactants have sparked a worldwide alert about their use, disposal, and the need for alternative surfactants with the least harmful environmental impact. Biosurfactants, which are tensioactive micellar biomolecules generated by plants and bacteria, have therefore emerged as nontoxic surfactants [2]. These are membrane-attached metabolites that help microorganisms grow on water-insoluble substrates by reducing tension at the boundary and increasing substrate availability for absorption. Asia Pacific region is the booming marketplace for the biosurfactants, owing to these regions' technological advancements and rising demand for renewable, innovative, less harmful, and biodegradable biosurfactant products [3]. The major types of

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microorganisms that produce biosurfactants and their potential application in food are summarized in Table 1.

Biosurfactants can be utilized in food because they offer unique features not found in other additives, such the capacity to serve as antibacterial, antiadhesive, and antioxidant activities. By inhibiting or reducing contamination and microbial adherence, it can improve not only the safety of materials, but also overall quality [5]. Biosurfactants have been used in food formulations for a variety reasons, including improving viscosity, inhibiting the development of harmful bacteria, increasing resistance to oxidation, prolonging product shelf life, and lowering the calorie value by substituting fat. It can also improve the texture of cakes, bread, and cookies, as well as the stability of salad dressings [13]. The efficiency of an antioxidant in foods is determined by its chemical reactivity as well as its physical qualities, which can

Table 1 Biosurfactant-producing microorganisms and application of biosurfactants in food industry.

Biosurfactant type	Microorganism	Applications	References
Rhamnolipid	*Pseudomonas aeruginosa* ATCC-10145	Antimicrobial	[4]
Sophorolipid	*Candida glabrata* CBS138	Emulsifier, antimicrobial	[5]
Glycolipid	*Bacillus* sp. MTCC 5877	Antiadhesive, antimicrobial	[2]
Lipopeptide	*Nesterenkonia* sp.	Emulsifier, antioxidant, antibiofilm, antimicrobial	[1]
Glycolipopeptide	*Lactobacillus pentosus*	Emulsifier	[1]
Lipopeptide	*Bacillus subtilis* SPB1	Antioxidant	[6]
Glycolipid	*Pseudomonas* spp.	Antibiofilm	[7]
Surfactin	*Bacillus* spp.	Antimicrobial	[8]
Phospho-lipopeptide	*Bacillus licheniformis* VS-16	Antibiofilm	[1]
Carbohydrate-lipid-protein complex	*Candida utilis*	Emulsifier	[9]
Glycolipoprotein	*Acinetobacter indicus* M6	Antibiofilm, antimicrobial	[1]
Rufisan	*Candida lipolytica* UCP 0988	Antiadhesive, antimicrobial	[10]
Lipopeptide	*Bacillus subtilis* C19	Antimicrobial	[11]
Sophorolipid	*Candida albicans* SC5314	Emulsifier, antimicrobial	[12]

influence how the antioxidant partitions in its environment. These substances have no negative impacts on health systems because they are naturally derived and have negligible toxicity when generated by Generally Recognized as Safe (GRAS) listed microorganisms.

Biosurfactants have antioxidant potential; for example, mannosylerythritol lipids (MELs) and *Bacillus subtilis* RW-I are efficient biosurfactants with a wide range of biochemical and interfacial characteristics. In cells, MEL has the strongest protective and antioxidant properties, suggesting that it might be used as an antiaging skin care product [3]. The *Klebsiella*-derived polysaccharide emulsifier was also exhibited to have significant control of soybean oil autooxidation by inhibiting peroxidation through encapsulation. In France, this polysaccharide was being developed as a precursor of rhamnose for the production of a flavoring agent, namely furaneol [14]. This chapter discusses the features of biosurfactants and the factors affecting biosurfactant production mostly focusing on food industries. The key aspects and functions of biosurfactants as antioxidants in food are summarized.

2 General characteristics of biosurfactant as food additives

Biosurfactants are becoming more popular for commercial use due to the ever-increasing array of chemicals available. The unique characteristics of biosurfactants as food additives include (i) surface and interface activity; (ii) pH, temperature, and ionic strength; (iii) emulsion forming and emulsion breaking; (iv) low toxicity; (v) biodegradability; and (vi) antimicrobial activity. The roles of different classes of biosurfactants, as well as their major advantages, are depicted in Fig. 1. By contracting the liquid surface of the biosurfactant, the surface and interfacial action helps to resist an external force. At a concentration of 20 mM, certain biosurfactants may minimize the water surface tension from 72 mN/m to 27 mN/m [15]. Many biosurfactants such as sophorolipids, rhamnolipids, and others can be effectively functioning at a wide range of pH (5–10) under autoclaving conditions (120°C and 15 lb. pressure). The emulsion can be stabilized (emulsified) or destabilized (de-emulsified) by biosurfactants. *Torulopsis bombicola*–derived sophorolipids have been proven to lower surface tension; however they are not effective emulsifiers. Liposan, on the other hand, has been effectively used to emulsified edible oils despite not lowering the surface tension. The polymeric surfactants provide further benefits by coating oil droplets and generating stable emulsions. This feature comes in handy for producing oil-in-water emulsions for food and the cosmetics sector [5].

Biosurfactants are particularly well suited for bioremediation and waste treatment due to their biodegradability. Bacteria and other microorganisms in water and soil readily breakdown them. Other significant attributes of biosurfactants are their low toxicity, ease of digestibility, and biocompatibility, which make them ideal for usage in food and cosmetics. The target product demonstrates that biosurfactants may be made from readily accessible sustainable raw materials as well as industrial waste as a substrate. Biosurfactants have specialized effects as the complex organic

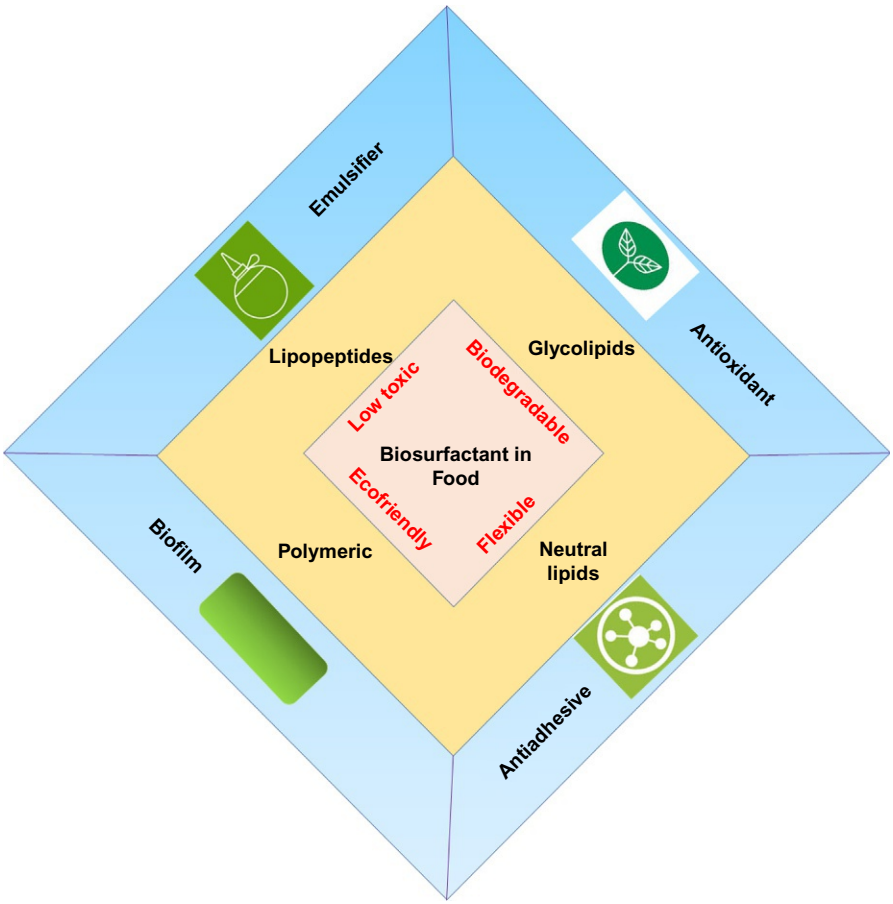

FIG. 1

Pictorial representation of functions of different types of biosurfactants and their key benefits.

compounds with the distinct functional groups, which are of great interest in detoxification of certain contaminants or in specific applications in the food sector [16].

3 Factors affecting biosurfactants production

Biosurfactants differ from synthetic counterparts in that they can be categorized according to their physical and chemical characteristics, molecular weight, and mode of activity. The lower molecular mass biosurfactants, such as lipopeptides and glycolipids, have a lipid hydrophobic moiety and may effectively reduce liquids surface tension and generate less stable emulsions. *Bacillus* and *Pseudomonas* species are well known for creating biosurfactants with a lower molecular mass [8].

Biosurfactants created by *Acinetobacter* with a high molecular weight, on the other hand, are made up of biopolymers such as polysaccharides, proteins, or lipopolysaccharides and are related to stable emulsions, while forming ineffective surface tension in liquids [6]. Biosurfactants are generated by a variety of bacteria, including *Pseudomonas aeruginosa*, *Serratia marcescens*, *Bacillus* species, *Acinetobacter bouvetii*, and fungus such as *Candida* species, *Penicillium* sp., *Aspergillus* sp., and others. Several factors influence microbial biosurfactant production, including physiological, genetic, nutritional, and environmental parameters. Under specific environmental and nutritional conditions, these biosurfactants can achieve maximal effectiveness [7].

The agitation and aeration improve dissolved oxygen delivery and mass transfer throughout the fermentation processes. But, intense agitation and aeration cause immoderate foaming, which disrupts the fermentation process. Joshi et al. controlled the rate of agitation (300 rpm) and aeration (1.0 vvm) to get a high yield of biosurfactant using *Bacillus licheniformis* R2 [17]. Another work by Yao et al. investigated the upshot of agitation and aeration in surfactin production and found that higher biosurfactant production is acquired at an agitation and aeration of 150 rpm and 1 vvm, respectively[18]. The concentration of salt has an effect on biosurfactant synthesis in some circumstances. For example, the maximal biosurfactant was formed in 3% w/v NaCl concentration and maintained about 80% of their activities in 12% w/v NaCl concentration [9].

The quality and yields of the biosurfactant manufacturing are influenced by the carbon substrate, such as crude oil, coconut oil, olive oil, fructose, glucose, maltose, sucrose, *n*-hexadecane, and so on [16]. The vegetable oils have developed as a promising and economical alternative for traditional carbon sources. Nayarisseri et al. investigated that sucrose employed that was accompanied by crude oil as effective carbon source to improve biosurfactant production from marine *B. subtilis* species [19]. The inorganic and organic nitrogen components applied in biosurfactant production include urea peptone, yeast extract, malt extract, ammonium nitrate, sodium nitrate, ammonium sulfate, and tryptone. Because enzyme and protein syntheses require nitrogen, it is critical for microbial growth during biosurfactant formation. *Nocardiopsis* sp. B4 preferred phenylalanine as a nitrogen source for biosurfactant synthesis [20], but sodium nitrate and yeast extract increased biosurfactant output in *B. subtilis* [19].

The amount of production and properties of biosurfactants are greatly related to the environmental factors such as pH, temperature, agitation, and aeration. The pH of the culture medium has been shown to impact biosurfactant synthesis in a similar way as the incubation temperature. It has been reported that pH disrupts formation of hydrogen bonds and alters the ionization of the amino acid groups, promoting variations in their structure, protein folding, as well as denaturation and, ultimately, the loss of activity. The majority of investigations on bacterial biosurfactants were done at neutral pH, whereas acidic conditions were required for yeast and fungus [11]. The advantages and disadvantages of utilizing biosurfactants in food are illustrated in Table 2.

Table 2 Merits and demerits of various biosurfactants in food industry.

Merits	Demerits
Biosurfactants may be made from a wide variety of low-cost raw sources.	Biosurfactants are notable for their poor productivity. This is due to the rarity of overproducing strains and recombinant stains.
Surface tensions are effectively reduced by biosurfactants.	At 37°C, the hemolytic activity of several biosurfactants may break erythrocytes.
Biosurfactants are less harmful than synthetic surfactants.	Obtaining pure biosurfactants necessitates a number of procedures, each of which comes at a cost.
Biosurfactants have a specialized activity and so perform a specific purpose.	The production of thick foam obstructs the increase in manufacturing yield.
Biosurfactants breakdown easily and quickly in the environment.	Biosurfactant production on a wide scale requires a lot of money.
Many biosurfactants can withstand high salinity, pH, and temperature.	
Biosurfactants may be made from industrial waste and also agroresidues, allowing for more cost-effective manufacture.	
Biosurfactants are employed in medicines, cosmetics, and the food industry because of their compatibility.	

4 Biosurfactant as antioxidant

Antioxidant potential is considered valuable in the food sector, as antioxidant-rich foods lower the cardiovascular disease risk, fight degenerative illnesses, and extend the shelf life of materials by delaying or inhibiting oxidation processes. As a result, food compositions with these attributes benefit sales, as consumers have become increasingly concerned about eating healthy meals. The biosurfactants can be utilized in two ways in the food processing industry: (a) indirectly to treat food contact surfaces and (b) directly as food additives. The abundance of unsaturated fatty acid linkages in biosurfactants may contribute to its high antioxidant activity [13]. After introducing antioxidant properties, biosurfactants play a vital role, which can provide a range of advantages, as described in the subsections below.

4.1 Control of lipid autoxidation

Antioxidant activity is influenced by a number of parameters, including the type of antioxidants, the state of oxidation, and qualities of the substrates being oxidized. The antioxidant has the potentials to determine the composition and qualities of ingredients for autooxidation of lipids. Auto-oxidation and peroxidation of fatty

acids are two parameters responsible for antibacterial activity of fatty acid. The formation of oxygen and hydrogen peroxide during the peroxidation reaction of fatty acids impedes the physiological functions of bacteria. Unsaturated fatty acids, but from the other side, promote autooxidation, producing oxylipins and low-molecular-weight aldehydes with antibacterial characteristics. The specific method through which the fatty acids reveal its antimicrobial effects by either preventing growth of bacteria or inducing microbial death will be determined by the composition and concentration of the fatty acids, as well as the kind of bacteria, temperature, pH, and binding sites [21].

4.2 Oxidative stress suppression

Oxidative stress is a state in which the cell's antioxidant-pro-oxidant equilibrium is disrupted, resulting in apoptosis, protein denaturation, and lipid peroxidation, all of which compromise the survival of the cells. Probiotics can create antioxidant metabolites including folate, glutathione, and butyrate, among others. Folate is a vitamin that takes one-carbon unit from donor entities and participates in a number of metabolic processes. Probiotics have long been recognized to offer a variety of health benefits, and studies have shown that strain-specific probiotics can have antioxidant activity and prevent oxidative damage whether consumed alone or with meals. A broad range of enzymes are effectively inhibited by free fatty acids. Unsaturated fatty acids have a higher enzyme inhibitory impact than saturated fatty acids. The enzymes required for biofilm development, molecular signaling, and cell division can be inhibited by fatty acids. The absorbed fatty acids might impact the microorganisms' own fatty acid production, altering the structural compositions of the cell membrane and, as a result, the cell membrane's permeability and fluidity, leading to cell death [22].

4.3 Enhancement of activity of other food additives

Many qualities are conferred by additives, including gelling, thickening, stability, and emulsification. Synthetic emulsifiers such as carboxymethyl cellulose and glyceryl monostearate are commonly utilized in the food sector. Biosurfactants help to enhance the activities of other food additives for improving their constancy, delay staling, and emulsify flavor oils in ice creams and bread compositions; they are also agents in fat and oil cookery. The inclusion of rhamnolipid surfactants improves the constancy of dough, quality, texture, and preservation of bread items. Rhamnolipids might potentially be used to enhance the qualities of the butter cream and preserved confectionary goods, according to the study. L-Rhamnose has a lot of potential as a flavoring precursor. The approval for the employ of biosurfactants in the food industry and quality analysis of these goods varied depending on the country. The Food and Drug Administration in the Brazilian and the United States Health Regulatory Agency are responsible for maintaining the quality of foods [23].

4.4 Improvement of free radical scavenging activities

Tocopherols, plant phenolics, and butylated hydroxytoluene are free radical scavengers that prevent the oxidation of lipids by converting alkoxyl and peroxyl radicals into constant molecules. The free radical scavengers that can block chain propagations and the fatty acids division through these routes, reducing the generation of rancidity-causing volatile fatty acid. To examine the free radical scavenging activity of biosurfactants, several simple one-dimensional assays have been devised that employ a vast variety of environments, the oxidants, and methodologies to assess oxidation end points. The hydrogen atom transfer reactions and electron transfer assays are two types of free radical scavenging potential tests. When a compound is tested using the first type of assay, it contends for peroxyl radicals with the probe, preventing probe oxidation and allowing the test compound's free radical scavenging activity to be assessed. The test substance decreases the probe radicals in the second type of assay, resulting in a color shift that may be employed to measure free radical scavenging activity. The ability of each compound to prevent lipid oxidation process in oil-in-water emulsions and fried ground beef samples was investigated to see whether their free radical scavenging activity could be utilized to forecast their potential to prevent oxidation of lipids in complicated food systems [24,25].

5 Future perspectives

Biosurfactants are emerging metabolites in the food industry with a wide variety of functional uses. Prior to incorporating biosurfactants into food products, numerous safety characteristics such as toxicity and digestion process should be considered. Despite the fact that there are numerous studies on the biological and functional properties of biosurfactants in food industry applications, only a handful has reached industrial extent. As a result, more research is needed to uncover the precise interactions or processes of biosurfactants shown by microorganisms when digesting food. If certain revolutionary strategies used in the biomedical and cosmetic domains are applied to the food industry, we will have a simple approach to understand how biosurfactants act in our bodies. The consistently faced problems associated with biosurfactant production engineering are the commercial manufacturing, structural analysis, and the economic inputs. Food industry professionals are looking for natural replacement to the various synthetic compounds already utilized in food compositions in response to consumer demand for completely natural goods. However, biosurfactant production is limited due to difficulties in maximizing the manufacturing process, insufficient quantities, purifying expenses, and changes in the quality of products, all of which impact the cost and make these natural goods less viable when compared with the artificial surfactants. As a result, biotechnology has taken important steps in enhancing the procurement of these biomolecules.

The sensory impacts of adding biosurfactants to the food compositions, as well as their effect in the food ingredients, are other issues that need to be explored. Despite

these challenges, biosurfactants have the unquestionable potential to substitute synthetic surfactants while also providing health and environmental advantages. Such biomolecules will definitely be used in the food processing sector as formulation components and surface altering additives in the near future. It is necessary to consider continuous optimization in order to get cost effective isolation methods and the deliberate use of sophisticated analysis techniques for biosurfactants characterization. The effect of antioxidant properties of biosurfactants in many other parameters such as mechanism of adhesion, mechanical stability, texture profile analysis, and so on will also be investigated to control the food safety and quality items.

6 Conclusions

Environmental and health concerns, as well as the desire to be green, are pushing demand for synthetic surfactants to be replaced by some surfactants originated from microbes. Biosurfactants have recently acquired popularity and have been used in a variety of industries, notably food processing due to their promising qualities including foaming, solubilizing, emulsifying, antimicrobial, antioxidant, and wetting agents. The major roles of biosurfactants in food are to control lipid autooxidation, suppression of oxidative stress, enhancement of free radical scavenging, and activities of other food additives. In future, genetic engineering will be extended to design superactive biosurfactants for the production of active and intelligent food additives. Biosurfactants are ideal alternatives to synthetic correlatives because of their biodegradability, low toxicity, remarkable physicochemical properties, bioavailability, and biosynthesis from renewable sources.

References

[1] I.F. da Silva, N.A. Neves, B.M. Pereira e Silveira, K.A.C. Vespermann, M.E.R. Valente, Microbial biosurfactants for contamination of food processing, in: Microbial Biosurfactants Preparation, Properties and Applications, 2021, pp. 11–30, https://doi.org/10.1007/978-981-15-6607-3_1.

[2] A.A. Jimoh, J. Lin, Biosurfactant: a new frontier for greener technology and environmental sustainability, Ecotoxicol. Environ. Saf. 184 (2019) 109607, https://doi.org/10.1016/j.ecoenv.2019.109607.

[3] S.J. Nadaf, V.M. Kumbar, A.I. Torvi, J.H. Hoskeri, A.K. Shettar, Antioxidant biosurfactants, in: Microbial Biosurfactants Preparation, Properties and Applications, Springer, Singapore, 2021, pp. 31–64, https://doi.org/10.1007/978-981-15-6607-3_3.

[4] G. Rawat, A. Dhasmana, V. Kumar, Biosurfactants: the next generation biomolecules for diverse applications, in: Environmental Sustainability, Springer, Singapore, 2020, pp. 353–369, https://doi.org/10.1007/s42398-020-00128-8.

[5] J.M. Campos, T.L. Montenegro Stamford, L.A. Sarubbo, J.M. de Luna, R.D. Rufino, I.M. Banat, Microbial biosurfactants as additives for food industries, Biotechnol. Prog. 29 (2013) 1097–1108, https://doi.org/10.1002/btpr.1796.

[6] R. Jahan, A.M. Bodratti, M. Tsianou, P. Alexandridis, Biosurfactants, natural alternatives to synthetic surfactants: physicochemical properties and applications, Adv. Colloid Interface Sci. 275 (2020) 102061, https://doi.org/10.1016/j.cis.2019.102061.
[7] R. Singh, B.R. Glick, D. Rathore, Biosurfactants as a biological tool to increase micronutrient availability in soil: a review, Pedosphere 28 (2018) 170–189, https://doi.org/10.1016/S1002-0160(18)60018-9.
[8] S.S. Giri, E.C. Ryu, V. Sukumaran, S.C. Park, Antioxidant, antibacterial, and anti-adhesive activities of biosurfactants isolated from *Bacillus* strains, Microb. Pathog. 132 (2019) 66–72, https://doi.org/10.1016/j.micpath.2019.04.035.
[9] S.K. Satpute, S.S. Zinjarde, I.M. Banat, Recent updates on biosurfactants in the food industry, in: Microbial Cell Factories, 2018, pp. 1–20, https://doi.org/10.1201/b22219-1.
[10] R.D. Rufino, J.M. de Luna, G.M. de Campos Takaki, L.A. Sarubbo, Characterization and properties of the biosurfactant produced by *Candida lipolytica* UCP 0988, Electron. J. Biotechnol. 17 (2014) 34–38, https://doi.org/10.1016/j.ejbt.2013.12.006.
[11] E.O. Fenibo, S.I. Douglas, H.O. Stanley, A review on microbial surfactants: production, classifications, properties and characterization, J. Adv. Microbiol. 18 (2019) 1–22,- https://doi.org/10.9734/jamb/2019/v18i330170.
[12] V.K. Gaur, R.K. Regar, N. Dhiman, K. Gautam, J.K. Srivastava, S. Patnaik, M. Kamthan, N. Manickam, Biosynthesis and characterization of sophorolipid biosurfactant by *Candida* spp.: application as food emulsifier and antibacterial agent, Bioresour. Technol. 285 (2019) 121314, https://doi.org/10.1016/j.biortech.2019.121314.
[13] B.G. Ribeiro, J.M.C. Guerra, L.A. Sarubbo, Biosurfactants: production and application prospects in the food industry, Biotechnol. Prog. 36 (2020) 1–39, https://doi.org/10.1002/btpr.3030.
[14] D.J. McClements, C.E. Gumus, Natural emulsifiers—biosurfactants, phospholipids, biopolymers, and colloidal particles: molecular and physicochemical basis of functional performance, Adv. Colloid Interface Sci. 234 (2016) 3–26, https://doi.org/10.1016/j.cis.2016.03.002.
[15] M.S. Yeh, Y.H. Wei, J.S. Chang, Enhanced production of surfactin from *Bacillus subtilis* by addition of solid carriers, Biotechnol. Prog. 21 (2005) 1329–1334, https://doi.org/10.1021/bp050040c.
[16] M. Fakruddin, Biosurfactant: production and application, J. Pet. Environ. Biotechnol. 03 (2012) 3–5, https://doi.org/10.4172/2157-7463.1000124.
[17] S.J. Joshi, S.J. Geetha, S. Yadav, A.J. Desai, Optimization of bench-scale production of biosurfactant by *Bacillus Licheniformis* R2, APCBEE Proc. 5 (2013) 232–236, https://doi.org/10.1016/j.apcbee.2013.05.040.
[18] S. Yao, S. Zhao, Z. Lu, Y. Gao, F. Lv, X. Bie, Control de las tasas de aireación y agitación en la producción de surfactina en un cultivo alimentado (fed-batch) en espuma desbordante con fermentación industrial, Rev. Argent. Microbiol. 47 (2015) 344–349, https://doi.org/10.1016/j.ram.2015.09.003.
[19] A. Nayarisseri, P. Singh, S.K. Singh, Screening, isolation and characterization of biosurfactant producing *Bacillus subtilis* strain ANSKLAB03, Bioinformation 14 (2018) 304–314.
[20] A. Khopade, R. Biao, X. Liu, K. Mahadik, L. Zhang, C. Kokare, Production and stability studies of the biosurfactant isolated from marine *Nocardiopsis* sp. B4, Desalination 285 (2012) 198–204, https://doi.org/10.1016/j.desal.2011.10.002.
[21] V.D. Kancheva, O.T. Kasaikina, Lipid oxidation in homogeneous and micro-heterogeneous media in presence of prooxidants, antioxidants and surfactants, Lipid Peroxidation (2012) 31–62.

[22] Y. Wang, Y. Wu, Y. Wang, H. Xu, X. Mei, D. Yu, Y. Wang, W. Li, Antioxidant properties of probiotic bacteria, Nutrients 9 (2017), https://doi.org/10.3390/nu9050521.
[23] S. Vijayakumar, V. Saravanan, Biosurfactants-types, sources and applications, Res. J. Microbiol. 10 (2015) 181–192, https://doi.org/10.3923/jm.2015.181.192.
[24] J. Alamed, W. Chaiyasit, D.J. McClements, E.-A. Decker, Relationships between free radical scavenging and antioxidant activity in foods, J. Agric. Food Chem. 57 (2009) 2969–2976.
[25] A. Sedaghat Doost, J. Van Camp, K. Dewettinck, P. Van der Meeren, Production of thymol nanoemulsions stabilized using Quillaja Saponin as a biosurfactant: antioxidant activity enhancement, Food Chem. 293 (2019) 134–143, https://doi.org/10.1016/j.foodchem.2019.04.090.

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CHAPTER

20 Applications of biosurfactants produced from food-related waste as substrate and their role in renewable resource utilization

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1 Introduction

Over the last few years, pollution by heavy metals such as Lead, Cadmium, and Mercury has been discovered to be a grave environmental concern. Numerous technologies are being found for clearing them out from the environment. Biosurfactants were shown to be a sustainable and environment-friendly solution for this purpose. Surfactants of microbial origin are usually known as Biosurfactants. Most of the microbes synthesize biosurfactants extracellularly and in large quantities. They are amphiphilic molecules containing both hydrophilic and hydrophobic domains, which can amass at the transition between fluid layers [1].

This forms emulsion by mitigating the surface and interfacial tension, and they have an extensive application in our day-to-day lives. Biosurfactants exist either in anionic or neutral form. These biosurfactants have sparked a lot of attention in recent years due to their unique qualities such as surface activity, pH, temperature and ionic strength tolerance, specificity, low toxicity, biodegradability, antimicrobial activity, and relative ease of synthesis. [2].

Biosurfactants are of great attention to a broad range of industrial areas, ranging from petroleum to pharmaceuticals. These microbial compounds demonstrate useful properties such as antimicrobial, antiadhesive, and emulsifying agents in the food industry. In the near future, the development of these biomolecules can play a significant part for the food industry. Because of their capacity to reduce surface

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tension, their outstanding emulsification capabilities, and their role in biofilm development, biosurfactants stand out in food and dairy preparations. [2].

Biosurfactants are also used to improve oil recovery and hydrocarbon bioremediation because of their biodegradability and low critical micelle level. Biosurfactants can be used to replace synthetic surfactants in a variety of industrial processes, including lubricating, wetting, fixing dyes, softening, producing emulsions, stabilizing dispersions, foaming, preventive foaming, and bioremediation of organic or inorganic contaminated areas. [3]. In the field of agriculture, Biosurfactants contribute in hydrocarbon bioremediation, metal detoxifying and/or removal, and soil washing technologies. Apart from these, Biosurfactants are also used in other industries such as laundry detergents, textile, paper, leather, mining, nanotechnology, and bioprocessing [2].

The most important biosurfactant types are Glycolipids and Rhamnolipids. They are produced by an organism *Pseudomonas aeruginosa*. Glycolipids are biosurfactants of low molecular weight produced from hydrocarbons, industrial waste, frying and olive oil wastes. They could be used to make oil/water emulsions for application in food, environment, and cosmetics [1]. Rhamnolipids are a type of Glycolipids, which can be extensively applied in industries such as food, petroleum, pharmaceutical, agriculture, bioremediation, etc.

Another type of biosurfactants, Sophorolipids are among the most promising biosurfactants produced by bacteria *Arthrobacter* sp. and *Mycobacterium* sp. They are also produced by yeast fungi *Candida bombicola* and are used as emulsifiers and antimicrobial agents. Phospholipids produced from *Acinetobacter* sp. are also used as emulsifiers. *Pseudomonas* sp. produces a type of biosurfactant called Trehalose lipid, which plays a key role in the medical field due to its surface tension properties.

Another form of biosurfactant that has been employed to improve oil recovery in harsh environmental circumstances is lipopeptide. They include Surfactin, produced from *Bacillus subtilis*, which is a powerful surfactant commonly used as an antibiotic [3].

2 Circular economy and its significance

Circular economy within the food system entails minimizing waste created in the food system, reusing food, using by-products and food waste, nutrient recycling, and dietary modifications toward more diversified and effective food patterns [4].

It is based on three principles:

- Plot out waste and pollution
- Products and materials should be in use
- Renew natural systems

Large volumes of waste produced in the food sector create major management concerns from both a social and an environmental standpoint. Food waste and by-products are created throughout the Food Life Cycle, from agriculture to industrial

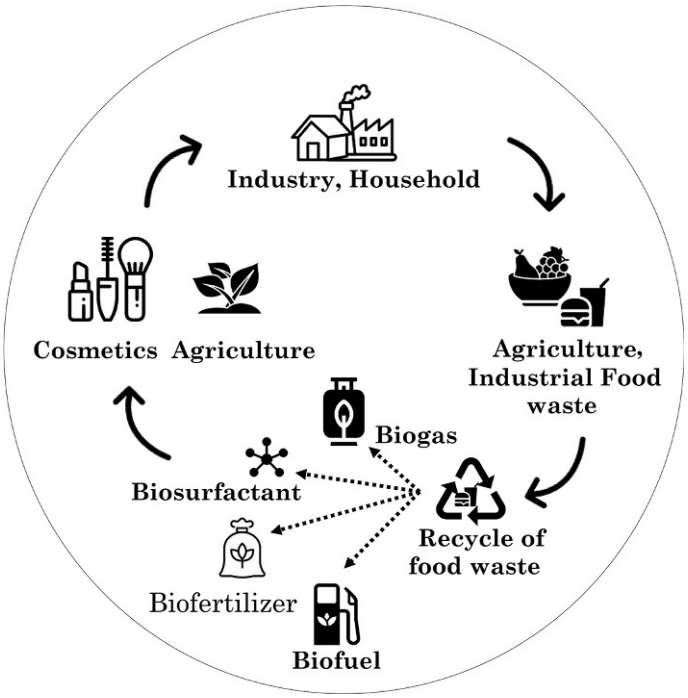

FIG. 1

Circular economy.

production, food manufacturing and sales, and finally family consumption. The nutrition loop associated with the food system, in general, can be closed [4] (Fig. 1).

With chemically processed liquids with a high organic load, significant pH fluctuations, and a relatively substantial load of suspended particles, whey is the major organic residue from the dairy industry. White hard cheese manufacturing produces sweet whey as a by-product. Cottage/cream cheese and Greek yogurt are all made using acid whey as a by-product [5]. Sergiy Smetana compared the potential of waste-to-food applications and Technology Readiness Levels (TRLs) for diverse biomass sources (e.g., insects, microalgae) to traditional animal-derived commodities in his research (e.g., chicken, whey concentrate) [6].

Residue streams are created during the industrial preparation of food. These residual streams appear during manufacturing as well as after consumption, for example, as packing material. Concerning food waste, it is necessary to distinguish between preventable and unavoidable food waste, since this affects priorities as well as management and preventive methods [4] (Fig. 2).

Goods derived from processing stages are frequently less nutritious than raw products. Proteins, fiber, and micronutrients are depleted to produce animal feed, while fats and sugars are retained [7].

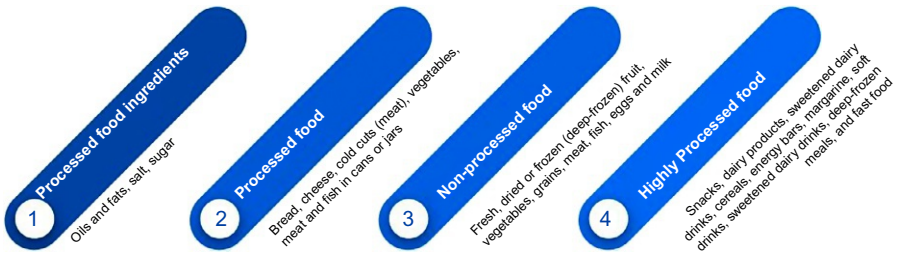

FIG. 2

Types of food in terms of processing.

A circular economy is one in which goods and services are exchanged. As a result, resource reliance and consumption, especially energy, are reduced, lowering manufacturing costs and limiting market vulnerability. It would identify ecosystem health by restoring resource flows, biodiversity, and nutrient cycles, as well as biorefineries to diversify income and contribute to the replacement of emission-intensive fossil-based production systems with bio-based chemicals, ingredients, and products. When all other alternatives for valorization have been tried, energy recovery from food waste would provide best optimization of food resources from manufacturing through consumption and disposal.

3 Types of biosurfactants

Microorganisms produce a wide range of biosurfactants. Some of them are mentioned in Table 1.

4 Strategies for biosynthesis of biosurfactants

4.1 Conversion of food wastes into biosurfactants

Different strategies are involved in the production of biosurfactants. Biosurfactants have environmentally beneficial features, such as minimal toxicity, biodegradability, and the capacity to be made from affordable and renewable basic resources [20]. Various bioreactors are used for studying the production of surfactants from food waste.

1. *Solid-state fermentation* (SSF) is one of the processes where microorganisms arise on a substrate in the absence of open fluid. The principle aim of SSF is to get the greatest supplement consideration from the substrate for aging by utilizing microorganisms [21]. Solid substrates such as grains, organic wastes, the liquid medium can be used as substrates in SSF, which brings down the cost of the end product. Das and Mukherjee carried out a study to produce biosurfactants

Table 1 Types of biosurfactants.

Biosurfactants	Subtype	Microorganism	Application	References
Glycolipids	Rhamnolipid	*Pseudomonas aeruginosa*	Removal of oil and petroleum contamination	
		Renibacterium salmoninarum	Bioremediation	
		Cellulomonas cellulans	ND	
		Nocardioides sp.	Crude oil and biodiesel	
		Burkholderia sp.	Degradation of biodiesel	
	Trehalolipid	*Arthrobacter*	ND	
		Corynebacterium	Biodegradation of crude oil	
		Mycobacterium	Biological activities	
		Rhodococcus erythropolis	Food and pharma industries	
		Rhodococcus ruber	Oil removal activity	
	Sophorolipid	*Candida bombicola*	Antiviral property	
		Yarrowia lipolytica	Extending shelf life off food products	
		Candida apicola	ND	
		Candida bogoriensis	ND	
	Mannosylerythritol lipid	*Pseudozyma hubeiensis*	Cosmetic application	
		Pseudozyma antarctica	Environmental application	
		P. rugulosa	Industrial application	
		Ustilago scitaminea	Oil recovery	
	Cellobiose lipid	*Ustilago maydis*	Polymer and pharma industry	
		Cryptococcus humicola	Fungicidal property	
		P. graminicola	ND	
		P. aphidis	ND	[1]

Continued

Table 1 Types of biosurfactants. *Continued*

Biosurfactants	Subtype	Microorganism	Application	References
Lipopeptides and lipoproteins	Surfactin	*P. aeruginosa*	ND	[8]
		Bacillus subtilis 20B	Antifungal activity	
	Iturin	*B. subtilis* RB14	Control of damping-off	[9]
	Fengycin	*B. subtilis* 9407	Antifungal property	[10]
	Bacitracin	*B. subtilis*	Antibiotic property	[11]
		B. licheniformis	Pharmaceutical industry, oil degradation	[12]
	Polymyxins	*B. polymyxa*	Used for treatment against multidrug resistance microbes	[13]
		B. brevis		
	Kurstakin	*B. thuringiensis* sub sp.	Antifungal and agricultural application	[14]
		B. cereus		[15]
	Viscosin	*Pseudomonas fluorescens*	Emulsification of *n*-hexadecane, biofilm dispersal property	[16]
				[17]
	Lichenysin	*Bacillus licheniformis*	Antiadhesion, antibiofilm activities	[18]
	Serrawettin	*Serratia marcescens*	Anticancer activities	[19]
Fatty acids		*Acinetobacter* sp.	Biodegradation of biodiesel, Crude oil	
Phospholipid and neutral lipid	Phosphatidylethanolamine	*Rhodococcus erythropolis*	ND	
	Lecithins	ND	Preparation of food products	

Polymeric biosurfactans	Emulsan	*Acinetobacter calcoaceticus*	Emulsifying agent for hydrocarbons in water.	
	Liposan	*Candida lipolytica*	Water soluble emulsifier and emulsifier in food and cosmetic industry.	
	Alasan	*Acinetobacter radioresistens*	Emulsification activity and treatment with specific proteases.	
Particulate biosurfactans		*Acinetobacter* sp.	Degradation of oil.	
Complex carbon lipid proteins		*Candida utilis*	Food industry	
Glycoproteins		*Lactobacillus pentosus*	Remediation of hydrocarbon contaminated sites.	
		Lactobacillus acidophilus		
		L. paracasei subsp. *paracasei*	Antimicrobial property, Stabilizing agent.	
		L. rhamnosus	Emulsifier and forth forming activities.	
		L fermentum	Antiadhesive capacity.	
		Rhodotorula mucilaginosa	Dispersive power against astaxanthin.	
		Geobacillus stearothermophilus	Environmental application with emulsifying properties.	

from *B. subtilis* strain DM-03 and DM-04 using potato peel as medium. Lipopeptide produced was 102 mg/g ds [22].

2. *Submerged fermentation* is another process used by the majority of ventures for high yield, minimal expenses, and defilement. Additionally, inconvenience like actual space or water prerequisites. This aging cycle is simpler to design by scientists on account of the simplicity of interaction control and disinfection. Jin et al. used *B. subtilis* 3-10 to produce 0.60 g/L iturin using rapeseed waste [23].
3. In *fed-batch cultivation* grouping of substrates and the development, the rate can be restrained by feeding. This may prompt compelling interaction that is accomplishing a high measure of biosurfactant by fermentation [24]. This option would be more financially reasonable than basic batches, because of the concealment of a few stages in regard to the inoculum planning, sterilization, and conclusion of investigation through the production of extended periods. The principle objective is to estimate durable fed-batch experiments to biosurfactants production [25]. Shah et al. produced 34 g/L Sophorolipids by fed-batch using restaurant oil waste by *C. bombicola* [26].
4. *Stirred tank reactors* are also used for surfactants, but in this sort of reactor, there is a probability of froth formation, which can be a serious issue. This kind of reactor can be used for low foaming surfactants such as Sophorolipids. de Lima et al. investigated biosurfactants in soybean frying oil waste by *P. aeruginosa* pac 1 strain, which produced 3.3 g/L of rhamnose [27].

5 Types of food waste

Food waste is the most common organic waste produced everywhere around the world. Food waste from various sources such as domestic, agricultural, and industrial sources has increased as a result of changing lifestyles and rising urbanization of the global population. Food waste valorization is a preferred technique for food waste management.

Few of the types of food waste are mentioned in Table 2.

5.1 Catering waste

(a) Vegetable oil

Based on fatty acids with carbon chain 8–24, several triacylglycerols exist in plant oils. Vegetable oil production has expanded worldwide as its consumption accelerated during the recent decade, which turned into a huge market within the food industry. Palm oil, sunflower oil, rapeseed oil, soybean oil are a few oils with primary growth [33].These oils produce plenty of waste in oil industries such as residual oils, frying oils, marine oils, etc. The abundance of nutrients in such waste can be utilized as a substrate for production of biosurfactants [34].

Table 2 Types of food waste.

Types of food waste	Description	Source	References
By products and vegetable oil processing waste	Vegetable oils wastes are produced from plant residues and oils. By-products are high amount of fats, oils.	Olive oil, soya bean oil, palm oil, peanut oil, sunflower seed oil	[28,29]
Frying oil waste	The major components of used frying oil are triglycerides, monoglycerides, diglycerides, and free fatty acids.	Coconut oil, rice bran oil, sunflower oil, canola oil	[29,30]
Fruits and vegetable waste	Inedible parts of fruits and vegetables that are discarded during collection, handling, transportation, and processing.	Cashew apple juice, peels from banana, orange, pineapple, lime	[29]
Starch rich waste	Extraction of starch from various crops produces wastewater rich in husk and starch.	Corn, rice, Cassava, wheat and potato	[29,31]
Lignocellulosic waste	Lignocellulose is one of the richest sources of carbon and has cellulosic content derived from plants.	Corn stalk, vascular plants, rice straw, bagasse	[29,32]
Animal and dairy waste	Meat processing industries produce large amount of animal and dairy waste such as dairy shed effluent, dairy manure, poultry litter, and other wastes from livestock operations.	Cattle, poultry, seafish	[29]

Surface tension values determine the quantity of production of biosurfactants, which are generated at the time of microbial activities in food waste. The microbial environment is enhanced by controlling carbon-nitrogen quantities, temperature, pH, O_2 rate, etc. Rhamnolipids have a prominent role in biosurfactant production. *P. aeruginosa* is the main microbe that produces glycolipids called Rhamnolipids [35].

Rhamnolipids contain one or two rhamnose units (mono-rhamnolipids or di-rhamnolipids) associated with 1-3 β hydroxy fatty acid units. Rhamnolipids can be produced from the most common industrial waste, which consists of fatty acids such as olive oil waste effluent, whey waste, frying oil, etc. Rhamnolipids have good antimicrobial properties and also act as an emulsifier [36].

Wong, J et al. produced Rhamnolipids with help of a shake flask where *Acinetobacter calcoaceticus* BU-03 is infused in nutrient broth as a medium of cultivation. At a speed of agitation of 150 rpm and temperature of 55°C for 16 h, bacteria were cultured with nutrient broth as substrate. Followed by seed fermentation and then passed onto first-stage fermenter along with some nutrient broth carried out for a period of 24 h. In addition to this, food waste digestate (without any pretreatment) was added in second-stage fermentation.

Next, it is moved onto the production-stage fermenter, which is performed for 43 h long. All the fermenters were sterilized at 121°C in association with food digestate for 1 h.

Later on, these fermenters are cooled down using water at room temperature, Subsequently, batch fermenters were governed aerobically at pH of 7.5 and a temperature of 55°C. No amount of broth was presumed to have been lost by evaporation through the fermentation process. After the completion of the fermentation process, downstream operations come to the next extraction of Rhamnolipids from food waste digestate. The production of Rhamnolipids from food waste during laboratory investigation was found out to be 10 g/L. The biomass and impurities were removed using centrifugation and filtration processes.

It is followed by extraction of rhamnolipid from the supernatant through hexane in the ratio of 1:3 v/v. Extraction was executed thrice to obtain the whole rhamnolipid. The distillation process is carried out to remove the hexane present in the rhamnolipid foam mixture. Finally, rhamnolipid is freeze-dried to gain solid Rhamnolipids. LC-MS was used to see the characterization of Rhamnolipids [37] (Fig. 3 and Table 3).

(b) Frying Oil

Frying oil is the most common waste seen in households and restaurants produced in large quantities. This waste is regarded as problematic and pollution-causing waste [62]. Nutrient content in these cooking oils depends on how frequently the oil is utilized and the product type used. The stability of frying oil is achieved by reducing the number of unsaturation along with the finest quality and less metal trace. Thereby traces are eliminated from hydrogenating oil in retreatment. Triglycerides and diglycerides are the two prime constituents of used frying oils, along with monoglycerides and fatty acids [29].

Waste frying oils can be utilized as an alternative for fresh oil as it is affordable. Oil is subjected to reactions such as hydrolysis, peroxidation, polymerization while frying, as a result, peroxides, ketones, and aldehydes are formed. This makes the oil worthless for frying.

S·D Wadekae et al. used *P. aeruginosa* to obtain rhamnolipid from frying oil residues in shake flask experiments, and with TLC and HPLC structural details were analyzed. They collected samples of waste frying oil as well as fresh oil at a local diner. This oil had a peroxide value of PV102 and PV0, respectively.

The oil was analyzed for different tests in accordance with the AOCS official analytical methods. At a temperature of 90°C under reduced pressure, 50 g of frying oil was heated up.

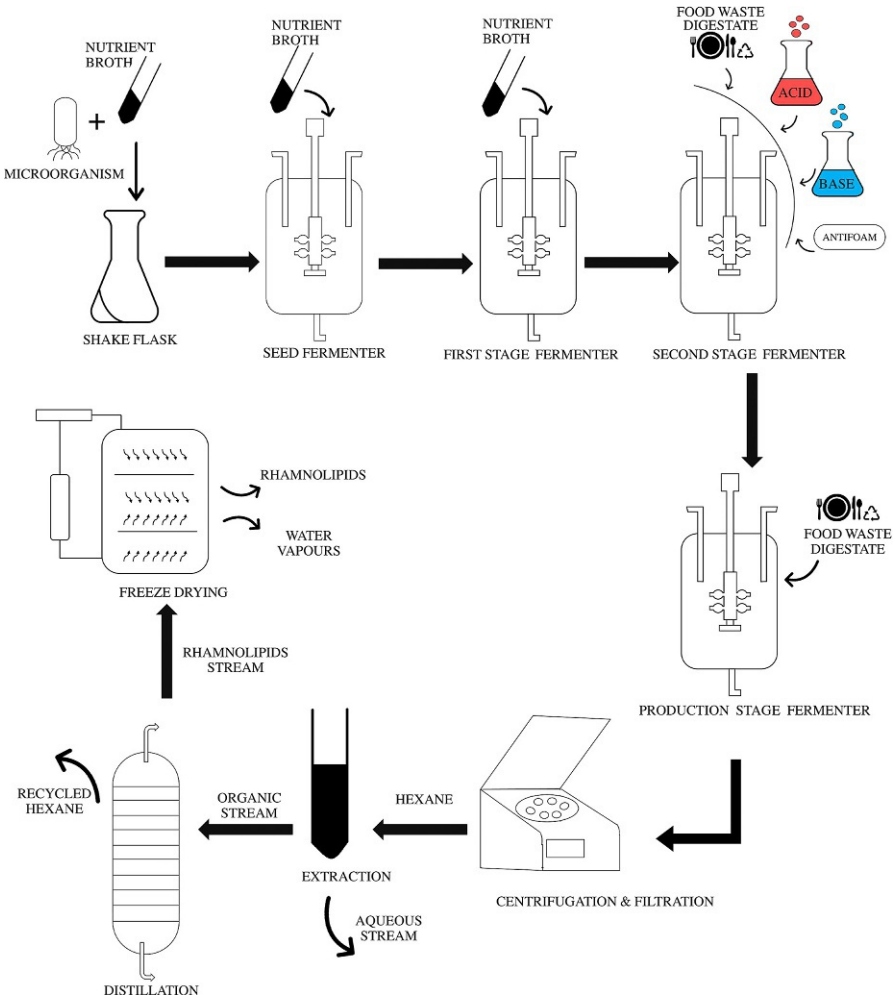

FIG. 3

Production of rhamnolipids from vegetable oil waste.

Followed by treating heated oil along with a predefined amount of activated earth (Tonsil Optimum 210FF) and charcoal, further adsorbents were separated by sintered glass crucible filtration. This oil was now considered treated oil.

P. aeruginosa was retained on nutrient agar slants. A loop of bacteria cells in conjugation with 50 mL nutrient broth was added to the Erlenmeyer flask. Through centrifugation, cells were separated and then resuspension occurred with a sterile mineral salt medium as a source of carbon.

All three different oils were autoclaved individually before adding to the medium, followed by centrifugation for the removal of biomass. A separating funnel

Table 3 Biosurfactants produced from vegetable oil waste.

Biosurfactants produced	Microorganism used	Substrate	Yield of biosurfactants and their ST		References
Lipopeptide	*Bacillus subtilis* 3-10	Untreated rapeseed meal as N_2 source	0.60 g/L	ND	[23]
Rhamnolipid	*Pseudomonas aeruginosa* AT10	Soybean oil residues	9.5 g/L	ST-26.8 mN/m	[38]
Glycolipid	*Candida antarctica* ATCC 20509 and *Candida apicola* ATTC 96134	Growth of yeast in two undefined oil residues	7.3–13.4 g/L	ST-35.6 mN/m	[39]
Glycolipid	*Penicillium citrinum*	Mineral medium—1% olive oil	0.54 p/s	ND	[40]
Biosurfactant	*Candida strains*	*n*-Hexadecane, soybean oil, groundnut oil, corn steep liquor, and glucose	ND	ST-(34–54) mN/m	[41]
Rhamnolipid	*Pseudomonas aeruginosa* AB4	Deoiled cakes of mahua, karanj, jatropha, and neem	40 g/L	ST-40 mN/m	[42]
Sophorolipid	*Starmerella bombicola* MTCC 1910	Shake flask level, Fermenter level Sunflower oil refinery waste	41.6–51.5 g/L	ST-35.5 mN/m	[43]
Surfactin	*Bacillus subtilis* KB1 *Bacillus subtilis* #309	Solid-state fermentation—Rapeseed cake	0.011 g/L	ND	[44]
Sophorolipid	*Starmerella bombicola* ATCC 22214	Solid-state fermentation—Winterization oil cake, sugar beet molasses	179 mg/g	ND	[45]
Biosurfactant	*Trametes versicolor*	Solid-state fermentation—Two-phase olive mill waste	373.6 mg/100 g	ST-34.5 mN/m	[46]
Biosurfactant	*Candida sphaerica* UCP0995	Groundnut oil refinery waste	21 g/L	ST-27 mN/m	[47]
Rhamnolipid	*Pseudomonas aeruginosa* 424	Vegetable oil	8.5 g/L	ND	[48]
Rhamnolipid	*Pseudomonas aeruginosa* 47T2	Olive oil mill effluents	0.058 g/g	ST-30 mN/m	[49]
Surfactin	*Bacillus subtilis* N1	Olive oil mill waste	0.0031 g/L	ST-46.6 mN/m	[50]

Rhamnolipid	*Pseudomonas aeruginosa* PAO1	Olive oil mill waste	0.00878 g/L	ST-33 mN/m	[50]
Rhamnolipid	*Pseudomonas aeruginosa* LBI	Batch shake flask—Oil refinery waste of corn, soyabean, palm, babassu	11.7 g/L	ST-26.9 mN/m	[51]
Biosurfactant	*Pseudomonas alcaligenes*	Palm oil	2.3 g/L	ND	[52]
Rhamnolipid	*Pseudomonas aeruginosa* PAO1	Palm oil agricultural refinery waste	0.43 g/L	ST-29 mN/m	[53]
Lipopeptide	*Candida lipolytica*	Groundnut oil, soybean oil residue	4.5 g/L	ST-25.3 mN/m	[54]
Rhamnolipid	*Pseudomonas aeruginosa* UW-1	Mixture of canola, soybean oil with glucose	24.3 g/L	ND	[55]
Glycolipid	*Candida sphaerica* UCP0995	By culture medium—5% groundnut oil residues, 2.5% corn steep liquor in distilled water	4.5 g/L	ST-26 mN/m	[56]
Sophorolipid	*Candida bombicola*	Fermentation—Soy molasses	55 g/L	ND	[57]
Biosurfactant	*Pseudomonas aeruginosa* A41	2% vegetable oil	6.58 g/L	ND	[58]
Sophorolipid	*Pichia anomala* PY1	4% soyabean oil	0.2 g/L	ST-28 mN/m	[59]
Glycolipid	*Tsukamurella* sp.	Sunflower oil rich in oleic acid	30 g/L	ST-30 mN/m	[60]
Lipopeptide	*Bacillus subtilis* SPB1	Solid-state fermentation-Olive oil mill effluents	30.67 mg/g	ND	[61]

with 75 mL of broth solution was taken along with *n*-hexane for eliminating oil residues in *n*-hexane extract. Ethyl acetate was added to extract Rhamnolipids from the aqueous layer and under vacuum evaporation crude, Rhamnolipids were extracted. Subsequently, distilled water was used for diluting crude Rhamnolipids. HPLC, TLC, etc., were used to determine the characteristics and analysis of Rhamnolipids [63] (Fig. 4 and Table 4).

5.2 Organic waste

Organic waste includes fruits and vegetable wastes, lignocellulose waste, starch-rich contents that can be utilized to produce surfactants. The agro-based industrial waste too serves as a basis of nutrients for the growth of microbes, as they are the origin of carbon. Lignocellulosic waste such as sugar bagasse, grape marc, walnut shells, etc., acts as a great supplement for the growth of microbes. Starch-rich waste such as potato effluents, Cassava waste, rice straw, etc., helps in producing biosurfactants [77] (Fig. 5).

Seba George and K. Jayachandran used *P. aeruginosa* to extract Rhamnolipids from orange feel, with the basal medium. Nutrient agar slants were used for the cultivation of bacteria under 4°C. The inoculum was prepared by nutrient broth and 5% cell suspension in 0.9% saline solution.

Fruit waste such as orange peel, lemon peel, carrot peel, banana peel, coconut oil cake was dried and crushed into fine granules. The Basal solution having a composition of glycerol, $NaNO_3$, $NaHPO_4$, KH_2PO_4, $MgSO_4{\cdot}7H_2O$, and yeast extract was converted into aliquots of liquid medium. The powdered carbon source replaced glycerol in the medium.

In the next step, 250 mL Erlenmeyer flasks were used along with the inoculum and 4 days' incubation at 37°C, followed by centrifugation at 1000 rpm for about 20 min. On the basis of high yield production, a carbon source was selected, which was found out to be from orange peel. The further step was deproteinization of cell-free broth at 110°C, followed by the addition of 2 N HCl to acidify the cell broth. Finally, Rhamnolipids were extracted with ethyl acetate. A Rotary evaporator was used for solvent evaporation and recovery of Rhamnolipids. Subsequently, purification of biosurfactants took place [78] (Fig. 6 and Table 5).

5.3 Animal waste

Large volumes of waste animal fat are generated in the meat processing industry and are utilized to make biosurfactants. Fish wastage, such as skin, bones, head, red flesh, and viscera, accounts for 40%–60% of the overall mass of the fish produced by fish processing plants.

These wastes, which have excessive organic carbon, suspended particles, and nitrogen, are likely to pollute the environment and perhaps cause health issues.

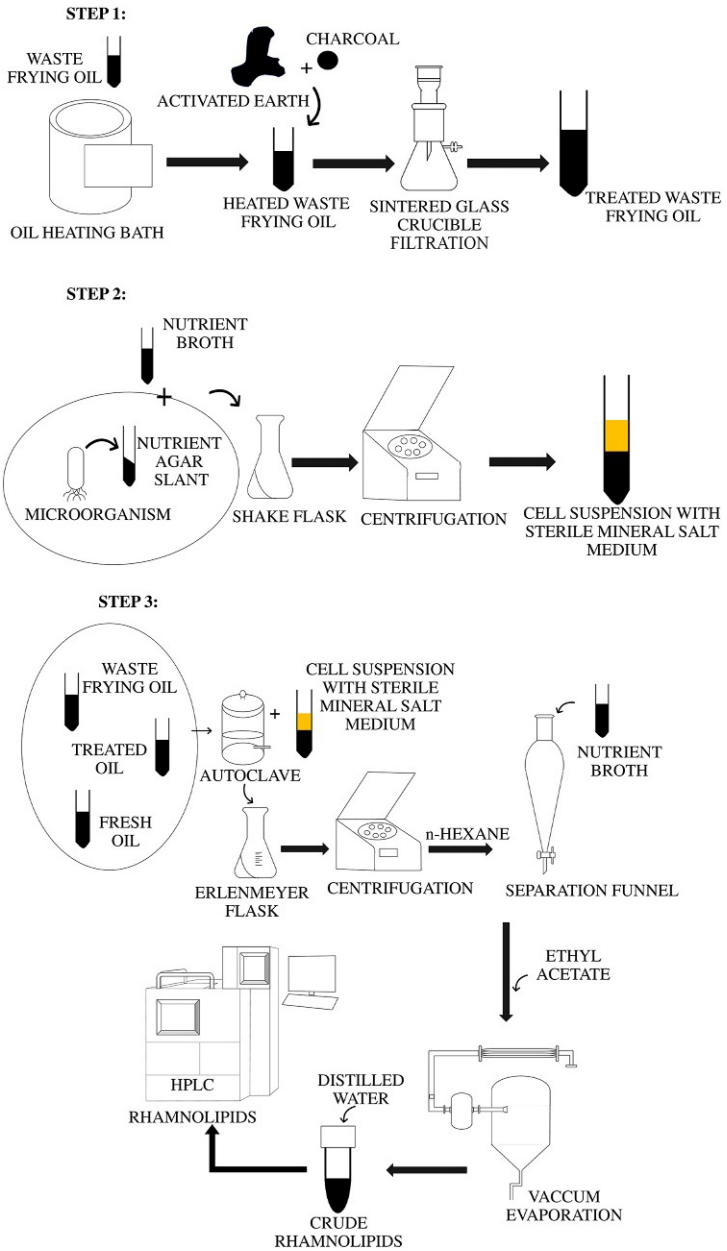

FIG. 4

Production of rhamnolipids from frying oil waste.

Table 4 Biosurfactants produced from frying oil waste.

Biosurfactants produced	Microorganism used	Substrate	Yield of biosurfactants and their ST		References
Sophorolipid	*Candida bombicola*	Batch and fed batch fermentation—40 g/L oil residues, 100 g/L glucose, 1 g/L urea	34 g/L	ND	[26]
Rhamnolipid	*Pseudomonas aeruginosa* PACL	Submerged fermentation, stirred tank reactor—soybean oil frying oil	3.3 g/L	ST-26 mN/m	[27]
Sophorolipid	*Candida bombicola* ATCC 22214	Bioreactor shake flask—Frying oil waste	5–10 g/L	ND	[64]
Rhamnolipid	*Pseudomonas aeruginosa*	Coconut frying oil waste	3.55 g/L	ST-53 mN/m	[65]
Rhamnolipid	*Pseudomonas* sp.	Sunflower oil and olive oil used for frying	2.7 g/L	ST-36 m/Nm	[66]
Glycolipid	*Mucor circinelloides*	Culture medium—5% waste frying oil	12.3 g/L	ST-26 mN/m	[67]
Lipopeptide	*Bacillus stratosphericus* FLU5	Basal medium—1%waste frying oil	0.05 g/L	ST-72-28 mN/m	[68]
Rhamnolipid and polyhydroxyalkanoates	*Burkholderia thailandensis* E264	4% sunflower-derived cooking oil	2.2 g/L	ST-37.7 mN/m	[69]
Biosurfactant	*Dietzia* sp. S-JS-1	Waste frying oil and paraffin	3.78 g/L	ND	[70]
Rhamnolipid	*Pseudomonas aeruginosa* OG1	Surface methodology and cultivation	13.3 g/L	ND	[71]
Glycolipid	*Rhodococcus erythropolis* 16 LM	USTHB—Batch culture Sunflower frying oil	ND	ST-31.9 mN/m	[72]
Glycoprotein	*Streptomyces* sp. DPUA 1559	Strain cultivation—1% Soybean frying oil waste	1.74 g/L	ST-25.3 mN/m	[73]

Lipoprotein	*Streptomyces* sp. DPUA 1559	10 g/L soybean frying oil waste 20 g/L corn steep liquor	1.9 g/L	ST-28 mN/m	[73]
Biosurfactant	*Pseudomonas cepacia* CCT6659	Basal medium—2%waste frying canola oil 3% corn steep liquor, 0.2% $NaNO_3$	40.5 g/L	ST-29 mN/m	[74]
Surfactin	*Bacillus subtilis* MTCC 2423	By culture medium—50 g/L frying oil waste, 5 g/L yeast extract, 1.7 g/L-mineral salts	2.1 g/L	ST-29 mN/m	[75]
Rhamnolipid	*Pseudomonas aeruginosa* DG30	Bacterial broth culture—5% w/v vegetable oil	15.6 g/L	ST-40 mN/m	[76]
Rhamnolipid	*Pseudomonas aeruginosa*	Bioreactor—frying oil residues	20 g/L	ND	[62]

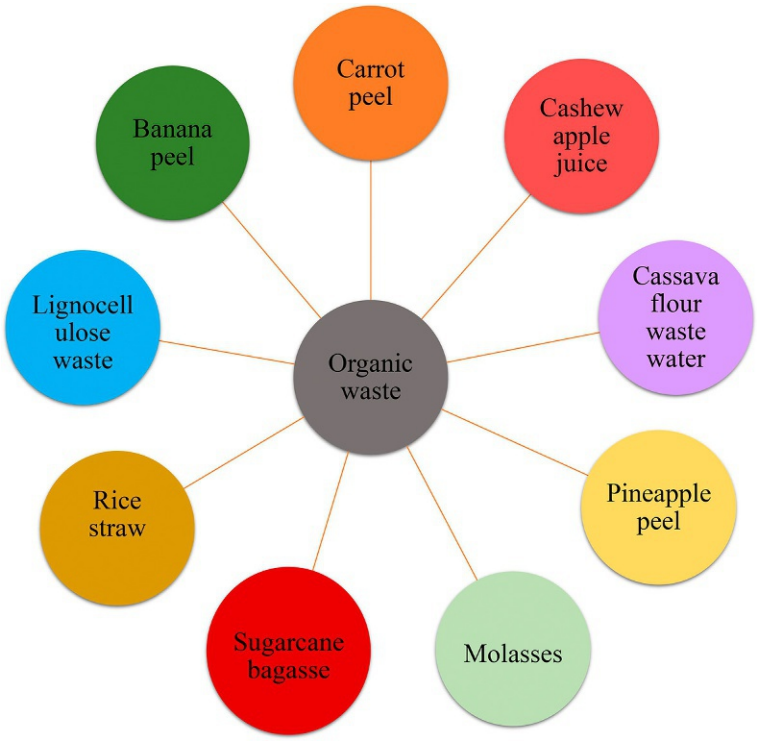

FIG. 5

Organic waste obtained from the food industry.

The tuna family includes the Bonito (*Katsuwonus pelamis*), a ray-finned, medium-sized predatory fish in the Scombridae family. They can be easily caught and hence are commonly used to produce fish products.

Tuna fish wastage was refined utilizing the process of enzymatic hydrolysis to produce fish peptones in order to meet the goals. These fish peptones were used as the basis for the production of biosurfactants. *B. subtilis* was chosen as a characteristic producer of lipopeptides. Ningbo Today Food Co. provided tuna red meat waste samples. Each sample was minced three times in a food processor for 120 s at medium speed. Prior to storage at 20°C for further experiments, fresh tuna red meat samples were obtained for composition analysis.

The red meat was taken from tuna fish and subjected to proximate composition analysis, accounting for 13%–15% of the total weight of the fish. The culture media was either nutrient agar medium or nutrient broth. The temperature in the growth environment was 30°C. Milk-white bacterial colonies can be visible after 24 h of culture on solid media. For preservation and activation, *B. subtilis* should be placed in a freeze-drying tube in a cryogenic refrigerator set to 80°C.

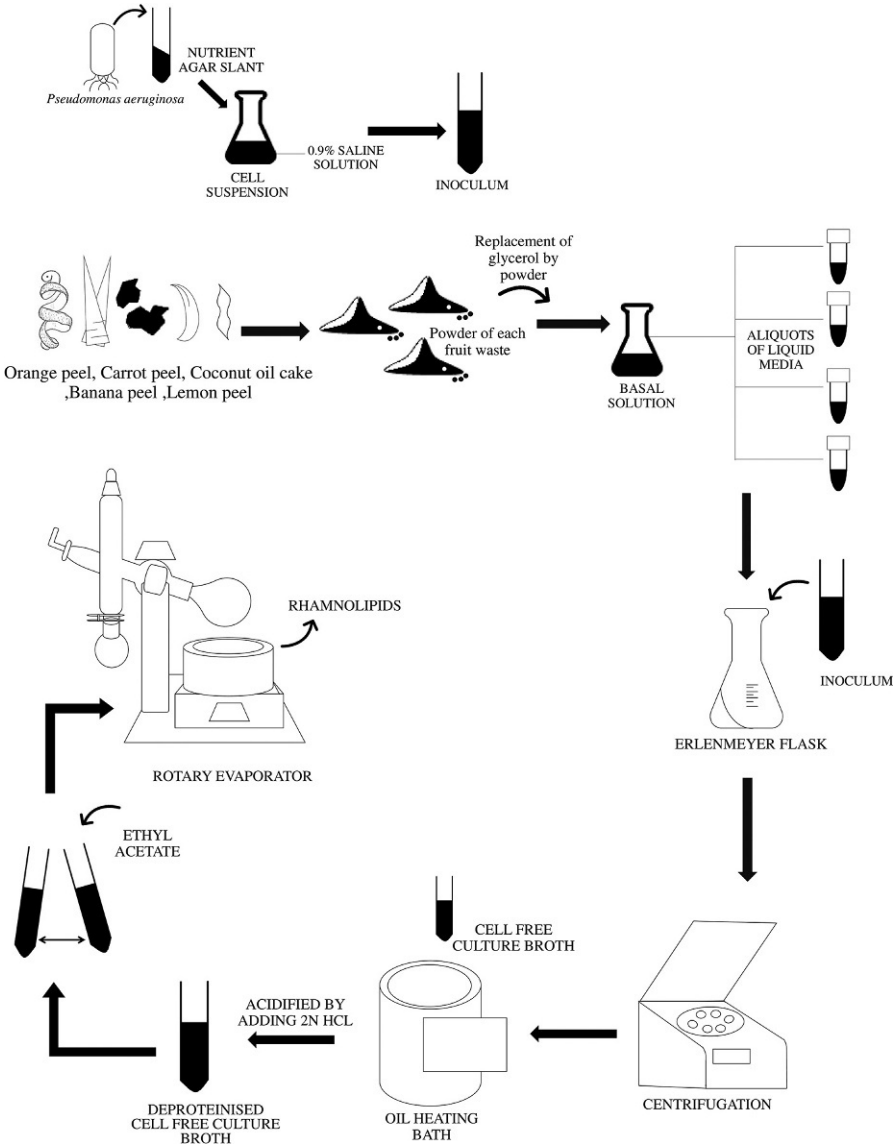

FIG. 6

Production of rhamnolipids from organic waste.

Every sample of concentrated head hydrolysate of tuna fish required 3–4 days of freeze-drying. The freeze-drying procedure resulted in an 89% weight decrease, indicating that 22.96% of tuna red flesh can be converted to fish peptone via an enzymatic process. The analysis of hydrolyzed peptones can be useful in the

Table 5 Biosurfactants produced from organic waste.

Biosurfactants produced	Microorganism used	Substrate	Yield of biosurfactants and their ST		References
Lipopeptide	*Bacillus subtilis* DM03	Submerged fermentation-Potato peel	102 mg/g ds	ND	[22]
Biosurfactant	*Lactobacillus pentosus*	Hydrolyzed distilled grape marc Corn steep liquor, Yeast extract	0.0048 g/L	ND	[32]
Rhamnolipid	*Pseudomonas aeruginosa* MTCC 2297	Banana waste, Orange waste, Fruit peelings, Carrot waste, Lime peelings, Coconut oil cake	9.2 g/L	ST-31.3 mN/m	[78]
Biosurfactant	*Pantoea* sp.	Pineapple peel juice and Corn oil	ND	ST-36 mN/m	[79]
Biosurfactant	*Serratia marcescens* UCP1549	Cassava flour wastewater	ND	ST-25.9 mN/m	[80]
Surfactin	*Bacillus subtilis* ATCC 21332	Cassava flour wastewater	2.4 g/L	ST-27 mN/m	[81]
Lipopeptide	*Halobacteriaceae archaeon* AS65	Mineral salt medium—Banana peel Monosodium glutamate	5.3 g/L	ST-25.5 mN/m	[82]
Biosurfactant	*Bacillus tequilensis* ZSB 10	Culture broths—Mexican brines	1.52 g/L	ST-38.6 mN/m	[83]
Rhamnolipid and PHAs	*Pseudomonas aeruginosa*	Cassava flour wastewater, glycerol, and frying oil	0.660 g/L	ST-30 mN/m	[84]
Surfactin	*Bacillus subtilis* LB5a	Cassava flour wastewater	0.02707 g/L	ND	[85]
Mannosylerythritol lipid	*Bacillus subtilis* LB5a	Cassava flour wastewater	1.26 g/L	ST-26 mN/m	[86]

Surfactin	*Bacillus subtilis* LAMI005	Cashew apple juice	0.319 g/L	ST-30 mN/m	[87]
Biosurfactant	*Bacillus subtilis* ATCC 21332	Potato	ND	ST-71.3–28.3 mN/m	[88]
Biosurfactant	*Yarrowia lipolytica*	3% v/v—Caj or glycerol, 0.5 g/L Yeast extract, 10 g/L-$(NH_4)_2SO_4$	6.9 g/L	ST-18 mN/m	[89]
Biosurfactant	*Rhizopus arrhizus*	Rice bran husks and corn steep liquor	9.1 g/L	ST-26.5 mN/m	[90]
Surfactin	*Bacillus subtilis* MTCC 2423	Rice mill polishing residue	4.17 g/kg	ST-27 mN/m	[91]
Biosurfactant	*Candida glabrata* UCP 1002	Vegetable fat waste	ND	ST-24 mN/m	[92]
Biosurfactant	*Klebsiella* sp. RJ-03	Corn powder, Potato peel powder, Sugarcane bagasse, *Madhuca indica*	9.6 g/L	ST-45.46 mN/m	[93]
Sophorolipid	*Starmerella bombicola* NBRC 10243	Corn cob	49.2 g/L	ND	[94]
Lipopeptide	*Bacillus licheniformis* KC710973	Orange peel, Banana peel, Potato peel and two commercial extracts having citrus peel	1.8 g/L	ND	[95]
Sophorolipid	*Cutaneotrichosporon mucoides* UFMG-CM-Y6148	Sugarcane bagasse hemicellulose hydrolysate	12.5 g/L	ND	[96]
Surfactin	*Bacillus* sp. LB5a	Cassava flour wastewater	3 g/L	ST-26.6 mN/m	[97]
Surfactin	*Bacillus subtilis* ATCC 21332	Cassava flour wastewater	2.2 g/L	ST-25.9 mN/m	[98]
Glycolipid	*Mucor indicus*	Rice husks	0.78 g/100 mL	ND	[99]
Lipopeptide	*Bacillus subtilis* KP7 *Bacillus subtilis* I′-1a	Mineral medium—two different brewery waste waters, Beet molasses, Apple peel extracts and Carrot	0.4287 g/L	ND	[100]

Continued

Table 5 Biosurfactants produced from organic waste. *Continued*

Biosurfactants produced	Microorganism used	Substrate	Yield of biosurfactants and their ST		References
		peel extract along with yeast extract and peptone			
Sophorolipid	*Candida bombicola* NRRL-Y17069	Mango kernel fat waste—SSF Submerged fermentation	17.48 g/100 g 5.8 g/100 g	ND	[101]
Lipopeptide	*Bacillus subtilis* ANR88	Orange peel, Banana peel, Potato peel and bagasse, Whey molasses	0.241–0.746 g/L	ND	[102]
Rhamnolipid	*Pseudomonas aeruginosa* ATCC 10145	Cashew apple juice	3.8 g/L	ST-29.5 mN/m	[103]
Surfactin	*Bacillus subtilis* LAMI008	Mineral medium Fermentation—Cashew apple juice	0.0035 g/L	ST-38.1 mN/m	[104]
Sophorolipid	*Candida bombicola* ATCC 22214	Lignocellulosic hydrolysates from sweet sorghum bagasse and corn fiber	84.6 g/L	ND	[105]
Glyco-lipopeptide and glycoprotein	*Lactobacillus paracasei* A20	Cellulosic sugars from vineyard pruning waste	ND	ST-24.8 mN/m	[106]
Surfactin	*Bacillus subtilis* 21332	Potato process effluent high solids and low solids Corn steep liquor	0.44 g/L	ST-26.2 mN/m	[107]
Lipopeptide	*Bacillus amyloliquefaciens* XZ-173	Soybean flour, Raw straw	50.01 mg/g ds	ND	[108]

production process. The hydrolyzed peptones had components that might be used as a general substrate. The biosurfactant synthesis was monitored using ST and CMD.

The production was assessed using ESI-MS and HPLC. To achieve system scale-up, three production scales, i.e., 20 mL of Bench scale, 7 L of Batch scale, and 100 L of Pilot scale were used. The optimal production conduction was regularly examined in a 100 L pilot-scale and 7 L batch-scale fermenter, with the goal of determining if large-scale Surfactin production is feasible.

B. subtilis was found to be capable of producing Surfactin from peptones found in fish waste. In the pilot-scale studies, the maximum Surfactin productivity was 274 mg/L [109] (Fig. 7 and Table 6).

5.4 Dairy waste

Raw milk is administered into a variety of products such as butter, cheese, cream, yogurt, ghee, condensed milk, milk powder, desserts, and other by-products such as buttermilk, whey, and skim milk by the dairy sector. Whey manufacturing is constantly increasing, and the high organic content of the product is a significant environmental and health concern. As a result, proper management of this by-product is necessary.

Recent research has focused on the efficient use of whey as a low-cost fermentation medium for manufacturing biosurfactants. Whey is a lactose-rich liquid by-product of cheese manufacturing that also contains other organic water-soluble components [34]. The food sector may be interested in biosurfactants produced by lactic acid bacteria because of its suitable emulsifying and antiadhesive properties.

Strain SR17 of *P. aeruginosa* was recovered from soil sample contaminated with hydrocarbons and found to be capable of producing Rhamnolipids from Cottage cheese whey. In Guwahati, Assam, India, samples were gathered in sealed polythene bags from a nearby automobile junkyard.

In 500-mL Erlenmeyer flasks, 1 g of collected soil was mixed with 100 mL MSM containing crude oil (2% w/v) as the sole carbon source and incubation is done at 35°C at 150 rpm in a rotary shaker. One milliliter of the solution was transferred to new MSM and incubated after 7 days. After another 7 days, 1 mL of culture media was used for serial dilution, and 100 L of diluted samples was placed on nutrient agar plates and incubated for 24 h at 35°C. Colonies that were structurally dissimilar were selected and streaked individually on nutrient agar surface to establish a pure culture of the bacterial isolates. The isolates were cultured on nutrient agar slant medium at 4°C.

The various strains were cultivated for 24 h in nutrient broth culture in a vortex mixer at 35°C, which is utilized as embryo culture in subsequent investigations. Five milliliter of each isolated inoculum was put in 100 mL of MSM (glucose-supplemented MSM) (2% w/v) as the source of carbon and grown for 4 days at 35°C with 150 rpm stirring. Bacterial isolates that were able to drop the surface

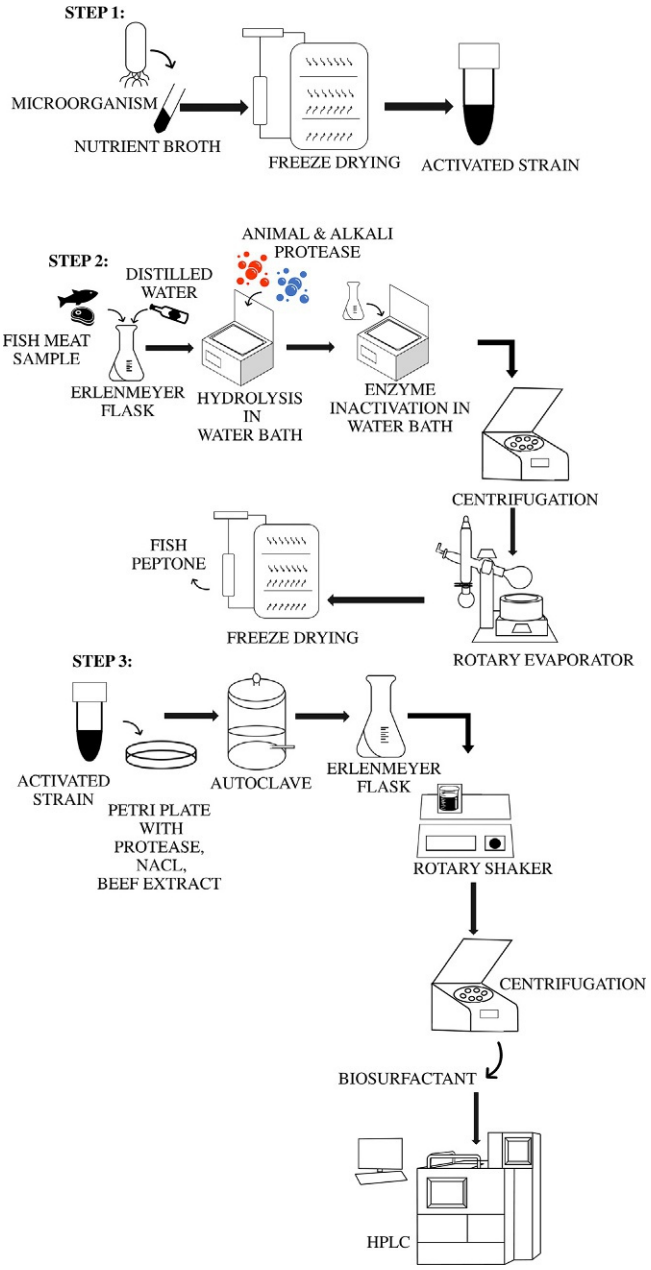

FIG. 7

Production of lipopeptides from animal waste.

Table 6 Biosurfactants produced from animal waste.

Biosurfactants produced	Microorganism used	Substrate	Yield of biosurfactants and their ST		References
Lipopeptide	*Bacillus subtilis*	Tuna fish waste	0.274 g/L	ST-28 mN/m	[109]
Biosurfactant	*Bacillus subtilis* PF1	Chicken feather meal, Potato peel, Rapeseed cake	0.82 g/L	ND	[110]
Rhamnolipid	*Pseudomonas aeruginosa* ATCC 9027 *Pseudomonas aeruginosa* ATCC 101045	Floating grease waste at poultry Residual brewery yeast and Ammonium nitrate	3.84 g/L	ST-27.5 mN/m	[111]
Biosurfactant	*Corynebacterium aquaticum* *Corynebacterium* sp. CCT 1968	Sugar bagasse fish waste crude glycerol	ND	ST-28.5 mN/m	[112]
Sophorolipid	*Candida bombicola*	Fermentation—Animal fat	120 g/L	ND	[113]
Biosurfactant	*Pseudomonas stutzeri* strain	Shrimp shell waste	4–6 g/L	40 mN/m	[114]
Sophorolipid	*Starmerella bombicola* ATCC 22214	Noodles, Chicken meat, Fat, Bones	0.33 g/g	ND	[115]
Sophorolipid	*Candida bombicola* ATCC 22214	Chicken fat, Sunflower oil, Sugarcane molasses	39.8 g/L	ST-35 mN/m	[116]
Biosurfactant	*Yarrowia lipolytica* MTCC 9520	Chicken tallow	4.4 g/L	37 mN/m	[117]
Glycolipid	*Candida lipolytica* UCP0988	Bovine fat, Corn steep liquor	0.7 g/L	ST-28 mN/m	[118]

tension below 35 mN/m were examined for their potential to produce biosurfactant from Cottage Cheese whey.

Freshly produced Cottage cheese whey was collected for this purpose in sterile plastic containers from a neighborhood dairy farm in Guwahati, Assam, India. The composition of the obtained whey was determined using a number of characteristics such as pH, COD, color, acid content, etc. One hundred milliliters of whey was placed into Erlenmeyer flasks of 500 mL, and the whey pH was changed from 5.2 to 5.5 to 7.0 ± 0.2 using 1 N NaOH, followed by autoclave sterilization.

After that, 5 mL of each isolated inoculum was put into flasks, which contained 100 mL sterilized whey and maintained at a temperature of 35°C in a shaker at 150 rpm. At each 24-h interval, till 72 h, the OD and surface tension of the whey media were monitored. A K11 Tensiometer was used to measure the medium's surface tension, and a UV-vis spectrophotometer was used to calculate the optical density at 600 nm.

The bacterial isolate that grew in Cottage cheese whey and when compared with the other isolates, the one that decreased the surface tension to the least value was chosen as the potential strain and given the name SR17. The impact of yeast extract, glucose, and mineral salts in Cottage cheese whey on SR17's biosurfactant production was also investigated.

The preparation of media composition is mentioned below:

(i) M0, which is Cottage cheese whey without added ingredients.
(ii) M1, which is made up of 2% glucose in Cottage cheese whey.
(iii) M2, with 0.1% yeast extract in Cottage cheese whey, and
(iv) M3, with the abovementioned composition of mineral salt and 2% glucose in waste of Cottage cheese whey.

All of the prepared media's growth and biosurfactant yields were estimated. For the extraction and purifying the biosurfactant, centrifugation of a 48-h-old culture of an effective bacterium producing biosurfactant cultivated in Cottage cheese whey medium at 10,000 rpm (15,120 g) and cooled at 4°C for 20 min separated the biomass of the cell from the culture medium, preceded by deproteinization of the supernatant for nearly 15 min at 110°C.

After cooling, acidification with 6 N HCl was used to achieve a pH of 2.0, followed by storage in a refrigerator at a temperature of 4°C overnight. The biosurfactant was then extracted using ethyl acetate as the solvent medium. The biosurfactant generated by SR17 from Cottage cheese whey waste was preserved at 4°C in a glass beaker for later analysis. The biosurfactant yield obtained was 2.7 g/L, which was increased to 4.8 g/L when 2% glucose and mineral salts were added.

The recovered biosurfactant is a mixture of mono- and di-rhamnolipid constituents, according to biochemical, FTIR, and LC-MS analyses. It was discovered that the CMC was 110 mg/L. In the mouse fibroblast cell line L292, the Rhamnolipid was found to be nontoxic [119] (Fig. 8 and Table 7).

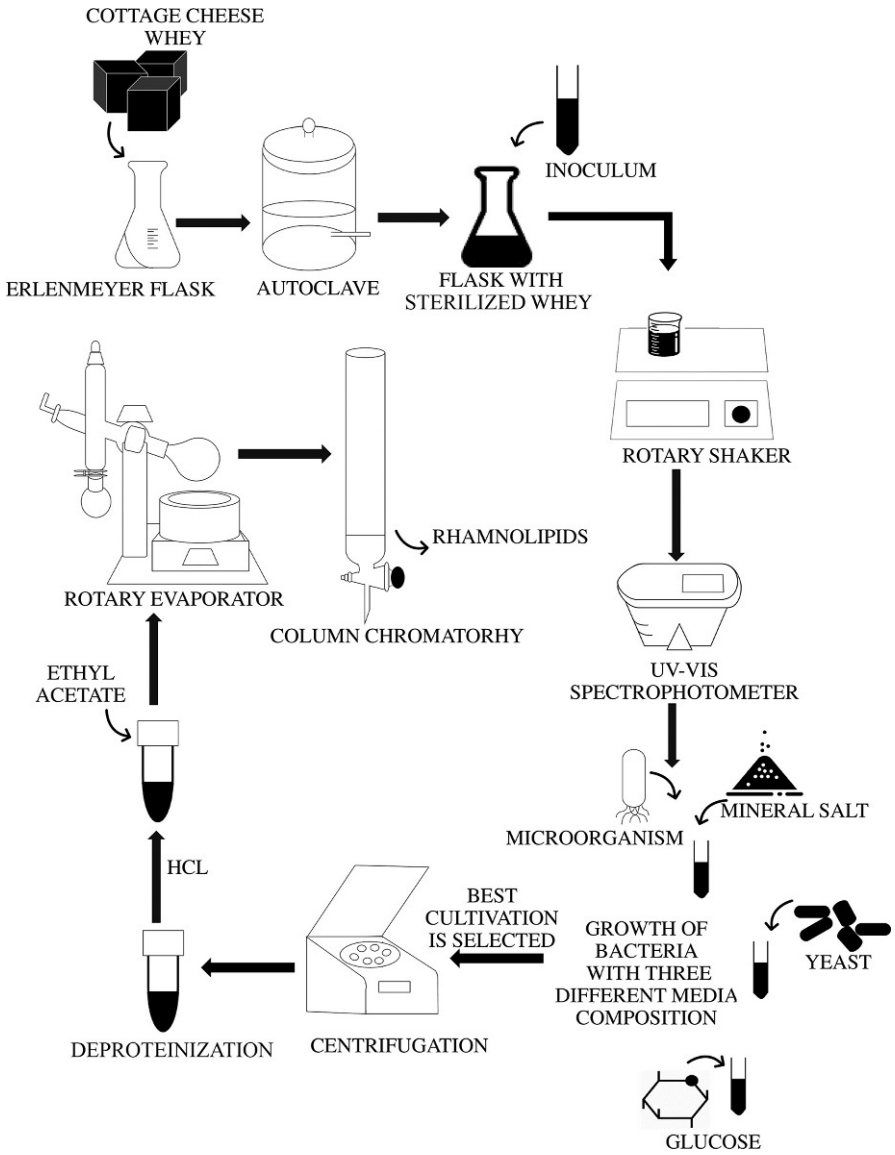

FIG. 8

Production of rhamnolipids from dairy waste.

6 Analysis of various biosurfactants produced from food waste

6.1 Applications

In the conversion of food wastes to biosurfactants, mainly three biosurfactant compounds were found. They are ***Rhamnolipids***, ***Sophorolipids***, and ***Lipopeptides***.

Table 7 Biosurfactants produced from dairy waste.

Biosurfactants produced	Microorganism used	Substrate	Yield of biosurfactants and their ST		References
Surfactin	*Bacillus subtilis* BS5	Molasses	1.12 g/L	ND	[120]
Sophorolipid	*Cryptococcus curvatus* ATCC 20509	Deproteinized whey	422 g/L	ST-30 mN/m	[121]
Sophorolipid	*Candida bombicola*	Deproteinized whey and glucose	34 g/L	ST-34.18 mN/m	[122]
Biosurfactant	*Pseudomonas aeruginosa* BS2	Distillery and whey wastes	0.97 g/L	ST-27 mN/m	[123]
Biosurfactant	*Bacillus licheniformis* K51, *Bacillus subtilis*	Molasses and cheese whey	ND	ST-29.5 mN/m	[124]
Rhamnolipid	*Pseudomonas* sp.	Distillery and whey waste	1.85 g/L	ND	[125]
Glycolipid	*Trichosporon montevideense*	Effluent of Dairy industry, Sunflower oil	20 g/L	ST-44.9 mN/m	[126]
Rhamnolipid	*Pseudomonas aeruginosa* B20	Sugar beet molasses	0.32 g/L	ND	[127]
Rhamnolipid	Bacterial strain	10% Soy molasses	9.09 g/L	ND	[128]
Rhamnolipid	*Pseudomonas aeruginosa* EBN-8	Blackstrap molasses	1.45 g/L	ND	[129]
Biosurfactant	*Lactococcus lactis* 53 *Streptococcus thermophilus*	Cheese whey and molasses	1.4 g/L	ND	[130]

The applications of these biosurfactant compounds have been explained below:

6.1.1 Rhamnolipids

In recent times, Rhamnolipids produced by *P. aeruginosa* are considered to be one of the most promising classes of biosurfactants [131]. Rhamnolipids can be considered as a substitute for chemical agents for food preservation. Its physicochemical and high antimicrobial properties contribute to the increase in the shelf life of food, without adding harmful synthetic preservatives. They are emulsifiers and surface-active substances found in nature. Rhamnolipids are preferred as they are originated from natural sources, with high purity and low toxicity [131] (Fig. 9).

The various applications of Rhamnolipids are:

In baking industry—Rhamnolipids are used in bread, pizza, rolls, croissants, and cakes to increase the consistency of their batter, volume, texture, and shape. They are also utilized in enhancing the properties of icing cream, butter cream,

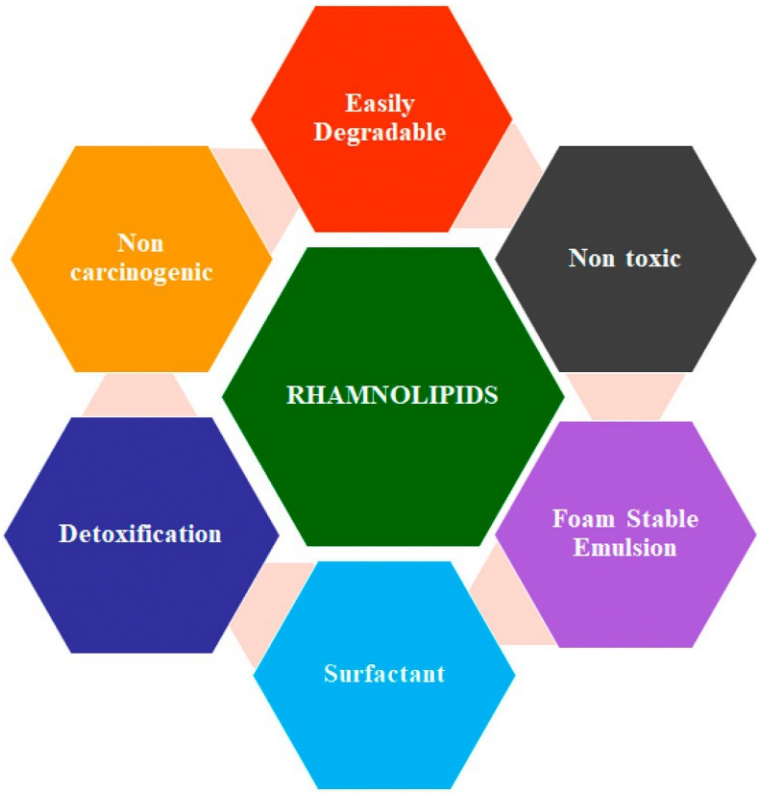

FIG. 9

Properties of rhamnolipids.

and croissants and pastries with cream-filling, and various confectionery products. [131].

Rhamnolipids can be utilized to manage texture, prevent moisture absorption, dissolve flavor oils, stabilize fats, and decrease discoloration in ice cream and pastry compositions [131]. Rhamnolipids can be administered to limit the adhesion of separate and mixed cultures of food-borne microorganisms, as well as to eliminate biofilms.

Rhamnose, which is utilized in the manufacture of culinary flavors, is also found in rhamnolipids. They can be deployed as food additives in the food business [132].

They serve a major role in lowering the surface and interfacial tension. They improve the shelf life and texture of starch-based products, limit fat globule agglomeration, stabilize aerated systems, adjust the rheological characteristics of wheat dough, and improve the consistency, texture, and stability of oils and fat-based products, while preventing separation [133]. They facilitate in the blending of ingredients and can help prevent mold and germs from growing in foodstuff (Fig. 10).

Due to their extensive application area and importance, research on Rhamnolipids has shown that these glycolipids have the potential to replace a variety of synthetic chemical surfactants [133]. In light of the social and technical contexts, the use of biosurfactants that are both ecologically friendly and highly purposeful, such as Rhamnolipids, has become increasingly significant.

6.1.2 Sophorolipids

Microbial BSs provide a dependable and long-term substitution for extracting compounds from plants or animals or manufacturing them through organochemical biosynthesis. Natural surfactants are also employed in the manufacture of a variety of

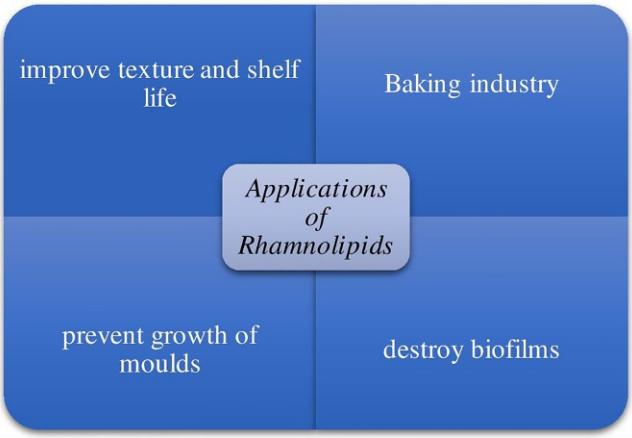

FIG. 10

Applications of rhamnolipids.

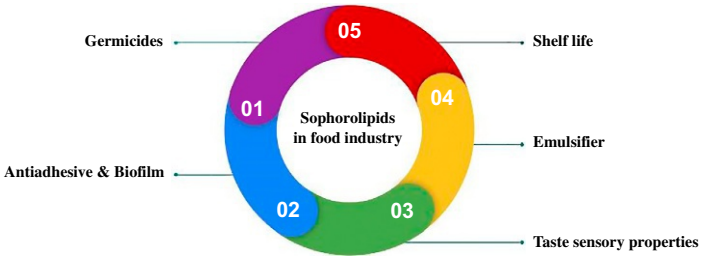

FIG. 11

Applications of sophorolipids.

culinary products such as appetizers, salad creams, mayonnaise, desserts, and so on [13].

Sophorolipids are glycolipids, which can be produced in huge quantities through fermentation [134]. Yeast strains such as *C. bombicola* are the primary producers of Sophorolipids [135] (Fig. 11).

SLP as emulsifier—SLP's emulsifying feature may be utilized in the food industry to improve the physical qualities such as texture, volume, etc., of goods based on vegetable fat emulsions or starch, as well as animal feeding items [135]. Xue et al. established the feasibility of SLP for use as an emulsifying ingredient in food items by formulating and characterizing O/W emulsions that included the oxidatively steady structured lipid [136].

SLP as antibacterial agent—SLP has antibacterial properties in addition to being an emulsifier. Fruits are very fragile goods, postharvest treatment, shipping, storage, and marketing all have an impact on durability.

The major cause of illnesses and rotting of fruits during transportation and storage is pathogenic fungus such as *Penicillium*, *Aspergillus flavus*, *Aspergillus niger* isolated from roots [135].

SLP prevents mycelium from multiplying, germinating, and spreading. SLP from *Wickerhamiella domercqiae* Y2A has antibacterial action against these pathogenic fungi [135]. Sleiman et al. evaluated the antibacterial activity of SLPs against *Escherichia coli*, *P. aeruginosa*, *Staphylococcus aureus*, *Staphylococcus pneumoniae*, *Proteus mirabilis*, and *Klebsiella pneumoniae* with that of conventional antibiotics.

Under the right circumstances, Zhang et al. and Olanya et al. found that SLPs have antibacterial action against food pathogens such as *Salmonella enterica*, *Listeria monocytogenes*, and *E. coli* O157-H7. Erick K. Nishio found *Clostridium perfringens* and *Campylobacter jejuni*, two poultry diseases, to have antibacterial action in Sophorolipids and Sophorolipids, in conjunction with lactic acid, which has the capability to be a naturally derived disinfectant for poultry applications [137].

SLP with taste sensory properties—Solaiman et al. used a vitro test system, where SLPs were shown to bind to taste receptors, resulting in the stimulation of taste-sensing cascade processes [136].

SLP as antiadhesive and biofilm—Biofilms are a major problem for the food business because they may be a source of chronic contamination, resulting in food deterioration and disease transmission [138]. Mad et al. found SLP has antibacterial characteristics and the capacity to prevent and disrupt the development of biofilms by both Gram-positive (*Ralstonia eutropha* ATCC 17699) and Gram-negative bacteria (*B. subtilis* BBK006) [135].

SLP as shelf life increasing agent—SLP, when combined with wheat flour or perhaps a product using it, alters the end product's quality. Bread made with SLP-enriched wheat flour has a higher volume, texture, and shelf life than bread made with regular wheat flour [135].

Due to its antiadhesive property, which has captivated attention as a novel strategy to prevent and disrupt biofilm development in food industries, biosurfactants exhibit a variety of possible qualities that might be beneficial in a variety of food-related applications [139] (Fig. 12).

6.1.3 Lipopeptides

Lipopeptides are distinguished in the food business sector because of their antiadhesive, antimicrobial, antiviral, and antitumor activities, which gives them important roles in pharmaceutical and cosmetic industries as well. Lipopeptide is a "halo-alkali and thermal tolerant biosurfactant," which is one of the main reasons for its usage in the food industry [140]. Lipopeptides are utilized generally in the procurement of raw materials as emulsifying agents (Fig. 13).

The three families of lipopeptides, Surfactin, Iturin, and Fengycin, have numerous applications in the food industry. Iturin and Fengycin have antifungal properties,

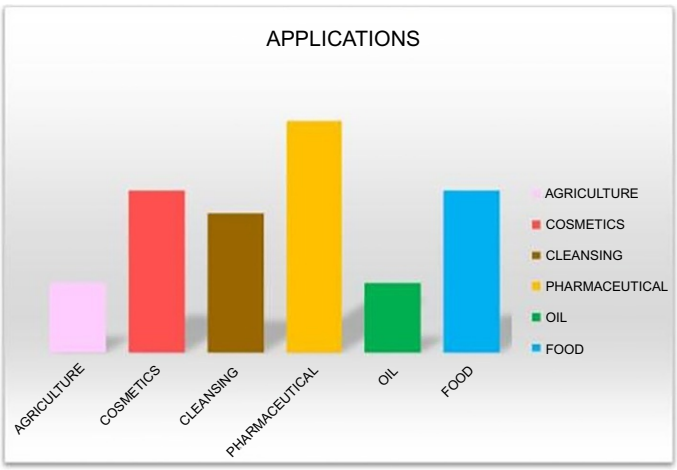

FIG. 12

Amount of usage of sophorolipids in different industries.

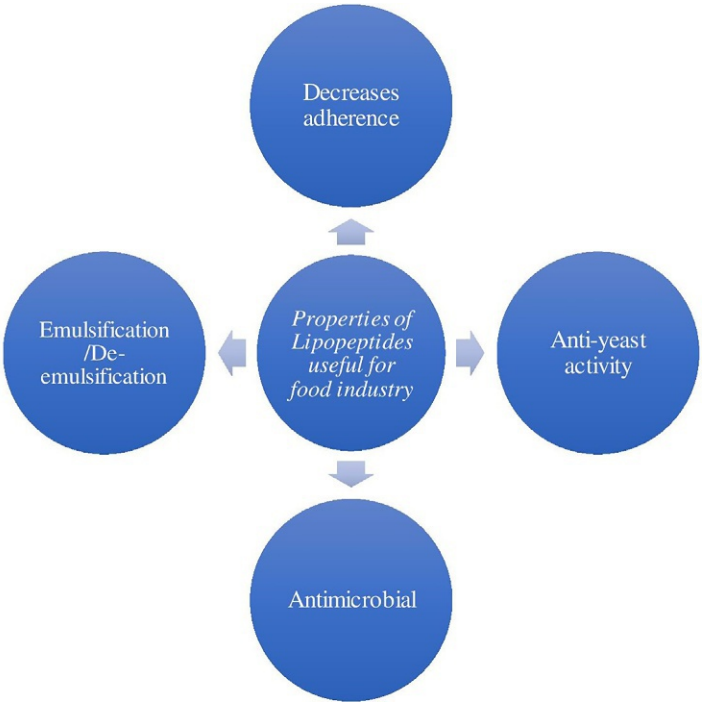

FIG. 13

Properties of lipopeptides.

whereas Surfactin has antibacterial properties and can also be used as a larvicidal agent [141].

Some of the applications are mentioned below:

In baking industry—Surfactin is used to preserve the texture, volume, and stability and help in the emulsification of fat to control the aggregation of fat globules [142]. Lipopeptides such as Surfactin have the ability to alter the surface hydrophobicity of bacteria and adhesion to solid surfaces; this effect is dependent on the initial bacterial hydrophobicity and the type of lipopeptide and its concentration and results in an increase or decrease in bacterial surface hydrophobicity [142].

Surfactin is a nontoxic and biodegradable product that can be utilized in the dairy business for a variety of purposes, including as an emulsifier, antimicrobial, and anti-adhesive properties.

Compared with other natural and synthetic emulsifiers, Surfactin has great ability to form and stabilize oil–water emulsions. [143]. According to Pecci et al. [144], a lipopeptide obtained from *Bacillus licheniformis* V9T14 improves the bactericidal action of silver ions against *E. coli* biofilms, lowering the population of biofilms. This synergistic interaction could be advantageous for cleaning surfaces and preventing bacterial colonization and spread of disease.

Lipopeptides are one of the bio preservatives used by food manufacturers to prevent rapid food spoilage as they can control food poisoning microbes due to their antimicrobial properties. Fengycin is incorporated in the food production chain for food preservation due to its antimicrobial activities [145].

The anti-yeast activity of Iturin provides a major application in controlling yeast spoilage in food. Iturin, which is obtained naturally, has a great chance to replace chemical preservatives. Some lipopeptides isolated from *Enterobacteriaceae* are known for their high emulsifying properties at an acidic pH at increased viscosity [146].

In spite of all these applications, the use of lipopeptides is limited in food preservation because of their implicit sensitivity to proteases. This can be prevented by using cyclic-ring structure lipopeptides [147]. Only if lipopeptides meet low-cost substrates and industrial-scale production can they be used as a substitute for chemical surfactants. It is likely that lipopeptides will occupy a significant percentage of food additives in the market in near future.

7 Conclusion and future perspectives

Biosurfactant is a term for a wide variety of biomolecules that range in molecular weight from low-molecular-mass glycolipids to high-molecular-mass compounds such as extracellular polymeric material or lipopolysaccharides. Basic factors for selecting an appropriate agro-industrial waste for production of biosurfactants include material availability and transportation costs, minimizing or avoiding pretreatment stages, and avoiding the inclusion of processed raw materials to the manufacturing media. On the other hand, more study is needed to identify resistant bacteria that can consume agro-industrial waste or residues in such an environment and produce greater biosurfactant titers [29].

Many of these biochemically produced substances and their manufacturing procedures have been patented in recent years, but only a few have been marketed. In conclusion, biosurfactant-based functional products, food-processing surface-cleaning formulations, the production of edible-oil-based emulsions for formulation, and the widespread usage of renewable biosurfactants could all be hot subjects in the next years [3].

Abbreviations

AOCS American Oil Chemists' Society
BS biosurfactant
CMD critical micelle dilution
COD chemical oxygen demand
ESI-MS electrospray ionization-mass spectrometry
FTIR Fourier-transform infrared spectroscopy
HPLC high-performance liquid chromatography

LC-MS liquid chromatography-mass spectrometry
MSM mineral salt medium
ND not defined
OD optical density
SLP sophorolipid
SSF solid-state fermentation
ST surface tension
TLC thin layer chromatography

References

[1] S. Vijayakuma, V. Saravanan, Biosurfactants-types, sources and applications, Res. J. Microbiol. 10 (2015) 181–192, https://doi.org/10.3923/jm.2015.181.192.

[2] R.S. Reis, G.J. Pacheco, A.G. Pereira, D.M.G. Freire, Biosurfactants: production and applications, in: Biodegradation-Life of Science, 2013, pp. 31–61, https://doi.org/10.5772/56144.

[3] N. Ranasalva, R. Sunil, G. Poovarasan, Importance of biosurfactant in food industry, IOSR J. Agric. Vet. Sci. 7 (5) (2014) 6–9. e-ISSN: 2319-2380.

[4] A. Jurgilevich, T. Birge, J. Kentala-Lehtonen, K. Korhonen, J. Pietikäinen, L. Saikku, H. Schösler, Transition towards circular economy in the food system, Sustainability 8 (2016) 69, https://doi.org/10.3390/su8010069.

[5] J. Gregg, J. Jürgens, M. Happel, N. Strøm-Andersen, A. Tanner, S. Bolwig, A. Klitkou, Valorization of bio-residuals in the food and forestry sectors in support of a circular bioeconomy: a review, J. Clean. Prod. 267 (2020) 122093, https://doi.org/10.1016/j.jclepro.2020.122093.

[6] N. Unger, F. Razza, Food Waste Management (Sector) in a Circular Economy, 2018, https://doi.org/10.1007/978-3-319-66981-6_15.

[7] T. Rood, H. Muilwijk, H. Westhoe, Food for the Circular Economy, PBL Netherlands Environmental Assessment Agency, 2017.

[8] R. Argôlo-Filho, L. Loguercio, *Bacillus thuringiensis* is an environmental pathogen and host-specificity has developed as an adaptation to human-generated ecological niches, Insects 5 (2013) 62–91, https://doi.org/10.3390/insects5010062.

[9] A. Bayoume, A. Tayel, N. Elguindy, Production of new rhamnolipids Rha C16-C16 by *Burkholderia* sp. through biodegradation of diesel and biodiesel, Beni-Suef Univ. J. Basic Appl. Sci. 7 (2018), https://doi.org/10.1016/j.bjbas.2018.05.003.

[10] J. Becker, H. Hosseinpour Tehrani, P. Ernst, L. Blank, N. Wierckx, An optimized *Ustilago maydis* for itaconic acid production at maximal theoretical yield, J. Fungi 7 (2020) 20, https://doi.org/10.3390/jof7010020.

[11] N. Christova, B. Tuleva, Z. Lalchev, A. Jordanova, B. Jordanov, Rhamnolipid biosurfactants produced by *Renibacterium salmoninarum* 27BN during growth on n-hexadecane, Z. Naturforsch. C J. Biosci. 59 (1–2) (2004) 70–74, https://doi.org/10.1515/znc-2004-1-215. 15018056.

[12] H. Fan, J. Ru, Y. Zhang, Q. Wang, Fengycin produced by *Bacillus subtilis* 9407 plays a major role in the biocontrol of apple ring rot disease, Microbiol. Res. 199 (2017), https://doi.org/10.1016/j.micres.2017.03.004.

[13] J. Guerra, I. Banat, L. Sarubbo, Natural and microbial biosurfactants' use in the food industry, in: Microbial Biosurfactants and Their Environmental and Industrial Applications, 2019, https://doi.org/10.1201/b21950-10.
[14] A. Moldes, X. Vecino, L. Rodríguez-López, M. Rincón-Fontán, C. Freire, J. Manuel, Microbial Glycoprotein and Lipopeptide Biosurfactants: Production, Properties and Applications, 2019, https://doi.org/10.1201/b21950-6.
[15] C. Madihalli, M. Doble, Microbial Production and Applications of Mannosylerythritol, Cellobiose and Trehalose Lipids, 2019, https://doi.org/10.1201/b21950-5.
[16] B. Ribeiro, B. Veras, J. Aguiar, J. Guerra, L. Sarubbo, Biosurfactant produced by *Candida utilis* UFPEDA1009 with potential application in cookie formulation, Electron. J. Biotechnol. 46 (2020), https://doi.org/10.1016/j.ejbt.2020.05.001.
[17] T. RengaThavasi, I. Banat, Introduction to microbial biosurfactants, in: Microbial Biosurfactants and Their Environmental and Industrial Applications, 2019, https://doi.org/10.1201/b21950-1.
[18] A. Roy, A review on the biosurfactants: properties, types and its applications, J. Fundam. Renew. Energy Appl. 8 (2017), https://doi.org/10.4172/2090-4541.1000248.
[19] N. Saborimanesh, C. Mulligan, Biosurfactants and their applications in petroleum industry, in: Microbial Biosurfactants and Their Environmental and Industrial Applications, 2018, https://doi.org/10.1201/b21950-9.
[20] P. Marcelino, F. Gonçalves, I. Jimenez, B. Carneiro, B. Santos, S. Silva, Sustainable Production of Biosurfactants and Their Applications, 2020, https://doi.org/10.1002/9781119568858.ch8.
[21] P. Kumar, S. Kumar, P. Chawla, J. Duhan, Fermentation: a boon for production of bioactive compounds by processing of food industries wastes (by-products), Molecules 23 (2018), https://doi.org/10.3390/molecules23102560.
[22] K. Das, A. Mukherjee, Comparison of lipopeptide biosurfactants production by *Bacillus subtilis* strains in submerged and solid state fermentation systems using a cheap carbon source: some industrial applications of biosurfactants, Process Biochem. 42 (2007) 1191–1199, https://doi.org/10.1016/j.procbio.2007.05.011.
[23] H. Jin, X. Zhang, K. Li, Y. Niu, M. Guo, C. Hu, X. Wan, Y. Gong, F. Huang, Direct bioutilization of untreated rapeseed meal for effective iturin A production by *Bacillus subtilis* in submerged fermentation, PLoS One 9 (2014) 1–7, https://doi.org/10.1371/journal.pone.0111171.
[24] J. Beuker, C. Syldatk, R. Hausmann, Bioreactors for the Production of Biosurfactants, 2014, https://doi.org/10.1201/b17599-10.
[25] F. Kronemberger, C. Borges, D. Freire, Fed-batch biosurfactant production in a bioreactor, Int. Rev Chem. Eng. 2 (2010).
[26] V. Shah, M. Jurjevic, D. Badia, Utilization of restaurant waste oil as a precursor for sophorolipid production, Biotechnol. Prog. 23 (2007) 512–515, https://doi.org/10.1021/bp0602909.
[27] C.J. de Lima, E.J. Ribeiro, E.F. Sérvulo, M.M. Resende, V.L. Cardoso, Biosurfactant production by *Pseudomonas aeruginosa* grown in residual soybean oil, Appl. Biochem. Biotechnol. 152 (2009) 156–168, https://doi.org/10.1007/s12010-008-8188-1.
[28] M.J. Dumont, S.S. Narine, Soapstock and deodorizer distillates from North American vegetable oils: review on their characterization, extraction and utilization, Food Res. Int. 40 (2007) 957–974, https://doi.org/10.1016/j.foodres.2007.06.006.
[29] Á. Rivera, M. Martínez-Urbina, V. López, Advances on research in the use of agro-industrial waste in biosurfactant production, World J. Microbiol. Biotechnol. 35 (2019), https://doi.org/10.1007/s11274-019-2729-3.

[30] L. Martino, M.V. Cruz, A. Scoma, F. Freitas, L. Bertin, M. Scandola, M.A.M. Reis, Recovery of amorphous polyhydroxybutyrate granules from *Cupriavidus necator* cells grown on used cooking oil, Int. J. Biol. Macromol. 71 (2014) 117–123, https://doi.org/10.1016/j.ijbiomac.2014.04.016.
[31] T. Graeme, G. Sansonetti, New Light on a Hidden Treasure: International Year of the Potato 2008, an End-of-Year Review, FAO, Rome, 2009.
[32] O.M. Portilla-Rivera, A.B. Moldes, A.M. Torrado, J.M. Domínguez, Lactic acid and biosurfactants production from hydrolyzed distilled grape marc, Process Biochem. 42 (2007) 1010–1020, https://doi.org/10.1016/j.procb io.2007.03.011.
[33] M. Nitschke, S. Costa, Biosurfactants in food industry, Trends Food Sci. Technol. 18 (2007) 252–259, https://doi.org/10.1016/j.tifs.2007.01.002.
[34] R.S. Makkar, S.S. Cameotra, I.M. Banat, Advances in utilization of renewable substrates for biosurfactant production, AMB Express 1 (1) (2011), https://doi.org/10.1186/2191-0855-1-5.
[35] B.Y. Zhang, G.H. Huang, B. Chen, Q. Guo, G.M. Zeng, Production of biosurfactant in batch reactor for food waste composting, WIT Trans. Ecol. Environ. 56 (2002), https://doi.org/10.2495/WM020141.
[36] M. Henkel, C. Syldatk, R. Hausmann, The Prospects for the Production of Rhamnolipids on Renewable Resources: Evaluation of Novel Feedstocks and Perspectives of Strain Engineering, 2014, https://doi.org/10.1201/b17599-8.
[37] R. Patria, J. Wong, D. Johnravindar, K. Uisan, R. Kumar, G. Kaur, Food waste digestate-based biorefinery approach for rhamnolipids production: a techno-economic analysis, Sustain. Chem. 2 (2021) 237–253, https://doi.org/10.3390/suschem2020014.
[38] A. Abalos, A. Pinazo, M.R. Infante, M. Casals, F. García, A. Manresa, Physiochemical and antimicrobial activities of new rhamnolipids produced by *Pseudomonas aeruginosa* AT10 from soybean oil refinery wastes, Langmuir 17 (2001) 1367–1371, https://doi.org/10.1021/la0011735.
[39] W. Bednarski, M. Adamczak, J. Tomasik, M. Płaszczyk, Application of oil waste in the biosynthesis of glycolipids by yeast, Bioresour. Technol. 95 (2004) 15–18, https://doi.org/10.1016/j.biortech.2004.01.009.
[40] M.M. Camargo-de-Morais, S.A.F. Ramos, M.C.B. Pimentel, M. de Morais Jr., J.L. Lima Filho, Production of an extracellular polysaccharide with emulsifier properties by *Penicillium citrinum*, World J. Microbiol. Biotechnol. 19 (2003) 191–194, https://doi.org/10.1023/A:1023299111663.
[41] C. Coimbra, R. Rufino, J. Luna, L. Sarubbo, Studies of the cell surface properties of *Candida* species and relation to the production of biosurfactants for environmental applications, Curr. Microbiol. 58 (2008) 245–251, https://doi.org/10.1007/s00284-008-9315-5.
[42] C. Hazra, D. Kundu, P. Ghosh, S. Joshi, N. Dandi, A. Chaunarajanri, Screening and identification of *Pseudomonas aeruginosa* AB4 for improved production, characterization and application of a glycolipid biosurfactant using low-cost agro-based raw materials, J. Chem. Technol. Biotechnol. 86 (2011) 185–198, https://doi.org/10.1002/jctb.2480.
[43] J.V. Jadhav, A.P. Pratap, S.B. Kale, Evaluation of sunflower oil refinery waste as feedstock for production of sophorolipid, Process Biochem. 78 (2019) 15–24, https://doi.org/10.1016/j.procbio.2019.01.015.
[44] P. Jajor, D. Piłakowska-Pietras, A. Krasowska, M. Łukaszewicz, Surfactin analogues produced by *Bacillus subtilis* strains grown on rapeseed cake, J. Mol. Struct. 1126 (2016) 141–146, https://doi.org/10.1016/j.molstruc.2016.02.014.

[45] P. Jiménez-Peñalver, T. Gea, A. Sánchez, X. Font, Production of sophorolipids from winterization oil cake by solid-state fermentation: optimization, monitoring and effect of mixing, Biochem. Eng. J. 115 (2016) 93–100, https://doi.org/10.1016/j.bej.2016.08.006.
[46] L.A. Lourenço, M.D. Alberton-Magina, L.B.B. Tavares, S.M.A. Guelli-Ulson de Souza, M. García-Román, D. Altmajer-Vaz, Biosurfactant production by *Trametes versicolor* grown on two-phase olive mill waste in solid-state-fermentation, Environ. Technol. 39 (2018) 3066–3076, https://doi.org/10.1080/09593330.2017.1374471.
[47] J. Luna, R. Rufino, A. Jara, P. Brasileiro, L. Sarubbo, Environmental applications of the biosurfactant produced by *Candida sphaerica* cultivated in low-cost substrates, Colloids Surf. A. Physiochem. Eng. Asp. 480 (2015) 413–418, https://doi.org/10.1016/j.colsurfa.2014.12.014.
[48] A. Mishra, R. Trivedi, Synthesis and characterization of biosurfactant using waste from oil processing industry as substrate by *Pseudomonas aeruginosa* (MTCC 424), Rasayan J. Chem. 12 (2019) 1011–1021, https://doi.org/10.31788/RJC.2019.1225073.
[49] M.E. Mercadé, M.A. Manresa, M. Robert, M.J. Espuny, C. de Andrés, J. Guinea, Olive oil mill effluent (OOME). New substrate for biosurfactant production, Bioresour. Technol. 43 (1993) 1–6, https://doi.org/10.1016/0960-8524(93)90074-L.
[50] I. Moya-Ramírez, K. Tsaousi, M. Rudden, R. Marchant, E. Jurado Alameda, M. García-Román, I.M. Banat, Rhamnolipid and surfactin production from olive oil mill waste as sole carbon source, Bioresour. Technol. 198 (2015) 231–236, https://doi.org/10.1016/j.biortech.2015.09.012.
[51] M. Nitschke, S. Costa, R. Haddad, L. Gonçalves, M. Eberlin, J. Contiero, Oil wastes as unconventional substrates for rhamnolipid biosurfactant production by *Pseudomonas aeruginosa* LBI, Biotechnol. Prog. 21 (2005) 1562–1566, https://doi.org/10.1021/bp050198x.
[52] F. Oliveira, L. Vazquez, N. Campos, F. França, Production of rhamnolipids by a *Pseudomonas alcaligenes* strain, Process Biochem. 44 (2009) 383–389, https://doi.org/10.1016/j.procbio.2008.11.014.
[53] M.N. Radzuan, I.M. Banat, J. Winterburn, Production and characterization of rhamnolipid using palm oil agricultural refinery waste, Bioresour. Technol. 225 (2017) 99–105, https://doi.org/10.1016/j.biortech.2016.11.052.
[54] R. Rufino, L. Sarubbo, G.M.J. Campos-Takaki, Enhancement of stability of biosurfactant produced by *Candida lipolytica* using industrial residue as substrate, World J. Microbiol. Biotechnol. 23 (2007) 729–734, https://doi.org/10.1007/s11274-006-9278-2.
[55] L. Sim, O. Ward, Z.-Y. Li, Production and characterisation of a biosurfactant isolated from *Pseudomonas aeruginosa* UW-1, J. Ind. Microbiol. Biotechnol. 19 (1997) 232–238, https://doi.org/10.1038/sj.jim.2900450.
[56] H. Sobrinho, R. Rufino, J. Luna, A. Salgueiro, G. Campos Takaki, L. Leite, L. Sarubbo, Utilization of two agroindustrial by-products for the production of a surfactant by *Candida sphaerica* UCP0995, Process Biochem. 43 (2008) 912–917, https://doi.org/10.1016/j.procbio.2008.04.013.
[57] D. Solaiman, R. Ashby, J. Zerkowski, T. Foglia, Simplified soy molasses-based medium for reduced-cost production of sophorolipids by *Candida bombicola*, Biotechnol. Lett. 29 (2007) 1341–1347, https://doi.org/10.1007/s10529-007-9407-5.
[58] J. Thaniyavarn, A. Chongchin, N. Wanitsuksombut, S. Thaniyavarn, P. Pinphanichakarn, N. Leepipatpiboon, M. Morikawa, S. Kanaya, Biosurfactant production by *Pseudomonas*

aeruginosa A41 using palm oil as carbon source, J. Gen. Appl. Microbiol. 52 (2006) 215–222, https://doi.org/10.2323/jgam.52.215.

[59] J. Thaniyavarn, T. Chianguthai, P. Sangvanich, N. Roongsawang, K. Washio, M. Morikawa, S. Thaniyavarn, Production of sophorolipid biosurfactant by *Pichia anomala*, Biosci. Biotechnol. Biochem. 72 (2008) 2061–2068, https://doi.org/10.1271/bbb.80166.

[60] E. Vollbrecht, U. Rau, S. Lang, Microbial conversion of vegetable oils into surface-active di-, tri-, and tetrasaccharide lipids (biosurfactants) by the bacterial strain *Tsukamurella* spec, Lipid/Fett 101 (1999) 389–394, https://doi.org/10.1002/(SICI)1521-4133(199910)101:10<389::AID-LIPI389>3.0.CO;2-9.

[61] R. Zouari, S. Ellouze-Chaabouni, D. Ghribi-Aydi, Optimization of *Bacillus subtilis* SPB1 biosurfactant production under solid-state fermentation using by-products of a traditional olive mill factory, Achieve Life Sci. 8 (2014) 162–169, https://doi.org/10.1016/j.als.201.

[62] Y. Zhu, J.-j. Gan, G.-l. Zhang, B. Yao, W.-j. Zhu, Q. Meng, Reuse of waste frying oil for production of rhamnolipids using *Pseudomonas aeruginosa* zju.u1M, J. Zheijang Univ. Sci. A 8 (2007) 1514–1520, https://doi.org/10.1631/jzus.2007.A1514.

[63] S.D. Wadekar, S.B. Kale, A.M. Lali, D.N. Bhowmick, A.P. Pratap, Microbial synthesis of rhamnolipids by *Pseudomonas aeruginosa* (ATCC 10145) on waste frying oil as low cost carbon source, Prep. Biochem. Biotechnol. 42 (3) (2012) 249–266, https://doi.org/10.1080/10826068.2011.603000. 22509850.

[64] S. Fleurackers, On the use of waste frying oil in the synthesis of sophorolipids, Eur. J. Lipid Sci. Technol. 108 (2006) 5–12, https://doi.org/10.1002/ejit.200500237.

[65] S. George, K. Jayachandran, Production and characterization of rhamnolipid biosurfactant from waste frying coconut oil using a novel *Pseudomonas aeruginosa* D, J. Appl. Microbiol. 114 (2012) 373–383, https://doi.org/10.1111/jam.12069.

[66] E. Haba, M.J. Espuny, M. Busquets, A. Manresa, Screening and production of rhamnolipids by *Pseudomonas aeruginosa* 47T2 NCIB 40044 from waste frying oils, J. Appl. Microbiol. 88 (2000) 379–387, https://doi.org/10.1046/j.1365-2672.2000.00961.x.

[67] P. Hasani Zadeh, H. Moghimi, J. Hamedi, Biosurfactant production by *Mucor circinelloides*: environmental applications and surface-active properties, Eng. Life Sci. 18 (2018), https://doi.org/10.1002/elsc.201700149.

[68] D. Hentati, A. Chebbi, F. Hadrich, I. Frikha, F. Rabanal, S. Sayadi, A. Manresa, M. Chamkha, Production, characterization and biotechnological potential of lipopeptide biosurfactants from a novel marine *Bacillus stratosphericus* strain FLU5, Ecotoxicol. Environ. Saf. 167 (2019) 441–449, https://doi.org/10.1016/j.ecoenv.2018.10.036.

[69] K. Kourmentza, J. Costa, Z. Azevedo, C. Servin, C. Grandfils, V. Freitas, M. Reis, *Burkholderia thailandensis* as a microbial cell factory for the conversion of used cooking oil to polyhydroxyalkanoates and rhamnolipids, Bioresour. Technol. 247 (2018) 829–837, https://doi.org/10.1016/j.biortech.2017.09.138.

[70] J. Liu, X.-F. Huang, L.-J. Lu, J.-C. Xu, Y. Wen, D.-H. Yang, Q. Zhou, Comparison between waste frying oil and paraffin as carbon source in the production of bioemulsifier by Dietziasp S-JS-1, Bioresour. Technol. 100 (2009) 6481–6487, https://doi.org/10.1016/j.biortech.2009.07.006.

[71] M. Ozdal, S. Gurkok, O. Gur-Ozdal, Optimization of rhamnolipid production by *Pseudomonas aeruginosa* OG1 using waste frying oil and chicken feather peptone, 3 Biotech 7 (2017) 1–8, https://doi.org/10.1007/s13205-017-0774-x.

[72] Z. Sadouk, H. Hacene, A. Tazerouti, Biosurfactants production from low cost substrate and degradation of diesel oil by a Rhodococcus strain, Oil Gas Sci. Technol. 63 (2008) 747–753, https://doi.org/10.2516/ogst:2008037.

[73] A.P.P. Santos, M.D.S. Silva, E.V.L. Costa, R.D. Rufino, V.A. Santos, C.S. Ramos, L.A. Sarubbo, A.L.F. Porto, Production and characterization of a biosurfactant produced by *Streptomyces* sp. DPUA 1559 isolated from lichens of the Amazon region, Braz. J. Med. Biol. Res. 51 (2018) 1–10, https://doi.org/10.1590/1414-431X20176657.

[74] R.C.F. Soares da Silva, D.G. de Almeida, P.P.F. Brasileiro, R.D. Rufino, J.M. de Luna, L. A. Sarubbo, Production, formulation and cost estimation of a commercial biosurfactant, Biodegradation 30 (2019) 191–201, https://doi.org/10.1007/s10532-018-9830-4.

[75] N. Vedaraman, N. Murthy, Venkatesh., Production of surfactin by *Bacillus subtilis* MTCC 2423 from waste frying oils, Braz. J. Chem. Eng. 28 (2011) 175–180, https://doi.org/10.1590/S0104-66322011000200001.

[76] C. Zheng, Z. Luo, L. Yu, L. Huang, X. Bai, The utilization of lipid waste for biosurfactant production and its application in enhancing oil recovery, Pet. Sci. Technol. 29 (2011) 282–289, https://doi.org/10.1080/10916460903117586.

[77] I. Banat, S. Satpute, S. Cameotra, R. Patil, N. Nyayanit, Cost effective technologies and renewable substrates for biosurfactants' production, Front. Microbiol. 5 (2014) 697, https://doi.org/10.3389/fmicb.2014.00697.

[78] S. George, K. Jayachandran, Analysis of rhamnolipid biosurfactants produced through submerged fermentation using orange fruit peelings as sole carbon source, Appl. Biochem. Biotechnol. 158 (2008) 694–705, https://doi.org/10.1007/s12010-008-8337-6.

[79] F. Almeida, C. Lins, A.M. Vieira, C.J. Vilar Jr., M.C. Lins, G.M. Campos-Takaki, E.B. Tambourgi, Biosurfactant Production by Pantoea sp in Submerged Fermentation using Pineapple Peel as an Alternative Medium, 2012, https://doi.org/10.1142/9789814405041_0070.

[80] H. Araújo, R. Andrade, D. Montero Rodríguez, D. Rubio Ribeaux, C. Silva, G. Campos Takaki, Sustainable biosurfactant produced by *Serratia marcescens* UCP 1549 and its suitability for agricultural and marine bioremediation applications, Microb. Cell Fact. 18 (2019), https://doi.org/10.1186/s12934-018-1046-0.

[81] F. Barros, A. Ponezi, G. Pastore, Production of biosurfactant by *Bacillus subtilis* LB5a on a pilot scale using cassava wastewater as substrate, J. Ind. Microbiol. Biotechnol. 35 (2008) 1071–1078, https://doi.org/10.1007/s10295-008-0385-y.

[82] C. Chooklin, S. Maneerat, A. Saimmai, Utilization of banana peel as a novel substrate for biosurfactant production by *Halobacteriaceae archaeon* AS65, Appl. Biochem. Biotechnol. 173 (2014), https://doi.org/10.1007/s12010-014-0870-x.

[83] S. Cortés-Camargoa, N. Pérez, R. Oliveira, B. Barragan, J. Domínguezb, Production of biosurfactants from vine-trimming shoots using the halotolerant strain *Bacillus tequilensis* ZSB10, Ind. Crop Prod. 79 (2016) 258–266, https://doi.org/10.1016/j.indcrop.2015.11.003.

[84] S. Costa, F. Lépine, M. Sylvain, E. Déziel, M. Nitschke, J. Contiero, Cassava wastewater as a substrate for the simultaneous production of rhamnolipids and polyhydroxyalkanoates by *Pseudomonas aeruginosa*, J. Ind. Microbiol. Biotechnol. 36 (2009) 1063–1072, https://doi.org/10.1007/s10295-009-0590-3.

[85] C.J. de Andrade, L.M. de Andrade, M.L. Bution, M.A.H. Dolder, F.F.C. Barros, G.M. Pastore, Optimizing alternative substrate for simultaneous production of surfactin and 2,3-butanediol by *Bacillus subtilis* LB5a, Biocatal. Agric. Biotechnol. 6 (2016) 209–218, https://doi.org/10.1016/j.bcab.2016.04.004.

[86] C.J. de Andrade, L.M. de Andrade, S. Aparecida-Rocco, M.L. Sforça, G.M. Pastore, P. Jauregi, A novel approach for the production and purification of mannosylerythritol lipids (MEL) by *Pseudozyma tsukubaensis* using cassava wastewater as substrate, Sep. Purif. Technol. 180 (2017) 57–167, https://doi.org/10.1016/j.seppur.2017.02.045.

[87] D.W.F. de Oliveira, I.W.L. França, A.K.N. Félix, J.J.L. Martins, M.E.A. Giro, V.M.M. Melo, L.R.B. Gonçalves, Kinetic study of biosurfactant production by *Bacillus subtilis* LAMI005 grown in clarified cashew apple juice, Colloids Surf. B Biointerfaces 101 (2013) 34–43, https://doi.org/10.1016/j.colsurfb.2012.06.011.

[88] S.L. Fox, G.A. Bala, Production of surfactant from *Bacillus subtilis* ATCC 21332 using potato substrates, Bioresour. Technol. 75 (2000) 235–240, https://doi.org/10.1016/S0960-8524(00)00059-6.

[89] G.C. Fontes, N.M. Ramos, P.F.F. Amaral, M. Nele, M.A.Z. Coelho, Renewable resources for biosurfactant production by *Yarrowia lipolytica*, Braz. J. Chem. Eng. 29 (2012) 483–493, https://doi.org/10.1590/S0104-66322012000300005.

[90] M.M. Freitas-Silva, P.M. Souza, A.A. Antunes, A. Cardoso, C.I.M. Lins, A.C.L. Batista, T.C.M. Stamford, G.M. Campos-Takaki, Biosurfactant production by Rhizopus arrhizus using agro industrials substrates as alternative medium, in: A. Mendez-Vilas (Ed.), Microbes in Applied Research-Current Advances and Challenges, World Scientific, Singapore, 2012, pp. 353–357, https://doi.org/10.1142/9789814405041_0071.

[91] J. Gurjar, B. Sengupta, Production of surfactin from rice mill polishing residue by submerged fermentation using *Bacillus subtilis* MTCC 2423, Bioresour. Technol. 189 (2015) 243–249, https://doi.org/10.1016/j.biortech.2015.04.013.

[92] C. Gusmão, R. Rufino, L. Sarubbo, Laboratory production and characterization of a new biosurfactant from *Candida glabrata* UCP1002 cultivated in vegetable fat waste applied to the removal of hydrophobic contaminant, World J. Microbiol. Biotechnol. 26 (2010) 1683–1692, https://doi.org/10.1007/s11274-010-0346-2.

[93] R.M. Jain, K. Mody, N. Joshi, A. Mishra, B. Jha, Effect of unconventional carbon sources on biosurfactant production and its application in bioremediation, Int. J. Biol. Macromol. 62 (2013) 52–58, https://doi.org/10.1016/j.ijbiomac.2013.08.030.

[94] M. Konishi, Y. Yoshida, J. Horiuchi, Efficient production of sophorolipids by *Starmerella bombicola* using a corncob hydrolysate medium, J. Biosci. Bioeng. 119 (2015) 317–322, https://doi.org/10.1016/j.jbiosc.2014.08.007.

[95] A.P. Kumar, A. Janardhan, B. Viswanath, K. Monika, J.Y. Jung, G. Narasimha, Evaluation of orange peel for biosurfactant production by *Bacillus licheniformis* and their ability to degrade naphthalene and crude oil, 3 Biotech 6 (2016) 1–10, https://doi.org/10.1007/s13205-015-0362-x.

[96] P.R.F. Marcelino, G.F.D. Peres, R. Terán-hilares, F.C. Pagnocca, C.A. Rosa, T.M. Lacerda, J.C. dos Santos, S.S. da Silva, Biosurfactants production by yeast using sugarcane bagasse hemicellulosic hydrolysate as new sustainable alternative for lignocellulosic biorefneries, Ind. Crop Prod. 129 (2019) 212–223, https://doi.org/10.1016/j.indcrop.2018.12.001.

[97] M. Nitschke, G. Pastore, Cassava four wastewater as a substrate for biosurfactant production, Appl. Biochem. Biotechnol. 106 (2003) 295–301.

[98] M. Nitschke, G.M. Pastore, Biosurfactant production by *Bacillus subtilis* using cassava-processing effluent, Appl. Biochem. Biotechnol. 112 (2004) 163–172.

[99] O.A. Oje, V.E. Okpashi, J.C. Uzor, U.O. Uma, A.O. Irogbolu, I.N.E. Onwurah, Effect of acid and alkaline pretreatment on the production of biosurfactant from rice husk using Mucor indicus, Res. J. Environ. Toxicol. 10 (2016) 60–67, https://doi.org/10.3923/rjet.2016.60.67.

[100] K. Paraszkiewicz, P. Bernat, A. Kusmierska, J. Chojniak, G. Płaza, Structural identification of lipopeptide biosurfactants produced by *Bacillus subtilis* strains grown on the media obtained from renewable natural resources, J. Environ. Manage. 209 (2018) 65–70, https://doi.org/10.1016/j.jenvman.2017.12.033.

[101] V.J. Parekh, V.B. Patravale, A.B. Pandit, Mango kernel fat: a novel source for the fermentative production of sophorolipid biosurfactant using *Starmerella bombicola* NRRL-Y 17069, Ann. Biol. Res. 3 (2012) 1798–1803.

[102] A. Rane, V.V. Baikar, V. Ravi-Kumar, R.L. Deopurkar, Agro-industrial wastes for production of biosurfactant by *Bacillus subtilis* ANR 88 and its application in synthesis of silver and gold nanoparticles, Front. Microbiol. 8 (2017) 1–12, https://doi.org/10.3389/fmicb.2017.00492.

[103] M.V.P. Rocha, M.C.M. Souza, S.C.L. Benedicto, et al., Production of biosurfactant by *Pseudomonas aeruginosa* grown on cashew apple juice, Appl. Biochem. Biotechnol. 137 (2007) 185–194, https://doi.org/10.1007/s12010-007-9050-6.

[104] M.V.P. Rocha, R.V. Gomes-Barreto, V.M.M. Melo, L.R.B. Gonçalves, Evaluation of cashew apple juice for surfactin production by *Bacillus subtilis* LAMI008, Appl. Biochem. Biotechnol. 155 (2009) 63–75, https://doi.org/10.1007/s12010-008-8459-x.

[105] A. Samad, J. Zhang, D. Chen, Y. Liang, Sophorolipid production from biomass hydrolysates, Appl. Biochem. Biotechnol. 175 (2015) 2246–2257, https://doi.org/10.1007/s12010-014-1425-x.

[106] X. Vecino, L. Rodríguez-López, E.J. Gudiña, J.M. Cruz, A.B. Moldes, L.R. Rodriges, Vineyard pruning waste as an alternative carbon source to produce novel biosurfactants by *Lactobacillus paracasei*, J. Ind. Eng. Chem. 55 (2017) 40–49, https://doi.org/10.1016/j.jiec.2017.06.014.

[107] D.N. Thompson, S.L. Fox, G.A. Bala, Biosurfactants from potato process effluents, Appl. Biochem. Biotechnol. 84 (2000) 917–930, https://doi.org/10.1385/ABAB:84-86:1-9:917.

[108] Z. Zhu, G. Zhang, Y. Luo, W. Ran, S. Qirong, Production of lipopeptides by *Bacillus amyloliquefaciens* XZ-173 in solid state fermentation using soybean four and rice straw as the substrate, Bioresour. Technol. 112 (2012) 254–260, https://doi.org/10.1016/j.biortech.2012.02.057.

[109] J. Hu, J. Luo, Z. Zhu, B. Chen, X. Ye, P. Zhu, B. Zhang, Multi-scale biosurfactant production by *Bacillus subtilis* using tuna fish waste as substrate, Catalysts 11 (2021) 456, https://doi.org/10.3390/catal11040456.

[110] K. Bhange, V. Chaturvedi, R. Bhatt, Simultaneous production of detergent stable keratinolytic protease, amylase and biosurfactant by *Bacillus subtilis* PF1 using agro industrial waste, Biotechnol. Rep. 10 (2016) 94–104, https://doi.org/10.1016/j.btre.2016.03.007.

[111] W.S. Borges, V.L. Cardoso, M.M. Resende, Use of a greasy effluent floater treatment station from the slaughterhouse for biosurfactant production, Biotechnol. Appl. Biochem. 59 (2012) 238–244, https://doi.org/10.1002/bab.1018.

[112] P. Chaves-Martins, V. Guimarães-Martins, Biosurfactant production from industrial wastes with potential remove of insoluble paint, Int. Biodeter. Biodegr. 127 (2018) 10–16, https://doi.org/10.1016/j.ibiod.2017.11.005.

[113] M. Deshpande, L. Daniels, Evaluation of sophorolipid biosurfactant production by *Candida bombicola* using animal fat, Bioresour. Technol. 54 (1995) 143–150, https://doi.org/10.1016/0960-8524(95)00116-6.

[114] D. Kadam, D. Savant, Biosurfactant production from shrimp shell waste by *Pseudomonas stutzeri*, Indian J. Geo-Mar. Sci. 48 (2019) 1411–1418.

[115] G. Kaur, H. Wang, M.H. To, S. Roelants, W. Soetaert, C. Lin, Efficient sophorolipids production using food waste, J. Clean. Prod. 232 (2019), https://doi.org/10.1016/j.jclepro.2019.05.326.

[116] T. Minucelli, R.M. Ribeiro-Viana, D. Borsato, G. Andrade, M.V. Torres-Cely, M.R. de Oliveira, C. Baldo, M.E.P. Colabone-Celligoi, Sophorolipids production by *Candida bombicola* ATCC 22214 and its potential application in soil bioremediation, Waste Biomass Valoriz. 8 (2017) 743–753, https://doi.org/10.1007/s12649-016-9592-3.

[117] R. Panjanathan, P. Suhazsini, K. Prabhu, A. Jayakumar, R. Kandasamy, Chicken tallow, a renewable source for the production of biosurfactant by *Yarrowia lipolytica* MTCC9520, and its application in silver nanoparticle synthesis, J. Surfactant Deterg. 23 (2019), https://doi.org/10.1002/jsde.12357.

[118] D. Santos, R. Rufino, J. Luna, V. Santos, A. Salgueiro, L. Sarubbo, Synthesis and evaluation of biosurfactant produced by *Candida lipolytica* using animal fat and corn steep liquor, J. Petrol. Sci. Eng. 105 (2013) 43–50, https://doi.org/10.1016/j.petrol.2013.03.028.

[119] R. Patowary, K. Patowary, M.C. Kalita, S. Deka, Utilization of paneer whey waste for cost-effective production of rhamnolipid biosurfactant, Appl. Biochem. Biotechnol. 180 (3) (2016) 383–399, https://doi.org/10.1007/s12010-016-2105-9. Epub 2016 May 3 27142272.

[120] A. Abdel-Mawgoud, N. Hassouna, Optimization of surfactin production by *Bacillus subtilis* isolate BS5, Appl. Biochem. Biotechnol. 150 (2008) 305–325, https://doi.org/10.1007/s12010-008-8155-x.

[121] H.J. Daniel, M. Reuss, C. Syldatk, Production of sophorolipids in high concentration from deproteinized whey and rapeseed oil in a two stage fed batch process using *Candida bombicola* ATCC 22214 and *Cryptococcus curvatus* ATCC 20509, Biotechnol. Lett. 20 (1998) 1153–1156.

[122] A. Daverey, K. Pakshirajan, Sophorolipids from *Candida bombicola* using mixed hydrophilic substrates: production, purification and characterization, Colloids Surf. B Biointerfaces 79 (2010) 246–253.

[123] K. Dubey, A. Juwarkar, Distillery and curd whey wastes as viable alternative sources for biosurfactant production, World J. Microbiol. Biotechnol. 17 (2001) 61–69, https://doi.org/10.1023/A:1016606509385.

[124] S. Joshi, C. Bharucha, A. Desai, Production of biosurfactant and antifungal compound by fermented food isolate *Bacillus subtilis* 20B, Bioresour. Technol. 99 (2008) 4603–4608.

[125] S. Maneerat, Production of biosurfactants using substrates from renewable resources Songklanakarin, J. Sci. Technol. 27 (2005) 675–683.

[126] A.S. Monteiro, J.O. Coutinho, A.C. Júnior, C.A. Rosa, E.P. Siqueira, V.L. Santos, Characterization of new biosurfactant produced by *Trichosporon montevideense* CLOA 72 isolated from dairy industry effluents, J. Basic Microbiol. 49 (6) (2009) 553–563, https://doi.org/10.1002/jobm.200900089. 19810042.

[127] D. Onbasli, B. Aslim, Determination of rhamnolipid biosurfactant production in molasses by some *Pseudomonas* spp, New Biotechnol. 25 (2009), https://doi.org/10.1016/j.nbt.2009.06.569.

[128] H. Rashedi, M.M. Assadi, B. Bonakdarpour, E. Jamshidi, Environmental importance of rhamnolipid production from molasses as a carbon source, Int. J. Environ. Sci. Technol. 2 (2005) 59–62.

[129] Z.A. Raza, M.S. Khan, Z.M. Khalid, Physicochemical and surface-active properties of biosurfactant produced using molasses by a *Pseudomonas aeruginosa* mutant, J. Environ. Sci. Health A Tox. Hazard. Subst. Environ. Eng. 42 (2007) 73–80.

[130] L.R. Rodrigues, J.A. Teixeira, R. Oliveira, Low-cost fermentative medium for biosurfactant production by probiotic bacteria, Biochem. Eng. J. 32 (2006) 135–142.

[131] J.P. Sinumvayo, N. Ishimwe, Agriculture and food applications of rhamnolipids and its production by *Pseudomonas aeruginosa*, J. Chem. Eng. Process Technol. 6 (2015) 223, https://doi.org/10.4172/2157-7048.1000223.

[132] R. Kumar, A.J. Das, Application of rhamnolipids in agriculture and food industry, in: Rhamnolipid Biosurfactant, Springer, Singapore, 2018, https://doi.org/10.1007/978-981-13-1289-2_9.

[133] M. Irfan-Maqsood, M. Seddiq-Shams, Rhamnolipids: well-characterized glycolipids with potential broad applicability as biosurfactants, Ind. Biotechnol. (2014) 285–291, https://doi.org/10.1089/ind.2014.0003.

[134] R. Ashby, D. Solaiman, X. Fan, M. Olanya, Antimicrobial Potential of Sophorolipids for Anti-Acne, Anti-Dental Caries, Hide Preservation, and Food Safety Applications, 2018, https://doi.org/10.1021/bk-2018-1287.ch010.

[135] M. Oliveira, A. Magri, C. Baldo, D. Camilios-Neto, T. Minucelli, M. Antonia, M.A. Celligoi, Review: sophorolipids a promising biosurfactant and it's applications, Int. J. Adv. Biotechnol. Res. 6 (2015) 161–174.

[136] S. Roelants, D. Solaiman, R. Ashby, S. Lodens, L. Van Renterghem, W. Soetaert, Production and Applications of Sophorolipids, 2019, https://doi.org/10.1016/B978-0-12-812705-6.00003-4.

[137] V.A. Silveira, E. Nishio, C. Freitas, I. Amador, R. Kobayashi, F. Macedo Jr., M.A. Celligoi, Production and antimicrobial activity of sophorolipid against *Clostridium perfringens* and *Campylobacter jejuni* and their additive interaction with lactic acid, Biocatal. Agric. Biotechnol. 21 (2019) 101287, https://doi.org/10.1016/j.bcab.2019.101287.

[138] I. Mnif, D. Ghribi, Glycolipid biosurfactants; main properties and potential applications in agriculture and food industry, J. Sci. Food Agric. 96 (2016), https://doi.org/10.1002/jsfa.7759.

[139] V. Sharma, D. Sharma, Microbial Biosurfactants: Future Active Food Ingredients, 2018, https://doi.org/10.1007/978-981-13-0053-0_14.

[140] G.S. Kiran, S. Priyadharsini, A. Sajayan, G.B. Priyadharsini, N. Poulose, J. Selvin, Production of lipopeptide biosurfactant by a marine *Nesterenkonia* sp. and its application in food industry, Front. Microbiol. 8 (2017) 1138, https://doi.org/10.3389/fmicb.2017.01138. 28702002. PMC5488535.

[141] K.R. Meena, S.S. Kanwar, Lipopeptides as the antifungal and antibacterial agents: applications in food safety and therapeutics, Biomed. Res. Int. 2015 (2015) 473050, https://doi.org/10.1155/2015/473050. Epub 2015 Jan 6 25632392. PMC4303012.

[142] I. Mnif, D. Ghribi, Review lipopeptides biosurfactants: mean classes and new insights for industrial, biomedical, and environmental applications, Biopolymers 104 (3) (2015) 129–147, https://doi.org/10.1002/bip.22630. 25808118.

[143] M. Nitschke, L.V. Araújo, S.G. Costa, R.C. Pires, A.E. Zeraik, A.C. Fernandes, D.M. Freire, J. Contiero, Surfactin reduces the adhesion of food-borne pathogenic bacteria to solid surfaces, Lett. Appl. Microbiol. 49 (2) (2009) 241–247, https://doi.org/10.1111/j.1472-765X.2009.02646.x.Epub. 2009 May 22 19486287.

[144] Y. Pecci, F. Rivardo, M.G. Martinotti, G. Allegrone, LC/ESI-MS/MS characterisation of lipopeptide biosurfactants produced by the Bacillus licheniformis V9T14 strain, J. Mass Spectrom. 45 (2010) 772–778.

[145] L.Z. Lin, Q.W. Zheng, T. Wei, Z.Q. Zhang, C.F. Zhao, H. Zhong, Q.Y. Xu, J.F. Lin, L. Q. Guo, Isolation and characterization of fengycins produced by *Bacillus amyloliquefaciens* JFL21 and its broad-spectrum antimicrobial potential against multidrug-resistant foodborne pathogens, Front. Microbiol. 11 (2020) 579621, https://doi.org/10.3389/fmicb.2020.579621. 33391199. PMC7775374.

[146] J. Shi, X. Zhu, Y. Lu, H. Zhao, F. Lu, Z. Lu, Improving iturin a production of *bacillus amyloliquefaciens* by genome shuffling and its inhibition against *Saccharomyces cerevisiae* in orange juice, Front. Microbiol. 9 (2018) 2683, https://doi.org/10.3389/fmicb.2018.02683. 30467499. PMC6236126.

[147] C.F. Carolin, P.S. Kumar, P.T. Ngueagni, A review on new aspects of lipopeptide biosurfactant: types, production, properties and its application in the bioremediation process, J. Hazard. Mater. 407 (2021) 124827, https://doi.org/10.1016/j.jhazmat.2020.124827. Epub 2020 Dec 11 33352424.

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CHAPTER

21

Characterization and screening of biosurfactants from microorganisms with a wide application in food industries

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1 Introduction

Surfactants are substances having hydrophilic and hydrophobic sections that oppose each other. They play a crucial role in everyday life as well as the food, pharmaceutical, cosmetic, and chemical sectors. Surfactants have unique properties, such as the ability to reduce surface tension between immiscible molecules and even create microemulsions from nonpolar molecules. Surfactants are formed or synthesized by microorganisms' metabolic processes. Biosurfactants are produced by microorganisms, notably Gram-negative bacteria. Among the production systems are bioreactors for submerged, solid, and biphasic fermentation of organic substrates [1].

Organic substrates include a wide range of agricultural and industrial waste, as well as waste polluted with oil and other contaminants. Agro-industrial leftovers offer a carbon and nitrogen source, as well as a place for bacteria to multiply and compete for space [2]. The bacteria create a variety of useful compounds, including biosurfactants, and can even be employed to lower aflatoxin levels produced by dangerous fungus during spontaneous fermentation [3]. Solid fermentation for the formation of biosurfactants relies on lignocellulosic wastes, which are complex and difficult to digest, necessitating a number of techniques to improve carbon and

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nitrogen availability [4]. The extraction of biosurfactants following fermentation, on the other hand, remains a difficulty, and green-technology-based techniques are choices that should be developed over time [5].

Biosurfactants are produced in the stationary phase of microbial growth, forming several conglomerates with other biomolecules (sugars, lipids, fatty acids, etc.). Among all biosurfactants, a class of glycolipids, known as rhamnolipid, produced from *Pseudomonas* spp., has been widely studied and has practical applications in the food, pharmaceutical, and petrochemical industries [6].

Microorganisms that create biosurfactants have been screened using a variety of methodologies and approaches. Different conventional and modern methods help to reduce costs, increase assertiveness and efficiencies of the strains. Along with modern genetic engineering techniques, researchers can produce new, more efficient, and productive microbial strains [6]. This review intends to give a concise and focused examination of biosurfactant screening approaches, as well as extraction, purification, and quantification analytical techniques. This review also considers the unique instance of rhamnolipid and its potential for food-related applications.

2 Screening of biosurfactant-producing microorganisms

Bioprospecting research can be used to investigate and assess biological materials to discover new genetically valuable items with intriguing biochemical features. Biological products are used in a variety of sectors, including pharmaceuticals, agrochemicals, cosmetics, aromas, and perfumes.

Biosurfactants are industrially important substances having tensoactive qualities and emulsifying activities that can increase oil recovery, crude oil drilling lubricants, and water-insoluble pollution bioremediation [7,8]. The interest in biosurfactants stems from their ability to mitigate most of the negative effects of chemical surfactants in a variety of applications, including bioremediation of water and soils, cleaning of oil reservoirs, and therapeutic applications as antimicrobials, antivirals, and antitumors, as well as their potential use in the formulation of herbicides and pesticides, and in the food sector, it is applied as an emulsifier, and it's also used in the pharmaceutical industry.

According to Marchant and Banat [9], they are amphiphilic compounds because their chemical structures feature hydrophilic and hydrophobic ends, allowing them to interact at the aqueous-nonaqueous interface. Bacteria, yeasts, and fungi produce it from a variety of sources, including sugars, oils, and alkanes.

Phospholipids, substituted fatty acids, and other biosurfactants are the six main categories: lipopolysaccharides, neutral lipids, glycolipids, lipopeptides, and lipoproteins [10]. Microbial biosurfactants are gaining popularity due to their application properties that are considered advantages, such as low environmental toxicity, significant ecological and biodegradability when compared with synthetics, which has boosted their use in agriculture, food industry, and other industries [10].

When a negative result is obtained, it may be due to the culture medium's low initial surface tension, which indicates the presence of surfactant in the medium's composition. As a result, even if the microbe created the metabolite, it would have no effect on the medium's surface tension [11]. According to Freitas de Oliveira et al. [12], there is no linear link between an increase in biosurfactant concentration and a decrease in surface tension, remaining constant from a given concentration, for the use of different cultivation substrates for *Bacillus pumilus*. According to the same author, in an alternate culture medium of cashew juice mixed with ammonium sulfate, *Bacillus subtilis* LAMI005 developed a biosurfactant. This enables us to refer to it as an economic option, as one of the major impediments to the growth of the biosurfactants sector is a lack of funding.

According to Neves et al. [13], *B. subtilis* is an attractive microbe for this application since it appears to be ideal for producing biosurfactant due to its lack of pathogenicity and the potential use of its products in the food and pharmaceutical sectors. Water-soluble substrates are believed to be less expensive than hydrocarbons, and that single-phase fermentation is easier than two-phase fermentation. Some of the main methods available for screening biosurfactant producers are hemolytic assay (HA), hydrocarbon overlay agar (HOA), asymmetric droplet shape analysis (ADSA), turbidity test, tensiometric measures, such as surface energy and interfacial tension, usage of *N*-Cethyl-*N*,*N*,*N*-trimethylammonium bromide agar plate (CTAB). There are six other ways that might be mentioned: bacterial adhesion to hydrocarbons (BATH), microbial adhesion to hydrocarbons (MATH), drop collapse method (DCM), oil scattering method (OSM), and emulsification activity slanted glass slide evaluation.

According to Varjani et al. [14], the evaluation of the screening of biosurfactant producers using a varied number of isolates showed good results for some isolates in reducing surface tension, but unpromising results about the emulsification activity obtained, so the authors mention that it is possible to consider that not all bioemulsifier producers are also biosurfactant producers, but the reverse order always happens. As a result, it is indicated that the emulsification activity approach can only indicate the examined isolate's potential for producing biosurfactants. Surface tension assessment and interfacial tension decrease have a high negative association, according to the same author. Once the drop collapse method is used to discover biosurfactant producers and the oil dispersion approach is used to assess biosurfactant concentration, the outcome is a simple and fast process for tracking and measuring biosurfactant synthesis. The blood agar lysis process achieves erroneous negative and positive results, as well as a poor relationship with surface tension ($rs = 0.15$), indicating that it is not a reliable method for determining biosurfactant formation [15]. Blood agar lysis is a common method for assessing biosurfactant makers, and it may also be used as a preliminary screening method. The lysis produced was used to measure surfactin and rhamnolipids. The temperature used for microbe growth, according to the same author, has a considerable influence on hemolysis generation, which could explain the disparity in hemolysis seen across data [15].

3 Extraction, purification, and characterization of biosurfactants

Characterization procedures have been thoroughly evaluated with precipitation, solvent filtration extraction, centrifugation, and foam fractionation [16]. Simple to complex chromatographic and/or spectrophotometric analytical procedures can be used to characterize extracted and purified biosurfactants. Thin layer chromatography (TLC), gas chromatography (GC), Fourier-transform infrared spectroscopy (FTIR), gas chromatography–mass spectrometry (GC-MS), electrospray ionization–mass spectrometry (ESI-MS), high-performance thin-layer chromatography (HP-TLC), high-performance liquid chromatography (HPLC), liquid chromatography–mass spectrometry (LC-MS), and fast atom bombardment [16].

The work of Martin et al. [17] developed a biosurfactant from *Streptomyces* sp. DPUA1559 obtained by Amazonian lichens, the product was also characterized, being the culture medium that used mineral containing 1% of soy residues, in the production of the biosurfactant, it was observed a decrease in the surface tension of the culture medium from 60 to about 27.14 mN/m. The final product x stability in pH, temperature, and in terms of surface tension decrease and resistance to concentrated saline solution. Biochemical investigation of the biosurfactant revealed that it is an acidic glycoprotein. The results point to future potential for this new biosurfactant in industrial applications.

Meena et al. [18] tested *Bacillus velezensis* cell-free broth KLP2016 against various organic solvents. At the best micellar concentration, the surface tension of the broth grown without *B. velezensis* cells was significantly reduced at 35°C. Still in the work carried out by Meena et al. [18], *B. velezensis* KLP2016 was used to isolate and describe the surfactin biosurfactant, a lipopeptide family member. Biosurfactants demonstrated lower surface tension, emulsifying activity, and motor oil degrading capability than conventional surfactants. This biosurfactant-based approach to engine oil deterioration appears to be very promising, with the potential to drastically reduce soil and water pollution in the near future.

San Martín et al. [19] looked at the potential of *Pseudomonas* sp. biosurfactant to remove pollutants from soil that had been artificially contaminated. The biosurfactant was identified as a rhamnolipid using spectroscopic and chromatographic testing, and 240 mg L^{-1} rhamnolipid was employed to achieve a maximum elimination of 85.5%. We know that biosurfactants have hydrophilic "heads" and hydrophobic "tails" as a structure, usually hydrocarbon chains that reduce the surface tension between the liquid phases, as in the air-water and oil-water interfaces, where based on this structure provide many applications [20]. At high concentrations of carbon and nitrogen, SM3 produced the ideal biosurfactant, reducing ST. Meanwhile, at low concentrations, the highest results for SMRG-5 were obtained. In one reactor, the ideal conditions for the production of biosurfactant were improved. The existence of two distinct lipopeptides was discovered using chromatographic techniques [21].

The potential of the biosurfactant was investigated using *Bacillus* strains. Twelve strains were found and examined, and those capable of decreasing the surface tension of water and hexadecane were chosen. The final product has outstanding pH, temperature, and salinity stability, as well as emulsifying capability in a variety of hydrocarbons. The created biosurfactant was able to form moderately stable emulsions for up to 96 hours, allowing it to be used in a variety of applications, according to phase diagram analyses. The cyclic lipopeptide biosurfactant with a molecular structure like surfactin did not show toxicity and can be used for remediation of contaminated soil or in food applications [22,23].

Low toxicity, biodegradability, biocompatibility, digestibility, high selectivity, and stability in extreme pH, temperature, and ionic strength, antimicrobial activity, acting as wetting agents, emulsifiers, foaming and dispersing agents, in addition to production conditions, are some of the characteristics that make biosurfactants products of great technological interest [24]. We cannot fail to mention the possibilities achieved with the consolidation of micelle stability, namely: the expansion of biosurfactant applications, producing nanoemulsion formulations, and drug delivery systems for the treatment of various diseases, due to the amphiphilic structural characteristics of some biosurfactants [24]. They can be used in a variety of fields, including biomedicine, cosmetics, agriculture, and procedures for removing toxins from soils, such as petroleum and heavy metals, as well as the removal of various types of oils in the environment, based on these qualities [25].

4 Rhamnolipid in *Pseudomonas* spp. for food applications

Rhamnolipids are well-studied glycolipids with substantial industrial potential, and *Pseudomonas* strains are the top producers of glycolipids comprising rhamnose and 3-hydroxylated fatty acids. However, *Pseudomonas aeruginosa* was discovered to be the best microorganism for making two forms of rhamnolipids with high surface activity: monorhamnolipids and dirhamnolipids. Numerous researchers have sought to boost its production because of its qualities and application potential, making it one of the most studied biosurfactants [26].

Composition and distribution of rhamnolipid homologues are important for understanding rhamnolipids; these parameters are regulated by the bacterial strain in question, its growth mode, and the medium composition. The raw material becomes onerous, and the yields produced are minimal, making large-scale rhamnolipid manufacture economically untenable. Strain engineering, on the other hand, is now a realistic way for achieving precise manufacturing, making large-scale rhamnolipid synthesis feasible [27].

Rhamnolipids have a diverse range of uses, making them good candidates for industrial usage, also the rhamnolipids have antibacterial properties that are effective against both fungus and bacteria [28], they have interesting properties including detergency, emulsification, metal sequestration, demulsification, solubilization, wetting, thickening, foaming, environmental compatibility, vesicle formation, and phase

dispersion, in addition to acting as potential and current biosurfactants. All these characteristics are linked to rhamnolipid molecules' amphiphilic nature, which allows them to aggregate across fluid phases, lowering surface and interfacial tensions. Rhamnolipids have a wide range of potential applications, including environmental biodegradation, agricultural, cosmetic, and medicinal applications. However, we will cover your application in the field of food processing [29].

Food manufacturers employ emulsifiers to lower to improve the properties of food products, allowing for better mixing during the preparation. In general, the food and beverage sectors use the emulsifier lecithin, which is derived from egg and soy, as well as various synthetic emulsifiers. The food industry's rapid expansion necessitates the use of natural, environmentally friendly emulsifiers. Rhamnolipids are an alternative emulsifier for the food sector because of their unique qualities, such as emulsification activity and ecofriendliness [30].

Natural surfactants such as milk proteins, sorbitan esters, lecithin, and sucrose esters are commonly used in food formulations such as salad dressings, mayonnaise, food emulsions, and desserts, according to Sharma [31], but their use in food and feed formulations has been halted due to toxicity and fate concerns. How biosurfactants work: They're a nontoxic, safe chemical with antiadhesive, antibacterial, emulsifying, and additive capabilities that might extend the shelf life of food. For example, salad dressings include rhamnolipid biosurfactants, which can help prevent mold growth [31].

Rhamnolipid biosurfactants can be also utilized to increase the volume, shape, and stability of dough in bread items including pizza, croissants, and baguettes, as well as in the production of cakes [31]. Some of the properties of rhamnolipids, such as emulsion formation and stability, as well as antiadhesive and antibacterial properties, make them desirable as food components. Rhamnolipids have been proposed to have a range of different roles in food, in addition to their role as surface-active agents. Rhamnolipid surfactants have been used to enhance dough stability, texture, volume, and preservation in bakery goods, while other authors have proposed adding rhamnolipid to improve buttercream, croissants, and frozen confectionery items [32].

De-emulsification is the process of dissolving a stable emulsion, which is crucial in the food sector, especially when working with fat and oil products. Rhamnolipids could both stabilize (emulsify) and destabilize (demulsify) an emulsion [28]. To measure emulsion stability, ternary phase diagrams were developed, which varied according to OE used. In vitro tests were performed on the antibacterial activity of the EOs and emulsions. The antimicrobial activity of essential oils enhanced when they were emulsified. Furthermore, rhamnolipids' surface characteristics facilitate oil dispersion, increasing their availability and antibacterial activity. As a result, rhamnolipid-based emulsions appear to be a potential avenue for creating oil delivery methods. Furthermore, rhamnolipid is used in sugar production after the hydrolysis process. L-Rhamnose is a sugar approved as a food additive by the Food and Drug Administration (FDA). As a result, acquiring industrial quantities of rhamnolipids to supply a source of L-rhamnose, which already has applications in the food sector, is of tremendous importance [32].

In the study by Dilarri et al. [33] to suppress microbial growth, several fruit washing techniques were tested. Regularly, the fruits were cleaned and inoculated in culture media to assess and count the colonies on their surfaces. The degree of degradation of the fruits was also measured after each treatment. The rhamnolipids therapy was shown to be the most effective at inhibiting fungal and bacterial growth. Initially, electrolyzed water appeared to be quite successful in bacterial inhibition, but it had no inhibitory impact in the end. In addition, the electrolyzed water was inefficient at removing fungus. The least effective treatment was carried out with running water. Rhamnolipids were the only treatment that proved to be more durable, increasing the usable life expectancy up to 2 days. Rhamnolipids are the most suggested way for fruit cleanliness, according to these authors.

By reducing spore germination and mycelial development, rhamnolipids, a kind of surfactant, can protect vines from Botrytis cinerea assault. Rhamnolipids and rhamnolipid surfactants have antifungal efficacy against a variety of fungal plant diseases, including *Botrytis* sp., *Rhizoctonia* sp., and *Pythium* sp. [33]. According to Valle Gomes and Nitschke [34], it can be used not only to remove existing biofilms, but also to provide a therapy capable of altering surface attributes and reducing contamination.

5 Concluding remarks

Screening and characterization procedures are critical for acquiring high-quality biosurfactants in sufficient quantities for industrial manufacturing. The best researched biosurfactant, rhamnolipid, may be utilized as a model among biosurfactants, making exploratory research and application paths in the food business easier. Currently, there are constraints to commercial biosurfactant manufacturing in terms of cost and microorganisms capable of large-scale production. However, we believe that in the coming years, they will be able to leverage the production of biosurfactants, making the process profitable and economically viable, due to improved production studies, the use of renewable substrates, the validation of multistep processes, and modern genetic engineering techniques.

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Conflict of interest

All authors agree that there are no conflicts of interest.

References

[1] J.R. Barbosa, S.B. da Silva, R.N. de Carvalho Junior, Biosurfactant production by solid-state fermentation, submerged fermentation, and biphasic fermentation, in: A.M. Inamuddin, A. Asiri, K. Suvardhan (Eds.), Green Sustainable Process for Chemical and Environmental Engineering and Science, Elsevier, 2021, pp. 155–171.

[2] J.R. Barbosa, S.B. da Silva, L.H. da Silva Martins, F.W.F. Bezerra, L.C. Freitas, M.C.R. Ferreira, R.N. de Carvalho Junior, Microbial degradation of food products, in: M. Inamuddin, I. Ahamed, R. Prasad (Eds.), Recent Advances in Microbial Degradation, Springer, Singapore, 2021, pp. 155–172.

[3] S.B. da Silva, J.R. Barbosa, L.H. da Silva Martins, V.S.V. Moraes, C.M.G. Bichara, F.R. S. Sousa, A.S. Lopes, Microbial degradation of aflatoxin, in: M. Inamuddin, I. Ahamed, R. Prasad (Eds.), Recent Advances in Microbial Degradation, Springer, Singapore, 2021, pp. 1–18.

[4] L.C. Freitas, J.R. Barbosa, A.L.C. da Costa, F.W.F. Bezerra, R.H.H. Pinto, R.N. de Carvalho Junior, From waste to sustainable industry: how can agro-industrial wastes help in the development of new products? Resour. Conserv. Recycl. 169 (2021) 105466.

[5] F.W.F. Bezerra, P.D.N. Bezerra, V.M.B. Cunha, M.D.L.A.R. Salazar, J.R. Barbosa, M.P. da Silva, R.N. de Carvalho Junior, Supercritical green solvent for Amazonian natural resources, in: Inamuddin, A.M. Asiri (Eds.), Advanced Nanotechnology and Application of Supercritical Fluids, Springer, Cham, 2020, pp. 15–31.

[6] R.V. Prasad, R.A. Kumar, D. Sharma, A. Sharma, S. Nagarajan, Sophorolipids and rhamnolipids as a biosurfactant: synthesis and applications, in: Inamuddin, A.M. Asiri, S. Kanch (Eds.), Green Sustainable Process for Chemical and Environmental Engineering and Science, Elsevier, 2021, pp. 423–472.

[7] S.L. Fox, G.A. Bala, Production of surfactant from *Bacillus subtilis* ATCC 21332 using potato substrates, Bioresour. Technol. 75 (2000) 235–240.

[8] S.K. Satpute, G.A. Plaza, A.G. Banpurkar, Biosurfactants' production from renewable natural resources: example of innovative and smart technology in circular bioeconomy, Manag. Syst. Prod. Eng. 25 (2017) 46–54.

[9] R. Marchant, I.M. Banat, Biosurfactants: a sustainable replacement for chemical surfactants? Biotechnol. Lett. 34 (2012) 1597–1605.

[10] H. Zhao, D. Shao, C. Jiang, J. Shi, Q. Li, Q. Huang, et al., Biological activity of lipopeptides from *Bacillus*, Appl. Microbiol. Biotechnol. 101 (2017) 5951–5960.

[11] S.B. Batista, A.H. Mounteer, F.R. Amorim, M.R. Tótola, Isolation and characterization of biosurfactant/bioemulsifier-producing bacteria from petroleum contaminated sites, Water Sci. Technol. Water Supply 34 (2006) 187–194.

[12] D.W.F. de Oliveira, I.W.L. França, A.K.N. Félix, J.J.L. Martins, M.E.A. Giro, V.M.M. Melo, L.R.B. Gonçalves, Kinetic study of biosurfactant production by *Bacillus subtilis* LAMI005 grown in clarified cashew apple juice, Colloids Surf. B Biointerfaces 101 (2013) 34–43.

[13] L.C.M. das Neves, K.S. de Oliveira, M.J. Kobayashi, T.C.V. Penna, A. Converti, Biosurfactant production by cultivation of *Bacillus atrophaeus* ATCC 9372 in semidefined glucose/casein-based media, Appl. Biochem. Biotechnol. 137 (2007) 539–554.

[14] S.J. Varjani, D.P. Rana, S. Bateja, M.C. Sharma, V.N. Upasani, Screening and identification of biosurfactant (bioemulsifier) producing bacteria from crude oil contaminated sites of Gujarat, India, Int. J. Innov. Res. Sci. Eng. Technol. 3 (2014) 9205–9213.

[15] N.H. Youssef, K.E. Duncan, D.P. Nagle, K.N. Savage, R.M. Knapp, M.J. McInerney, Comparison of methods to detect biosurfactant production by diverse microorganisms, J. Microbiol. Methods 56 (2004) 339–347.
[16] S.J. Varjani, V.N. Upasani, Critical review on biosurfactant analysis, purification and characterization using rhamnolipid as a model biosurfactant, Bioresour. Technol. 232 (2017) 389–397.
[17] G.D. Martin, L.T. Tan, P.R. Jensen, R.E. Dimayuga, C.R. Fairchild, C. Raventos-Suarez, W. Fenical, Marmycins A and B, cytotoxic pentacyclic C-glycosides from a marine sediment-derived actinomycete related to the genus Streptomyces, J. Nat. Prod. 70 (2007) 1406–1409.
[18] K.R. Meena, R. Dhiman, K. Singh, S. Kumar, A. Sharma, S.S. Kanwar, A.K. Mandal, Purification and identification of a surfactin biosurfactant and engine oil degradation by *Bacillus velezensis* KLP2016, Microb. Cell Fact. 20 (2021) 1–12.
[19] Y.B. San Martín, H.F.T. León, A.A. Rodríguez, A.M. Marqués, M.I.S. López, Rhamnolipids application for the removal of vanadium from contaminated B sediment, Curr. Microbiol. 78 (2021) 1949–1960.
[20] A.R. Markande, D. Patel, S. Varjani, A review on biosurfactants: properties, applications and current developments, Bioresour. Technol. 330 (2021) 124963.
[21] N.S. Rosas-Galván, F. Martínez-Morales, S. Marquina-Bahena, R. Tinoco-Valencia, L. Serrano-Carreón, B. Bertrand, M.R. Trejo-Hernández, Improved production, purification, and characterization of biosurfactants produced by *Serratia marcescens* SM3 and its isogenic SMRG-5 strain, Biotechnol. Appl. Biochem. 65 (2018) 690–700.
[22] A.K.N. Felix, J.J. Martins, J.G.L. Almeida, M.E.A. Giro, K.F. Cavalcante, V.M.M. Melo, R.S. de Santiago Aguiar, Purification and characterization of a biosurfactant produced by *Bacillus subtilis* in cashew apple juice and its application in the remediation of oil-contaminated soil, Colloids Surf. B Biointerfaces 175 (2019) 256–263.
[23] I. Ruba, B. Zain, C. Ildikó, Regulatory status quo and prospects for biosurfactants in pharmaceutical applications, Drug Discov. Today 26 (2021) 1929–1935.
[24] E.B. Manga, P.A. Celik, A. Cabuk, I.M. Banat, Biosurfactants: opportunities for the development of a sustainable future, Curr. Opin. Colloid Interface Sci. 56 (2021) 101514.
[25] S. Mishra, Z. Lin, S. Pang, Y. Zhang, P. Bhatt, S. Chen, Biosurfactant is a powerful tool for the bioremediation of heavy metals from contaminated soils, J. Hazard. Mater. 418 (2021) 126253.
[26] A. Zgola-Grzeskowiak, E. Kaczorek, Isolation, preconcentration and determination of rhamnolipids in aqueous samples by dispersive liquid–liquid microextraction and liquid chromatography with tandem mass spectrometry, Talanta 83 (2011) 744–750.
[27] M.M. Muller, J.H. Kugler, M. Henkel, M. Gerlitzki, B. Hormann, M. Pohnlein, R. Hausmann, Rhamnolipids-next generation surfactants? J. Biotechnol. 162 (2012) 366–380.
[28] E. Haba, S. Bouhdid, N. Torrego-Solana, A.M. Marqués, M.J. Espuny, M.J. García-Celma, A. Manresa, Rhamnolipids as emulsifying agents for essential oil formulations: antimicrobial effect against *Candida albicans* and methicillin-resistant *Staphylococcus aureus*, Int. J. Pharm. 476 (2014) 134–141.
[29] M. Nitschke, S.G. Costa, J. Contiero, Rhamnolipids and PHAs: recent reports on *Pseudomonas*-derived molecules of increasing industrial interest, Process Biochem. 46 (2011) 621–630.
[30] R. Kumar, A.J. Das, Application of rhamnolipids in agriculture and food industry, in: R. Kumar, A.J. Das (Eds.), Rhamnolipid Biosurfactant, Springer, Singapore, 2018, pp. 97–109.

[31] D. Sharma, Applications of biosurfactants in food, in: D. Sharma (Ed.), Biosurfactants in Food, Springer, Cham, 2016, pp. 43–80.
[32] M.G. Rikalovic, M.M. Vrvic, I.M. Karadzic, Rhamnolipid biosurfactant from *Pseudomonas aeruginosa*: from discovery to application in contemporary technology, J. Serb. Chem. Soc. 80 (2015) 279–304.
[33] G. Dilarri, V.L.D. Silva, H.B. Pecora, R.N. Montagnolli, C.R. Corso, E.D. Bidoia, Electrolytic treatment and biosurfactants applied to the conservation of *Eugenia uniflora* fruit, J. Food Sci. Technol. 36 (36) (2016) 456–460.
[34] M.Z. do Valle Gomes, M. Nitschke, Evaluation of rhamnolipid and surfactin to reduce the adhesion and remove biofilms of individual and mixed cultures of food pathogenic bacteria, Food Control 25 (2012) 441–447.

Further reading

M. Nitschke, G.M. Pastore, Production and properties of a surfactant obtained from *Bacillus subtilis* grown on cassava wastewater, Bioresour. Technol. 97 (2006) 336–341.

CHAPTER

22 Plant saponin biosurfactants used as soap, hair cleanser, and detergent in India

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1 Introduction

The biosurfactants derived from plants, microbes, and animals are bestowed with many characteristic features such as natural abundance, renewability, sustainability, lower cost, superior stability at variable conditions (temperature, pH, ionic strength, salinity), biocompatibility, biodegradability, low critical micellar concentration, lesser environmental and human toxicity, and lower human allergenicity, in comparison with synthetic surfactants. In addition, their manufacture does not diminish the inadequate petroleum reservoirs, and they exhibit an array of biological activities [1–3]. The saponins are one class of biosurfactants derived from various parts of the plants as secondary metabolites and broadly categorized as triterpenoid and steroid saponins. The saponins are commonly available from different plant species, characterized by bitterness, facile extraction, and amenable for mass production when compared with microbial biosurfactants. They exhibit distinctive properties of surfactants such as wetting, foaming, emulsifying, and detergent properties [1,4].

The presence of saponins in plant extracts is qualitatively detected by foam test based on surfactant property, Rosenthaler test based on violet color formation by pentacyclic saponins and Liebermann Burchard test based on pink/red color formation by triterpenicgenins and blue/green color formation by steroid saponins. The thin layer chromatography technique is also used to qualitative identification saponins based on the separation of saponins on silica gel stationary phase and chloroform/ethanol/water as mobile phase. The saponins are visualized by spraying of different chromogenic agents such as p-anisaldehyde, vanillin, antimony trichloride, silver nitrate, and iodine and heating. The compounds, including digitonin, hecogenin, diosgenin, and cholesterol, are employed as positive reference standards. The saponins are quantitatively determined by spectrophotometry at 430 nm against

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the calibration curve of digitonin, using p-anisaldehyde and sulfuric acid reagents [5]. The concentration of extracted saponins is also determined by high-pressure liquid chromatography [6].

The present review gives a brief account of the importance of plant saponin biosurfactants, their quantification, and commonly used saponin yielding plants and trees employed as soap, hair cleanser, and detergent in India. In India, most of the saponin-containing plants belong to various families, including Sapindaceae (*Sapindus* sp), Fabaceae (*Acacia concinna, Trigonella foenum-graecum, Vigna radiata, Cicer arietinum, Albizia lebbeck*), Rutaceae (*Citrus* sp), Malvaceae (*Hibiscus rosa sinensis*), Phyllanathaceae (*Phyllanthus emblica*), Asphodelaceae (*Aloe vera*), Asteraceae (*Eclipta alba*), Asparagaceae (*Agave* sp), Dioscoreaceae (*Dioscorea esculenta*), etc. The various plant parts such as leaves, bark, stem tubers, fruits, fruit pericarp, seeds, etc., are utilized as bathing soap, washing detergent, utensil cleaner, hair cleanser, hair, shampoo, hair oil, etc. [5]. Notably, some of the saponiferous fruits (*Sapindus* sp., *Acacia concinna, P. emblica*) were found in the archeological excavations carried out in Punjab and Haryana by Archeological Survey of India, India. The utilization of herbal detergent and shampoo in pre-Harappan civilization (2750–2500 BCE) indicates the advancement in health and hygienic practices [7].

2 *Sapindus sp.*

The various species of the genus *Sapindus* such as *Sapindus mukorossi*, *S. emarginatus*, and *S. trifoliatus* are commonly known as *kunkudu, phenila, reetha, aritha, boondi kottai*, etc., in various Indian languages and belong to the family Sapindaceae. They are tropical, deciduous/evergreen trees and known as soapnut, soapberry, and washnut in English. The species are native to tropical Asia, China, India, Myanmar, Indonesia, Philippines, Malaysia, Thailand, Cambodia, and Vietnam. They are usually seen in the warm plains of South and Central India. The tree's fruits are classified as drupes and are 1–3 lobed and 1–3 cm in diameter. The young drupes are fleshy, leathery skinned, velvety, and yellowish, and the dried drupes are blackish brown, hard, and smooth with 1–3 spherical seeds (Fig. 1A) [1,5,8]. The pericarps of the fruit are rich

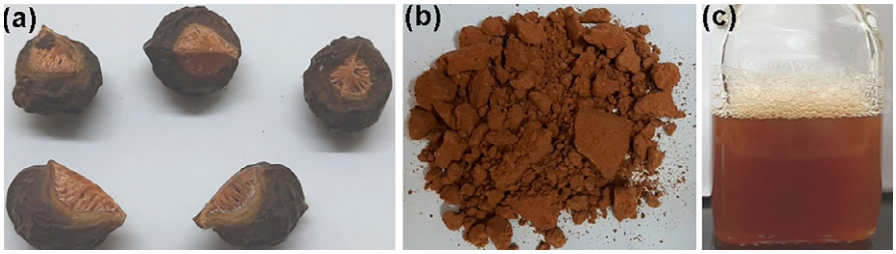

FIG. 1

Photographic images of *Sapindus emarginatus* (A) fruits, (B) pericarp powder, and (C) aqueous pericarp extract.

source of triterpenoid saponins (6%–11.5%), sugars (10%), and mucilage. The isolated saponins are categorized as oleanane (sapindoside A-C, sapinmusaponin K-N, mukorozisaponin G, E1), dammarane (sapinmusaponin A-E, O, P), and tirucullane (sapinmusaponin F-J, Q, R, sapinmukoside A-J) types [6,8–10].

In Asian countries such as India, powder and the cold and aqueous pericarp extracts are routinely used as traditional, natural hair cleanser, shampoo, bathing soap, body cleanser, etc., due its natural lather (Fig. 1B and C). The abundant vitamins (A, D, E, K) present in the extracts serve as a natural conditioner for the skin and hair and provide smoothness and shine for hair. The insecticidal action of the extract is also traditionally employed for the removal and killing of head lice from the infected, itchy scalp. The extract also exhibits antidermatophytic action against various fungi (*Epidermophyton floccosum*, *Trichophyton mentagrophytes*, *T. rubrum, Sabouraudites canis, Candida albicans*), which are responsible for various skin infections including dandruff. Thus, it is an essential ingredient in many herbal soaps, shampoos, cleansers, hair dye, hair oil, and skincare products. The extracts and powders of pericarp are also exploited as a laundry detergent for cotton, synthetic, silk, and woolen fabrics; dish washer for kitchen utensils, polisher for silver and gold ornaments, etc. The superior emulsifying action of natural *Sapindus* saponins when compared with synthetic surfactant, sodium dodecyl sulfate makes it an ideal, ecofriendly, commercial, economical candidate of laundry detergent [1,9–12].

The fruit pericarps are used for treating asthma, migraine, skin diseases (eczema, psoriasis, pimples, freckles, dandruff), and psychiatric disorders (hysteria, epilepsy) in *Ayurveda* and folklore medicine. The pericarp extracts are known to exhibit an array of biological activities, including antibacterial, antifungal, antidermatophytic, antiprotozoan, antitrichomonial, antioxidant, anticancer, antiinflammatory, anxiolytic, hepatoprotective, insecticidal, spermicidal, piscicidal, molluscicidal activities [1,8,10–13].

3 *Acacia concinna*

The plant *Acacia concinna* is a wild, woody climbing shrub or small tree belonging to the family Fabaceae. It is normally known as *shikakai*, *kantavalli*, *bahuphenarasa*, etc., in various Indian languages. It is native to tropical Asia, China, India, Myanmar, Indonesia, Philippines, Malaysia, Thailand, Cambodia, and Vietnam. It is generally found in the warm plains of South and Central India. The fruits of the plants are classified as pods, and the dried pods are thick, hard, flat, wrinkled, stalked, sized around 5–8 × 1.5–1.8 cm (Fig. 2A) [5,14–16]. The pods are rich source of different secondary metabolites such as flavonoids, alkaloids, saponins, tannins, phenolics, phytosterols, and gums. The saponins present in the pod extract serve as foaming agents [14,15,17–19]. The dried or powdered pods are soaked either in cold or warm water, and the obtained extract has been widely utilized as a traditional, natural shampoo in the Indian subcontinent for centuries (Fig. 2B and C). The extract is mild, low in pH, and serves as a gentle, low foaming hair cleanser and conditioner without stripping

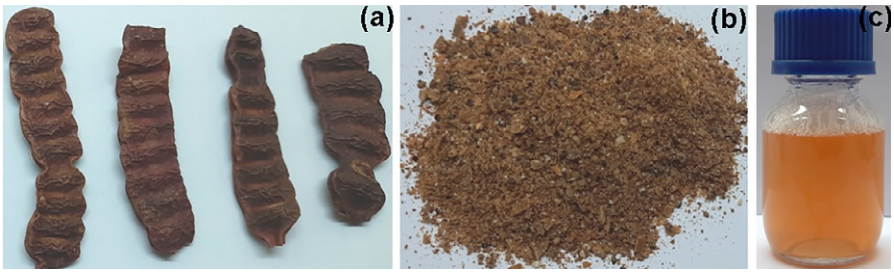

FIG. 2

Photographic images of *Acacia concinna* (A) fruits, (B) fruit powder, and (C) aqueous fruit extract.

the natural oils of the hair and scalp. The extract promotes hair growth, strengthens hair roots, and controls dandruff and skin diseases [5,16–20]. The powdered pods, pod paste, and pod extract/decoction are also used as a detergent for washing ordinary, delicate and silk cloths, cleaning vessels, silver, and gold jewelry. It is a crucial ingredient in many commercially marketed, herbal hair cleaning products such as shampoo, soap, hair care products, etc. It is also employed as a marine fish poison, purgative, laxative, immunomodulant, etc. [5,17–19].

The aqueous pod extract exhibited antibacterial action toward *Escherichia coli, Pseudomonas aeruginosa, Klebsiella pneumoniae, Staphylococcus aureus*, *and Bacillus subtilis*, and the inhibition zone ranged from 5.5 to 10.2 mm. It also showed antifungal action against *C. albicans, Aspergillus niger, and Penicillium* sp., and the inhibition zones ranged from 8.2 to 12.5 mm [14]. The extract also showed antidermatophytic activity against various dermatophytic fungi such as *Trichophyton rubrum*, *T. mentagrophytes*, *T. violaecum*, *Microsporum nanum*, and *Epidermophyton floccosum* and the minimum inhibitory concentration (MIC) and minimum fungicidal concentration (MFC) were found to be 62.5 μg/mL [20]. The extract showed antioxidant (DPPH and ABTS radical scavenging; linoleic acid peroxidation inhibition) and antityrosinase activities implying its application in skin repair and skin whitening cosmetic products [15].

4 *Trigonella foenum-graecum*

The plant *T. foenum-graecum* is called as fenugreek and belongs to the family Fabaceae. It is an annual, erect, cultivated herb across the Asian and African continents. It is commonly known as *menthi, methi*, *methika, bahuparni,* etc., in various Indian languages and has been grown extensively in most Indian states. The seeds are cuboid in shape, yellow to light brown in color, and each pod contains around 10–20 seeds (Fig. 3A) [5,21]. The seeds are abundant in steroid saponins such as diosogenin, yamogenin, gitogenin, neogitogenin, homorientin saponaretin, neogigogenin, protodioscin, and trigogenin; in addition to flavonoids, alkaloids,

FIG. 3

Photographic images of *T. foenum-graecum* (A) seeds, (B) seed paste; and (C) application of seed paste on hair and scalp.

polysaccharides, coumarins, proteins, and vitamins [21–24]. The seeds are soaked overnight, ground into a paste, and applied on the hair and scalp (Fig. 3B and C). It functions as a conditioner by smoothening the hair, reducing the hair fall, and promoting the hair growth by blood circulation stimulating the hair follicles. It is also utilized for controlling dandruff and dry itchy scalp [23–27]. The seed extract is used for making hair care products such as shampoo, conditioner, soap, hair tonic, hair gel, hair oil, etc. [26,28,29]. The antifungal action of methanolic seed extract was studied toward various fungi, including *Botrytis cinerea*, *Fusarium graminearum*, *Alternaria* sp., *Pythium aphanidermatum,* and *Rhizoctonia solani*. The mycelial growth inhibition ranged from 58.5% to 87.21%. The MIC value toward *Alternaria* sp. and *Rhophitulus solani* was 60 μg/mL [30]. Thus, the study suggests the development of antifungal drugs against dandruff-causing dermatophytic fungi.

The leaves and seeds are utilized as vegetables and spices in Indian cuisine, respectively. The seeds are employed as laxative, carminative, expectorant, stomachic, etc., in Indian traditional medicine. The seeds are used to reduce blood glucose, blood pressure, cholesterol; treat indigestion, wounds, bruises, skin, scars, acne, blackheads; and promote appetite and serve as a functional food. The seeds exhibit restorative, nutritive, hypocholesterolemic, antidiabetic, hypotriglyceridemic, antiulcerogenic, antipyretic, lactation stimulating, antioxidant, antimelanogenic, antineoplastic, antitumor, antigenotoxic**,** antiinflammatory, immunomodulatory, antibacterial, and antifungal activities [5,21,22,31–34].

5 *Vigna radiata*

The plant *V. radiata* is called as green gram and belongs to the family Fabaceae. It is called as *pesara, mung*, *moong,* etc., in various Indian languages. This annual food crop legume is cultivated in East Asia, Southeast Asia, and the Indian subcontinent. The seeds are light green in color, globose/ellipsoid/cuboid in shape, and sized

around 2.5–4 × 3 mm (Fig. 4A) [35,36]. The dried hulled or dehulled seeds are used as food ingredients in many culinary preparations/dishes of Indian cuisine. The seeds are abundant in carbohydrate (50%–60%), protein (20%–24%), and other biomolecules including flavonoids, phenolics, alkaloids, terpenoids, saponins, organic acids, and lipids [37–39].

The saponin-rich seed flour gives foam after mixing with water [35]. The flour made from dried seeds known as *nalugu pindi* in Telugu language is used as soap, body scrub, bathing powder, face pack, and hair cleanser after making it as a paste (Fig. 4B). Mainly, it is used for the delicate skin and hair of newborn babies and infants by applying oil, followed by homemade bathing powder, *nalugu pindi* as soap, scrub, and hair cleanser. It is a common practice in Telugu states (Andhra

FIG. 4

Photographic images of *V. radiata* (A) seeds and (B) seed powder.

Pradesh, Telangana) of India and the elders and grandmothers make this nonallergic, skin- and hair-friendly *nalugu pindi* by grinding dried rice, green gram seeds, sweet orange peels, and rose petals. It serves as an exfoliating and hair-removing agent for babies and gives the skin and hair a smooth and silky appearance without depleting the natural oils [40]. The seeds exhibit antioxidant, antibacterial, antifungal, antimelanogenic, antiinflammatory, immunomodulatory, antidiabetic, hypolipidemic, hepatoprotective, antihypertensive, antitumor, antisepsis effects [36,39]. The seeds constituents are known inhibit tyrosinase, a key enzyme in the melanogenesis, thus confirming its long-standing application in skin tone improvement [36]. The seed extract has inhibitory action against *Salmonella typhi, P. aeruginosa,* and *Citrobacter freundi* in terms of inhibition zones (2.8–3 mm), and the MIC value was 2 mg/mL [41]. Also, the antibacterial, antifungal, and antioxidant properties of the seed biomolecules imply its multipurpose dermatological applications [36,41,42].

6 *Cicer arietinum*

The plant *C. arietinum* is known as Bengal gram and chickpea and belongs to the family Fabaceae. It is called as *sanaga*, *chana, chole,* etc., in various Indian languages. The food crop legume is largely cultivated in Indian subcontinent and exists as larger, light-colored, smooth seed coated, Mediterranean and Middle East-based *kabuli*; and smaller, dark-colored, rough seed coated Indian *desi* varieties [43,44]. The seeds are light green in color, globose/ellipsoid/cuboid in shape, and sized around 2.5–4 × 3 mm (Fig. 5A). The seeds are abundant in carbohydrates, proteins, fats, dietary fiber, minerals, vitamins, and other secondary metabolites and saponins (sapogenol A and sapogenol B) (0.2%–0.3%) [44,45]. The seeds show antiinflammatory, anticancer, etc., activities. The decorticated split seed flour known as *sanaga pindi/besan* and its paste are used as face mask, body cleanser, soap, hair cleanser, and also for cleaning utensils, silver, and gold jewelry (Fig. 5B) [44].

7 *Albizia lebbeck*

The plant *A. lebbeck* is a tall, deciduous, timber-yielding forage tree belonging to the family Fabaceae. The native, widely cultivated, and naturalized tree is commonly seen in the tropical and subtropical regions of India, Pakistan, Bangladesh, Sri Lanka, Myanmar, China, Thailand, Malaysia, Indonesia, New Guinea, and Australia. The tree is commonly known as *siris, sirisha, dirisena, vaagai,* etc., in various Indian languages [5,46,47]. The tree's bark is rough, flakey, grayish brown with an inner reddish tinge and rich in saponins known as albiziasaponins and tannins (Fig. 6A) [48–51]. The bark and its extract are used as detergent and soap in India states of Tamil Nadu, Karnataka, Kerala, Goa, and Lakshadweep (Fig. 6B) [5].

In traditional Indian medicine, the bark is employed for treating erysipelas, skin diseases, wounds, allergy, inflammation, arthritis, asthma, bronchitis, infectious

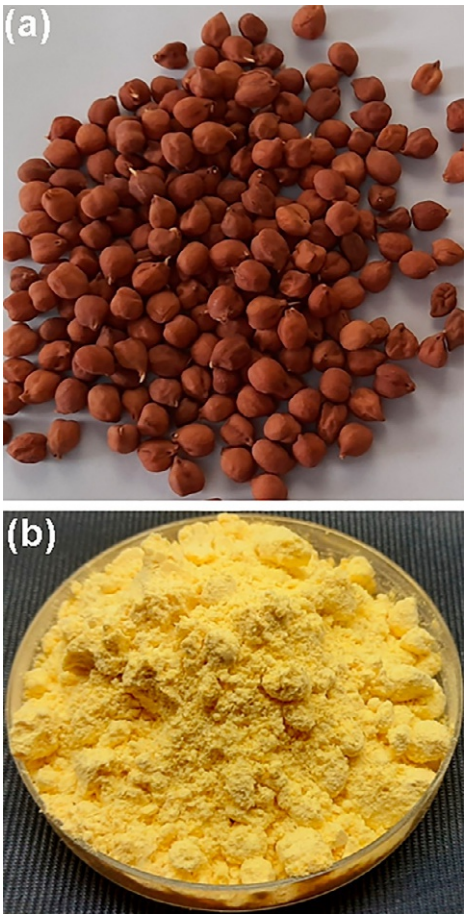

FIG. 5

Photographic images of *C. arietinum* (A) seeds and (B) decorticated split seed powder.

diarrhea, abdominal tumors, and anxiety [47,51]. The ethyl acetate bark extract showed antidermatophytic activity against *Trichophyton rubrum*, a common etiologic agent of human dermatophytosis. The inhibition zone and MIC values were 18 mm and 100 µg/mL, respectively [52]. The ethyl acetate extract of bark showed inhibitory action against various bacteria including *S. aureus*, *B. subtilis*, *S. typhi*, *Vibrio mimicus,* and *Shigella flexneri* in the range of 11–14 mm at 300 µg/disc. The antifungal activity ranged from 7 to 10 mm against *Saccharomyces cerevisiae, C. albicans*, *A. niger,* and *C. arrizae*. At 100 µg/mL, the extract showed scavenging action of 81.3% and 67.4% toward DPPH and hydrogen peroxide, respectively [46]. The administration of methanolic bark extract arrested spermatogenesis in male rats [48]. The ethanolic bark extract at 400 mg/kg of body weight exhibited 68.5%

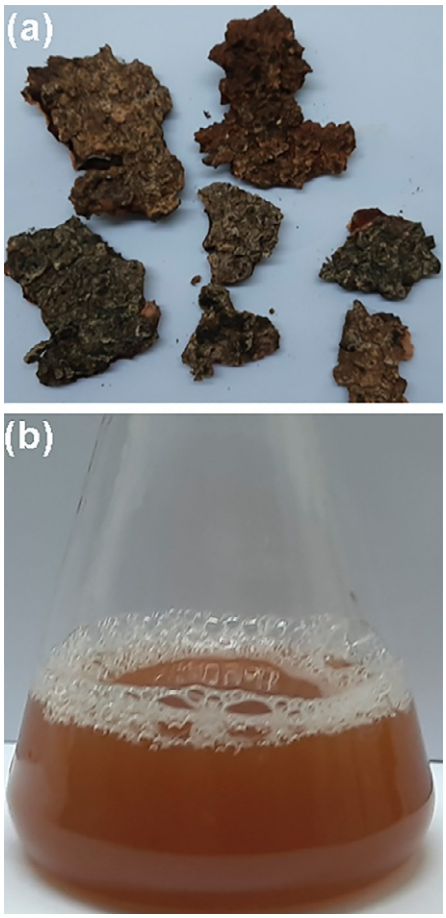

FIG. 6

Photographic images of *A. lebbeck* (A) bark and (B) aqueous bark extract.

inhibition of inflammation induced by Freund's adjuvant in the rat model [51]. The saponin-rich methanolic bark extract demonstrated anticancer activity toward hepato, colon, larynx, cervical, and breast carcinoma cell lines [47].

8 *Citrus sp.*

The various species of the genus *Citrus,* such as *Citrus limon* (lemon), *Citrus limetta* (sweet lime), and *C. reticulata* (sweet orange), belong to the family Rutaceae. They are small to medium-sized, evergreen, spiny trees, and local to tropical and subtropical regions of Asia such as India, Myanmar**,** China, Philippines, Iran, etc. The tree's

fruit is classified as hesperidium with a leathery pericarp/rind, fibrous mesocarp, and juicy endocarp. They are oval, green, yellow, and orange in color based on the species (Fig. 7A–C) [53–57]. The pericarps/peels of citrus fruits are an abundant source of flavonoids, phenolics, alkaloids, steroids, essential oils, saponins, carotenoids, organic acids, dietary fibers, and pectin [57,58]. The peel extracts exhibited antihyperglycemic [56], antiinflammatory [59], antifungal [57], antibacterial, antioxidant, and anticholinesterase activities [60]. In addition to its application in treating indigestion, obesity, urinary disorders, gum diseases, [57], the citrus peels are employed in the making of skin and hair care product *nalugu pindi*, a natural soap, scrub, and hair cleanser (Fig. 7D–F) [40].

9 *Hibiscus rosa sinensis*

It is scientifically known as *Hibiscus rosa sinensis* and belongs to the family Malvaceae. It is called as *mandara*, *rudra pushpa, gudhal, dasani,* etc., in various Indian languages. It is an ornamental, tropical, evergreen, small tree or bushy shrub. The leaves are simple, ovate/ovate lanceolate, acute, serrated, reticulately venated and sized around 3.5–12 × 2–5.5 cm (Fig. 8A). The leaves contain various compounds such as phlobatannins, glycosides, saponins, tannins, phenolics, flavonoids, and terpenoids [61–63].

The leaves are used for treating hair fall, premature gray hair, alopecia, baldness, dandruff, dry hair, etc. [64–66]. The fresh or dried leaves are soaked either in cold or

FIG. 7

Photographic images of fruits of (A) *C. limon* (lemon), (B) *C. limetta* (sweet lime), (C) *C. reticulata* (*sweet orange*); and fruit peels of (D) lemon, (E) sweet lime, and (F) sweet orange.

FIG. 8

Photographic images of *Hibiscus rosa sinensis* (A) leaves and (B) aqueous leaf extract.

warm water, and the obtained extract is used as a hair cleanser and conditioner in India due to its frothing and lusturing qualities attributed to saponin content (Fig. 8B). It is mainly used with either soapnut or *shikakai* extracts [67,68]. The leaf extract is well known for its hair growth promoting activity in terms of increased hair length, hair density, hair follicle population, and hair follicle length [61,65,66,69]. The leaves and flowers are used in commercial herbal hair care products such as conditioner, hair oil, hair dye, etc. [65]. The methanolic leaf extract showed inhibitory activity against bacterial skin pathogens including *K. pneumoniae, E. coli, P. aeruginosa, Enterobacter aerogenes*, *S. aureus,* and *Streptococcus pyogenes,* and the inhibition zones ranged from 12 to 13 mm [62]. In another study, the zones ranged from 15.2 to 18.8 mm toward *E.coli, S. aureus, and B. subtilis* [63]. The antifungal activity of leaf extract was evident against dermatophytic yeasts (*C. albicans, Candida parapsilosis, C. glabrata*) and other fungi such as *A. niger* and *A. flavus* [61,62]. The leaf extracts also demonstrated antioxidant activity in terms of DPPH, hydrogen peroxide, superoxide, and nitric oxide radical scavenging [61,70]. The leaf extract showed an array of biological properties, including antibacterial [62–64], antifungal [62], anticancer, immunomodulatory, antidiabetic, hypocholesterolemic, gastroprotective, hepatoprotective, neuroprotective, cardioprotective, wound healing, antiinflammatory, antipyretic, and antifertility activities [61].

10 *Phyllanthus emblica*

P. emblica is an average-sized, deciduous tree belonging to the family Phyllanathaceae. It is native, tropical tree of India, Sri Lanka, Pakistan, Bangladesh, China, Uzbekistan, and Malaysia. The Indian gooseberry is commonly known as *amla,*

amlaki, amlaka, rathi usiri, etc., in various Indian languages [71]. The fruits are categorized as drupe, globose, wider, light greenish yellow, vertically striped, and sized around 15–20 × 18–25 mm (Fig. 9A) [72]. The fruits are abundant in tannins, flavonoids, phenolics, saponins, terpenoids, alkaloids, ascorbic acid, carbohydrates, and minerals. In Indian traditional medicine, the fresh and dried fruits are extensively used for treating various ailments, including anemia, fever, dyspepsia, constipation, hemorrhage, diarrhea, diabetes, dysentery, and rheumatism [71–74]. It is the crucial constituent of *ayurvedic* formulations *triphala* and *chyawanprash* [71,74].

As it is abundant in vitamin C, it moisturizes and detoxifies the skin; enhances skin complexion, boosts immunity, and protects from skin infections. The repeated use of dried fruits, powders, and extracts of the fruit makes the hair shiny, retards the

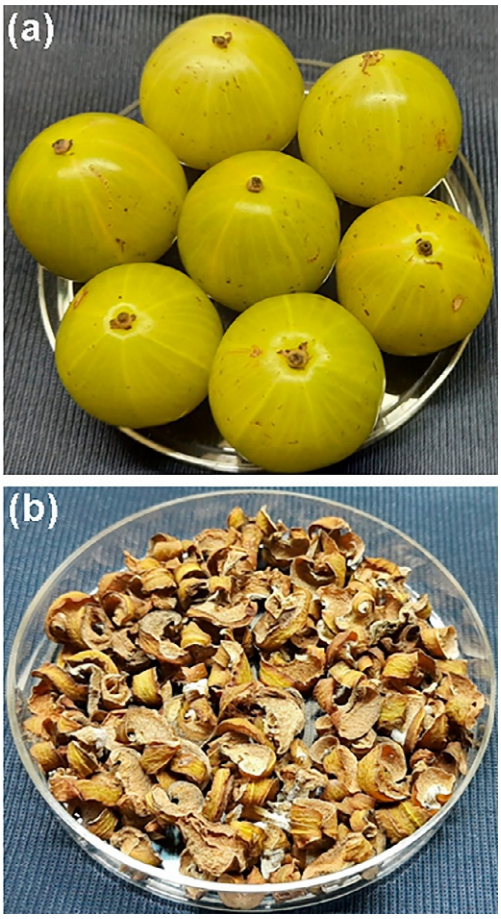

FIG. 9

Photographic images of *P. emblica* (A) fruits and (B) dried fruit slices.

premature graying, maintains the dark color, and darkens the natural color (Fig. 9B). It strengthens the hair follicles via improved calcium absorption, promotes hair growth, and maintains a fuller looking. Thus, it an important ingredient in herbal shampoos, hair oils, hair dyes, and skincare products [71,74,75].

The methanolic fruit extract exhibited antibacterial action toward *E. coli, K. pneumoniae, Proteus vulgaris, Micrococcus luteus, B. subtilis, Enterococcus faecalis, Streptococcus faecalis* at 100 mg/mL, and the respective inhibition zones were 32, 36, 24, 27, 31, 36, and 26 mm [73]. The ethanolic fruit extract exhibited antifungal action toward Fusarium *equiseti* and *C. albicans,* and the corresponding inhibition zone values were at 9.5 and 11 mm at 0.5 mg/disc [72]. The fruit extracts show a wide range of properties, including antioxidant, antidiabetic, antiinflammatory, antiulcerogenic antidepressant, antimutagenic, anticancer, anti-HIV, hypolipidemic, hepatoprotective, radioprotective, wound healing, immunomodulatory, insecticidal, larvicidal, and mosquitocidal activities [71,74].

11 *Aloe vera*

A. vera is an evergreen, perennial, succulent, xerophytic, ornamental plant belonging to the family Asphodelaceae. It is extensively distributed in tropical, semitropical, and arid climates and local to southern and eastern Africa**.** The leaves are thick, fleshy, green to gray green, serrated, sized around 30–50 × 10 cm, and arranged as a rosette (Fig. 10A). The leaf is abundant in many secondary metabolites, including tannins, flavonoids, lignins, saponins, anthraquinones, glycoproteins, sterols, vitamins, amino acids, enzymes, and saccharides [76,77]. The leaf extracts demonstrate wound healing, antiaging, antinflammatory, antiulcer, antitumor, antihypercholesterolemic, antidiabetic, antiasthmatic, antioxidant, antibacterial, antifungal, antiviral, antihelminthic, antiseptic, cleansing, moisturizing, emollient, and laxative activities. Thus, the leaf extracts/gels are widely used for treating minor cuts,

FIG. 10

Photographic images of plants (A) *A. vera* and (B) *Agave* sp.

minor burns, sunburns, wounds, dry skin, skin rashes, acne, etc., and an important ingredient in various topical skincare lotions, ointments; bathing soaps, shampoos, hand sanitizers, and hair oils [76–78].

12 *Eclipta alba*

It is an erect or prostrate annual herb, belonging to the family Asteraceae and grows in tropical and warm temperate areas of India, Nepal, Thailand, and China. It is commonly known as *bringaraj, gunta galagara, karisalankanni,* etc., in various Indian languages. The leaves are used in Indian medicine of Ayurveda, Unani, and Siddha. The leaves are green, oblong, lanceolate, and sized around 2–10 × 0.8–2 cm. The leaves are abundant in alkaloids, terpenoids, sterols, steroids, flavonoids, saponins, volatile oils, phenolic acids, etc. The leaf paste/extract/oil is used for treating various skin (pimples, eczema, wounds, cuts, burns, ringworm, allergy) and hair (dandruff, alopecia, baldness, hair fall, lice) problems. The leaves are extensively used for preventing premature graying, promoting growth, rejuvenating, and cleaning of hair. The leaves are the key ingredient in commercial, herbal hair, and skin formulations such as hair oil, hair tonic, conditioner, and shampoo. The leaves exhibit various properties, including hair growth promoting, hepatoprotective, antidiabetic, analgesic, antiinflammatory, antioxidant, antibacterial, antifungal, antihelminthic, antimalarial, immunomodulatory, anticancer, and antiulcer activities [79,80].

The saponin rich parts of other plants such as leaves of *Agave* sp. (Fig. 10B) and root tubers of *D. esculenta* are also utilized either directly or as an ingredient in shampoo, hair conditioner, hair oil, hand sanitizers, detergents, etc. [5].

13 Conclusions

As India is rich in biodiversity, various parts of saponiferous plants are widely used either directly or in the preparation of bathing soap, washing detergent, utensil cleaner, hair cleanser, hair, shampoo, hair oil, and hair dye. These natural saponins are bestowed with important features such as biocompatibility, biodegradability, lesser environmental and human toxicity, and lower human allergenicity; compared with the synthetic surfactant counterparts. Utilization of various plant parts including leaves, bark, stem tubers, fruits, fruit pericarp, and seeds for health and hygiene is known for many centuries especially in Indian civilization [7] and other Asian countries [81]. However, further in-depth studies are needed on bulk raw material production, methodology development for isolation and purification of saponins, development of skin and hair care formulations with mixed saponins, toxicological evaluation of the developed products, and economic viability.

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References

[1] A.J. Kora, Antibacterial and antioxidant activities of aqueous extract of soapnuts (*Sapindus mukorossi*), Curr. Trends Biotechnol. Pharm. 14 (4) (2020) 388–395.
[2] N. Ranasalva, R. Sunil, G. Poovarasan, Importance of biosurfactant in food industry, IOSR J. Agric. Vet. Sci. 7 (5) (2014) 06–09.
[3] S. Vijayakumar, V. Saravanan, Biosurfactants-types, sources and applications, Res. J. Microbiol. 10 (5) (2015) 181–192.
[4] Z. Liu, Z. Li, H. Zhong, G. Zeng, Y. Liang, M. Chen, et al., Recent advances in the environmental applications of biosurfactant saponins: a review, J. Environ. Chem. Eng. 5 (2017) 6030–6038.
[5] K. Bhaskar, Ethbotany and conservation status of saponin rich plants of Ganegetic plain having both medicinal and cleansing properties, Plant Arch. 18 (1) (2018) 81–97.
[6] M. Du, S. Huang, J. Zhang, J. Wang, L. Hu, J. Jiang, Isolation of total saponins from *Sapindus mukorossi* Gaerth, Open J. For. 4 (1) (2014) 24–27.
[7] R.S. Bisht, C. Dorje, A. Banerji, Paleobotanical and pollen analytical investigations, in: Indian Archaeology 1993–1994—A Review, Archaeological Survey of India, Government of India, New Delhi, 2000, pp. 143–145.
[8] B.N. Suhagia, I.S. Rathod, S. Sindhu, *Sapindus mukorossi (Areetha)*: an overview, Int. J. Pharm. Sci. Res. 2 (8) (2011) 1905–1913.
[9] R. Ghagi, S.K. Satpute, B.A. Chopade, A.G. Banpurkar, Study of functional properties of *Sapindus mukorossi* as a potential bio-surfactant, Indian J. Sci. Technol. 4 (5) (2011) 530–533.
[10] A. Upadhyay, D.K. Singh, Pharmacological effects of *Sapindus mukorossi*, Rev. Inst. Med. Trop. Sao Paulo 54 (5) (2012) 273–280.
[11] M. Ibrahim, A.A. Khan, S.K. Tiwari, M.A. Habeeb, M.N. Khaja, C.M. Habibullah, Antimicrobial activity of *Sapindus mukorossi* and *Rheum emodi* extracts against *H. pylori*: *in vitro* and *in vivo* studies, World J. Gastroenterol. 12 (44) (2006) 7136–7142.
[12] Y. Tamura, K. Mizutani, T. Ikeda, K. Ohtani, R. Kasai, K. Yamasaki, et al., Antimicrobial activities of saponins of pericarps of *Sapindus mukurossi* on dermatophytes, Nat. Med. 55 (1) (2001) 11–16.
[13] M.D. Köse, O. Bayraktar, Extraction of saponins from soapnut (*Sapindus mukorossi*) and their antimicrobial properties, World J. Res. Rev. 2 (5) (2016) 89–93.
[14] S.S. Todkar, V.V. Chavan, A.S. Kulkarni, Screening of secondary metabolites and antibacterial activity of *Acacia concinna*, Res. J. Microbiol. 5 (2010) 974–979.
[15] W. Poomanee, W. Chaiyana, N. Intasai, P. Leelapornpisid, Biological activities and characterization of the pod extract from sompoi (*Acacia concinna* Linn) grown in Northern Thailand, Int J Pharm Pharm Sci 7 (5) (2015) 237–241.
[16] R. Antil, Priyanka, P. Dahiya, N. Singh, Phytochemical and antibacterial potential of acetone leaf extract of *Acacia concinna* (Willd.) DC, Int. J. Pharmacogn. 3 (3) (2016) 161–166.

[17] M. Akram, A. Hamid, A. Khalil, A. Ghaffar, N. Tayyaba, A. Saeed, et al., Review on medicinal uses, pharmacological, phytochemistry and immunomodulatory activity of plants, Int. J. Immunopathol. Pharmacol. 27 (3) (2014) 313–319.

[18] Plants *A. Acacia concinna* (Willd.) DC. (saptala) -Medicinal Uses, Morphology, Images, Side Effects, Pharmacology, 2019. [updated 06 February 2019; cited 2021 14th Jul 2021]. Available from: https://www.indianmedicinalplants.info/herbs/index.php/a/1312-acacia-concinna-willd-dc-saptala-medicinal-uses-morphology-images-side-effects-pharmacology.

[19] R. Kukhetpitakwong, C. Hahnvajanawong, P. Homchampa, V. Leelavatcharamas, J. Satra, W. Khunkitti, Immunological adjuvant activities of saponin extracts from the pods of *Acacia concinna*, Int. Immunopharmacol. 6 (11) (2006) 1729–1735.

[20] V. Natarajan, S. Natarajan, Antidermatophytic activity of *Acacia concinna*, Glob. J. Pharmacol. 3 (1) (2009) 06–07.

[21] U.C.S. Yadav, N.Z. Baquer, Pharmacological effects of *Trigonella foenum-graecum* L. in health and disease, Pharm. Biol. 52 (2) (2014) 243–254.

[22] P.R. Petit, Y.D. Sauvaire, D.M. Hillaire-Buys, O.M. Leconte, Y.G. Baissac, G.R. Ponsin, et al., Steroid saponins from fenugreek seeds: extraction, purification, and pharmacological investigation on feeding behavior and plasma cholesterol, Steroids 60 (10) (1995) 674–680.

[23] K. Medizin, C. Schoen, S. Bielfeldt, J. Reimann, Fenugreek+micronutrients: efficacy of a food supplement against hair loss, Cosmetic Med. 27 (4) (2006) 176–179.

[24] K.C. Nagulapalli Venkata, A. Swaroop, D. Bagchi, A. Bishayee, A small plant with big benefits: fenugreek (*Trigonella foenum-graecum* Linn.) for disease prevention and health promotion, Mol. Nutr. Food Res. 61 (6) (2017).

[25] Renuka, Fenugreek Benefits for Hair: Everything You Need to Know Hyderabad, Sresta Natural Bioproducts Pvt Ltd, 2021. [cited 2021 14 July 2021]. Available from: https://www.24mantra.com/blogs/health-and-nutrition/fenugreek-benefits-for-hair-everything-you-need-to-know/.

[26] G.D. Noudeh, F. Sharififar, P. Khazaeli, E. Mohajeri, J. Jahanbakhsh, Formulation of herbal conditioner shampoo by using extract of fenugreek seeds and evaluation of its physicochemical parameters, Afr. J. Pharm. Pharmacol 5 (22) (2011) 2420–2427.

[27] Healthline, Are Fenugreek Seeds Good for Your Hair?, Healthline Media, 2021. [cited 2021 14th Jul 2021]. Available from: https://www.healthline.com/nutrition/fenugreek-for-hair.

[28] K. Chitra, R. Kandasamy, K.G. Lalitha, M. Kumar, Regupathi., Formulation and evaluation of herbal hair gel for hair growth potential, J. Pharmacol. Clin. Res. 2 (2) (2017) 555581.

[29] L. Purwal, S.P.B.N. Gupta, S.M. Pande, Development and evaluation of herbal formulations for hair growth, E-J. Chem. 5 (1) (2008) 34–38.

[30] R. Haouala, S. Hawala, A. El-Ayeb, R. Khanfir, N. Boughanmi, Aqueous and organic extracts of *Trigonella foenum-graecum* L. inhibit the mycelia growth of fungi, J. Environ. Sci. 20 (12) (2008) 1453–1457.

[31] S. Goyal, N. Gupta, S. Chatterjee, Investigating therapeutic potential of *Trigonella foenum-graecum* L. as our defense mechanism against several human diseases, J. Toxicol. 2016 (2016), 1250387.

[32] T. Kawabata, M.Y. Cui, T. Hasegawa, F. Takano, T. Ohta, Anti-inflammatory and anti-melanogenic steroidal saponin glycosides from fenugreek (*Trigonella foenum-graecum* L.) seeds, Planta Med. 77 (7) (2011) 705–710.

[33] T.K. Basu, A. Srichamroen, Health benefits of fenugreek (*Trigonella foenum-graecum* leguminosse), in: R.R. Watson, V.R. Preedy (Eds.), Bioactive Foods in Promoting Health, Academic Press, San Diego, 2010, pp. 425–435 (Chapter 28).
[34] R. Randhir, Y.T. Lin, K. Shetty, Phenolics, their antioxidant and antimicrobial activity in dark germinated fenugreek sprouts in response to peptide and phytochemical elicitors, Asia Pac. J. Clin. Nutr. 13 (3) (2004) 295–307.
[35] *Vigna radiata* - (L.) R.Wilczek, 2021. [cited 2021 15th Jul 2021]. Available from: https://pfaf.org/user/Plant.aspx?LatinName=Vigna+radiata.
[36] D. Hou, L. Yousaf, Y. Xue, J. Hu, J. Wu, X. Hu, et al., Mung bean (*Vigna radiata* L.): bioactive polyphenols, polysaccharides, peptides, and health benefits, Nutrients 11 (6) (2019).
[37] A. Kataria, B.M. Chauhan, D. Punia, Antinutrients in amphidiploids (black gram x Mung bean): varietal differences and effect of domestic processing and cooking, Plant Foods Hum. Nutr. 39 (3) (1989) 257–266.
[38] M.R. Lee, J.S. Lee, J.C. Wang, G.R. Waller, Structural determination of saponins from mungbean sprouts by tandem mass spectrometry, in: G.R. Waller, K. Yamasaki (Eds.), Saponins Used in Food and Agriculture, Springer US, Boston, MA, 1996, pp. 331–337.
[39] D. Tang, Y. Dong, H. Ren, L. Li, C. He, A review of phytochemistry, metabolite changes, and medicinal uses of the common food mung bean and its sprouts (*Vigna radiata*), Chem. Cent. J. 8 (1) (2014) 4.
[40] Anupama, Sunni-Pindi: The Traditional Bathing Powder for All, 2019, [updated 27th Jun 2019; cited 2021 15th Jul 2021]. Available from: https://www.momspresso.com/parenting/the-imperfect-mother/article/sunnipindi-the-magical-herbal-bathing-powder.
[41] A.J. Kayode, A.S. Awe, F. Banji-Onisile, A.A. Agboola, O.V. Oyetayo, The phytochemical content and antimicrobial activity of mung bean (*Vigna radiata* l.) extract against some selected pathogens of medical importance, World J. Adv. Healthcare Res. 4 (3) (2020) 147–155.
[42] A. Jaya Prakash Priya, G. Yamini SudhaLakshmi, B. Fouzia, S. Gopalakrishnan, P. Dhanalakshmi, E. Sagadevan, et al., Phytochemical screening and antibacterial activity of *Vigna radiata* L. against bacterial pathogens involved in food spoilage and food borne diseases, J. Acad. Ind. Res. 1 (6) (2012) 355–359.
[43] A.J. Kora, Applications of sand roasting and baking in the preparation of traditional Indian snacks: nutritional and antioxidant status, Bull. Natl. Res. Cent. 43 (1) (2019) 158.
[44] Ministry of Food Processing Industries, in: Ministry of Food Processing Industries (Ed.), Reading Manual for Besan Flour Under PMFME Scheme, National Institute of Food Technology Entrepreneurship and Management, Sonipat, India, 2021, pp. 1–31.
[45] R.P. Srivastava, H. Vasishtha, Saponins and lectins of Indian chickpeas (*Cicer arietinum*) and lentils (*Lens culinaris*), Indian J. Agric. Biochem. 25 (1) (2012) 44–47.
[46] M.T. Ali, S.T. Haque, M.L. Kabir, S. Rana, M.E. Haque, A comparative study of *in vitro* antimicrobial, antioxidant and cytotoxic activity of *Albizia lebbeck* and *Acacia nilotica* stem bark, Bull. Fac. Pharm. Cairo Univ. 56 (1) (2018) 34–38.
[47] T.H. Desai, S.V. Joshi, Anticancer activity of saponin isolated from *Albizia lebbeck* using various *in vitro* models, J. Ethnopharmacol. 231 (2019) 494–502.
[48] R.S. Gupta, R. Chaudhary, R.K. Yadav, S.K. Verma, M.P. Dobhal, Effect of saponins of *Albizia lebbeck* (L.) Benth bark on the reproductive system of male albino rats, J. Ethnopharmacol. 96 (1) (2005) 31–36.
[49] B.C. Pal, B. Achari, K. Yoshikawa, S. Arihara, Saponins from *Albizia lebbeck*, Phytochemistry 38 (5) (1995) 1287–1291.

[50] E.O.R. Viana, M.F.S.J. Cruz, M.J. da Silva, G.M. Pereira, B.P. da Silva, L.W. Tinoco, et al., Structural characterization of a complex triterpenoid saponin from *Albizia lebbeck* and investigation of its permeability property and supramolecular interactions with membrane constituents, Carbohydr. Res. 471 (2019) 105–114.
[51] N.P. Babu, P. Pandikumar, S. Ignacimuthu, Anti-inflammatory activity of *Albizia lebbeck* Benth., an ethnomedicinal plant, in acute and chronic animal models of inflammation, J. Ethnopharmacol. 125 (2) (2009) 356–360.
[52] T.S. Devi, I.J. Padmaja, Y.N. Sastry, A. Nagamani, A study on anti dermatophytic potential of selected ethno medicinal plants against *Trichophyton rubrum*, a common etiologic agent in and around Visakhapatnam region (India), Asian Pac. J. Trop. Biomed. 2 (3, Supplement) (2012) S1874–S1878.
[53] A.M.G.N. Mamede, C.C.S. Coelho, O. Freitas-Silva, H.T.G. Barboza, A.G. Soares, Lemon, in: A.K. Jaiswal (Ed.), Nutritional Composition and Antioxidant Properties of Fruits and Vegetables, Academic Press, 2020, pp. 377–392 (Chapter 23).
[54] S. Karmakar, S. De, Pectin removal and clarification of juices, in: C.M. Galanakis (Ed.), Separation of Functional Molecules in Food by Membrane Technology, Academic Press, 2019, pp. 155–194 (Chapter 5).
[55] S. Mandal, M. Mandal, Tangerine (*Citrus reticulata* L. var.) oils, in: V.R. Preedy (Ed.), Essential Oils in Food Preservation, Flavor and Safety, Academic Press, San Diego, 2016, pp. 803–811 (Chapter 91).
[56] S. KunduSen, P.K. Haldar, M. Gupta, U.K. Mazumder, P. Saha, A. Bala, et al., Evaluation of antihyperglycemic activity of *Citrus limetta f*ruit peel in streptozotocin-induced diabetic rats, ISRN Endocrinol. 2011 (2011), 869273.
[57] J.M. Ehiobu, M.E. Idamokoro, A.J. Afolayan, Phytochemical content and antioxidant potential of leaf extracts of *Citrus limon* (L.) Osbeck collected in the eastern Cape Province, South Africa, S. Afr. J. Bot. 141 (2021) 480–486.
[58] C.A. Ezeabara, C.U. Okeke, B.O. Aziagba, C.V. Ilodibia, A.N. Emeka, Determination of saponin content of various parts of sx *Citrus* species, Int. Res. J. Pure Appl. Chem. 4 (1) (2014) 137–143.
[59] Y.-S. Huang, S.-C. Ho, Polymethoxy flavones are responsible for the anti-inflammatory activity of citrus fruit peel, Food Chem. 119 (3) (2010) 868–873.
[60] M.R. Loizzo, R. Tundis, M. Bonesi, F. Menichini, D. De Luca, C. Colica, et al., Evaluation of *Citrus aurantifolia* peel and leaves extracts for their chemical composition, antioxidant and anti-cholinesterase activities, J. Sci. Food Agric. 92 (15) (2012) 2960–2967.
[61] A. Missoum, An update review on *Hibiscus rosa sinensis* phytochemistry and medicinal uses, J. Ayurvedic Herb. Med. 4 (3) (2018) 135–146.
[62] K. Hemarana, K.V. Jeyashree, M. Babu, M. Kannan, Preliminary bioactive compounds screening and antibacterial activity of methanolic extract of *Hibiscus rosasinensis* against selected skin pathogens, Res. J. Pharm., Biol. Chem. Sci. 5 (2014) 1210–1218.
[63] I.J. Udo, M. Ben, C. Etuk, A.I. Tiomthy, Phytochemical, proximate and antibacterial properties of *Hibiscus rosa-sinensis* L. leaf, J. Med. Plants Stud. 4 (2016) 193–195.
[64] C. Amita Paul, B. Karpagam, N. Krishnaveni, Phytochemical screening and antibacterial evaluation of the aqueous extracts of *Hibiscus rosa-sinensis'* leaves against *Aeromonas hydrophila*, Int. J. Adv. Sci. Res. 1 (8) (2016) 21–23.
[65] S.S. Pathak, M.Y.N. Gala, Evaluation of hair growth potentiation activity of Hibiscus rosa sinensis in disturbed circadian rhythm, Int. J. Curr. Res. 10 (03) (2018) 67151–67155.

[66] N. Singh, M. Tailang, A.K. Pathak, Evaluation of hair growth activity of Hibiscus rosa-sinensis and *Calotropis gigantea* leaves on stress induced alopecia, World J. Pharm. Res. 6 (17) (2017) 477–484.
[67] Hibiscus- Reduce Hair Fall and Promote a Healthy Scalp, 2019. [cited 2021 15th Jul 2021]. Available from: https://www.nowiamnappy.com/bloghair-tips/2019/3/19/hibiscus-reduce-hair-fall-and-promote-a-healthy-scalp.
[68] S. Raj, This Tea Can Stop Hair Fall and Strengthen your Hair, The Times of India, 2020 (28th Nov 2020;Sect. Beauty Tips).
[69] I.B. Putra, N.K. Jusuf, I.B. Sumantri, The potency of *Hibiscus rosa-sinensis* Linn. leaves ethanol extract as hair growth, Open Access Maced. J. Med. Sci. 8 (A) (2020) 89–92.
[70] F.R.A. Ghaffar, I.A. El-Elaimy, *In vitro*, antioxidant and scavenging activities of *Hibiscus rosa sinensis* crude extract, J. Appl. Pharm. Sci. 02 (01) (2012) 51–58.
[71] M.R. Hasan, M.N. Islam, M.R. Islam, Phytochemistry, pharmacological activities and traditional uses of *Emblica officinalis*: a review, Int. Curr. Pharm. J. 5 (2) (2016) 14–21.
[72] M. Hossain, K. Mazumder, M. Hossen, T.T. Tanmy, M.J. Rashid, *In vitro* antibacterial and antifungal activities of *Emblica officinalis*, Int. J. Pharm. Sci. Res. 3 (4) (2012) 1124–1127.
[73] K.S. Jyothi, B.S. Rao, Screening of antibacterial activity of *Emblica officinalis* L. fruits, Pharmacologyonline 3 (2011) 848–852.
[74] E. Singh, S. Sharma, A. Pareek, J. Dwivedi, S. Yadav, Phytochemistry, traditional uses and cancer chemopreventive activity of Amla (*Phyllanthus emblica*): the sustainer, J. Appl. Pharm. Sci. 2 (2012) 176–183.
[75] S. Patel, R.K. Goyal, Emblica officinalis Geart.: a comprehensive review on phytochemistry, pharmacology and ethnomedicinal uses, Res. J. Med. Plant 6 (2012) 6–16.
[76] P.K. Sahu, D.D. Giri, R. Singh, P. Pandey, S. Gupta, A.K. Shrivastava, et al., Therapeutic and medicinal uses of *Aloe vera*: a review, Pharmacol. Pharm. 4 (2013) 1–13.
[77] Z. López, G. Núñez-Jinez, G. Avalos-Navarro, G. Rivera, J. Salazar-Flores, J.A. Ramírez, et al., Antioxidant and cytotoxicological effects of *Aloe vera* food supplements, J. Food Qual. 2017 (2017) 7636237.
[78] Narsih, S. Kumalaningsih, Wignyanto, S. Wijana, Identification of aloin and saponin and chemical composition of volatile constituents from *Aloe vera* (L.) Peel, J. Agric. Food Technol. 2 (5) (2012) 79–84.
[79] R. Sollepura Boregowda, N. Murali, A.C. Udayashankar, S.R. Niranjana, O.S. Lund, H. S. Prakash, Antifungal activity of *Eclipta alba* metabolites against *Sorghum* pathogens, Plan. Theory 8 (3) (2019) 72.
[80] R. Jahan, A. Al-Nahain, S. Majumder, M. Rahmatullah, Ethnopharmacological significance of *Eclipta alba* (L.) Hassk. (Asteraceae), Int. Sch. Res. Notices 2014 (2014), 385969.
[81] J. Wisetkomolmat, A. Inta, C. Krongchai, S. Kittiwachana, K. Jantanasakulwong, P. Rachtanapun, et al., Ethnochemometric of plants traditionally utilised as local detergents in the forest dependent culture, Saudi J. Biol. Sci. 28 (5) (2021) 2858–2866.

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CHAPTER

Antibiofilm activities against food pathogens

23

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Abbreviations

AI	autoinducer
AIPs	auto-inducting peptides
CDC	Center for Disease Control and Prevention
EFSA	European Food Safety Authority
EHEC	enterohemorrhagic *Escherichia coli*
EPS	extra cellular polymeric substance
FDA	Food and Drug Administration
IUPAC	International Union of Pure and Applied Chemistry
LLO	listeriolysin O
MRSA	methicillin-resistant *S. aureus*
PVC	polyvinyl chloride
QS	quorum sensing
SAMB	self-assembling marine biofilms
SEs	staphylococcal enterotoxins
SFP	staphylococcal food poisoning
Stx	shiga toxin
WHO	World Health Organization

1 Introduction

Biofilms formed by pathogenic microbes are a consortium of microbial cells enveloped and protected by extracellular polymeric substance (EPS) matrix and have become a worldwide challenge because of its great resistance to antimicrobials and other common disinfectants used in the industries and to the immune system of host [1]. Worldwide research teams developed various antibiofilm strategies, of which most of the attention is being paid to microbial surfactants (biosurfactants).

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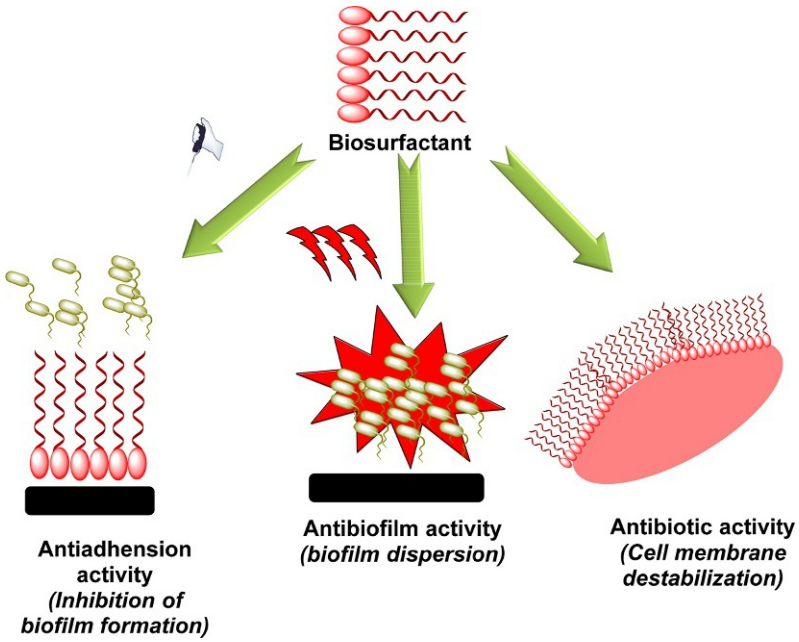

FIG. 1

Activities of biosurfactants.

Many of the biosurfactants reported to have antiadhesive, biofilm disruption and antimicrobial properties (Fig. 1) [2,3].

2 The nature and functions of biosurfactants

Biosurfactants are surface-active agents synthesized by living cells (i.e., yeast or bacteria or fungi) from various substrates such as oils, carbohydrates, hydrocarbons, wastes, etc. The world of microbes gives different sources of new energy and more diverse green chemicals, in which biosurfactant is one group of green chemicals given by microorganisms. Biosurfactants have more advantages over chemical surfactants in terms of biodegradability, production of renewable resources, less chemical toxicity, work in extreme conditions of pH and temperature [4]. Biosurfactants are amphiphilic compounds having both hydrophilic part (head) and hydrophobic part (tail). Hydrophobic tail is composed of unsaturated, saturated, or hydroxylated fatty acids. Hydrophilic head is made up of peptides or amino acids, anions, or cations, mono-, di-, or polysaccharides [2]. Biosurfactants are mostly anionic or neutral; however, cationic and amphoteric biosurfactants are also present (Fig. 3).

Biosurfactants were classified as two types based on their molecular weight, i.e., low-molecular-weight and high-molecular-weight biosurfactants; former consists of glycolipids, lipopeptide, phospholipids, and high-molecular-weight biosurfactants

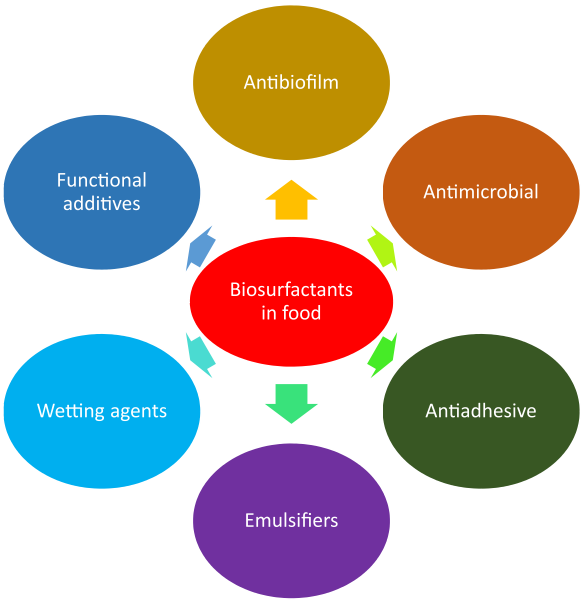

FIG. 2

Biosurfactants' functions in the food industry.

contain polymeric and particulate surfactants. Based on the hydrophilic part of biosurfactant, naming was given by IUPAC, i.e., those carrying sophorose are called as sophorolipids, rhamnose are rhamnolipids, and glycolipids are those usually having a carbohydrate moiety along with rhamnose and sophorose.

Biosurfactants have been widely used in many industries, as emulsifiers, antiadhesives, antimicrobial, antibiofilm, antioxidants, wetting agents, and functional additives, especially in food industry (Fig. 2) due to their environment-friendly nature such as less chemical toxicity and easy biodegradability. By having the distinct characteristics, well-defined structures, and also growing consumer concern for green above chemically synthesized constituents, Biosurfactants emerged as innovative functional additives in the food formulations.

Food products contamination is the foremost problem in the industry of food, which is pushing human life in threat. As per World Health Organization (WHO) records, every year one in 10 people is becoming sick by eating contaminated food and causing around 420,000 deaths. Although there are wide range of sources for food contamination by pathogens, biofilm has been considered as the principal source [5] and associated with many food-borne outbreaks. Biofilm is an aggregation of microbial cells associated to surfaces, and these microbial cells are enclosed in the extracellular polymeric substance (EPS) matrix. The ability of food-borne pathogens to form biofilm is considered as significant factor for their persistence in the food processing industry [6].

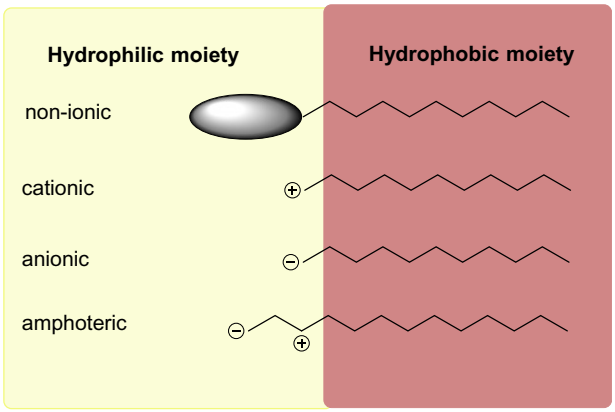

FIG. 3

General structures of surfactants.

Biofilms became serious problem in the food industry, in particular areas involving poultry, animal slaughter, dairy processing, and red meat [7]. The bacteria present in the biofilm show increased resistance to antibiotics, disinfectants, phagocytosis. Hence, it will become difficult to eradicate this biofilm, which substantially leads to health problems and economic issues. The prevention of biofilm formation and its growth has the great importance to reduce the risk of intoxication and to safeguard the consumers' health. All features of biofilm control, monitoring, measurement, inhibition, and dispersal are flattering subject of growing importance.

New cognizant intuition on physiology of biofilm is now allowing researchers to plan most effective bacterial dispersal or inhibition strategies. Biofilm inhibition can be possible mostly by two ways, based on novel antibiofilm compounds formulations and the building of surfaces that are hostile to biofilm [8]. Most auspicious candidates to inhibit the formation and growth of biofilm came from biosurfactants (biological surface-active agents). Many of the biosurfactants are described to have biofilm disruption, antiadhesive and antimicrobial properties [9]. Here in this chapter, we will discuss biofilm characteristics and antibiofilm activities of biosurfactant against food pathogens (Fig. 3).

3 The process of biofilm formation and related aspects in food industry

Biofilm formation is natural protection strategy of microorganisms toward environmental challenges and external toxic factors, and its formation depends on the several factors such as surface of cell and area of attachment chemical and physical properties and supporting medium composition. The microbial biofilm formation is a complicated process. Firstly, on the equipment surfaces, organic molecules present in the

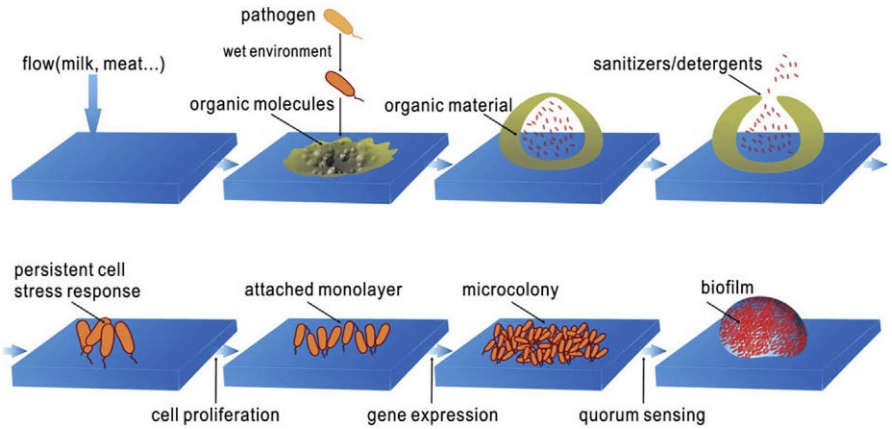

FIG. 4

Sequence of steps involved in the formation of biofilm.

From X. Shi, X. Zhu, Biofilm formation and food safety in food industries, Trends Food Sci. Technol. 20(9) (2009) 407–413. https:/doi.org/10.1016/j.tifs.2009.01.054.

food are deposited. Secondly, microorganisms, which are bioactive, are attracted toward the surfaces. Thirdly, even after regular cleaning processes, some microbial cells remain present and initiate growth. Lastly, is the development of biofilm with the help of quorum sensing (QS) and gene expression (Fig. 4).

In biofilm, microorganisms have the ability to communicate with all the cells present in same species and different species and act as population by cell-cell signaling, called QS, regulate and coordinate behavior of bacteria in the environment, e.g., antibiotic production, biofilm formation, bioluminescence. This behavior has been documented in biofilm formations [10] and depends on environment or nutrition and maturation stage of the microorganisms. QS is based on pheromones or autoinducers (AI), which are chemical signal molecules. Homoserine lactones (AHLs) usually play a signaling molecule role in the Gram-negative bacteria, whereas oligopeptides (AIPs, auto-inducting peptides) play a role of signaling molecules in the Gram-positive bacteria. Signaling molecules will be released into the environment and accumulated, as the function of the cell density (signaling molecules) concentration increases. When a threshold stimulatory concentration is obtained, the microorganisms are going to sense it and respond inclusively, e.g., by altering gene expression [11]. All the above-described characteristics of sessile bacteria make eradication of biofilms difficult.

Microbial biofilms show a different bacterial physiology, which is basically different from their planktonic bacteria. In one of two states (planktonic and sessile), generally bacteria exist in the nature and also in the food systems: in bulk solution, they exist as free cells, planktonic cells, and cells that exist as composite are sessile cells, which are generally observed in the biofilm. The attachment and detachment ability of microorganisms toward surfaces are a general characteristic feature of

microorganisms; however, it is the advantage character and of course it is necessary character of microorganisms to survive in the environment, this attachment ability also favors the formation of biofilm causing serious risk in the food industry. Corrosion, unsightliness, product spoilage, equipment failure, infection, unpleasant odors, product efficiency reduction, and biofouling are examples of biofilm harmful effects.

Primarily, they develop resistance to stress conditions, e.g., pH changes, nutrient deprivation, oxygen limitation. Secondly, they exhibit distinct virulence factors and show increased resistance to the host immune system and antimicrobial agents [12]. Also we can observe improved organic compounds degradation [13].

The natural environment biofilms, which are made up of one species of microorganisms, were relatively few, rather they are found in the complex of multispecies communities [14]. Prior to now, dissimilarities between biofilm and planktonic physiologies appeared insignificant. Majority of the tests for bacterial inhibition were exclusively built on the physiology of planktonic bacteria but not on the biofilm physiology; in the natural environment, these conditions are found not readily. The standard physiology of planktonic bacteria is usually characterized by a free living single bacterium that has optimal gas exchange, nutrition, and agitation (typically 250 rpm). Technol and Bueno [15] contrast that the physiology of biofilm had multicellular communication, internal architecture, rudimental fluid transport system, and multicellular differentiation [16]. Most importantly in vitro analysis, biofilms had no or little agitation, variable levels of nutrients, gas exchange, and therefore deliberate development. This variance in physiology of planktonic bacteria could be serious mainly in clinical circumstances where pathogens produce high virulence factors, e.g., *Pseudomonas aeruginosa* [17]. In contrast to the planktonic counterparts, the pathogens in the biofilm physiology could be one to three orders of magnitude tough to dispersal by usual inhibition/chemotherapy [16]. Some biofilms are likely to undergo phenotypic changes as outcome of chemotherapy confirming in the increased resistance.

Bacterial biofilms were the mostly studied biofilms from the molecular and morphological perspectives. Both yeast and filamentous fungi can also form biofilms, among these yeast biofilms are predominantly explored compared with that of fungal biofilms. Among yeasts, *Candida albicans* biofilm formation was the mostly studied model, and it shows similar process of development as bacterial biofilms. *Paracoccidioides brasiliensis* forms a biofilm, which is related with adhesins, gene expression, and enzymes such as aspartyl proteinase, GP43, and GADPH; this was characterized by de Cássia Orlandi Sardi et al. [18]. In recent times, studies increased considerably on fungal biofilms. Costa-Orlandi and team in their review discussed and compared the formation of biofilm by filamentous fungi *Trichophyton mentagrophytes* and *Trichophyton rubrum* (dermatophytes) and *C. albicans* [19]. Authors proposed that there are six stages in the fungal biofilm formation: (1) propagule adsorption—sporangia or hyphal fragments, spores contact the surface; (2) active adhesion—spores secret the adhesins; (3) microcolony formation—hyphal branching and elongation, monolayer formation with the release of extracellular matrix (ECM); (4) initial maturation and formation of second microcolony—ECM covers

the hyphae network and formation of channels or pores; (5) final maturation—further survivor structures and fruiting bodies are formed; and (6) planktonic phase or dispersion—releases hyphae fragments and the beginning of new cycle. Hydrophobins (small proteins) are released by filamentous fungi. For the hyphae adhesion and in the process of biofilm formation proteins are involved. In this also described the structures of polymicrobial biofilms. Microorganisms living in the complex communities are called as polymicrobial biofilms.

Extracellular polymeric substance (EPS) plays a structural role and makes these biofilms strong persistence in the food industry. The EPS matrix envelops the biofilm and is made up of polysaccharide, for example, proteins, cellulose, and exogenous DNA. This EPS could be bind to biological structures, such as bones, meat, fruits, vegetables, etc., and to hard surfaces such as equipment of food industry, dispensing, soil, storage surfaces, transport, etc. [12]. The EPS gives rise to complex gradients in regard to oxygen diffusion and nutrients, having extracellular enzymes, which are utilized for the purpose of nutrition, allows transfer of communication molecules of a cell, and protects the cells that are embedded from the toxic compounds. In brief, biofilm formation gives out to more precedence to microbial cells in the environment of food industry, for instance, chemical protection (resistance to disinfectants, chemicals, and antimicrobials utilized in the industry), mechanical resistance (resistance to fluid streams in pipelines), physical resistance (resistance to desiccation) [12].

In the food industry, formation of biofilm can occur rapidly. Initial two steps in the biofilm formation are the conditioning of material's surface and cells binding to those surfaces reversibly; later this reversible binding turns to irreversible and the microcolony development occurs. Finally, a complex ecosystem is produced by forming the tridimensional structure of biofilm and ready for dispersion [20]. Some of the microbial biofilms forming in the food industry are human pathogens; for this reason, antibiofilm strategies in the food industry have got particular importance. These pathogens can able to form biofilms on various artificial substrates, which are regularly used in the food industry, for instance, polyethylene, polypropylene, wood, rubber, glass, stainless steel, etc. [21]. The effects associated with the formation of biofilm (metal surface corrosion, pathogenicity, organoleptic properties alteration because of secretion of proteases or lipases) gain a lot of importance in certain industries such as dairy, where there are numerous structures and processes (pipelines, packing tools, butter centrifuges, tanks of raw milk and cheese) will behave as substrates for the development of biofilm with various colonizing species and at various temperatures. Some pathogenic species *Listeria monocytogenes*, *Salmonella enterica*, *Aeromonas hydrophila*, form biofilms on fish products, so that causes economic issues and affects human health (Fig. 5) [22].

Key genes expressed in the biofilm formation and responsible for biofilm characteristics can undergo genomic variations, which lead to the formation of entirely different biofilms at different conditions. Along with this complexity, affected environments, high diversity, and various types of species of colonizing bacteria make the biofilm more complicated to be eradicate from the food industry.

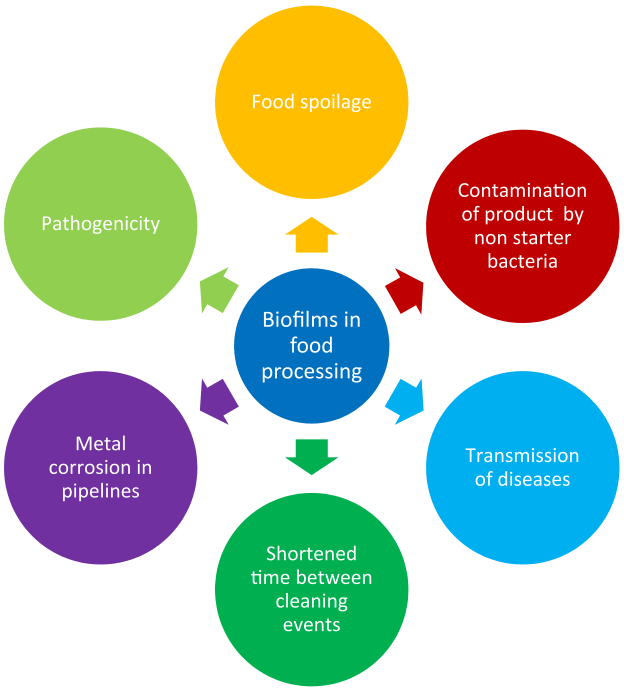

FIG. 5

Consequences of biofilm formation in food processing area.

4 Biofilm formation by various bacterial pathogens

Due to the formation of biofilms on food factory equipment or food matrixes, food-borne illness or diseases might arise through infections or intoxication. Toxins could be produced by the biofilm, which is formed on the processing plants in the food industry, and from here it can contaminate the whole food matrix, leading to individual intoxications or also multiple intoxications (if outbreak happens). In both the cases, there is a risk for the human health. Based on the tridimensional living system forming bacterial species, the seriousness of the risk varies. For the formation and development of biofilms, main locations are dependent on the type of the factory; however, it might include milk, water, and other fluid pipelines, tables, pasteurizer plates, packing material, contact surfaces, animal carcasses, dispensing tubing, additives, employee gloves, etc. [23]. Most importantly, there are five bacterial pathogens that are food-borne, capable of developing biofilm on various substrates; those are *Bacillus cereus*, *Enterohemorrhagic Escherichia coli*, *S. enterica*, *L. monocytogenes*, *Staphylococcus aureus*.

4.1 *Bacillus cereus*

B. cereus are spore-forming and anaerobic Gram-positive bacteria, commonly seen in food; it has resistant toward chemical treatments, radiation, and heat; besides, it can grow in various environments and temperatures of wide range (4–50°C) [24]. Due to the formation of endospores in this bacterium, it is also resistant to pasteurization process in the industry. Some of these strains produce emetic toxin and diarrheal enterotoxins, former one produces by heat-stable cereulide, produces vomiting symptoms, while the latter one causes abdominal pain and diarrhea; all of these strains of this bacterium cause food poisoning. This food-borne pathogen produces hemolysin, which has a great importance among all since it causes extreme dehydration and often death [25]. The emetic syndrome is due to the toxic activity of cereulide in the mitochondria, in which it behaves as potassium ionophore, leading to immunomodulatory effects and cellular damage [26]. Diarrhea related to this bacterium also includes watery diarrhea and abdominal cramps after 8–16 h of ingestion; if this is the case, the infective dose needed is in between 10^5 and 10^8 CFU [26].

B. cereus biofilms are also related with other microorganisms, exopolysaccharides, extracellular DNA, and compound matrix of proteins favors this relation; these are important for bonding on various substrates [27]. *B. cereus* initially attaches to the surfaces of food manufacturing industries and facilitates other bacterial species for their faster attachment to those surfaces so that they cannot be removed by milk streams, water flow, and other physical activities performed by these industries [28]. In the dairy industry, *B. cereus* mostly grow biofilms at the interface between air and liquid and on the submerged surfaces such as on pipes, stainless steel equipment; in the both of these cases, biofilm development is because of flagellar motility [29]. *B. cereus* is a contaminant present in raw milk. A study performed in the pasteurized milk producing plant divulged that 5.5% of these milk products hold 10^5 CFU/mL *B. cereus* and enterotoxins present in around 4% of this milk products lead to food-borne diseases. After the storage of the pasteurized milk, *B. cereus* can produce enterotoxins within 7–8 days. For products having high butterfat and those pasteurized with high temperatures, short-time treatments will have high count of *B. cereus* [30].

In a study, Saleh-Lakha and team isolated and characterized 17 *B. cereus* from pasteurized milk; from these isolates they identified the patterns of 5 toxigenic genes, all of which have coding genes for diarrheal toxins. All four genes for diarrheal enterotoxin also contained in one strain [30]. This learning revealed the significance of scrutinizing food-borne pathogens in the pasteurized products even at high temperatures. This pathogen importance has been highlighted by the truth that, because of this pathogen, 6657 suffered due to food-borne intoxications during the years 2007–14 in the EU (of which 505 only in 2014), allied to outbreaks of 413, and reported no casualties. The food matrices involved in those outburst were products of cereals (10.9%), poultry meat (5.3%), red meat (8%), juices or vegetables (4.6%), and the buffet meals (27.6%) [31]. In this review, among the other pathogens that are included, *B. cereus* stood next to *S. enterica* (of all cases 26.2%).

Table 1 Formation of biofilm on various food processing areas.

Place	Biofilm-forming microorganisms	References
Feeding unit or Conveyor belt	*Shigella* sp., *Listeria monocytogenes*	[32]
Dairy equipment	*Listeria monocytogenes*	[33]
Dairy processing plant and pasteurization lines	*Staphylococcus aureus*, *Shigella* sp., *Escherichia coli*, *Bacillus cereus*	[34]
Floor drain	*Leuconostoc citreum*, *Pseudomonas fragi*, *Listeria monocytogenes*, *Pseudomonas mucidolens*, *Acetobacter tropicalis*	[35]
Floors	*Listeria monocytogenes*, *Bacillus* sp., *Pseudomonas fluorescens* E9	[36]
Meat processing Environments	*Salmonella* sp.	[37]
Packing device	*Bacillus* sp.	[38]
Stainless steel	*Listeria monocytogenes*	[36]
Stainless steel pipes	*Proteus mirabilis*, *Serratia marcescens*, *Enterococcus faecalis*, *Staphylococcus epidermidis*, *S. hominis*, *S. aureus*, *Bacillus cereus*	[39]
Ultrafiltration Membrane	*Klebsiella oxytoca*	[40]
Whey processing line	*Paenibacillus glucanolyticus*, *Bacillus subtilis*, *cereus*, *licheniformis*, *pumilus*, and *thuringiensis*	[4]

4.2 Enterohemorrhagic *Escherichia coli* (EHEC)

Most of the strains of the *E. coli* are present in intestinal microbiota of human, but they cause no health issues. However, some of the strains are virulent food-borne pathogens, which cause a serious health risk, imparted by drinking water, vegetables, and fruits (melons, lettuce, parsley, tomatoes, spinach, cilantro, etc.), fresh meat, and raw milk. At the origin itself, the products might be contaminated or during the process of food manufacturing (Table 1). This contamination may take place both preharvest and postharvest environments of the food industry. By the usage of contaminated water for vegetables cultivation, the contamination of food takes place in the preharvest period, which will still present after processing of raw material (vegetables, caracasses, etc.), and also this bacterial contamination undergoes faster growth because of storage temperatures in the postharvest environments [41].

Many species of *E. coli* interact or attach with various surfaces containing stainless steel, glass, PVC, polypropylene, Teflon, polystyrene, biotic surfaces, and other microbes by the formation of curli fimbriae. The surface material hydrophobicity and the temperature play a crucial role in the formation of biofilm by this species. For

instance, formation of well-built biofilm on stainless steel surfaces and the borosilicate glass and the little or no biofilm production on the surfaces of polypropylene by the species of *E. coli* O157:H7, it was due to the difference in the hydrophobic nature of the surfaces [41]. For instance, these species were incubated on the surfaces of beef for 7 days at 15°C, than there is increase in the character of adherent and planktonic cells [42], it proves that temperature also plays a major role in the formation of biofilm by the species of *E. coli*. This factor poses serious threat for the processing plants of beef, where the processing takes place at regular temperature of 15°C.

Like *B. cereus*, other species also included in the biofilm formation by *E. coli* O157:H7, which results arise in ecological benefits. For instance, biofilm formation by *E. coli* O157:H7 along with *Burkholderia caryophylli* or *Ralstonia insidiosa* increases its survival rates up to sixfold on the surfaces of food industry.

E. coli survival and abilities to from biofilm are dependent on the serotype; hence, it is serotype-dependent bacterium. For example, if we compare the serotypes O157:H7, O111:H-, O103:H25, O26, or O145, among all these, O157:H7 serotype shows greater resistance to high pressure, high temperatures, and the common disinfectants applied in the food industry [43].

E. coli O157:H7 is a frequent EHEC, which we encounter mostly, after this serotype *E. coli* O26 is called as the second EHEC found in the major cases of food-borne diseases in Australia. Most of the strains of *E. coli* O26 are well known for their resistance toward acid, it can withstand utmost acid conditions, which are there in the food processing industry involving citric acid (used as preservative in fruit juice), acetic acid (used in the products of canned vegetables), lactic and propionic acids (used in meat products and fermented dairy) [44].

With respect to clinical issues, *E. coli*, which produces Stx (shiga toxin), secrets toxins Stx1 and Stx2 and leads to enterohemorrhagic gastroenteritis that causes feces with blood and watery diarrhea, contamination of fresh products, sauces, salads, soups, ground beef, cooked chickens with strains will cause to food-borne outbursts and often death [45].

Another major issue with strains of *E. coli* O157:H7 is biofilm, which is even in low grade also can cause serious health issues, and to control this, strong measures are to be taken in place of contamination in the food industry, because as an infectious dose, these strains need ingested CFUs less than 50 only [46].

4.3 *Listeria monocytogenes*

L. monocytogenes is the Gram-positive food-borne bacterium. Food-borne outbreaks caused by *L. monocytogenes* are a major health concern due to its high death rate. It causes severe disease called listeriosis, which has 20%–30% mortality rate approximately [47]. Spontaneous abortion or fetus damage can be caused by this disease (listeriosis) in the pregnant women [48]. Gastroenteritis can be caused by this pathogen even in the healthy humans.

This pathogen can cause serious illness in pregnant women, neonates, immunocompromised individuals. Motility of this pathogen is temperature-dependent; at low

temperatures (<28°C), this pathogen is highly motile, whereas at high temperatures (37°C), this pathogen is nonmotile. For flagellar motility, *L. monocytogenes* genes are involved, which are important for the formation of biofilm (food products such as dairy products, sea food, ice creams, candied apples, poultry, soft cheeses, frozen vegetables, etc.) are known to spread this pathogen [49].

This pathogen can form biofilm composed mainly of teichoic acid, on rubber, steel, polypropylene, and glass surfaces in the food industry, from here it can reach to food batches, and at refrigeration temperature, this pathogen replicates [50]. The foods that are commonly contaminated by this pathogen are fresh cheese, cold cuts, and smoked fish.

Together with replication ability at low temperature, this pathogen increases its hydrophilicity, so that enhancing its adherence to surfaces in the food industry and also increasing its capacity to withstand the regular cleaning process in the industry of food [51]. *L. monocytogenes* has the resistance toward treatments even at 60°C; this behavior makes this pathogen more complicated to be eradicated from the food industry [52]. After the cell invasion of host followed by ingestion, phospholipases (PlcB and PlcA) and listeriolysin O (LLO) are used by *L. monocytogenes* to enter the human cell cytosol [53]. LLO is a cytolysin, and it is cholesterol-dependent, which invades into the cell membranes of host, through the formation of pores and vacuolar compartments breaking, enters into the cytolysin space. Then, the host cell actin-assembly machinery is activated by this bacterium using surface protein ActA, which enhances this bacterium to spread cell-cell and also facilitates bacterial movements intracellularly [53].

The confirmed cases of listeriosis in the humans have been increasing continuously; in the year 2012, the confirmed cases were 1720, and in the year 2016, the confirmed cases in the humans were 2536 [54]. Among all the illness deaths occurred in the year 2016, 8.0% of deaths occurred by this pathogen. In all the above cases, biofilm is the principal source of food contamination, so it is important to study how to control and monitor the formation and development of biofilm.

4.4 *Staphylococcus aureus*

S. aureus is a nonspore-forming, facultative anaerobic, nonmotile, Gram-positive bacterium. This bacterium is called as human opportunistic bacterium. Between the temperatures 10°C and 46°C, this pathogen when its growing in the food produces characteristic enterotoxins (staphylococcal enterotoxin),which is the common reason for food poisoning (staphylococcal food poisoning) (Table 1). The species of this bacterium have the capability to multiply on the food handler's skin and also on mucous membranes, which is the significant issue in the food industry [55]. The preformed staphylococcal enterotoxins (SEs) by this pathogen when its growing in the food are heat stable and cannot be removed by the normal disinfectants; hence, the eradication of growth of *S. aureus* and formation of toxin has the great importance to reduce the risk of intoxication.

Earlier believed that based on *S. aureus* colony forming units present in the products of food risk analysis of staphylococcal food poisoning (SFP) is estimated, which is disproven recently. Instead now researchers believe that impending threat appearing from unpredicted SEs secreted under stress conditions [56], in order to assess the risk, it is critical to interpret role of stressors on the SEs expression. This bacterium shows a higher growth in the food with high concentrations of salt or sugar and low pH.

S. aureus is a common pathogen of biofilm formation in wide range of environments, and worldwide it is considered as a common food-borne pathogen, the biofilm forming abilities of this pathogen increase its persistence at the contaminated areas or infected sites. The bacteria present in the biofilm show increased resistance to antibiotics, disinfectants, phagocytosis. So it will become difficult to eradicate this biofilm, which substantially leads to health problems and economic issues. Most of the cleaning agents used in the food processing industries as disinfectants containing ethanol; several studies reported that alcohols in low concentrations increase the biofilm formation ability by the staphylococcus species. *S. aureus* biofilm growth can be increased by some common processing procedures used in the food processing industries, such as combination of glucose and salt or improper disinfection, suboptimal temperatures [57].

In the year 2015, the EFSA reported 434 registered staphylococcus food poisoning outbreaks, and it is more than half of total bacterial toxins caused food-borne outbreaks. The CDC in the United States estimated that in each year, 240,000 cases of staphylococcus food poisoning can occur, leading to 6 fatalities and 1000 hospitalizations. After the intake of staphylococcus enterotoxins, shortly patients might suffer from violet vomiting and nausea, which is later accompanied by prostration, moderate fever, and watery diarrhea; generally within 24 h, symptoms subside, but it may be fatal in the elderly and in children with 4.4% of fatality rate. The genes expressed during the formation of *S. aureus* biofilm play a role in toxin secretion (*hla* and *hlb*), adhesion (fnbA, fnbB, can, icaD), transcriptional regulation (sarA and agr) [58].

4.5 *Salmonella enterica*

Salmonella spp. are Gram-negative, food-borne, most important pathogen, worldwide. It was the first reported biofilm forming pathogen on food surfaces; this fact demonstrates its significance as a food-borne pathogen (Table 1). It is the common bacterium related to gastroenteritis in human [59]. This pathogen is described to develop biofilm on abiotic and biotic surfaces. The biofilms of salmonella contain two important matrix components: curli and cellulose. Curli (also called Tafi, thin aggregative fimbriae) are amyloid fibers, involved in cell aggregation, surface adhesion, development of biofilm, and environmental persistence [60]. *S. aureus* serovar Enteriditis is common serotype causing abdominal pain, vomiting, diarrhea, nausea, and fever as the major symptoms [61]. In the processed food, a common reservoir for this bacterium was poultry meat. The rate of formation of biofilm is enhanced with increase in pH and temperature. The cells adhered to the surfaces decreased after the 24 h of increased temperature, and there was no difference between pH 6 and 7. The properties of electron-accepting or -donating change with temperature and pH; however, these changes were not

matched up with the ability of biofilm formation. *S. enterica* is adept of forming biofilm on the surfaces of stainless steel, resulting in 3D structure with multilayered cells, based on the nutrients available, morphologies may vary. For instance, when it's grown in TSB (tryptic soy broth), it generates an articulate shape. Some strains of this bacterium show more preference for attachment on the surfaces of stainless steel rather than the acrylic surfaces in all the tested conditions [61]. The surface attachment is initially influenced by the topography. For instance, mechanically sanded and untreated steel was colonized very easily. However, brightalum finished and electropolished steel was less colonized by this pathogen [62].

S. enterica can easily attach and form biofilm on food matrixes and meat, subsequently causing cross-contamination in food manufacturing plant and also at supermarket between the food batches [63], which causes serious food-borne outbreaks. In the year 2011, 95,548 reported cases of salmonellosis in human; this pathogen is most often reported for its food-borne disease spread in the EU. In the year 2013, several food-borne outbreaks and recalls, which are reported by CDC and FDA in the United States, were related to salmonella contaminations.

5 Conventional strategies for the prevention of biofilm

Cleaning and sanitization measures are very much important, but the general cleaning procedures followed in the food industry can remove 90% of the microbial cells, do not remove all the sessile cells present in the biofilm, and thus do not decrease the biofilm development [64]. It was observed commonly that sessile cells have greater ability to survive at high amount of disinfectants than the planktonic cells. Hence, more amount of disinfectants may be required than the concentrations prescribed in the areas of food formulations in order to confirm the concentrations of biocide [65]. That various microbial pathogens capable of forming biofilms developed resistance to antimicrobials and other common disinfectants.

Other alternative approaches to control the biofilm are the use of enzymes, cell wall inhibitors, and different proteolytic cell membranes, interface of cell signaling, and bacteriophages can shrink the biofilm development. It is better to prevent or control vigorous formation of biofilm than to fight formed biofilm.

However, using conventional antimicrobials and various control measures failed to eliminate the bacteria that are attached to the equipment and utensils in the food processing industry, notably showing the need of developing novel approaches for the prevention of biofilms [66].

6 Biosurfactants as antibiofilm agents

6.1 Lipopeptides

Lipopeptides being one of the major categories of microbial surfactants can efficiently disperse the biofilms. This lipopeptide group includes polymyxins, surfactins, fusaricidins, and fengycins [67]. Structurally, these lipopeptides are made up

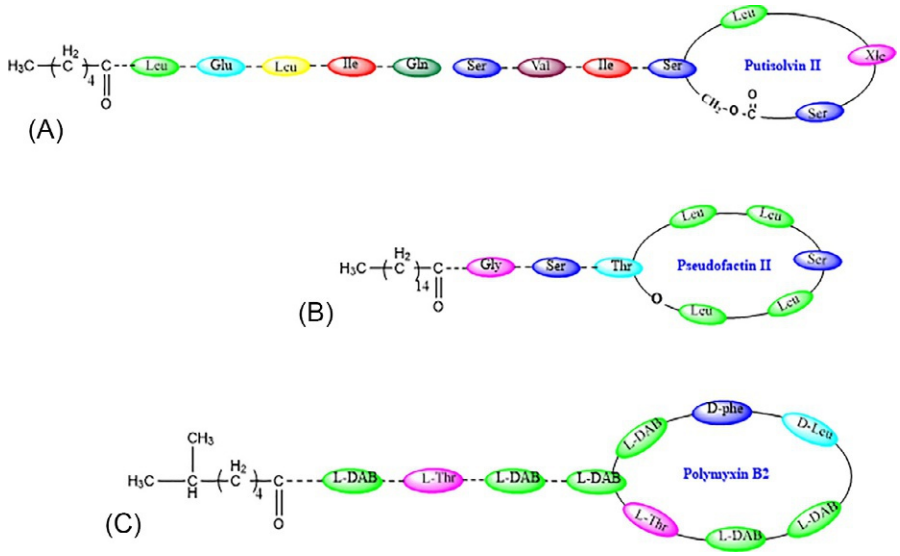

FIG. 6

Structure of biosurfactants. (A) Putisolvin II, (B) Pseudofactin II, (C) Polymyxin B2.

of hydrophilic peptide liked to hydrophobic fatty acid or lipid. The branched, aliphatic, or cyclic peptides and lipids differ in conformations and length ensuring broad diverse structures. Many of the reported lipopeptides disperse or inhibit the microbial biofilms derived from *Paenibacillus* or *Bacillus* [68].

Polymyxins: Polymyxins belongs to the nonribosomally produced class of cyclic lipopeptides. Generally these are secondary metabolites produced by *Bacillus* or similar species [69]. The distinctive shape of polymyxin is that hydrophobic fatty acid tail linked to hydrophilic head of cyclic polypeptide. They can also have (DAB) 2,4-diaminobutyric acid like foreign bacterial amino acids (Fig. 6A). There are several polymyxin formulations, which are commercially available including polymyxin E (colistin), polymyxin B are provided as neosporin and polymyxin B sulfate [70]. These are prescribed as topically or intestinally as powders or creams in majority cases of multidrug-unaffected *P. aeruginosa*, *Klebsiella pneumoniae*, *Acinetobacter baumannii* [71]. Polymyxin is reported to eradicate *P. aeruginosa* biofilms at 24 μg/mL concentrations by 99% in time period of 12 h and almost totally reduced over 24 h [72]. Anyway these results are on the basis of bacterial viability but in their dispersal. Although it was recorded that bacterial cells showed altered morphology. Polymyxin E (colistin) is recommended as an early aggressive therapy to delay the onset of chronic *P. aeruginosa* infection (which frequently forms biofilms) or intermittent colonization in cystic fibrosis patients, a combination of oral ciprofloxacin with colistin inhalation [2].

Polymyxin D1 was found in undefined ratios of furanisidin and surfactin [68], effective in opposition to mixed bacterial biofilms. It was reported that polymyxin

inhibits the formation of Gram-positive bacterial biofilm such as *Streptococcus bovis*, *Micrococcus luteus*, *S. aureus*, and *Bacillus subtilis* and Gram-negative bacterial biofilm such as *P. aeruginosa*. More interestingly, these biosurfactants inhibit the formation of SAMB (self-assembling marine biofilm) biofilm of mixed species in assays of coincubation by 99.3% and disrupt already formed SAMB by 72.4% [68]. Polymyxin mode of action on microbial biofilms remains indeterminate. However, the mode of action on planktonic cells is based on its high affinity to liposaccharides [73].

6.2 Fengycin-like lipopeptides

These are cyclic lipopeptides having beta hydroxy fatty acid attached to 8–10 amino acids (Fig. 6B) and are derived from *Bacillus licheniformis* and *B. subtilis*. Fengycin-like lipopeptides inhibit the biofilm formation (Xu et al2013) of both Gram-positive bacteria such as *S. aureus* causing dispersion up to 90% and Gram-negative bacteria such as *E. coli* causing dispersion up to 97% [74].

6.3 Putisolvin

These are cyclic lipodepsipeptides derived from *Pseudomonas putida*. Putisolvin typical structure consists of hydrophilic cyclic peptide attached with hydrophobic α-amino acids. These have been referred as Putisolvin I and Putisolvin II (Fig. 6).

These biosurfactants have cyclic four-member peptide and the residue of valine in Putisolvin I, whereas in Putisolvin II, isoleucine and leucine are present (Fig. 6C) [75]. Although it's been involved in the *P. putida* biofilm development, these surfactants also act as a successful dispersion agent before and after addition to other strains of *Pseudomonas* sp. biofilms [76].

6.4 Pseudofactin

Pseudofactin isolated from *Pseudomonas fluorescens* is a cyclic lipopeptide. Its structure depends on the attached palmitic acid at the terminal amino acid of the peptide chain of eight amino acids. Lactone formation takes place between the carboxylic group at C-terminal of last amino acid and the third amino acid (threonine) hydroxyl group. It was reported that Pseudofactin II is involved in the inhibition of bacterial biofilms adhesion by 36%–90%, such as *E. coli*, *Proteus mirabilis*, *Staphylococcus epidermidis*, *Enterococcus faecalis*, and *Enterococcus hirae* on the surfaces of silicone, polystyrene, and glass. Similar inhibition was also reported that Pseudofactin II involved in the inhibition (92%–99%) of yeast biofilms adhesion such as *C. albicans* at 0.5 mg/mL concentrations [77].

These was reported to have 26%–70% active dispersal on preexisting biofilms developing on unprocessed surfaces and also showed its effect of marked inhibition on *E. coli*, *C. albicans*, *E. hirae*, *and E. faecalis* initial adhesion to silicone urethral catheters. Almost complete inhibition of *S. epidermidis* growth was observed at 0.5 mg/mL tested in highest concentrations, which in case of other bacteria causes

inhibition partially, i.e., 18%–37%, which also causes inhibition of growth 8%–9% in *C. albicans* yeast and adhesion prevention up to 99% [77].

6.5 Surfactin

Surfactins are powerful microbial surfactants originally derived from *B. subtilis*. Surfactin consists of acyclic heptamer peptide attached to β-hydroxy fatty acid chain of 13–15 carbons (Fig. 7). Due to surfactin interactions with cell membranes, woefully these are also cytotoxic to hemolytic activities [78]. Surfactins inhibit the biofilm growth of *Salmonella* sp. growing on urethral catheters and PVC microtiter wells [79]. Conditioning of polystyrene surfaces using high concentrations of surfactin inhibits the *L. monocytogenes* and *P. fluorescens* adhesion (up to 54%).

This effect in the mature biofilms was not same, since at high concentrations (0.50%), only inhibits adhesion (32%) of *P. fluorescens* ATCC 13525 [80]. Another lipopeptide, Lichenysin from the family of surfactin isolated from the cultures of *B. licheniformis*, was reported that it effectively inhibits the formation of biofilm by *C. albicans* (74.4%) and MRSA (methicillin-resistant *S. aureus*) (87%), with values of ED_{50} 17.2 and 8.3 μg/mL, respectively. It was also observed that *Lichenysin* effective in post treatment of surfaces at biofilm disruption of *Yersinia enterocolitica* by 51.5% and MRSA by 57% with an ED_{50} values of 4.1 and 2.8 μg/mL, respectively [81].

6.6 Mixture of lipopeptides

Mixture of fengycin and iturin extracted from *Bacillus amyloliquefaciens* AR strain reduced surface cell hydrophobicity of *C. albicans*, decreased the mRNA expression of ALS3 and HWP1 hyphae specific genes, and hindered the formation of germ tube. Subject to the type of *Candida* strain used and the concentrations, lipopeptide mixtures inhibit the formation of biofilm by 46%–100% and dislodged the preformed biofilm by 25%–100% from the surfaces of polystyrene plates [82].

Another mixture of lipopeptides, fusaricidin B1 (Fig. 7B), polymyxin D1, and some traces of surfactin isolated from *Paenibacillus polymyxa PPE* showed that they inhibit the biofilm formation (87%–98%) by several Gram-positive bacteria such as *B. subtilis*, *S. bovis*, *S. aureus*, *M. luteus*, and also few Gram-negative bacteria such as *P. aeruginosa* [68]. These mixtures of lipopeptide biosurfactants inhibit the biofilm formation (99%) by mixed microbial strains and preestablished biofilm up to 74%.

6.7 Glycolipids

Glycolipid biosurfactants consists of carbohydrate moiety attached to the fatty acid chain. This group of biosurfactants comprises sophorolipids, rhamnolipids, cellobiose lipids, mannosylerythritol lipids (MELs), and trehalolipids. Most of the glycolipids reported that they are capable of forming pores and destabilize the biological membranes.

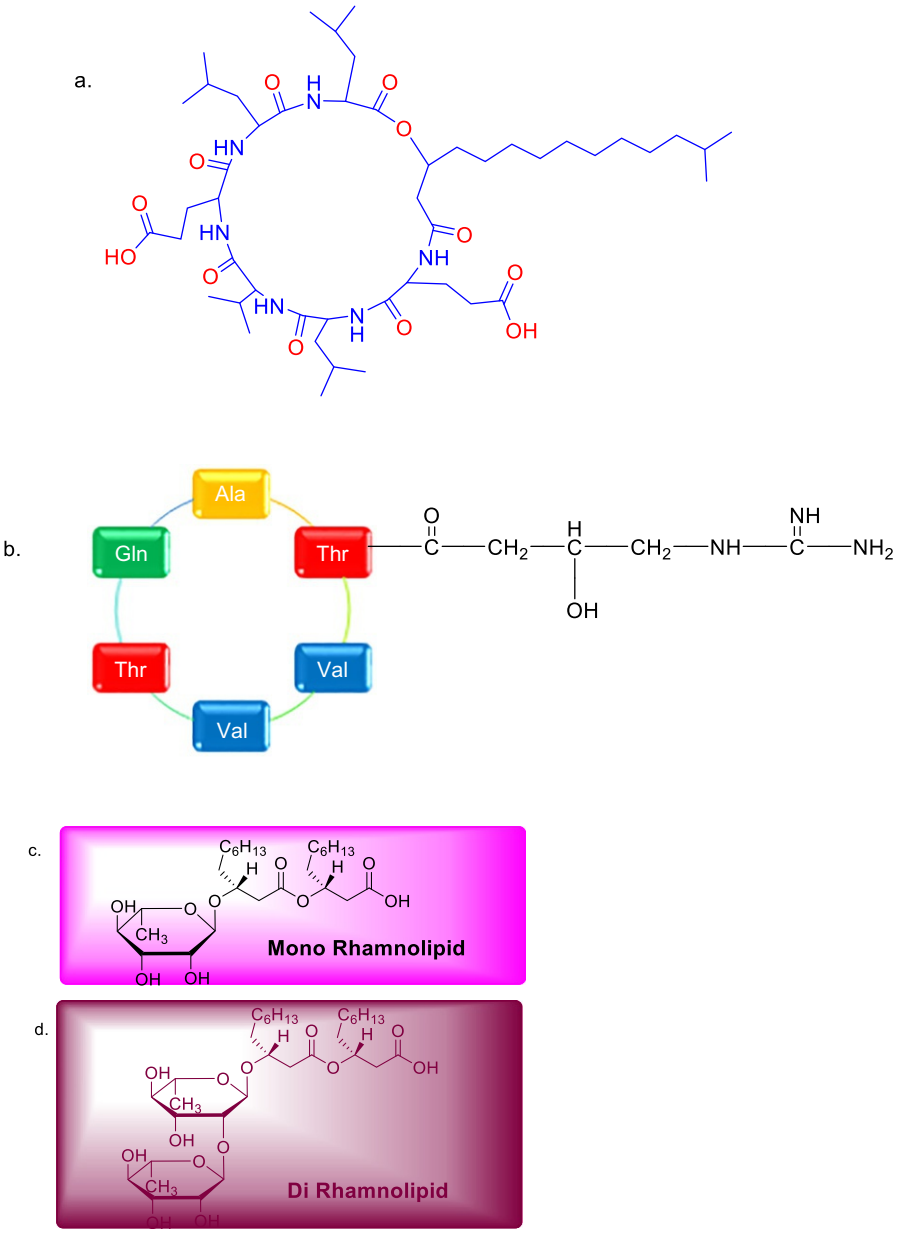

FIG. 7

Biosurfactants: (A) Surfactin, (B) fusaricidin B1, (C) mono rhamnolipid, (D)di-rhamnolipid.

6.8 Rhamnolipids

Rhamnolipids secreted as extracellular secondary metabolites mostly by various strains of *Pseudomonas* (originally by *P. aeruginosa*) and also by *Tetragenococcus koreensis*, *Nocardioides*, *Cellulomonas cellulans*, *Renibacterium salmoninarum* [83]. Rhamnolipids consist of rhamnose (mono or di) sugars linked to fatty acids (Fig. 7C and D). Rhamnolipids acts as antiadhesive and biofilm-disrupting agent against various pathogenic microorganisms. Rhamnolipids inhibit the biofilm formation (67%) and preexisted biofilm of *Yarrowia lipolytica* [84]. These also disrupt the *C. albicans* biofilm established on the surfaces of polystyrene [85] and also act as antiadhesive agent against *C. albicans*. Mono (Rha-C10-C10) rhamnolipids and di (Rha-C14-C14) rhamnolipids derived from *P. aeruginosa* ATCC 9027 and nonpathogenic *Burkholderia thailandensis* E264, respectively, interrupt the *B. subtilis* BBK006 biofilm [80].

Pure rhamnolipids with the concentrations of 0.50 (w/v) and 0.05% (w/v) reduce the formation of biofilms of *L. monocytogenes* ATCC 7644 (74%) and *P. fluorescens* ATCC 13525 (79%), respectively, on the surfaces of polystyrene. Crude rhamnolipids act against adhesion of *P. fluorescens* ATCC 13525 and *monocytogenes* ATCC 19112 inhibiting 34% and 30%, respectively.

Due to the presence of carboxyl and rhamnosyl groups, rhamnolipids also act as anionic biosurfactants. Rhamnolipids in solution can form micelle, vesicles, or lamellas based on the pH and the ions present in the solution [86]. Rhamnolipids conditioned polystyrene surfaces showed increase in the acid character and reduced hydrophobicity; these factors suggest the decreased formation of biofilm by various microorganisms, which do not form biofilm on the less hydrophobic surfaces.

6.9 Sophorolipids

Sophorolipids are produced from nonpathogenic yeast (e.g., *Candida apicola* and *Candida bombicola*). Sophorolipids consist of dimer sugar sophorose attached to fatty acid (long chain). Sophorolipids biosurfactant activity of biofilm disruption was investigated against both the strains of gram bacteria;(Table 2) results revealed that these biosurfactants at 5% *v*/v concentrations inhibited the growth of biofilm of *B. subtilis* BBK006 and *Cupriavidus necator* ATCC 17699 and also the biofilm disruption of *S. aureus* ATCC 9144 and *B. subtilis* BBK006 [99].

Sophorolipids inhibit the biofilm formation of *C. albicans* and decrease the viability of preexisted biofilm [100]. The mixture of fluconazole or amphotericin with sophorolipids acted collectively against biofilms of *Candida* sp.

6.10 Other glycolipids

Novel glycolipids secreted by *Lysinibacillus fusiformis* strain S9 are known to inhibit the formation of biofilm by *Streptococcus mutans* and *E. coli* pathogenic strains. This biosurfactant does not show bactericidal activity at 40 μg/mL, but it completely

Table 2 Biosurfactant-mediated control of biofilm formation in the food industry.

Biosurfactant/ biosurfactant-producing microorganism	Applications	Food or related surfaces	References
Bacillus licheniformis	Antibiofilm	Methicillin resistant *Staphylococcus aureus*	[81]
Bacillus sp.	Antimicrobial	*Listeria monocytogenes*, *Staphylococcus aureus*	[87]
Bacillus subtilis	*Listeria monocytogenes*	–	[88]
Pseudomonas aeruginosa and *Bacillus subtilis*	*Listeria monocytogenes*, *Staphylococcus aureus*, *Salmonella enteritidis*	Polystyrene Surface	[85]
Bacillus subtilis subsp. *subtilis*	*Listeria monocytogenes*	–	[88]
Enterococcus faecium MRTL9	*Bacillus cereus*	Polystyrene Surface	[89]
Lactobacillus acidophilus and *Lactobacillus fermentum*	*Enterococcus faecalis*	Glass surface	[90]
Lactobacillus helveticus	Antibiofilm	Stainless steel	[91]
Lactobacillus helveticus MRTL91	*Bacillus cereus*	Polystyrene Surface	[89]
Lactobacillus paracasei subsp. *paracasei* A20	*Staphylococcus epidermidis*, *Staphylococcus aureus*, *Streptococcus agalactiae*	Polypropylene Surface	[104]
Pseudomonas aeruginosa	Antibiofilm against *Listeria monocytogenes*, *Enterobacter sakazakii*, *Salmonella enteritidis*	Polypropylene surface and Stainless steel	[92]
Pseudomonas fluorescens	Antibiofilm	Stainless steel	[91]
Pseudomonas fluorescens	*Listeria monocytogenes*	PTFE and Stainless steel	[93]
Pseudomonas fluorescens	Anticorrosion	Stainless steel	[94]
Rhamnolipids	Postharvest control	*Alternaria alternata*	[45]

Table 2 Biosurfactant-mediated control of biofilm formation in the food industry. *Continued*

Biosurfactant/ biosurfactant-producing microorganism	Applications	Food or related surfaces	References
Surfactin and Rhamnolipids	*Listeria monocytogenes*	Polystyrene Surface	[80]
Sophorolipids	Antibiofilm	*Escherichia coli*, *Pseudomonas aeruginosa*	[95]
Sophorolipids	Antimicrobial	*Escherichia coli* O157:H7	[96]
Streptococcus thermophilus	*Streptococcus* sp.	Heat-exchanger plates in pasteurizers	[97]
Surfactin and rhamnolipid	*Listeria monocytogenes*, *Staphylococcus aureus*, *Micrococcus luteus*	Polystyrene Surface	[98]
Surfactin and rhamnolipid	*Enterobacter sakazakii*, *Listeria monocytogenes*, *Salmonella enteritidis*	Polypropylene and polystyrene	[92]

inhibited the formation of biofilm on both hydrophobic and hydrophilic surfaces [101].

Glycolipid biosurfactants based on palmitic acid and glucose secreted by *Serratia marcescens* was effectual in inhibiting the *Bacillus pumilus* biofilms and also *P. aeruginosa* PAO1 and *C. albicans* adhesion [84]. Other glycolipids complex produced by *Brevibacterium casei* MSA19 reported to considerably inhibit and disrupt mixed culture and individual biofilms of fish and human at 30 mg/mL concentrations [102].

Glycolipids having antibiofilm property were derived from *Lactobacillus* [103]. Glycolipid biosurfactants produced from *Lactobacillus paracasei* A20 inhibit the biofilm formation of yeast, filamentous fungi, and both Gram-positive and Gram-negative bacteria [104]. Against the pathogenic *S. epidermidis*, *S. aureus*, *C. albicans*, *Streptococcus agalactiae*, and *E. coli*, this glycolipid biosurfactant exhibited antiadhesive activity. Plants produced glycolipids also have been recorded to inhibit biofilms. This includes hydroxyproline-rich biosurfactant from datucin eradicating biofilms of *C. albicans*, which is antifungal-resistant [105].

6.11 Fungal biosurfactants

Biosurfactants that have antibiofilm activity have been seen in *C. bombicola* (fungi). This gives rise to sophorolipid biosurfactant that inhibits *Vibrio cholerae* biofilm [106]. Lunasan produced from *Candida sphaerica* [107] is effective in adhesion

inhibition of *P. aeruginosa*, *Streptococcus sanguinis*, and *S. agalactiae* up to 92%. Similarly, *Candida lipolytica* produced rufisan inhibits the formation of biofilm by *S. agalactiae*, *S. aureus*, and *S. mutans*, at concentrations equal to or greater than 0.75 μg/mL [108].

7 Conclusion

Biofilms of food-borne pathogens are significant in food-borne outbreaks and intoxications, posing serious threat to consumer life. These problems are enhanced by the formation of biofilms resistant to regular disinfectants in the food processing industry; for these problems biosurfactants represent emerging solution. Antibiofilm activity of biosurfactants inhibits the formation of biofilm and also preexisted biofilms by food-borne pathogens Therefore, biosurfactants are great potential compounds on food, health, and biomedical-related issues. It will be fascinating to continue study on microbial surfactants as coating agents to build novel biomaterials, which can able to reduce initial attachment of microorganisms.

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References

[1] S.M. Ribeiro, M.R. Felício, E.V. Boas, S. Gonçalves, F.F. Costa, R.P. Samy, N.C. Santos, O.L. Franco, New frontiers for anti-biofilm drug development, Pharmacol. Ther. 160 (2016) 133–144, https://doi.org/10.1016/j.pharmthera.2016.02.006.

[2] I.M. Banat, M.A.D. De Rienzo, G.A. Quinn, Microbial biofilms: biosurfactants as antibiofilm agents, Appl. Microbiol. Biotechnol. 98 (24) (2014) 9915–9929, https://doi.org/10.1007/s00253-014-6169-6.

[3] K. Paraszkiewicz, M. Moryl, G. Płaza, D. Bhagat, K.S. Satpute, P. Bernat, Surfactants of microbial origin as antibiofilm agents, Int. J. Environ. Health Res. (2019) 1–20, https://doi.org/10.1080/09603123.2019.1664729.

[4] D. Sharma, Biosurfactants in food, in: Surfactant Science Series, 2016, https://doi.org/10.1007/978-3-319-39415-2.

[5] X. Shi, X. Zhu, Biofilm formation and food safety in food industries, Trends Food Sci. Technol. 20 (9) (2009) 407–413, https://doi.org/10.1016/j.tifs.2009.01.054.

[6] L.K. Vestby, T. Møretrø, S. Langsrud, E. Heir, L.L. Nesse, Biofilm forming abilities of *Salmonella* are correlated with persistence in fish meal- and feed factories, BMC Vet. Res. 5 (2009) 1–6, https://doi.org/10.1186/1746-6148-5-20.

[7] M. Simões, L.C. Simões, M.J. Vieira, A review of current and emergent biofilm control strategies, LWT Food Sci. Technol. 43 (4) (2010) 573–583, https://doi.org/10.1016/j.lwt.2009.12.008.
[8] S. Galié, C. García-Gutiérrez, E.M. Miguélez, C.J. Villar, F. Lombó, Biofilms in the food industry: health aspects and control methods, Front. Microbiol. 9 (MAY) (2018) 1–18, https://doi.org/10.3389/fmicb.2018.00898.
[9] L. Rodrigues, I.M. Banat, J. Teixeira, R. Oliveira, Biosurfactants: potential applications in medicine, J. Antimicrob. Chemother. 57 (4) (2006) 609–618, https://doi.org/10.1093/jac/dkl024.
[10] D. Davies, Understanding biofilm resistance to antibacterial agents, Nat. Rev. Drug Discov. 2 (2003) 114–122, https://doi.org/10.1038/nrd1008.
[11] S.T. Rutherford, B.L. Bassler, Bacterial quorum sensing: its role in virulence and possibilities for its control, Cold Spring Harb. Perspect. Med. 2 (11) (2012), https://doi.org/10.1101/cshperspect.a012427.
[12] H.C. Flemming, J. Wingender, U. Szewzyk, P. Steinberg, S.A. Rice, S. Kjelleberg, Biofilms: an emergent form of bacterial life, Nat. Rev. Microbiol. 14 (9) (2016) 563–575, https://doi.org/10.1038/nrmicro.2016.94.
[13] C. Delcaru, I. Alexandru, P. Podgoreanu, M. Grosu, E. Stavropoulos, M.C. Chifiriuc, V. Lazar, Microbial biofilms in urinary tract infections and prostatitis: etiology, pathogenicity, and combating strategies, Pathogens 5 (4) (2016), https://doi.org/10.3390/pathogens5040065.
[14] P. Stoodley, K. Sauer, D.G. Davies, J.W. Costerton, Biofilms as complex differentiated communities, Annu. Rev. Microbiol. 56 (2002) 187–209, https://doi.org/10.1146/annurev.micro.56.012302.160705.
[15] J.M.B. Technol, J. Bueno, Anti-biofilm drug susceptibility testing methods: looking for new strategies against resistance mechanism, Microb. Biochem. Technol. (2014), https://doi.org/10.4172/1948-5948.S3-004.
[16] L.P. Girard, H. Ceri, A.P. Gibb, M. Olson, F. Sepandj, MIC versus MBEC to determine the antibiotic sensitivity of *Staphylococcus aureus* in peritoneal dialysis peritonitis, Perit. Dial. Int. 30 (2009) 652–656, https://doi.org/10.3747/pdi.2010.00010.
[17] G. Croda-García, V. Grosso-Becerra, A. Gonzalez-Valdez, L. Servín-González, G. Soberón-Chávez, Transcriptional regulation of *Pseudomonas aeruginosa* rhlR: role of the CRP orthologue Vfr (virulence factor regulator) and quorum-sensing regulators LasR and RhlR, Microbiology 157 (2011) 2545–2555, https://doi.org/10.1099/mic.0.050161-0.
[18] J. de Cássia Orlandi Sardi, N. de Souza Pitangui, A.R. Voltan, J.D. Braz, M.P. Machado, A.M. Fusco Almeida, M.J.S. Mendes Giannini, In vitro *Paracoccidioides brasiliensis* biofilm and gene expression of adhesins and hydrolytic enzymes, Virulence 6 (6) (2015) 642–651, https://doi.org/10.1080/21505594.2015.1031437.
[19] C.B. Costa-Orlandi, J.C.O. Sardi, N.S. Pitangui, H.C. de Oliveira, L. Scorzoni, M.C. Galeane, K.P. Medina-Alarcón, W.C.M.A. Melo, M.Y. Marcelino, J.D. Braz, A.M. Fusco-Almeida, M.J.S. Mendes-Giannini, Fungal biofilms and polymicrobial diseases, J. Fungi 3 (2) (2017) 1–24, https://doi.org/10.3390/jof3020022.
[20] Y.A. Nikolaev, V.K. Plakunov, Biofilm—"city of microbes" or an analogue of multicellular organisms? Microbiology 76 (2) (2007) 125–138, https://doi.org/10.1134/S0026261707020014.
[21] M. Abdallah, O. Khelissa, A. Ibrahim, C. Benoliel, L. Heliot, P. Dhulster, N.E. Chihib, Impact of growth temperature and surface type on the resistance of *Pseudomonas*

aeruginosa and *Staphylococcus aureus* biofilms to disinfectants, Int. J. Food Microbiol. 214 (2015) 38–47, https://doi.org/10.1016/j.ijfoodmicro.2015.07.022.

[22] M.F.R. Mizan, I.K. Jahid, S.D. Ha, Microbial biofilms in seafood: a food-hygiene challenge, Food Microbiol. 49 (2015) 41–55, https://doi.org/10.1016/j.fm.2015.01.009.

[23] A.C. Camargo, J.J. Woodward, D.R. Call, L.A. Nero, *Listeria monocytogenes* in food-processing facilities, food contamination, and human Listeriosis: the Brazilian scenario, Foodborne Pathog. Dis. 14 (11) (2017) 623–636, https://doi.org/10.1089/fpd.2016.2274.

[24] E.J. Bottone, *Bacillus cereus*, a volatile human pathogen, Clin. Microbiol. Rev. 23 (2) (2010) 382–398, https://doi.org/10.1128/CMR.00073-09.

[25] A.C. López, J. Minnaard, P.F. Pérez, A.M. Alippi, A case of intoxication due to a highly cytotoxic *Bacillus cereus* strain isolated from cooked chicken, Food Microbiol. 46 (2015) 195–199, https://doi.org/10.1016/j.fm.2014.08.005.

[26] A. Soni, I. Oey, P. Silcock, P. Bremer, Bacillus spores in the food industry: a review on resistance and response to novel inactivation technologies, Compr. Rev. Food Sci. Food Saf. 15 (2016) 1139–1148, https://doi.org/10.1111/1541-4337.12231.

[27] V. Sébastien, J.M. Pretorius, T. Jacques, V.S. Brözel, DNA as an adhesin: *Bacillus cereus* requires extracellular DNA to form biofilms, Appl. Environ. Microbiol. 75 (2009) 2861–2868, https://doi.org/10.1128/AEM.01317-08.

[28] S. Marchand, J. De Block, V. De Jonghe, A. Coorevits, M. Heyndrickx, L. Herman, Biofilm formation in milk production and processing environments; influence on milk quality and safety, Compr. Rev. Food Sci. Food Saf. 11 (2) (2012) 133–147, https://doi.org/10.1111/j.1541-4337.2011.00183.x.

[29] H. Hayrapetyan, M. Tempelaars, M. Nierop Groot, T. Abee, *Bacillus cereus* ATCC 14579 RpoN (Sigma 54) is a pleiotropic regulator of growth, carbohydrate metabolism, motility, biofilm formation and toxin production, PLoS One 10 (2015), https://doi.org/10.1371/journal.pone.0134872, e0134872.

[30] S. Saleh-Lakha, C.G. Leon-Velarde, S. Chen, S. Lee, K. Shannon, M. Fabri, G. Downing, B. Keown, A study to assess the numbers and prevalence of *Bacillus cereus* and its toxins in pasteurized fluid milk, J. Food Prot. 80 (7) (2017) 1085–1089, https://doi.org/10.4315/0362-028X.JFP-16-521.

[31] EFSA Panel on Biological Hazards (BIOHAZ), Risks for public health related to the presence of *Bacillus cereus* and other *Bacillus* spp. including *Bacillus thuringiensis* in foodstuffs, EFSA J. 14 (7) (2016), https://doi.org/10.2903/j.efsa.2016.4524.

[32] G.T. Gunduz, G. Tuncel, Biofilm formation in an ice cream plant, Anton. Leeuw. Int. J. Gen. Mol. Microbiol. 89 (3–4) (2006) 329–336, https://doi.org/10.1007/s10482-005-9035-9.

[33] M.J. Piercey, P.A. Hingston, L. Truelstrup Hansen, Genes involved in *Listeria monocytogenes* biofilm formation at a simulated food processing plant temperature of 15 °C, Int. J. Food Microbiol. 223 (2016) 63–74, https://doi.org/10.1016/j.ijfoodmicro.2016.02.009.

[34] M. Sharma, S.K. Anand, Biofilms evaluation as an essential component of HACCP for food/dairy processing industry—a case, Food Control 13 (6–7) (2002) 469–477, https://doi.org/10.1016/S0956-7135(01)00068-8.

[35] M. Dzieciol, E. Schornsteiner, M. Muhterem-Uyar, B. Stessl, M. Wagner, S. Schmitz-Esser, Bacterial diversity of floor drain biofilms and drain waters in a *Listeria monocytogenes* contaminated food processing environment, Int. J. Food Microbiol. 223 (2016) 33–40, https://doi.org/10.1016/j.ijfoodmicro.2016.02.004.

[36] B. Carpentier, D. Chassaing, Interactions in biofilms between *Listeria monocytogenes* and resident microorganisms from food industry premises, Int. J. Food Microbiol. 97 (2) (2004) 111–122, https://doi.org/10.1016/j.ijfoodmicro.2004.03.031.

[37] J. Wang, R. Li, L. Hu, X. Sun, J. Wang, J. Li, Development of a quantitative fluorescence single primer isothermal amplification-based method for the detection of *Salmonella*, Int. J. Food Microbiol. 219 (2016) 22–27, https://doi.org/10.1016/j.ijfoodmicro.2015.12.005.

[38] M. Kolari, J. Nuutinen, M.S. Salkinoja-Salonen, Mechanisms of biofilm formation in paper machine by *Bacillus* species: the role of *Deinococcus geothermalis*, J. Ind. Microbiol. Biotechnol. 27 (6) (2001) 343–351, https://doi.org/10.1038/sj.jim.7000201.

[39] A. Cherif-Antar, B. Moussa-Boudjemâa, N. Didouh, K. Medjahdi, B. Mayo, A.B. Flórez, Diversity and biofilm-forming capability of bacteria recovered from stainless steel pipes of a milk-processing dairy plant, Dairy Sci. Technol. 96 (1) (2016) 27–38, https://doi.org/10.1007/s13594-015-0235-4.

[40] F. Tang, H.Y. Hu, L.J. Sun, Y.X. Sun, N. Shi, J.C. Crittenden, Fouling characteristics of reverse osmosis membranes at different positions of a full-scale plant for municipal wastewater reclamation, Water Res. 90 (2016) 329–336, https://doi.org/10.1016/j.watres.2015.12.028.

[41] M.Q. Carter, J.W. Louie, D. Feng, W. Zhong, M.T. Brandl, Curli fimbriae are conditionally required in *Escherichia coli* O157:H7 for initial attachment and biofilm formation, Food Microbiol. 57 (2016) 81–89, https://doi.org/10.1016/j.fm.2016.01.006.

[42] D. Dourou, C.S. Beauchamp, Y. Yoon, I. Geornaras, K.E. Belk, G.C. Smith, G.J.E. Nychas, J.N. Sofos, Attachment and biofilm formation by *Escherichia coli* O157:H7 at different temperatures, on various food-contact surfaces encountered in beef processing, Int. J. Food Microbiol. 149 (3) (2011) 262–268, https://doi.org/10.1016/j.ijfoodmicro.2011.07.004.

[43] A. Álvarez-Ordóñez, O. Alvseike, M.K. Omer, E. Heir, L. Axelsson, A. Holck, M. Prieto, Heterogeneity in resistance to food-related stresses and biofilm formation ability among verocytotoxigenic *Escherichia coli* strains, Int. J. Food Microbiol. 161 (2013) 220–230, https://doi.org/10.1016/j.ijfoodmicro.2012.12.008.

[44] S.A. Lajhar, J. Brownlie, R. Barlow, Survival capabilities of *Escherichia coli* O26 isolated from cattle and clinical sources in Australia to disinfectants, acids and antimicrobials, BMC Microbiol. 17 (1) (2017) 1–10, https://doi.org/10.1186/s12866-017-0963-0.

[45] Y. Yang, M. Mikš-Krajnik, Q. Zheng, S.B. Lee, S.C. Lee, H.G. Yuk, Biofilm formation of *Salmonella enteritidis* under food-related environmental stress conditions and its subsequent resistance to chlorine treatment, Food Microbiol. 54 (2016) 98–105, https://doi.org/10.1016/j.fm.2015.10.010.

[46] N. Marouani-Gadri, O. Firmesse, D. Chassaing, D. Sandris-Nielsen, N. Arneborg, B. Carpentier, Potential of *Escherichia coli* O157:H7 to persist and form viable but non-culturable cells on a food-contact surface subjected to cycles of soiling and chemical treatment, Int. J. Food Microbiol. 144 (1) (2010) 96–103, https://doi.org/10.1016/j.ijfoodmicro.2010.09.002.

[47] S.R. Milillo, E.C. Friedly, J.C. Saldivar, A. Muthaiyan, C. O'Bryan, P.G. Crandall, M. G. Johnson, S.C. Ricke, A review of the ecology, genomics, and stress response of *Listeria innocua* and *Listeria monocytogenes*, Crit. Rev. Food Sci. Nutr. 52 (8) (2012) 712–725, https://doi.org/10.1080/10408398.2010.507909.

[48] V. Ferreira, M. Wiedmann, P. Teixeira, M.J. Stasiewicz, *Listeria monocytogenes* persistence in food-associated environments: epidemiology, strain characteristics, and

implications for public health, J. Food Prot. 77 (1) (2014) 150–170, https://doi.org/10.4315/0362-028X.JFP-13-150.
[49] CDC, 2017, https://www.cdc.gov/listeria/outbreaks/index.html.
[50] S. Silva, P. Teixeira, R. Oliveira, J. Azeredo, Adhesion to and viability of *Listeria monocytogenes* on food contact surfaces, J. Food Prot. 71 (7) (2008) 1379–1385, https://doi.org/10.4315/0362-028X-71.7.1379.
[51] T. Zhang, D. Bae, C. Wang, *Listeria monocytogenes* DNA glycosylase AdlP affects flagellar motility, biofilm formation, virulence, and stress responses, Appl. Environ. Microbiol. 82 (17) (2016) 5144–5152, https://doi.org/10.1128/AEM.00719-16.
[52] T. Møretrø, B.C.T. Schirmer, E. Heir, A. Fagerlund, P. Hjemli, S. Langsrud, Tolerance to quaternary ammonium compound disinfectants may enhance growth of *Listeria monocytogenes* in the food industry, Int. J. Food Microbiol. 241 (2017) 215–224, https://doi.org/10.1016/j.ijfoodmicro.2016.10.025.
[53] M. Kanki, H. Naruse, K. Kawatsu, Comparison of listeriolysin O and phospholipases PlcA and PlcB activities, and initial intracellular growth capability among food and clinical strains of *Listeria monocytogenes*, J. Appl. Microbiol. 124 (2018) 899–909, https://doi.org/10.1111/jam.13692.
[54] European Food Safety Authority, European Centre for Disease Prevention and Control, The European Union summary report on trends and sources of zoonoses, zoonotic agents and food-borne outbreaks in 2016, EFSA J. 15 (2017) e05077, doi:10.2903/j.efsa.2017.5077.
[55] E. Giaouris, E. Heir, M. Hébraud, N. Chorianopoulos, S. Langsrud, T. Møretrø, O. Habimana, M. Desvaux, S. Renier, G.J. Nychas, Attachment and biofilm formation by foodborne bacteria in meat processing environments: causes, implications, role of bacterial interactions and control by alternative novel methods, Meat Sci. 97 (3) (2014) 298–309, https://doi.org/10.1016/j.meatsci.2013.05.023.
[56] J. Schelin, Y.B. Susilo, S. Johler, Expression of staphylococcal enterotoxins under stress encountered during food production and preservation, Toxins 9 (12) (2017), https://doi.org/10.3390/toxins9120401.
[57] M. Slany, J. Oppelt, L. Cincarova, Formation of *Staphylococcus aureus* biofilm in the presence of sublethal concentrations of disinfectants studied via a transcriptomic analysis using transcriptome sequencing (RNA-seq), Appl. Environ. Microbiol. 83 (24) (2017), https://doi.org/10.1128/AEM.01643-17.
[58] D.C.V. De Oliveira, A. Fernandes Júnior, R. Kaneno, M.G. Silva, J.P. Araújo Júnior, N. C.C. Silva, V.L.M. Rall, Ability of *Salmonella* spp. to produce biofilm is dependent on temperature and surface material, Foodborne Pathog. Dis. 11 (6) (2014) 478–483, https://doi.org/10.1089/fpd.2013.1710.
[59] S. Zhang, R.A. Kingsley, R.L. Santos, H. Andrews-Polymenis, M. Raffatellu, J. Figueiredo, J. Nunes, R.M. Tsolis, L.G. Adams, A.J. Ba, Molecular pathogenesis of *Salmonella enterica* serotype, Infect. Immun. 71 (1) (2003) 1–12, https://doi.org/10.1128/IAI.71.1.1.
[60] S.K. Collinson, S.C. Clouthier, J.L. Doran, P.A. Banser, W.W. Kay, *Salmonella enteritidis* agfBAC operon encoding thin, aggregative fimbriae, J. Bacteriol. 178 (1996) 662–667, https://doi.org/10.1128/jb.178.3.662-667.1996.
[61] H.D.N. Nguyen, Y.S. Yang, H.G. Yuk, Biofilm formation of *Salmonella* Typhimurium on stainless steel and acrylic surfaces as affected by temperature and pH level, LWT Food Sci. Technol. 55 (1) (2014) 383–388, https://doi.org/10.1016/j.lwt.2013.09.022.

[62] D.B. Schlisselberg, S. Yaron, The effects of stainless steel finish on *Salmonella* Typhimurium attachment, biofilm formation and sensitivity to chlorine, Food Microbiol. 35 (1) (2013) 65–72, https://doi.org/10.1016/j.fm.2013.02.005.

[63] H. Wang, S. Ding, G. Wang, X. Xu, G. Zhou, In situ characterization and analysis of *Salmonella* biofilm formation under meat processing environments using a combined microscopic and spectroscopic approach, Int. J. Food Microbiol. 167 (3) (2013) 293–302, https://doi.org/10.1016/j.ijfoodmicro.2013.10.005.

[64] T. Møretrø, E. Heir, L.L. Nesse, L.K. Vestby, S. Langsrud, Control of Salmonella in food related environments by chemical disinfection, Food Res. Int. 45 (2012) 532–544, https://doi.org/10.1016/j.foodres.2011.02.002.

[65] D.E. Norwood, A. Gilmour, The growth and resistance to sodium hypochlorite of *Listeria monocytogenes* in a steady-state multispecies biofilm, J. Appl. Microbiol. 88 (2000) 512–520, https://doi.org/10.1046/j.1365-2672.2000.00990.x.

[66] E. Torlak, D. Sert, Combined effect of benzalkonium chloride and ultrasound against *Listeria monocytogenes* biofilm on plastic surface, Lett. Appl. Microbiol. 57 (3) (2013) 220–226, https://doi.org/10.1111/lam.12100.

[67] Y. Pecci, F. Rivardo, M.G. Martinotti, G. Allegrone, LC/ESI-MS/MS characterisation of lipopeptide biosurfactants produced by the *Bacillus licheniformis* V9T14 strain, J. Mass Spectrom. 45 (2010) 772–778, https://doi.org/10.1002/jms.1767.

[68] G.A. Quinn, A.P. Maloy, S. McClean, B. Carney, J.W. Slater, Lipopeptide biosurfactants from *Paenibacillus polymyxa* inhibit single and mixed species biofilms, Biofouling 28 (10) (2012) 1151–1166, https://doi.org/10.1080/08927014.2012.738292.

[69] N.P.J. Price, A.P. Rooney, J.L. Swezey, E. Perry, F.M. Cohan, Mass spectrometric analysis of lipopeptides from *Bacillus* strains isolated from diverse geographical locations, FEMS Microbiol. Lett. 271 (1) (2007) 83–89, https://doi.org/10.1111/j.1574-6968.2007.00702.x.

[70] J. He, K.R. Ledesma, W.-Y. Lam, D.A. Figueroa, T.-P. Lim, D.S.-L. Chow, V.H. Tam, Variability of polymyxin B major components in commercial formulations, Int. J. Antimicrob. Agents 35 (2010) 308–310, https://doi.org/10.1016/j.ijantimicag.2009.11.005.

[71] M.E. Falagas, S.K. Kasiakou, Colistin: the revival of polymyxins for the management of multidrug-resistant gram-negative bacterial infections, Clin. Infect. Dis. 40 (2005) 1333–1342, https://doi.org/10.1086/429323.

[72] I. Aleksic, M. Petkovic, M. Jovanovic, D. Milivojevic, B. Vasiljevic, J. Nikodinovic-Runic, L. Senerovic, Anti-biofilm properties of bacterial di-rhamnolipids and their semi-synthetic amide derivatives, Front. Microbiol. 8 (DEC) (2017) 1–16, https://doi.org/10.3389/fmicb.2017.02454.

[73] M.M. Domingues, R.G. Inacio, J.M. Raimundo, M. Martins, M.A.R.B. Castanho, N.C. Santos, Biophysical characterization of polymyxin B interaction with LPS aggregates and membrane model systems, Biopolymers 98 (4) (2012) 338–344, https://doi.org/10.1002/bip.22095.

[74] F. Rivardo, M.G. Martinotti, R.J. Turner, H. Ceri, The activity of silver against *Escherichia coli* biofilm is increased by a lipopeptide biosurfactant, Can. J. Microbiol. 56 (3) (2010) 272–278, https://doi.org/10.1139/W10-007.

[75] J.-F. Dubern, B.J.J. Lugtenberg, G.V. Bloemberg, The ppuI-rsaL-ppuR quorum-sensing system regulates biofilm formation of *Pseudomonas putida* PCL1445 by controlling biosynthesis of the cyclic lipopeptides putisolvins I and II, J. Bacteriol. 188 (2006) 2898–2906, https://doi.org/10.1128/JB.188.8.2898-2906.2006.

[76] I. Kuiper, E.L. Lagendijk, R. Pickford, J.P. Derrick, G.E.M. Lamers, J.E. Thomas-Oates, B.J.J. Lugtenberg, G.V. Bloemberg, Characterization of two *Pseudomonas putida* lipopeptide biosurfactants, putisolvin I and II, which inhibit biofilm formation and break down existing biofilms, Mol. Microbiol. 51 (2004) 97–113, https://doi.org/10.1046/j.1365-2958.2003.03751.x.

[77] T. Janek, Ł. Marcin, A. Krasowska, Antiadhesive activity of the biosurfactant pseudofactin II secreted by the Arctic bacterium *Pseudomonas fluorescens* BD5, BMC Microbiol. 12 (24) (2012) 1–9, https://doi.org/10.1186/1471-2180-12-24.

[78] L. D'Auria, M. Deleu, S. Dufour, M.P. Mingeot-Leclercq, D. Tyteca, Surfactins modulate the lateral organization of fluorescent membrane polar lipids: a new tool to study drug: membrane interaction and assessment of the role of cholesterol and drug acyl chain length, Biochim. Biophys. Acta Biomembr. 1828 (9) (2013) 2064–2073, https://doi.org/10.1016/j.bbamem.2013.05.006.

[79] J.R. Mireles 2nd, A. Toguchi, R.M. Harshey, *Salmonella enterica* serovar typhimurium swarming mutants with altered biofilm-forming abilities: surfactin inhibits biofilm formation, J. Bacteriol. 183 (2001) 5848–5854, https://doi.org/10.1128/JB.183.20.5848-5854.2001.

[80] L.V. de Araujo, C.R. Guimarães, R.L.S. da Marquita, V.M.J. Santiago, M.P. de Souza, M. Nitschke, D.M.G. Freire, Rhamnolipid and surfactin: anti-adhesion/antibiofilm and antimicrobial effects, Food Control 63 (2016) 171–178, https://doi.org/10.1016/j.foodcont.2015.11.036.

[81] J. Coronel-León, A.M. Marqués, J. Bastida, A. Manresa, Optimizing the production of the biosurfactant lichenysin and its application in biofilm control, J. Appl. Microbiol. 120 (1) (2016) 99–111, https://doi.org/10.1111/jam.12992.

[82] R. Rautela, A. Kumar, S. Abha, Lipopeptides from *Bacillus* strain AR2 inhibits biofilm formation by *Candida albicans*, Antonie Van Leeuwenhoek 105 (2014) 809–821, https://doi.org/10.1007/s10482-014-0135-2.

[83] A.M. Abdel-Mawgoud, Rhamnolipids: diversity of structures, microbial origins and roles, Appl. Microbiol. Biotechnol. 86 (5) (2010) 1323–1336, https://doi.org/10.1007/s00253-010-2498-2.

[84] D.H. Dusane, Y.V. Nancharaiah, S.S. Zinjarde, V.P. Venugopalan, Rhamnolipid mediated disruption of marine *Bacillus pumilus* biofilms, Colloids Surf. B Biointerfaces 81 (1) (2010) 242–248, https://doi.org/10.1016/j.colsurfb.2010.07.013.

[85] M. Zezzi do Valle Gomes, M. Nitschke, Evaluation of rhamnolipid and surfactin to reduce the adhesion and remove biofilms of individual and mixed cultures of food pathogenic bacteria, Food Control 25 (2) (2012) 441–447, https://doi.org/10.1016/j.foodcont.2011.11.025.

[86] B. Dahrazma, C.N. Mulligan, M.P. Nieh, Effects of additives on the structure of rhamnolipid (biosurfactant): a small-angle neutron scattering (SANS) study, J. Colloid Interface Sci. 319 (2) (2008) 590–593, https://doi.org/10.1016/j.jcis.2007.11.045.

[87] M.J. Torres, G. Petroselli, M. Daz, R. Erra-Balsells, M.C. Audisio, *Bacillus subtilis* subsp. *subtilis* CBMDC3f with antimicrobial activity against Gram-positive foodborne pathogenic bacteria: UV-MALDI-TOF MS analysis of its bioactive compounds, World J. Microbiol. Biotechnol. 31 (6) (2015) 929–940, https://doi.org/10.1007/s11274-015-1847-9.

[88] D.C. Sabaté, M.C. Audisio, Inhibitory activity of surfactin, produced by different *Bacillus subtilis* subsp. *subtilis* strains, against *Listeria monocytogenes* sensitive and

bacteriocin-resistant strains, Microbiol. Res. 168 (3) (2013) 125–129, https://doi.org/10.1016/j.micres.2012.11.004.

[89] D. Sharma, B.S. Saharan, N. Chauhan, S. Procha, S. Lal, Isolation and functional characterization of novel biosurfactant produced by *Enterococcus faecium*, Springer Plus 4 (1) (2015) 1–14, https://doi.org/10.1186/2193-1801-4-4.

[90] A.N. Madhu, S.G. Prapulla, Evaluation and functional characterization of a biosurfactant produced by *Lactobacillus plantarum* CFR 2194, Appl. Biochem. Biotechnol. 172 (4) (2014) 1777–1789, https://doi.org/10.1007/s12010-013-0649-5.

[91] T. Meylheuc, C. Methivier, M. Renault, J.M. Herry, C.M. Pradier, M.N. Bellon-Fontaine, Adsorption on stainless steel surfaces of biosurfactants produced by gram-negative and gram-positive bacteria: consequence on the bioadhesive behavior of *Listeria monocytogenes*, Colloids Surf. B Biointerfaces 52 (2) (2006) 128–137, https://doi.org/10.1016/j.colsurfb.2006.04.016.

[92] M. Nitschke, L.V. Araújo, S.G.V.A.O. Costa, R.C. Pires, A.E. Zeraik, A.C.L.B. Fernandes, D.M.G. Freire, J. Contiero, Surfactin reduces the adhesion of food-borne pathogenic bacteria to solid surfaces, Lett. Appl. Microbiol. 49 (2) (2009) 241–247, https://doi.org/10.1111/j.1472-765X.2009.02646.x.

[93] T. Meylheuc, C.J. Van Oss, M.N. Bellon-Fontaine, Adsorption of biosurfactant on solid surfaces and consequences regarding the bioadhesion of *Listeria monocytogenes* LO28, J. Appl. Microbiol. 91 (5) (2001) 822–832, https://doi.org/10.1046/j.1365-2672.2001.01455.x.

[94] C. Dagbert, T. Meylheuc, M.N. Bellon-Fontaine, Corrosion behaviour of AISI 304 stainless steel in presence of a biosurfactant produced by *Pseudomonas fluorescens*, Electrochim. Acta 51 (24) (2006) 5221–5227, https://doi.org/10.1016/j.electacta.2006.03.063.

[95] M.A. Díaz De Rienzo, I.M. Banat, B. Dolman, J. Winterburn, P.J. Martin, Sophorolipid biosurfactants: possible uses as antibacterial and antibiofilm agent, N. Biotechnol. 32 (6) (2015) 720–726, https://doi.org/10.1016/j.nbt.2015.02.009.

[96] X. Zhang, X. Fan, D.K.Y. Solaiman, R.D. Ashby, Z. Liu, S. Mukhopadhyay, R. Yan, Inactivation of *Escherichia coli* O157:H7 in vitro and on the surface of spinach leaves by biobased antimicrobial surfactants, Food Control 60 (2016) 158–165, https://doi.org/10.1016/j.foodcont.2015.07.026.

[97] H.J. Busscher, M. Van Der Kuijl-Booij, H.C. Van Der Mei, Biosurfactants from thermophilic dairy streptococci and their potential role in the fouling control of heat exchanger plates, J. Ind. Microbiol. 16 (1) (1996) 15–21, https://doi.org/10.1007/BF01569916.

[98] A.E. Zeraik, M. Nitschke, Biosurfactants as agents to reduce adhesion of pathogenic bacteria to polystyrene surfaces: effect of temperature and hydrophobicity, Curr. Microbiol. 61 (6) (2010) 554–559, https://doi.org/10.1007/s00284-010-9652-z.

[99] M.A.D. De Rienzo, P.J. Martin, Effect of mono and di-rhamnolipids on biofilms preformed by *Bacillus subtilis* BBK006, Curr. Microbiol. 73 (2016) 183–189, https://doi.org/10.1007/s00284-016-1046-4.

[100] F. Haque, M. Sajid, S.S. Cameotra, M.S. Battacharyya, Anti-biofilm activity of a sophorolipid-amphotericin B niosomal formulation against *Candida albicans*, Biofouling 33 (9) (2017) 768–779, https://doi.org/10.1080/08927014.2017.1363191.

[101] A.K. Pradhan, N. Pradhan, L.B. Sukla, P.K. Panda, B.K. Mishra, Inhibition of pathogenic bacterial biofilm by biosurfactant produced by *Lysinibacillus fusiformis* S9,

Bioprocess Biosyst. Eng. 37 (2) (2014) 139–149, https://doi.org/10.1007/s00449-013-0976-5.

[102] G.S. Kiran, B. Sabarathnam, J. Selvin, C.J. Selvin, Biofilm disruption potential of a glycolipid biosurfactant from marine *Brevibacterium casei*, FEMS Immunol. Med. Microbiol. 59 (2010) 432–438, https://doi.org/10.1111/j.1574-695X.2010.00698.x.

[103] E. Zakaria Gomaa, Antimicrobial and anti-adhesive properties of biosurfactant produced by lactobacilli isolates, biofilm formation and aggregation ability, J. Gen. Appl. Microbiol. 59 (2013) 425–436, https://doi.org/10.2323/jgam.59.425.

[104] E.J. Gudiña, J.A. Teixeira, L.R. Rodrigues, Isolation and functional characterization of a biosurfactant produced by *Lactobacillus paracasei*, Colloids Surf. B Biointerfaces 76 (1) (2010) 298–304, https://doi.org/10.1016/j.colsurfb.2009.11.008.

[105] S.M. Mandal, A novel hydroxyproline rich glycopeptide from pericarp of *Datura stramonium*: proficiently eradicate the biofilm of antifungals resistant *Candida albicans*, Pept. Sci. 98 (2012) 332–337, https://doi.org/10.1002/bip.22083.

[106] R. Mukherji, A. Prabhune, Novel glycolipids synthesized using plant essential oils and their application in quorum sensing inhibition and as antibiofilm agents, Sci. World J. 2014 (2014) 1–7, https://doi.org/10.1155/2014/890709.

[107] J.M. Luna, R.D. Rufino, L.A. Sarubbo, L.R.M. Rodrigues, J.A.C. Teixeira, G.M. de Campos-Takaki, Evaluation antimicrobial and antiadhesive properties of the biosurfactant Lunasan produced by *Candida sphaerica* UCP 0995, Curr. Microbiol. 62 (2011) 1527–1534, https://doi.org/10.1007/s00284-011-9889-1.

[108] R.D. Rufino, J.M. Luna, L.A. Sarubbo, L.R.M. Rodrigues, J.A.C. Teixeira, G.M. Campos-Takaki, Antimicrobial and anti-adhesive potential of a biosurfactant Rufisan produced by *Candida lipolytica* UCP 0988, Colloids Surf. B Biointerfaces 84 (2011) 1–5, https://doi.org/10.1016/j.colsurfb.2010.10.045.

CHAPTER

Applications of biosurfactants in dairy industry

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Over the centuries, surfactant compounds have been used extensively for manufacturing variety of food products. Fermented and nonfermented dairy products, baked foods, breweries often use surfactants that are artificially synthesized or extracted from natural sources. In many dairy products, such as milk, curd, cheese, the composition of a food-based surfactants used is of a permissible nature. Other properties offered by the biosurfactants are fat flavor enhancement, improvement of bakery products, and formulation of milk products [1,2].

1 Applications of biosurfactants in dairy industry as emulsifiers/deemulsifiers

Food emulsions are intricate systems that are stable kinetically but with no balance. Emulsions consist of two immiscible liquids usually oil and water dispersed into each other by the incorporation of an emulsifier or by employing an emulsification technique. The stability of an emulsion is highly dependent on the variables such as the properties of emulsifier, composition of liquid phases, processing conditions, etc. [3]. Emulsifiers are amphiphiles having lower molecular weight and play significant role in the stability of food emulsions by rapidly adsorbing on oil-water interface. They are surface-active agents that reduce the interfacial tension between the immiscible liquid phases, prevent coagulation, consequently leading to the formation of stable emulsion with improved texture and consistency [4,5]. Currently,

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biosurfactants found an extensive application as an emulsifier and deemulsifier in food and beverage industry in wide range of products [6].

Biosurfactants are surface-active agents that are produced by the microbial cells such as bacteria, fungi, and yeast and reduce the interfacial tension. Biosurfactants exhibit low toxicity, increased biodegradability, and their ability to remain unaffected even at higher salt concentration and pH in comparison to synthetic surfactants. They possess extensive market potential in food industry due to their positive environmental impact along with nontoxic and biodegradable nature [7].

Examples of biosurfactants include glycolipids such as rhamnolipids, trehalose lipids, and sophorolipids; lipoproteins such as fengycin and surfactin; and lipopeptides. These microbial surfactants can be extracted from plant-based natural sources such as plant or by fermentation process using microorganisms. However, in both the scenarios, biosurfactants are first isolated from intricate mixture and then purified before their incorporation into food [8]. The applications of surfactants and emulsifiers have gained importance in the past few years. Their isolation and identification have been reported in various studies. Owing to consumer preference, the demand for the use of natural emulsifiers and surfactants as compared with synthetic ones in the food industries has increased significantly. Synthetic emulsifiers and surfactants have less popularity due to economic and health-related aspects. Hence, the use and demand of emulsifiers from natural sources have increased to a considerable extent [9].

Lipoprotein biosurfactants produced by certain *Bacillus* spp. including LB5a and rhamnolipids manufactured by *Pseudomonas* spp. including aeruginosa 47T2 help in the formation of sable emulsions with coconut fat and soybean oil. These applications make them useful for their use as emulsifiers in the food industries [10].

Rhamnolipid biosurfactants, also known as RL biosurfactants, manufactured from *Pseudomonas* spp. are among the most efficient surface-active substances. Some of their significant roles in food industries include applications as emulsifiers and antimicrobial agents. Emulsions are important components of food systems serving their roles as quality products including mayonnaise, margarine, and salad dressings. On the contrary, deemulsification is a process of breaking a stable emulsion, and this application is widely being used in food industries particularly oil and fat processing industries. RL biosurfactants have been reported to serve both functions, i.e., stabilizing and destabilizing of emulsions [11].

Moreover, these biosurfactants showed no toxic effects and have high emulsifying potential, thus can be used to enhance properties of food products. A study tested the antimicrobial efficiency of rhamnolipids derived from *Pseudomonas aeruginosa* p.a.28. The bacteria were isolated from contaminated soil and are among the best producers of rhamnolipids with high emulsifying properties. The inhibitory properties of this biosurfactant toward certain bacteria and yeasts were tested. Results showed maximum inhibition against *Bacillus subtilis* with inhibition zone of 30 mm, whereas the inhibition against *Micrococcus roseus* was found to be minimum with 10 mm inhibition zone. No significant toxic effects were observed with human blood and sensory attributes, i.e., taste, flavor, and appearance of mayonnaise and ice cream were found to be improved significantly [12].

In another study, emulsions having various fat fractions (MF13, 27, and 42) and anhydrous milk fat were developed at oil-in-water ratios of 1:9, 5:5, 7:3, and 3:7. These emulsions were prepared using rhamnolipids as emulsifying agents. These emulsions were assessed for their storage stability and physicochemical properties including color, zeta potential, rheological aspects, and particle size. Emulsions with an oil-in-water ratio of 7:3 revealed a condensed configuration irrespective of the type of fat used. The emulsion prepared from 1:9 oil-in-water ratio showed highest stability during 12-week storage period due to smaller particle size and reduced collision among droplets. Moreover, the presence of salts such as NaCl and calcium chloride resulted in lower stability of emulsion.

The polymeric configuration of biosurfactants comprising polysaccharides and protein structures enhances their emulsifying potential and increases the ability to stabilize emulsions. Bioemulsifiers produced by some yeast spp. particularly *Candida* and *Yarrowia* spp., come under GRAS category, which serves an important basis for their applications in food industries [13]. Ice cream consists of fat particles, air bubbles, and crystals of ice evenly scattered in sticky aqueous point making it an emulsion and foam. The surface activity of emulsifiers in ice cream is far greater as compared with milk proteins. Emulsifiers have the potential to displace proteins from the fat droplet area, disrupt the integrity of emulsion, and enhance the morphology of ice cream. The integration of surfactants to ice cream provided increased whipping ability, consistency, and resulted in smoother body and texture. A study was conducted to optimize pulsed electric field parameters following the isolation of natural saponins from Chubak root. Results exhibited a significant effect of voltage on the foaming ability and emulsion stability. An increase in voltage from 1 to 4 kV/cm decreased foam density and stability due to an increased isolation of saponins. This in turn resulted in an increased water retention, enhanced whipping ability, and greater fat globule stabilization [14].

2 Applications of biosurfactants in dairy industry as antimicrobials

Active surfactants such as sorbitan esters and their ethoxylate derivatives are being utilized in various food products [15,16]. Lipopeptides such as surfactant and other biosurfactants produced by *Bacillus* sp. such as fengycin, iturin, and bacillomycins are among the most popular biosurfactants that exhibit antimicrobial potential. Although, the surfactant exhibits hemolytic action alongside red blood cells [17]. The type of certain material of biosurfactants attained from pathogenic bacteria (such as *Pseudomonas aeruginosa*) can promote safety concerns in industry, and such biosurfactants are not used on direct food contact surfaces. In addition, further efforts have been made regarding the use of biosurfactants to change bacteria in foods. Huang et al. [18] stated that the amount of *Salmonella enteritidis* was remarkably dense in milk when 10 μg/mL combination of surfactin/polylysine compound was used in 1:1. The patent also proves the use of

rhamnolipids in food commodities such as milk, fruit, and vegetable as purifiers [19]. Biosurfactants produced by yeast can represent other safer substitutes in nutrition processing sector.

Antimicrobial exploitation of biosurfactants is centered on the disruption of membrane-bounding phospholipid layer, resulting in interruption of biological functions such as altering the membrane penetration into ion channels, through natural cation adducts, and through solubilizing membrane [20,21]. Rhamnolipids have been found effectual in inhibiting planktonic culture and *Listeria monocytogenes* [22]. Magalhães and Nitschke [23] examined the antimicrobial activity of rhamnolipids against different dietary variants of pathogenic *Listeria monocytogenes* and demonstrated strong susceptibility profile with minimum inhibitory concentration ranging from 78.1 to 2500 μg/mL. The utilization of biosurfactants in combination with bacteriocin such as nisin exhibited the potent bacteriostatic activity. It was proposed that the supposed synergism may be due to the fact that both molecules actively interact with molecules in the cell membrane. Considering the potent antimicrobial potential of nisin, it has found its application in biosurfactants system; for instance, their mix with bacteriocins can advance the improvement of dynamic antimicrobial bundling with dynamic fixing that upgrades their glue, antimicrobial, and synergistic potential.

3 Applications of biosurfactants in dairy industry as antibiofilm formation

Bacteria have the ability to remain attached to the industrial surfaces and result in formation of biofilms. Food processing surfaces can provide a variety of environments, which favors the development of biofilms. Biofilms are protective extracellular matrix composed of glycoprotein and polysaccharides formed by microorganisms to survive hostile environmental conditions such as temperature and chemicals. These are composed of microbial cells and extracellular polysaccharides commonly formed over the surfaces and equipment in food industry resultantly serve as a major source of contamination. Bacterial species such as *Bacillus cereus, Listeria monocytogenes, Escherichia coli* O157:H7, *Salmonella, Campylobacter jejuni,* and *Pseudomonas* are serious concern in food industry due to the formation of biofilm. In food industries, "real biofilms" are formed in closed surfaces such as hoses, water, or food piping systems such as metal, glass, plastic, or rubber surfaces.

In dairy industry, *B. cereus* can seriously affect the product safety and quality by producing extracellular enzymes, which can decrease the organoleptic properties of milk and milk products. Some spoilage psychrophilic microbes (*Pseudomonas*) can multiply exponentially at low temperature and form biofilms on inside surfaces of milk cooling tanks and pipelines [24].

Besides forming biofilm, these bacteria also release heat-stable lecithinases, photolytic and lipolytic enzymes may result in milk spoilage. In a study conducted on a

dairy plant in New Zealand dairy, six biofilm-producing cultures were isolated from internal surface of raw milk tanker, which can result in proteolysis. Lipolysis was induced by heat-stable lipases due to improperly cleaned milk tanker [25]. Lactic acid bacteria (LAB) are serious concern for food industry as they produce undesirable changes in food by forming biofilm. For example, *Lactobacillus curvatus* forms biofilm and produces isomers responsible for calcium lactate crystals in cheese [26].

In dairy industry, the biofilm fundamentally contains parts of natural milk. This first stage happens between the initial 5 and 10 s after the situation of a perfect region in any case turning into a wetland [27]. This condition additionally changes the physicochemical properties of the surface, such as changes in hydrophobicity, electrostatic charges, and surface free energy, which might influence resulting request of bacterial occasions [28]. Whole milk contains natural surfactants and phospholipids, both of the facial chemicals that can be returned to milk fat globules.

Surfactants are nontoxic compounds of biodegradable nature that have found extensive application in dairy industry as [29]:

1. as an emulsifier
2. antimicrobials
3. antiadhesive/antisticking agents.

Amphiphilic compounds of microbial origin, with the ability to alter hydrophobic properties of bacterial surface, are called biosurfactants. Biosurfactants work by affecting building and adhesion characteristics of biofilms. For example, *B. licheniform* produce cyclic nonribosomal lipopeptide, lichenysin, which on industrial scale is used in different concentrations against different bacteria. Adhesion properties of methicillin-resistant *Staphylococcus aureus* to 50% at a level of 8.3 mg/mL, for *Campylobacter jejuni* at 188.5 mg/mL, while microbes such as *Yersinia enterocolitica*, *Candida albicans* can be controlled at a level of 16.1 mg/mL and 17.2 mg/mL, respectively [30]. Compounds such as fengycin, iturin produced by *B. amyloliquefaciens* and surfactin produced by *B. subtilis* reduce surface tension, thus altering binding properties of target surface, entering the cell membrane, increasing membrane permeability by disrupting it, resulting in cell swelling and ultimate death [31].

The effectiveness of polysaccharides and proteases enzymes in removing *Bacillus* spp. biofilms was reported to be improved by incorporating buffer enzyme solution containing surfactants, decongestant, and decongestant agents [32]. Thus, the combination of enzymes targeting several components of extracellular polymeric substances, decongestant, decongestant agents along with surfactants would be suitable for application as chemical cleaning agents.

Yeh et al. [33] conducted research and found that the biosurfactants exhibit good emulsion strength when incorporated into ice cream at a concentration of 0.16 g/L with lecithin (1 g/L). These also improves organoleptic properties, quality, and overeating in ice cream. Huang et al. [34] conducted a research to investigate the antimicrobial potential of surfactin against various type of *Salmonella enteritidis* in milk. Surfactin at a concentration of 6.25 μg/mL exhibited a clear zone of inhibition against *Salmonella enteritidis* in agar well diffusion test method. Moreover, addition

of surfactin into milk at a dose of 6.25 μg/mL contaminated with *Salmonella enteritidis* reduced its count by 6 logs. Nayak et al. [35] in their study proved that the addition of biosurfactant at 0.1% concentration into the detergents reduced the residual biofilms of *Listeria monocytogenes* incidence by 95% and *Salmonella enteritidis* by 35%. Meena and Kanwar [36] have shown that surfactant provides adhesion function by preventing the formation of biofilm by two selected pathogenic strains of *Salmonella*. In conclusion, biosurfactants are produced by bacteria and fungi, mainly by bacteria. Biosurfactants are becoming more industrial reality than they were in the past. Synthetic surfactants are toxic and do not decompose that's why, biosurfactants are used because they are biodegradable and nontoxic. Biosurfactants can be produced from dairy industry waste such as whey cheese. They have dairy and other industrial applications to avoid environmental pollution.

4 Applications of biosurfactants in dairy industry as antiadhesive agent

Food contamination and adsorption of spoilage or pathogenic microorganisms in food surfaces, food processing materials, pipelines, and equipment can be prevented by implementing the biosurfactants to solid surfaces. In dairy industry, microbial adhesion and fouling are major problems encountered. The role of biosurfactants as antiadhesives has been studied. *Streptococcus thermophilus* releases a surfactant used to control the fouling in heat exchanger plates of pasteurizer by retarding the colonization of thermophilic strains of *Streptococcus* that may cause fouling [37]. In order to remove the *Staphylococcus aureus* biofilms established to polystyrene surfaces, a biosurfactant produced by *Pseudomonas aeruginosa*, Rhamnolipids is found to be effective up to 88% even at low temperature of 4°C to moderate 37°C. Their application in dairy industry has been suggested where low temperature is usually applied [38].

In order to reduce the adhesion, surface conditioning is also done by biosurfactants. Silicon rubber pipes and polystyrene packaging are widely used in dairy industry. Silicon is ideal material used for dairy tubing and has been approved by Food and Drug Administration as safe and hygienic tubing material. Molded polystyrene is used in dairy industry for the production of disposable cups and packaging material. By modifying the physicochemical properties of surface, biosurfactants can inhibit the formation of biofilm and inhibit the adhesion of microorganisms on biomaterials. Trehalose lipid, a biosurfactant produced by *Rhodococcus fascians* BD8, shows significant antiadhesive properties against Gram-negative, Gram-positive bacterial strains and some fungal strains, for example, *Candida albicans*. It has been reported that trehalose lipid has 70%–95% antiadhesive properties to polystyrene against *Candida albicans, Escherichia coli, Pseudomonas mirabilis, Enterococcus hirae*, and 40%–44% against *Enterococcus faecalis*, *Pseudomonas vulgaris,* and *Staphylococcus epidermidis* [39]. Silicon rubber pretreatment by *Streptococcus thermophilus* released surfactant which

inhibited the *Candida albicans* adhesion by 85% [40]. Biosurfactant released by *Lactobacilli* strains inhibits the adhesion of probiotic microorganisms [41]. As a new strategy to reduce the adhesion, surface bioconditioning by microbial biosurfactants is an emerging technique. Rhamnolipids at 100 mM concentration prevent bacterial adhesion and biofilm formed by *B. pumilus* [42]. *Candida lipolytica* UCP0988 produces a biosurfactant known as rufisan possessing antiadhesive activity at the concentration of 3–50 mg/L against various pathogenic bacteria, yeast, and molds [43]. It has been suggested that preconditioning of stainless-steel surfaces with biosurfactants can improve their antiadhesive properties, thereby contamination can be prevented. In the industry, preconditioning of stainless-steel plates or surfaces by *Pseudomonas fluorescence* biosurfactant inhibits the adhesion of *Listeria monocytogenes* L028 strain [29]. *Lactobacillus plantarum* biosurfactants show more than 50% antiadhesive property and behave as antimicrobial agents as well at concentration 4–25 mg/L against *Escherichia coli*, *Staphylococcus aureus and Salmonella typhi* [44]. Bacterial biofilm formation is a serious issue in dairy and food industry. The first step in biofilm formation is bacterial or any other organic matter that gains adherence and accumulation over any surface depending on many factors such as type of microorganism, environmental conditions, production of extracellular polymer by microorganism, and hydrophobicity. The adherence of microorganisms to any surface can be prevented by altering the hydrophobicity of surface. The biosurfactants are utilized to alter the hydrophobicity of surface, which in turn affects the adherence of microorganisms. In dairy industries, control of fouling of heat exchangers' plates is important to increase their nonstop operation and life. Biosurfactant addition in detergents at the rate of 0.1% concentration can inhibit the adhesion and biofilm formation of *Salmonella enteritidis* by 35.5% and *Listeria monocytogenes* by 95.9% [7]. For dairy vats and cans, surfactin from *Bacillus* sp. (AMS-H2O-1) gives antiadhesive and antibiofilm activity against common pathogenic microbial strains of *Escherichia coli* by 90% and *Staphylococcus aureus* by 97%, respectively [36]. Iturin from *B. subtilis* has shown to exhibit antiadhesive activity against fungal strains [45].

Although biosurfactants have shown the antiadhesive effects against various microorganisms but antiadhesive effects depend on type of microorganism and concentration of biosurfactants used. Biosurfactant isolated from *Candida lipolytica* showed antiadhesive property proportional to concentration of biosurfactants used. In case of *Candida lipolytica*, biosurfactants antiadhesive value of 81% were observed against *Lactobacillus* at minor concentration of 0.75 mg/L. At minor concentration, major antiadhesive property was observed against *Lactobacillus casei* with values ranging from 91% to 99% [43]. Thus antiadhesive effect of biosurfactants has much significance in controlling microbial accumulation, colonization, and ensuring the safety of food products in dairy industry. In future, the development of edible films and coatings comprising biosurfactants and natural polymers is a perspective research area. Fig. 1 shows the antiadhesive effect of biosurfactants that include inhibiting the biofilm formation or antifouling effect. Table 1 exhibits antiadhesive application of biosurfactants in dairy.

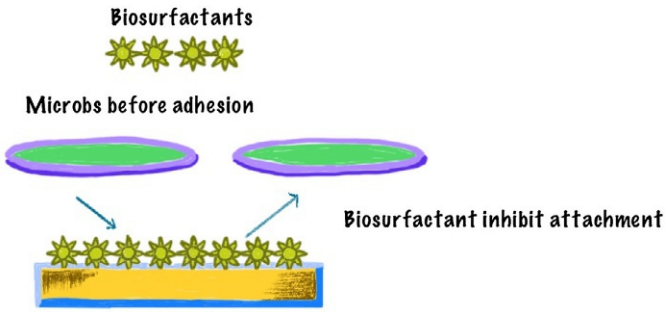

FIG. 1

Antiadhesive effect of biosurfactants (inhibiting the biofilm formation or antifouling effect).

Table 1 Antiadhesive application of biosurfactants in dairy.

Biosurfactant	Source	Antiadhesive effect	Reference
Streptococcus thermophilus biosurfactant	*Streptococcus thermophilus*	Antibiofouling effect on heat exchanger plates against thermophilic strains of *Streptococcus*, *Candida albicans* adhesion on silicon rubber pipes	[37,40]
Rhamnolipids	*Pseudomonas aeruginosa*	*Staphylococcus aureus* on polystyrene surfaces	[38]
Trehalose lipid	*Rhodococcus fascians* BD8	*Escherichia coli, Candida albicans, Pseudomonas mirabilis, Enterococcus hirae*, *Enterococcus faecalis*, *Pseudomonas vulgaris* and *Staphylococcus epidermidis*	[39]
Glycolipid	Lactobacilli strains	Probiotics	[41]
Rhamnolipids	*Pseudomonas aeruginosa*	*Staphylococcus epidermidis*, *B. pumilus*, *Staphylococcus aureus*	[42,46]
Rufisan	*Candida lipolytica* UCP0988	*Lactobacillus* pathogenic yeast and mold	[43]
Viscosin	*Pseudomonas fluorescence*	*Listeria monocytogenes* L028 strain	[29,41]
Glycoprotein like biosurfactant	*Lactobacillus plantarum*	*Escherichia coli*, *Staphylococcus aureus, and Salmonella typhi*	[44]
Surfactin	*Bacillus* sp. (AMS-H2O-1)	*Escherichia coli*, *Staphylococcus aureus, Pseudomonas aeruginosa* CI 3, *Enterococcus faecalis* CI 068	[36]
Iturin	*B. subtilis* (K1)	Antiadhesive antifungal	[45]

5 Applications of biosurfactants in dairy industry as preservatives

In food industry, the most important factors include food safety and quality, which are directly related to maintenance, storage, and provenience of product safety for consumer's health. In this context biosurfactants derived from living organisms or microorganisms are better choice than chemical ones being biodegradable, nontoxic, and environment-friendly. Food products are mainly spoiled by microbial action and environmental reagents or through oxidation reactions. Biosurfactants possess strong antioxidant activity, which is due to the unsaturated fatty acids, they contain in their molecules [47]. Biosurfactant obtained from *Lactobacillus* at the concentration of 5 mg/mL has shown best (77%) radical sequestering ability [48]. Lactic acid bacteria (LAB), the genera of beneficial bacteria, have been widely utilized for fermentation of dairy products. LAB have been commercially utilized for dairy fermentation and also the part of commensal flora of human intestine. These bacteria are known to synthesize certain antimicrobial compounds such as bacteriocins, biosurfactants, carbon peroxide, organic acids, antimicrobial substances, and hydrogen peroxide. These features make them ideal for their application for food preservation. Probiotic strains of LAB are also potentially used in food systems for establishing the competitive environment for pathogenic microbes. *Lactobacillus* species is known to have antioxidative activity in food, thereby decreasing the risk of oxidative deterioration and acting as food preservative [49]. Some of the biosurfactant-producing probiotic strains include *Lactobacillus pentosus, Lactococcus paracasei* subsp. *paracasei* A20*, Lactobacillus fermenti, Lactobacillus reuteri, Lactobacillus rhamnosus,* and *Lactobacillus brevis* [50]. Rhamnolipids are another type of biosurfactants recently permitted for food use by Environmental Protection Agency United States. Rhamnolipids in combination with niacin prolong the shelf life of cottage cheese, UHT processed soymilk and salads by retarding the growth of molds and bacteria. Rhamnolipids are also employed in ice cream and bakery products to solubilize the flavor, stabilize fat, and retard the staleness [51]. Rhamnolipids biosurfactants can be utilized as bio preservative for the preservation and shelf life extension of dairy products while surfactin nano emulsion has shown reduction of microbial load in milk, raw chicken, and apple juice [52,53]. As an attempt to control the microbial load in milk, the milk was treated with 10 μg/mL of surfactin and poly lysine combination (1:1), and consequently microbial population of *Salmonella enteritidis* was found to be significantly reduced in milk [18]. Bacteriocins have shown synergistic preservative effect when used in combination with bacteriocin, nisin, which could be due to the action of both molecules on cell membrane. There is possibility to use nisin along biosurfactants for application as food bio-preservative and development of antimicrobial active packaging with synergistic preservative effect [53]. Fig. 2 shows the schematic representation of biosurfactants application in dairy industry.

Surfactant is a chemical compound commonly located between different phases with respect to various degrees for hydrogen bonds and polarity [54]. Relying upon the origin, these surfactants are may be natural or synthetical, the first of which is

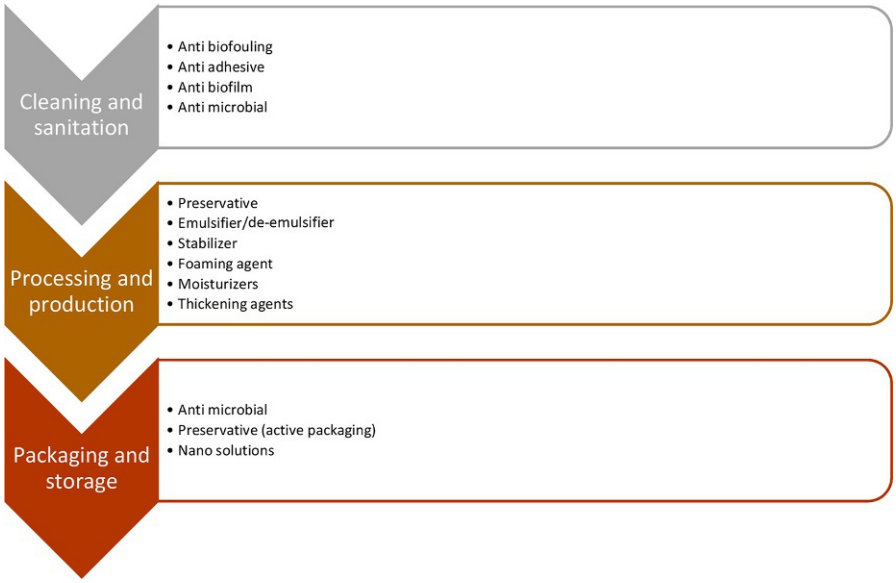

FIG. 2

Schematic representation of biosurfactants application in dairy industry.

influenced biosurfactant that is composed of amphipathic molecule with in hydrophobic and hydrophilic amounts. The portion of polar region can be ionic (anionic/cationic), nonionic, or either amphoteric, whereas the portion containing nonpolar side usually composed of hydrocarbons chain [55]. These properties give capacity to biosurfactants to minimize surface tensions between the interfaces and form emulsions, which are water-solubilized hydrocarbons and vice versa [56].

In recent years, food and dairy industry has seen a significant increase in the use of these products.

6 Applications of biosurfactants in dairy industry as stabilizers

Surfactants are nontoxic and biodegradable chemicals that have stabilizing, antimicrobial, and antiadhesive characteristics that can be utilized in the dairy industry. Other than that, biosurfactants are becoming of a great interest in formulations of dairy items due to their potential to reduce surface tensions, improve emulsification characteristics, minimize biofilm formation, and last but not the least antimicrobial activities [10].

Sharma [57] studied biosurfactants, characterized from thermophilic *Streptococci*, which were found effective in regulating the fouling of plate heat exchangers. As the fouling of heat exchangers, pipelines, and membrane filtration is a massive trouble encountered in dairy industry. Probiotic dairy culture, i.e., *Lactobacillus acidophilus* is utilized by humans due to its suitability to be used as an additive in

functional foods resulting in the desirable characteristics. Among most of the dairy-related items, such as curd, milk, cheese, formulation of creams, food grade emulsifiers/surfactants are always admissible [58].

To successfully homogenize dairy fat, emulsifiers/stabilizers play significant role to facilitate the initial formation of lipids droplet during homogenization or enhancing droplets stability during production of milk. Stabilizers enhance the fundamental life of products and reduce phase separation by preventing the clustering in globules, improving the stability for heterogeneous systems [7].

In some of the literature, addition of biosurfactant as emulsifier in formulation of ice cream having the aim to improve the creaminess, texture, and quality of the item was studied. As compared with lecithin (1 g/L), incorporation of 0.16 g/L of biosurfactant in ice-cream mix improves the stability of emulsion. It enhances overrun, quality, and organoleptic properties. This improvement is associated with the potential of biosurfactants to improve consistency and preserve physical properties to solubilize the aromatic oils that are incorporated for intensification of flavor. Partial coalescence in ice cream is the significant parameter during processing or the biosurfactant must formulate a thin, easily penetrable layer in spite of strongly affecting resistance to this coalescence. Hence, it doesn't entirely obstruct the fat crystals from entering the liquid parts for the droplets [44].

Biosurfactants having low molecular weight have also been applied as stabilizing agents in intricate emulsions. As in ice cream, it reduces the surface tension and lowers the period required to attain the emulsion balance as compared with proteins. Likewise, there is a possibility to use low-molecular-weight mixture of proteins and biosurfactants in a proportion that results in a synergic influence rather than adsorption competition between phase interference [59].

Processed milks, for example, drinking yogurts are seen as a decent transporter for probiotics, and they are at the first spot on the list for stomach well-being items in the European market of food. The fusion of different convergences of a biosurfactant separately characterized from alcohol of corn steep into a probiotic drinking yogurt impacts the *Lactobacillus casei* biomass. Furthermore, the usage of sugars and its change into lactic corrosive by *Lactobacillus casei* have not seen any sort of issue by the consolidation of the biosurfactant. No adjustment of the fermentative pathway of *Lactobacillus casei* was noticed. This examination has opened the entryway for utilization of a characteristic biosurfactant extricate, obtained from corn, in the probiotic food things to further develop its practical qualities as it impacted emphatically the development of probiotic microorganisms [60].

7 Applications of biosurfactants in dairy industry as moisturizers

Role of biosurfactants as antibacterial, antifungal, and antiviral actions together with its solubilizing/moisturizing characteristics makes them the important molecules in food and cosmetic industries [61]. Biosurfactants procured by liposan (i.e., *Candida lipolytica*) and *Saccharomyces cerevisiae* were applied in order to stabilize the

emulsification of vegetable fat in dairy item and water sequentially. Other than that, sophorolipids procured by *Candida bombicola* utilized as nonionic biosurfactants, having wetting, low foaming, emulsifying, and massive deterging characteristics are now having great application in food and dairy industries [62].

8 Applications of biosurfactants in dairy industry for the preparation of nano-solutions

Currently, the application of biosurfactants also known as microbial surfactants appeared as emergent technology for the nanomaterial synthesis as well as the formation of nano-emulsions [57]. The application of biosurfactants in the field of nanotechnology reduces waste, minimizes adverse environmental impacts, and improves energy efficiency. Nanoparticle synthesis mediated by biosurfactants is a more suitable method as it reduces the possibility of clustering or clump formation between the particles, thus promoting the uniform surface morphology of nanoparticles [63].

Moreover, biosurfactants also possess the distinguished property of self-assembling and mediating van der Waals interactions and hydrogen bonding that are usually lacked by synthetic surfactants [57]. Therefore, utilization of biosurfactants in replacement of synthetic surfactants for the synthesis of metal nanoparticles and nano-emulsion-based nanoparticles is attributed as the green route of synthesis. Various kinds of biosurfactants such as lipoproteins, glycolipids, and lipopeptides are mainly composed of protein, sugars, and lipids. During nanoparticle formation, they are transformed into nontoxic, environmentally friendly, and biologically compatible parts in contrast to synthetic surfactants [63].

In nanoparticles synthesis, biosurfactants play multiple roles as capping agents that act as a stabilizer and provide colloidal stability by preventing coagulation and terminate nanoparticles overgrowth; coating agent; dispersing agent, and reducing agent [63,64]. Moreover, due to these properties, biosurfactant-mediated nanoparticles exhibit some additional bioactivities such as enhanced antimicrobial properties. Glycolipid biosurfactants are the most employed biosurfactant for nanoparticles synthesis after lipopeptide biosurfactants based upon their microbiological origin and chemical properties. Rhamnolipids are most utilized glycolipid biosurfactants for the synthesis of nickel oxide, silver, and zinc sulfide nanoparticles producing stable nanoparticles with size ranging from 1 to 300 nm. While surfactin is a lipopeptide that widely acts as a coating agent during nanoparticles synthesis providing particles with small size (3–160 nm) [63,65]. Due to its unique characteristics, it is has been utilized extensively in the dairy industry as an antimicrobial, antibacterial, and emulsifier. Lecithin is the emulsifier of plant or animal origin and is also commercially utilized in the dairy industry. But in contrast to surfactin, it is less likely employed in the dairy sector [7]. Concisely, the data on the application of glycoproteins, glycolipopeptides, and glycopeptides for nanoparticles synthesis is still very limited [63,65].

Emulsions have found numerous applications in the industrial division such as food, pharmaceutical, textile, cosmetics, and petroleum industries. Nano-emulsions are kinetically stable heterogeneous liquid-liquid dispersion and have droplet size less than 200 nm. The smaller size of nano-emulsion facilitates targeted drug delivery, enhances the bioavailability, improves physical stability, surface rheology, and appearance. Nano-emulsions are preferentially considered more beneficial than conventional emulsions. Biosurfactants also play a significant role in the development of stable nano-emulsions that have extended applications in foods, pharmaceuticals, and cosmetics. Nano-emulsions are widely stabilized by biosurfactants that have replaced synthetic surfactants and present numerous biological properties [65,66].

Joe et al. [52] developed the surfactin-based nano-emulsions from various cooking oils. The lowest droplet size of 72.52 nm was shown by the sunflower-oil-based nano-emulsion. During antimicrobial studies, surfactin-sunflower oil-based nanoemulsion displayed fungicidal activity against, *Aspergillus niger, Rhizopus nigricans*, and *Penicillium* sp. It has also inhibited the growth of *Salmonella typhi* to a greater extent followed by *Listeria monocytogenes* and *Staphylococcus aureus*. Moreover, the antimicrobial potential of these nano-emulsions was also studied against various food samples. The results shown a lessening in the bacterial and fungal count in milk, apple juice, chicken, and mixed vegetables. Therefore, surfactin-based nano-emulsions can be employed in the formulation as preservatives for the dairy and meat industry [52].

Recently, surfactin-based nano-emulsions were also reported to be utilized for the encapsulation of active ingredients including curcumin, vitamin C and E. Nano-based drug delivery system exhibited elevated bioavailability and improved stability up to 6 months. So it also showed wide applications in various fields such as dermatology and cosmetology [67].

9 Applications of biosurfactants in the dairy industry as food additive

Food formulations such as mayonnaise, desserts, salad dressings, and food emulsions mainly utilized naturally occurring surfactants (lecithin, sucrose esters, milk proteins). But their application in food formulations is limited due to associated toxicity and their chemical properties. Alternatively, biosurfactants proved to be nontoxic and widely utilized in various food preparations as antimicrobial, emulsifier, antiadhesive, stabilizer, and thickening agent, functional food additive, and also as a preservative. The synergistic effect of rhamnolipids used in combination with Natamycin and Nisin was exploited. It appeared to improve the storage life of cottage cheese by the growth inhibition of bacteria and mold, which cause spoilage. Therefore, they can be employed as a food additive in various food formulations to prevent food contamination or as a detergent for the decontamination of food contact surfaces. Moreover, biosurfactants are also incorporated as additives for the improvement of dairy cream or nondairy cream to be utilized efficiently in bakery and

confectionery formulations [41]. Rhamnolipids are also added into ice-cream preparations to improve consistency and reduce spattering. Biosurfactants are incorporated for the preservation of various foods such as raw chicken, vegetables, milk, and milk products. Conclusively, biosurfactants act as a thickening agent, stabilizing agent, gelling agent, and mainly as an emulsifying agent. Thus, it considerably improves the consistency and texture of food [41,57,58].

10 Conclusion

Dairy industry is a major food industry, which produces a diverse range of milk-based products. For the production of these products, different processing operations and wide variety of minor ingredients are used for various technological benefits. Biosurfactants are the surface-active agents, which are produced by the microbial cells such as bacteria, fungi, and yeast and reduce the interfacial tension. Different types of biosurfactants are produced and used in the dairy industry for different technological advantages such as for emulsification, foaming, antimicrobial properties, biofilm inhibition, and texturization. These biosurfactants also act as antiadhesives, stabilizers, moisturizers, thickening agents, and help in preparation of nano-solutions. Biosurfactants include glycolipids such as rhamnolipids, trehalose lipids, and sophorolipids; lipoproteins such as fengycin and surfactin; and lipopeptides, which are used as emulsifiers and antimicrobial agents. As an emulsifier in the formulation of ice cream, biosurfactants improve its creaminess, texture, and overall acceptability. Biosurfactants affect the building and adhesion characteristics of biofilms as *B. licheniform* produces cyclic nonribosomal lipopeptide, lichenysin, which is used against different bacteria on industrial scale. Recently, the application of biosurfactants has appeared as emergent technology for the nanomaterial synthesis as well as the formation of nano-emulsions. These compounds exhibit low toxicity, increased biodegradability, and their ability to remain unaffected even at higher salt concentration and pH in comparison to the synthetic surfactants. These possess extensive market potential in food industry due to the positive environmental impact along with nontoxic and biodegradable nature.

References

[1] N. Kosaric, Biosurfactants and their Applications for soil bioremediation, Food Technol. Biotechnol. 39 (2001) 295.
[2] N. Kosaric, F.V. Sukan, Biosurfactants: Production and Utilization—Processes, Technologies and Economics, CRC Press, 2014. 03-Nov—Science—389 pages.
[3] J.M. Campos, T.L. Montenegro Stamford, L.A. Sarubbo, J.M. De Luna, R.D. Rufino, I. M. Banat, Microbial biosurfactants as additives for food industries, Biotechnol. Prog. 29 (2013) 1097–1108.
[4] B. Burghoff, Foam fractionation applications, J. Biotechnol. 161 (2012) 126–137.

[5] A. Santamaria-Echart, I.P. Fernandes, S.C. Silva, S.C. Rezende, G. Colucci, M.M. Dias, M.F. Barreiro, New Trends in Natural Emulsifiers and Emulsion Technology for the Food Industry, IntechOpen, 2021.
[6] P.R.F. Marcelino, F. Gonçalves, I.M. Jimenez, B.C. Carneiro, B.B. Santos, S.S. Da Silva, Sustainable production of biosurfactants and their applications, in: Lignocellulosic Biorefining Technologies, Wiley Online Library, 2020, pp. 159–183.
[7] M. Sanjana, K. Shivalkar Yadav, L. Malashree, R. Prabha, Bacterial biosurfactants-A boon to dairy industry, Int. J. Curr. Microbiol. App. Sci. 6 (2017) 608–612.
[8] R. Parthasarathi, K. Subha, Prospective biosurfactant applications in food industry-opportunities and challenges, Int J Curr Res Life Sci. 7 (2018) 1560–1565.
[9] T. Lukondeh, N.J. Ashbolt, P.L. Rogers, Evaluation of Kluyveromyces marxianus FII 510700 grown on a lactose-based medium as a source of a natural bioemulsifier, J. Ind. Microbiol. Biotechnol. 30 (2003) 715–720.
[10] E.J. Gudiña, L.R. Rodrigues, Research and production of biosurfactants for the food industry, in: Bioprocessing for Biomolecules Production, Wiley Online Library, 2019, pp. 125–143.
[11] P. Eslami, H. Hajfarajollah, S. Bazsefidpar, Recent advancements in the production of rhamnolipid biosurfactants by Pseudomonas aeruginosa, RSC Adv. 10 (2020) 34014–34032.
[12] A.K. Al-Asady, A. Al-Waelyw, H. Majeedg, Study of rhamnolipids cytotoxicity, inhibitory effect on some microorganisms and applying in food products, Int. J. Bio-Technol. Res. (IJBTR) 6 (2016) 1–8.
[13] J.M. Campos, T.L. Stamford, L.A. Sarubbo, Production of a bioemulsifier with potential application in the food industry, Appl. Biochem. Biotechnol. 172 (2014) 3234–3252.
[14] S. Khosrow Shahi, Z. Didar, M.A. Hesarinejad, M. Vazifedoost, Optimized pulsed electric field-assisted extraction of biosurfactants from Chubak (Acanthophyllum squarrosum) root and application in ice cream, J. Sci. Food Agric. 101 (9) (2021) 3693–3706, https://doi.org/10.1002/jsfa.11000.
[15] G.L. Hasenhuettl, Synthesis and commercial preparation of food emulsifiers, in: G.L. Hasenhuettl, R.W. Hartel (Eds.), Food Emulsifier and Their Applications, 2008 (Chapter 2).
[16] T.F. Tadros, Emulsions: Formation, Stability, Industrial Applications, Published by Walter de Gruyter GmbH and Co K, 2016.
[17] L. D'Auria, M. Deleu, S. Dufour, M.P. Mingeot-Leclercq, D. Tyteca, Surfactins modulate the lateral organization of fluorescent membrane polar lipids: a new tool to study drug: membrane interaction and assessment of the role of cholesterol and drug acyl chain length, Biochim. Biophys. Acta 1828 (2013) 2064–2073.
[18] X. Huang, J. Suo, Y. Cui, Optimization of antimicrobial activity of surfactin and polylysine against Salmonella enteritidis in milk evaluated by a response surface methodology, Foodborne Pathog. Dis. 8 (3) (2011) 439–443.
[19] Q. Meng, G. Zhang, Application of Rhamnolipid as Biological Cleaning Agent. Chinese Patent CN 102399644, 2012.
[20] A. Ortiz, J.A. Teruel, M.J. Espuny, A. Marques, A. Manresa, F.J. Aranda, Interaction of Rhodococcus sp. biosurfactant trehalose lipid with phosphatidylethanolamine membranes, Biochim. Biophys. Acta 1778 (2008) 2806–2813.
[21] A. Zaragoza, F.J. Aranda, M.J. Espuny, J.A. Teruel, A. Marques, A. Manresa, A. Ortiz, Mechanism of membrane permeabilization by a bacterial trehalose lipid biosurfactant produced by Rhodococcus sp, Langmuir 26 (2009) 8567–8572.

[22] L.V. Araujo, F. Abreu, U. Lins, L.M.M. Santa Anna, M. Nitschke, D.M.G. Freire, Rhamnolipid and surfactin inhibit Listeria monocytogenes adhesion, Food Res. Int. 44 (2011) 481–488.
[23] L. Magalhães, M. Nitschke, Antimicrobial activity of rhamnolipids against Listeria monocytogenes and their synergistic interaction with nisin, Food Control 29 (1) (2013) 138–142, https://doi.org/10.1016/j.foodcont.2012.06.009.
[24] G.B. Oliveira, L. Favarin, R.H. Luchese, D. McIntosh, Psychrotrophic bacteria in milk: how much do we really know? Braz. J. Microbiol. 46 (2) (2015) 313–321.
[25] K.H. Teh, S. Flint, J. Palmer, P. Andrews, P. Bremer, D. Lindsay, Proteolysis produced within biofilms of bacterial isolates from raw milk tankers, Int. J. Food Microbiol. 157 (2012) 28–34.
[26] S. Agarwal, K. Sharma, B.G. Swanson, G. Ünlü, Nonstarter lactic acid bacteria biofilms and calcium lactate crystals in Cheddar cheese, J. Dairy Sci. 89 (5) (2006) 1452–1466.
[27] M.W. Mittelman, Structure and functional characteristics of bacterial biofilms in fluid processing operations, J. Dairy Sci. 81 (10) (1998) 2760–2764.
[28] J.S. Dickson, M. Koohmaraie, Cell-surface charge characteristics and their relationship to bacterial attachment to meat surfaces, Appl. Environ. Microbiol. 55 (4) (1989) 832–836.
[29] M. Krishnaswamy, G. Subbuchettiar, T.K. Ravi, S. Panchaksharam, Biosurfactants properties, commercial production, and application, Curr. Sci. 94 (2008) 736–747.
[30] J. Coronel-León, A.M. Marqués, J. Bastida, A. Manresa, Optimizing the production of the biosurfactant lichenysin and its application in biofilm control, J. Appl. Microbiol. 120 (2016) 99–111, https://doi.org/10.1111/jam.12992.
[31] H. Zhao, D. Shao, C. Jiang, J. Shi, Q. Li, Q. Huang, Biological activity of lipopeptides from bacillus, Appl. Microbiol. Biotechnol. 101 (2017) 5951–5960, https://doi.org/10.1007/s00253-017-8396-0.
[32] Y. Lequette, G. Boels, M. Clarisse, C. Faille, Using enzymes to remove biofilms of bacterial isolates sampled in the food-industry, Biofouling 26 (4) (2010) 421–431.
[33] M.S. Yeh, Y.H. Wei, J.S. Chang, Enhanced production of surfactin from Bacillus subtilis by the addition of solid carriers, Biotechnol. Prog. 2 (1) (2005) 1329–1334.
[34] X.F. Huang, D.H. Yang, Q. Zhou, Evaluation of screening methods for demulsifying bacteria and characterization of lipopeptide bio-demulsifier produced by Alcaligenes sp, Bioresour. Technol. 100 (3) (2009) 1358–1365.
[35] A.S. Nayak, M.H. Vijaykumar, T.B. Karegoudar, Characterization of biosurfactant produced by Pseudoxanthomonas sp. PNK-04 and its application in bioremediation, Int. Biodeterior. Biodegrad. 63 (2009) 73–79.
[36] K.R. Meena, S.S. Kanwar, Lipopeptides as the antifungal and antibacterial agents: applications in food safety and therapeutics, Biomed. Res. Int. 2015 (2015).
[37] H.J. Busscher, M. Van der Kuijl-Booij, H.C. Van der Mei, Biosurfactants from thermophilic dairy streptococci and their potential role in the fouling control of heat exchanger plates, J. Ind. Microbiol. 16 (1) (1996) 15–21.
[38] S.S. Silva, J.W.P. Carvalho, C.P. Aires, M. Nitschke, Disruption of Staphylococcus aureus biofilms using rhamnolipid biosurfactants, J. Dairy Sci. 100 (2017) 1–10.
[39] T. Janek, A. Krasowska, Ż. Czyżnikowska, M. Łukaszewicz, Trehalose lipid biosurfactant reduces adhesion of microbial pathogens to polystyrene and silicone surfaces: an experimental and computational approach, Front. Microbiol. 9 (2018) 2441.
[40] H.J. Busscher, C.G. Van Hoogmoed, G.I. Geertsema-Doornbusch, M. Van der Kuijl-Booij, H.C. Van der Mei, Streptococcus thermophilus and its biosurfactants inhibit

adhesion by Candida spp. on silicone rubber, Appl. Environ. Microbiol. 63 (10) (1997) 3810–3817.
[41] M. Nitschke, S.G.V.A.O. Costa, Biosurfactants in food industry, Trends Food Sci. Technol. 18 (5) (2007) 252–259.
[42] D.H. Dusane, Y.V. Nancharaiah, S.S. Zinjarde, V.P. Venugopalan, Rhamnolipid mediated disruption of marine Bacillus pumilus biofilms, Colloids Surf. B Biointerfaces 81 (1) (2010) 242–248.
[43] R.D. Rufino, J.M. Luna, L.A. Sarubbo, L.R.M. Rodrigues, J.A.C. Teixeira, G.M. Campos-Takaki, Antimicrobial and anti-adhesive potential of a biosurfactant Rufisan produced by Candida lipolytica UCP 0988, Colloids Surf. B Biointerfaces 84 (1) (2011) 1–5.
[44] B.G. Ribeiro, J.M. Guerra, L.A. Sarubbo, Biosurfactants: production and application prospects in the food industry, Biotechnol. Prog. 36 (5) (2020), e3030.
[45] E. Arrebola, R. Jacobs, L. Korsten, Iturin A is the principal inhibitor in the biocontrol activity of Bacillus amyloliquefaciens PPCB004 against postharvest fungal pathogens, J. Appl. Microbiol. 108 (2) (2010) 386–395.
[46] S. Vijayakuma, V. Saravanan, In vitro cytotoxicity and antimicrobial activity of biosurfactant produced by Pseudomonas aeruginosa strain PB3A, Asian J. Sci. Res. 8 (4) (2015) 510–518.
[47] G.S. Kiran, S. Priyadharsini, A. Sajayan, G.B. Priyadharsini, N. Poulose, J. Selvin, Production of lipopeptide biosurfactant by a marine Nesterenkonia sp. and its application in food industry, Front. Microbiol. 8 (2017) 1138.
[48] A. Merghni, I. Dallel, E. Noumi, Y. Kadmi, H. Hentati, S. Tobji, A.B. Amor, M. Mastouri, Antioxidant and antiproliferative potential of biosurfactants isolated from Lactobacillus casei and their anti-biofilm effect in oral Staphylococcus aureus strains, Microb. Pathog. 104 (2017) 84–89.
[49] T. Kullisaar, M. Zilmer, M. Mikelsaar, T. Vihalemm, H. Annuk, C. Kairane, A. Kilk, Two antioxidative lactobacilli strains as promising probiotics, Int. J. Food Microbiol. 72 (3) (2002) 215–224.
[50] S. Nalini, R. Parthasarathi, D. Inbakanadan, Biosurfactant in food and agricultural application, Environ. Biotechnol. 2 (45) (2020) 75.
[51] I.P.H. Van Haesendonck, E.C.A. Vanzeveren, Rhamnolipids in Bakery Products. International Application Patent (PCT) WO, 2004. pp.2004-040984.
[52] M.M. Joe, K. Bradeeba, R. Parthasarathi, P.K. Sivakumaar, P.S. Chauhan, S. Tipayno, A. Benson, T. Sa, Development of surfactin based nanoemulsion formulation from selected cooking oils: evaluation for antimicrobial activity against selected food associated microorganisms, J. Taiwan Inst. Chem. Eng. 43 (2) (2012) 172–180.
[53] M. Nitschke, S.S.E. Silva, Recent food applications of microbial surfactants, Crit. Rev. Food Sci. Nutr. 58 (4) (2018) 631–638.
[54] A.R. Markande, D. Patel, S. Varjani, A review on biosurfactants: properties, applications and current developments, Bioresour. Technol. (2021) 124963.
[55] S. Akbari, N.H. Abdurahman, R.M. Yunus, F. Fayaz, O.R. Alara, Biosurfactants—a new frontier for social and environmental safety: a mini review, Biotechnol. Res. Innov. 2 (1) (2018) 81–90.
[56] A. Roy, Review on the biosurfactants: properties, types and its applications, J. Fundam. Renew. Energy Appl. 8 (2017) 1–14.
[57] D. Sharma, Applications of biosurfactants in food, in: Biosurfactants in Food, Springer, 2016.

[58] S.K. Satpute, S.S. Zinjarde, I.M. Banat, Recent updates on biosurfactant/s in food industry, in: Microbial Cell Factories, Taylor & Francis, 2018.
[59] S.K. Satpute, G.A. Płaza, A.G. Banpurka, Biosurfactants' production from renewable natural resources: Example of innovativeand smart technology in circular bioeconomy, Manag. Syst. Prod. Eng 25 (1) (2017) 46–54, https://doi.org/10.1515/mspe-2017-0007.
[60] A. López-Prieto, L. Rodríguez-López, M. Rincón-Fontán, A.B. Moldes, J.M. Cruz, Effect of biosurfactant extract obtained from the corn-milling industry on probiotic bacteria in drinkable yogurt, J. Sci. Food Agric. 99 (2) (2019) 824–830.
[61] N.A. Kaloorazi, M.F.S. Choobari, Biosurfactants: properties and applications, J Biol Today's World 2 (2013) 235–241.
[62] G. Rawat, A. Dhasmana, V. Kumar, Biosurfactants: the next generation biomolecules for diverse applications, Environ. Sustain. (2020) 1–17.
[63] X. Vecino, L. Rodríguez-López, M. Rincón-Fontán, J.M. Cruz, A.B. Moldes, Nanomaterials synthesized by biosurfactants, in: Comprehensive Analytical Chemistry, Elsevier, 2021.
[64] R. Javed, M. Zia, S. Naz, S.O. Aisida, N. Ul Ain, Q. Ao, Role of capping agents in the application of nanoparticles in biomedicine and environmental remediation: recent trends and future prospects, J. Nanobiotechnol. 18 (2020) 1–15.
[65] M. Nitschke, C.A. Marangon, Microbial surfactants in nanotechnology: recent trends and applications, Crit. Rev. Biotechnol. (2021) 1–17.
[66] L. Bai, D.J. Mcclements, Formation and stabilization of nanoemulsions using biosurfactants: rhamnolipids, J. Colloid Interface Sci. 479 (2016) 71–79.
[67] A. Lewińska, M. Domżał-Kędzia, A. Jaromin, M. Łukaszewicz, Nanoemulsion stabilized by safe surfactin from Bacillus subtilis as a multifunctional, custom-designed smart delivery system, Pharmaceutics 12 (2020) 953.

CHAPTER

Application of biosurfactants in juice industry

25

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1 Introduction

Biosurfactants are important for the maximum adaptable set of synthetic compounds possibly utilized in different productions systems, which incorporate cleansers, paper items, drugs, paints, beauty care products, oil, food, and wastewater treatment. At present, surfactants are produces by petrochemicals, creature fats, microorganisms, and plants. Research studies had shown that greater part of the creation markets depended on petrochemicals [1,2].

Surfactants are surface dynamic mixtures with both hydrophilic and hydrophobic spaces. They are suitable for decreasing surface and interfacial pressure at the surface and interface among fluids, solids, and gases. Basically biosurfactants have been classified into two sets: low-atomic-weight particles named biosurfactants, and high-sub-atomic-mass biosurfactants or bioemulsifiers. The significance of biosurfactants has continuously expanded throughout the previous years [3,4]. Biosurfactants are better than chemical substance surfactants due to their organic beginning, biodegradability, and low degree of poisonousness [5]. Therefore, biosurfactants have been widely considered for application in food handling, beauty care products

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details, microbial upgraded oil recuperation, soil washing, emulsifying materials, and bioremediation specialists [6,7].

Biosurfactants are normally blended by a variety of bacteria on various materials such as sweeties, fats, and waste material. The assorted utilitarian characteristics of biosurfactants such as frothing, purging, stage partition, surface movement, and decrease in thickness of substantial fluids, such as unrefined petroleum, make them suitable for some, modern and homegrown application purposes [8]. The flexibility and variety of these particles have prompted expanded interest in modern and natural applications such as bioremediation, soil washing, improved oil recuperation, and additional oil handling, and associated ventures. Biosurfactants might possibly displace essentially any manufactured surfactant and also some existing stimulating physicocompound properties. Presently, the vital application is for improvement of oil recuperation and hydrocarbon bioremediation because of their biodegradability and low basic micelle absorption [9].

During the last few years, biosurfactants production system from different microorganisms has been researched broadly. Apparently, not many research activities have been made to portray the innovative work methodologies of making the biosurfactants production process less expensive and economically alluring. Fundamental point of this survey article is to underscore the double-dealing of modest and effectively accessible agro-modern waste as substrate for business creation of biosurfactants alongside squandering the executives and their possible applications in food and agribusiness field [10].

In the last few years, biosurfactants formation from different types of microorganisms has been examined broadly. To the best of our knowledge, very few attempts have been made to describe the research and development ideas of creating the biosurfactants making process inexpensive and attractive material in markets. In this chapter, we look at the several forms of biosurfactants that are created and used in the juice production market for frothing, antibacterial characteristics, biofilm suppression, and emulsifying. Such biosurfactants additionally serve as antiadhesives, modifiers, conditioners, and additives, as well as assisting in the creation of nano preparations.

Generally, biosurfactants are excellent composites having both hydrophilic and hydrophobic proportions exclusively. In fluid, the relations of hydrophobic gatherings are used as lyophobic and hydrophilic structures as lyophilic. Surfactants are dynamic blends found in chemicals, and they decrease the outer strain or interfacial endure the air–water and oil boundaries. Biosurfactants can be gotten from various minimal expense materials accessible in huge amounts. They have a few benefits over engineered compound inferred surfactants, which incorporate low harmfulness, bioavailability, biodegradability, high frothing, climate cordial, minimal expense as far as feedstock accessibility in nature, pH, and saltiness. In this way, they are protected and better option in contrast to compound surfactants, especially in food, drug, restorative, and eatable oils [11]. Essentially, the developing biological trouble about artificially determined surfactants starts another devotion on normal biosurfactants because of their weak attributes [12–15].

2 Factors affecting the production of biosurfactants

The formation or progress of different material of biosurfactant does not simply depend upon the manufacturer strain and life condition, by this system of the carbon and nitrogen basis, nitrogen extent, nutritious limitations, blend and actual boundaries, for instance, temperature, pH, and air flow affect the proportion of biosurfactant along with the sort of polymer conveyed [16] (Fig. 1).

2.1 Carbon sources

Large amounts of carbon-based materials have been used after examining for the formation of biosurfactant. Crude petrol, glucose, diesel, sucrose have been represented as a good source of carbon substrate for biosurfactant formation. Basically carbon substrate has an essential impact in biosurfactant production system, but different carbon sources in the medium affected the synthesis of the biosurfactant production; however, substrate with different chain sizes showed no effect on the chain length of unsaturated fat moieties in glycolipid biosurfactants.

2.2 Natural elements

Natural components are basic in the production of the biosurfactants. To obtain great amount of biosurfactants, it is essential to elevate the bioprocess by alterations in temperature, pH, air flow, or disconcerting speed. Different biosurfactants are performed well in a temperature from 25°C to 300°C. The influence of pH on elements

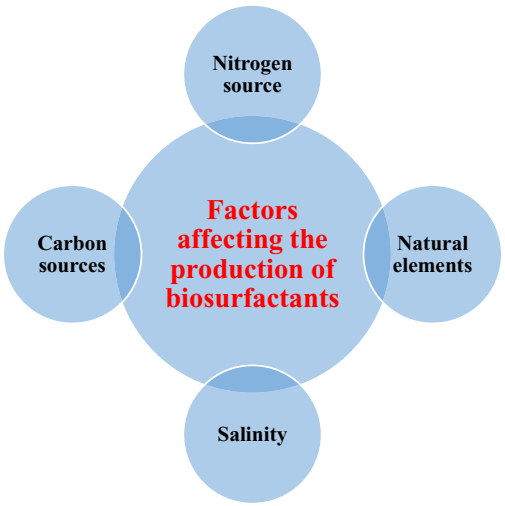

FIG. 1

Sources for the production of biosurfactant.

biosurfactant transferred was considered by Zinjarde and Pant [17], who reported that the best formation occurred after the pH was extent at 8.0, that is, the standard pH of sea water, which is *Y. lipolytica* normal regular environmental factors. With *Pseudomonas* sp., rhamnolipids formation was at its generally outrageous at a pH from 6 to 6.5 and decreased distinctly above pH 7.

2.3 Salinity

Salinity of a particular medium also expects basic part in the biosurfactant formation. Regardless, inverse insights were seen for some biosurfactant things, which were not impacted by obsessions about 10% (weight/volume) but small diminishments in the CMC were recognized [15].

3 Industrialized uses of biosurfactants

Biosurfactants have an extensive extent of usages in different modern applications such as magnificence care items, pharmaceutics, food, petroleum, wastewater, articles of clothing industry, cultivation, painting, and various undertakings. Moreover, these combinations are considering in different functional components such as settling, moistening, antibacterial. As of late, biosurfactants achieved an expected significance for regular industrial uses in remediation toxic substance, mostly in significant metal ejection from earth and marine, supportive, drug things, and advanced oil recuperation. The applications of biosurfactants in various advanced cycle are shown in Table 1.

4 Application of biosurfactants in juice industry

Production economy is the significant portion in the biosurfactants formation. The measure of unrefined substance can contribute notably to the production rate. The greater part of the study depends on different sources of glycerol, sucrose, ethanol, and glucose as fermentation material. It is assessed that unrefined substances represent from 10% to 30% of the complete creation rate in many biosurfactants-related technological processes. To decrease this formation rate, research in the space of biosurfactants has extended a considerable amount as of late because of their likely use in various regions, such as the food, horticulture, drug and oil industry, and bioremediation of toxins. The one chance examined broadly can be the consumption uncertain and food-based natural substance as substrate for biosurfactants formation. Biosurfactants effortlessly will be delivered from inexhaustible properties through microbial maturation, having an extra benefit over engineered surfactants. The significant difficulties for the viable production of biosurfactants incorporate exceptional revenues such as alternative minimal expense substrates reasonable bioprocesses.

Table 1 Major applications of biosurfactants in different sectors.

Industry	Application	Role of biosurfactant
Pharmaceutical and cosmetic	Soaps, hair care items, creams, eye shadow, skin break out therapy, lip tone, antiperspirants, hostile to wrinkle items, skin smoothing, antimicrobial specialist, antifungal specialist, against glue specialists, hostile to disease specialist, antiviral specialist, frothing specialists	Keeping up with skin hindrance work and keeping overabundance water misfortune from the skin
Foodstuff production	Antibacterial, and anti-adhesive biofilm substances, food protection stuffs	**1.** Interface with hydrocarbons **2.** Solubilize flavor oils, control consistency, retard staling
Agricultural science	Soil remediation specialist, fungicides, bactericides, insecticide, bulky metal evacuation, plant virus expulsion specialist, root colonization	**1.** Assistance of biocontrol components of organisms like parasitism, antibiosis, rivalry, Induced fundamental obstruction and hypo virulence, emulsifier **2.** Prevent hardening during stockpiling

4.1 Biosurfactants in juice industry as emulsifier/de-emulsifiers

Biosurfactants play a significant role in our daily life. Mainly biosurfactants are used agriculture and wastewater treatment as fertilizer and cleaning agents. But the range of applications also covers the juice industry. Among yields of these industries, biosurfactants account for 52% of the ingredients. Dairy products, juices, creams, and many other bakery products would not be imaginable without the use of biosurfactants as emulsifiers. Especially in food sector, biosurfactant as a preserver plays a major role [18]. Biosurfactants cause less chemical reaction as compared with conventional surfactants, which justifies their use in the juice industry. Many industrially processed foods contain emulsifiers that enable homogeneous mixing of hydrophilic and hydrophobic ingredients, e.g., in bakery products, dairy products, and juices. Emerging toward a responsible nutrition, consumer care more about the source of ingredients in their food and demand for a transparent tagging of the used elements. A developing demand for biologically produced food additives will help biosurfactants to establish in the food sector. In pharmaceutical applications, surfactants are used as emulsifiers for active pharmaceutical elements but also as adjuvants in vaccines. Some biosurfactants show antimicrobial or antiviral properties making them potential candidates for future use in therapeutics substitute as emulsifier and active ingredient at the same time [19].

4.2 Biosurfactants in juice industry as antimicrobial agents

The search for new antimicrobial based atoms is of expanding distress right now for the recently arisen antimicrobial harmless microorganisms to existing regular antitoxins. Metabolites of microbial beginning can be recognized as a vital origin of atoms of original mechanical characteristics with important natural actions. The major inactivation of infections advances that constrain activity is presumably because of associations between the covering of the infection and the biosurfactants. A few research studies have revealed important antimycoplasma actions by different biosurfactants alike Surfactin. Contaminating because of the mycoplasma in cell growth research centers is a generally emerging significant issue when it influences the cell lines [20]. The property such as antifungal ability of biosurfactants is extensive while being discovered. Current microbial hindrance approaches are set up on planktonic bacterial cell lead composition that has insufficient practicality to the extent improvement restriction of biofilm masses. Biosurfactants have attention as threatening to biofilm experts because of their ability to scatter microbial organizations. The above analyzed employments of biosurfactants there are one or two regions to discover biosurfactants, for instance, food and feed dealing with, dynamic food components [21].

4.3 Biosurfactants in juice industry as antibacterial

The disruption of the surface phospholipid layer by biosurfactants causes organic purposes to be affected, such as modifying external surface penetration into ion channels, by natural ion precursors, and by emulsifying the surface. Biosurfactants can break in between different layer of plasma, in this manner eventually controlling the chains growth of hydrocarbon furthermore changing the plasma film viscosity. Biosurfactant intervened layer disturbances are expansive range in method of activity and are advantageous for disorder of plasma films of pathogenic microbes. Biosurfactants got from different microorganisms showed likely antimicrobial properties. Antimicrobial activities of rhamnolipids inaccessible from the *P. aeruginosa* LBI have been demonstrated besides a diversity of bacteriological and fungiform pathogens, were assessed by determining the growth progress of numerous microorganisms. Several infections were found to be effective against the production of biosurfactant produced from A20 and ssp. paracasei types [22]. The initiation of hostile to microbials for quelling bacterial microorganisms is seen as one of the principal upgrades in contemporary prescription. Anti-infection use and maltreatment during the last 10–20 years have come about in a large portion of clinical diseases creating protection from a few antitoxins. Therefore, new antimicrobials that are successful in controlling irresistible sicknesses brought about by multidrug-safe microbe microorganisms are required. Biosurfactants are turning out to be all the more generally perceived as an expected new class of antibacterial synthetic compounds [23]. They are a practical option in contrast to conventional antimicrobials and could be utilized as protected and powerful helpful specialists, particularly for clinical contaminations that are multidrug safe. One of the central questions in controlling

shrewd illnesses is the development of multidrug obstruction creatures. A new glycolipid, xylolipid biosurfactant delivered from *Lactobacillus lactis*, was examined for its antibacterial properties against multidrug-safe living beings. Filtered *L. lactis* biosurfactants had antibacterial movement against *E. coli and S. aureus*. Accordingly, *L. lactis* biosurfactant may be viewed as a wide range antimicrobial. Biosurfactant of *L. lactis* is ok for restorative specialist/food added substance/in beauty care products, because of its probiotic beginning. Biosurfactants produced from LAB have antimicrobial properties, which could prompt biosurfactants-based elective medicines for the avoidance of nosocomial diseases.

4.4 Biosurfactants in juice industry as preserver and stabilizer

It has been recommended that a few types of *Lactobacilli* and *Streptococcus* are displayed to create biosurfactants as their optional metabolite during the way of evolution. These microorganisms are viewed as GRAS (generally viewed as protected) and have direct food application. The utilization of biosurfactants delivered by *Lactobacilli* strains property is extremely encouraging since these microorganisms are normally existing in nutrition. Lactobacillus surfactants kinds are regularly along with different surfactants such as *Streptococcus* are utilized as corrosive and essence makers in the juice business. Lactic corrosive microbes, for example, *Latococcus lactis* and *Streptococcus thermophilus* protection the nutritive characteristics of food material for a shell time span of usability by inhibiting food waste and development of pathogenic microorganisms [24]. It has been reported that metabolites from lactic corrosive microscopic organisms can be taken advantage of as natural additives in juice packing materials. Lactic corrosive microorganisms and their juice-related products give a range of significant dietary and helpful advantages in people such as restraint of pathogenic creature, improvement of microbial equilibrium in the digestive tract, lightening of lactose prejudice, and decrease of blood cholesterol. *Lactobacillus pentosus*, a lactic corrosive microscopic organisms developed on whey at 31°C viably bringing down the external pressure of moderate between 54 $mN\,m^{-1}$ and 45 $mN\,m^{-1}$. Much more research is required aside, to know the influence of Lactobacilli type surfactants in stopping microbe colonization [25].

5 Biosurfactants' sustainability goals and consumer perspective

According to a consumer point of view, biosurfactants have some promising and progressive properties. The interest for sustainable elements in results of the regular life is growing frequently. In food industry items such as dairy product, juice, bakery products, synthetic biosurfactants could difficulty in the preservation and stabilization properties. Proven by field trial of novel definitions for food products, the utilization of biosurfactants in food industry could be profitable for each distinctive individual as biosurfactants are milder and moderate to the juice and dairy products than synthetic surfactants. Although, the end consumer requirements to understand

the advantage of a natural ingredient replacing a chemical-based product in everyday products. Terms such as bioeconomy, biosurfactants, or supportable production lead to entirely unexpected relationship among people. This can be handled by instructive awareness campaigns clarifying the requirement for a capable utilization of normal resources. Once settled in the market, ecological agreeable biosurfactants show further fields of use in agriculture or in remediation. Hydrophobic natural mixtures or substantial metal toxins could be eliminated from food by biosurfactants showing their wide applicability in various industries. According to the modern perspective, some facts support the change from chemical-based to biosurfactants. Most organizations welcome to enhance their asset providers. Administrative guidelines and their individual awareness of certain expectations push the business toward a decrease the uses of chemical as a preserver in food industry. Organizations frequently make an interpretation of these guidelines into sustainability objectives for their long-term policy. One of the requirements for new manufacture processes is frequently to diminish the general chemicals use for the ideal item. Despite the fact that biosurfactants are low-volume and high execution elements, the change of chemical based toward biobased feedstock materials in the surfactant formation assists with diminishing the general chemical use of relating products results such as juice and dairy industries [26].

6 Future perspectives

The future idea of the microbial derived biosurfactants for the finding of different atoms in way of cost effective production, classification, properties, and the features of novel and present biosurfactants must be highlighted so that the microbial surfactants contribute important role in juice industry to be certified as safe for juice-related products. A capable approach is by all accounts the exploitation of inoculants of biosurfactants creating microscopic organisms as preserver and stabilizer hydrocarbon in juice industry to work on the proficiency of this process. The use of biosurfactants in juice industry has a huge scope expects test to distinguish their likely harmful impact on food industry. Although the biosurfactants are considered to be eco-friendly, a few analyses demonstrated that in specific situations they can be poisonous to the climate. However, careful and well-ordered utilization of these attractive surface dynamic atoms will unquestionably help in the improved preserver property of the harmful natural toxins and also provide new opportunities for juice and food-related industries.

7 Conclusion

Biosurfactants have various elements that could be utilized in food business, such as dairy items, bread shop things, and juices. The biosurfactants have been gaining considerable attention in food sector because of their vast applications and easy

fabrication. Biosurfactants have great potential in food handling, skin health management items definitions, microbial improved oil recuperation, water treatment, emulsifying materials, and drug area. The accomplishment of biosurfactants development relies upon the development of less expensive techniques in light of the utilization of minimal expense unrefined substances, which represent 10%–30% of the general cycle cost. Biosurfactants are excellent composites blended from plants and microbes. Biosurfactants having important characteristics in juice productions contain the uses as emulsifiers and antibacterial constituents. In this chapter, we have reviewed the developments, and applications of biosurfactants in juice industry. Specifically, critical role of biosurfactants in production of juices has been discussed. The limited usage of biosurfactants in the juice industry appears to be due to the lack of information about toxicity, as well as expensive production costs. Biosurfactants have broad market efficiency in food market because of their progressive ecological effect alongside nonharmful and decomposable nature.

References

[1] I.M. Banat, R.S. Makkar, S.S. Cameotra, Potential commercial applications of microbial surfactants, Appl. Microbiol. Biotechnol. 53 (5) (2000) 495–508.
[2] K. Gautam, V. Tyagi, A review of microbial surfactant, J. Oleo Sci. 55 (4) (2006) 155–166.
[3] Y. Aşçı, M. Nurbaş, Y.S. Açıkel, Investigation of sorption/desorption equilibria of heavy metal ions on/from quartz using rhamnolipid biosurfactant, J. Environ. Manag. 91 (3) (2010) 724–731.
[4] Z. Ahmad, et al., Biosurfactants for sustainable soil management, Adv. Agron. 150 (2018) 81–130.
[5] K. Muthusamy, et al., Biosurfactants: properties, commercial production and application, Curr. Sci. (2008) 736–747.
[6] Y. Al-Wahaibi, et al., Biosurfactant production by Bacillus subtilis B30 and its application in enhancing oil recovery, Colloids Surf. B: Biointerfaces 114 (2014) 324–333.
[7] D.G. Almeida, et al., Response surface methodology for optimizing the production of biosurfactant by Candida tropicalis on industrial waste substrates, Front. Microbiol. 8 (2017) 157.
[8] C.N. Mulligan, Environmental applications for biosurfactants, Environ. Pollut. 133 (2) (2005) 183–198.
[9] E.Z. Ron, E. Rosenberg, Natural roles of biosurfactants: Minireview, Environ. Microbiol. 3 (4) (2001) 229–236.
[10] H.P. Kaur, B. Prasad, S. Kaur, A review on applications of biosurfactants produced from unconventional inexpensive wastes in food and agriculture industry, World J. Pharm. Res. 4 (2015) 827–842.
[11] L. Fracchia, et al., Biosurfactants and bioemulsifiers biomedical and related applications—present status and future potentials, in: Biomedical Science, Engineering and Technology, vol. 14, IntechOpen, 2012, pp. 326–335.
[12] F.V. Sukan, Biosurfactants produced by genetically manipulated microorganisms: challenges and opportunities, in: Biosurfactants, CRC Press, 2014, pp. 60–83.

[13] H. Frumkin, J. Hess, S. Vindigni, Energy and public health: the challenge of peak petroleum, Public Health Rep. 124 (1) (2009) 5–19.
[14] D.G. De Almeida, et al., Biosurfactants: promising molecules for petroleum biotechnology advances, Front. Microbiol. 7 (2016) 1718.
[15] S.S. Bhadoriya, et al., Biosurfactants: a new pharmaceutical additive for solubility enhancement and pharmaceutical development, Biochem. Pharmacol. 2 (2) (2013) 113.
[16] S. Vijayakumar, V. Saravanan, Biosurfactants—types, sources and applications, Res. J. Microbiol. 10 (5) (2015) 181.
[17] S.S. Zinjarde, A. Pant, Emulsifier from a tropical marine yeast, *Yarrowia lipolytica* NCIM 3589, J. Basic Microbiol. 42 (1) (2002) 67–73.
[18] A. Belligno, et al., Effects of industrial orange wastes on soil characteristics and on growth and production of durum wheat, Agron. Sustain. Dev. 25 (1) (2005) 129–135.
[19] S. George, K. Jayachandran, Analysis of rhamnolipid biosurfactants produced through submerged fermentation using orange fruit peelings as sole carbon source, Appl. Biochem. Biotechnol. 158 (3) (2009) 694–705.
[20] D. Hancock, et al., The global epidemiology of multiresistant Salmonella enterica serovar Typhimurium DT104, in: Emerging Diseases of Animals, Wiley Online Library, 2000, pp. 217–243.
[21] P. Singh, S.S. Cameotra, Potential applications of microbial surfactants in biomedical sciences, Trends Biotechnol. 22 (3) (2004) 142–146.
[22] K. Sambanthamoorthy, et al., Antimicrobial and antibiofilm potential of biosurfactants isolated from lactobacilli against multi-drug-resistant pathogens, BMC Microbiol. 14 (1) (2014) 1–9.
[23] A. Abalos, et al., Physicochemical and antimicrobial properties of new rhamnolipids produced by Pseudomonas aeruginosa AT10 from soybean oil refinery wastes, Langmuir 17 (5) (2001) 1367–1371.
[24] V.C. Akubude, et al., Application of biosurfactant in food industry, in: Green Sustainable Process for Chemical and Environmental Engineering and Science, Elsevier, 2021, pp. 109–125.
[25] L. Rodrigues, et al., Biosurfactants: potential applications in medicine, J. Antimicrob. Chemother. 57 (4) (2006) 609–618.
[26] M. Müller, Biosurfactants—Trends and Perspectives Networking, in: Biotechnology Creating Sustainability, 2021.

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APPLICATIONS OF NEXT GENERATION BIOSURFACTANTS IN THE FOOD SECTOR

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Explores the mechanisms, applications, and methodologies of biosurfactant-induced antimicrobial effects against food pathogens

Applications of Next Generation Biosurfactants in the Food Sector provides detailed information on the sustainable approach to the utilization of biosurfactants as the next-generation green biotechnology to mitigate various problems encountered in the food industry. These biosurfactants help reduce risks such as food spoilage, food poisoning, and postharvest losses of fruits, vegetable, grains, tubers, cereals, and pulses.

This resource will benefit academics, R&D professionals, and postgraduate students in food science and related fields as they explore recent trends in the application of these green biosurfactants and the many uses they can provide.

Key Features

- Provides examples of mathematical modeling, metabolomics, bioinformatics, metabolic engineering, systems biology, and computer technology for solving real-life food challenges using biosurfactants
- Presents biosurfactants as an innovative, green, biotechnological solution to resolving food safety issues and improving human health
- Includes applications of biosurfactants for the prevention of mycotoxins and as a biodegradable material with a wide variety of uses

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